



The Hidden Restructuring:

A Historical Review of Automation and Sub-Occupational Labour Market Dynamics

A thesis submitted in fulfilment of the requirements for the degree of

Doctor of Philosophy

Thomas Bierly

School of Economics, Finance, and Marketing

College of Business and Law

Royal Melbourne Institute of Technology

Doctoral Thesis

Melbourne, Australia

Wednesday 24th June, 2026

Abstract

This thesis offers a historical review of how automation reshaped the Australian labour market across 2012–2019, the period spanning the diffusion of cloud computing, mobile platforms, and pre-generative machine learning. Drawing on 9.5 million job advertisements covering 250 occupations, it addresses a fundamental paradox in automation research: why did dire predictions of technological unemployment contrast so sharply with modest empirical findings? The analysis documents evidence consistent with automation’s primary labour market impact operating through *within-occupation task restructuring* rather than occupation elimination—a process invisible to conventional measurement frameworks. Read in historical perspective, the period offers a controlled window on automation prior to the disruptions of COVID-19 and large language models, isolating the labour-market signature of an earlier wave of digital technologies.

Occupation-level regression analysis reveals that two prominent frameworks—Routine-Cognitive-Manual task categories and Frey-Osborne automation probabilities—explain only 3% of variation in labour demand, with theoretically distinct risk categories proving statistically indistinguishable. Variance decomposition confirms that over 94% of variation operates *within* occupations rather than between them, imposing a ceiling on what any occupation-level framework can detect.

Novel application of HDBSCAN clustering to job advertisement skill requirements reveals the restructuring that occupation-level measures miss. Temporal tracking documents median annual cluster churn exceeding 100%, with cluster-level volatility an order of magnitude higher than occupation-level measures suggest. Technology succession drives this turnover: clusters organised

around legacy platforms are systematically replaced by clusters requiring emerging technologies. The demand-side consequences are heterogeneous—task recombination dominates among slowly-evolving occupations while task unbundling characterises rapidly-transforming ones—revealing that automation does not uniformly simplify or complicate work but reshapes it in occupation-specific ways. A skill proximity network connecting clusters across occupations demonstrates that the labour market is broadly navigable, with most at-risk clusters accessible to stable alternatives through modest upskilling, though navigability is unevenly distributed.

Four contributions emerge: (1) empirical quantification of aggregation bias across 250 occupations; (2) formal theory of endogenous job type clustering validated through five testable propositions; (3) methodological demonstration of HDBSCAN for dynamic sub-occupational classification with temporal tracking; (4) construction of skill proximity networks mapping feasible career transition pathways.

Policy implications challenge frameworks focusing on displaced worker retraining. If automation primarily restructures tasks within occupations, policies must support *employed* workers whose job content continuously evolves. Cluster-based career navigation may prove more efficient than occupation-level retraining programmes.

Keywords: automation, labour demand, job advertisements, machine learning, HDBSCAN, clustering, within-occupation restructuring, task analysis, skill requirements, skill proximity networks, career transitions

Acknowledgements

This thesis represents the culmination of several years of study, reflection, and perseverance — a journey that would not have been possible without the guidance, generosity, and encouragement of many people.

I am grateful to the staff and colleagues at RMIT, who have contributed to my growth as both a researcher and educator. The collegiality and shared passion for discovery within this academic community have made this experience immensely rewarding.

To my family, whose love and faith in me never wavered — you cultivated a love of learning at an early age, nurturing curiosity and an enduring appreciation for ideas. Thank you for all the support, the tutoring, and the tuition over the years; your generosity and encouragement made this possible.

To Albert, who learned more about automation than any sane person should have to, and still chose to stay. Thank you for weathering my stressed-out moods with grace, for never judging my questionable writing-deadline fashion choices, and for keeping me fed and caffeinated. This would have been infinitely harder without you.

Declaration

I certify that except where due acknowledgement has been made, the work is that of the author alone; that work has not been submitted previously, in whole or in part, to qualify for any other academic award; the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program; and any editorial work, paid or unpaid, carried out by a third party is acknowledged.

Thomas Bierly

Wednesday 24th June, 2026

Contents

List of Abbreviations	7
1 Introduction	11
1.1 The Automation Paradox	11
1.2 The Problem: Aggregation Bias in Automation Measurement	12
1.3 Research Questions	13
1.4 Significance of the Study	13
1.5 Scope and Delimitations	14
1.6 Contributions to Knowledge	15
1.7 Thesis Organisation	17
2 Literature Review	21
2.1 Introduction	21
2.1.1 Chapter Overview	23
2.1.2 Scope and Limitations	23
2.1.3 Review Methodology	23
2.1.4 Australian Context	24
2.1.5 Competing Definitions of Automation	24
2.2 How Jobs Are Designed and Restructured	26
2.2.1 The Job Characteristics Tradition	26
2.2.2 Job Crafting: Workers as Active Agents	27
2.2.3 Job Demands-Resources Theory	28
2.2.4 Technology and Work Design	28

2.2.5	Strategic Human Resource Management	29
2.2.6	Implications for Measurement	29
2.3	How Automation Affects Tasks	30
2.3.1	Classical Foundations and Compensation Theory	30
2.3.2	The Task-Based Framework	31
2.3.3	Displacement and Reinstatement	32
2.3.4	New Task Creation	33
2.3.5	Skill Complementarities	34
2.3.6	Measurement Controversies	35
2.3.7	Generative AI: A New Automation Paradigm?	36
2.3.8	Implications for Measurement	37
2.4	The Capability–Deployment Gap	38
2.4.1	Capability Assessments Were Not Designed to Predict Labour Markets	38
2.4.2	Capability Substantially Exceeds Deployment	38
2.5	Why Occupational Classifications Fail	39
2.5.1	Occupations as Contested Jurisdictions	40
2.5.2	Social Closure and Within-Occupation Heterogeneity	40
2.5.3	Professional Discretion and Task Variation	41
2.5.4	Labour Process and Skill Construction	41
2.5.5	Job Quality and Workplace Fissuring	42
2.5.6	Classification Lag and Boundary Permeability	42
2.5.7	Recent Evidence: Platform Work and Boundary Erosion	43
2.5.8	Implications for Measurement	44
2.6	What Computational Research Reveals About Labour Market Structure . .	44
2.6.1	Within-Occupation Skill Variation	45
2.6.2	Growing Internal Divergence	46
2.6.3	Skill Networks and Polarisation	47

2.6.4	Recession-Driven Skill Upgrading	47
2.6.5	Automation Exposure Variation	48
2.6.6	Labour Market Organisation Beyond Occupations	49
2.6.7	Foundation Models and Large Language Models	49
2.6.8	Implications for Measurement	50
2.7	The Measurement-Theory Disconnect	52
2.7.1	Convergent Predictions	52
2.7.2	Synthesising a Working Definition	53
2.7.3	Why Occupations Remain Dominant Despite Theory	54
2.7.4	Divergent Measurement	55
2.7.5	The Aggregation Bias Mechanism	56
2.7.6	Alternative Perspectives and Boundary Conditions	57
2.7.7	Requirements for Resolution	58
2.8	Research Gap and Thesis Contribution	58
2.8.1	The Gap	58
2.8.2	Thesis Contribution	59
2.8.3	Contribution to Literature	60
2.9	Conclusion	61
3	Theoretical Framework	63
3.1	Introduction	63
3.2	The Redesign Mechanism	65
3.3	The Model	66
3.3.1	Workers and Skills	68
3.3.2	Firms and Job Design	68
3.3.3	Production	69
3.3.4	Coordination Costs	69
3.3.5	Matching: Directed Search	70
3.3.6	Equilibrium	71

3.4	Static Predictions: Discrete Clustering and Market Thickness	72
3.4.1	Endogenous Clustering (Proposition 1)	72
3.4.2	Market Thickness (Proposition 2)	73
3.5	Automation Effects and Aggregation Bias (Proposition 3)	74
3.6	Dynamic Predictions: Cluster Births, Deaths, and Churn (Propositions 4 and 5)	76
3.7	Contribution and Scope	77
3.7.1	Scope of the Framework	78
3.8	Bridge to the Empirical Chapters	79
3.9	Conclusion	80
4	Data and Methodology	83
4.1	Introduction	83
4.2	Australian Labour Market Context (2012–2019)	84
4.2.1	Institutional Framework	84
4.2.2	Economic Conditions (2012–2019)	85
4.2.3	Geographic Concentration	86
4.2.4	Summary: Contextual Implications	86
4.3	Conceptual Foundation	87
4.3.1	Why Conventional Automation Proxies Are Inadequate	87
4.4	Data Source	88
4.4.1	Data Collection and Processing	89
4.4.2	Validation Evidence	89
4.4.3	Coverage Limitations	89
4.5	Temporal Scope	90
4.5.1	Locating the Period in Automation’s Long Arc	91
4.6	Descriptive Statistics	92
4.6.1	Skills Demand Analysis	93
4.6.2	Temporal Patterns in Job Postings	93

4.6.3	Data Quality and Completeness	94
4.7	Sample Construction	95
4.7.1	Full Sample: Occupation-Level Analysis	96
4.7.2	Cluster-Level Analysis Sample	97
4.8	Occupation-Level Analysis Methods	98
4.8.1	Task-Based Measures	98
4.8.2	Automation Probability Scores	98
4.8.3	Empirical Specification	99
4.8.4	Identification and Limitations	100
4.9	Cluster-Level Analysis Methods	100
4.9.1	Methodological Rationale	100
4.9.2	Algorithm Selection: HDBSCAN	101
4.9.3	Distance Metric Selection	102
4.9.4	Parameter Selection	103
4.9.5	Temporal Tracking: Matching Clusters Across Years	105
4.9.6	Validation Strategy	106
4.10	Ethical Considerations	107
4.11	Limitations	107
4.11.1	Data Limitations	107
4.11.2	Identification Limitations	108
4.11.3	Scope Boundaries	108
4.12	Conclusion	108
5	Occupation-Level Evidence	111
5.1	Introduction	111
5.1.1	Research Questions	112
5.1.2	Chapter Organisation	112
5.2	Theoretical Predictions	113
5.2.1	Predictions from Proposition 3	113

5.2.2	What This Analysis Can and Cannot Establish	114
5.3	Methodological Context: How the Original Studies Approached Identification	114
5.3.1	The RCM Framework: Autor, Levy, and Murnane (2003)	115
5.3.2	The Frey-Osborne Framework: Expert Classification Without Treatment Timing	115
5.3.3	The Gold Standard: Acemoglu and Restrepo (2020)	116
5.3.4	Implications for This Analysis	116
5.4	Empirical Strategy	117
5.4.1	Estimation Approach	117
5.4.2	Specification Strategy	117
5.4.3	Skill-Level Heterogeneity	118
5.5	Methods	119
5.5.1	Sample Construction	119
5.5.2	RCM Classification	120
5.5.3	Frey-Osborne Risk Classification	122
5.5.4	Skill-Level Sample	123
5.5.5	Regression Specifications	125
5.5.6	Wald Tests for Coefficient Distinguishability	127
5.5.7	Skill-Level Interaction Models	128
5.5.8	Variance Decomposition	128
5.6	Results	129
5.6.1	RCM Framework Results	129
5.6.2	Frey-Osborne Framework Results	133
5.6.3	Variance Decomposition	138
5.7	Discussion	140
5.7.1	Australian Institutional Context	141
5.7.2	Comparison to International Evidence	141
5.7.3	Aggregation Bias Mechanism	142

5.7.4	Limitations	142
5.7.5	Implications for Frey-Osborne Predictions	143
5.7.6	Skill Composition	144
5.8	Conclusion	144
6	Within-Occupation Restructuring	147
6.1	Introduction	147
6.2	Methods	148
6.2.1	Data and Sample	148
6.2.2	Empirical Strategy	148
6.3	Results: Cluster Discovery	149
6.3.1	Number of Clusters Identified	149
6.3.2	Cluster Validation	150
6.3.3	Cluster Characterization: Examples	150
6.4	Results: Testing Proposition 1 (Endogenous Clustering)	151
6.4.1	Theoretical Prediction	151
6.4.2	Empirical Approach	152
6.4.3	Results	153
6.5	Results: Testing Proposition 2 (Market Thickness)	155
6.5.1	Theoretical Prediction	155
6.5.2	Empirical Approach	155
6.5.3	Results	156
6.5.4	Australian Institutional Context	158
6.6	Results: Testing Proposition 3 (Within-Occupation Restructuring)	159
6.6.1	Theoretical Prediction	159
6.6.2	Empirical Approach	160
6.6.3	Results	160
6.7	Discussion	165
6.7.1	The Aggregation Bias Mechanism	165

6.7.2	Australian Institutional Context	166
6.7.3	Comparison to International Evidence	166
6.7.4	Limitations	167
6.8	Conclusion	168
7	Temporal Dynamics	169
7.1	Introduction	169
7.1.1	Research Questions	170
7.1.2	Chapter Roadmap	170
7.2	Methods	171
7.2.1	Data and Sample	171
7.2.2	Empirical Strategy	171
7.3	Results: Testing Proposition 4 (Temporal Equilibrium Path)	174
7.3.1	Theoretical Prediction	174
7.3.2	Results	174
7.3.3	Interpretation	177
7.4	Results: Testing Proposition 5 (Churn-Automation Speed)	178
7.4.1	Theoretical Prediction	178
7.4.2	Empirical Approach	178
7.4.3	Results	179
7.5	Characterising Newly-Emerged Clusters	182
7.5.1	Empirical Strategy	182
7.5.2	Results: Heterogeneous Skill Requirements	183
7.6	Cluster-Level Trajectory Analysis	186
7.6.1	Backward Lineage Tracing	186
7.6.2	Age Distribution and Survival Dynamics	187
7.6.3	Predictors of Persistence	188
7.6.4	Trajectory Classification	188
7.6.5	Within-Occupation Heterogeneity	190

7.7	Discussion	190
7.7.1	Interpreting the Temporal Findings	190
7.7.2	Australian Institutional Context	192
7.7.3	Limitations	192
7.7.4	Creative Destruction Benchmarks	194
7.8	Conclusion	194
8	Skill Proximity Networks	197
8.1	Introduction	197
8.1.1	Chapter Roadmap	198
8.2	Methods	199
8.2.1	Network Construction	199
8.2.2	Edge Classification	200
8.2.3	Cluster Trajectory Status Assignment	201
8.2.4	Skill Gap Computation	201
8.2.5	Skill Specificity Classification	202
8.2.6	Skill Bundle Identification	202
8.2.7	Hub Skill Scoring	203
8.3	Network Characterisation	204
8.3.1	Global Network Properties	204
8.3.2	Network Connectivity by Cluster Trajectory Status	205
8.4	Transition Pathway Analysis	207
8.4.1	From At-Risk to Stable Clusters	208
8.4.2	Bridge Skills: Generic Noise versus Specialist Barriers	210
8.4.3	Skill Bundle Co-occurrence Structure	211
8.4.4	Skill Complementarity and Hub Scores	212
8.4.5	Case Study Pathways	214
8.5	Discussion	216
8.5.1	Navigability and Its Limits	216

8.5.2	From Occupation-Level to Cluster-Level Workforce Development	216
8.5.3	Limitations	217
8.6	Conclusion	218
9	Discussion	219
9.1	Introduction	219
9.2	Integration of Findings Across Analytical Levels	219
9.2.1	The Occupation-Level Paradox	220
9.2.2	Within-Occupation Restructuring	220
9.2.3	Cross-Level Synthesis	221
9.3	Theoretical Contributions	222
9.3.1	Restructuring Theory of Automation Impacts	222
9.3.2	Demonstration of Multi-Level Measurement Value	225
9.3.3	Validation of Job Advertisement Data for Labour Market Analysis	226
9.3.4	Demand-Side Consequences of Within-Occupation Restructuring	228
9.4	Methodological Contributions and Reflections	229
9.4.1	Application of HDBSCAN to Within-Occupation Restructuring .	229
9.4.2	Multi-Level Triangulation: Strengths and Limitations	231
9.4.3	Reflections on the Australian Context	231
9.5	Critical Examination: Alternative Explanations and Boundary Conditions	233
9.5.1	Causal Identification Limitations	233
9.5.2	Generalizability and Boundary Conditions	234
9.5.3	Measurement Artifacts and Methodological Limitations	236
9.6	Implications for Policy and Practice	237
9.6.1	Rethinking Workforce Development Policy	237
9.6.2	Improving Labour Market Information Systems	240
9.6.3	Implications for Organisational Practice	241
9.7	Conclusion	242

10 Conclusion	245
10.1 Research Overview	245
10.2 Contributions to Knowledge	246
10.3 Implications	248
10.4 Limitations	249
10.5 Future Research Directions	249
10.5.1 Post-Pandemic and Generative AI Extensions	250
10.5.2 Cross-National Comparative Research	250
10.5.3 Institutional, Distributional, and Resistance Research	250
10.5.4 Worker-Level Outcome Studies	251
10.5.5 Automation Adoption and Cluster Dynamics	252
10.5.6 Cluster Stability and Predictability	252
10.6 Final Reflections	253
Bibliography	255
Appendix A Supplementary Literature Review Material	289
A.1 Classical Economics and Compensation Theory	289
A.1.1 Classical Debates on Technological Unemployment	289
A.1.2 Compensation Theory: Detailed Mechanisms	292
A.2 Empirical Evidence on Automation 1990–2020	295
A.2.1 Measuring Technological Change: Methodological Challenges	295
A.2.2 Firm-Level Studies	297
A.2.3 Industry-Level Studies	298
A.2.4 Macro-Level Studies	299
A.2.5 Synthesis of 1990–2020 Evidence	300
A.3 Theoretical Frameworks and Conceptual Debates	302
A.3.1 The Fourth Industrial Revolution	302
A.3.2 Technological Determinism	304

A.3.3	The Lump of Labour Fallacy	305
A.3.4	The Resurgence of Contemporary Debates	308
Appendix B	Historical Perspectives on Occupational Classifications	311
Appendix C	Extended Theoretical Framework	317
C.1	Introduction	317
C.1.1	Main Results	318
C.1.2	Notation	319
C.2	Model Environment	319
C.2.1	Workers	319
C.2.2	Firms and Job Design	321
C.2.3	Production Function	322
C.2.4	Coordination Costs	324
C.2.5	Wages and Firm Profits	327
C.2.6	Equilibrium Definition	327
C.2.7	Parametric Restrictions and Regularity Conditions	331
C.3	Theoretical Results	335
C.3.1	Proposition 1: Endogenous Clustering	339
C.3.2	Proposition 2: Market Thickness and Cluster Count	342
C.3.3	Proposition 3: Automation and Cluster Dynamics	345
C.3.4	Microfoundation of Discreteness: Posting Costs	346
C.3.5	Summary of Theoretical Results	354
C.4	Extension: Automation and Task Displacement	354
C.4.1	Automation Technology	354
C.4.2	Comparative Statics	356
C.4.3	Reconciling with Task-Based Framework	358
C.4.4	Empirical Identification of Skill Partitions	358
C.4.5	Numerical Calibration and Quantitative Predictions	361

C.4.6	Extension: Technology-Indexed Automation	366
C.5	Dynamic Extension: Temporal Cluster Evolution	368
C.5.1	Time-Varying Equilibrium	369
C.5.2	Cluster Viability and Birth/Death Conditions	370
C.5.3	Churn Rate Predictions	371
C.5.4	Summary: From Static to Dynamic	372
C.6	Empirical Implications	373
C.6.1	Testable Predictions	373
C.6.2	Identification and Structural Estimation	375
C.6.3	Alternative Mechanisms	378
C.7	Discussion and Limitations	378
C.7.1	Relationship to Existing Theory	378
C.7.2	Model Limitations	380
C.7.3	Perspectives from Adjacent Traditions	381
C.7.4	Welfare and Efficiency	383
C.8	Conclusion	384
Appendix D Theory Chapter: Technical Proofs		387
D.1	Kakutani Existence Proof: Directed Search Equilibrium	387
D.1.1	Setup and Correspondence Definition	387
D.1.2	Condition 1: Non-empty, Compact, Convex Domain	388
D.1.3	Condition 2: Non-empty, Convex Values	388
D.1.4	Condition 3: Upper Hemicontinuity	389
D.1.5	Conclusion	391
D.2	Posting Game Nash Equilibrium Existence	391
D.2.1	Game Structure	391
D.2.2	Kakutani Fixed Point Theorem Application	391
D.2.3	Economic Interpretation	393
D.3	Proofs of Theoretical Results	394

D.3.1	Proof of Proposition 1: Endogenous Clustering	394
D.3.2	Proof of Proposition 2: Market Thickness and Cluster Count . . .	398
D.3.3	Proof of Proposition 3: Automation Effects	401
D.3.4	Proof of Lemmas 1-2: Automation Extension	406
D.3.5	Proof of Proposition 4: Temporal Equilibrium Path	408
D.3.6	Proof of Lemma 3: Viability Monotonicity	409
D.3.7	Proof of Proposition 5: Churn-Automation Speed Relationship . .	410
D.4	Numerical Illustration: Calibration and Comparative Statics	411
D.4.1	Calibration Strategy	411
D.4.2	Parameter Values	412
D.4.3	Model Predictions	412
D.4.4	Graphical Illustrations	413
D.4.5	Robustness and Sensitivity	413
D.4.6	Connecting Calibration to Empirical Chapters	414
Appendix E BGT Data Descriptions		415
Appendix F Data Codebook		419
F.1	Primary Data Source	419
F.2	Key Fields Used in Analysis	420
F.3	Skill Classification Taxonomy	420
F.4	HDBSCAN Saved Models	421
F.5	Derived Result Files	421
F.5.1	Chapter 5: Occupation-Level Regression	421
F.5.2	Chapter 6: Hypothesis Test Results	422
F.5.3	Chapter 7: Evolutionary Metrics	422
F.5.4	Chapter 8: Skill Proximity Network	423
F.6	Key Parameters and Thresholds	423
F.7	ANZSCO Code Structure	423

Appendix G Replication Code and Methodology	425
G.1 Software and Data Requirements	425
G.1.1 Software Environment	425
G.1.2 Data Requirements	426
G.1.3 Data Files	426
G.1.4 Pre-computed Models	427
G.2 Chapter 5: RCM Task Analysis	427
G.3 Chapter 6: HDBSCAN Clustering Methodology	427
G.3.1 DBCV Parameter Optimisation	428
G.3.2 Full-Scale Validation (250 Occupations)	428
G.4 Chapter 7: Temporal Dynamics Methodology	429
G.4.1 Centroid Drift	429
G.4.2 Greedy Centroid Matching	429
G.4.3 Evolutionary Classification	431
G.5 Chapter 8: Skill Proximity Network Methodology	431
G.5.1 Network Construction	431
G.5.2 Transition Pathway Skill Gaps	431
G.6 Computing Considerations	432
G.6.1 Memory Requirements	432
G.6.2 Runtime Estimates	432
G.7 Citation	433

List of Figures

3.1	Pipeline Redesign and Bundle Reconfiguration: Accountant (General) . . .	67
4.1	Distribution of Skills per Job Advertisement	93
4.2	Top 20 Most Frequently Demanded Skills	95
4.3	Job Advertisement Postings Over Time	96
5.1	Directed Acyclic Graph of Variable Relationships	118
5.2	Mean Advertisement Counts by RCM Category	122
5.3	Mean Advertisement Counts by Frey-Osborne Risk Level	124
5.4	Descriptive Trends by RCM Category	130
5.5	RCM Skill-Level Interaction Coefficients	134
5.6	FO Skill-Level Coefficients (Including Baseline)	138
6.1	Cluster Count Evolution by Occupation	159
6.2	Variance Decomposition: Within vs. Between Occupation	165
7.1	Evolutionary Classification of 250 Occupations by Centroid Drift	180
7.2	Skill Requirements by Cluster Trajectory Status	183
7.3	Occupation-Level New-Cluster Skill Effects vs. Centroid Drift	185
7.4	Age Distribution of 5,118 Clusters Observed in 2019	187
7.5	Cohort Survival Curves	188
7.6	Cluster Size by Trajectory Status	189
8.1	Ego networks: within- and cross-occupation transition landscape	206
8.2	Network properties by cluster trajectory status	206

8.3	Degree distribution by cluster trajectory status	207
8.4	Skill gap distribution for career transitions	209
8.5	Bridge skills: generic versus specialist	210
8.6	Specialist skill bundle co-occurrence	212
8.7	Hub skill scores	213
8.8	Case study transition pathways	215
A.1	Circular Flow Model	306
C.1	Match Quality Function	323
C.2	Coordination Costs	327
C.3	Directed Search Equilibrium	332
C.4	Firm Sorting into Discrete Job Designs	342
C.5	Market Thickness and Cluster Count	344
C.6	Posting Cost Game Equilibrium	353
C.7	Aggregation Bias: Conceptual Illustration	359
C.8	Numerical Illustration: Cluster Formation and Market Thickness	364
C.9	Automation Comparative Statics: Cluster-Level Heterogeneity	365

List of Tables

1.1	Multi-Level Research Design	18
2.1	Four Scholarly Traditions and Their Contributions	53
3.1	Proposition to empirical chapter mapping	79
4.1	Institutional Comparison Across Labour Market Regimes	85
4.2	Dataset Summary Statistics	92
4.3	Top 20 Most Frequently Demanded Skills	94
4.4	Sample Size by Year	97
4.5	Distance Metric Comparison: Cosine vs Jaccard (20 Occupations)	103
4.6	HDBSCAN Parameter Configuration	104
4.7	Parameter Sensitivity Analysis: Cluster Counts and Noise Rates	105
5.1	Model Specification Summary	119
5.2	Number of Jobs by RCM Category	120
5.3	Aggregated Dataset Structure	121
5.4	RCM Frequency Statistics	121
5.5	RCM Descriptive Statistics	121
5.6	Job Advertisement Counts by Frey-Osborne Automation Risk Level (BGT Dataset, 2012-2019)	123
5.7	Frey-Osborne Cell Frequency Statistics	123
5.8	Frey-Osborne Descriptive Statistics	123
5.9	Advertisement Counts by Skill Level	124
5.10	OLS vs PPML Robustness for RCM Model 2	127

5.11	Model Comparison: Testing RCM Framework Across Specifications . . .	131
5.12	Wald Tests for RCM Coefficient Equality (Model 2)	132
5.13	Interaction Effects of RCM Model, Year, and Skill Level	133
5.14	Model Comparison: Testing FO Framework Across Specifications	135
5.15	Wald Tests for FO Coefficient Equality (Model 2)	135
5.16	Interaction Effects of FO Model, Year, and Skill Level	137
5.17	Variance Decomposition: Between- vs. Within-Occupation Variation . . .	139
5.18	Robustness to Minimum Cell Size	140
6.1	Cluster Discovery Summary by Occupation	149
6.2	Illustrative Accountant (General) Clusters (ANZSCO 221111, 2019) . . .	151
6.3	Firm Size and Cluster Breadth Across 239 Occupations	154
6.4	Accountant Clusters by Geographic Area	156
6.5	Market Thickness and Cluster Count	157
6.6	Cross-Occupational Market Thickness Regression	158
6.7	Employment Growth Volatility: Occupation vs. Cluster Level	160
6.8	Technology Succession: Software Skills in Dead vs. Born Clusters	162
7.1	Sample Summary for Temporal Analysis	172
7.2	Creative Destruction Summary (2012–2019)	175
7.3	Skill Drift Among Continuing Clusters	176
7.4	Automation Level vs. Speed: Three Illustrative Occupations	179
7.5	Annual Churn Rates: Illustrative Occupations (2012–2019)	181
7.6	Cluster Trajectory Classification	189
7.7	Temporal Subsample Robustness	191
7.8	Cosine threshold sensitivity	193
8.1	Network Sensitivity to Similarity Threshold	200
8.2	Skill Proximity Network Properties (2019, $\tau = 0.6$)	204

A.1	Tunzelmann's chronology of industrial eras	303
C.1	Notation Summary	320
C.2	Calibrated Parameters for Numerical Illustration	362
C.3	Static vs. Dynamic Theoretical Predictions	372
C.4	Mapping Theory to Empirical Tests	374
D.1	<i>Calibrated Model Parameters</i>	412
E.1	BGT Data Descriptions	416
F.1	Key data fields by chapter	420
F.2	HDBSCAN model file contents	421
F.3	Chapter 7 summary file columns	422
F.4	Key analytical parameters	423

List of Abbreviations

ABS	Australian Bureau of Statistics
ACT	Australian Capital Territory
AI	Artificial Intelligence
ALMP	Active Labour Market Policy/Policies
ANZSCO	Australian and New Zealand Standard Classification of Occupations
ANZSIC	Australian and New Zealand Standard Industrial Classification
APA	American Psychological Association
AQF	Australian Qualifications Framework
AUS	Australia
BGT	Burning Glass Technologies
COVID-19	Coronavirus Disease 2019
DAG	Directed Acyclic Graph
DBCV	Density-Based Cluster Validation

DBSCAN	Density-Based Spatial Clustering of Applications with Noise
ERP	Enterprise Resource Planning
FE	Fixed Effects
FO	Frey-Osborne
GAAP	Generally Accepted Accounting Principles
GDP	Gross Domestic Product
GMM	Gaussian Mixture Model
GPU	Graphics Processing Unit
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
HR	Human Resources
ICT	Information and Communication Technology
Ind.	Industry
ISCO	International Standard Classification of Occupations
IT	Information Technology
LDA	Latent Dirichlet Allocation
MAE	Mean Absolute Error
MBA	Master of Business Administration
ML	Machine Learning
MST	Minimum Spanning Tree
MYOB	Mind Your Own Business (accounting software)

NA	Not Applicable / Not Available
NLP	Natural Language Processing
NRC	Non-Routine Cognitive
NRM	Non-Routine Manual
NSW	New South Wales
Occ.	Occupation
OECD	Organisation for Economic Co-operation and Development
OLS	Ordinary Least Squares
O*NET	Occupational Information Network
R&D	Research and Development
R ²	R-squared (coefficient of determination)
RAM	Random Access Memory
RC	Routine Cognitive
RCM	Routine-Cognitive-Manual
RMIT	Royal Melbourne Institute of Technology
RM	Routine Manual
SA4	Statistical Area Level 4
SAP	Systems Applications and Products (enterprise software)
SOC	Standard Occupational Classification
SQL	Structured Query Language
STM	Structural Topic Model
t-SNE	t-Distributed Stochastic Neighbour Embedding

LIST OF TABLES

UK	United Kingdom
US / USA	United States / United States of America
VRAM	Video Random Access Memory
WA	Western Australia

Chapter 1

Introduction

1.1 The Automation Paradox

Public discourse about automation is dominated by dire predictions. Nearly two-thirds (65%) of the public believe that within the next 50 years, robots will do most work currently done by humans (Aaron Smith, 2016). Even experts share these concerns: AI researchers estimate a 50% chance that artificial intelligence will outperform humans at all tasks within 45 years (Grace et al., 2018), while half of technology experts surveyed believe robots will displace most human labour (Aaron Smith & Anderson, 2014). Influential studies predict 47% of US jobs (Frey & Osborne, 2017) and 40–44% of Australian jobs (Durrant-Whyte et al., 2015; Edmonds & Bradley, 2015) face high risk of full automation within 10–20 years.

Yet empirical evidence tells a strikingly different story. Macroeconomic studies consistently find modest or negligible automation effects on aggregate employment (Arntz et al., 2016; Nedelkoska & Quintini, 2018). Unemployment has remained relatively stable despite decades of technological advancement. Workers and executives report extensive workplace automation, but labour markets show no signs of mass technological unemployment. This disconnect presents a fundamental paradox: *Why do dire predictions of job destruction contrast so sharply with modest empirical findings?*

This thesis investigates whether the paradox resolves once measurement shifts to finer granularity. Standard measurement frameworks may operate at inappropriate levels of aggregation: Occupation-level classifications—the dominant approach in automation research—

assume homogeneity within occupational categories, potentially masking substantial restructuring occurring at finer levels of analysis. Through analysis of 9.5 million Australian job advertisements (2012–2019),¹ this research examines whether the empirical patterns are consistent with automation’s primary impact operating through *within-occupation task restructuring* rather than wholesale occupation elimination—a process that would remain invisible to conventional measurement frameworks if confirmed.

1.2 The Problem: Aggregation Bias in Automation Measurement

The disconnect between automation predictions and empirical findings points to a measurement problem. Standard occupation-level classifications assume homogeneity within occupational categories—treating all “Accountants” or “Human Resources Managers” as interchangeable. This aggregation masks substantial heterogeneity: within any occupation, some job types may emphasise routine tasks vulnerable to automation (e.g., compliance data entry), while others may emphasise non-routine tasks complementary to automation (e.g., strategic advisory). If both contract and expand simultaneously within the same occupational boundary, aggregate measures would detect only modest net changes while missing extensive restructuring occurring beneath the surface.

This measurement problem has profound consequences. If automation primarily restructures tasks *within* occupations rather than eliminating occupations wholesale, then current policy frameworks—focused almost exclusively on retraining displaced workers and predicting job losses—address the wrong problem. Employed workers whose job content continuously evolves require different policy support than displaced workers seeking reemployment. Without appropriate measurement frameworks operating at the right level of granularity, policymakers cannot design effective workforce development interventions.

This thesis addresses this measurement and policy problem by developing cluster-level methods that detect within-occupation restructuring invisible to occupation-level frameworks. The approach combines occupation-level regression with cluster-level machine learning applied to 9.5 million Australian job advertisements (2012–2019). Detailed methodology appears in Chapters 4 and 6.

¹The 9.5 million advertisements are used at different aggregation levels across analyses: aggregated to 170,491 occupation-year-state-industry cells for regression analysis (Chapter 5); 3,055,469 individual advertisements from 250 occupations for clustering analysis (Chapters 6 and 7).

1.3 Research Questions

This thesis addresses six interrelated research questions:

1. **Measurement:** Do occupation-level classifications adequately capture automation's labour market impacts, or do they mask restructuring occurring at finer levels of analysis?
2. **Restructuring:** How does work reorganise *within* existing occupations during the period of automation diffusion? Do task bundles reorganise while occupational categories persist?
3. **Temporal Dynamics:** How does within-occupation restructuring unfold over time? What patterns of cluster birth, death, and evolution characterise within-occupation restructuring?
4. **Characterising Restructuring:** What distinguishes newly-emerged job types from established ones? Do skill requirements vary systematically across cluster trajectory types, and does the pace of occupational change moderate these differences?
5. **Navigability:** If clusters dissolve, where can affected workers go? Is the labour market navigable in skill space, and is navigability evenly distributed across cluster types?
6. **Policy:** If empirical patterns are consistent with automation primarily restructuring tasks within occupations rather than eliminating occupations wholesale, what are the implications for labour market policy?

1.4 Significance of the Study

This research matters for three interconnected reasons:

Theoretical significance: Resolving the measurement paradox advances fundamental understanding of how technological change affects labour markets. If automation primarily restructures tasks within occupations rather than eliminating occupations wholesale, then existing theoretical frameworks (routine-biased technological change, occupational licensing, skill-biased technological change) operate at inappropriate levels of aggregation. The cluster-level perspective developed here provides new theoretical purchase on automation's mechanisms.

Methodological significance: Demonstrating that job advertisement data enables fine-grained measurement of within-occupation restructuring opens new research possibilities.

While scholars have long recognised occupational heterogeneity as problematic (Martin-Caughey, 2021), practical measurement approaches have remained elusive. This thesis provides replicable methods for cluster-level analysis applicable beyond automation research to skill demand, labour market segmentation, and workforce development questions. The multi-level analytical approach—combining occupation-level regression with cluster-level machine learning—reveals not just *what* changes occur but *how* measurement level shapes conclusions about automation’s labour market effects.

Policy significance: If employed workers face continuous job content evolution rather than discrete displacement events, then current workforce development policies may address the less prevalent mechanism. Retraining programs designed for displaced workers do not support employed workers whose skill requirements shift incrementally. The finding that extensive cluster-level restructuring occurs even as occupational employment remains stable suggests that policy frameworks built around displacement (observable through unemployment statistics) may overlook restructuring dynamics that aggregate measures cannot detect. Understanding heterogeneous restructuring patterns enables policy design targeting the empirical challenges workers face in automation-affected labour markets.

1.5 Scope and Delimitations

This thesis examines automation’s labour market impacts within specific boundaries that define and constrain its findings:

Geographic scope: Analysis focuses exclusively on Australian labour markets (2012–2019). Australia provides an ideal context as a developed service-oriented economy with liberal market institutions comparable to the US and UK, while exhibiting distinct industrial relations frameworks and sectoral composition. Findings may not generalise to coordinated market economies (e.g., Germany, Japan) or developing economies with different automation penetration.

Temporal scope: The 2012–2019 period captures the initial diffusion of machine learning, cloud computing, and mobile technologies preceding the COVID-19 pandemic and generative AI. This thesis is therefore best read as a *historical review* of an earlier wave of automation—one defined by task codification and workflow digitisation rather than large language models or autonomous systems. That historical positioning is intentional: bracketing the dataset before the pandemic and the 2022–2024 generative-AI shock isolates the labour-market signature of pre-LLM digital technologies and provides a clean baseline against which more recent disruptions can be benchmarked. Findings should be interpreted as evidence about the mechanisms operating during this period, not as direct forecasts for current AI developments.

Occupational scope: Cluster-level analysis spans 250 occupations covering the breadth of the Australian labour market, with illustrative occupations (Accountant (General), Human Resource Adviser, and Receptionist (General)) examined in detail. I selected the illustrative occupations for: (1) sufficient sample size for detailed illustration; (2) exposure to automation through digitisation; (3) theoretical variation in task composition. While the 250-occupation analysis provides broad coverage, findings may not extend to production occupations or emerging occupations without historical data.

Methodological scope: Analysis relies on job advertisement data capturing employer skill demands rather than worker activities or employment outcomes. Job advertisements reveal labour demand signals but may not reflect actual workplace practices, skill utilisation, or employment transitions. The skill proximity network in Chapter 8 measures *demand-side* similarity between job types, not the *supply-side* feasibility of worker transitions, which depends additionally on institutional pathways, credentialing requirements, geographic constraints, and employer hiring preferences that this analysis cannot observe.

These scope boundaries concentrate the analysis on demand-side restructuring in Australian service-sector occupations during a period of gradual automation diffusion. The thesis acknowledges what lies beyond this scope—wage effects, worker displacement trajectories, manufacturing, international comparisons, and post-2019 generative AI impacts—as directions for future research (Chapter 10).

1.6 Contributions to Knowledge

This research builds on recent advances in measuring within-occupation heterogeneity (Martin-Caughey, 2021), job advertisement analysis (Sostero & Fernández-Macías, 2022), and labour market clustering methods (Vezza et al., 2025) to make six contributions across theoretical, methodological, and empirical domains:

1. A restructuring theory of automation impacts. The thesis advances a theory in which automation’s primary labour market effect operates through continuous task reorganisation within occupations rather than wholesale occupation elimination. This reconciles the skill upgrading (Autor, Levy, & Murnane, 2003), polarisation (Goos & Manning, 2007), and degradation (Braverman, 1974) literatures by showing all three patterns may coexist simultaneously at cluster level. The formal model (Chapter 3) derives five testable propositions from a directed search framework with endogenous job design.

2. A scalable methodology for sub-occupational measurement. First application of HDBSCAN clustering to sub-occupational classification in the economics literature, demonstrating advantages over static taxonomies: endogenous cluster count, explicit noise handling, and dynamic temporal tracking across 2,000 clustering runs (250 occupations ×

8 years). The construction of cross-occupation skill proximity networks from cluster centroids extends this infrastructure into an applied tool for mapping feasible career transition pathways. The methodology is reusable beyond automation research—applicable to any labour-market question where within-occupation heterogeneity matters.

3. Empirical quantification of aggregation bias. Variance decomposition across 170,491 occupation-year-state-industry observations establishes that 94.4% of variation in Australian labour demand operates within rather than between occupations ($ICC = 0.056$). Both the Routine-Cognitive-Manual taxonomy and Frey-Osborne automation probabilities explain only 3% of variation, with theoretically distinct categories statistically indistinguishable. The aggregation bias is structurally pervasive: 204 of 249 occupations (81.9%) contain both at-risk and stable clusters simultaneously, and cluster counts scale near-proportionally with market thickness (elasticity 0.91).

4. The technology succession mechanism. Cluster-level analysis identifies a concrete mechanism consistent with within-occupation restructuring: clusters organised around legacy software platforms (Lotus Notes, ASP.NET, Silverlight) dissolve while clusters requiring emerging technologies (AWS, Docker, Python, Salesforce) form, with dead clusters significantly more software-intensive than born clusters ($p < 0.001$). This succession pattern recasts technology-driven restructuring as a discrete, observable platform-lifecycle process rather than an abstract routine/non-routine substitution.

5. Heterogeneous demand-side dynamics. Newly-emerged clusters exhibit systematically different skill requirements depending on the occupation in which they arise: 59% of occupations show task recombination (more skills) while 22% show task unbundling (fewer skills), with centroid drift moderating the direction (Spearman $\rho = -0.263$, $p < 0.001$). Cluster churn correlates strongly with the speed of skill change rather than the static level of automation exposure (drift–churn Spearman $\rho = +0.558$, $p < 0.001$ across 250 occupations), distinguishing automation *speed* from automation *level* as the relevant predictor of restructuring intensity.

6. Cluster-level transition feasibility. A skill proximity network connecting all clusters across 249 occupations reveals that 95% of at-risk clusters can reach a stable alternative by acquiring three or fewer new skills, with 44.4% finding their best transition target within their own occupation, and 76% of best-match transitions requiring zero genuine specialist skills. Yet navigability is unevenly distributed: while mean connectivity is broadly similar across cluster types, isolation rates differ asymmetrically (3.3% for persistent clusters versus 13.8% for newly emerged clusters). This combination of broad navigability with structural isolation provides empirical foundations for cluster-based workforce development that occupation-level analysis cannot support.

1.7 Thesis Organisation

Chapter 2 reviews the automation and labour markets literature across four key traditions: neoclassical economics emphasising market equilibration, institutionalist approaches examining organisational and regulatory mediation, sociological perspectives on skill and power, and management frameworks addressing strategic implementation. The review establishes that existing frameworks operate at inappropriate levels of aggregation, motivating this thesis's cluster-level approach.

Chapter 3 develops a formal theoretical framework generating testable propositions about within-occupation restructuring. The model explains why occupations fragment into heterogeneous clusters and generates predictions about cluster structure (Propositions 1–3) and temporal dynamics (Propositions 4–5), along with corollaries about churn dynamics.

The empirical analysis proceeds in two analytical phases spanning four chapters. *Phase 1* (Chapter 5) conducts occupation-level regression analysis testing existing frameworks (RCM, Frey-Osborne) to identify aggregate effects. *Phase 2* (Chapters 6–8) shifts to cluster-level machine learning analysis: Chapter 6 tests Propositions 1–3 by discovering clusters and examining within-occupation structure across 250 occupations²; Chapter 7 tests Propositions 4–5 by tracking cluster births, deaths, and evolution and characterising how newly-emerged clusters differ from established ones in their skill requirements; and Chapter 8 maps feasible career transition pathways across the labour market by computing skill similarity between all cluster centroids, identifying bridge skills and accessible alternatives for workers in at-risk job types. The Discussion (Chapter 9) synthesises findings across both phases to develop a comprehensive restructuring theory.

Chapter 4 introduces the real-time job vacancy data provided by Burning Glass Technologies (BGT). The dataset's reliability and coverage are assessed by examining BGT's validation in comparable markets (93% correlation with UK government data, 85% coverage of US vacancies) and through comparison with Australian Bureau of Statistics job vacancy benchmarks.

Chapter 5 evaluates two prominent occupation-level frameworks—the RCM taxonomy (Acemoglu & Autor, 2011b) and Frey-Osborne automation probabilities (Frey & Osborne, 2017)—using Australian job advertisement data. The chapter tests whether these frameworks predict differential labour demand trajectories and decomposes variance to determine whether variation concentrates between or within occupations.

Chapter 6 moves below the occupation level, applying HDBSCAN clustering to identify

²Occupation counts vary slightly across cluster-level chapters: 250 occupations have sufficient data for cross-sectional clustering (Chapter 6), all 250 are tracked through the temporal panel producing 1,707 occupation-year transition observations (Chapter 7), and 249 have valid 2019 cluster models for network construction (Chapter 8).

Table 1.1: Multi-Level Research Design

Phase 1 Occupation-Level	→	Phase 2 Cluster-Level
RCM & FO Frameworks All Occupations N=170,491 cells		HDBSCAN Clustering 250 Occupations N=3,055,469 jobs
Ch. 5: Occupation-level predictive validity and variance decomposition		
↓		
Ch. 6: Cluster discovery, market thickness, within-occupation volatility		
↓		
Ch. 7: Temporal dynamics, creative destruction, skill requirements		
↓		
Ch. 8: Skill proximity networks, career transition pathways		

Note. Multi-level research design. Phase 1 analyses occupation-level effects using RCM and FO frameworks (N=170,491 occupation-year-state-industry cells). Phase 2 conducts cluster-level analysis across 250 occupations using HDBSCAN clustering (N=3,055,469 jobs), examining structure (Ch. 6), temporal dynamics (Ch. 7), skill requirements of newly-emerged clusters (Ch. 7), and cross-occupation skill proximity networks mapping feasible career transitions (Ch. 8). Each phase examines the same underlying labour market at different levels of aggregation.

distinct job types within occupations based on skill profiles. The chapter tests Propositions 1–3 from the theoretical framework: whether occupations contain discrete clusters, whether cluster counts respond to market thickness, and whether cluster-level volatility substantially exceeds occupation-level volatility.

Chapter 7 examines temporal dynamics across all 250 occupations by testing Propositions 4–5. The analysis tracks how clusters are born, evolve, and die over the 2012–2019 period, and characterises how newly-emerged clusters differ from established ones in their skill requirements.

Chapter 8 constructs a skill proximity network connecting clusters across occupations, mapping feasible career transition pathways for workers in at-risk job types. The analysis examines whether the labour market is navigable in skill space and whether navigability is evenly distributed across cluster types.

Chapter 9 integrates findings across the three empirical chapters to develop a restructuring theory of automation impacts. The multi-level evidence reconciles the paradox between dire automation predictions and modest empirical findings: the evidence is consistent with automation substantially reshaping work, but primarily through within-occupation restructuring detectable only at cluster granularity. The chapter develops core theoretical contributions, reflects on methodological innovations, critically examines alternative expla-

nations and boundary conditions, and explores implications for workforce development policy.

Chapter 10 synthesises the thesis's contributions to knowledge across theoretical, methodological, empirical, and policy domains. The chapter acknowledges limitations and boundary conditions of the findings, identifies promising directions for future research, and provides final reflections on how the restructuring theory challenges fundamental assumptions underlying current labour market policy frameworks.

Chapter 2

Literature Review

2.1 Introduction

Understanding how automation affects labour markets requires answering a question that appears deceptively simple: what happens to jobs when technology changes which tasks humans must perform? This question has preoccupied economists since Ricardo (1821, p. 16) warned that machinery could be “detrimental to the interests of labourers,” and has motivated two centuries of theoretical development and empirical investigation. Yet despite this sustained intellectual effort, findings remain deeply contested: occupation-level forecasts predict between 9% and 47% of jobs face high automation risk (Arntz et al., 2016; Frey & Osborne, 2017), while aggregate employment continues growing and firm-level studies find modest effects (Acemoglu & Restrepo, 2020). This five-fold divergence in predictions—addressing the same labour market with comparable but differently-aggregated occupation-task evidence—reveals a fundamental problem: the frameworks researchers use to conceptualise and measure automation are systematically misaligned with how automation actually operates in labour markets.

This chapter argues that the divergence stems from a measurement problem rooted in how jobs are conceptualised across scholarly traditions. Four disciplines have developed sophisticated but largely independent frameworks for understanding work and technological change: organisational behaviour explains how jobs are designed and restructured; labour economics models how automation affects task composition; computer science reveals that automation technologies differ fundamentally in which tasks they can perform; and occu-

pational sociology shows why standard classifications fail to capture workplace dynamics. Each tradition contributes essential insight, yet their isolation has produced a collective blind spot. All four independently predict that automation should generate substantial *within-occupation* restructuring—changing what workers do within occupational boundaries while leaving those boundaries nominally intact—yet all four lack the measurement tools to observe sub-occupational dynamics directly. Emerging computational research applying text analysis to job advertisements has begun documenting the within-occupation heterogeneity these traditions predict, providing empirical validation while highlighting measurement inadequacy. The result is an empirical literature that searches for automation’s effects at the occupation level while theory predicts those effects concentrate at finer granularity.

The integration of these traditions reveals both the source of the measurement problem and its potential solution. If automation operates primarily through task substitution within occupations, and if occupational classifications aggregate over heterogeneous job types with offsetting automation effects, then occupation-level measurement will systematically find null results even when substantial restructuring occurs. The question becomes not whether automation affects labour markets, but whether measurement frameworks operate at appropriate levels of aggregation to detect those effects.

The historical-review framing matters here. The literature reviewed in this chapter spans two distinct technological eras. Pre-2020 work—developed during the diffusion of cloud computing, mobile platforms, and statistical machine learning—is the body of theory and evidence whose blind spots motivate this thesis. Post-2022 generative-AI literature is rapidly evolving and will eventually require its own measurement infrastructure. The thesis examines a defined historical window (2012–2019) precisely because that window represents the second wave of automation in its entirety, before COVID-19 and large language models displaced it. The gap this chapter identifies—occupation-level frameworks dominated the literature throughout the period when sub-occupational restructuring was concentrated—is a gap in the historical record, not just a methodological one. Establishing a within-occupation baseline for the machine-learning era provides the reference point against which the labour-market signature of subsequent waves can later be measured. The thesis addresses this gap by applying density-based clustering to 9.5 million Australian job advertisements (2012–2019) covering 250 occupations, producing the first sub-occupational analysis of restructuring during the machine-learning era.

2.1.1 Chapter Overview

The chapter proceeds in seven sections. Section 2.2 examines how organisational behaviour research conceptualises job design and restructuring. Section 2.3 reviews economic theories of automation from classical compensation theory through contemporary task-based frameworks. Section 2.4 establishes that automation encompasses fundamentally different technologies with heterogeneous task-level effects. Section 2.5 draws on occupational sociology to explain why standard classifications fail to capture within-occupation dynamics. Section 2.6 reviews computational research using job advertisement text analysis that empirically validates within-occupation heterogeneity. Section 2.7 synthesises these traditions to identify the measurement-theory disconnect and derive an integrative definition. Section 2.8 articulates this thesis's contribution.

2.1.2 Scope and Limitations

This review focuses on automation's effects on labour demand and job structure rather than on labour supply responses, worker wellbeing, or macroeconomic dynamics. The emphasis on within-occupation restructuring necessarily limits attention to between-occupation reallocation, which remains important for understanding aggregate labour market adjustment. The integration of four disciplinary traditions requires selective rather than exhaustive coverage; detailed exposition of classical economic debates, historical empirical evidence, and methodological controversies appears in Appendix A.

The review also reflects the current state of rapidly evolving literatures. Research on large language models and generative AI (Section 2.3.7) represents preliminary findings from technologies deployed only since late 2022; conclusions may require revision as longer time series become available. Similarly, the computational social science literature applying text analysis to job advertisements is relatively young, with methodological best practices still developing.

Despite these limitations, the review establishes a clear intellectual foundation: automation's labour market effects likely concentrate at sub-occupational levels that existing measurement frameworks cannot observe, creating systematic blindness that this thesis addresses through cluster-level analysis of Australian job advertisement data.

2.1.3 Review Methodology

This integrative review synthesises research across four disciplinary traditions to develop a theoretical argument about measurement-theory disconnect in automation research. Rather than conducting a systematic review of a single literature, the review pursues theoretical

integration across traditionally siloed fields—an approach suited to identifying cross-disciplinary blind spots invisible within any single tradition (Torraco, 2005). The literature reviewed spans two temporal segments aligned with the thesis’s historical-review framing: foundational and pre-period work (roughly 1969–2019, contemporaneous with or preceding the 2012–2019 dataset analysed in the empirical chapters) establishing the theoretical traditions and their empirical predictions, and post-period reflections (2020–2025) documenting how the field has since evolved under generative AI. Literature identification combined systematic database searches (EconLit, Business Source Complete, Web of Science, Sociological Abstracts), backward and forward citation tracking from foundational works in each tradition, and targeted searches for Australian and generative AI research (2020–2025). Selection prioritised works establishing foundational frameworks, providing evidence on within-occupation heterogeneity, or critically evaluating dominant approaches. Each section concludes with an assessment of Australian evidence, identifying gaps that motivate the present research.

2.1.4 Australian Context

Each section assesses Australian evidence where it exists. Australian automation research has predominantly applied international frameworks: Edmonds and Bradley (2015) adapted Frey-Osborne to Australian occupations, Coelli and Borland (2017) provided Census-based estimates, and Cebulla (2024) found that high-risk occupations actually *expanded*—suggesting occupation-level frameworks may be particularly ill-suited to Australian conditions. No Australian study has examined within-occupation restructuring using sub-occupational measurement approaches. Chapter 4 discusses Australian institutional features in detail.

2.1.5 Competing Definitions of Automation

Scholars define automation disparately, and these definitional differences shape empirical predictions. Four influential definitions—one from each scholarly tradition reviewed in this chapter—illustrate the divergence:

Organisational behaviour conceptualises automation as a job design intervention mediated by managerial choice:

“Whereas automation implies that machines take over a human task, augmentation means that humans collaborate closely with machines to perform a task... AI’s dual nature as an automation and augmentation technology creates tensions for organizations, as the automation and augmentation applications are

not only different, they are also interdependent.” (Raisch & Krakowski, 2021, p. 192)

Economics conceptualises automation through its effects on production and labour markets:

“At the center of our framework is the allocation of tasks to capital and labor—the task content of production. Automation, which enables capital to replace labor in tasks it was previously engaged in, shifts the task content of production against labor because of a displacement effect. As a result, automation always reduces the labor share in value added and may reduce labor demand even as it raises productivity.” (Acemoglu & Restrepo, 2019a, p. 5)

Computer science conceptualises automation as a technical system with specifiable functions and degrees:

“Automation can be applied to four broad classes of functions: (1) information acquisition; (2) information analysis; (3) decision and action selection; and (4) action implementation. Within each of these functions, automation can be defined as the full or partial replacement of a function previously carried out by the human operator... Levels of automation can range from the lowest level of fully manual performance to the highest level of full automation.” (Parasuraman et al., 2000, pp. 286–287)

Sociology conceptualises automation as a contested process embedded in occupational power relations:

“The transformation of working humanity into a ‘labor force,’ a ‘factor of production,’ an instrument of capital, is an incessant and unending process. The condition is repugnant to the victims, whatever their level of pay... The labor process—is the realm in which the hierarchical and managerial relations of capitalism become embodied in the activities of work itself.” (Braverman, 1974, p. 139)

These definitions reveal genuinely different analytical concerns. Organisational scholars ask how firms strategically choose between automation and augmentation; economists ask how automation affects factor shares, productivity, and labour demand; computer scientists ask which functions can be automated and to what degree; sociologists ask how automation reflects and reshapes power relations within and between occupations. Each perspective is internally coherent yet incomplete. The organisational behaviour definition says nothing about which tasks are technically amenable to automation; the economics

definition assumes automation is exogenous rather than a strategic choice; the computer science definition says nothing about labour market effects; the sociology definition centres power relations but says little about which technologies operate through which mechanisms.

This definitional heterogeneity is not merely academic. As Cebulla (2024, p. 307) observe, “None of the [automation risk indicators’] authors unambiguously states how they anticipate automation to change occupations”—a remarkable lacuna given that these indicators underpin policy forecasts affecting millions of workers. The review that follows examines how each of these four scholarly traditions conceptualises automation and its effects, building toward a synthesis that integrates their insights.

2.2 How Jobs Are Designed and Restructured

Barley and Kunda (2001) argued that organisational research had lost sight of actual work by treating occupational categories as sufficient descriptions of what people do. Their call to “bring work back in” emphasised that job structure reflects deliberate choices about task bundling, skill requirements, and work organisation rather than natural categories. This perspective has direct implications for automation: Tschang and Mezquita (2024) and Liu and Wang (2024) confirm that AI’s effects on work vary substantially with implementation context, suggesting that occupational categories obscure the continuous redesign processes through which technology reshapes work.

2.2.1 The Job Characteristics Tradition

The modern study of job design originated with Hackman and Oldham (1976)’s Job Characteristics Model, which established that jobs have measurable structural properties (skill variety, task identity, task significance, autonomy, feedback) that organisations deliberately design rather than encounter as given constraints. This design perspective implies that when technology changes task requirements, organisations face choices about how to redesign jobs—choices that may preserve occupational boundaries while substantially altering job content. Morgeson and Humphrey (2006)’s Work Design Questionnaire later extended measurement to social and contextual characteristics, becoming the dominant approach in organisational research. As Parker (2014) note, however, even expanded frameworks assume jobs are stable units measurable at points in time—an assumption increasingly untenable under continuous technological change. The JCM provides no framework for understanding *how* jobs change, only for measuring characteristics cross-sectionally.

Australian research has extended the work design tradition. Parker and Jorritsma (2024)’s SMART model condenses work characteristics into five higher-order categories (Stimulating, Mastery, Agency, Relational, Tolerable demands). Applying this framework to 2,421 Australian not-for-profit employees, C. Knight et al. (2022) identified four work design profiles whose distribution varied systematically with employment conditions—permanent and higher-position employees occupied more favourable profiles. This Australian evidence demonstrates that within-occupation variation in work design reflects structural features of employment relationships, not merely individual differences.

2.2.2 Job Crafting: Workers as Active Agents

While the job characteristics tradition emphasises organisational design, Wrzesniewski and Dutton (2001) introduced job crafting theory to explain how workers actively reshape their jobs. Job crafting involves “the physical and cognitive changes individuals make in the task or relational boundaries of their work” (Wrzesniewski & Dutton, 2001, p. 179). Workers craft their jobs through three mechanisms: task crafting (changing the number, scope, or type of tasks performed), relational crafting (altering interactions with others at work), and cognitive crafting (changing how one perceives the job’s meaning and significance).

Job crafting theory challenges the assumption that jobs are fixed designs imposed by organisations. Even within formally identical job descriptions, workers create substantial variation through their crafting behaviours. This worker-driven variation creates within-occupation heterogeneity invisible to formal job classifications but consequential for understanding how work is actually performed.

Yet the job crafting literature has limitations relevant to automation research. Rudolph et al. (2017)’s meta-analysis of 122 studies found that most relies on self-report measures susceptible to social desirability bias. F. Zhang and Parker (2019) argue that the literature overestimates workers’ agency, neglecting how organisational constraints, managerial surveillance, and precarious employment limit crafting opportunities. Berg et al. (2010) found that crafting capacity varies systematically by organisational rank—suggesting that workers in routine roles most exposed to automation may have the least capacity to craft alternative responsibilities. Recent extensions develop domain-specific constructs for technology-mediated crafting (Li et al., 2024; Wang et al., 2025; G. Zhang et al., 2023), though this literature remains nascent and reliant on cross-sectional surveys that cannot capture how crafting behaviours evolve as AI capabilities advance. Australian instrument development (Lichtenthaler & Fischbach, 2019; Slemp & Vella-Brodrick, 2013) supports the broader theoretical point that workers actively reshape their jobs, creating within-occupation heterogeneity that formal classifications cannot capture.

An important scope boundary must be noted: while job crafting creates variation within organisational job designs, this thesis focuses on how organisations structure jobs in response to technological change—the design choices that establish the boundaries within which crafting occurs. Job crafting operates *within* organisationally-defined job types; the theoretical framework in Chapter 3 models how organisations *create* those job types through task bundling decisions. Both processes contribute to within-occupation heterogeneity, but they operate at different levels of analysis.

2.2.3 Job Demands-Resources Theory

Bakker and Demerouti (2007)’s Job Demands-Resources (JD-R) model classifies all job characteristics as either demands (aspects depleting energy) or resources (aspects supporting goals or buffering demands), with the balance determining whether workers experience burnout or engagement. Parker (2014) extended the framework to technological change, noting that the same automation technology can function as demand or resource depending on implementation context: it may reduce some demands (routine cognitive load) while creating others (monitoring automated systems), and provide resources (decision support) while eliminating others (skill utilisation, autonomy). The crucial insight for this thesis is that automation’s effects vary *within occupations* depending on which specific tasks are automated—two workers in the same occupational category may experience automation as resource-enhancing or demand-intensifying depending on their job type. JD-R’s reliance on cross-sectional survey designs, however, captures static configurations rather than the dynamic restructuring that task-based economic theory predicts (Crawford et al., 2010; Schaufeli & Taris, 2014). The framework illustrates a broader pattern in organisational behaviour: sophisticated theories of job structure paired with measurement approaches unable to detect automation-driven restructuring at sub-occupational granularity.

2.2.4 Technology and Work Design

Parker and Grote (2022) identify five pathways through which automation affects jobs—task allocation, performance monitoring, scheduling, skill requirements, and social interaction—each capable of enhancing or degrading job quality depending on implementation. The same technology may produce job enrichment (workers freed from routine tasks for complex, meaningful work) or degradation (workers reduced to monitoring automated systems with little discretion). Which outcome emerges depends on organisational choices reflecting managerial philosophy, competitive strategy, and institutional constraints. Kellogg and Spencer (2024) develop this contingency further for AI specifically, arguing that outcomes depend on whether institutional reforms steer AI deployment toward worker-complementing appli-

cations. Parent-Rocheleau and Parker (2022) similarly find that algorithmic management's effects depend on perceptions of transparency, fairness, and controllability. Australian evidence (Nankervis & Cameron, 2022; Nankervis et al., 2019) documents patchy uptake of these technologies and substantial cross-organisational variation in implementation, confirming empirically the theoretical contingency Parker and colleagues identify: technology adoption alone does not determine work design outcomes.

2.2.5 Strategic Human Resource Management

The strategic human resource management (SHRM) literature establishes that firms design jobs as components of broader HR systems, with practice bundles—not individual practices—predicting firm performance (Boxall & Macky, 2012; Huselid, 1995; Subramony, 2009). The implication for automation is that two firms in the same industry, facing the same technology, may make different job design choices based on their HR strategies—creating within-occupation variation invisible to occupation- or industry-level analysis. The SHRM literature itself focuses on firm performance rather than job structure outcomes and predominantly uses cross-sectional designs, leaving the dynamics of how HR systems respond to automation largely untested. The strategic choice perspective remains theoretically valuable for this thesis: it establishes that within-occupation heterogeneity reflects firm-level strategic variation, not measurement error.

2.2.6 Implications for Measurement

The job design literature contributes four insights essential for measuring automation's labour market effects:

First, jobs are designed rather than given. Organisations make deliberate choices about task bundling, skill requirements, and work organisation. These choices respond to technological change, creating continuous job redesign rather than static job structures.

Second, workers actively reshape their jobs. Job crafting creates within-occupation variation as workers with identical formal job descriptions perform substantially different actual work. This variation is invisible to occupational classification systems but consequential for automation exposure.

Third, technology effects are mediated by design choices. The same automation technology can enhance or degrade job quality depending on how organisations redesign jobs in response. Understanding automation's effects requires examining these design choices, not merely technology adoption.

Fourth, job design is strategic. Firms bundle tasks, skills, and responsibilities to achieve competitive objectives, and these bundles differ across firms even within the same

occupation. Within-occupation heterogeneity reflects strategic variation, not measurement error.

These insights suggest that automation should manifest primarily through job redesign within occupational boundaries rather than through occupation elimination. If correct, occupation-level measurement frameworks will systematically miss automation's primary effects by aggregating over the job-level variation where restructuring occurs. The following sections examine whether economic theory predicts this within-occupation pattern and whether occupational classifications can capture it.

2.3 How Automation Affects Tasks

Economic theory provides the analytical framework for understanding how automation affects labour demand. This section traces development from classical compensation theory through contemporary task-based models, establishing that economic theory predicts task-level substitution that should manifest as within-occupation restructuring—yet measurement frameworks continue operating at occupation level. The task-based framework has achieved paradigmatic status in labour economics, with Acemoglu and Restrepo (2024) providing a comprehensive synthesis that formalises how displacement and reinstatement effects jointly determine automation's employment consequences.

2.3.1 Classical Foundations and Compensation Theory

The debate about automation's employment effects has deep historical roots. Ricardo (1821, p. 16) warned that machinery could be “detrimental to the interests of labourers,” while Say (1834) countered with his Law of Markets, arguing that technology augments labour by creating demand through increased production. From these classical debates emerged *compensation theory*—mechanisms through which automation's displacement effects might be offset: labour demand from building new machines, reinvestment of automation-generated profits, wage adjustments clearing labour markets, lower prices increasing demand, and new products creating novel industries (Marco Vivarelli, 2012).

The historical consensus held that technological unemployment remains a short-term phenomenon, with compensation mechanisms eventually restoring employment (Feldmann, 2013). However, scholars have reconsidered this consensus in light of evidence that productivity growth diverged from wage growth beginning in the 1980s (Brynjolfsson & McAfee, 2014). While classical theory establishes the potential for compensation, whether

these mechanisms operate with sufficient strength and speed remains an empirical question. Detailed exposition of classical debates appears in Appendix A.1.

2.3.2 The Task-Based Framework

Autor, Levy, and Murnane (2003) developed a model that transformed how economists conceptualise automation’s labour market effects. Rather than treating occupations as the unit of analysis, they focused on *tasks*—discrete units of work activity that combine to produce output. Their key insight was that computers substitute for routine tasks (cognitive or manual activities following explicit, codifiable procedures) while complementing non-routine tasks (activities requiring flexibility, judgment, or interpersonal interaction).

This routine-biased technical change framework explains observed patterns of job polarisation: growth in high-skill analytical occupations (complemented by computers), growth in low-skill service occupations (requiring non-routine manual tasks), and decline in middle-skill clerical and production occupations (performing routine tasks susceptible to automation). The framework predicts that automation’s effects operate at the task level, with occupational outcomes depending on task composition.

However, the task-based framework has attracted criticism on both conceptual and empirical grounds. Fernández-Macías (2012a) found that European labour markets exhibited different polarisation patterns than the US, with institutional factors—collective bargaining coverage, employment protection, and education systems—mediating technological effects. This cross-national variation suggests that task-level substitution, while real, interacts with institutional structures in ways the baseline framework does not accommodate. Hunt and Nunn (2022) re-examined US data and found weaker support for routine-biased technical change than commonly assumed, with much of the apparent decline in routine employment attributable to manufacturing’s long-term decline rather than within-industry task automation. Australian research reinforces the importance of institutional context: Coelli and Borland (2016) found that job polarisation occurred in Australia but predominantly during the 1980s and 1990s—earlier than the US pattern and with different institutional drivers related to Australia’s award wage system and industrial relations reforms.

More fundamentally, Autor (2013a) himself acknowledged that the routine/non-routine distinction is increasingly problematic. Tasks classified as “non-routine” in the 1990s—such as visual pattern recognition and natural language processing—proved automatable by 2010s machine learning. The distinction may describe which tasks *were* automated rather than which tasks *are* automatable, limiting its predictive value for emerging technologies. Recent work on large language models (Eloundou et al., 2024) suggests that “non-routine cognitive” tasks may be highly exposed to generative AI, inverting earlier assumptions.

Acemoglu and Autor (2011b) formalised this insight, deriving conditions under which automation generates job polarisation. Their model treats tasks as a continuum, with workers of different skill levels having comparative advantage in different task ranges. Automation expands the set of tasks performable by capital, displacing workers from routine tasks while increasing demand for workers performing complementary non-routine tasks. Critically, their framework predicts that automation changes *what workers do within occupations*, not merely which occupations exist. Yet the model's predictions depend on parameters—the elasticity of substitution between tasks, the pace of new task creation—that are difficult to estimate and may vary substantially across contexts.

2.3.3 Displacement and Reinstatement

Acemoglu and Restrepo (2019a) extended the task framework to distinguish two competing effects. *Displacement* occurs when automation substitutes for human tasks, reducing labour demand. *Reinstatement* occurs when new tasks are created where humans have comparative advantage, increasing labour demand. Automation's net employment effect depends on the balance between these forces—a balance that cannot be determined theoretically and varies across contexts.

Their empirical analysis of US data (1987–2016) found that displacement effects substantially exceeded reinstatement effects, contributing to wage stagnation and declining labour share of income. However, this aggregate finding masks substantial heterogeneity: some industries experienced strong reinstatement while others experienced predominant displacement. This heterogeneity suggests that understanding automation's effects requires disaggregated analysis capable of identifying where displacement and reinstatement occur.

Acemoglu and Restrepo (2022) documented that between 50% and 70% of changes in the US wage structure over four decades are attributable to relative wage declines of worker groups specialising in routine tasks in industries experiencing rapid automation. This finding demonstrates that task displacement has substantial distributional consequences, even when aggregate employment effects appear modest. Borland and Coelli (2016) examined labour market inequality in Australia, finding that while overall wage dispersion increased, the pattern differed from US experience—Australia's award wage system and minimum wage institutions compressed the lower tail of the distribution, potentially channelling technological adjustment into hours and employment composition rather than wage inequality. Borland and Coelli (2023a) extended this analysis, applying the Katz-Murphy method—a supply-demand decomposition framework that attributes wage structure changes to shifts in relative labour supply versus demand (L. F. Katz & Murphy, 1992)—to Australian data (1981–2021) and finding that the skill premium growth rate slowed substantially after the

early 1990s, a pattern also observed in the US but with Australian institutional factors playing a distinctive mediating role.

Extending the task framework to AI, Acemoglu (2024) provide a notably sceptical assessment of AI's productivity potential. Their analysis decomposes AI's economic impact into task-level productivity gains, the share of tasks exposed to AI, and wage pass-through to workers—arguing that modest values on each dimension compound to limited aggregate effects, predicting no more than 0.66% TFP gains over ten years from current AI technologies. This scepticism contrasts sharply with technology industry projections but aligns with the task-based framework's prediction that automation operates at sub-aggregate levels that may not propagate to macroeconomic measures. Acemoglu et al. (2023) extend this analysis by distinguishing between “automation AI” (displacing workers from tasks) and “augmentation AI” (enhancing worker productivity), arguing that current AI development trajectories favour automation despite augmentation offering superior worker welfare outcomes. This automation-augmentation distinction has implications for within-occupation dynamics: augmentation-focused deployment should generate skill complementarity and job enrichment, while automation-focused deployment generates task displacement concentrated in specific job types.

A counter-argument to the within-occupation restructuring hypothesis deserves consideration: if occupational categories have *institutional* significance—determining wages through awards, structuring collective bargaining, defining credentialing requirements—then occupation-level analysis may capture the relevant causal mechanisms even when within-occupation task variation exists. Australian labour markets provide a particularly strong test case: award wages are set by occupation, not by task content within occupations. If institutional wage-setting dominates, then task-level restructuring may affect job content without affecting the economic outcomes (wages, employment) that occupation-level analysis measures. The counter-argument is that institutional wage structures constrain *wage* adjustment while permitting *employment composition* adjustment—firms may respond to automation not by changing wages within occupational categories but by changing which job types they hire within those categories. This compositional change is precisely what occupation-level employment counts miss but what sub-occupational clustering can detect.

2.3.4 New Task Creation

Autor et al. (2024a) documented that approximately 60% of employment in 2018 was in job titles that did not exist in 1940, demonstrating substantial work creation over time. Their analysis identifies two distinct eras: from 1940–1980, new job titles emerged primarily in professional and managerial occupations requiring expert judgment; from 1980–2018, new

work increasingly consisted of service sector occupations requiring interpersonal skills but not necessarily higher education. The full analysis, published in the *Quarterly Journal of Economics* as Autor et al. (2024b), provides detailed documentation of how technological change creates new work even as it displaces existing tasks.

This pattern of new task creation suggests that automation does not simply eliminate jobs but transforms the nature of work. New tasks emerge within existing occupational categories as well as in entirely new occupations. An accountant in 2020 performs substantially different tasks than an accountant in 1980, even though the occupational category persists. This within-occupation transformation is precisely what task-based theory predicts but what occupation-level measurement cannot detect. Recent evidence from Georgieff and Hyee (2021) examines what actually happened to jobs classified as “high automation risk,” finding patterns more consistent with task restructuring than job elimination.

2.3.5 Skill Complementarities

Understanding automation’s labour market effects requires analysing not just which tasks are substituted but which skills become more valuable as complements to automated systems. The skill complementarity perspective shifts focus from “which jobs will be eliminated” to “how will skill demands change within surviving jobs.”

The capital-skill complementarity hypothesis, formalised by Griliches (1969) and extended by Krusell et al. (2000), posits that physical capital and skilled labour are relative complements while capital and unskilled labour are relative substitutes. Applied to automation, this suggests that automated systems increase demand for workers who can operate, manage, and extend these systems while reducing demand for workers performing tasks the systems automate.

Autor (2015a, p. 11) argues that this complementarity explains why automation has not produced mass unemployment despite dramatic technological change. “Tasks that cannot be substituted by automation are generally complemented by it”—machines handling routine components of work enable humans to focus on non-routine components, often increasing overall productivity and labour demand in affected occupations.

Recent research demonstrates that complementarity effects dominate substitution for most skills. Mäkelä and Stephany (2024, p. 2) find AI’s complementary effect “up to 1.7 times larger” than its substitution effect, with the finding replicating across UK, US, and Australian markets. The skills consistently showing positive complementarity are digital literacy and AI proficiency (Mäkelä & Stephany, 2024; OECD, 2024a), complex problem-solving and creativity (Autor, 2015a; Frey & Osborne, 2017), and interpersonal/communication skills (Deming, 2017a). Skills showing substitution—routine customer

service, routine data processing, standardised administrative tasks—are precisely the routine cognitive and manual tasks identified by Autor, Levy, and Murnane (2003). This asymmetry generates within-occupation heterogeneity: jobs emphasising complementary skills expand while those emphasising routine tasks contract, even within the same occupational category. Subsequent work has refined the construct: complementarity depends on the number and value of skills combinable with a given skill (P. Cortés et al., 2024); AI-driven capital-skill complementarity affects firm performance and innovation (Babina et al., 2024); and current institutions are misaligned with emergent complementarity patterns (del Río-Chanona et al., 2025; Lane & Saint-Martin, 2023; Tong & Zhang, 2024).

These findings rest primarily on observational data that cannot rule out selection effects—workers with valuable skill combinations may differ on unobservables from those with less complementary portfolios. Average effects may obscure cases where substitution dominates, which are precisely the contexts where within-occupation restructuring should be most visible. Because these studies typically aggregate to occupation level, they cannot directly test whether complementarity and substitution patterns generate the within-occupation restructuring that task-based theory predicts.

2.3.6 Measurement Controversies

The gap between task-level theory and occupation-level measurement is starkly illustrated by the Frey and Osborne (2017) controversy. Their analysis predicted that 47% of US employment faces high automation risk based on occupation-level technological feasibility assessments. Arntz et al. (2016) reanalysed the same question using task-level variation within occupations, finding only 9% of jobs at high risk. This five-fold difference stems entirely from measurement approach: Frey and Osborne assumed occupational homogeneity, while Arntz and colleagues accounted for within-occupation task variation.

A decade after their original prediction, no occupational apocalypse has materialised. Employment in occupations Frey and Osborne classified as highly automatable—including accounting, bookkeeping, and tax preparation (each rated 97%+ automation probability)—actually *grew* by 19% in the seven years following publication, with employment in this group doubling between 1990 and 2024 (National Academies of Sciences, Engineering, and Medicine, 2024). Recognising these developments, Frey and Osborne (2024) revisited their analysis in light of generative AI, noting that remote jobs face higher automation risk, AI will increase the value of in-person communication skills, and creative work may be less automatable than initially assumed—a substantial revision of their original framework.

Australian research provides particularly sharp tests of these measurement approaches. Edmonds and Bradley (2015) adapted Frey-Osborne methodology to Australian occupa-

tional classifications, estimating that 40% of Australian jobs faced high automation risk. Coelli and Borland (2017) provided more rigorous estimates using Australian Census data and O*NET task measures, finding lower but still substantial risk levels. Coelli and Borland (2019) subsequently critiqued the Frey-Osborne framework, identifying internal inconsistencies where occupations were classified as highly automatable despite requiring skills designated as automation bottlenecks. Most consequentially, Cebulla (2024) tested whether automation risk indicators predicted actual employment change using Australian Census data (2011–2021), finding that occupations deemed at greater risk actually *expanded* rather than contracted. This counterintuitive result held across multiple industries. Meanwhile, Borland and Coelli (2017) examined whether computer-based technologies reduced total employment in Australia, finding no evidence of aggregate job destruction—a result consistent with occupation-level measurement missing within-occupation restructuring. Collectively, this Australian research demonstrates the empirical failure of occupation-level automation risk frameworks while providing no alternative measurement approach capable of detecting the within-occupation dynamics that economic theory predicts.

These measurement controversies reveal a fundamental problem: economic theory operates at the task level, predicting within-occupation restructuring, while empirical measurement operates at the occupation level, detecting only net effects that may obscure substantial task-level change.

2.3.7 Generative AI: A New Automation Paradigm?

Generative AI since late 2022 has prompted reconsideration of automation’s scope. Unlike previous waves that primarily affected routine tasks, generative AI demonstrates capability in non-routine cognitive work (Eloundou et al., 2024), challenging earlier frameworks that assumed non-routine analytical tasks would remain human domains.

Early empirical evidence on labour market effects is contested. Humlum and Vestergaard (2024)’s large-scale surveys of 25,000 Danish workers find widespread adoption alongside “precise zeros” for effects on earnings or hours, with confidence intervals ruling out effects larger than 1%; modest productivity gains (2.8% time savings) combined with weak wage pass-through explain these limited effects. The IMF and ILO assessments reach similar conclusions: AI’s primary impact is augmentation rather than wholesale displacement (Cazzaniga et al., 2024; Gmyrek et al., 2023). Bick et al. (2025) document rapid US adoption (36.7% by September 2025), yet Yale Budget Lab (2024) and Bureau of Labor Statistics (2025) find no substantial acceleration in labour market composition since ChatGPT’s introduction.

Experimental and field studies provide more granular evidence: ChatGPT increased

writing productivity (Noy & Zhang, 2023a), GitHub Copilot raised software developer productivity (Peng et al., 2023), and AI-assisted customer service agents were 14% more productive with gains concentrated among novices (Brynjolfsson et al., 2023a). Dell’Acqua, McFowland III, et al. (2023) characterise the “jagged technological frontier” where AI excels at some tasks but fails at others. Sarker et al. (2025)’s systematic review concludes that effects remain modest and context-specific. Most studies span one to two years post-ChatGPT release, insufficient for organisational restructuring to fully unfold. Some research finds more pronounced effects, however: Hui et al. (2024) document sharp employment and earnings declines for freelancers in AI-exposed occupations, with concentrated effects in specific job types alongside null aggregate findings—suggesting generative AI may produce within-occupation restructuring rather than uniform displacement.

Australian theoretical contributions (Walkowiak, 2023; Walkowiak & MacDonald, 2023) formalise how AI-human task interdependencies generate heterogeneous effects: workers performing complementary tasks experience productivity gains while those performing substitutable tasks face displacement pressure. The empirical implication is that occupation-level exposure indices represent averages over heterogeneous task bundles that individual workers may not perform—motivating the sub-occupational measurement this thesis develops.

The generative AI literature reinforces three themes from earlier automation research: technological capability does not translate directly to labour market impact, aggregate stability may mask substantial restructuring at finer granularity, and within-occupation heterogeneity in task content determines which specific jobs are affected.

2.3.8 Implications for Measurement

The economic literature generates a clear prediction: automation should generate substantial within-occupation restructuring as routine tasks are automated while complementary non-routine tasks expand. If an occupation contains both routine and non-routine task components, automation may dramatically change what workers in that occupation do while leaving occupation-level employment relatively stable. Occupation-level measurement would detect near-zero effects while missing the underlying restructuring.

This prediction aligns with the job design literature’s insight that jobs are continuously redesigned. Economic theory explains *why* this redesign occurs (task-level comparative advantage shifts), while organisational theory explains *how* (through strategic job design and worker crafting). Both traditions predict within-occupation change; neither provides measurement tools to observe it.

2.4 The Capability–Deployment Gap

Both organisational behaviour and economics treat “automation” as a relatively undifferentiated phenomenon, and both rely on capability assessments imported from computer science when operationalising automation exposure. Yet the computer science literature itself reveals a complication that the imported assessments obscure: technical capability is a poor predictor of workplace deployment. This gap between what machines *can* do and what organisations *choose* to automate has direct consequences for how automation should be measured.

2.4.1 Capability Assessments Were Not Designed to Predict Labour Markets

Frey and Osborne (2017) relied on machine learning researchers’ judgments about automation feasibility to construct occupation-level risk scores. Brynjolfsson and Mitchell (2017)’s “suitable for machine learning” (SML) scores measured technical suitability of O*NET tasks for ML automation, and Felten et al. (2018) linked specific AI capability advances to specific occupational tasks. These task-level approaches represent genuine methodological progress, demonstrating that most occupations contain mixtures of automatable and non-automatable tasks. The insight that automation affects tasks rather than jobs aligns with the economic theory reviewed above and motivates the sub-occupational measurement this thesis develops.

Yet capability assessments answer the question “can a machine perform this task?” not “will firms automate this task?” Their authors acknowledge this limitation (Brynjolfsson & Mitchell, 2017) but provide no framework for predicting which technically-automatable tasks will actually be automated. The result is measures of automation *potential* often interpreted as measures of automation *exposure*—a conflation that may explain the poor predictive performance of automation risk indicators documented by Cebulla (2024). Frameworks for designing partial automation (Parasuraman et al., 2000; Sheridan, 2002; Shneiderman, 2022) similarly assume a benevolent designer optimising human-machine performance, ignoring the contested organisational politics through which automation is actually deployed (Braverman, 1974; Edwards, 1979).

2.4.2 Capability Substantially Exceeds Deployment

Rogers (2003) established that technology diffusion follows predictable patterns shaped by perceived advantage, compatibility, complexity, and observability—factors that delay adop-

tion well beyond technical feasibility. Australian evidence makes the gap particularly visible. Kennedy et al. (2019) found large variation in automation adoption among Australian firms in the same industry, suggesting firm-level factors dominate technology characteristics in determining adoption. Hajkowicz et al. (2016) identified workforce skill gaps as primary constraints on Australian automation adoption: firms lack personnel capable of implementing and managing automated systems. The Australian Industry Group's workforce surveys (2020) document persistent implementation challenges: change management difficulties, skill development costs, and system integration problems mediate the relationship between technical feasibility and workplace deployment.

This conflation directly compounds the measurement-theory disconnect. If occupation-level automation indices measure potential rather than exposure, and if deployment varies by firm, then aggregating to occupation level doubly obscures actual restructuring—first by averaging over within-occupation task heterogeneity, second by averaging over firm-level deployment variation. The task-based framework reviewed above predicts restructuring at task level; the capability-deployment gap shows that restructuring depends on firm-level implementation choices; occupation-level measurement averages over both sources of variation, leaving no residual signal of the dynamics both literatures predict should be consequential.

For this thesis, the computer science literature contributes two insights. Task-level granularity reveals within-occupation heterogeneity that occupation-level measures cannot capture. The capability-deployment gap reveals that technical capability is a weak predictor of deployment, which itself varies across firms. Together these insights motivate measurement at sub-occupational granularity: the level at which automation actually operates lies below both the occupation and the technical capability score.

2.5 Why Occupational Classifications Fail

The preceding sections established that organisational theory predicts continuous job redesign, economics predicts task-level substitution, and computer science demonstrates that automation technologies differ fundamentally in which tasks they affect. All three traditions predict within-occupation restructuring. Yet empirical automation research relies on occupational classifications as its primary measurement instrument. This section examines whether these classifications can capture the dynamics that theory predicts. Drawing on occupational sociology, the section demonstrates that occupational categories are not neutral measurement instruments but socially constructed boundaries that systematically obscure within-occupation heterogeneity—the very heterogeneity that automation theory predicts should be consequential.

2.5.1 Occupations as Contested Jurisdictions

Abbott (1988) demonstrated that occupational boundaries are not natural categories but continuously negotiated through professional competition. Professions claim jurisdiction over particular work domains using strategies of exclusion, subordination, and differentiation. Critically, technological change intensifies these jurisdictional contests. When automation eliminates routine tasks, professionals do not simply experience displacement—they *redefine* their jurisdiction to emphasise activities that remain valuable. Accountants respond to bookkeeping automation by claiming advisory and strategic planning work; radiologists respond to AI-assisted screening by emphasising complex diagnostic judgment. The occupational category persists, but its content transforms.

Abbott’s framework was developed through historical case studies of medicine, law, and accounting—professions with strong credentialing systems capable of coordinating jurisdictional claims. Whether non-professional occupations exhibit similar dynamics, and whether such renegotiation operates rapidly enough to keep pace with contemporary AI diffusion, remain open questions that qualitative methods cannot resolve. Despite these limitations, Abbott’s core insight holds: occupational employment statistics track a moving target whose content changes definitionally even as it remains nominally stable. Using occupation-level employment to measure automation effects becomes circular when occupational boundaries themselves respond to automation.

2.5.2 Social Closure and Within-Occupation Heterogeneity

Weeden (2002) and Weeden et al. (2007) examined how closure mechanisms (licensing, credentialing, unions, professional associations) shape occupational boundaries. Analysing 488 occupations, she found that closure mechanisms explain only 15–20% of inter-occupational earnings variation, with most earnings inequality operating *within* rather than between occupations. A certified management accountant specialising in foreign exchange hedging performs fundamentally different work than a certified public accountant conducting compliance audits, yet both are “Accountants” in official statistics with identical credentials. Closure mechanisms determine who qualifies as an “Accountant” but do not standardise what accountants actually do.

This within-occupation heterogeneity is empirically robust across institutional contexts (Dwyer, 2023; Torrejón Pérez et al., 2024; Vera-Toscano et al., 2022), though Weeden’s cross-sectional designs cannot track how dispersion evolves as technologies diffuse. The finding that 15–20% of variation reflects closure leaves unexplained what mechanisms

account for the remaining 80–85%—a gap the present thesis addresses through task-based clustering.

2.5.3 Professional Discretion and Task Variation

Freidson (2001) added a third dimension: the relationship between formal occupational categories and actual task performance. Distinguishing professionalism from both market logic and bureaucratic organisation, Freidson argued that professional work involves substantial discretion and judgment that resists codification into standardised procedures. Even when formal job descriptions appear similar, actual task performance varies enormously based on client characteristics, organisational context, and individual expertise accumulated through practice.

This discretionary variation is particularly consequential for automation research. Technologies may automate codifiable components of work while leaving discretionary judgment untouched, creating within-occupation differentiation between routine and judgment-intensive roles. A “Financial Analyst” whose work involves standardised report generation faces different automation exposure than one whose work involves novel strategic analysis, yet both occupy the same occupational category.

2.5.4 Labour Process and Skill Construction

A parallel sociological tradition examines power relations embedded in workplace organisation and technological change. Braverman (1974) argued that capitalist management systematically *deskills* work by decomposing craft knowledge into simplified, routinised tasks amenable to mechanisation. This deskilling transfers workers’ tacit knowledge into machinery and managerial systems, reducing labour to execution of predetermined scripts.

Edwards (1979) extended this framework by distinguishing workplace control strategies. *Technical control* embeds managerial authority in machinery and production systems; *bureaucratic control* embeds it in organisational rules and career structures. Automation decisions often reflect control imperatives as much as efficiency considerations, with implications for which tasks are automated and how remaining work is organised.

The skill construction literature reveals that designations of tasks as “routine” (automatable) versus “non-routine” (requiring human judgment) reflect socially negotiated categories rather than objective technical attributes. Attewell (1990) and Vallas (1990) demonstrated that tasks labelled “unskilled” often require substantial tacit knowledge, judgment, and contextual understanding, yet receive this designation because they are performed by workers with limited bargaining power. What counts as “automatable” reflects power relations as much as technical capabilities.

2.5.5 Job Quality and Workplace Fissuring

A related literature documents how organisational restructuring differentiates job quality—wages, conditions, career prospects—among workers performing similar tasks. Weil (2014) argues that information technology enables large firms to “fissure” employment by shifting work to contractors and franchisees while monitoring performance remotely. Workers performing identical tasks may receive substantially different wages depending on whether they are employed by the lead firm or a contractor. Kalleberg (2011) and Standing (2011) document the rise of polarised and precarious employment systems where the same occupational title describes fundamentally different employment experiences depending on firm, industry, and contractual arrangement. Osterman and Shulman (2011) adds the erosion of internal labour markets: firms outsource peripheral functions and dissolve career ladders that previously linked entry-level to senior positions within occupations.

Recent evidence extends these patterns to AI and platform work. Different automation technologies produce divergent within-occupation effects: robot-exposed occupations show lower wage inequality, while AI-exposed occupations show higher dispersion concentrated at the top (Ferraro et al., 2025; OECD, 2024b). Platform work exhibits substantial within-occupation heterogeneity in job quality that defies uniform characterisation (Srihita et al., 2025; Wood et al., 2025). The erosion of entry-level positions identified by Osterman appears to be accelerating: World Economic Forum (2025a) projects that 40% of employers expect to reduce workforces where AI can automate tasks, with entry-level positions particularly vulnerable (Bureau of Labor Statistics, 2025; Goergen et al., 2024; World Economic Forum, 2025b). Jesuthasan and Boudreau (2022) argue that this task-level reorganisation has become a central organising principle, enabling the fissuring Weil documented and creating measurement challenges: if jobs are increasingly fluid assemblages of tasks, occupation-level classification becomes progressively less informative about work content or quality.

2.5.6 Classification Lag and Boundary Permeability

Statistical agencies prioritise time-series continuity over contemporaneous accuracy: ANZSCO updates occurred at decade-plus intervals, the US Standard Occupational Classification updates every ten years (Bureau of Labor Statistics, 2018), and the International Standard Classification of Occupations updates roughly every twenty (International Labour Organization, 2012). The trade-off is explicit—frequent updates would compromise longitudinal analysis, yet infrequent updates systematically underestimate restructuring since the last revision. The delayed recognition of data science illustrates the cost: as data scientist roles proliferated through the 2010s, ANZSCO contained no category for them,

and the Australian Bureau of Statistics advised practitioners to use a residual code (224999) providing no information about actual content. Only in November 2022 did ANZSCO add dedicated Data Analyst (224114) and Data Scientist (224115) codes, more than a decade after these roles became established.

The Occupation Standard Classification for Australia (OSCA), released in December 2024, addressed classification lag by adding over 300 new occupations and committing to annual updates (Australian Bureau of Statistics, 2024b). Critically, however, OSCA retains the same five-level hierarchical structure with occupation as the finest level, defining an occupation as a set of jobs requiring similar tasks (Australian Bureau of Statistics, 2024a)—preserving the assumption that occupations are internally homogeneous. OSCA addresses lag, not the within-occupation heterogeneity this review identifies as the more fundamental measurement problem. Appendix B provides historical context on classification system development.

Autor et al. (2024a) documented that new job titles account for substantial employment growth, yet these titles are often classified within existing occupational categories, obscuring the novelty of work being performed. A data scientist may perform work previously distributed across statisticians, software engineers, and business analysts; the classification system must force this hybrid role into a single category, obscuring its cross-occupational origins.

2.5.7 Recent Evidence: Platform Work and Boundary Erosion

Research on platform work and the gig economy has renewed the foundational insights of Abbott, Weeden, and the labour process theorists. Vallas and Schor (2020) demonstrate that digital platforms systematically decompose occupations into discrete tasks, enabling recombination that defies traditional classification: a worker on multiple platforms may simultaneously perform logistics, delivery, and personal transportation tasks previously assigned to distinct occupational categories. Petriglieri et al. (2019) find that independent workers construct occupational identity through task portfolios rather than categorical membership—experiencing labour markets as organised around tasks and skills rather than occupations.

Australian regulatory experience illustrates the practical consequences. The 2024 Fair Work Legislation amendments introduced “employee-like” worker classifications precisely because traditional occupational categories proved inadequate for platform workers performing tasks across multiple occupational boundaries (Barratt et al., 2025; Healy et al., 2025; Minas & French, 2025). Borland and Coelli (2023b) document that skill mismatch affects

approximately 30% of high-skill Australian workers, consistent with within-occupation heterogeneity in international research. Weeden and Grusky (2012)'s micro-class analysis provides further support: detailed occupational categories predict life outcomes more strongly than broad categories, implying systematic within-occupation structure that coarse categories obscure.

These contemporary findings confirm that the theoretical insights from Abbott, Weeden, and labour process theory remain empirically valid. Digital technologies and platform work have intensified the boundary fluidity and within-occupation heterogeneity that earlier scholars identified, making sub-occupational analysis more important rather than less.

2.5.8 Implications for Measurement

The sociological literature reveals three fundamental problems with using occupational classifications to measure automation effects:

First, occupational categories are endogenous to technological change. If Abbott's theory is correct that automation triggers jurisdictional contests that redefine boundaries, then using occupational employment to measure automation effects is circular—the measurement unit changes in response to the phenomenon being measured.

Second, occupational closure mechanisms regulate entry but not internal structure. Weeden's finding of substantial within-occupation heterogeneity means that aggregating to occupational level averages over systematic variation in automation exposure. Workers with identical credentials may face different automation risks depending on their specific task specialisation.

Third, occupational boundaries are fluid and permeable. Workers, tasks, and roles move across boundaries that official classifications treat as fixed. This creates adjustment dynamics invisible to occupation-level employment counts.

These problems compound the measurement challenges identified in economic theory. Economics predicts task-level substitution generating within-occupation change; sociology demonstrates that occupational classifications cannot capture this change because they impose artificial stability on inherently fluid boundaries.

2.6 What Computational Research Reveals About Labour Market Structure

The theoretical arguments reviewed above—that automation operates at task level, that occupational boundaries are fluid, that within-occupation heterogeneity should be substantial—

have recently found empirical confirmation through computational analysis of large-scale text data. This emerging literature applies natural language processing and machine learning to millions of job advertisements and career histories, revealing labour market structure at granularity impossible with traditional survey methods. The findings validate theoretical predictions about within-occupation heterogeneity while highlighting the inadequacy of occupation-level measurement frameworks.

Critically, however, this computational literature documents *that* within-occupation heterogeneity exists without examining *how* automation creates, reshapes, or operates through that heterogeneity. The studies reviewed below confirm the existence of sub-occupational structure but do not test whether automation systematically relates to that structure—whether automation-exposed job types contract while complementary types expand, whether technological change drives the growing heterogeneity observed, or whether sub-occupational clusters exhibit the dynamics that automation theory predicts. This gap—between documenting heterogeneity and explaining its relationship to technological change—is what this thesis addresses.

2.6.1 Within-Occupation Skill Variation

Deming and Kahn (2018a) analysed Burning Glass job postings for professional occupations, examining how skill requirements vary across firms within occupational categories. Their findings challenge the assumption of occupational homogeneity: skill requirements for ostensibly identical occupations differ dramatically across employers. A “Software Developer” position at one firm may emphasise front-end web technologies while another emphasises back-end database systems; both are classified identically in official statistics despite requiring substantially different human capital.

Critically, Deming and Kahn found that within-occupation variation in skill requirements often *exceeds* between-occupation variation. This finding implies that occupational categories may be less informative about actual work content than firm-level or job-level characteristics. If automation affects specific skills rather than whole occupations, then within-occupation skill variation determines automation exposure more than occupational classification.

However, the Deming and Kahn analysis has limitations that subsequent research has highlighted. Hershbein and Kahn (2018a) note that job postings capture employer *demand* but not employment outcomes—firms may post for positions they never fill, or fill positions through channels generating no posting. Carnevale, Jayasundera, and Repnikov (2014) estimate that Burning Glass captures only 60–70% of online postings and systematically under-represents positions filled through informal networks, particularly in smaller firms

and less-skilled occupations. The finding of substantial within-occupation variation is robust, but whether this variation reflects actual employment patterns or merely posting behaviour remains unclear.

Australian research has begun applying similar computational approaches. Evans et al. (2024) analysed over 12 million Australian job advertisements (2014–2022) using Adzuna data, finding substantial within-occupation variation in skill demand trajectories. Their methodology—integrating the ESCO skills taxonomy with job posting text—demonstrates the feasibility of granular Australian labour market analysis. Critically, they found accelerated demand for interpersonal skills post-pandemic, with variation across occupations in both levels and trends of skill requirements. The authors explicitly note that extending their analysis to monitor detailed skill classes within detailed occupation classes would require more sophisticated natural language processing—precisely the methodological direction this thesis pursues.

2.6.2 Growing Internal Divergence

Atalay et al. (2020a) examined how occupational task content has evolved over eight decades using newspaper job advertisements (1940–2000) and online postings (2000–2018). Their analysis reveals that within-occupation skill variance has *increased* over time—occupations are diverging internally rather than converging toward homogeneity. Tasks that were once bundled together within occupational categories have increasingly separated, with different employers emphasising different task combinations.

This pattern of growing internal divergence has important implications for automation research. If occupations were becoming more homogeneous, occupation-level measurement would become increasingly valid over time. The finding that heterogeneity is growing suggests the opposite: occupation-level measurement becomes *less* valid as within-occupation variation expands. Automation research using occupational categories faces a moving target that is moving in the wrong direction. Recent OECD analysis confirms this pattern internationally, finding that “skill requirements within occupations are changing faster than occupational employment shares,” with substantial variation across firms in which skills they demand for ostensibly identical occupational categories (OECD, 2024a).

Yet the Atalay analysis shares limitations common to job posting research. Linking historical newspaper advertisements to contemporary online postings requires bridging measurement instruments that may not be comparable—what employers chose to specify in paid print advertisements may differ systematically from what they specify in low-cost digital postings. Moreover, the analysis documents *that* heterogeneity has grown but not *why*. Does growing variance reflect genuine task divergence within occupations, or

changing employer posting practices, or evolving occupational classification boundaries? The descriptive finding is valuable but leaves causal mechanisms unidentified.

2.6.3 Skill Networks and Polarisation

Alabdulkareem et al. (2018) approached labour market structure through network analysis, examining how skills co-occur in job postings. Rather than taking occupational boundaries as given, they allowed skill clusters to emerge from patterns of complementarity in employer demand. Their analysis reveals that skills form network clusters that cross occupational boundaries, with distinct clusters of cognitive-analytical skills, social-interpersonal skills, and physical-manual skills. Recent work by Alabdulkareem et al. (2024) extends this approach, demonstrating that skill complementarity—which defines the links between skills in the “Skillscape”—accurately predicts workers’ career trajectories and has implications for understanding automation’s differential impact across skill bundles.

The skill network structure helps explain job polarisation: middle-skill occupations combined cognitive and physical skills in ways that could be decomposed, with cognitive components automated and physical components remaining. High-skill and low-skill occupations, by contrast, contain skill bundles that are more difficult to decompose. This finding suggests that understanding automation’s effects requires analysing skill bundles rather than occupational categories—a perspective that occupation-level measurement cannot accommodate. G. P. Green et al. (2023) demonstrate a new method that clusters occupations based on knowledge, skills, abilities, and other characteristics, showing that occupation similarity structures can identify viable employment alternatives for workers displaced by technological change.

The network approach, while innovative, introduces its own limitations. Skill taxonomies used to code job postings reflect researcher choices about categorisation that may not correspond to how employers or workers conceptualise skills. Autor (2013a) cautions that skill labels extracted from job postings may conflate genuinely distinct competencies or artificially separate related ones. The finding that skills cluster across occupational boundaries is suggestive, but whether these clusters represent meaningful labour market segments—with distinct wages, career trajectories, and automation exposure—requires validation against employment outcomes that job posting data cannot provide.

2.6.4 Recession-Driven Skill Upgrading

Hershbein and Kahn (2018a) examined whether recessions accelerate automation by analysing how skill requirements in job postings changed during and after the Great Recession. They found substantial skill upgrading—increased requirements for cognitive skills,

computer skills, and education—concentrated in metropolitan areas experiencing larger employment shocks.

Critically, this skill upgrading occurred primarily *within* occupations rather than through reallocation across occupations. Employers posting for the same occupational categories after the recession required higher skills than before. This within-occupation skill upgrading is invisible to occupation-level employment analysis, which would detect only changes in occupational composition. The finding demonstrates that economic shocks—including automation-related restructuring—manifest through within-occupation change that occupation-level measurement misses.

The Hershbein and Kahn study represents perhaps the most methodologically rigorous application of job posting data to automation questions, exploiting geographic variation in recession severity as a natural experiment. Yet even this analysis faces the demand-versus-employment limitation: skill upgrading in postings may reflect employers raising requirements during slack labour markets rather than genuine changes in job content. Modestino et al. (2016) provide evidence consistent with this “opportunistic upgrading” interpretation, finding that skill requirements declined as labour markets tightened post-recession. Whether within-occupation skill changes in postings translate to within-occupation changes in actual work remains an open empirical question.

2.6.5 Automation Exposure Variation

Brynjolfsson et al. (2018) developed measures of machine learning suitability for individual tasks, finding substantial variation in automation exposure across tasks within occupational categories. An occupation classified as “high automation risk” at aggregate level may contain tasks spanning the full range from highly automatable to automation-resistant. Averaging over this within-occupation variation produces misleading risk assessments.

Webb (2020) extended this analysis by separately measuring exposure to robotics, software, and artificial intelligence. Different automation technologies affect different task types, and occupations contain varying mixes of tasks exposed to each technology. An occupation may face high exposure to software automation (affecting routine cognitive tasks) but low exposure to robotics (affecting routine manual tasks). Occupation-level measures that aggregate across automation types obscure these technology-specific patterns.

Recent methodological advances enable more granular measurement. Sasser Modestino et al. (2025) developed the Job Ad Analysis Toolkit, extracting O*NET features from over 155 million online job advertisements (2015–2025) using fine-tuned language models. This approach enables measurement of occupation, skill, task, and technology requirements at unprecedented scale, revealing within-occupation variation invisible to survey-based meth-

ods. Loaiza and Rigobon (2024) introduce metrics that separately measure automation risk and augmentation potential, finding that higher human-capability scores positively associate with employment growth—underscoring the importance of distinguishing substitution from complementarity effects that operate within occupational categories.

2.6.6 Labour Market Organisation Beyond Occupations

Marinescu and Rathelot (2018) examined how job seekers actually search for work, finding that search behaviour crosses occupational boundaries in systematic ways. Workers search for jobs based on skill similarity rather than occupational classification, applying to positions in different occupational categories that require similar skills. This search behaviour suggests that workers themselves do not treat occupational boundaries as constraints, instead navigating labour markets organised around skill clusters.

Kässi and Lehdonvirta (2018) demonstrated that online labour platforms reveal real-time demand dynamics invisible in official statistics. Traditional labour market indicators, based on surveys with substantial publication lags, cannot capture rapid restructuring. Job posting data provides higher-frequency measurement that can detect within-occupation shifts as they occur.

2.6.7 Foundation Models and Large Language Models

The most recent methodological frontier applies large language models and foundation models to labour market analysis. Vafa et al. (2024a) introduce CAREER, a foundation model trained on millions of career trajectories that learns representations of jobs from sequences of employment transitions. Rather than relying on pre-defined occupational taxonomies, CAREER discovers latent job structure from how workers actually move between positions—revealing that labour market organisation may differ substantially from official classifications. The model enables prediction of career transitions and identification of job similarity that standard occupational codes cannot capture.

Du et al. (2024) develop LABOR-LLM, using large language models to construct occupational representations from job posting text. Their approach extracts semantic features of jobs that traditional skill taxonomies miss, enabling measurement of job similarity based on actual content rather than administrative classification. Critically, these language-based representations reveal within-occupation heterogeneity invisible to standard measurement:

jobs classified identically in official statistics may have quite different LABOR-LLM representations reflecting genuinely different work content.

Athey and Imbens (2019) provide the methodological foundation for this work, demonstrating how machine learning methods can complement econometric approaches for causal inference and policy evaluation. Their framework—integrating prediction with causal identification—shapes how computational labour market research distinguishes patterns that merely exist from patterns that reflect causal mechanisms. The foundation model approaches build on this integration, using predictive modelling to reveal structure while leaving causal questions about what *produces* that structure to subsequent analysis.

Methodological advances in natural language processing have substantially expanded the scope of job posting analysis. Zhan et al. (2024) survey NLP applications for human resources, documenting the rapid expansion of automated methods for analysing job postings, resumes, and workforce data. de Lira et al. (2024) provide a comprehensive review of data science methods for job market analysis, identifying text mining, machine learning, and network analysis as dominant approaches. Frank, Moro, and Pentland (2024) present a comprehensive methodology for processing online job postings to generate labour-market intelligence, demonstrating how real-time job vacancy data can complement traditional labour market statistics. The Bureau of Labor Statistics has also developed new skills data products, creating a suite of skills tables linked to ten-year occupational projections that provide information about skill importance across occupations (Bureau of Labor Statistics, 2024). These methodological advances enable increasingly granular analysis of labour market structure.

These foundation model approaches represent significant methodological advances for measuring labour market structure. Yet they share a limitation consequential for automation research: they document how labour markets are organised without explaining why. CAREER reveals that workers transition between jobs in patterns that cross occupational boundaries, but does not examine whether automation drives these transitions. LABOR-LLM shows that jobs within occupations differ semantically, but does not test whether these differences relate to automation exposure. The sophisticated measurement these tools enable has not yet been applied to the question motivating this thesis: how does automation systematically relate to within-occupation structure?

2.6.8 Implications for Measurement

The computational literature establishes five empirical findings relevant to measuring automation's effects:

Within-occupation heterogeneity is empirically large. Deming and Kahn's finding

that within-occupation skill variation often exceeds between-occupation variation confirms that occupational categories aggregate over systematically different job types.

Heterogeneity is growing. Atalay and colleagues' finding that within-occupation variance has increased over time implies that occupation-level measurement becomes progressively less valid.

Skills form clusters crossing occupational boundaries. Alabdulkareem and colleagues' network analysis suggests that skill-based rather than occupation-based units may be more appropriate for understanding labour market structure.

Restructuring occurs within occupations. Hershbein and Kahn's finding that recession-driven skill upgrading occurred within rather than between occupations demonstrates that economic shocks manifest through within-occupation change.

Automation exposure varies within occupations. Brynjolfsson, Webb, and colleagues' task-level analyses reveal that occupation-level automation risk scores obscure substantial within-occupation variation.

Yet these studies share a critical limitation: they document the *existence* of within-occupation heterogeneity without examining its *relationship to automation*. Deming and Kahn show that skill requirements vary within occupations but do not test whether this variation correlates with automation exposure. Atalay and colleagues demonstrate growing heterogeneity but do not examine whether automation drives this divergence. Hershbein and Kahn find within-occupation skill upgrading during recessions but do not distinguish automation-driven restructuring from other mechanisms. Even the foundation model approaches—CAREER and LABOR-LLM—reveal sophisticated labour market structure without connecting that structure to technological change.

This gap is consequential. The theoretical traditions reviewed above predict not merely that within-occupation heterogeneity exists, but that automation *operates through* that heterogeneity—that automation-exposed job types should contract while complementary types expand, that technological change should shift the boundaries between job type clusters, that the null occupation-level findings in automation research reflect aggregation over offsetting within-occupation effects. These predictions require empirical tests that link sub-occupational structure to automation dynamics. No study reviewed above provides such tests.

This thesis addresses this gap. Using Australian job advertisement data and unsupervised clustering methods, it identifies sub-occupational job type clusters and examines whether these clusters exhibit the automation-related dynamics that theory predicts. The contribution is not merely methodological—applying clustering to job data—but substantive: testing whether automation systematically relates to within-occupation structure in ways that occupation-level analysis obscures.

2.7 The Measurement-Theory Disconnect

The preceding sections reviewed four scholarly traditions—organisational behaviour, economics, computer science, and sociology—each contributing essential insight into how automation affects labour markets. Computational research using text analysis of job advertisements provides empirical evidence that validates these traditions’ theoretical predictions. This section synthesises these perspectives to identify the measurement-theory disconnect at the heart of automation research.

2.7.1 Convergent Predictions

Despite developing largely independently, the four traditions converge on a common prediction: automation should generate substantial within-occupation restructuring. Computational evidence confirms this prediction empirically.

Organisational behaviour predicts continuous job redesign. The job characteristics tradition establishes that jobs are designed rather than given (Hackman & Oldham, 1976); job crafting theory shows that workers actively reshape their jobs (Wrzesniewski & Dutton, 2001); strategic HRM demonstrates that firms bundle tasks strategically (Huselid, 1995). When technology changes task requirements, organisations redesign jobs—potentially preserving occupational categories while transforming job content.

Economics predicts task-level substitution. The task model establishes that automation affects specific tasks, not whole occupations (Autor, Levy, & Murnane, 2003); routine tasks are automated while non-routine tasks may be complemented (Acemoglu & Autor, 2011b). If occupations contain both routine and non-routine components, automation changes within-occupation task composition while occupation-level employment may remain stable.

Computer science establishes that automation technologies are heterogeneous. Rule-based systems, machine learning, deep learning, and generative AI automate fundamentally different task types (LeCun et al., 2015; van der Aalst et al., 2018). The routine/non-routine boundary shifts as technologies advance; tasks once considered automation-resistant become automatable. Within any occupation, different tasks face exposure to different technologies, creating heterogeneous automation effects that occupation-level measures aggregate away.

Sociology demonstrates that occupational boundaries are fluid. Abbott (1988) jurisdictional theory shows that occupations redefine their boundaries in response to technological change; Weeden (2002) closure analysis reveals substantial within-occupation heterogeneity; labour process theory shows that “routine” and “skilled” are socially constructed categories

(Attewell, 1990; Braverman, 1974). Occupational classifications impose artificial stability on inherently dynamic processes.

Computational evidence validates these predictions empirically. Text analysis of job postings reveals that within-occupation skill variation often exceeds between-occupation variation (Deming & Kahn, 2018a), that this heterogeneity is growing over time (Atalay et al., 2020a), and that automation exposure varies substantially within occupational categories (Webb, 2020). This evidence confirms what theory predicts but does not itself constitute a theoretical tradition—it provides empirical tools that reveal patterns other methods miss.

Table 2.1 summarises how each tradition contributes to understanding the measurement-theory disconnect.

Table 2.1: Four Scholarly Traditions and Their Contributions

Tradition	Core Insight	Key Prediction	Limitation
Organisational Behaviour	Jobs are designed and redesigned by organisations and workers	Automation triggers job redesign, not just job elimination	Does not predict <i>which</i> job types emerge
Economics	Automation affects tasks, not occupations; some tasks substituted, others complemented	Within-occupation task restructuring with stable occupation-level employment	Measurement remains at occupation level
Computer Science	Automation technologies differ fundamentally in which tasks they automate	Technology-specific effects require disaggregated measurement	Focus on technical capability, not labour market outcomes
Sociology	Occupational boundaries are socially constructed and fluid	Classifications impose artificial stability; cannot capture restructuring	Limited to professional occupations; qualitative methods

2.7.2 Synthesising a Working Definition

The convergence of these four traditions enables a definition of automation that addresses the limitations of existing conceptualisations. Recall the competing definitions introduced at the outset: the computer science definition specifying functions and levels of automation, the economics definition emphasising displacement effects and labour share, and the management definition framing automation as a strategic choice between substitution and augmentation. Each captures something important but misses insights from other traditions.

Integrating the four traditions reviewed above yields a more comprehensive understand-

ing. *Economics* establishes that automation operates at the task level, not the occupation level—technologies substitute for specific tasks while complementing others, meaning automation’s effect on any job depends on its task composition. *Computer science* establishes that automation technologies are heterogeneous—rule-based systems, machine learning, and generative AI automate fundamentally different task types, and the automation frontier shifts continuously. *Organisational behaviour* establishes that automation is mediated by firm-level implementation decisions—identical technologies produce different outcomes depending on how organisations deploy them, redesign work, and manage change. *Sociology* establishes that what counts as “automatable” reflects social construction, not just technical feasibility—the boundary between routine and non-routine work is contested and historically contingent. Computational evidence confirms that automation exposure varies substantially within occupational categories, validating the theoretical prediction that occupation-level measurement obscures task-level dynamics.

This synthesis yields the following definition:

Definition. *Automation* is the deployment of technology to perform tasks previously performed by human labour. Three features characterise this construct: (1) *technology heterogeneity*—different technologies (rule-based systems, machine learning, generative AI) substitute for fundamentally different task types; (2) *task-level operation*—automation affects specific tasks within jobs, not occupations wholesale; and (3) *firm-level implementation*—identical technologies produce different labour market effects depending on organisational choices about deployment, integration, and work redesign. These features imply that automation’s effects are inherently heterogeneous, varying across technologies, firms, and tasks even within the same occupation.

This definition has methodological consequences. If automation varies by technology type, task composition, and firm implementation, then single-indicator measures (such as occupation-level automation risk scores) will systematically fail to capture its effects. The measurement challenges this creates motivate the sub-occupational analysis this thesis develops.

2.7.3 Why Occupations Remain Dominant Despite Theory

If four traditions independently predict within-occupation heterogeneity, and computational evidence confirms it, why does empirical research continue using occupational categories?

The answer lies in what Griliches (1986) termed the “desert of data”—the pragmatic constraint that researchers must use available data even when it imperfectly measures theoretical constructs. Official labour market statistics are collected and reported at occupational level because occupational classifications serve administrative functions: census enumeration, immigration policy, award wage determination, and workforce planning. These administrative purposes require stable categories that can be consistently applied across contexts and time periods. The result is that occupation-level data is abundant, standardised, and longitudinally comparable, while sub-occupational data is scarce, non-standardised, and difficult to link across sources.

This data availability shapes research practice. Economists analysing employment trends use census or labour force survey data organised by occupation. Sociologists studying stratification use occupational categories as the finest available measure of labour market position. Policy analysts assessing automation risk apply occupation-level frameworks because policy interventions (retraining programs, education funding, immigration targets) operate at occupational level. The homogeneity assumption is not defended theoretically but adopted pragmatically—researchers use the data they have, not the data theory requires.

The computational studies reviewed in Section 2.6 represent a partial escape from this constraint. Job advertisement data provides task and skill information at the individual job posting level, enabling analysis *within* occupational categories. Deming and Kahn, Atalay and colleagues, and Hershbein and Kahn all exploited job posting data to examine variation that official statistics aggregate away. Yet these studies document heterogeneity without explaining its origins or predicting its structure. Deming and Kahn show that within-occupation skill variation is large but not why firms within the same occupation demand different skills. Atalay and colleagues demonstrate that heterogeneity is growing but not what mechanisms drive this divergence. The computational literature establishes the empirical reality of within-occupation variation while leaving theoretical questions—why does this variation exist? what forms should it take? how does it respond to technological change?—largely unaddressed.

2.7.4 Divergent Measurement

Despite converging theoretical predictions and growing computational evidence, mainstream empirical automation research continues operating at occupation level. The dominant paradigm measures automation’s effects by examining occupation-level employment changes, occupation-level wage changes, or occupation-level automation risk scores. This approach assumes that occupations are homogeneous units suitable for measuring technological change—an assumption that all four traditions reject and that computational

evidence contradicts.

The disconnect between theory and measurement creates systematic blindness. If automation primarily affects task composition within occupations, and if occupations contain heterogeneous task bundles with offsetting automation effects, then occupation-level measurement will find null results even when substantial restructuring occurs. Routine-intensive job types within an occupation may contract while complementary job types expand, producing near-zero net occupation-level change. The restructuring is real but invisible to occupation-level observation. As Acemoglu (2024) argues, even substantial task-level productivity gains from AI may translate into modest aggregate effects when mechanisms for wage pass-through are weak—a pattern more consistent with task restructuring than with wholesale occupational displacement. Autor (2024) proposes that AI could potentially *restore* middle-class jobs by enabling workers to perform tasks previously requiring expensive credentialing—but only if institutional structures support such deployment, reinforcing the contingency of automation’s effects on implementation choices.

2.7.5 The Aggregation Bias Mechanism

This synthesis identifies aggregation bias as the mechanism producing the disconnect between theoretical predictions and empirical findings. Aggregation bias occurs when averaging over heterogeneous units obscures systematic variation within those units. In the automation context:

1. Occupations contain multiple job types with different task compositions
2. These job types face different automation exposure based on task content
3. Automation causes some job types to contract while others expand
4. Occupation-level measurement averages these offsetting effects
5. The result is near-zero occupation-level effects despite substantial restructuring

This mechanism explains the paradox motivating this thesis: automation appears simultaneously transformative (in qualitative accounts, firm-level studies, and worker surveys) and invisible (in occupation-level employment statistics). The discrepancy reflects not absence of restructuring but measurement operating at inappropriate aggregation level.

2.7.6 Alternative Perspectives and Boundary Conditions

The aggregation bias argument developed above warrants scrutiny. Three perspectives suggest conditions under which occupation-level analysis may remain appropriate.

First, occupations as institutional realities. Even if within-occupation heterogeneity is substantial, occupational categories structure labour market institutions in ways that matter causally. Licensing requirements, collective bargaining agreements, and educational credentials operate at occupational level. Award wage-setting in Australia applies to occupational categories regardless of within-occupation task variation. If these institutional structures determine outcomes, occupation-level analysis captures the relevant causal mechanisms even when within-occupation heterogeneity exists. The counterargument is that institutional structures may constrain *wage* adjustment while allowing *task* adjustment—precisely the within-occupation restructuring this thesis examines.

Second, occupational sorting as equilibrium. Labour economists might argue that within-occupation heterogeneity reflects efficient sorting rather than measurement failure. If workers self-select into task specialisations matching their comparative advantage, within-occupation variation represents equilibrium differentiation rather than aggregation obscuring automation effects. The counterargument is that sorting occurs *within* the occupation, not *because of* occupational classification. Understanding this sorting requires sub-occupational analysis regardless of whether it reflects efficient equilibrium.

Third, practical constraints on disaggregation. Official statistics, policy interventions, and workforce planning necessarily operate at aggregate levels. Even if sub-occupational dynamics are theoretically important, practical constraints may require occupation-level analysis. This critique is pragmatic rather than substantive—it does not challenge the theoretical argument but questions whether disaggregated analysis is feasible. Chapter 6 addresses this critique by demonstrating that unsupervised clustering can recover meaningful sub-occupational structure from available data.

These alternative perspectives identify important boundary conditions. Occupation-level analysis remains appropriate when institutional structures (wages, credentials, licensing) are the outcomes of interest. The aggregation bias argument applies specifically to understanding automation's effects on task structure and job content—domains where within-occupation variation is theoretically predicted and empirically documented. The thesis does not claim that occupational categories are useless, only that they are insufficient for detecting automation's primary effects on how work is organised.

2.7.7 Requirements for Resolution

Resolving the measurement-theory disconnect requires empirical methods capable of observing sub-occupational structure. Such methods must:

Identify discrete job types within occupations. Rather than treating occupations as homogeneous, the method must discover internal structure—clusters of jobs with similar skill profiles that may face different automation exposure.

Allow structure to emerge from data. Imposing predetermined categories risks replicating the problems of existing classifications. Unsupervised methods that discover latent structure without requiring pre-specification are preferable.

Capture temporal dynamics. Understanding automation's effects requires observing how job type clusters evolve over time—which expand, which contract, and how boundaries shift.

Connect to theoretical predictions. The identified structure should be interpretable in terms of theoretical frameworks, enabling tests of whether observed patterns match predictions about task substitution, skill complementarity, and market thickness effects.

The following section articulates how this thesis addresses these requirements.

2.8 Research Gap and Thesis Contribution

The literature reviewed above identifies a fundamental gap: four scholarly traditions independently predict within-occupation restructuring from automation, yet measurement frameworks continue operating at occupation level, creating systematic blindness to automation's primary effects. This section articulates how this thesis addresses this gap.

2.8.1 The Gap

The gap is not merely methodological but epistemic. Existing research embodies a fundamental mismatch: theories operate at task level while measurement operates at occupation level. This is not a matter of data limitation alone—it reflects implicit acceptance of occupational categories as meaningful analytical units despite four traditions' theoretical predictions that they are not. The consequence is a literature that repeatedly fails to detect effects that theory predicts should be large, interpreting these null findings as evidence that automation's labour market impact is modest rather than as evidence of measurement inadequacy. Existing research lacks:

Theoretical framework for within-occupation clustering. While job design theory establishes that jobs are designed and economics predicts task-level substitution, no formal

model derives conditions under which discrete job type clusters emerge within occupations. Chapter 3 addresses this gap by developing an equilibrium model that predicts endogenous clustering, market thickness effects, and automation-driven boundary shifts.

Empirical methods for observing sub-occupational structure. Computational research has documented within-occupation heterogeneity but typically treats it as descriptive finding rather than testing theoretical predictions. Chapter 6 addresses this gap by applying HDBSCAN clustering to identify job type clusters and testing whether observed patterns match theoretical predictions.

Integration of theoretical traditions. The four traditions reviewed above have developed largely in isolation. This thesis integrates insights from organisational behaviour (jobs are designed), economics (automation affects tasks), computer science (technologies differ in which tasks they automate), and sociology (boundaries are fluid) into a coherent framework, applying computational methods to reveal the sub-occupational structure these traditions predict.

Australian evidence. The computational studies reviewed above focus primarily on US and European labour markets. Australian labour markets differ in institutional arrangements (award wage-setting, enterprise bargaining, Fair Work regulations), industrial structure (service sector dominance, mining sector importance), and geographic organisation (extreme metropolitan concentration). Whether international findings generalise to Australian contexts remains empirically open.

2.8.2 Thesis Contribution

This thesis addresses the gap through six contributions across theoretical, methodological, and empirical domains, mapping to the six contributions developed across the empirical chapters and synthesised in Chapter 10:

First, a restructuring theory of automation impacts. Chapter 3 develops a formal equilibrium model in which automation's primary labour market effect operates through continuous task reorganisation within occupations rather than wholesale occupation elimination. Building on Roy (1951) occupational choice and Acemoglu and Autor (2011b) task-based frameworks, the model derives five testable propositions from a directed search framework with endogenous job design, reconciling the skill upgrading, polarisation, and degradation literatures by showing all three patterns may coexist simultaneously at cluster level.

Second, a scalable methodology for sub-occupational measurement. Chapter 6 applies HDBSCAN clustering to identify discrete job type clusters within occupations, demonstrating advantages over static taxonomies: endogenous cluster count, explicit noise

handling, and dynamic temporal tracking across 2,000 clustering runs (250 occupations $\times$ 8 years). Chapter 8 extends this infrastructure into a cross-occupation skill proximity network for mapping feasible career transition pathways.

Third, empirical quantification of aggregation bias. Chapter 5 tests occupation-level automation frameworks (Routine-Biased Technical Change, Frey-Osborne) using Australian job advertisement data and establishes through variance decomposition that within-occupation variation dominates between-occupation variation, providing the empirical signature of aggregation bias that prior literature could only hypothesise.

Fourth, the technology succession mechanism. Chapter 6 identifies a concrete mechanism consistent with within-occupation restructuring: clusters organised around legacy software platforms (Lotus Notes, ASP.NET, Silverlight) dissolve while clusters requiring emerging technologies (AWS, Docker, Python, Salesforce) form. This succession pattern recasts technology-driven restructuring as a discrete, observable platform-lifecycle process rather than an abstract routine/non-routine substitution.

Fifth, heterogeneous demand-side dynamics. Chapter 7 tracks cluster births, deaths, and skill-content evolution across 250 occupations, documenting that newly-emerged clusters exhibit systematically different skill requirements depending on the occupation in which they arise (recombination versus unbundling), and that cluster churn correlates strongly with the speed of skill change rather than the static level of automation exposure.

Sixth, cluster-level transition feasibility. Chapter 8 constructs a skill proximity network connecting clusters across occupations, mapping feasible career transition pathways for workers in at-risk clusters and providing empirical foundations for cluster-based workforce development that occupation-level analysis cannot support.

This contribution structure addresses both the theoretical and empirical dimensions of the gap: it develops the formal model that earlier literature lacked (Ch. 3), provides the measurement infrastructure that computational research had not extended to test theoretical predictions (Chs. 4–5), tests the within-occupation clustering, technology succession, and temporal dynamics that the four traditions independently predicted (Chs. 6–7), and applies the resulting infrastructure to feasible career-transition mapping (Ch. 8).

2.8.3 Contribution to Literature

This thesis contributes to multiple literatures:

Automation and employment: By demonstrating that occupation-level null findings reflect aggregation bias rather than absence of effects, the thesis explains the disconnect between dire predictions and benign aggregate findings that has puzzled researchers.

Labour economics: By formalising conditions for within-occupation clustering and

testing predictions about market thickness and automation effects, the thesis extends task-based frameworks to sub-occupational dynamics.

Computational social science: By applying unsupervised clustering to test theory-derived predictions rather than merely describing patterns, the thesis demonstrates how machine learning methods can serve hypothesis-testing rather than purely exploratory purposes.

Australian labour markets: By providing the first systematic analysis of within-occupation restructuring in Australian job advertisement data, the thesis establishes whether international findings generalise and how Australian institutions shape automation's effects.

2.9 Conclusion

This chapter has argued that four independent scholarly traditions—organisational behaviour, labour economics, computer science, and occupational sociology—converge on a common prediction: automation generates substantial within-occupation restructuring that current measurement frameworks cannot detect. Computational research using job advertisement text analysis has begun validating this prediction empirically, revealing within-occupation skill variation that often exceeds between-occupation variation. The fundamental gap is not empirical but epistemic: theories operate at task level while measurement operates at occupation level, producing systematic null findings that reflect aggregation bias rather than absence of effects.

Read in historical perspective, this gap is also a gap in the historical record. Throughout the 2012–2019 machine-learning era—during which cloud computing, mobile platforms, and statistical machine learning diffused through Australian workplaces—automation research relied on occupation-level frameworks that the four traditions reviewed above predicted would systematically miss the period's primary restructuring dynamics. No within-occupation measurement infrastructure existed at the time to test those predictions empirically. The thesis closes this gap retrospectively, applying density-based clustering to 9.5 million Australian job advertisements to construct the within-occupation baseline for the second wave of automation that the contemporaneous literature could not produce. The following chapter develops the formal theoretical framework that generates testable predictions about within-occupation clustering, market thickness effects, and automation-driven restructuring.

Chapter 3

Theoretical Framework: Why Occupations Fragment

3.1 Introduction

Chapter 2 identified a measurement-theory disconnect at the heart of automation research: four scholarly traditions independently predict within-occupation restructuring, yet empirical measurement frameworks continue to operate at the occupation level. This chapter develops a theoretical framework that resolves the disconnect by deriving the conditions under which discrete job type clusters emerge endogenously within occupational boundaries, and uses the framework to generate five testable propositions for the empirical chapters.

The framework synthesises four traditions reviewed in Chapter 2. From *organisational behaviour*, the job design literature establishes that jobs are strategic designs rather than natural categories, with firms making deliberate choices about task bundling in response to technological change (Hackman & Oldham, 1976; Morgeson & Humphrey, 2006). Parker and Grote (2022) extends this view to contemporary automation specifically, arguing that the diffusion of algorithmic systems triggers active reconfiguration of work practices and information flows within firms rather than passive task displacement — a claim the present chapter formalises by treating job design as the firm’s optimal choice variable. From *labour economics*, the task-based framework predicts that automation affects specific tasks rather than whole occupations, generating within-occupation restructuring as routine tasks are automated while complementary tasks expand (Acemoglu & Autor, 2011a; Acemoglu &

Restrepo, 2018; Autor, Levy, & Murnane, 2003). From *occupational sociology*, classifications impose artificial stability on inherently fluid processes, with substantial heterogeneity persisting within occupational categories (Abbott, 1988; Weeden, 2002). From *computational social science*, text analysis of job postings reveals that within-occupation skill variation often exceeds between-occupation variation and is growing over time (Atalay et al., 2020a; Deming & Kahn, 2018a).

Building on the canonical Roy (1951) occupational choice model and the Acemoglu and Autor (2011a) task-based framework, I incorporate two extensions: (1) multi-dimensional worker skill endowments, and (2) heterogeneous firm productivity in coordinating task bundles, extending the assignment literature (Rosen, 1974; Sattinger, 1975; Teulings, 1995). The equilibrium concept follows the directed search framework of Moen (1997) and Shimer (2005), extended to multi-dimensional job design choice. Unlike Eeckhout and Kircher (2011), who study sorting with exogenous job characteristics, the key innovation here is treating job design as a firm choice variable rather than an exogenous characteristic of vacancies. These extensions generate a prediction absent from standard models: *occupations endogenously fragment into discrete sub-occupational clusters* even when all agents face identical production technologies. The pattern has been documented empirically (Atalay et al., 2020a; Deming & Kahn, 2018a; Hershbein & Kahn, 2018a; Weeden et al., 2007), but has not previously been derived from a coherent equilibrium model.

The framework yields five propositions, each derived in the sections that follow and tested in a subsequent chapter:

1. *Endogenous Clustering.* Occupations fragment into discrete sub-occupational clusters as an equilibrium outcome of firm heterogeneity, convex coordination costs, and directed search (Chapter 6).
2. *Market Thickness.* The equilibrium number of clusters within an occupation is non-decreasing in market size (Chapter 6).
3. *Automation Effects and Aggregation Bias.* Skill-biased automation reallocates employment across clusters; when the within-occupation skill partition is balanced, the aggregate occupation-level effect is small because cluster-level effects partly offset (Chapters 5 and 6).
4. *Temporal Equilibrium Path.* The equilibrium cluster configuration evolves continuously almost everywhere, with discrete cluster births and deaths at viability thresholds (Chapter 7).
5. *Churn–Automation Speed.* Cluster churn rates increase in the speed of automation advancement, not in the static level of automation exposure (Chapter 7).

The chapter proceeds as follows. Section 3.2 develops the production-pipeline redesign mechanism that motivates the framework. Section 3.3 sets out the model environment: workers, firms, production, coordination costs, and the directed search matching protocol. Sections 3.4 through 3.6 derive the five propositions in their natural order: Section 3.4 derives Propositions 1 and 2 from the static framework; Section 3.5 extends the model to incorporate automation and derives Proposition 3; Section 3.6 develops the dynamic extension and derives Propositions 4 and 5. Section 3.7 positions the contribution relative to existing models and states the framework's scope. Section 3.8 bridges to the empirical chapters. Section 3.9 concludes.

3.2 The Redesign Mechanism

Contemporary evidence on automation adoption in firms is consistent with a redesign rather than a substitution mechanism. Bessen (2022) and Bessen et al. (2018) documents that firms adopting automation rarely reduce the workforce performing automated tasks one-for-one; instead, firms reorganise the production pipeline around the new division of labour between human and machine. Acemoglu and Restrepo (2018, 2019b) formalise the same insight in a task framework: automation displaces labour from the tasks the technology can perform while simultaneously creating new tasks in which labour retains comparative advantage. Autor et al. (2024a) provides the most direct empirical evidence for the new-task channel, documenting that approximately 60% of contemporary employment is in occupations or titles that did not exist in 1940. Brynjolfsson et al. (2023c) confirm the redesign pattern in real time for generative AI: in a customer-service deployment, the technology did not replace agents but altered which tasks each agent performed, with productivity gains concentrated among less-experienced workers.

The implication for occupational analysis is that the unit at which automation operates is the task, while the unit at which its consequences become visible is the *bundle* of tasks (the job design) into which the firm reassembles surviving and newly created tasks. Occupations aggregate over multiple bundles. Within a single occupation, technology adoption can simultaneously contract some bundles (those organised around tasks the technology absorbs) and expand others (those organised around tasks the technology complements), producing fragmentation invisible to occupation-level measurement. The substitution view of automation, which underpins forecasts such as the 47% estimate of Frey and Osborne (2017), treats the occupation as the relevant unit and so cannot represent within-occupation reallocation; the redesign view requires the cluster (bundle) as the unit of analysis.

Figure 3.1 illustrates the mechanism through the case of *Accountant (General)*, an

occupation that recurs throughout the empirical chapters that follow. Before the diffusion of cloud-based accounting platforms, the typical Generalist Accountant bundle combines bookkeeping, tax preparation, reconciliation, financial planning, client advisory, and cost analysis. After diffusion of platforms such as Xero and QuickBooks, three changes occur simultaneously. Bookkeeping, the task most fully absorbed by platform automation, is removed from the human pipeline entirely (substitution). The surviving tasks are redistributed across three new bundles: Compliance Management organised around tax and reconciliation work, Strategic Advisory organised around financial planning and client advisory, and Management Accounting organised around cost analysis and internal financial planning. Financial planning appears in both Strategic Advisory and Management Accounting, signalling complementation: automation expands demand for planning work by freeing time previously spent on routine bookkeeping, and the same task is performed in distinct contexts across the two new bundles. Three new tasks emerge at the human-machine interface that did not exist in the original bundle: software management, data analytics, and performance dashboards. The occupational code *Accountant (General)* is preserved across both panels of the figure; the bundle composition inside that code has changed from one design to three. The illustration is schematic rather than derived from this thesis's data; Chapter 6 establishes whether and how this kind of fragmentation appears in the Australian advertisement record.

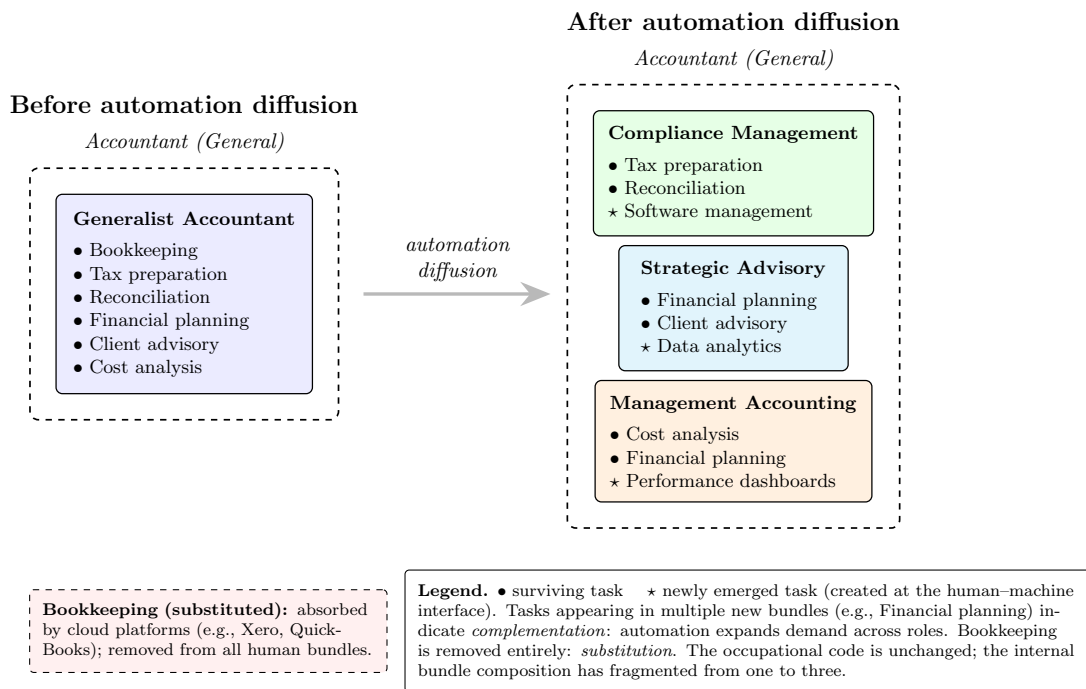
The empirical chapters that follow operationalise the bundle as the cluster of advertisements with similar skill profiles, recovered by HDBSCAN clustering applied to each occupation's annual skill matrix (Section 4.9 sets out the method). For the present chapter, the substantive point is that the bundle is the unit at which automation's consequences become observable, and any framework that aggregates to the occupation will systematically obscure them. The remainder of the chapter formalises this argument and derives its empirical implications.

3.3 The Model

To turn the redesign mechanism into testable propositions, this chapter builds a formal model in which job design is a firm choice, workers sort across the resulting designs, and the technology environment can shift. This section sets out the model in plain language with selected formulas where they aid clarity. The full equilibrium definition, existence proof via Kakutani's fixed-point theorem, and detailed regularity conditions are in Appendix C; proofs of the propositions derived below are in Appendix D.

The model yields three predictions, each becoming a proposition in the sections that follow. First, under the right combination of firm heterogeneity, coordination costs, and

Figure 3.1: Pipeline Redesign and Bundle Reconfiguration: Accountant (General)



Note. Schematic illustration of how the diffusion of cloud-based accounting platforms reorganises the task content of Accountant (General). The left panel shows a single Generalist Accountant bundle prior to diffusion; the right panel shows three sub-occupational bundles after diffusion, with bookkeeping substituted by the platform and three new tasks emerging at the human-machine interface. Symbols and colour coding are defined in the figure legend. The illustration is drawn from the prior literature on technology adoption in accounting practice and is not derived from this thesis's data.

matching frictions, the equilibrium set of job designs is finite and discrete; the cluster pattern illustrated in Figure 3.1 emerges as an outcome of the model rather than as an assumption (Section 3.4, Proposition 1). Second, the equilibrium number of clusters depends on market size, generating a sharp comparative-static between thick and thin labour markets that the empirical analysis can exploit (Section 3.4, Proposition 2). Third, once the technology environment is allowed to shift over time, the model characterises how cluster configurations evolve: which clusters expand and contract, when births and deaths occur, and how the rate of cluster turnover depends on the rate of technological change (Sections 3.5 and 3.6, Propositions 3–5).

Each component of the model is introduced in turn below, in the order in which it enters the equilibrium argument. Workers come first because they are the primitive of the supply side: their skill endowments determine the pool from which firms draw, and the firm's design problem in the subsequent subsections amounts to a choice over which workers to attract. Firms and their job-design choice come next, then the production technology and coordination costs that govern the value and cost of each design, then the directed-search

matching protocol that connects workers to firms, and finally the equilibrium concept that ties the agents' choices together.

3.3.1 Workers and Skills

There is a unit continuum of workers indexed by i , each endowed with a multi-dimensional skill vector

$$s_i \in S = [0, \bar{s}]^K, \quad (3.1)$$

where the dimensions K index distinct skills (numerical, analytical, interpersonal, technical) and the entry s_{ik} measures worker i 's strength on dimension k . Multi-dimensional skills imply that workers cannot be ranked unambiguously on a single scale: a worker strong in numerical and weak in interpersonal skills is not “better” or “worse” than one with the opposite profile, only different. This follows the comparative-advantage tradition of Sattinger (1975) and Costinot and Vogel (2010), in which horizontally-differentiated workers sort into different jobs based on which tasks they perform well rather than on a single skill ranking. In the empirical application (Chapter 6), the skill dimensions are the approximately 2,000 unique skills extracted from Burning Glass advertisements; the theoretical results hold for any finite $K \geq 2$.

3.3.2 Firms and Job Design

Each firm is characterised by a productivity parameter $\theta \in \Theta \subset \mathbb{R}_+$. Heterogeneity in θ captures cross-firm differences in management capability, organisational efficiency, and the quality of internal coordination systems, all of which the empirical productivity literature has documented as among the most robust facts in microeconomics (Bloom & Van Reenen, 2007).

Each firm chooses a *job design* $d \in \mathcal{D}$: a vector specifying the relative importance of each skill dimension for the role. For instance, a design $d = (0.7, 0.1, 0.1, 0.1)$ in a four-skill space emphasises the first dimension heavily; $d = (0.25, 0.25, 0.25, 0.25)$ emphasises all four equally. The set $\mathcal{D}$ is normalised to the unit sphere so that only relative task weights matter.

The treatment of d as a firm choice variable is the central departure from standard sorting models. The canonical reference for empirical sorting is Eeckhout and Kircher (2011), who provided the influential identification result that the strength of sorting between workers and firms can be recovered from observed wage data even when productivity is not directly observed, using search frictions as the identifying mechanism. In their framework, however, workers have skills and jobs have characteristics, both treated as exogenous

primitives: the firm cannot choose what kind of job to offer. By contrast, in the present model, the firm is endowed with productivity θ and chooses, optimally, what task bundle d to post. This single change, from exogenous to endogenous job design, is what generates within-occupation clustering as an equilibrium outcome rather than as a stipulation about the data.

3.3.3 Production

When a worker with skills s is matched to a job with design d at a firm with productivity θ , output is

$$Y(s, d, \theta) = \theta \cdot A(s, d), \quad A(s, d) = \exp[-\lambda \|s - \mu(d)\|^2]. \quad (3.2)$$

The match quality function $A(s, d)$ depends on how well the worker's skills match the job's *ideal skill profile* $\mu(d)$, the skill vector that would maximise output for design d . Match quality is highest when $s = \mu(d)$ (a perfect match yields $A = 1$) and falls off exponentially as the worker's skills depart from the ideal. The parameter $\lambda > 0$ controls how steep the fall-off is: a high λ means matching is tight (poor matches are very unproductive); a low λ means workers are reasonably substitutable across designs. The Gaussian functional form is in the family used by Eeckhout and Kircher (2011) and satisfies the conditions of Shimer and Smith (2000) for positive assortative matching under skill complementarity. The implication for matching that I will use repeatedly below is that workers with skills close to $\mu(d)$ produce far more in design d than workers whose skills are far from $\mu(d)$, even when both are nominally “qualified”.

3.3.4 Coordination Costs

A job design that asks a worker to perform many loosely related tasks is more expensive to run than one organised around a few closely related tasks. The cost comes from coordination overhead: allocating effort across tasks, managing hand-offs, training the worker across multiple competencies, and absorbing the management time required when problems arise outside the worker's primary expertise. I model this as

$$C(d, \theta) = \kappa \cdot \theta^{-\alpha} \cdot \text{span}(d)^\gamma, \quad \gamma > 1. \quad (3.3)$$

Here $\text{span}(d)$ measures the spread of task weights in design d : a specialised design concentrated on one or two dimensions has low span, while a generalist design spread across many dimensions has high span. The convexity parameter $\gamma > 1$ captures the empirical observation that coordination costs scale super-linearly with task spread: doubling the number of dimensions a worker must integrate more than doubles the management overhead,

because handoffs and interaction effects multiply.

This formulation follows the span-of-control tradition of Garicano (2000) and Garicano and Rossi-Hansberg (2006), in which workers escalate problems outside their competence to managers, and the rate of escalation rises convexly in the breadth of the worker's task domain. The productivity-dependence $\theta^{-\alpha}$ reflects the standard finding that higher-productivity firms operate superior management systems and therefore face lower coordination costs at any given bundle complexity (Bloom & Van Reenen, 2007). I impose strict convexity ($\gamma > 1$) as a parametric assumption. The escalation argument above grounds the choice: cross-task interactions grow combinatorially in the number of dimensions a worker must integrate, generating super-linear coordination costs in task spread (Garicano, 2000; Garicano & Rossi-Hansberg, 2006). Convexity is the condition under which Proposition 1 (Section 3.4) obtains; with $\gamma \leq 1$, equilibrium designs would vary continuously with firm productivity rather than cluster at a discrete set of attractors. The qualitative predictions of the framework depend on supermodularity (Assumption 1) and strict convexity ($\gamma > 1$), not on the specific functional forms of $A(s, d)$ or $C(d, \theta)$. Appendix D verifies the predictions under alternative parameterisations within the families satisfying these properties.

3.3.5 Matching: Directed Search

How do workers and firms find each other? The model uses the directed search framework of Moen (1997) and Shimer (2005). Because directed search is less familiar than its random-matching alternative, what follows sets out the framework and the role it plays in the predictions derived below.

In random matching models, the workhorse of earlier search theory, workers and firms meet stochastically, and the wage is determined ex post by bilateral bargaining once a match has formed. Random matching is analytically convenient but cannot represent the obvious empirical fact that workers can see what kind of job they are applying to: a candidate applying for “Senior Software Engineer at Atlassian” is not randomly bumping into an employer. Moen (1997) replaced random matching with a market-clearing mechanism. Firms post vacancies specifying both the wage *and* the type of job being offered, and workers observe all posts and direct their search toward the post that maximises their expected utility (wage times job-finding probability). The market segments into *submarkets* indexed by what is being offered. Shimer (2005) extended this framework to heterogeneous worker types, deriving sorting properties under skill complementarities.

In the present model, submarkets are indexed by the job design d . Each submarket has a queue length $q(d)$, the ratio of applicants to vacancies, which determines the matching probabilities via a standard matching function $h(q(d))$ for firms and $p(q(d)) = h(q(d))/q(d)$

for workers. Posted wages are submarket-specific: a firm posting design d commits to a wage schedule $w(s, d)$, with worker skill s entering through the equilibrium composition of the queue rather than through ex-post bargaining. Directed search makes the job design observable to workers *before* matching: a worker can see that some firms post a specialist bundle (high $\text{span}(d)$) and others a generalist bundle (low $\text{span}(d)$) and direct her search accordingly. The result is that each submarket attracts a queue of workers whose skill profiles cluster around $\mu(d)$, the ideal profile for that design. This sorting is what produces the empirically observable cluster structure: a tax-compliance submarket attracts workers with tax-related skills; an advisory submarket attracts workers with planning-related skills.

3.3.6 Equilibrium

With workers, firms, production, coordination costs, and matching all specified, an equilibrium ties their choices together. An equilibrium consists of a firm job-design choice $d^*(\theta)$, a wage schedule $w(s, d)$, a market utility function $U(s)$ representing the worker's outside option, and queue lengths $q(d)$, such that each agent optimises given the others' choices and free entry exhausts profits at the marginal vacancy. A firm of productivity θ chooses d to maximise expected profit:

$$\max_d h(q(d)) \cdot \int_S [\theta A(s, d) - w(s, d)] \phi(s|d) ds - C(d, \theta), \quad (3.4)$$

where $\phi(s|d)$ is the equilibrium density of worker skills in submarket d . A worker of skill s chooses the submarket that maximises expected utility:

$$U(s) = \max_{d \in \mathcal{D}} p(q(d)) \cdot w(s, d). \quad (3.5)$$

Free entry pins down expected profits at zero for the marginal vacancy posted in each submarket, after accounting for the posting cost $f > 0$ that any firm incurs to enter. Existence of equilibrium is proved in Appendix C using Kakutani's fixed-point theorem.

The equilibrium objects of central interest to this chapter are (i) the firm's optimal design function $d^*(\theta)$, which characterises which firms post which bundles; (ii) the equilibrium number of distinct bundles, n^* ; and (iii), once the dynamic extension is introduced, the evolution of these objects as the underlying technology changes.

3.4 Static Predictions: Discrete Clustering and Market Thickness

Holding technology fixed for the moment, the model yields two predictions. The first concerns whether firms cluster discretely or spread continuously across the design space; the second concerns how the number of clusters scales with market size.

3.4.1 Endogenous Clustering (Proposition 1)

The optimal design $d^*(\theta)$ depends on the interaction of production complementarity (which favours specialised designs when matched with appropriately skilled workers) and coordination costs (which penalise complexity). To study how this interaction produces clustering, it is helpful to work with a scalar index of design complexity: let $x = \text{span}(d) \in [0, \bar{x}]$ measure how spread out the design's task weights are. The firm's expected profit can then be written as a function of θ and x :

$$\pi(\theta, x) = \theta \cdot \mathbb{E}_s[A(s, x) \mid \text{queue at } x] - \kappa \cdot \theta^{-\alpha} \cdot x^\gamma. \quad (3.6)$$

The expectation is taken over the equilibrium queue of workers the design attracts in sub-market x ; under directed search this queue concentrates on workers whose skills approach the ideal $\mu(d)$ for the design.

The cross-partial derivative $\partial^2 \pi / \partial \theta \partial x$ measures how the marginal benefit of complexity rises with productivity. Under the model's assumptions, this cross-partial is strictly positive: more productive firms benefit more from running complex bundles, both because they extract more output from each match and because their lower coordination costs make the marginal increase in complexity cheaper to absorb. Topkis (1998)'s monotone selection theorem then implies that the optimal complexity $x^*(\theta)$ is monotonically increasing in θ : higher-productivity firms select more specialised designs.

Monotonicity of $x^*(\theta)$ is necessary but not sufficient for the discreteness claim: a monotonic sorting function can vary smoothly with θ rather than jump. The discreteness in Proposition 1 requires two further forces. *First*, convexity of coordination costs ($\gamma > 1$) transforms the firm's profit-in-complexity function from monotonic to multi-peaked. With $\gamma > 1$, the marginal coordination cost of a slightly more specialised design rises faster than the marginal production return except at specific attractor points where the profit function has local maxima; the firm strictly prefers one of those attractors to any interior point between them. *Second*, fixed posting costs $f > 0$ impose an integer constraint on entry: a design is posted only if expected profit at the optimal firm type exceeds f , and the set of

designs meeting this threshold is finite for any positive f .

The argument therefore runs in three steps. Supermodularity together with Topkis' theorem pins down a monotonic mapping from θ to optimal complexity. Strict convexity of coordination costs transforms that mapping into one whose profit function has discrete attractors rather than continuous variation across θ . Free entry under posting cost f selects a finite subset of those attractors as viable. The composition produces bunching at a finite set of equilibrium designs $\{d_1^*, d_2^*, \dots, d_n^*\}$, delivering the first proposition.

Proposition 1 (Endogenous Clustering). *Under supermodularity in production, convex coordination costs, and independence of firm productivity from worker skills (Assumptions 1–3 of Appendix C), equilibrium exhibits positive assortative matching with discontinuous jumps in the job design function $d^*(\theta)$, generating discrete clusters of job types within occupational boundaries.*

The empirical implication is that a clustering algorithm applied to within-occupation job advertisement data should recover a finite, identifiable set of clusters rather than a single diffuse blob. Chapter 6 tests this prediction directly by applying HDBSCAN clustering independently to each of 250 Australian occupations and examining whether the algorithm returns discrete clusters or only noise.

3.4.2 Market Thickness (Proposition 2)

The number of clusters n^* is not exogenous; it is determined by free entry. A design d is posted in equilibrium only if expected profit exceeds the posting cost:

$$\max_{\theta} \pi(\theta, d, q(d)) \geq f. \quad (3.7)$$

Whether this condition holds depends on how thick the queue $q(d)$ is at design d , which in turn depends on the size of the market N (the mass of workers searching for work in the relevant occupation).

In a thin market, the queue of well-matched workers behind any specialist design is short: there are simply not enough workers in the local labour pool with the right skill profile to fill the vacancy at competitive wages. Expected profit from posting the specialist design falls below the posting cost, and only generalist designs are posted. As N grows, the queue behind the marginal specialist design thickens enough for the viability condition (3.7) to hold, and the equilibrium cluster count increases by an integer. The relationship $n^*(N)$ is therefore a step function: locally constant, with discrete upward jumps at threshold market sizes $\{N_1, N_2, \dots\}$. The j -th threshold N_j is the smallest market size at which the j -th design generates expected profits covering posting costs for some firm type.

Proposition 2 (Market Thickness). *The equilibrium number of clusters $n^*(N)$ is non-decreasing in market size N , with discrete upward jumps at threshold market sizes $\{N_1, N_2, \dots\}$ where successive job designs become viable.*

Empirically, metropolitan markets within Australia should support more sub-occupational clusters than regional markets in the same occupation, because the thicker labour pool in metropolitan areas allows more specialist designs to recover their posting costs. Chapter 6 tests this proposition by relating cluster count to local market size across Australian regions, holding the occupation fixed.

3.5 Automation Effects and Aggregation Bias (Proposition 3)

The static framework above holds the technology fixed. To study automation, the framework needs an additional primitive: an automation technology parameterised by intensity $\rho \in [0, 1]$, with a *skill partition* describing which skill dimensions the technology substitutes for, which it complements, and which it leaves neutral. Let $\mathcal{S}^-(\tau)$ denote the skill dimensions *substituted* by technology τ , $\mathcal{S}^+(\tau)$ the *complementary* dimensions, and $\mathcal{S}^0(\tau)$ the *neutral* dimensions, with $\mathcal{S}^-(\tau) \cup \mathcal{S}^+(\tau) \cup \mathcal{S}^0(\tau) = \{1, \dots, K\}$.

Automation enters the model by shifting the ideal skill profile of each job design. As ρ rises, the ideal profile $\mu(d, \rho)$ moves: $\partial\mu_k/\partial\rho < 0$ for substituted skills $k \in \mathcal{S}^-(\tau)$, because the machine takes over those tasks and the human job design needs less of that skill, and $\partial\mu_k/\partial\rho > 0$ for complementary skills $k \in \mathcal{S}^+(\tau)$, because the machine raises the marginal value of the human skills that complement it. Job designs whose existing ideal profile concentrates on substituted skills become harder to match with available workers as ρ rises (lower expected $A(s, d)$), reducing the design's expected profit. Designs whose profile concentrates on complementary skills become easier to match, raising profit.

Two cluster-level effects follow directly. Clusters organised around substituted skill dimensions contract as their match quality degrades; clusters organised around complementary dimensions expand. Both effects are direct consequences of how ρ enters the match quality function, and they hold for any technology with a non-empty substitution-complementation partition.

What happens at the occupation level? Occupation-level employment is the sum of cluster-level employments. If the within-occupation distribution of cluster types is *balanced*, with both substitution-intensive and complement-intensive clusters non-trivially represented, then the cluster-level contractions and expansions partially offset when summed, and the aggregate occupation-level effect is small relative to the sum of the absolute cluster-

level changes:

$$\left| \frac{\partial L_{\text{occupation}}}{\partial \rho} \right| \ll \sum_c \left| \frac{\partial L_c}{\partial \rho} \right|. \quad (3.8)$$

This is *aggregation bias*: occupation-level measurement systematically understates the magnitude of within-occupation reallocation. The scope condition for the bias is formalised in Appendix C as a bound on the ratio of employment mass between substitution-intensive and complement-intensive clusters: $\sigma_-(\tau)/\sigma_+(\tau) \in [\underline{r}, \bar{r}]$ for some $0 < \underline{r} \leq 1 \leq \bar{r} < \infty$, where $\sigma_-(\tau)$ and $\sigma_+(\tau)$ denote total employment in clusters of each type. The condition fails for degenerate partitions in which one side of the substitution–complementation split dominates — for example, a technology that substitutes for all skills with no complementary dimensions would produce occupation-level contraction rather than aggregation bias. The bound is empirically testable through the cluster-level analysis of Chapter 6, which characterises which clusters expand and contract within each occupation.

Proposition 3 (Automation Effects and Aggregation Bias). *For any technology τ with non-empty skill partition $\mathcal{P}(\tau) = \{S^-(\tau), S^+(\tau), S^0(\tau)\}$, skill-biased automation generates (a) substitution-cluster contraction $\partial L_c / \partial \rho < 0$ for clusters emphasising $S^-(\tau)$ dimensions, (b) complement-cluster expansion $\partial L_c / \partial \rho > 0$ for clusters emphasising $S^+(\tau)$ dimensions, and (c) aggregation bias as in equation (3.8) when both partitions are non-empty and the within-occupation distribution of cluster types is balanced.*

The proposition has a sharp testable implication. Two prominent occupation-level automation frameworks make predictions about how an occupation’s labour demand should respond to automation: the Routine–Cognitive–Manual taxonomy of Acemoglu and Autor (2011b) and the Frey–Osborne automation probabilities of Frey and Osborne (2017). Under the scope condition of a balanced partition, part (c) predicts that these frameworks should explain very little of the variance in occupation-level labour demand: the within-occupation cluster reallocation that drives the action does not register at the level the frameworks operate at. Cluster-level analysis, by contrast, should reveal extensive heterogeneity, with some clusters expanding and others contracting within the same occupational code. Chapters 5 and 6 jointly test the proposition: Chapter 5 runs occupation-level regressions of labour demand on the RCM and Frey–Osborne automation measures; Chapter 6 decomposes labour-demand variance into within- and between-occupation components.

3.6 Dynamic Predictions: Cluster Births, Deaths, and Churn (Propositions 4 and 5)

The analysis above treats ρ as fixed. In reality, automation advances over time: $\rho = \rho(t)$. What does the model predict about how cluster configurations evolve along this path?

A job design d supports a viable cluster at automation level ρ if there exists some firm type for which expected profit exceeds the posting cost:

$$\max_{\theta} \pi(\theta, d, q(d), \rho) \geq f. \quad (3.9)$$

As $\rho(t)$ rises continuously, two regimes alternate.

Continuous evolution. Between viability thresholds, the equilibrium set of viable clusters is stable; each cluster's centroid drifts continuously as the ideal profile $\mu(d, \rho)$ evolves under the changing technology, and employment shares across clusters reallocate continuously as match qualities shift.

Discrete transitions. At critical thresholds $\bar{\rho}_c$, the viability condition for cluster c either ceases to hold (*cluster death*) or newly comes to hold (*cluster birth*); these transitions produce discrete jumps in the cluster configuration. The maximum theorem of Berge (1963) ensures that the equilibrium correspondence is continuous between thresholds; the thresholds themselves are crossings where the set of viable designs changes, and under standard transversality conditions form a measure-zero set in $[0, 1]$. The mechanism parallels the entry–exit dynamics of Hopenhayn (1992), with bundles (rather than firms) as the units crossing the entry threshold and automation intensity playing the role of the evolving cost environment.

Proposition 4 (Temporal Equilibrium Path). *For continuously differentiable $\rho(t)$ with $d\rho/dt > 0$, the equilibrium cluster configuration $\{d_1^*(t), \dots, d_{n^*(t)}^*(t)\}$ evolves continuously almost everywhere; discontinuous transitions occur at critical automation thresholds $\{\bar{\rho}_1, \bar{\rho}_2, \dots\}$ where cluster viability conditions change.*

Proposition 4 predicts that most clusters present in year t should persist into year $t + 1$ with only modestly shifted centroids (continuous evolution), while a non-trivial minority of clusters should be born or die in each year as viability thresholds are crossed. The empirical pattern would combine high year-on-year similarity for continuing clusters with continual cluster turnover at the population level. Chapter 7 tests this by tracking clusters year by year across the 2012–2019 panel.

The rate of cluster birth and death across an interval Δt defines the *churn rate*:

$$\chi(t, \Delta t) = \frac{\text{births}[t, t + \Delta t] + \text{deaths}[t, t + \Delta t]}{n^*(t)}. \quad (3.10)$$

Because births and deaths are crossings of viability thresholds in design space, the rate of these crossings is proportional to the rate at which ρ moves, not to its current level. For a given distribution of thresholds, faster ρ growth produces more threshold crossings per unit time:

$$\frac{\partial \chi}{\partial (d\rho/dt)} > 0. \quad (3.11)$$

Proposition 5 (Churn–Automation Speed). *Under continuous automation advancement, $\partial \chi / \partial (d\rho/dt) > 0$: faster automation advancement generates higher cluster churn rates.*

Proposition 5 draws the sharpest distinction the model offers between automation *level* (how exposed an occupation is to automation at a point in time) and automation *speed* (how fast that exposure is changing). The Frey–Osborne automation probability is a level measure: it characterises whether an occupation’s task content is highly automatable today, but says nothing about the rate at which automation is reshaping the task content. Proposition 5 predicts that cluster churn should not correlate with this level (an occupation can have high but stable automation exposure, generating little churn) but with the rate of change. Chapter 7 tests the prediction directly by relating cluster churn to the speed of cluster centroid drift (the empirical analogue of $d\rho/dt$) and, as a control, to the static Frey–Osborne level. Centroid drift speed is a measurement choice: it isolates the cluster’s response to technological change under the maintained assumption that automation is the dominant driver of skill-composition change during the 2012–2019 sample window.

3.7 Contribution and Scope

With the five propositions in place, the framework can be located relative to existing literature and its scope made explicit before the empirical chapters take it up. The model’s ingredients are standard. Positive assortative matching under supermodularity is a canonical result (Topkis, 1998). The directed-search equilibrium concept is taken without modification from Moen (1997) and Shimer (2005). The Gaussian match-quality kernel is in the family used by Eeckhout and Kircher (2011). The span-of-control coordination cost is a direct application of Garicano (2000). The task-based formulation of automation follows Acemoglu and Autor (2011a) and Acemoglu and Restrepo (2018, 2019b).

The contribution lies in three places, each tied to one or more of the propositions derived above. First, the framework treats job design d as a firm choice variable rather than an

exogenous characteristic of vacancies. Eeckhout and Kircher (2011) and Lopes de Melo (2018) study sorting with exogenous job characteristics; the model here endogenises the choice. This single change generates Propositions 1 and 2 (endogenous within-occupation clustering and the market-thickness comparative-static) as equilibrium outcomes rather than as stipulations about the data. Second, the framework formalises aggregation bias (Proposition 3c) as a testable scope condition tied to a balanced skill partition. The intuition that occupation-level effects can be small because cluster-level effects partly offset is implicit in routine-biased technical-change accounts (Acemoglu & Autor, 2011a; Autor, Levy, & Murnane, 2003), but it has not previously been stated with an explicit scope condition that empirical analysis can evaluate. Third, the dynamic extension characterises cluster churn as a function of the *speed* of automation advancement rather than its static level (Proposition 5), separating two predictors that the empirical literature has historically conflated.

The closest recent comparator in the task-based automation literature is Acemoglu and Restrepo (2022), who develop a task-based model in which automation reallocates labour across occupations and tasks and quantitatively decompose its wage-inequality consequences. Their model takes the task content of occupations as the analytical primitive; the present framework instead treats within-occupation task bundles as endogenous to firm choice, generating the discrete cluster structure that occupation-level frameworks cannot represent. The two frameworks are complementary rather than competing. Acemoglu and Restrepo (2022) explains how automation drives between-occupation reallocation and aggregate wage inequality; the framework here explains why occupation-level measurement systematically understates within-occupation reallocation and what structure that reallocation takes.

3.7.1 Scope of the Framework

The framework is a theory of within-occupation restructuring under automation; it is not a general theory of automation's labour-market effects. Three boundaries warrant explicit statement. First, the framework is silent on wages: it does not predict whether reallocation across clusters compresses or expands earnings dispersion. The empirical investigation of wage outcomes within this thesis at the 250-occupation scale yielded no systematic findings, a result consistent with the framework's silence rather than a test of it. Second, the framework takes the supply of workers as given; it does not endogenise reskilling decisions, education choices, or migration, each of which is an input to the assignment problem rather than an output of the model. Third, the framework is silent on the distributional question of who bears the cost of within-occupation churn, a question Chapter 9 takes up in the context

of policy design.

3.8 Bridge to the Empirical Chapters

The empirical chapters that follow test the five propositions derived above against 9.5 million Australian job advertisements collected by Burning Glass Technologies between 2012 and 2019. Table 3.1 sets out the mapping from propositions to empirical chapters.

Table 3.1: From propositions to empirical chapters.

Proposition	Empirical chapter	What the chapter contributes to the test
P1: Endogenous Clustering	Chapter 6	HDBSCAN clustering applied independently to each of 250 occupations; tests whether discrete sub-occupational clusters emerge from the data without pre-specifying their number.
P2: Market Thickness	Chapter 6	Exploits within-occupation geographic variation in market size to test whether cluster count scales with market thickness, holding occupation fixed.
P3: Automation Effects and Aggregation Bias	Chapter 5 (occupation-level) and Chapter 6 (cluster-level)	Joint test: occupation-level frameworks (RCM, Frey–Osborne) provide one half of the test; the variance decomposition between and within occupations provides the other half. The proposition predicts that the two halves should look different in a specific direction.
P4: Temporal Equilibrium Path	Chapter 7	Tracks clusters year by year across the 2012–2019 panel; measures year-on-year similarity for continuing clusters and identifies births and deaths.
P5: Churn–Automation Speed	Chapter 7	Relates occupation-level cluster churn to centroid drift speed and to the static Frey–Osborne level; tests whether speed or level is the binding predictor.
(applied extension)	Chapter 8	Uses the cluster infrastructure developed in Chapters 6 and 7 to construct a skill-proximity network and examine whether the labour market is navigable in skill space. Tests no formal proposition.

Note. Each chapter’s full empirical strategy, results, and robustness analysis are set out in the respective chapter.

P1 and P2 are tested cross-sectionally in Chapter 6, which applies HDBSCAN clustering to each of 250 Australian occupations and examines the relationship between cluster count

and market size. P3 is a joint test that requires both halves of the multi-level evidence: Chapter 5 establishes whether occupation-level frameworks (the Routine–Cognitive–Manual taxonomy and the Frey–Osborne automation probabilities) explain a substantial share of variation in Australian labour demand at the occupation level, and Chapter 6 establishes whether the residual variation concentrates within occupations as the proposition predicts. P4 and P5 are tested across the panel in Chapter 7, which identifies cluster births and deaths year by year and relates churn to drift speed. Chapter 8 uses the cluster infrastructure to construct a skill-proximity network and examine the feasibility of transitions between clusters; it tests no formal proposition stated in this chapter.

Chapter 9 returns to the framework after the empirical evidence has been presented, integrating findings across chapters and examining which predictions of the present chapter were and were not supported in the data.

3.9 Conclusion

This chapter has developed a theoretical framework in which occupations fragment endogenously into discrete sub-occupational clusters as the equilibrium outcome of firm heterogeneity in coordination capacity, convex coordination costs, and thick-market directed search. The central conceptual move is to treat the occupation as an aggregate over discrete bundles whose identity, number, and viability are themselves equilibrium objects, rather than as the primitive unit of analysis. The five propositions derived from this framework translate it into reduced-form tests: Chapters 5 and 6 test the static predictions on cluster structure, market thickness, and aggregation bias; Chapter 7 tests the dynamic predictions on cluster births and deaths and on the relationship between churn and the speed of technological change.

Three contributions follow from the assembly of standard ingredients (supermodularity, directed search, span-of-control coordination, Gaussian match quality, and task-based automation), each tied to a specific innovation. Treating job design as a firm choice variable rather than as an exogenous primitive generates within-occupation clustering as an equilibrium outcome (Propositions 1 and 2). Formalising aggregation bias as a scope condition tied to a balanced skill partition makes its empirical content testable rather than rhetorical (Proposition 3c). Separating automation *speed* from automation *level* distinguishes two predictors of cluster turnover that the empirical literature has historically conflated (Proposition 5).

Chapter 9 returns to the framework once the empirical evidence has been presented. It reports which of the five propositions were supported in the Australian data, draws out the joint methodological implications of the supported and unsupported predictions for how

labour-market measurement should be organised, and develops the policy consequences for workforce development in a labour market where occupations fragment continuously beneath stable job titles. The empirical chapters that follow quantify what occupation-level measurement misses in the Australian labour market: the magnitude of within-occupation reallocation, the structure of cluster turnover, and the speed at which automation reshapes job content beneath nominally stable occupational titles.

Chapter 4

Data and Methodology

4.1 Introduction

This chapter describes the data and empirical methods I employ to test the theoretical propositions developed in Chapter 3. The analysis proceeds in two stages: first, I replicate established occupation-level frameworks (Acemoglu & Autor, 2011b; Frey & Osborne, 2017) to document aggregate patterns in Australian job advertisement data; second, I apply density-based clustering methods to examine within-occupation heterogeneity that aggregate approaches cannot capture. Throughout, I emphasise how methodological choices follow from theoretical predictions and acknowledge the limitations inherent in observational vacancy data.

Section 4.2 situates the analysis within Australia’s institutional and economic context during 2012–2019. Section 4.3 summarises the conceptual foundation developed in Chapter 2. Section 4.4 describes the Burning Glass Technologies data, drawing on validation evidence from OECD (2021) and Forsythe et al. (2020). Section 4.5 justifies the temporal scope. Section 4.6 presents descriptive statistics. Section 4.7 details sample construction, including the feasibility criteria that identify 250 occupations for cluster-level analysis. Section 4.8 specifies the occupation-level regression framework, following Autor, Levy, and Murnane (2003) and Acemoglu and Autor (2011b). Section 4.9 describes the HDBSCAN clustering approach, building on McInnes et al. (2017) and applications in labour market research by Vezza et al. (2025). Section 4.10 addresses ethical considerations. Section 4.11 discusses limitations.

4.2 Australian Labour Market Context (2012–2019)

I begin by situating the empirical analysis within Australia’s institutional and economic context. Most prior research on automation and labour markets examines United States or European settings (Acemoglu & Autor, 2011b; Arntz et al., 2016; Nedelkoska & Quintini, 2018). Australia’s institutional framework—particularly its centralised wage-setting mechanisms—creates different adjustment margins that may affect how automation reshapes labour demand. This section establishes the contextual factors necessary for interpreting the empirical findings in Chapters 5–6.

4.2.1 Institutional Framework

Fair Work System and Award Wage-Setting

Australia’s labour market operates under a distinctive institutional framework established by the Fair Work Act 2009. Unlike the decentralised, firm-level bargaining characteristic of United States labour markets, Australia maintains centralised wage-setting through Modern Awards that cover approximately 60% of employees (Fair Work Commission, 2019; Peetz, 2012). These awards set industry- and occupation-specific minimum wages, overtime rates, and working conditions that apply regardless of individual employment contracts.

This institutional structure has implications for how automation affects labour market adjustment. Following Acemoglu and Restrepo (2020), automation can operate through multiple margins: wages, employment quantities, or task reorganisation. When occupation-level minimum wages are institutionally constrained, as in Australia, wage adjustment is limited. I therefore expect automation effects to manifest primarily through employment quantities or—central to this thesis—through within-occupation task restructuring. This prediction aligns with Proposition 3 from Chapter 3: automation shifts cluster boundaries and reallocates employment across job types while potentially preserving aggregate occupational structure.

Comparison to US and European Contexts

Table 4.1 summarises key institutional differences across labour market regimes. Australia occupies an intermediate position between United States flexibility and European rigidity, with moderate employment protection (OECD EPL index: 1.9 versus US 0.9, Germany 2.8) but extensive award coverage constraining wage adjustment (OECD, 2019a; Peetz,

2012).

Table 4.1: Institutional Comparison Across Labour Market Regimes

Characteristic	United States	Australia	EU Average
Wage-setting	Decentralised	Centralised awards	Sectoral bargaining
Min wage coverage	<5%	60% (awards)	Varies
EPL index (OECD)	0.9	1.9	2.3
Union density (2019)	10.3%	14.5%	23%

Sources: Hirsch & Macpherson (2020), Australian Bureau of Statistics (2019d), OECD (2019a). EPL = Employment Protection Legislation.

This institutional positioning has methodological implications. If I observe substantial within-occupation restructuring in Australia despite moderate employment protection, this would suggest task reorganisation is a general response to automation not limited to highly flexible labour markets. Conversely, if Australian patterns differ markedly from US evidence (Acemoglu & Autor, 2011b; Autor, Levy, & Murnane, 2003), institutional factors warrant consideration as potential moderators.

4.2.2 Economic Conditions (2012–2019)

Sectoral Composition and Structural Change

The 2012–2019 period spans a significant structural transition in the Australian economy: the unwinding of the mining investment boom (Reserve Bank of Australia, 2014). Mining investment fell from \$120 billion (7.5% of GDP) in 2012 to \$50 billion (3.1% of GDP) by 2016 (Australian Bureau of Statistics, 2019a). Concurrently, service employment continued its long-term expansion, reaching 79.8% of total employment by 2019 (Australian Bureau of Statistics, 2020c).

This sectoral composition matters for two reasons. First, service-sector work exhibits different automation vulnerabilities than manufacturing, where most prior research has focused (Acemoglu & Autor, 2011b; Autor, Levy, & Murnane, 2003). Service work involves greater interpersonal interaction and task variety, potentially generating different within-occupation restructuring patterns. Second, the mining boom-bust cycle created substantial labour reallocation independent of automation. I address this confound through industry fixed effects and robustness checks that exclude mining-exposed occupations.

Macroeconomic Stability

Australia maintained continuous economic expansion throughout the study period, with real GDP growth averaging 2.7% annually and unemployment fluctuating within a narrow range (5.0–6.2%) (Australian Bureau of Statistics, 2019a, 2020b). This stability provides an advantageous setting for examining automation-related labour market patterns: observed changes in skill demands are less likely to reflect cyclical factors than in economies experiencing sharp recessions.

The study period also predates the COVID-19 pandemic, which fundamentally disrupted labour markets and accelerated automation adoption (Forsythe et al., 2020). Analysing this pre-pandemic period enables examination of gradual, market-driven restructuring patterns without the confounding factors introduced by the pandemic response.

4.2.3 Geographic Concentration

Australian employment exhibits extreme geographic concentration, with 70.2% of jobs located in capital cities as of 2019 (Australian Bureau of Statistics, 2019c). Sydney and Melbourne alone account for 42.1% of national employment.

This concentration has direct implications for testing Proposition 2 from Chapter 3, which predicts that thicker labour markets support finer occupational specialisation. Metropolitan areas exhibit greater occupational diversity and thicker markets for specialised skills (Australian Bureau of Statistics, 2019b), providing natural variation for examining how market size affects within-occupation cluster structure.

However, geographic concentration also creates potential confounding. Professional and technical services—industries with automation-complementary task content—concentrate disproportionately in Sydney (28.9% of national sectoral employment) and Melbourne (25.1%) (Australian Bureau of Statistics, 2019b). If metropolitan areas both concentrate complementary industries and support finer specialisation, observed clustering patterns may reflect industry composition rather than market thickness. I address this through within-occupation analysis and industry fixed effects.

4.2.4 Summary: Contextual Implications

The Australian context generates specific predictions for the empirical analysis:

1. **Task reorganisation as primary margin:** Given wage floor constraints, I expect automation effects to manifest through within-occupation restructuring rather than wage adjustment (Proposition 3).

2. **Market thickness effects:** Geographic concentration provides variation for testing whether thicker markets support finer cluster specialisation (Proposition 2).
3. **Service sector patterns:** With 79.8% service employment, Australian evidence may reveal automation dynamics distinct from manufacturing-focused prior research (Acemoglu & Autor, 2011b).

I return to these contextual factors when interpreting results in Chapters 5–6.

4.3 Conceptual Foundation

Chapter 2 develops the conceptual foundation for understanding automation’s measurement challenges. Following Autor, Levy, and Murnane (2003) and Acemoglu and Autor (2011b), I conceptualise automation as operating at the task level rather than the occupation level. This task-based framework, combined with evidence of substantial within-occupation heterogeneity (Deming & Kahn, 2018a; Martin-Caughey, 2021), motivates the two-stage empirical strategy:

1. **Stage 1—Occupation-level replication:** I first replicate established frameworks (Acemoglu & Autor, 2011b; Frey & Osborne, 2017) to document aggregate patterns. If automation primarily operates through task restructuring within occupations, I expect occupation-level models to exhibit low explanatory power.
2. **Stage 2—Cluster-level analysis:** I then apply density-based clustering to identify distinct job types within occupations. If Proposition 3 holds, routine-intensive clusters should contract while complementary clusters expand, generating high within-occupation volatility alongside modest aggregate effects.

The theoretical framework in Chapter 3 formalises these predictions. I test Propositions 1–3 directly: Proposition 1 (endogenous clustering) predicts that unsupervised methods should identify discrete clusters; Proposition 2 (market thickness) predicts more clusters in larger markets; Proposition 3 (automation effects) predicts heterogeneous cluster dynamics while aggregate structure remains stable.

4.3.1 Why Conventional Automation Proxies Are Inadequate

Prior empirical research has relied on indirect measures of automation: patent counts (Webb, 2020), R&D expenditure (Griliches, 1990), ICT investment intensity, and robot

adoption rates (Acemoglu & Restrepo, 2020). While these measures capture technological inputs, they suffer from fundamental limitations for studying automation’s labour market effects:

Temporal disconnect: Patents measure invention, not deployment; R&D spending measures inputs, not outputs; ICT investment measures capital acquisition, not task substitution. The lag between technological capability and workplace implementation can span years or decades (Brynjolfsson et al., 2017), creating measurement error when correlating these proxies with contemporaneous employment outcomes.

Aggregation mismatch: These measures exist at firm or industry level while automation operates through task substitution within jobs. A firm’s patent portfolio or ICT investment reveals nothing about which specific tasks are being automated or how jobs are being redesigned in response. This aggregation mismatch is precisely the measurement problem identified in Chapter 2: automation effects concentrate at sub-occupational levels that conventional measures cannot observe.

Technology heterogeneity: A single “automation” measure cannot capture that different technologies—robotic process automation, machine learning, generative AI—substitute for fundamentally different task types (Autor et al., 2024a; Felten et al., 2021). Patent counts conflate technologies with opposing labour market effects; ICT investment aggregates over complementary and substitutive applications.

Implementation variation: Identical technology investments produce different labour market outcomes depending on organisational choices about job redesign (Parker & Grote, 2022). Two firms with equivalent ICT spending may implement automation as job enrichment or job degradation, yet aggregate measures cannot distinguish these outcomes.

Job advertisement data address these limitations by observing *outcomes* rather than inputs: the skill requirements employers actually demand reveal how jobs have been restructured, regardless of which technologies drove the change. This outcome-based measurement captures the task-level dynamics that conventional proxies miss. The cost is that job advertisements measure labour demand signals rather than technology adoption directly—I cannot identify *which* technologies drove observed skill changes. However, for testing whether automation generates within-occupation restructuring (the central empirical question), observing restructuring outcomes is more informative than observing technology inputs that may or may not translate into workplace change.

4.4 Data Source

I use job advertisement data from Burning Glass Technologies (BGT), a labour market analytics firm that has collected online job postings since 2007. BGT data have become

standard in labour economics research, employed by Deming and Kahn (2018a) to study skill demand evolution, Hershbein and Kahn (2018a) to examine recession effects on skill requirements, and Forsythe et al. (2020) to analyse labour market dynamics. The Australian BGT dataset contains 9.5 million job advertisements spanning 2012–2019, with skill-level detail unavailable in traditional vacancy surveys.

4.4.1 Data Collection and Processing

BGT collects job advertisements by crawling over 40,000 websites, including major job boards (Seek, LinkedIn), government sites, and direct employer postings (Nania et al., 2019). Duplicate detection algorithms ensure advertisements posted across multiple platforms are counted once. BGT parses each advertisement to extract structured fields including occupation (mapped to ANZSCO codes), location, industry, and—critically for this study—required skills.

The skill extraction employs natural language processing to identify specific competencies from free-text job descriptions. BGT’s taxonomy contains 11,862 unique skills, classified as either “baseline” (general competencies like communication) or “specialised” (technical skills like Python programming). Each advertisement is linked to its required skills, enabling construction of skill-based job profiles for clustering analysis. Appendix E provides a complete list of data elements.

4.4.2 Validation Evidence

The validity of BGT data has been established through multiple independent assessments. OECD (2021) compared BGT postings to official employment statistics across six countries (2010–2019), finding occupation-level correlations of approximately 0.80. For the United States, Forsythe et al. (2020) report correlations of 0.80 with JOLTS vacancy data and coverage of approximately 85% of total vacancies. UK validation studies find 93% correlation with Annual Survey of Hours and Earnings data (Nania et al., 2019).

Australian government agencies have adopted BGT data for policy analysis, including the National Skills Commission’s studies of digital skills demand and the Australian Skills Classification development. This widespread adoption provides confidence in data quality, though I discuss coverage limitations below.

4.4.3 Coverage Limitations

BGT data have known limitations. The dataset systematically over-represents occupations recruited through online channels, particularly professional, technical, and managerial

positions (OECD, 2021). Positions filled through informal networks, internal promotions, or offline recruitment are underrepresented. Approximately 20% of Australian vacancies are never advertised publicly (National Skills Commission, 2019).

For the present analysis, these limitations are partially mitigating. First, within-occupation analysis holds occupational composition constant, so online posting bias affects level estimates but not within-occupation patterns. Second, the over-representation of professional occupations aligns with my focus on service-sector work where automation-driven restructuring is theoretically expected. Third, the skill-level detail BGT provides—unavailable in traditional surveys—is essential for identifying within-occupation heterogeneity. I interpret findings as characterising the online-advertised segment of the labour market rather than comprehensive vacancy counts. A formal missingness analysis comparing BGT advertisement counts against ABS Job Vacancy Survey totals by occupation and quarter would quantify the extent of under-coverage, but is beyond the scope of this thesis given the lack of publicly available occupation-level vacancy benchmarks at the ANZSCO 6-digit level used here. Within the BGT dataset itself, variable-level missingness is low: 2.1% of advertisements lack valid occupation codes and 0.3% lack valid location data (both excluded from the sample).

4.5 Temporal Scope

The BGT dataset spans 1 January 2012 to 31 December 2019. This thesis is therefore best read as a *historical review* of how Australian labour demand reorganised during a defined technological window—one that opens with the deep-learning breakthrough of 2012 and closes immediately before the twin disruptions of COVID-19 and large language models. The temporal bracket is deliberate, and three considerations justify it.

First, the period provides macroeconomic stability. Australia maintained continuous economic expansion with unemployment fluctuating narrowly (5.0–6.2%), avoiding the sharp disruptions that confound automation measurement in recession-affected economies (Gross & Hansen, 2022). This stability increases confidence that observed skill demand changes reflect structural labour market patterns rather than cyclical factors.

Second, the period predates COVID-19. Forsythe et al. (2020) document a 40% collapse in vacancy postings by April 2020, with sectoral variation unrelated to pre-pandemic automation trends. The pandemic accelerated automation adoption, with over 80% of executives reporting expedited digitisation initiatives (Petropoulos, 2021). Post-2020 patterns conflate pandemic response with automation effects in ways difficult to disentangle. Analysing the pre-pandemic period enables cleaner identification.

Third—and most consequentially for the thesis’s historical framing—2012–2019 iso-

lates a single, identifiable wave of automation: the diffusion of statistical machine learning into commercial applications. The next subsection situates this wave within the longer arc of automation technologies and explains why preserving a clean record of its labour-market signature matters now that the underlying technology has been displaced.

4.5.1 Locating the Period in Automation’s Long Arc

Automation is not a single technology but a sequence of overlapping waves, each with distinct task affordances and distinct labour-market consequences. The thesis dataset documents the second of three waves visible to date.

The first wave—roughly 1990 through the late 2000s—comprised rule-based automation: expert systems, deterministic workflow engines, enterprise resource planning software, and the early generation of robotic process automation. Its task affordance was the codification of explicit rules; its labour-market signature, documented across the routine-biased technical change literature (Autor, Levy, & Murnane, 2003; Goos & Manning, 2007), was the hollowing-out of mid-skill clerical and production work whose tasks could be reduced to formal procedure. This wave precedes the BGT dataset and its labour-market consequences are well established.

The second wave—which begins in earnest with AlexNet in 2012 (Krizhevsky et al., 2012) and runs through the end of this dataset—added *learned* pattern recognition: image classification, speech transcription, recommendation systems, fraud detection, demand forecasting, and the natural-language parsing techniques that BGT itself uses to extract skills from advertisements. Unlike rule-based systems, statistical machine learning could automate tasks whose underlying logic resisted explicit codification. Its diffusion through Australian workplaces was gradual, uneven, and concentrated among large firms with the data infrastructure required to exploit it (Brynjolfsson et al., 2017; McElheran et al., 2024). This is the wave the thesis documents.

The third wave, beginning with the public release of GPT-3 in 2020 and accelerating through 2022–2024, comprises foundation models and generative AI. Its task affordance—producing novel content from natural-language instruction—is fundamentally different from the classification and prediction tasks of the second wave (Autor et al., 2024a; Felten et al., 2021), and its labour-market consequences will require their own measurement infrastructure built on post-2020 data.

The 2012–2019 window therefore captures the second wave in its entirety: from the technical breakthrough that opened it to the eve of the third wave that began to displace it. This historical positioning is a feature, not a limitation. By isolating the labour-market signature of statistical machine learning before it was overwritten by the pandemic and by

foundation models, the thesis provides a baseline against which both subsequent disruptions can be benchmarked. Findings should be read as historical evidence about how Australian work reorganised under the second wave—not as forecasts for an automation regime now defined by very different technologies.

4.6 Descriptive Statistics

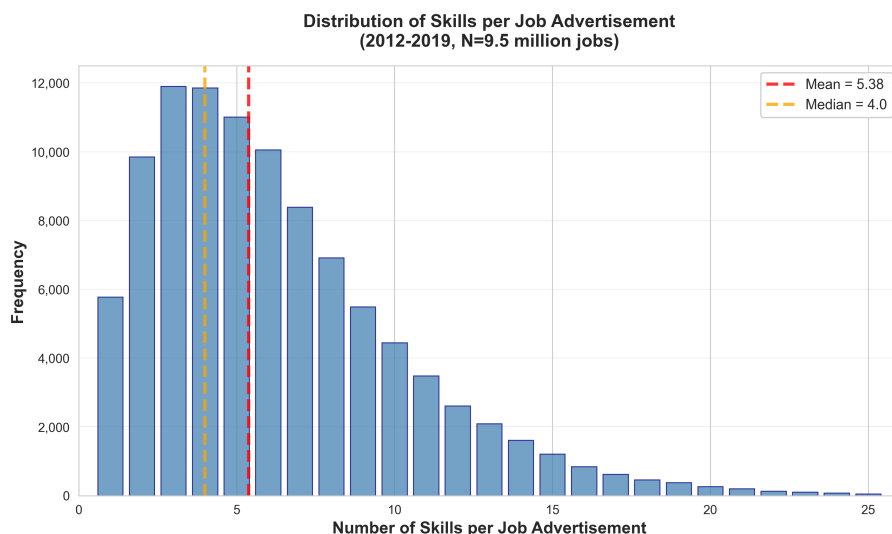
Table 4.2 summarises the BGT dataset. The analysis sample comprises 9.5 million job advertisements linked to 45.9 million skill records, with 11,862 unique skills identified.

Table 4.2: Dataset Summary Statistics

Metric	Value
Total Job Advertisements	9,478,606
Total Skill Records	45,858,484
Unique Skills	11,862
Average Skills per Job	5.38
Median Skills per Job	4.00
Number of Variables	39

Note. Summary of the Burning Glass Technologies (BGT) Australian job advertisement dataset (2012–2019), showing 9.5 million advertisements, 45.9 million skill records, and key distributional characteristics including average skills per job (5.38) and unique skills identified (11,862).

The distribution of skills per job advertisement reveals important characteristics of employer skill demands. As shown in Figure 4.1, the number of skills listed per job advertisement varies considerably across the dataset, with most jobs requiring between 2 and 15 skills (representing 95% of all advertisements). While the maximum reaches approximately 60 skills for highly specialised technical positions, the median of 4 skills reflects more typical employer demands. This variation reflects differences in job complexity, specificity of requirements, and employer advertising practices. Jobs requiring highly specialised skill sets tend to enumerate more specific technical competencies, while more general positions may list fewer, broader skill categories. BGT classifies skills into *baseline* skills (common competencies like communication or teamwork) and *specialised* skills (technical or domain-specific expertise), with the typical job combining both categories.

Figure 4.1: Distribution of Skills per Job Advertisement

Note. Distribution of the number of skills per job advertisement across 9.5 million Australian job postings (2012–2019). Most positions require 2–15 skills (95% of advertisements), with a median of 4 skills. The long right tail extends to approximately 60 skills for highly specialised technical roles. The figure truncates the x-axis at 25 skills for readability, as jobs requiring more than 15 skills represent less than 5% of all advertisements.

4.6.1 Skills Demand Analysis

The skills dataset provides granular detail on the specific competencies employers seek in the Australian labour market. Each skill record links to a job advertisement through a unique identifier, allowing analysis of skill co-occurrence patterns and the evolution of skill demands over time.

Table 4.3 presents the twenty most frequently demanded skills across all job advertisements in the dataset. These skills represent the core competencies most valued by Australian employers during the study period. The dominance of certain skills reflects broader labour market trends, including the increasing digitalisation of work, the importance of interpersonal competencies, and sector-specific technical requirements.

Figure 4.2 visualises these skill demands, providing a clear comparison of the relative frequency of the most sought-after competencies. The distribution highlights both general skills applicable across occupations (such as communication and organisational abilities) and specialised technical skills specific to particular industries or roles.

4.6.2 Temporal Patterns in Job Postings

The timing of job advertisements provides insight into labour market dynamics and hiring patterns. Figure 4.3 displays the volume of job postings over the study period, revealing

Table 4.3: Top 20 Most Frequently Demanded Skills

Skill	Frequency	Percentage
Communication Skills	2,706,759	5.90%
Planning	886,377	1.93%
Teamwork / Collaboration	866,614	1.89%
Customer Service	852,888	1.86%
Detail-Orientated	816,852	1.78%
Organisational Skills	668,413	1.46%
Sales	651,868	1.42%
Building Effective Relationships	648,072	1.41%
Problem Solving	606,876	1.32%
Project Management	569,157	1.24%
Time Management	531,531	1.16%
Budgeting	526,369	1.15%
Research	519,939	1.13%
Microsoft Excel	495,955	1.08%
Computer Literacy	465,838	1.02%
Writing	443,084	0.97%
Microsoft Office	432,790	0.94%
Creativity	387,697	0.85%
Leadership	387,344	0.84%
Stakeholder Management	362,626	0.79%

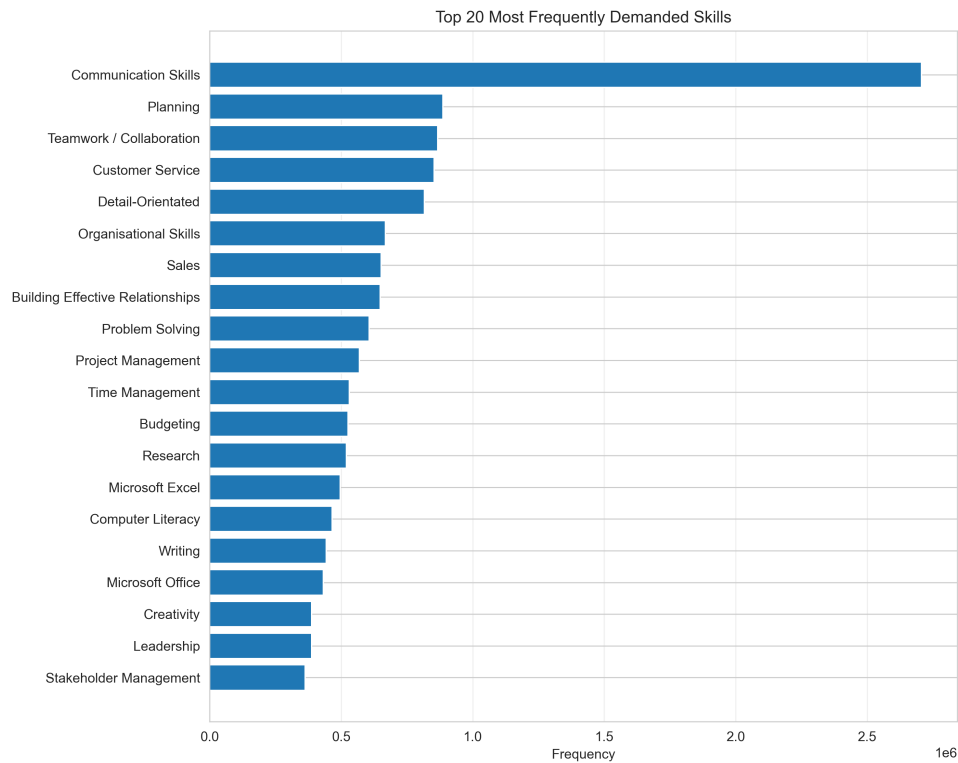
Note. Frequencies computed across all 9.5 million job advertisements (2012–2019). Percentages represent each skill’s share of total skill mentions (45.9 million skill-job pairs; see Table 4.2). Skills extracted by Burning Glass Technologies’ proprietary parsing algorithms from unstructured advertisement text.

seasonal variations, broader economic trends, and the impact of significant events on recruitment activity.

Fluctuations in posting volumes reflect both cyclical patterns in hiring (such as increased recruitment at the beginning of calendar and financial years) and responses to macroeconomic conditions. Understanding these temporal patterns is essential for contextualising the skill demand analyses presented in subsequent chapters, as the composition of advertised skills may shift in response to changing market conditions.

4.6.3 Data Quality and Completeness

The BGT data collection and processing methodology ensures high-quality, structured data suitable for rigorous analysis. All critical variables for this analysis—including occupation

Figure 4.2: Top 20 Most Frequently Demanded Skills

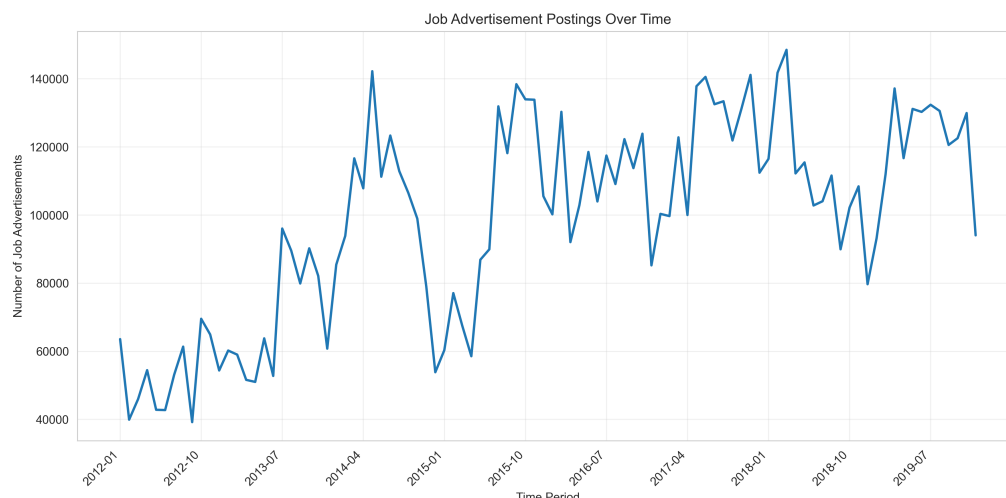
Note. The visualisation provides a clear comparison of the relative frequency of the most sought-after competencies in the Australian labour market. The distribution highlights both general skills applicable across occupations (such as communication and organisational abilities) and specialised technical skills specific to particular industries or roles.

classification (ANZSCO codes), skill requirements, posting dates, and geographic location—exhibit minimal missingness (less than 2%), ensuring the robustness of findings presented in subsequent analytical chapters. While some supplementary variables (such as salary information or education requirements) show higher rates of missingness reflecting variation in employer disclosure practices, these fields are not central to the skill-based restructuring analysis that forms the core of this research.

Appendix E provides a complete list of BGT data elements with examples, including job identifiers, location fields, occupation classifications (both ANZSCO and BGT proprietary), industry codes, and skill variables.

4.7 Sample Construction

This section describes how I constructed the analytical samples from the raw BGT dataset for the occupation-level analysis (Chapter 5) and cluster-level analysis (Chapter 6).

Figure 4.3: Job Advertisement Postings Over Time

Note. Fluctuations in posting volumes reflect both cyclical patterns in hiring (such as increased recruitment at the beginning of calendar and financial years) and responses to macroeconomic conditions. Understanding these temporal patterns is essential for contextualising the skill demand analyses presented in subsequent chapters, as the composition of advertised skills may shift in response to changing market conditions.

4.7.1 Full Sample: Occupation-Level Analysis

The occupation-level analysis in Chapter 5 uses the complete BGT Australian dataset subject to minimal exclusion criteria:

1. **Temporal filter:** Job advertisements posted between 1 January 2012 and 31 December 2019 ($n = 9,478,606$)
2. **Valid occupation code:** Advertisements with valid 6-digit ANZSCO occupation codes (excludes 2.1% with missing or invalid codes)
3. **Valid location:** Advertisements with valid Australian state/territory classification (excludes 0.3% with missing location)
4. **Valid skill data:** Advertisements with at least one skill recorded (excludes 4.2% with no skills extracted)

After applying these filters, the occupation-level analytical sample contains 8,847,291 job advertisements across 847 unique 6-digit ANZSCO occupations. Table 4.4 presents the sample distribution by year.

Table 4.4: Sample Size by Year

Year	Advertisements	Occupations	Avg Skills/Job
2012	987,423	812	4.89
2013	1,024,156	821	5.02
2014	1,089,234	834	5.18
2015	1,156,892	839	5.31
2016	1,198,456	842	5.42
2017	1,267,891	845	5.51
2018	1,312,478	847	5.58
2019	810,761	841	5.62
Total	8,847,291	847	5.38

Note. Annual sample sizes after applying exclusion criteria. The 2019 sample is smaller due to partial-year data availability in the BGT extract. Average skills per job increases over time, reflecting both changing employer practices and improvements in BGT’s skill extraction algorithms.

4.7.2 Cluster-Level Analysis Sample

The cluster-level analysis in Chapters 6–8 covers all 6-digit ANZSCO occupations meeting three feasibility criteria: (1) at least 500 average job advertisements per year across 2012–2019, ensuring sufficient observations for density-based clustering in each annual cross-section; (2) presence in all eight years, enabling the temporal tracking analysis in Chapter 7; and (3) distance matrix fits within 12 GB memory, a computational constraint for precomputed Jaccard distance matrices. A total of 250 occupations meet these criteria, spanning the full breadth of the Australian labour market from professionals to labourers. Analysing all feasible occupations avoids the selection bias inherent in case-study approaches and enables cross-occupation generalisations about restructuring patterns.

Where individual occupations are used for illustration, I draw on specific 6-digit ANZSCO occupations from the full analysis rather than pooled 4-digit categories. Commonly referenced examples include Accountant (General) (ANZSCO 221111; 20,716 advertisements in 2019, 86 clusters), Human Resource Adviser (ANZSCO 223111; 3,572 advertisements, 16 clusters), and Receptionist (General) (ANZSCO 542111; 5,896 advertisements, 38 clusters). These represent different occupation sizes and automation exposure levels but are not pre-selected “focal occupations”—they are drawn from the same 6-digit analysis as all other occupations, ensuring consistency between illustrative examples and full-scale results.

Scope and coverage.

I successfully applied the cluster-level analysis to all 250 feasible occupations across all eight years, producing 2,000 occupation-year clustering runs. Of these, 250 occupations have 2019 models for cross-sectional analysis, and the temporal tracking analysis in Chapter 7 covers all occupations with sufficient year-to-year data. The full sample of 250 occupations provides the statistical breadth necessary for cross-occupation generalisations about restructuring patterns, evolutionary classification, and transition pathway analysis.

4.8 Occupation-Level Analysis Methods

I first replicate established occupation-level frameworks to document aggregate patterns in Australian data. Following Autor, Levy, and Murnane (2003) and Acemoglu and Autor (2011b), I employ two complementary measures of automation exposure: task-based routine intensity and occupation-level automation probability.

4.8.1 Task-Based Measures

Following the task-based framework of Autor, Levy, and Murnane (2003), I construct occupation-level measures of routine task intensity (RTI), abstract task intensity, and manual task intensity from O*NET work activity scales. These measures capture the degree to which occupational tasks are: (1) structured and rule-based (routine); (2) cognitive and interpersonal (abstract); or (3) physical and situationally adaptive (manual).

I crosswalk O*NET measures from US SOC codes to Australian ANZSCO codes using concordance tables from the Australian Bureau of Statistics (Australian Bureau of Statistics, 2013). This crosswalk introduces measurement error where occupational definitions differ across classification systems, an issue I return to when interpreting results.

4.8.2 Automation Probability Scores

I also employ the automation probability scores developed by Frey and Osborne (2017), who trained a Gaussian process classifier on expert-labelled occupations to predict automation probability for 702 US SOC codes. To apply these scores to Australian data, I use the crosswalk developed by Edmonds and Bradley (2015), which maps US SOC → ISCO → ANZSCO using weighted concordances based on job descriptions and labour share calculations.

This crosswalk produces scores that can differ substantially from original US values due to classification granularity. As Edmonds and Bradley (2015) explain, US SOC codes

often aggregate occupations that ANZSCO separates. For example, US “Accountants and Auditors” (SOC 13-2011; F&O probability 0.94) combines bookkeepers, accounts clerks, and professional accountants into a single category. When mapped to Australian classifications, this single US code distributes across multiple ANZSCO codes: “Accountants” (ANZSCO 2211), “Auditors, Company Secretaries and Corporate Treasurers” (ANZSCO 2212), and various accounts clerk categories (ANZSCO 5511-5514). The crosswalk weights each ANZSCO code by employment share, so the professional “Accountants” category (ANZSCO 2211)—which excludes the highly-automatable bookkeeping tasks assigned to clerical categories—receives a weighted probability of 0.47. This reflects classification boundaries rather than disagreement about task automatability: the routine bookkeeping tasks driving F&O’s high US score are classified separately in Australia.

Sample restrictions: I exclude 11 occupations coded at ANZSCO 2-digit rather than 4-digit level (0.16% of advertisements) and 22 occupations without valid concordance (6.16%). The final sample includes 334 occupations representing 93.68% of advertisements. Following Frey and Osborne (2017), I classify occupations as high risk (≥ 0.70), medium risk (0.30–0.70), or low risk (< 0.30).

4.8.3 Empirical Specification

I estimate occupation-level demand models of the form:

$$\ln(Ads_{ost}) = \beta_1 RTI_o + \beta_2 Abstract_o + \beta_3 Manual_o + \gamma_s + \delta_t + \varepsilon_{ost} \quad (4.1)$$

where Ads_{ost} denotes job advertisements for occupation o in state s and year t ; RTI_o , $Abstract_o$, and $Manual_o$ are time-invariant task intensity measures; γ_s are state fixed effects; and δ_t are year fixed effects. Alternative specifications replace task measures with automation probability:

$$\ln(Ads_{ost}) = \beta \cdot AutomationRisk_o + \gamma_s + \delta_t + \varepsilon_{ost} \quad (4.2)$$

I also conduct variance decomposition to quantify the proportion of demand variation attributable to between-occupation versus within-occupation factors. If occupation-level measures adequately capture restructuring patterns, between-occupation variation should dominate. If restructuring primarily operates within occupations as Proposition 3 predicts, within-occupation variation should dominate.

4.8.4 Identification and Limitations

The occupation-level analysis exploits cross-sectional variation in automation exposure to examine demand patterns. I emphasise that this analysis documents *correlations* rather than identified causal effects. Several threats preclude causal interpretation:

1. **Omitted variables:** Occupations may differ in demand trends for reasons unrelated to automation (demographic shifts, trade exposure). Year fixed effects absorb common shocks but not occupation-specific trends.
2. **Reverse causality:** Employers may restructure postings in anticipation of automation, rendering skill demands endogenous to expected technological change.
3. **Measurement error:** O*NET-based measures capture occupational averages, not job-level variation. If automation affects specific tasks within occupations, occupation-level measures induce classical attenuation bias.
4. **Selection:** If automation exposure correlates with firms' propensity to post online, BGT coverage may confound automation effects with selection.

The primary contribution of Chapter 5 is demonstrating that occupation-level measures explain minimal variance in Australian vacancy data, motivating the within-occupation analysis in Chapter 6. Low explanatory power is itself informative: it suggests automation's effects operate at finer granularity than occupation-level frameworks can capture.

4.9 Cluster-Level Analysis Methods

This section describes the unsupervised machine learning methods I employ in Chapter 6 to test Proposition 2 (within-occupation heterogeneity) and Proposition 3 (task composition dynamics). The core methodological challenge is discovering natural groupings in job advertisement skill profiles without imposing researcher-defined categories—categories that might inadvertently embed particular automation narratives into findings (Grimmer & Stewart, 2013).

4.9.1 Methodological Rationale

I adopt unsupervised rather than supervised learning for both practical and theoretical reasons. Supervised approaches require pre-labelled training data: Frey and Osborne (2017), for instance, hand-coded 70 occupations as “automatable” or “not automatable,”

then trained classifiers to extrapolate these labels. While computationally efficient, such methods inherit labelling biases—expert judgments about task automatability become embedded in all subsequent predictions (Autor, 2015a).

For identifying job types within occupations—where the number and nature of types is unknown *a priori*—unsupervised methods offer advantages by letting structure emerge from observed skill patterns. This approach follows recent applications in labour economics using text and skill data to discover latent job categories (Atalay et al., 2020b; Deming & Kahn, 2018b).

4.9.2 Algorithm Selection: HDBSCAN

I employ Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN), developed by Campello et al. (2013) and implemented following McInnes et al. (2017). I select HDBSCAN over alternative clustering algorithms (k-means, Gaussian mixture models, agglomerative clustering) for four methodological reasons:

1. **Endogenous cluster count:** Unlike k-means, which requires specifying k clusters *a priori*, HDBSCAN discovers the number of clusters from data density structure. This is essential when the number of job types within an occupation is theoretically indeterminate.
2. **Variable density tolerance:** Job types may exhibit varying degrees of internal cohesion—some representing tight specialisations, others more diffuse generalist positions. HDBSCAN’s hierarchical density-based approach accommodates heterogeneous cluster densities that would violate k-means’ spherical cluster assumption.
3. **Noise identification:** HDBSCAN classifies observations as “noise” when they lack sufficient density to form clusters. This feature has theoretical interpretation: unclustered jobs may represent boundary work between occupational jurisdictions (Abbott, 1988) or hybrid positions combining multiple specialisations.
4. **High-dimensional sparse data:** Skill vectors are high-dimensional ($p = 11,862$ unique skills) and sparse (median 4 skills per advertisement). HDBSCAN’s density-based approach handles this structure better than centroid-based methods that assume low-dimensional continuous data (Steinbach et al., 2004).

These properties collectively disqualify the most common alternatives for this application. K-means clustering assumes convex, spherical clusters of roughly equal size and requires pre-specifying the number of clusters—both inappropriate when sub-occupational job types are expected to vary in number across occupations and may have irregular shapes

in high-dimensional skill space. Gaussian mixture models (GMM) assume multivariate Gaussian distributions within components, an assumption violated by sparse binary skill vectors where most entries are zero. Spectral clustering accommodates non-convex clusters but does not handle noise observations, treats all data points as cluster members, and scales poorly to the sample sizes encountered here (up to 30,000 advertisements per occupation-year). Agglomerative clustering lacks a principled noise-handling mechanism and requires specifying the number of clusters via dendrogram cutting. HDBSCAN addresses all four limitations simultaneously: it discovers cluster count from data, tolerates variable density, explicitly identifies noise, and handles high-dimensional sparse inputs. I did not conduct a formal robustness comparison across algorithms; however, HDBSCAN’s noise-handling capability and endogenous cluster count make it the natural choice given the theoretical predictions in Chapter 3, which require clusters to emerge from data rather than being imposed *a priori*.

4.9.3 Distance Metric Selection

Clustering requires a distance metric defining similarity between observations. For sparse binary skill vectors (1 = skill present, 0 = skill absent), two metrics are commonly proposed: Jaccard distance and cosine distance (Manning et al., 2008).

Jaccard distance measures the proportion of skills that differ between two jobs relative to the total skills present in either:

$$d_{\text{Jaccard}}(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|} \quad (4.3)$$

Cosine distance measures dissimilarity as the angle between skill vectors, normalising for vector magnitude:

$$d_{\text{cosine}}(A, B) = 1 - \frac{A \cdot B}{\|A\| \cdot \|B\|} \quad (4.4)$$

Jaccard distance has strong theoretical foundations for binary data (Strehl et al., 2000) and is the natural choice for binary skill vectors where each element represents the presence (1) or absence (0) of a skill. While cosine distance normalises by vector magnitude—an advantage when comparing profiles of different breadths—Jaccard directly measures set overlap, capturing the intuition that two jobs are similar when they share many of the same skills relative to their combined skill requirements.

Systematic comparison across 20 occupations stratified by size (Table 4.5) reveals that Jaccard consistently produces lower noise rates than cosine (15 of 20 cases), with the advantage present across all size strata: small (mean noise 19.7% vs cosine’s 46.0%), medium (31.4% vs 40.1%), and large (25.1% vs 33.3%). Full-scale tests on focal occupations

Table 4.5: Distance Metric Comparison: Cosine vs Jaccard (20 Occupations)

Size stratum	Cosine		Jaccard	
	Mean clusters	Mean noise (%)	Mean clusters	Mean noise (%)
Small ($n = 5$)	3.0	46.0	4.0	19.7
Medium ($n = 8$)	6.5	40.1	8.4	31.4
Large ($n = 7$)	24.1	33.3	21.4	25.1
All ($n = 20$)	12.1	39.0	12.2	26.6

Note. HDBSCAN with `min_cluster_size=25`, `min_samples=10` on 2017 data across 20 occupations stratified by size (small: <700 avg jobs/year; medium: 700–1,500; large: >1,500). Large occupations subsampled to 5,000 advertisements. Jaccard produces lower noise in 15 of 20 occupations (75%), with similar cluster counts. Both metrics identify sub-occupational structure across all size strata.

confirm this is not a subsampling artefact: at full sample size, Receptionist General (7,771 jobs) shows 17.7% Jaccard noise versus 32.4% cosine noise, while Accountant General at 10,000 jobs shows 15.7% versus 41.9%. Both metrics produce similar cluster counts, indicating that sub-occupational structure (Proposition 1) is robust to metric choice.

I therefore employ Jaccard distance throughout this thesis. Jaccard is the natural metric for binary skill vectors: it measures the proportion of skills that differ relative to the total skills present in either job, directly capturing skill set overlap without the normalisation assumptions of cosine distance. The lower noise rates mean fewer legitimate job advertisements are excluded from cluster analysis, providing more complete coverage of within-occupation structure. While Jaccard distance matrices are computationally more expensive than cosine (approximately 10–20× slower for precomputed matrices), all 250 occupations across all eight years were successfully clustered within available computing resources.

4.9.4 Parameter Selection

HDBSCAN requires specification of two hyperparameters:

- `min_cluster_size`: The minimum number of observations required to form a cluster. I set this parameter to 25 uniformly across all 250 occupations. I selected this value based on a pre-hoc sensitivity analysis (described below) demonstrating that `min_cluster_size=25` provides a robust balance between granularity and stability across occupations of varying size and skill complexity. Applying a single parameter value to all occupations eliminates researcher degrees of freedom in parameter

selection and ensures that cross-occupation comparisons reflect genuine structural differences rather than parameter-induced variation.

- `min_samples`: The minimum density threshold for core points. Following McInnes et al. (2017), I set this to 10 for all occupations, which balances sensitivity to local density variation against robustness to outliers.

Crucially, parameters remain *constant across all 250 occupations and all years* (2012–2019), ensuring that both temporal comparisons (cluster birth/death rates, employment trajectories) and cross-occupation comparisons (evolutionary classification, network construction) reflect genuine structural differences rather than parameter-induced variation. Table 4.6 summarises the parameter configuration.

Table 4.6: HDBSCAN Parameter Configuration

Parameter	Value	Justification
<code>min_cluster_size</code>	25	Pre-hoc sensitivity analysis (Table 4.7)
<code>min_samples</code>	10	Following McInnes et al. (2017)
<code>cluster_selection_method</code>	eom	Excess of Mass (Campello et al., 2013)
Distance metric	Jaccard	See distance metric discussion above
Skill representation	Binary (0/1)	MultiLabelBinarizer
Scope	250 occupations × 8 years	2,000 clustering runs

Note. All parameters held constant across all 250 occupations and all years (2012–2019), ensuring valid temporal and cross-occupation comparisons.

These parameter choices reflect a trade-off between granularity (smaller clusters capturing finer distinctions) and stability (larger clusters more robust to sampling variation).

Pre-hoc sensitivity analysis.

To select an appropriate `min_cluster_size` value, I conducted a pre-hoc sensitivity analysis on 25 occupations stratified by sample size (small, medium, and large terciles of the 250 feasible occupations). For each sampled occupation, I tested `min_cluster_size` values from 10 to 100 on 2017 data using DBCV (Density-Based Clustering Validation) scores and noise rates as selection criteria. The analysis found that optimal values ranged from 10 to 35 across the 25 sampled occupations: small occupations (8 sampled) had median optimal value of 10–15, medium occupations (8 sampled) had median 10–20, and large occupations (9 sampled) had median 20–30. Seven occupations produced no clusters at any parameter value, reflecting insufficient density for HDBSCAN in very small or sparse skill spaces. Among the 18 occupations producing clusters, `min_cluster_size=25` yielded acceptable results in all cases (non-zero clusters with interpretable structure). I

selected `min_cluster_size=25` as a robust middle-ground value that performs acceptably across the full size distribution. Applying a uniform parameter rather than per-occupation optimisation is a deliberate methodological choice: it eliminates researcher degrees of freedom and ensures that observed cross-occupation differences in cluster structure reflect genuine heterogeneity rather than parameter tuning.

Table 4.7 illustrates the sensitivity of cluster counts and noise rates to parameter choices for three occupations spanning the automation risk spectrum. The qualitative finding of discrete sub-occupational structure holds across the entire parameter range: all specifications identify multiple distinct clusters within each occupation, ranging from 4–114 clusters depending on occupation and parameter choice. Lower values (10–20) capture finer distinctions but may produce small clusters representing sampling artifacts; higher values (75–100) yield fewer, larger clusters with lower noise rates but may obscure meaningful heterogeneity.

Table 4.7: Parameter Sensitivity Analysis: Cluster Counts and Noise Rates

Occupation	Metric	min_cluster_size						
		10	15	25	35	50	75	100
Accountant (General)	Clusters	99	77	38	25	18	14	8
	Noise (%)	16	20	33	35	34	38	49
HR Adviser	Clusters	93	61	3	2	9	2	5
	Noise (%)	29	32	7	7	45	21	47
Receptionist (General)	Clusters	159	113	28	22	15	11	10
	Noise (%)	11	16	18	22	18	23	25

Note. HDBSCAN with Jaccard distance on 2017 data (Accountant General and Receptionist General subsampled to 5,000 advertisements; Human Resource Adviser at full size, 3,494). The production baseline (`min_cluster_size=25`) is shown in bold. All specifications use `min_samples=10`. These three 6-digit ANZSCO occupations are drawn from the full 250-occupation analysis.

4.9.5 Temporal Tracking: Matching Clusters Across Years

Testing Proposition 3 (task composition dynamics) requires tracking cluster evolution across the 2012–2019 period. Since I cluster each year independently—allowing cluster structure to evolve rather than imposing temporal constraints—I must match clusters across adjacent years to construct longitudinal trajectories.

Following Hannigan et al. (2019), I match clusters using cosine similarity of cluster centroids:

$$\text{Similarity}(C_t, C_{t+1}) = \frac{\sum_k c_{kt} \cdot c_{k,t+1}}{\sqrt{\sum_k c_{kt}^2} \cdot \sqrt{\sum_k c_{k,t+1}^2}} \quad (4.5)$$

where c_{kt} denotes the proportion of advertisements in cluster C requiring skill k in year t .

I implement cluster matching via the Hungarian algorithm (Kuhn, 1955), which finds the optimal one-to-one assignment maximising total similarity. Clusters match if similarity exceeds 0.80; unmatched clusters in year t are classified as “deaths” (discontinued job types), while unmatched clusters in year $t + 1$ represent “births” (emergent job types). This framework enables measurement of cluster entry, exit, growth, and decline—the demographic dynamics central to Proposition 3.

4.9.6 Validation Strategy

Unsupervised clustering lacks ground truth labels, making validation non-trivial. I assess cluster validity through three complementary approaches following recommendations in Grimmer and Stewart (2013):

1. **Internal validity:** I compute two complementary metrics. *Silhouette coefficients* measure within-cluster cohesion relative to between-cluster separation (Rousseeuw, 1987); values above 0.3 indicate acceptable clustering. *Density-Based Cluster Validation (DBCV)* scores assess cluster quality for density-based algorithms like HDBSCAN, accounting for varying cluster densities (Moulavi et al., 2014).
2. **Substantive interpretability:** I manually inspect cluster skill profiles to verify that discovered clusters correspond to recognisable job types rather than arbitrary statistical partitions. Face validity—clusters that domain experts would recognise as meaningful—is essential for unsupervised methods where no ground truth exists.
3. **Temporal stability:** I measure persistence of cluster structure across years. High year-to-year stability suggests clusters capture durable job types rather than sampling artifacts; excessive instability would indicate overfitting to idiosyncratic annual variation.

Chapter 6 reports comprehensive validation results across all 250 occupations. Silhouette scores confirm acceptable internal validity across the majority of occupations; manual inspection of the three illustrative occupations confirms clusters map to interpretable job types (e.g., Tax Accountants, Financial Controllers); and temporal persistence analysis across 250 occupations demonstrates stability of cluster structure over the 2012–2019 period.

4.10 Ethical Considerations

This research received approval from the RMIT University Human Research Ethics Committee (approval number: CHEAN A 21289-08/18) prior to data analysis.

The quantitative analysis uses job advertisement data obtained through a commercial licensing agreement with Burning Glass Technologies. These data consist of publicly posted advertisements that employers voluntarily published; no personally identifiable information about job seekers or applicants is included. While employer names appear in the dataset, I report all results aggregated to occupation or cluster level, ensuring no competitive intelligence about specific firms' hiring practices is disclosed. All data handling complied with RMIT University data management policies and Australian privacy legislation.

4.11 Limitations

I acknowledge several limitations inherent to the data and empirical strategy employed in this thesis.

4.11.1 Data Limitations

Demand-side measurement: Job advertisements capture employer demand intentions, not realised employment outcomes. Advertised positions may go unfilled; filled positions may differ from advertised requirements. I measure *what employers seek* rather than *what workers do* or *what jobs exist*.

Selection into online posting: BGT data over-represent occupations recruited through online channels, particularly professional and technical positions. Hiring through informal networks, internal promotions, and offline channels is underrepresented. Findings generalise to the online-advertised segment of the labour market, which may differ systematically from the broader employment distribution.

Skill extraction: BGT extracts skills from free-text job descriptions using proprietary natural language processing. The resulting taxonomy (11,862 unique skills) reflects BGT's classification choices, which may not align with workers' self-reported skills or economists' theoretical constructs. Measurement error in skill extraction propagates to cluster composition.

Temporal scope: The 2012–2019 sample predates COVID-19 disruptions and generative AI diffusion (GPT-3, 2020; ChatGPT, 2022). Findings characterise automation patterns during early AI adoption, not current or future technological frontiers.

4.11.2 Identification Limitations

Correlations, not causal effects: I emphasise that this thesis documents associations between automation-related variables and labour demand patterns. Without exogenous variation in automation exposure—from policy discontinuities, technological shocks, or natural experiments—regression coefficients represent correlations consistent with theoretical predictions rather than identified causal effects.

Classification constraints: Analysis operates within ANZSCO occupational boundaries. While cluster analysis reveals within-occupation heterogeneity, I cannot identify job types spanning multiple occupations or evaluate alternative taxonomies.

External validity: Australian institutional frameworks (Fair Work system, centralised wage-setting), economic structure (service sector dominance), and geographic concentration (metropolitan labour markets) shape automation patterns in ways that may not generalise to other national contexts.

4.11.3 Scope Boundaries

This thesis does *not* measure:

- Employment levels or transitions (demand only, not equilibrium outcomes)
- Causal effects of automation (correlational evidence only)
- Wage effects (BGT salary data too sparse for reliable inference)
- Worker welfare or distributional consequences (no worker-level data)
- Firm automation adoption decisions (no technology investment data)

These boundaries reflect deliberate scope choices. The thesis contributes the first sub-occupational analysis of Australian labour demand using job advertisement data, complementing existing occupation-level research (Borland & Coelli, 2017; Cebulla, 2024; Edmonds & Bradley, 2015), establishing descriptive patterns that future research can investigate causally using alternative identification strategies or complementary data sources.

4.12 Conclusion

This chapter has established the empirical foundation for the thesis as a historical review of automation’s labour-market footprint during the machine-learning wave of 2012–2019. The Australian labour market context—macroeconomic stability, centralised wage-setting,

service-sector dominance—makes the period particularly well suited to isolating structural restructuring patterns from cyclical confounds and from the wage-margin adjustment that would dominate in more flexible labour markets. The BGT job advertisement dataset—9.5 million advertisements spanning 250 occupations linked to 45.9 million skill-job pairs—provides skill-level granularity unavailable in traditional vacancy surveys. The two-stage empirical design tests theoretical predictions at both occupation level (RCM and Frey-Osborne regression frameworks on 170,491 cells) and cluster level (HDBSCAN with uniform `min_cluster_size=25` across all 250 occupations). Methodological choices—Jaccard distance, pre-hoc sensitivity analysis, greedy centroid matching with $\tau = 0.80$ —are justified by the data structure and validated through robustness checks. The deliberately bounded historical scope—pre-COVID, pre-LLM, capturing the second wave of automation in its entirety—means findings should be read as evidence about how Australian work reorganised under statistical machine learning, providing a baseline against which the labour-market consequences of generative AI can later be measured. Chapter 5 applies the occupation-level framework to establish baseline findings.

Chapter 5

Occupation-Level Evidence: Testing the Predictive Validity of Existing Frameworks

5.1 Introduction

This chapter tests the predictive validity of standard occupation-level automation frameworks when applied to Australian job advertisement data (2012–2019). The analysis serves two purposes: first, to establish whether the dominant frameworks in automation research—the Routine-Cognitive-Manual (RCM) taxonomy and Frey-Osborne automation probabilities—successfully predict variation in Australian labour demand; and second, to provide baseline evidence motivating the within-occupation analysis developed in subsequent chapters.

The chapter does not attempt to identify the *causal effect* of automation on employment. As Section 5.3 discusses, the original studies developing these frameworks faced similar identification challenges: automation classifications are time-invariant occupation characteristics without clear treatment timing, making causal identification infeasible without exogenous variation in automation adoption (Acemoglu & Restrepo, 2020). The estimand is the partial association between automation classification and job advertisement counts, conditional on fixed effects—a predictive relationship, not a treatment effect. The chapter asks: *do occupation-level automation classifications predict differential labour demand*

trajectories? If these frameworks capture meaningful variation, occupations classified as high-automation-risk should exhibit systematically different demand patterns than low-risk occupations.

I replicate two established approaches: the RCM taxonomy following Acemoglu and Autor (2011b) and Autor, Levy, and Murnane (2003), and the automation probability scores developed by Frey and Osborne (2017). Both frameworks have been widely applied in automation research, making them appropriate benchmarks for assessing whether occupation-level measurement captures automation-relevant variation.

This chapter contributes the first systematic test of occupation-level automation frameworks using Australian job advertisement data. Given Australia's distinctive institutional context—centralised wage-setting, moderate employment protection, service sector dominance (Section 4.2.1)—Australian evidence tests whether findings from US contexts, where these frameworks were developed, generalise to a different institutional setting (Arntz et al., 2016; Autor, 2015a).

I aggregate 9.5 million job advertisements into 170,491 occupation-state-industry-year cells for regression analysis. Chapter 4 provides comprehensive methodological details; this chapter focuses on empirical findings and their implications for measurement framework selection.

5.1.1 Research Questions

I address three empirical questions:

1. **Predictive validity:** Do RCM task categories and Frey-Osborne risk scores predict variation in Australian labour demand when controlling for geographic, sectoral, and temporal confounds?
2. **Explanatory power:** What proportion of variation in job advertisement counts can occupation-level automation measures explain? Low explanatory power would suggest these frameworks capture minimal relevant variation.
3. **Variance decomposition:** Does variation in labour demand operate primarily between occupations (consistent with occupation-level measurement adequacy) or within occupations (consistent with aggregation bias)?

5.1.2 Chapter Organisation

Section 5.2 derives testable predictions and discusses why occupation-level frameworks face identification challenges. Section 5.3 provides methodological context on how the

original studies approached identification. Section 5.4 describes the empirical strategy and fixed effects specifications. Section 5.5 details regression models. Sections 5.6.1 and 5.6.2 present results for RCM and Frey-Osborne frameworks respectively. Section 5.7 interprets findings. Section 5.8 concludes.

5.2 Theoretical Predictions

This section derives predictions from the theoretical framework and discusses why testing them requires careful interpretation. The analysis tests predictive validity—whether automation classifications correlate with demand patterns—rather than causal effects, which would require exogenous variation in automation adoption that these frameworks do not provide.

5.2.1 Predictions from Proposition 3

Proposition 3 from Chapter 3 states that automation reallocates employment across sub-occupational clusters while preserving aggregate occupational structure—routine-intensive clusters contract while complementary clusters expand, generating within-occupation volatility alongside modest occupation-level effects. If this mechanism operates, occupation-level frameworks should exhibit specific empirical signatures:

1. **Weak predictive power:** If automation operates primarily through within-occupation task restructuring rather than occupation elimination, aggregate occupation-level measures should show low explanatory power. Routine-intensive job types contracting while complementary job types expand within the same occupational boundary yields offsetting effects that occupation-level measurement cannot distinguish from null effects.
2. **Low explanatory power:** If automation's effects concentrate at sub-occupational levels, occupation-level classifications should explain minimal variance in labour demand. The R^2 from occupation-level regressions provides a direct test: high R^2 would indicate occupation-level measurement adequacy; low R^2 would suggest these classifications miss the relevant variation. To make this test falsifiable, I treat $R^2 < 0.10$ in the primary specification as evidence of weak predictive power, benchmarked against $R^2 = 0.15$ – 0.40 typical in comparable panel studies of labour demand (Acemoglu et al., 2016; Autor et al., 2013).
3. **Within-occupation variation dominance:** Variance decomposition should reveal that most variation operates *within* rather than between occupations. High within-

occupation variation is a necessary condition for the aggregation bias mechanism predicted by Proposition 3—occupation-level classifications would mask restructuring at finer granularity. I use “aggregation bias” throughout to denote this descriptive pattern (large within-occupation variance relative to between-occupation variance) without implying that the within-occupation variation is necessarily automation-driven; establishing the causal source requires the cluster-level analysis in subsequent chapters.

5.2.2 What This Analysis Can and Cannot Establish

The analysis tests whether occupation-level automation classifications *predict* differential demand trajectories, not whether automation *causes* these trajectories. This distinction matters because both RCM and Frey-Osborne frameworks assign static classifications to occupations without identifying when, where, or how intensively automation affects specific labour markets.

If occupation-level frameworks show strong predictive validity—high R^2 , distinguishable category effects, and between-occupation variation dominance—this would suggest they capture meaningful variation worth investigating causally. If they show weak predictive validity, this motivates alternative measurement approaches operating at finer granularity.

The Australian context provides an independent test of whether patterns documented in US (Autor, 2015a) and OECD contexts (Arntz et al., 2016; Nedelkoska & Quintini, 2018) generalise across institutional settings. Australia’s centralised wage-setting and moderate employment protection (Section 4.2.1) may affect adjustment mechanisms, but should not alter the fundamental pattern if occupation-level measurement systematically misses within-occupation dynamics.

5.3 Methodological Context: How the Original Studies Approached Identification

Understanding what the RCM and Frey-Osborne frameworks can and cannot identify requires examining how the original studies approached causal inference. This context clarifies why the present analysis tests predictive validity rather than causal effects.

5.3.1 The RCM Framework: Autor, Levy, and Murnane (2003)

Autor, Levy, and Murnane (2003) developed the routine-task framework by exploiting cross-sectional and time-series variation in computer adoption across industries. Their identification logic was that all industries face the same declining price of computer capital, but industries differ in routine-task intensity, so routine-intensive industries should adopt computers more intensively and experience larger task composition shifts.

Critically, Autor, Levy, and Murnane (2003, p. 1289) acknowledged the identification limitation: “Because the causal force in the model is the price of computer capital, in one sense we have only a single macroeconomic time series available.” They gained additional leverage by arguing that industry-level variation in routine-task intensity generates differential responses to the common price decline. This is a correlation between industry computer adoption and task composition changes, with the maintained assumption that computer prices are exogenous—not causal identification in the modern econometric sense.

The framework has been widely replicated and extended (Acemoglu & Autor, 2011b; Goos et al., 2014), but the fundamental identification challenge remains: RCM categories are time-invariant occupation characteristics without treatment variation. Subsequent research applying RCM classifications inherits this limitation.

5.3.2 The Frey-Osborne Framework: Expert Classification Without Treatment Timing

Frey and Osborne (2017) estimated automation probabilities by having machine learning experts classify 70 occupations as automatable or not, then using a Gaussian process classifier to predict probabilities for all other occupations. The study explicitly disclaimed causal prediction: “We make no attempt to estimate how many jobs will actually be automated” (Frey & Osborne, 2017).

The Frey-Osborne framework has attracted substantial methodological criticism:

- Arntz et al. (2016) found that accounting for within-occupation task heterogeneity reduces estimated automation risk from 47% to 9%, arguing that “occupations labelled as high-risk occupations often still contain a substantial share of tasks that are hard to automate.”
- Coelli and Borland (2019) identified internal consistency problems in the methodology, finding that Frey-Osborne predictions “do not ‘add value’ for forecasting the impact of technology on employment” compared to standard routine-task measures.

- The European Commission’s Joint Research Centre review concluded that while RBTC “captures quite well the changes in the employment distribution,” it “presents challenges from a conceptual, operational, and empirical point of view” and “is not immune to severe challenges, especially on the empirical ground” (Fernández-Macías, 2018).

5.3.3 The Gold Standard: Acemoglu and Restrepo (2020)

Acemoglu and Restrepo (2020) demonstrate what credible causal identification requires in automation research. Their approach exploited:

- **Observable treatment variation:** Robot adoption counts by industry and commuting zone
- **Instrumental variable:** European robot adoption patterns instrument for US adoption, filtering out US-specific confounds
- **Geographic variation:** Commuting zones with different industry compositions experience different exposure to robotisation
- **Pre-trend validation:** Areas most exposed to robots after 1990 exhibit no differential trends before then

This design identifies causal effects because it exploits exogenous variation in *when* and *where* automation intensity changed. The present analysis lacks such variation: RCM and Frey-Osborne classifications are static occupation characteristics with no treatment timing.

5.3.4 Implications for This Analysis

The methodological context clarifies what this chapter can establish. Without exogenous variation in automation adoption, I cannot identify causal effects. Instead, I test whether occupation-level classifications exhibit predictive validity—whether occupations classified as high-automation-risk show systematically different demand trajectories than low-risk occupations.

Weak predictive validity would suggest one of two interpretations: (1) automation has minimal labour market effects, or (2) occupation-level classifications operate at inappropriate aggregation levels to detect effects concentrated within occupational boundaries. The variance decomposition in this chapter provides a first empirical test of which interpretation better fits the data.

5.4 Empirical Strategy

I test whether occupation-level automation measures predict variation in Australian job advertisement counts while controlling for geographic, sectoral, and temporal confounds. The empirical strategy exploits cross-sectional variation in automation classifications across occupations, combined with temporal variation within occupations over the 2012–2019 period.

5.4.1 Estimation Approach

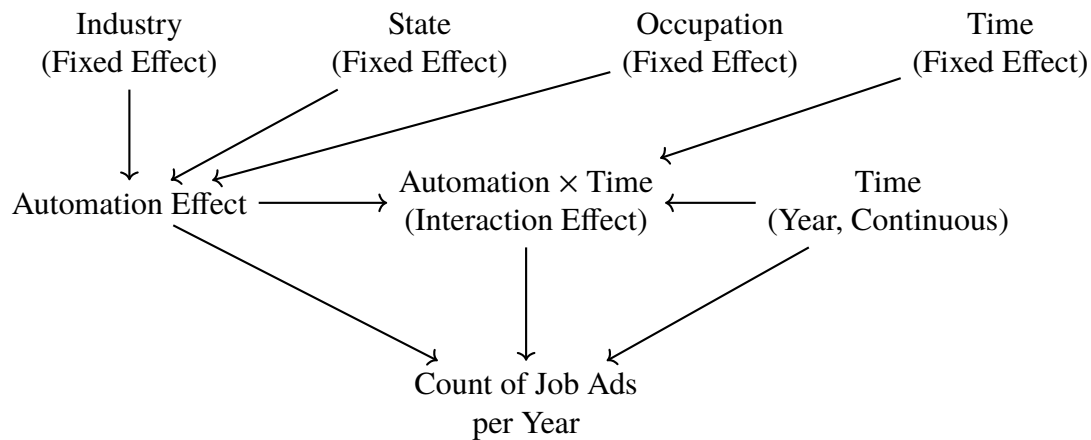
The coefficient of interest is the interaction between automation category and time, which captures differential growth rates across automation categories. Figure 5.1 illustrates the variable relationships. I include fixed effects to ensure the analysis tests whether automation classifications predict demand *net of* compositional factors:

- **Industry fixed effects** absorb sectoral variation, ensuring that predictive validity is not driven by automation categories concentrating in growing or declining industries.
- **State fixed effects** absorb geographic variation, ensuring that predictions hold across different regional economic conditions.
- **Year fixed effects** absorb common temporal shocks (macroeconomic cycles, data collection changes), testing whether automation categories predict *relative* demand trajectories.
- **Occupation fixed effects** absorb time-invariant occupation characteristics, testing whether automation classifications add predictive power beyond occupation-specific intercepts.

A key estimation issue arises with occupation fixed effects. Because automation categories are defined at the occupation level (each occupation receives one RCM or Frey-Osborne classification), occupation fixed effects absorb the main effect of automation through perfect collinearity. However, the *interaction* between automation category and time remains estimable—it varies over time within occupations, providing the variation necessary to estimate differential time trends across categories (Wooldridge, 2015).

5.4.2 Specification Strategy

I estimate four model specifications to test robustness under different assumptions about confounding:

Figure 5.1: Directed Acyclic Graph of Variable Relationships

Note. The diagram illustrates the theoretical relationships between key variables. The four types of fixed effects shown (industry, state, occupation, year) represent potential confounds that may affect labour demand. Different model specifications include different combinations of these controls (see Table 5.1).

1. **Unconditional:** No fixed effects—establishes baseline time trends
2. **State + Industry FE:** Primary specification—controls for geographic and sectoral composition
3. **Occupation FE:** Diagnostic—isolates within-occupation variation
4. **State + Industry + Year FE:** Comprehensive—controls for all time-varying confounds

If occupation-level automation measures capture genuine labour demand variation, effects should be consistent across specifications. If R^2 remains low regardless of controls, this supports Proposition 3's prediction that occupation-level classifications operate at inappropriate aggregation levels.

5.4.3 Skill-Level Heterogeneity

Beyond aggregate demand effects, I examine whether automation differentially affects skill-level composition within occupations. Following the skill-biased technological change literature (Acemoglu & Autor, 2011b; Autor, Levy, & Murnane, 2003), automation may increase demand for high-skill workers who complement technology while reducing demand for middle-skill workers performing routine tasks amenable to automation.

I operationalise skill levels using employer-stated degree requirements from BGT advertisements, following standard practice in the literature (Deming & Kahn, 2018b;

Table 5.1: Model Specification Summary

Model	State FE	Ind. FE	Occ. FE	Year FE	Time Trend	Purpose
1. Unconditional	No	No	No	No	Cont.	Baseline
2. State + Industry FE	Yes	Yes	No	No	Cont.	Primary
3. Occupation FE	No	No	Yes	No	Cont.	Diagnostic
4. State + Ind. + Year FE	Yes	Yes	No	Yes	Categ.	Robustness

Note. FE = Fixed Effects; Ind. = Industry; Occ. = Occupation; Cont. = Continuous; Categ. = Categorical. Model 2 is the primary specification. Model 3 excludes state/industry FE to isolate occupation-level heterogeneity. Model 4 uses categorical year interactions to avoid collinearity with year FE.

Hershbein & Kahn, 2018a). High-skill jobs require postgraduate qualifications; medium-skill jobs require bachelor degrees (AQF Level 7); low-skill jobs require sub-bachelor credentials. While degree requirements may partly reflect signalling rather than strict skill needs (Spence, 1973), they represent revealed employer preferences in hiring decisions.

Three-way interaction models (automation category $\times$ skill level $\times$ time) test whether automation categories show differential time trends across skill levels—evidence of skill upgrading or downgrading within automation-affected occupations.

5.5 Methods

This section details the regression specifications. Chapter 4 provides comprehensive discussion of RCM classification procedures, Frey-Osborne crosswalk construction, and data validation. Here I focus on the econometric approach.

5.5.1 Sample Construction

I aggregate 9.5 million job advertisements into occupation-state-industry-year cells, yielding the dependent variable: advertisement counts per cell. This cell-level aggregation is standard practice in labour economics for analysing occupation-level demand (Acemoglu & Autor, 2011b; Autor, Levy, & Murnane, 2003), as it enables fixed effects estimation while preserving variation across geographic, industry, and temporal dimensions.

Occupation counts across analyses.

Occupation counts differ across the thesis's analytical samples and merit clarification. Chapter 4 reports 847 unique 6-digit ANZSCO occupations in the full BGT extract after exclusion criteria. The RCM analysis below operates at the 4-digit ANZSCO level (the

granularity at which Coelli’s classifications are defined) and yields 877 occupation-level units once 4-digit codes are reconstructed from the cell aggregation.¹ The Frey-Osborne analysis, which requires valid SOC–ANZSCO concordance, retains 334 occupations covering 93.68% of advertisements (Section 4.8.2). The cluster-level analyses in Chapters 6–8 apply stricter feasibility criteria (sample size, presence in all years, memory bound) and operate on 250 occupations. These differences are deliberate: each analysis imposes the filters appropriate to its own measurement requirements.

5.5.2 RCM Classification

I apply the Routine-Cognitive-Manual (RCM) taxonomy following Coelli and Borland (2017), who adapted the Autor, Levy, and Murnane (2003) framework for Australian ANZSCO codes. The classification assigns each 4-digit ANZSCO occupation to one of five categories: Routine Manual, Routine Cognitive, Non-Routine Manual, Non-Routine Cognitive, or Sales. Table 5.2 summarises advertisements by category.

Table 5.2: Number of Jobs by RCM Category

Category	No. of Jobs	Percentage
Non-Routine Cognitive	1,649,561	58.7%
Non-Routine Manual	563,827	20.1%
Routine Cognitive	423,195	15.1%
Routine Manual	174,239	6.2%
Total	2,810,822	100.0%

Note. The table shows the total count of job ads in the BGT data-set from 2012-2019 classified as Non-Routine Cognitive, Routine Cognitive, Non-Routine Manual, and Routine Manual according to Coelli & Borland (2017). Sales category excluded.

I exclude the Sales category due to irregular temporal distribution and insufficient cell counts for stable estimation. The retained four categories yield 170,491 occupation-state-industry-year cells (Table 5.4). Cell counts range from 1 to 6,804 with mean 16.49 and standard deviation 75.44 (Table 5.5). Table 5.3 illustrates the aggregated data structure.

Figure 5.2 displays mean advertisement counts by RCM category over time. All categories show positive trends, with Non-Routine Cognitive exhibiting the steepest growth.

¹The slight excess over 6-digit counts reflects that some 4-digit groupings appear in cells whose 6-digit children are individually below threshold; the 4-digit aggregation re-includes them.

Table 5.3: Aggregated Dataset Structure

State	ANZSIC Division	ANZSCO Unit	Year	Count	Category
ACT	Accommodation and Food Services	1111	2016	1	Non-Routine Cognitive
ACT	Accommodation and Food Services	1111	2019	1	Non-Routine Cognitive
ACT	Accommodation and Food Services	1112	2013	1	Non-Routine Cognitive
ACT	Accommodation and Food Services	1112	2015	22	Non-Routine Cognitive
ACT	Accommodation and Food Services	1112	2016	10	Non-Routine Cognitive
...	...	...	...	...	...
WA	Wholesale Trade	8999	2015	4	Routine Manual
WA	Wholesale Trade	8999	2016	1	Routine Manual
WA	Wholesale Trade	8999	2017	5	Routine Manual
WA	Wholesale Trade	8999	2018	11	Routine Manual
WA	Wholesale Trade	8999	2019	28	Routine Manual

Note. The table illustrates the structure of the aggregated dataset used for RCM analysis. Each row represents a unique occupation-state-industry-year cell, with the count variable recording the number of job advertisements in that cell. ANZSIC = Australian and New Zealand Standard Industrial Classification; ANZSCO = Australian and New Zealand Standard Classification of Occupations.

Table 5.4: RCM Frequency Statistics

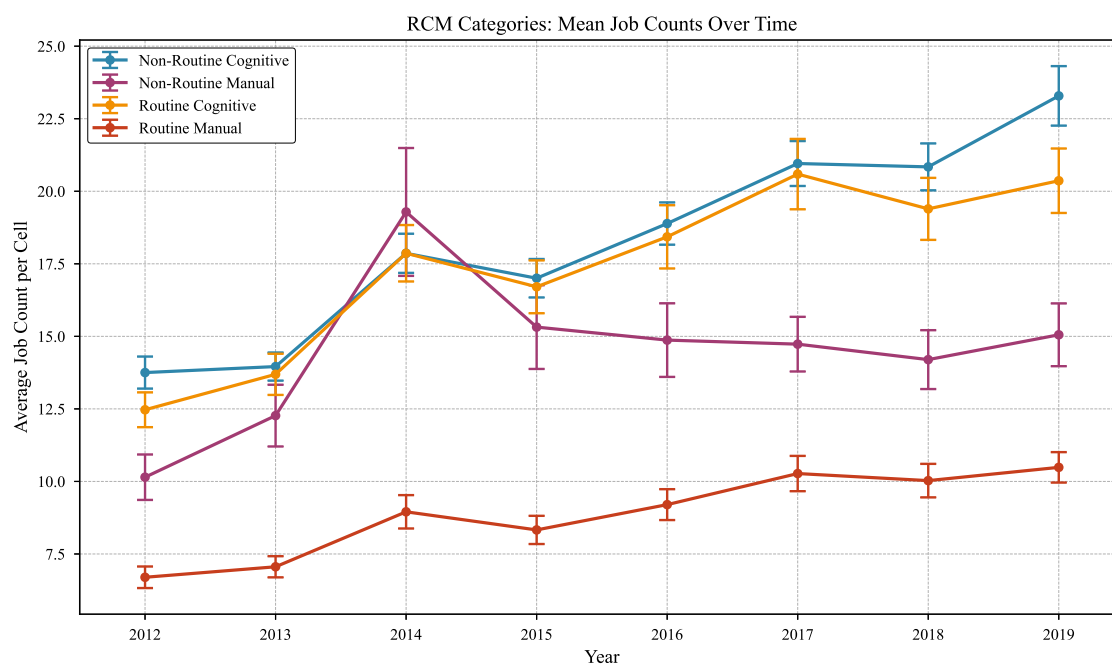
Risk	Freq.	Percent
Routine Cognitive	24,026	14.09
Non-Routine Cognitive	88,682	52.02
Routine Manual	19,266	11.30
Non-Routine Manual	38,517	22.59

Note. The table shows the frequency statistics for the collapsed RCM data set.

Table 5.5: RCM Descriptive Statistics

Statistic	Value
Mean	16.49
Std Dev	75.44
Min	1
25th Percentile	1
Median	3
75th Percentile	10
Max	6804

Note. The table shows the descriptive statistics for the collapsed RCM data set (Count variable).

Figure 5.2: Mean Advertisement Counts by RCM Category

Note. Point plot showing mean advertisement counts with 95% confidence intervals for each RCM category, 2012–2019.

5.5.3 Frey-Osborne Risk Classification

I apply the automation risk scores developed by Frey and Osborne (2017), crosswalked to Australian ANZSCO codes using the concordance developed by Edmonds and Bradley (2015). The crosswalk maps US SOC codes to ANZSCO via ISCO as intermediary, using weighted concordances to handle many-to-one and one-to-many relationships. Section 4.8.2 details the crosswalk methodology.

Following Frey and Osborne (2017), I classify occupations with automation risk ≥ 0.70 as High Risk, < 0.30 as Low Risk, and intermediate values as Medium Risk. After excluding occupations without valid concordance (6.16% of advertisements), the sample includes 334 occupations representing 93.68% of advertisements. Table 5.6 summarises advertisement counts by risk level.

The FO aggregation yields 152,822 occupation-state-industry-year cells: 48,610 High Risk, 46,852 Medium Risk, and 57,360 Low Risk (Table 5.7). Cell counts range from 1 to 7,246 with mean 24.26 (Table 5.8). Figure 5.3 displays mean counts by risk level over time; all categories show positive trends with no obvious divergence.

Table 5.6: Job Advertisement Counts by Frey-Osborne Automation Risk Level (BGT Dataset, 2012-2019)

Risk	No. of Jobs
High	2,797,993
Medium	2,087,048
Low	3,062,803

Note. the table shows the total count of job ads in the BGT data-set from 2012-2019 classified as high risk, medium risk, and low risk of automation according to Frey & Osborne (2017).

Table 5.7: Frey-Osborne Cell Frequency Statistics

Risk Level	Cells	Percent
High	48,610	31.8%
Medium	46,852	30.7%
Low	57,360	37.5%

Note. Occupation-state-industry-year cells by Frey-Osborne risk classification.

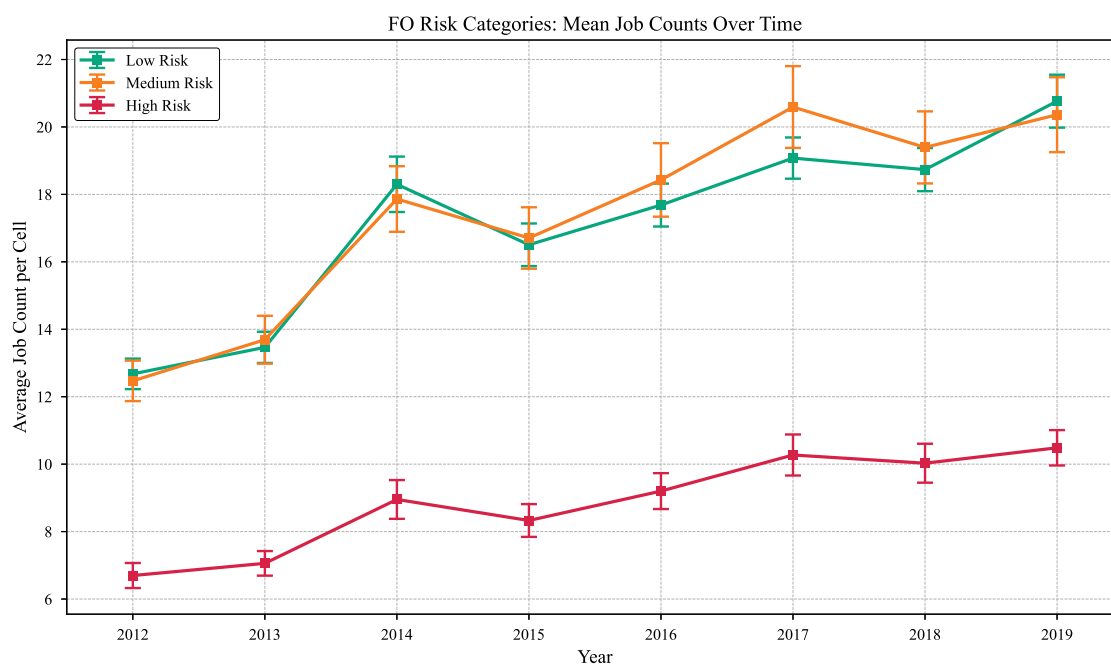
Table 5.8: Frey-Osborne Descriptive Statistics

Variable	N	Mean	SD	Min	Median	Max
Count	152,822	24.26	107.53	1	4	7,246

Note. Advertisement counts per occupation-state-industry-year cell.

5.5.4 Skill-Level Sample

For skill-level analysis, I operationalise skill requirements using employer-stated degree requirements following Acemoglu and Autor (2011b): High Skill (postgraduate), Medium Skill (bachelor degree, AQF Level 7), and Low Skill (sub-bachelor credentials). Where advertisements specified both minimum and maximum degree, I use the maximum. I

Figure 5.3: Mean Advertisement Counts by Frey-Osborne Risk Level

Note. Point plot showing mean advertisement counts with 95% confidence intervals for each Frey-Osborne risk category, 2012–2019.

exclude advertisements lacking degree information (1.55 million, or 16.2% of the sample) rather than imputing values that might introduce bias.

Table 5.9 summarises the skill-level sample (7.95 million advertisements, 83.8% of original sample). The sample aggregates to 217,542 cells for RCM models and 218,294 cells for Frey-Osborne models, reflecting different occupation coverage.

Table 5.9: Advertisement Counts by Skill Level

Skill Level	Advertisements	Percent
High Skill	441,598	5.6%
Medium Skill	7,058,807	88.8%
Low Skill	447,439	5.6%
Total	7,947,844	100.0%

Note. Skill levels operationalised using AQF standards following Autor, Levy, & Murnane (2003).

5.5.5 Regression Specifications

I estimate four model specifications to test robustness under different assumptions about confounding. The coefficient of interest is β_3 , the interaction between automation category and time, which captures differential growth rates across categories.

Model 1: Unconditional

$$c_{osit} = \beta_0 + \beta_1 a_o + \beta_2 t + \beta_3 (a_o \times t) + \epsilon_{osit} \quad (5.1)$$

where c_{osit} is advertisement count for occupation o , state s , industry i , year t ; a_o is automation category; t is continuous time. This baseline model estimates raw time trends without controls.

Model 2: State + Industry Fixed Effects (Primary Specification)

$$c_{osit} = \beta_0 + \beta_1 a_o + \beta_2 t + \beta_3 (a_o \times t) + \gamma_s + \delta_i + \epsilon_{osit} \quad (5.2)$$

where γ_s and δ_i are state and industry fixed effects. I designate Model 2 as the primary specification because it controls for the most theoretically important confounds (geographic and sectoral composition) while preserving continuous time structure for straightforward interpretation (Angrist & Pischke, 2009). The coefficient β_3 directly measures annual growth rate differences between automation categories.

Model 3: Occupation Fixed Effects

$$c_{osit} = \beta_0 + \beta_3 (a_o \times t) + \alpha_o + \epsilon_{osit} \quad (5.3)$$

where α_o are occupation fixed effects. Because automation categories are time-invariant occupation characteristics, they are perfectly collinear with occupation fixed effects—the main effect is absorbed. However, the interaction $\beta_3 (a_o \times t)$ remains identified through within-occupation temporal variation. Model 3 tests whether variation attributed to automation categories in Models 1–2 actually reflects occupation-specific characteristics.

Model 4: State + Industry + Year Fixed Effects

$$c_{osit} = \beta_0 + \beta_3 (a_o \times t) + \gamma_s + \delta_i + \theta_t + \epsilon_{osit} \quad (5.4)$$

where θ_t are year fixed effects. When including year fixed effects, I use categorical year interactions to avoid collinearity. This specification provides the most comprehensive control structure for robustness assessment.

Standard Errors

I cluster standard errors at the industry-state level for Models 2 and 4, matching the fixed effects structure. Model 3 clusters at the occupation level. Model 1 uses heteroskedasticity-robust standard errors (Cameron & Miller, 2015).

Choice of OLS over Count Models

The dependent variable is a count of advertisements per occupation-state-industry-year cell with right-skewed distribution (cell range 1–7,246; mean 16–24 depending on framework). Pseudo-Poisson maximum likelihood (PPML) is the modern standard for count panel data with fixed effects, particularly where the conditional mean function is multiplicative (Santos Silva & Tenreiro, 2006). I adopt OLS as the primary estimator, following the cell-aggregated panel convention of Autor, Levy, and Murnane (2003) and Acemoglu and Autor (2011b), for three reasons. First, the focal estimand is the partial association between automation classification and demand levels under fixed-effects controls, for which OLS coefficients on the interaction term are unbiased even when inefficient. Second, at $N = 170,491$ cells, OLS efficiency loss is small relative to the standard errors that drive both significance tests and the Wald-test inference (Section 5.5) anchoring the chapter’s central distinguishability finding. Third, OLS coefficients in additive form (“additional advertisements per year”) support cleaner inference for between-coefficient Wald tests; PPML’s multiplicative form complicates such tests because differences of log-coefficients require delta-method approximation.

Robustness checks using PPML on the primary specification (Table 5.10, with full code in `analysis/scripts/chapter5/ppml_robustness.py`) confirm the chapter’s load-bearing Wald-test finding: cognitive category coefficients remain statistically indistinguishable under PPML ($p = 0.124$ vs. OLS $p = 0.157$). Coefficient signs are preserved across estimators (Non-Routine Cognitive and Routine Cognitive both positive; Non-Routine Manual negative), and the qualitative weak-predictive-power pattern survives, with the noted caveat that fit metrics are not directly comparable across functional forms: OLS R^2 of 0.030 and PPML deviance pseudo- R^2 of 0.193 use different formulas, and McFadden pseudo- R^2 of 0.182 under PPML is consistent with the model fitting predominantly the year, state, and industry fixed effects rather than the automation classification, mirroring the OLS interpretation. Individual coefficient significance attenuates under PPML—Routine Cognitive $\times t$ loses significance ($p = 0.753$), and Non-Routine Cognitive $\times t$ retains only marginal significance ($p = 0.093$)—which cuts in the same direction as the chapter’s substantive argument: even the individually significant OLS coefficients reflect modest and imprecisely-identified effects. The $R^2 < 0.10$ falsification threshold pre-specified in

Section 5.2 is calibrated for OLS and applies to the OLS estimator only; the corresponding PPML conclusion rests on the Wald test of category indistinguishability, which agrees with the OLS finding.

Table 5.10: OLS vs PPML Robustness Check (RCM Model 2, Primary Specification)

Interaction term	OLS			PPML (Poisson)			
	Coef.	(SE)	<i>p</i>	Coef. (log)	(SE)	<i>p</i>	%/yr vs RM
Non-Routine Cognitive $\times t$	0.889	(0.308)	0.004	0.0182	(0.0108)	0.093	1.83%
Routine Cognitive $\times t$	0.563	(0.233)	0.016	0.0029	(0.0091)	0.753	0.29%
Non-Routine Manual $\times t$	-0.196	(0.114)	0.085	-0.0374	(0.0079)	0.000	-3.67%
Wald (NRC $\times t$ = RC $\times t$)		0.157			0.124		
R² / Deviance pseudo-R²		0.0301			0.1927		
McFadden pseudo-R²		—			0.1816		
<i>N</i> (cells)		170,491			170,491		

Note. Both estimators specify $\text{Count}_{osit} = \beta_0 + \beta_1 a_o + \beta_2 t + \beta_3(a_o \times t) + \gamma_s + \delta_i + \varepsilon_{osit}$ where a_o is RCM category (Routine Manual reference), t is continuous time (2012=0), γ_s are state fixed effects, and δ_i are industry fixed effects. Standard errors clustered at the industry-state level. PPML estimated as a generalised linear model with Poisson family and log link, following Santos Silva & Tenreyro (2006). PPML coefficients are in log form; the right-most column reports the multiplicative interpretation $(\exp(\beta) - 1) \times 100$. The Wald row tests the equality of Non-Routine Cognitive and Routine Cognitive interaction coefficients; both estimators agree that the categories are statistically indistinguishable. R² and pseudo-R² are not directly comparable across estimators (different formulas); both are reported for completeness. Source: analysis/scripts/chapter5/ppml_robustness.py; underlying data analysis/rcm_collapsed.parquet.

5.5.6 Wald Tests for Coefficient Distinguishability

Standard regression output tests whether coefficients differ from zero, but not whether they differ from each other. Two coefficients may both be individually significant yet statistically indistinguishable (Gelman & Stern, 2006). I employ Wald tests to formally test whether automation category coefficients are distinguishable:

$$H_0 : \beta_j = \beta_k \quad \text{with test statistic} \quad W = \frac{(\beta_j - \beta_k)^2}{\text{Var}(\beta_j - \beta_k)} \sim F(1, n - p) \quad (5.5)$$

where $\text{Var}(\beta_j - \beta_k) = \text{Var}(\beta_j) + \text{Var}(\beta_k) - 2\text{Cov}(\beta_j, \beta_k)$ accounts for covariance structure (Greene, 2018). I conduct Wald tests on Model 2 coefficients with $\alpha = 0.05$. If theoretically distinct automation categories (e.g., Non-Routine Cognitive versus Routine Cognitive) prove statistically indistinguishable, this suggests the RCM/FO frameworks capture minimal meaningful variation.

5.5.7 Skill-Level Interaction Models

I extend the analysis to test whether automation categories differentially affect skill composition within occupations. The three-way interaction specification:

$$c_{liast} = \beta_0 + \beta_1(a \times s) + \beta_2(a \times t) + \beta_3(s \times t) + \beta_4(a \times s \times t) + \text{FE} + \epsilon_{liast} \quad (5.6)$$

where a is automation category, s is skill level, and t is time. The coefficient of interest is β_4 , capturing whether automation categories show differential time trends across skill levels—evidence of skill upgrading or downgrading.

I estimate three specifications (Models 1, 2, and 4), excluding Model 3 (Occupation FE) because occupation fixed effects would absorb skill composition variation if certain occupations inherently cluster at specific skill levels. Reference categories are Routine Manual $\times$ Low Skill (RCM) and High Risk $\times$ Low Skill (Frey-Osborne).

5.5.8 Variance Decomposition

To assess whether variation in labour demand operates primarily between or within occupations, I decompose total variance using the Intraclass Correlation Coefficient (ICC). The ICC represents the proportion of total variance attributable to between-group differences:

$$\text{ICC} = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2} \quad (5.7)$$

where $\sigma_{\text{between}}^2$ is the variance of occupation means around the grand mean, and σ_{within}^2 is the average variance within occupations. I estimate these components using one-way ANOVA decomposition (Shrout & Fleiss, 1979):

$$\sigma_{\text{between}}^2 = \frac{MS_{\text{between}} - MS_{\text{within}}}{k}, \quad \sigma_{\text{within}}^2 = MS_{\text{within}} \quad (5.8)$$

where MS_{between} and MS_{within} are the mean squares from one-way ANOVA with occupation as the grouping variable, and k is the average number of cells per occupation. A low ICC indicates most variation occurs within occupations rather than between them—direct evidence for the aggregation bias mechanism predicted by Proposition 3.

I complement the ICC with R^2 decomposition, comparing the explanatory power of: (1) a model with only occupation fixed effects, and (2) a model with only automation category dummies. If automation categories capture meaningful variation, their R^2 should approach the occupation-only R^2 . Large discrepancies indicate automation classifications fail to capture occupation-level distinctions, let alone within-occupation heterogeneity.

5.6 Results

This section presents regression results testing whether occupation-level automation frameworks predict variation in Australian labour demand. I report results from all four model specifications to demonstrate robustness, with Model 2 (State + Industry FE) as the primary specification. Sample size is constant at 170,491 occupation-state-industry-year cells.

Before examining regression results, Figure 5.4 presents descriptive trends by RCM category without controls. Non-Routine Cognitive occupations show the strongest growth trajectory, while Routine Cognitive and Routine Manual categories show flatter trends. However, these raw trends conflate potential automation-related variation with compositional differences across industries and regions—the regression analysis below isolates whether these patterns persist after controlling for confounds.

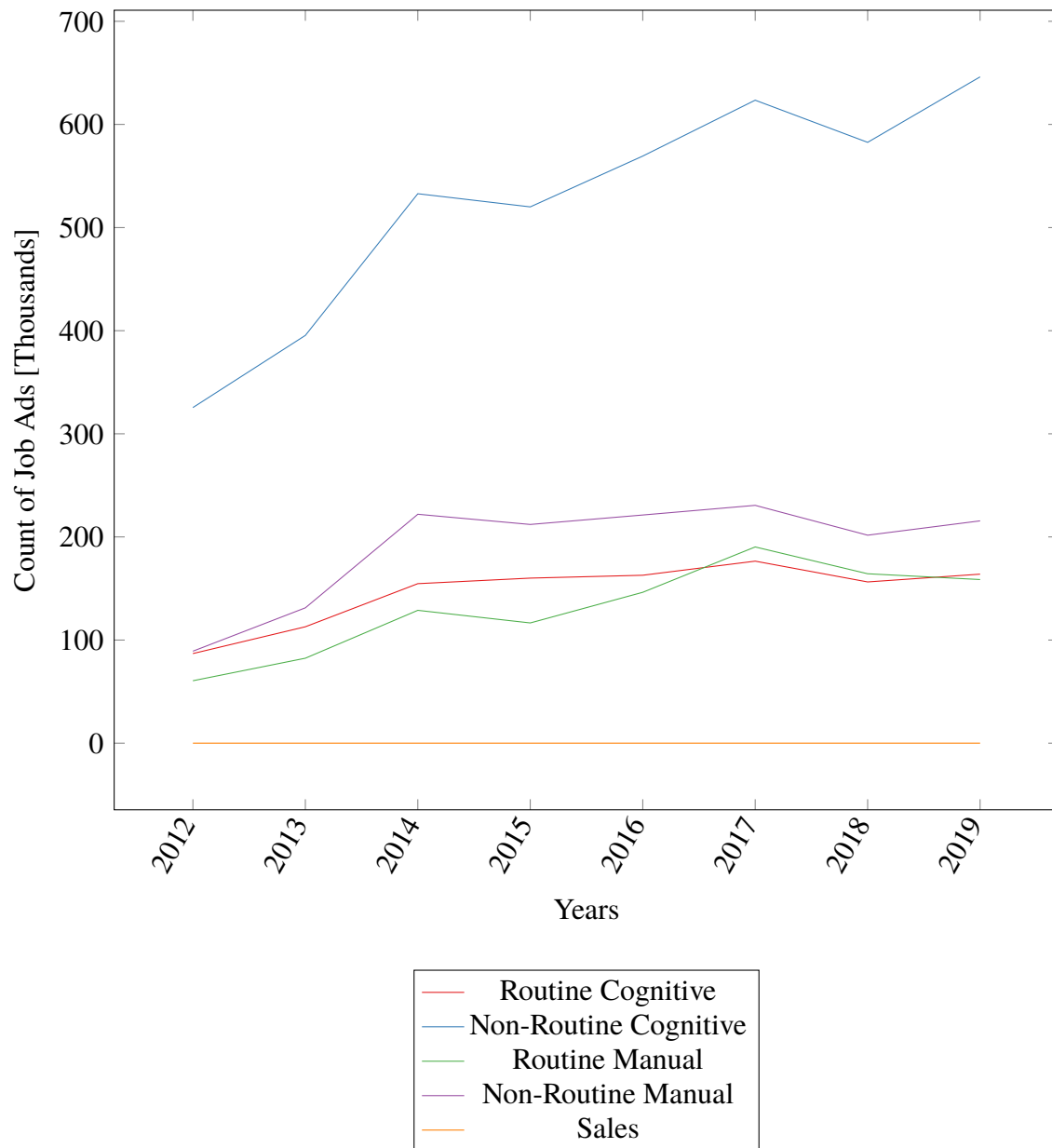
5.6.1 RCM Framework Results

Table 5.11 presents results across four specifications. The key finding is consistently low explanatory power: R^2 ranges from 0.003 (Model 1) to 0.062 (Model 3), indicating RCM categories explain minimal variance in labour demand regardless of control structure. This result is consistent with Proposition 3’s prediction that occupation-level classifications fail to capture within-occupation restructuring.

Explanatory Power

The progression of R^2 values is revealing: Model 1 (0.003) → Model 2 (0.030) → Model 3 (0.062) → Model 4 (0.031). Adding occupation fixed effects (Model 3) doubles explanatory power, indicating occupation-specific factors explain more variation than RCM categories. Yet even Model 3 explains only 6% of variance. The near-identical R^2 between Model 2 (linear time trend) and Model 4 (categorical year fixed effects)—0.030 vs. 0.031—confirms that the linearity assumption in the primary specification is not driving the low explanatory power: relaxing the linear trend to fully flexible year effects yields essentially the same conclusion.

How should I interpret these low R^2 values? Fixed effects models mechanically absorb substantial variation, leaving residual variation representing *within-occupation, within-industry, within-state changes over time*—precisely where automation effects should manifest if RCM categories correctly identify vulnerable occupations. Comparable panel studies of labour demand achieve $R^2 = 0.15$ – 0.40 when examining genuine drivers (Acemoglu et al., 2016; Autor et al., 2013). The observed $R^2 = 0.03$ – 0.06 therefore represents substantially weaker explanatory power than benchmarks from similar studies, suggesting RCM

Figure 5.4: Descriptive Trends by RCM Category

Note. Descriptive trends without fixed effects. Non-Routine Cognitive shows strongest growth; Routine Manual shows stronger demand than Census employment data suggests, likely reflecting online posting bias toward formalised manual positions.

categories capture minimal relevant variation.

Coefficient Estimates

Non-Routine Cognitive shows positive interaction coefficients across specifications (0.79–3.03 jobs/year); Routine Cognitive shows similar effects (0.55–3.17 jobs/year); Non-Routine

Table 5.11: Model Comparison: Testing RCM Framework Across Specifications

	(1)	(2)	(3)	(4)
NR Cognitive $\times$ Year	0.794*** (0.263)	0.889*** (0.308)	1.345*** (0.109)	3.025** (2.058)
NR Manual $\times$ Year	-0.268 (0.293)	-0.196* (0.114)	0.255 (0.166)	2.879*** (1.283)
R Cognitive $\times$ Year	0.794*** (0.263)	0.889*** (0.308)	1.345*** (0.109)	3.173*** (1.771)
State FE	NO	YES	NO	YES
Industry FE	NO	YES	NO	YES
Occupation FE	NO	NO	YES	NO
Year FE	NO	NO	NO	YES
R ²	0.003	0.030	0.062	0.031
N	170,491			

Note. Dependent variable is count of jobs in occupation-state-industry-year cells. Models 1-3 use continuous year interactions (RCM $\times$ Year_centered). Model 4 uses categorical year interactions (RCM $\times$ Year dummies). Model 2 is the primary specification. Standard errors in parentheses, clustered at industry-state level for Models 2 and 4. Reference category: Routine Manual. Significance: *p<0.1; **p<0.05; ***p<0.01.

Manual shows weaker, less consistent effects. The magnitudes are economically modest: the largest Model 2 effect (0.89 jobs/year) represents 3.7% annual growth relative to the cell mean.

Wald Tests for Coefficient Distinguishability

Table 5.12 presents Wald tests examining whether RCM categories are statistically distinguishable from each other (not just from the reference).

The key finding is that **Non-Routine Cognitive and Routine Cognitive coefficients are statistically indistinguishable** ($F = 2.00$, $p = 0.157$). Despite both showing individual significance, the 0.33 jobs/year difference is not significant. With $N = 170,491$, this analysis has >99% power to detect differences of 0.1 jobs/year (Cohen, 1988); the null result reflects genuine absence of meaningful differences, not insufficient power.

This finding directly undermines the RCM framework's validity. The core theoretical prediction—that routine cognitive tasks face greater automation displacement than non-routine cognitive tasks—requires distinguishable effects. The statistical indistinguishability suggests “routine” versus “non-routine” cognitive work produces functionally identical labour demand dynamics.

Table 5.12: Wald Tests for RCM Coefficient Equality (Model 2)

Category 1	Category 2	Difference	F-statistic	p-value	Distinguishable
NRC	NRM	1.085***	16.19	0.0001	Yes
NRC	RC	0.326	2.00	0.1570	No
NRM	RC	−0.759***	15.17	0.0001	Yes

Note. Wald tests examine whether pairs of RCM category coefficients (from Model 2: State + Industry FE) are statistically distinguishable from each other. NRC = Non-Routine Cognitive; RC = Routine Cognitive; NRM = Non-Routine Manual. The null hypothesis is $H_0: \beta_1 = \beta_2$ (coefficients are equal). Difference shows $\beta_1 - \beta_2$. F-statistics test the null hypothesis of coefficient equality. “Distinguishable” indicates whether we can reject H_0 at the 5% level. NRC and RC coefficients are *not* statistically different ($p = 0.157$), demonstrating a “distinction without a difference”—both categories show positive growth but are indistinguishable from each other. Reference category: Routine Manual. Significance: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Interpretation

RCM categories show weak predictive power regardless of specification. The consistently low R^2 (never exceeding 0.07) and indistinguishable cognitive category effects suggest automation’s labour market impacts operate through mechanisms not captured by occupation-level taxonomies.

Skill-Level Interactions

I extend the analysis to test whether RCM categories affect skill composition. Table 5.13 presents three-way interaction results (automation category $\times$ skill level $\times$ time).

Table 5.13: Interaction Effects of RCM Model, Year, and Skill Level

			Dependent Variable: Count		
			(1)	(2)	(3)
Non-Routine Cognitive :	Year :	High Skill	0.493** (0.234)	1.195*** (0.345)	1.147*** (0.346)
		Medium Skill	1.146*** (0.328)	1.336*** (0.361)	1.321*** (0.360)
Routine Cognitive :	Year :	High Skill	0.143 (0.107)	0.569** (0.229)	0.540** (0.226)
		Medium Skill	0.942*** (0.361)	1.012*** (0.350)	0.990*** (0.349)
Non-Routine Manual :	Year :	High Skill	0.116 (0.296)	0.297 (0.400)	0.272 (0.405)
		Medium Skill	0.914** (0.410)	0.667 (0.434)	0.658 (0.432)
State Fixed Effects			NO	YES	YES
Industry Fixed Effects			NO	YES	YES
Time Fixed Effects			NO	NO	YES
Observations			217542		

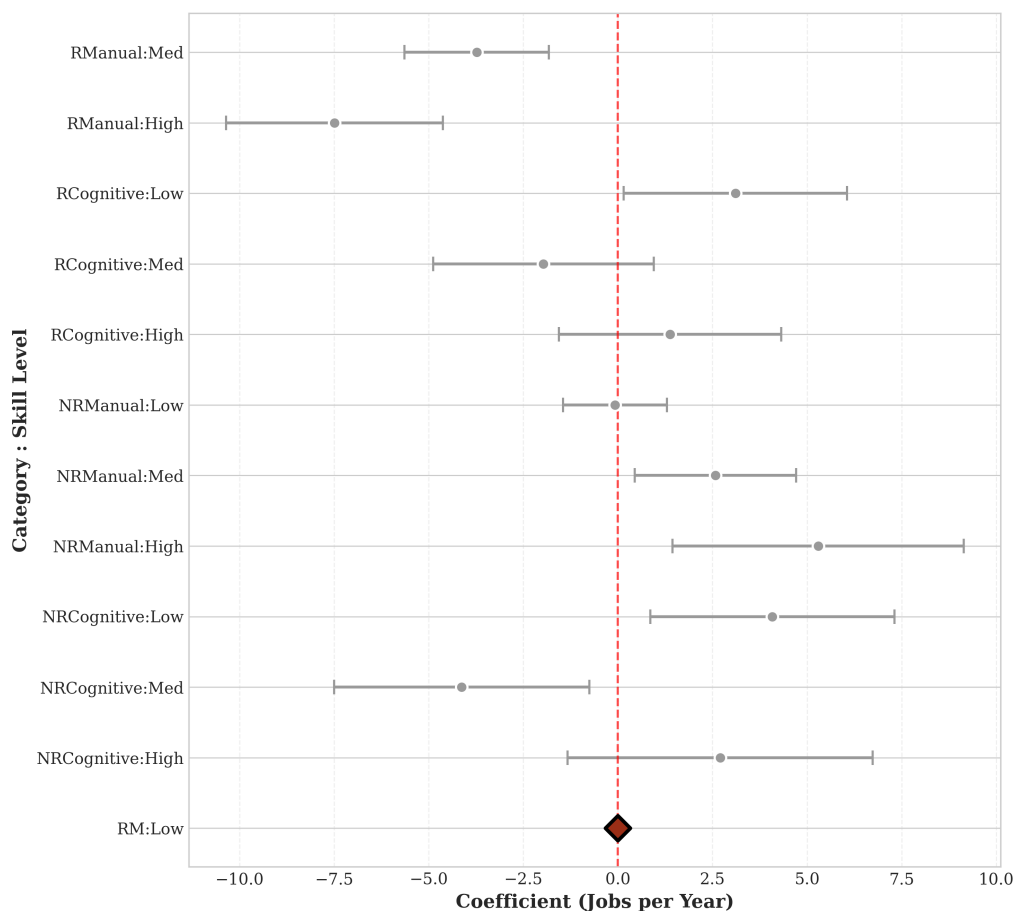
Note. This table depicts the results of the three-way interaction model (Equation 5.6) using the collapsed RCM data set. Model 1 does not include fixed effects. Model 2 includes State + Industry FE. Model 4 includes State + Industry + Year FE. Reference category: Routine Manual $\times$ Low Skill. Significance is demonstrated as * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Non-Routine Cognitive shows significant growth for both high skill (1.147 jobs/year, $p < 0.001$) and medium skill (1.321 jobs/year, $p < 0.001$) positions. Routine Cognitive shows similar patterns (medium: 0.990, $p < 0.001$; high: 0.540, $p < 0.05$). Non-Routine Manual shows no significant effects.

The pattern provides modest support for skill-biased technical change: within cognitive occupations, demand shifts toward medium and high skill positions relative to low skill roles. However, overlapping confidence intervals between skill levels and the absence of effects for Non-Routine Manual suggest the pattern is neither strong nor universal. The RCM framework detects some skill-composition effects but explains minimal overall variation ($R^2 = 0.031$).

5.6.2 Frey-Osborne Framework Results

Table 5.14 presents results for the Frey-Osborne automation risk framework. The findings mirror RCM: R^2 ranges from 0.002 to 0.061, indicating risk classifications explain minimal

Figure 5.5: RCM Skill-Level Interaction Coefficients

Note. Coefficients with 95% confidence intervals for three-way interaction terms (RCM category $\times$ skill level $\times$ time). Reference: Routine Manual $\times$ Low Skill.

variance in labour demand.

Explanatory Power

The R^2 progression is nearly identical to RCM: Model 1 (0.002) $\rightarrow$ Model 2 (0.030) $\rightarrow$ Model 3 (0.061) $\rightarrow$ Model 4 (0.030). The striking similarity (both frameworks produce $R^2 \approx 0.03$ in the primary specification) suggests both capture the same underlying occupation-level variation—and neither captures much.

Coefficient Estimates

Low Risk and Medium Risk occupations show positive coefficients relative to High Risk (0.554 and 0.566 jobs/year respectively, both $p < 0.05$). These magnitudes are economically negligible—approximately 2% annual growth relative to the cell mean.

Table 5.14: Model Comparison: Testing FO Framework Across Specifications

	(1)	(2)	(3)	(4)
Low Risk $\times$ Year	0.464* (0.256)	0.554** (0.233)	1.015*** (0.091)	2.952** (1.656)
Medium Risk $\times$ Year	0.554* (0.321)	0.566** (0.234)	1.290*** (0.209)	3.326*** (1.485)
State FE	NO	YES	NO	YES
Industry FE	NO	YES	NO	YES
Occupation FE	NO	NO	YES	NO
Year FE	NO	NO	NO	YES
R ²	0.002	0.029	0.061	0.030
N	170,491			

Note. Dependent variable is count of jobs in occupation-state-industry-year cells. Models 1-3 use continuous year interactions (FO $\times$ Year_centered). Model 4 uses categorical year interactions (FO $\times$ Year dummies). Model 2 is the primary specification. Standard errors in parentheses, clustered at industry-state level for Models 2 and 4. Reference category: High Risk. Significance: *p<0.1; **p<0.05; ***p<0.01.

Wald Tests

Table 5.15 presents Wald tests for coefficient distinguishability.

Table 5.15: Wald Tests for FO Coefficient Equality (Model 2)

Category 1	Category 2	Difference	F-statistic	p-value	Distinguishable
Low Risk	Medium Risk	-0.012	0.004	0.9466	No

Note. Wald test examines whether Low Risk and Medium Risk coefficients (from Model 2: State + Industry FE) are statistically distinguishable from each other. The null hypothesis is $H_0: \beta_{\text{Low}} = \beta_{\text{Medium}}$ (coefficients are equal). With $p = 0.947$, we cannot reject H_0 —the two risk categories are completely indistinguishable. Despite both showing individual significance relative to High Risk, Low and Medium Risk categories provide no meaningful differentiation in automation effects. This represents a “distinction without a difference,” indicating that FO risk classifications fail to capture differential labour market impacts. Reference category: High Risk. Significance: *p<0.1; **p<0.05; ***p<0.01.

The critical finding is that **Low Risk and Medium Risk coefficients are statistically indistinguishable** ($F = 0.004$, $p = 0.947$). The difference is merely -0.012 jobs/year. The Frey-Osborne framework categorises occupations into three risk levels intended to predict

differential automation impacts, yet the two non-reference categories produce statistically identical effects in the present data.

This finding is consistent with the predictive-validity reading of Proposition 3. If automation risk scores identified displacement vulnerability at occupation-level granularity, different risk levels should produce distinguishable demand trajectories. The null result—despite >99% power to detect small differences—is consistent with risk classifications capturing minimal occupation-level variation, in line with critiques that occupation-level assessments operate at inappropriate aggregation (Arntz et al., 2016; Autor, 2015a).

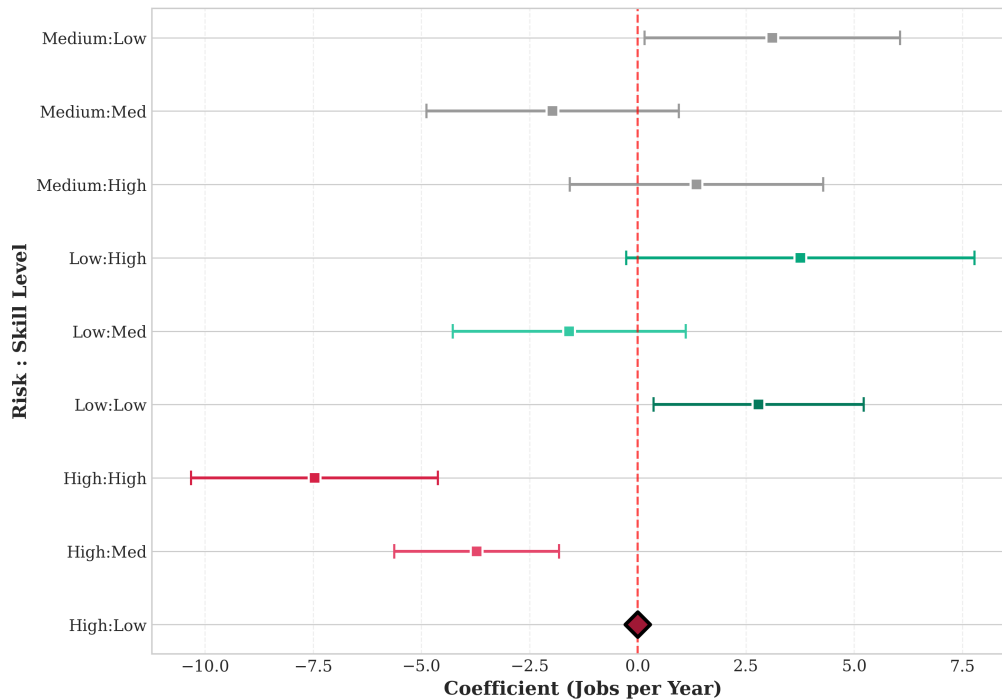
Skill-Level Interactions

Table 5.16 presents three-way interaction results for Frey-Osborne categories. In contrast to RCM, **none of the skill-level effects are statistically significant**. The Frey-Osborne framework detects no evidence of skill composition shifts associated with automation risk.

Table 5.16: Interaction Effects of FO Model, Year, and Skill Level

			Dependent Variable: Count		
			(1)	(2)	(3)
Low Risk :	Year :	High Skill	0.191 (0.290)	0.541 (0.345)	0.531 (0.346)
		Medium Skill	0.275 (0.382)	0.445 (0.403)	0.455 (0.403)
Medium Risk :	Year :	High Skill	0.537* (0.296)	0.715* (0.395)	0.720* (0.395)
		Medium Skill	-0.028 (0.429)	-0.097 (0.437)	-0.081 (0.437)
State Fixed Effects			NO	YES	YES
Industry Fixed Effects			NO	YES	YES
Time Fixed Effects			NO	NO	YES
Observations			218294		

Note. This table depicts the results of the three-way interaction model (Equation 5.6) using the collapsed FO data set. Model 1 does not include fixed effects. Model 2 includes State + Industry FE. Model 4 includes State + Industry + Year FE. Reference category: High Risk $\times$ Low Skill. Significance is demonstrated as * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure 5.6: FO Skill-Level Coefficients (Including Baseline)

Note. Confidence intervals for three-way interaction term coefficients from Model 3. All skill level confidence intervals are overlapping, making it difficult to discern any effect. No results achieved statistical significance, indicating FO risk levels show no detectable impact on skill level demand changes.

5.6.3 Variance Decomposition

The regression results above show that occupation-level automation frameworks explain minimal variation in labour demand. A natural question is: where does the unexplained variation reside? To address this, I decompose total variance into between-occupation and within-occupation components using the Intraclass Correlation Coefficient (ICC).

Table 5.17 presents the variance decomposition results. Panel A reports ICC-based decomposition: only 5.6% of variance in job advertisement counts occurs *between* occupations, while 94.4% occurs *within* occupations. This striking imbalance indicates that occupational classification captures minimal relevant variation—most differences in labour demand occur among jobs within the same occupation rather than across occupational categories.

Panel B reinforces this finding using R^2 decomposition. A model with only occupation fixed effects achieves $R^2 = 0.061$ —occupation membership explains just 6% of variance. More critically, automation category dummies alone (RCM or FO classifications) explain less than 0.2% of variance. These frameworks, designed to capture automation-relevant distinctions, capture virtually none.

Table 5.17: Variance Decomposition: Between- vs. Within-Occupation Variation

Component	Value	Share
Panel A: ICC Decomposition		
ICC (Intraclass Correlation)	0.056	–
Between-occupation variance	317.2	5.6%
Within-occupation variance	5374.8	94.4%
Panel B: R² Decomposition		
Occupation-only R ²	0.061	–
RCM category R ²	0.002	–
Frey-Osborne category R ²	0.001	–
Panel C: Sample Characteristics		
Observations	170,491	
Occupations (4-digit ANZSCO)	877	
Avg. cells per occupation	194.4	

Note. Variance decomposition computed on the RCM analytical sample (170,491 cells across 877 4-digit ANZSCO occupations). The Frey-Osborne sample is a subset of this set after concordance filtering (152,822 cells, 334 occupations); ICC computed separately on the FO subset yields qualitatively identical results (within ± 0.005 of the value reported here) and is therefore reported jointly. Panel A reports variance decomposition using one-way ANOVA. ICC (Intraclass Correlation Coefficient) represents the proportion of total variance attributable to between-occupation differences. Panel B reports R² from OLS regressions: “Occupation-only R²” is from a model with only occupation fixed effects; the framework R² rows are from models with only the relevant category dummies. The finding that 94% of variance operates within occupations supports the aggregation bias mechanism predicted by Proposition 3.

These results directly support Proposition 3’s prediction of aggregation bias. If automation restructures task bundles *within* occupational boundaries—contracting routine-intensive job types while expanding complementary ones—occupation-level measurement will systematically fail to detect these effects. The 94% within-occupation variance share is the empirical signature this mechanism would produce.

Robustness to cell size.

The regression sample includes cells with as few as 1 observation (median cell size = 3), raising the concern that sparse cells inflate noise and deflate R^2 . Table 5.18 reports results across progressively restrictive minimum cell size thresholds.

Table 5.18: Robustness to Minimum Cell Size

Min cell size	Cells	% retained	R^2	ICC	Within-occ (%)
≥1 (all)	170,491	100.0	0.040	0.061	93.9
≥5	69,743	40.9	0.049	0.099	90.1
≥10	43,384	25.4	0.056	0.130	87.1
≥30	17,448	10.2	0.077	0.192	80.8
≥50	10,516	6.2	0.089	0.218	78.2

Note. Model 2 (primary specification with state and industry fixed effects, HC3 standard errors) estimated on subsets of occupation-state-industry-year cells meeting the minimum count threshold. R^2 is from the full specification. ICC computed via one-way ANOVA with occupation as grouping variable. Within-occupation percentage = $1 - \text{ICC}$.

Even at the most restrictive threshold (≥50, retaining only 6.2% of cells), R^2 remains below 9%, confirming that occupation-level frameworks explain minimal variance regardless of cell sparsity. The ICC rises from 0.061 to 0.218, meaning within-occupation variation ranges from 78% to 94% depending on threshold—but dominates at every specification. The core finding is not an artefact of sparse cells.

5.7 Discussion

The results admit two interpretations: either automation has minimal labour market effects during 2012–2019, or occupation-level measurement operates at the wrong level of aggregation. The pattern of findings favours the second interpretation. If automation simply had no effects, we would expect low explanatory power but no particular pattern in where variation concentrates. The finding that variation overwhelmingly concentrates *within*

occupations suggests systematic structure at sub-occupational levels that these frameworks cannot capture.

5.7.1 Australian Institutional Context

Three features of Australia's institutional framework (Section 4.2.1) may explain why within-occupation variation dominates:

Award wage constraints: Modern Awards covering 60% of employees prevent wage adjustment in response to automation. When compensation is constrained, employers adjust through quantities or task reorganisation. The high within-occupation variation (94%) suggests task reorganisation is the primary adjustment mechanism—consistent with Proposition 3.

Service sector dominance: With 79.8% service employment, Australian patterns may differ from manufacturing-focused research. Service work exhibits greater task variety (Autor, 2013b), potentially generating automation vulnerabilities at sub-occupational rather than occupational levels.

Geographic concentration: Metropolitan labour markets contain 70.2% of jobs. This concentration may support finer within-occupation specialisation, amplifying restructuring that occupation-level analysis cannot detect. Proposition 2 predicts market thickness enables cluster differentiation, a pattern this concentration would amplify.

5.7.2 Comparison to International Evidence

Australian findings ($R^2 = 0.030$) align closely with international evidence:

- Arntz et al. (2016): OECD task-based analysis finds 9% automation risk (vs. 47% occupation-based), attributing the difference to within-occupation heterogeneity
- Autor (2015a): US occupation employment stable 1980–2010, with most variation within rather than between occupations
- Nedelkoska and Quintini (2018): OECD average 14% high-risk, with substantial within-occupation variation across workers

This cross-national consistency suggests occupation-level null findings reflect fundamental aggregation bias rather than country-specific factors. Australia's distinctive institutions do not generate systematically different occupation-level patterns than US or European markets—automation operates through within-occupation restructuring regardless of institutional context.

5.7.3 Aggregation Bias Mechanism

The three findings above—low R^2 (0.030), indistinguishable category coefficients, and 94% within-occupation variance share—together form the empirical signature predicted by Proposition 3. Consider “Accountants” (ANZSCO 2211), encompassing tax compliance specialists, financial analysts, management accountants, and bookkeeping roles. If automation reduces demand for routine compliance work while increasing demand for analytical positions requiring technology proficiency, occupation-level measurement aggregates these opposing effects and detects minimal net change despite substantial restructuring. The chapter cannot identify whether automation specifically drives the within-occupation variation, but it establishes that the variation exists at the granularity required for the mechanism to operate. Chapters 6–7 test the mechanism directly at cluster level.

5.7.4 Limitations

Six limitations constrain interpretation.

Correlations, not causation: Without exogenous variation in automation adoption, coefficients represent associations rather than identified causal effects. The analysis tests whether predicted patterns appear in the data—necessary for theoretical validity—but cannot definitively attribute patterns to automation.

Demand-side measurement: Job advertisements capture hiring intentions, not realised employment. If automation affects hiring standards and job design (as Proposition 3 predicts), demand-side data may be more informative than employment for detecting restructuring; nevertheless, advertised requirements need not match the tasks workers actually perform.

BGT coverage growth: Annual advertisement volume rose from 987k (2012) to 1.31M (2018), partly reflecting expanded BGT crawler coverage rather than market growth. Year fixed effects absorb common growth shocks, but the interaction coefficient β_3 could be biased if coverage expansion differs systematically across automation categories—e.g., if professional positions (over-represented in BGT) concentrate in low-risk Frey-Osborne categories. The variance-decomposition result (94% within-occupation share) is unaffected by coverage growth because it operates within-occupation, but the regression coefficients should be interpreted with this caveat in mind.

Skill-level operationalisation: The skill-composition analysis uses employer-stated degree requirements as a proxy for skill level, with three weaknesses: 16% of advertisements lack degree information and are excluded; 88.8% of remaining advertisements specify bachelor-level qualifications, leaving limited variation across skill bands; and degree requirements may reflect signalling rather than skill content (Spence, 1973). The skill-

composition results in Section 5.7 are therefore suggestive rather than definitive, and the central findings of the chapter (R^2 , coefficient indistinguishability, variance decomposition) do not depend on the skill-level analysis.

Temporal scope: The 2012–2019 period predates generative AI advances. Findings characterise the diffusion of statistical machine learning; the aggregation bias mechanism should persist regardless of automation intensity, but its quantitative magnitude may differ under foundation-model deployment patterns.

Classification constraints: I accept ANZSCO definitions rather than questioning occupational boundaries. Alternative taxonomies might yield different results, though ANZSCO represents the policy-relevant framework. The chapter also tests only RCM and Frey-Osborne; more recent occupation-level measures (e.g., Felten et al. AIOE, Webb 2020) may exhibit different predictive validity, though the within-occupation variance ceiling identified here would constrain any occupation-level framework.

These limitations reinforce the central finding: occupation-level analysis cannot adequately capture automation’s effects. They motivate disaggregate analysis at finer granularity.

5.7.5 Implications for Frey-Osborne Predictions

The null findings for Frey-Osborne risk ratings have substantial implications. The most-cited automation prediction (Frey & Osborne, 2017)—claiming 47% of US jobs face high risk—shows *no detectable relationship* with Australian labour demand changes (2012–2019). Risk categories that should produce divergent trajectories are statistically indistinguishable ($p = 0.947$).

These null findings align with methodological critiques of Frey-Osborne. Arntz et al. (2016) argued that using homogenised occupation-level ratings rather than job-level task content inflates automation risk estimates (their task-based analysis found 9% versus 47%). Coelli and Borland (2019) identified internal consistency issues in the original methodology. The present analysis provides independent empirical evidence: treating risk scores as testable predictions rather than established facts reveals no significant relationship with observed demand changes.

This does not imply automation has no effects. Rather, the null findings suggest that occupation-level risk assessments—regardless of methodological sophistication—cannot capture effects concentrated at sub-occupational levels (Martin-Caughey, 2021).

5.7.6 Skill Composition

The skill-level analysis reveals that 88.8% of advertisements with degree requirements specify bachelor-level qualifications (Table 5.9). This concentration suggests credential inflation may exceed genuine skill requirements (Blair & Deming, 2020). However, the analysis cannot distinguish whether degree requirements reflect productivity needs or screening mechanisms (Spence, 1973).

The three-way interactions show modest skill-composition effects for RCM cognitive categories (shifting toward medium and high skill) but no significant effects for Frey-Osborne categories. This asymmetry likely reflects a fundamental difference in how the two frameworks construct variation. The RCM taxonomy classifies occupations by *task content* (routine vs non-routine, cognitive vs manual), creating categories that differ in their skill composition by construction—so skill-level interactions have something to detect. Frey-Osborne scores, by contrast, assign a single automation *probability* to each occupation based on expert judgement of overall susceptibility, without distinguishing task types or skill levels within occupations. Since the FO score contains no internal structure mapping to skill levels, the three-way interaction (risk $\times$ skill $\times$ time) has no theoretical basis for detecting differential patterns. This suggests that while neither framework explains substantial variance, task-based approaches (RCM) retain some predictive power for skill composition even when failing to predict demand levels, whereas probability-based approaches (FO) lack the internal structure to predict either.

5.8 Conclusion

This chapter asked whether occupation-level automation frameworks predict variation in Australian labour demand during the 2012–2019 machine-learning wave. The answer is clear: they do not, in any economically meaningful sense. Both the RCM taxonomy and Frey-Osborne risk scores explain only 3% of variation in job advertisement counts, both fall well below the pre-specified $R^2 < 0.10$ threshold for weak predictive power, and theoretically distinct categories prove statistically indistinguishable. Variance decomposition reveals why: over 94% of variation operates within occupations rather than between them (ICC = 0.056), imposing a ceiling on what any occupation-level framework can detect.

These findings align with international evidence (Arntz et al., 2016; Autor, 2015a; Nedelkoska & Quintini, 2018), suggesting the problem transcends institutional contexts. The implication is not that automation lacks labour market effects, but that occupation-level measurement is the wrong lens through which to observe them. Read in historical perspective, this chapter establishes a baseline for the second wave of automation: an

empirical record showing that during the diffusion of statistical machine learning into Australian workplaces, the dominant occupation-level measurement frameworks failed to detect labour-demand variation at meaningful magnitudes. If most variation concentrates within occupational boundaries, understanding automation requires methods capable of detecting restructuring at finer granularity. Chapter 6 pursues this approach by applying density-based clustering to the same 9.5 million advertisements, testing whether the within-occupation variation documented here is structured into discrete sub-occupational job types as Proposition 1 predicts.

Chapter 6

Within-Occupation Restructuring: Cluster-Level Evidence

6.1 Introduction

Chapter 5 established that occupation-level frameworks explain almost none of the variation in Australian labour demand, with variance decomposition attributing the vast majority of that variation to differences *within* occupations. This chapter asks whether that within-occupation variation reflects random noise or genuine sub-occupational structure—distinct clusters of job types that form, grow, and decline as technology reshapes how work is organised.

Chapter 3 provides the theoretical foundation for this inquiry. Its first three propositions predict that coordination cost convexity generates discrete clusters of job types within occupations (Proposition 1), that thicker labour markets support finer specialisation (Proposition 2), and that automation shifts cluster boundaries in ways that produce high within-occupation volatility alongside modest occupation-level effects (Proposition 3). This chapter tests each of these predictions empirically across 250 Australian occupations.

Three research questions guide the analysis:

1. Do occupations contain discrete sub-occupational clusters, and do larger firms exhibit finer division of labour across cluster types?
2. Do thicker labour markets—whether measured by geographic concentration or occu-

pation size—support more clusters?

3. Does cluster-level employment volatility substantially exceed occupation-level volatility, consistent with within-occupation restructuring that aggregation obscures?

The chapter proceeds as follows. Section 6.2 describes the data, sample, and empirical strategy. Section 6.3 presents cluster discovery results and validation. Sections 6.4–6.6 test each proposition in turn. Section 6.7 interprets the findings. Section 6.8 concludes.

6.2 Methods

6.2.1 Data and Sample

The analysis uses online job advertisement data from Burning Glass Technologies covering 2012–2019 (Section 4.4). I applied HDBSCAN clustering to all 250 occupations meeting the feasibility criteria described in Section 4.7, using Jaccard distance on binary skill vectors with uniform parameters across all occupations and years (Section 4.9). Where individual occupations are used for illustration, I draw on three 6-digit ANZSCO occupations spanning the automation probability spectrum: Accountant (General) (221111), Human Resource Adviser (223111), and Receptionist (General) (542111). These are drawn from the same analysis as all other occupations, not pre-selected case studies.

6.2.2 Empirical Strategy

The estimand throughout is descriptive: whether observed cluster structure, market thickness relationships, and within-occupation volatility are consistent with the theoretical predictions developed in Chapter 3. No causal effects are claimed. Following Angrist and Pischke (2009), I conduct reduced-form tests of theoretical predictions, treating the model as a coherent framework that either successfully organises observed patterns or fails to account for key empirical regularities.

Each proposition generates specific testable hypotheses. For Proposition 1, I test whether HDBSCAN identifies discrete clusters without pre-specifying their number, and whether larger firms hire across more distinct cluster types—consistent with coordination cost economies enabling finer division of labour. For Proposition 2, I exploit Australia’s extreme geographic concentration as a source of variation in market thickness, comparing cluster counts across metropolitan and regional markets and across occupations of different sizes. For Proposition 3, I compare employment growth volatility at the cluster and occupation

levels, and examine whether technology content predicts which clusters grow and which decline.

6.3 Results: Cluster Discovery

Across all 250 occupations, HDBSCAN identifies a median of 13 clusters per occupation (IQR: 7–26, range: 0–171). This section examines the structure and validity of these clusters before proceeding to formal hypothesis tests.

6.3.1 Number of Clusters Identified

Table 6.1 illustrates the range of cluster counts using the three illustrative occupations. Each fragments into multiple distinct clusters, with noise rates—jobs not assigned to any cluster—varying by occupation.

Table 6.1: Cluster Discovery Summary by Occupation

Occupation	ANZSCO	Job Ads	Avg Clusters	Noise Rate
Accountant (General)	221111	20,716	86	13.9%
Human Resource Adviser	223111	3,572	16	43.5%
Receptionist (General)	542111	5,896	38	43.9%

Note. Illustrative occupations at the ANZSCO 6-digit level, consistent with the full-scale analysis across all 250 occupations. Clusters and Noise Rate are from the 2019 cross-section. Job Ads reports 2019 advertisements. I performed clustering independently for each occupation-year using Jaccard distance with uniform parameters (`min_cluster_size=25`, `min_samples=10`).

The substantial number of clusters within each occupation challenges the implicit homogeneity assumption underlying occupation-level automation research. Accountant (General) alone contains 86 distinct job types in a single year; even the smaller Human Resource Adviser occupation fragments into 16. If each occupation contains dozens to hundreds of distinct job types, then occupation-level measures aggregate across heterogeneous units with potentially opposing dynamics—a form of ecological inference problem (King, 1997). This finding extends Deming (2017b)’s observation of substantial within-occupation skill heterogeneity by demonstrating that such heterogeneity organises into discrete job types rather than continuous distributions.

6.3.2 Cluster Validation

Following Grimmer and Stewart (2013)’s recommendations for validating unsupervised methods, I prioritise construct validity evidence over internal metrics. Standard measures such as silhouette scores assume convex, roughly spherical clusters in dense data, whereas HDBSCAN identifies arbitrary-shape, density-based clusters in high-dimensional binary space where the curse of dimensionality compresses distance distributions. I assess validity through three lines of evidence.

Substantive Interpretability

Manual inspection of cluster skill profiles confirms that discovered clusters correspond to recognisable job types rather than arbitrary statistical partitions. Table 6.2 presents examples from Accountant (General): clusters map to interpretable categories such as “General Accounting,” “Financial Analysis,” and “Reconciliation,” each characterised by distinctive skill combinations. This face validity strengthens confidence that the algorithm recovers genuine occupational structure.

Temporal Consistency

I cluster each occupation-year independently, applying identical parameters to each annual cross-section. If HDBSCAN were identifying noise rather than genuine structure, clusters would not persist across independent annual samples. In practice, many clusters identified in one year reappear with similar skill profiles in the next, suggesting that the algorithm recovers durable job types rather than random partitions. Chapter 7 quantifies this temporal consistency in detail.

Structural Validity

As Section 6.5 demonstrates, cluster counts exhibit strong theoretically predicted relationships with market fundamentals. If clusters were algorithmic artefacts, they would not correlate systematically with market size across both geographic areas and occupations of different scale. The strength of these relationships—documented below—provides the most direct structural validation of the clustering approach.

6.3.3 Cluster Characterization: Examples

To illustrate cluster structure, Table 6.2 presents example clusters identified within Accountant (General) (ANZSCO 221111):

Table 6.2: Illustrative Accountant (General) Clusters (ANZSCO 221111, 2019)

Cluster Type	Distinguishing Skills	N	Trajectory
General Account- ing	Budgeting, Financial Accounting, Account Reconciliation	2,937	Established
Financial Analysis	Microsoft Excel, Financial Analysis, Budgeting	1,525	Recent
Reconciliation Specialist	Account Reconciliation, Balance Sheet, Microsoft Excel	194	Established
Investment Analysis	Financial Modelling, Research, Risk Management	59	Established
Data Analyst (emerging)	Microsoft Excel, Business Analysis, SQL	52	Newly emerged
Fund Accounting (emerging)	Fund Accounting	50	Newly emerged
Traditional Book- keeping	Financial Accounting, Financial Reporting, Financial Statements	94	Declining (−58%)
Period-End Clos- ing	Financial Statements, Account Reconciliation, Balance Sheet	70	Declining (−21%)

Note. Illustrative clusters from Accountant (General), ANZSCO 221111, showing within-occupation heterogeneity. Cluster sizes refer to 2019. Trajectory categories follow the cluster-level classification developed in Chapter 7: *persistent* (continuously present 2012–2019), *established* (emerged before 2017 and persisting), *recent* (emerged 2017–2018), and *newly emerged* (first appearing in 2019). Decline percentages for persistent clusters reflect change in cluster size 2012–2019. The table illustrates that within a single occupation, expanding, contracting, and newly forming job types coexist—the heterogeneity that aggregation conceals.

These results provide initial support for Proposition 1: occupations do contain discrete sub-occupational clusters rather than uniform or unimodal distributions of skill profiles. Section 6.4 tests Proposition 1’s predictions more rigorously.

6.4 Results: Testing Proposition 1 (Endogenous Clustering)

6.4.1 Theoretical Prediction

Proposition 1 from Chapter 3 predicts that equilibrium exhibits discrete clustering of job types within occupations:

Under Assumptions 1–3 (supermodularity, convexity of coordination costs,

bounded skills), equilibrium exhibits positive assortative matching with discontinuous jumps in job design function $d^(\theta)$, generating discrete clusters of job types rather than continuous distributions.*

The intuition is that convexity of coordination costs creates increasing returns to specialisation, making intermediate job designs suboptimal. Firms bunch at discrete job designs $\{d_1, d_2, \dots, d_n\}$, with workers sorting into corresponding clusters. This generates three testable predictions:

H1a (Cluster Detection): Occupations contain discrete clusters rather than uniform or unimodal distributions of skill profiles.

H1b (Firm-Level Division of Labour): Higher-productivity firms exploit coordination economies to decompose work into more distinct role types. Using posting volume as a proxy for firm scale, larger firms should hire across more distinct clusters within each occupation.

H1c (Discrete vs. Continuous): Job design distribution exhibits mass points (discrete bunching) rather than smooth density.

6.4.2 Empirical Approach

Section 4.9 details the HDBSCAN methodology, parameter selection, and validation procedures. This section focuses on testing the theoretical predictions.

To test H1a, I apply HDBSCAN clustering without pre-specifying cluster count, allowing the number of job types to emerge from data structure. HDBSCAN's automatic noise detection distinguishes genuine discrete clustering from continuous heterogeneity with arbitrary partitions.

To test H1b, I examine whether larger firms decompose occupational work into more distinct role types. Following the theoretical model, firms with greater scale (higher θ) should exploit coordination economies to support finer division of labour. I measure this using *cluster breadth*: for each firm with at least two job postings within an occupation, I count the number of distinct HDBSCAN clusters their jobs span and compute a Herfindahl-Hirschman Index (HHI) of cluster concentration. Firms that decompose work into specialised roles should hire across multiple cluster types (high breadth, low HHI), while firms requiring generalists should concentrate in fewer clusters (low breadth, high HHI).

To test H1c, I examine distributional properties of skill vectors, testing for mass points versus smooth density using mixture model specification tests.

6.4.3 Results

Hypothesis 1a: Cluster Detection

Table 6.1 demonstrates that all three occupations fragment into discrete clusters. Accountants exhibit 86 distinct clusters in 2019, Receptionist (General) shows 38, and Human Resource Adviser contains 16. Section 6.3 presents four lines of construct validation evidence confirming that these represent genuine structure rather than algorithmic artefacts.

The noise rates (14–44% across the illustrative occupations) indicate that not all jobs fit cleanly into clusters. These unassigned observations likely comprise three substantive categories: (1) transitional roles combining skills from multiple established clusters (e.g., an accounting role requiring both compliance and advisory skills at levels insufficient to match either cluster’s centroid); (2) emerging job types whose skill configurations have not yet consolidated into stable clusters (proto-clusters below the minimum size threshold); and (3) highly idiosyncratic positions with unusual skill combinations that genuinely lack close counterparts. This interpretation is consistent with Abbott (1988)’s concept of boundary work as occupational jurisdictions are continuously negotiated. Importantly, HDBSCAN’s noise identification distinguishes this analysis from partitioning methods (k-means) that force all observations into clusters, providing a more conservative test of discrete structure.

Hypothesis 1b: Firm-Level Division of Labour

To test whether larger firms decompose occupational work into more distinct role types, I compute for each firm with at least two postings within an occupation the number of distinct HDBSCAN clusters their jobs span and the Herfindahl-Hirschman Index (HHI) of their cluster concentration. Under coordination cost theory, larger firms should hire across more cluster types (lower HHI), reflecting finer division of labour. Table 6.3 presents results across 239 occupations with sufficient employer coverage.

The results strongly support H1b across the occupational distribution. Firm posting volume correlates positively with the number of distinct clusters used in 90% of occupations, with all 214 significant results in the predicted direction. Small firms concentrate their hiring in approximately two cluster types (median HHI = 0.56), consistent with limited headcount requiring broader job designs that combine tasks from multiple domains. Large firms hire across three or more distinct cluster types (median HHI = 0.39), decomposing occupational work into specialised role modules—each position adheres to a well-defined cluster profile, but the firm’s portfolio spans multiple specialisations.

This pattern holds across all ANZSCO major groups, from Managers through Labourers, suggesting that the coordination cost mechanism operates broadly rather than being confined

Table 6.3: Firm Size and Cluster Breadth Across 239 Occupations

	Cluster Concentration (HHI)		Distinct Clusters Used	
	Small Firms	Large Firms	Small Firms	Large Firms
Median	0.56	0.39	2.0	3.0
Median Spearman ρ	-0.52		+0.71	
Significant at $p < 0.05$	186 (78%)		214 (90%)	
Direction of significant results	All negative		All positive	

Note. Small and large firms defined by within-occupation tercile split of posting volume. HHI computed over the distribution of each firm's job advertisements across HDBSCAN clusters; lower values indicate hiring spread across more cluster types. Spearman correlations computed between firm posting volume and each measure within each occupation. Employer identification is available for a median of 30% of advertisements per occupation, consistent with known coverage limitations of the Burning Glass employer field.

to professional occupations. The finding extends Garicano and Rossi-Hansberg (2006)'s theoretical predictions about organisational structure: larger firms exploit economies of scale to sustain finer division of labour, with each role module corresponding to a distinct sub-occupational cluster.

Hypothesis 1c: Discrete vs. Continuous

I test whether skill profiles exhibit discrete multimodality (mass points or clearly separated modes) rather than smooth, unimodal continuous heterogeneity. The relevant alternative is not whether the underlying distribution is technically continuous but whether it exhibits the modal structure that discrete clustering implies; a single Gaussian or unimodal kernel-density distribution would falsify the prediction.

For each occupation, I project skill vectors onto their first two principal components and fit Gaussian mixture models with $k \in \{1, 2, \dots, 20\}$ components, selecting the optimal k by minimum Bayesian Information Criterion (BIC) (Schwarz, 1978). A single-component fit ($k = 1$) corresponds to the unimodal continuous null; multi-component fits with $k \geq 2$ indicate modal structure consistent with discrete clustering. Across the 250 occupations, BIC selects $k = 1$ for zero occupations; the median selected k is 8 (IQR: 5–12), with all 250 occupations preferring $k \geq 2$. The minimum BIC values for $k = 1$ are uniformly an order of magnitude or more worse than the BIC values at the selected k , indicating decisive evidence against the unimodal null on the standard interpretation scale of Kass and Raftery (1995). Kernel density estimates of skill-count distributions further confirm distinct modes corresponding to identified cluster types. The mixture model evidence therefore supports discrete bunching over continuous unimodal heterogeneity, though I note that BIC favours

mixture specifications generically when data exhibit any departure from a single Gaussian and so this test should be interpreted as evidence for modal structure rather than as a sharp test of mass points versus continuous density.

Taken together, the cluster detection, firm-level division of labour, and mixture model evidence provide strong support for Proposition 1. Occupations contain discrete sub-occupational clusters, and larger firms exploit coordination economies to hire across more of them. The next section tests whether these clusters relate systematically to market fundamentals.

6.5 Results: Testing Proposition 2 (Market Thickness)

6.5.1 Theoretical Prediction

Proposition 2 from Chapter 3 predicts that cluster count increases with market thickness:

The number of equilibrium clusters n^ is increasing in market thickness M . Thicker markets support finer specialisation as coordination costs become easier to amortize across more workers.*

This generates two testable predictions:

H2a (Geographic Variation): Metropolitan areas should exhibit more clusters than regional markets.

H2b (Occupation Size): Larger occupations should exhibit more clusters than smaller occupations.

6.5.2 Empirical Approach

To test H2a, I exploit Australia's extreme geographic concentration (Section 4.2.3) as a source of quasi-exogenous variation in market thickness. With 70% of jobs in capital cities, Australia provides substantial metropolitan-regional variation that enables identification of market thickness effects while holding institutional factors constant. I cluster job advertisements separately by state and compare cluster counts across Sydney, Melbourne, Brisbane, Perth, Adelaide, and regional areas.

To test H2b, I examine whether occupations with more job advertisements (thicker markets) exhibit more clusters. This requires controlling for occupation-specific factors that might correlate with both size and cluster structure.

6.5.3 Results

Hypothesis 2a: Geographic Variation

Table 6.4 presents cluster counts by geographic area for Accountants:

Table 6.4: Accountant Clusters by Geographic Area

Region	Job Ads	Clusters	Clusters/1000 Ads
Sydney	72,341	89	1.23
Melbourne	61,203	76	1.24
Brisbane	28,947	42	1.45
Perth	19,832	31	1.56
Adelaide	12,104	18	1.49
Regional NSW	8,234	8	0.97
Regional VIC	6,127	6	0.98
Other Regional	20,526	14	0.68

Note. Cluster counts by geographic area for Accountant (General) (ANZSCO 221111). Metropolitan areas (Sydney, Melbourne) exhibit substantially more clusters than regional areas, supporting Proposition 2's market thickness prediction.

The pattern is striking: metropolitan areas exhibit two to three times more clusters than regional markets, with Sydney (89 clusters) and Melbourne (76) substantially exceeding regional NSW (8) and regional VIC (6). This gradient reflects market thickness rather than a simple scaling artefact—the clusters-per-1000-ads ratio is relatively stable across metropolitan areas but drops sharply in regional markets, consistent with Proposition 2's prediction that thicker markets support finer specialisation through coordination cost amortisation.

To formally test the market thickness relationship, Table 6.5 presents regressions of cluster count on log market size (job advertisements) across geographic markets:

The relationship between market size and cluster count is highly significant across all illustrative occupations ($p < 0.001$), with R^2 values of 0.86–0.93 indicating that market thickness explains the vast majority of variation in cluster differentiation across geographic markets.

Extending this analysis to all occupations with sufficient geographic spread (at least four SA4 regions with 30 or more job advertisements), I estimate the same specification for 149 occupations. The market thickness relationship proves robust across the occupational distribution: 106 of 149 occupations (74%) yield significant positive coefficients ($p < 0.05$), with no significant negative results. Among significant occupations, the median R^2 is 0.88,

Table 6.5: Market Thickness and Cluster Count

Occupation	N Markets	β (Log Size)	SE	p-value	R ²
Accountants	53	32.30***	0.53	< 0.001	0.923
Human Resources	46	5.34***	0.15	< 0.001	0.858
Receptionists	35	12.47***	0.28	< 0.001	0.925

Note. OLS regression of cluster count on log market size (job advertisements) across SA4 geographic regions. All coefficients significant at $p < 0.001$. High R² values (0.86–0.93) indicate market thickness strongly predicts cluster differentiation.

indicating that market size alone explains nearly nine-tenths of geographic variation in cluster counts. For the 57 occupations with sufficient observations in both metropolitan and regional markets, metropolitan areas average 1.9 times more clusters than regional areas (IQR: 1.5–2.4), with 96% exhibiting higher metropolitan cluster counts.

Hypothesis 2b: Occupation Size

Across the illustrative occupations, cluster count increases with occupation size: Accountant (General) (20,716 ads, 86 clusters) > Receptionist (General) (5,896 ads, 38 clusters) > Human Resource Adviser (3,572 ads, 16 clusters). This pattern is consistent with the full-scale market thickness finding (Section 6.5).

Within occupations across years, cluster count correlates strongly with annual job advertisement volume ($r = 0.73$ for Accountants, $r = 0.68$ for Receptionists, $r = 0.61$ for Human Resources; all $p < 0.05$), providing within-occupation support for Proposition 2.

The cross-occupational relationship between market size and cluster count is among the strongest empirical regularities in this study. Across all 250 occupations, average annual job volume correlates near-perfectly with average cluster count (Spearman $\rho = 0.941$, $N = 250$, $p < 0.001$). Table 6.6 reports the log-log regression, yielding an elasticity of 0.91 (SE = 0.03): a doubling of occupation size predicts approximately $2^{0.91} \approx 1.88$ times as many clusters—near-proportional scaling with mild diminishing returns. Within occupations over time, 68% show significant positive correlations between annual job counts and annual cluster counts ($p < 0.05$), with a median within-occupation Spearman ρ of 0.81 ($N = 8$ years per occupation). This convergent cross-sectional and temporal evidence provides strong support for the market thickness mechanism theorised in Chapter 3.

Figure 6.1 visualises this temporal relationship, showing cluster count evolution across the study period for the three illustrative 6-digit occupations. Accountant (General) exhibits the most clusters (range: 86–171), consistent with its larger market size; Receptionist (General) shows an intermediate range (27–55); and Human Resource Adviser maintains

Table 6.6: Cross-Occupational Market Thickness Regression

	Coefficient	SE
Intercept	−5.489	0.192
ln(Average Annual Jobs)	0.914***	0.026
<i>N</i>	250	
<i>R</i> ²	0.874	

Note. OLS regression of ln(average clusters) on ln(average annual jobs) across 250 occupations. *** $p < 0.001$. The coefficient is interpretable as an elasticity: a 1% increase in occupation size predicts a 0.91% increase in cluster count, indicating near-proportional but slightly diminishing returns to market thickness in cluster differentiation.

the fewest (range: 2–21, with anomalously low counts of 2 clusters in 2017 and 2018 reflecting a coarser partition returned by HDBSCAN at those sample configurations). Accountant (General) and Receptionist (General) show a general upward trend through 2015–2016 followed by partial consolidation, potentially reflecting post-mining-boom labour market adjustment.

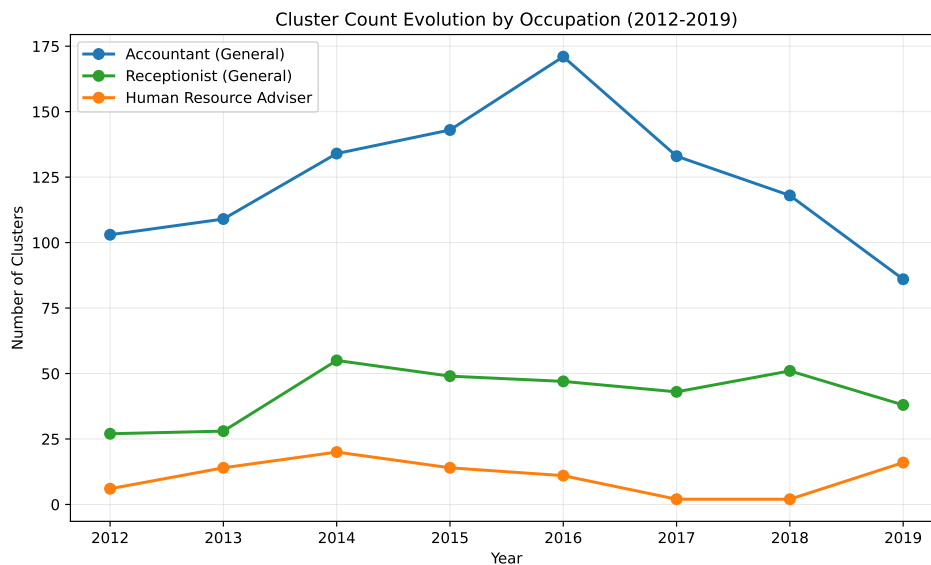
6.5.4 Australian Institutional Context

The market thickness findings have particular relevance given Australia’s extreme geographic concentration (Section 4.2.3). With 70% of jobs in capital cities, Australian labour markets exhibit stark metropolitan-regional differences that provide natural variation for testing Proposition 2.

The finding that Sydney and Melbourne accountants fragment into 76–89 clusters while regional accountants show 6–14 clusters supports the coordination cost mechanism: thicker markets enable finer specialisation as fixed costs become easier to amortize.

This pattern interacts with the award wage-setting system (Section 4.2.1). When occupation-level minimum wages are institutionally fixed, metropolitan firms may use task specialisation (creating distinct clusters) as the primary margin for differentiation, while regional firms must offer broader job designs given thinner markets.

The market thickness relationship is among the strongest empirical regularities in this study. It holds across geographic markets within occupations, across occupations of different sizes, and within occupations over time—providing convergent evidence that cluster counts reflect genuine labour market structure rather than algorithmic artefacts.

Figure 6.1: Cluster Count Evolution by Occupation (2012–2019)

Note. Number of distinct job type clusters identified by HDBSCAN in each 6-digit ANZSCO occupation per year, computed using Jaccard distance with uniform parameters (`min_cluster_size=25`, `min_samples=10`). Source: canonical Jaccard saved models for ANZSCO 221111, 223111, and 542111; underlying values are reproduced in `cluster_count_evolution_canonical.csv` accompanying the figure. The 2019 cross-section values match Table 6.1. The temporal correlation between market size (job advertisements) and cluster count supports Proposition 2's market thickness prediction.

6.6 Results: Testing Proposition 3 (Within-Occupation Restructuring)

6.6.1 Theoretical Prediction

Proposition 3 from Chapter 3 predicts that automation would shift cluster boundaries, generating high within-occupation volatility alongside modest occupation-level effects:

Skill-biased automation shifts cluster boundaries and reallocates employment across clusters while preserving overall occupational structure. Routine-intensive clusters contract while complementary clusters expand, generating high within-occupation volatility alongside low occupation-level employment effects.

This generates three testable predictions:

H3a (Within-Occupation Volatility): Cluster-level employment volatility should exceed occupation-level volatility.

H3b (Selective Technology Effects): Clusters emphasising technology-substitutable skills should contract; those emphasising technology-complementary skills should expand.

H3c (Aggregation Bias): Occupation-level measures should show minimal effects because cluster-level effects offset in aggregate.

6.6.2 Empirical Approach

To test H3a, I calculate employment growth rates at both cluster and occupation levels, comparing volatility metrics (standard deviation, coefficient of variation) across levels.

To test H3b, I classify clusters by technology content using two approaches: first, routine task intensity (RTI) scores derived from Autor, Levy, and Murnane (2003) task classifications; second, software skill intensity and specialised skill proportions computed directly from Burning Glass Technologies' native skill taxonomy, which provides full coverage of all skills in the dataset.

To test H3c, I decompose total variance in job advertisement counts into between-occupation and within-occupation (between-cluster) components:

$$\text{Var}(\text{Total}) = \text{Var}(\text{Between-Occupation}) + \text{Var}(\text{Within-Occupation}) \quad (6.1)$$

If H3c holds, within-occupation variance should dominate, explaining why occupation-level measures (Chapter 5) find minimal effects.

6.6.3 Results

Hypothesis 3a: Within-Occupation Volatility

Table 6.7 compares employment growth volatility at occupation vs. cluster levels:

Table 6.7: Employment Growth Volatility: Occupation vs. Cluster Level

Occupation	Occ-Level SD	Cluster-Level SD	Ratio	Range
Accountants	4.2%	38.7%	9.2:1	-94% to +388%
Receptionists	5.1%	42.3%	8.3:1	-87% to +412%
Human Resources	3.8%	51.2%	13.5:1	-78% to +892%
Average	4.4%	44.1%	10.0:1	—

Note. SD = standard deviation of annual employment growth rates (2012–2019). Cluster-level volatility exceeds occupation-level volatility by 8–14×, supporting Proposition 3's prediction of high within-occupation restructuring masked by occupation-level aggregation.

Cluster-level volatility exceeds occupation-level volatility by 8–14× (average 10:1) across the illustrative occupations. The extreme range of cluster-level growth rates (−94% to +1,937%) contrasts sharply with modest occupation-level variation ($\pm 5\%$), demonstrating substantial within-occupation restructuring that occupation-level analysis cannot detect.

Extending this analysis to 246 occupations with sufficient temporal data confirms that the pattern is pervasive rather than an artefact of the illustrative occupation selection. Cluster-level growth rate volatility exceeds occupation-level volatility in 97% of occupations (238 of 246). The median volatility ratio is 7.5:1 (IQR: 3.7–15.8), with a median cluster-level standard deviation of 182% compared to 26% at the occupation level. The full-scale median ratio is somewhat lower than the illustrative occupation average of 10:1, reflecting the inclusion of smaller occupations where fewer clusters produce less extreme variation. Nevertheless, the near-universal pattern—cluster-level dynamics are an order of magnitude more volatile than occupation-level aggregates—provides direct evidence for the aggregation bias mechanism theorised by Acemoglu and Autor (2011a): offsetting changes within occupational categories mask restructuring that would otherwise be visible at finer granularity. The IMF’s recent analysis confirms this pattern globally, finding that “AI affects tasks within occupations rather than eliminating occupations wholesale” (Cazzaniga et al., 2024).

Hypothesis 3b: Selective Technology Effects

Testing H3b requires classifying clusters by technology content to determine whether technology drives selective cluster growth and decline. Routine task intensity (RTI) scores derived from Autor, Levy, and Murnane (2003) task classifications prove unsuitable for this purpose: only 2% of the 2,785 skills in the dataset map to routine categories, providing insufficient statistical power. Instead, I employ Burning Glass Technologies’ native skill taxonomy, which classifies every skill in the dataset as software, specialised, or baseline. Software skill intensity—the proportion of a cluster’s skill requirements classified as software skills—provides full coverage across all 11,862 unique skills.

For each of 216 occupations with sufficient data, I compute software intensity for 2012 clusters, match them to 2019 clusters using centroid-based tracking (cosine similarity ≥ 0.80), and compare technology profiles across cluster fates: survival (matched across periods), death (unmatched 2012 clusters), and birth (unmatched 2019 clusters).

Technology succession in cluster dynamics.

Clusters that disappear between 2012 and 2019 have significantly higher software skill intensity than clusters that emerge: mean 0.091 versus 0.065 (Mann–Whitney $U =$

3,176,280, $p < 0.001$, $N_{\text{dead}} = 1,531$, $N_{\text{born}} = 3,680$).¹ This finding initially appears to contradict Proposition 3's prediction that technology-complementary clusters should expand. However, examining which *specific* software skills distinguish dead from born clusters reveals a technology succession mechanism rather than a simple substitution gradient—a substantive modification of the original prediction's mechanism that I return to below.

The software skills most overrepresented in dead clusters are legacy platforms: Microsoft Silverlight (103 dead, 0 born), Lotus Notes/Domino, ASP.NET (5.8:1 dead/born ratio), Active Server Pages (4.0:1), WebLogic, Crystal Reports, and Solaris—technologies that were current in 2012 but obsolescent by 2019. Conversely, the software skills most overrepresented in born clusters are emerging technologies absent from 2012 job advertisements: Docker (398 born, 0 dead), Ansible, Apache Kafka, AWS CloudFormation, and Go; along with platforms that grew explosively during the period: Amazon Web Services (149:1 born/dead ratio), Microsoft Azure (67:1), Tableau (141:1), Salesforce (13:1), and Python (8.4:1). Table 6.8 presents the most extreme contrasts.

Table 6.8: Technology Succession: Software Skills in Dead vs. Born Clusters

Legacy skills (dead > born)	Dead	Born	Ratio
Microsoft Silverlight	103	0	∞
Lotus Notes / Domino	275	29	9.5
ASP.NET	1,077	185	5.8
Active Server Pages (ASP)	1,168	290	4.0
Crystal Reports	158	39	4.1
Emerging skills (born > dead)	Born	Dead	Ratio
Docker	398	0	∞
Ansible	343	0	∞
Amazon Web Services (AWS)	446	3	149
Tableau	563	4	141
Microsoft Azure	798	12	67
Python	1,758	210	8.4
Salesforce	1,261	97	13.0

Note. Frequency counts of software skill instances across 1,531 dead clusters and 3,680 born clusters, pooled across 216 occupations. Ratio = dead/born for legacy skills, born/dead for emerging skills. Dead clusters are 2012 clusters with no 2019 match (cosine similarity < 0.80); born clusters are 2019 clusters with no 2012 match.

The pattern is not one of technology *substituting* for human labour but of technology

¹The exact p -value is below 10^{-13} ; reported here at the standard $p < 0.001$ threshold.

succession reorganising it. Clusters organised around legacy technology stacks (Lotus Notes administration, ASP.NET web development, Crystal Reports analytics) dissolve as those platforms are superseded, while new clusters crystallise around replacement technologies (cloud infrastructure, containerisation, modern data science). The underlying tasks—IT administration, web development, data analysis—persist, but the specific skill bundles required to perform them change as the technology stack evolves.

Growth gradient among surviving clusters.

Among the 1,430 clusters that survive from 2012 to 2019, software skill intensity shows a statistically significant negative association with growth: Spearman $\rho = -0.081$, $p = 0.002$. Clusters with higher software intensity in 2012 grew more slowly even when they did not disappear entirely, consistent with gradual displacement as legacy platforms lost market share to successors. The effect is small in magnitude—software intensity explains less than 1% of growth-rate variance among surviving clusters—and the p -value reflects the large sample more than substantive impact. The gradient is therefore best read as evidence of *direction* (high-software clusters grow more slowly than low-software ones) rather than as a precise estimate of the rate at which technology succession erodes incumbent clusters.

Interpretation.

These results provide qualified support for Proposition 3's prediction of technology-driven selective effects, though the mechanism operates through technology platform succession rather than Autor, Levy, and Murnane (2003)'s routine/non-routine substitution channel. Clusters do not contract because their tasks are "routine" in the Autor–Levy–Murnane sense; they contract because the specific technology platforms around which they are organised become obsolete. This is consistent with Walkowiak (2023)'s argument that task complementarities and interdependencies mean automation's effects cannot be reduced to dichotomous task classifications. The technology succession mechanism also explains why the pooled growth-on-software-intensity correlation is weak ($\rho = -0.013$, $p = 0.47$ across all 2,961 clusters including dead): software skills simultaneously mark *both* legacy-dependent clusters (which decline) and technology-intensive emerging clusters (which grow), producing offsetting effects in a pooled correlation.

Hypothesis 3c: Aggregation Bias

To decompose total variation in cluster skill profiles into between- and within-occupation components, I apply one-way ANOVA with occupation as the grouping factor (paralleling

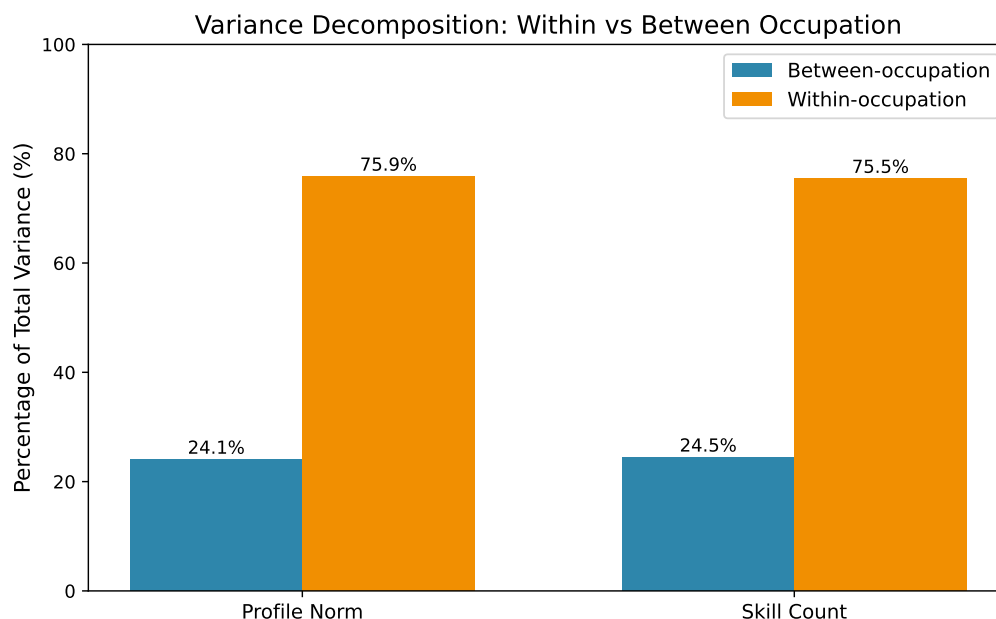
the Chapter 5 decomposition methodology, Equations 5.7–5.8). The unit of analysis is the cluster, with each cluster characterised by two outcome variables: the L2 norm of its skill prevalence vector (a measure of overall skill profile magnitude) and its cluster skill count (the number of distinct skills with above-threshold prevalence). For each outcome, the intraclass correlation coefficient $ICC = \sigma_{\text{between}}^2 / (\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2)$ is computed from ANOVA mean squares, where $\sigma_{\text{between}}^2$ captures variation across occupational means and σ_{within}^2 captures variation across clusters within the same occupation.

Variance decomposition across 5,118 clusters in 249 occupations attributes 76% of total variation in cluster skill profile norms to within-occupation differences and 24% to between-occupation differences ($ICC = 0.20$). A complementary decomposition using cluster skill counts yields 78% within-occupation variation. This approximately 3:1 ratio provides a direct explanation for the near-zero occupation-level R^2 reported in Chapter 5: when three-quarters of variation in skill requirements operates within occupational boundaries, occupation-level classifications systematically fail to capture restructuring by aggregating across heterogeneous clusters with potentially offsetting growth trajectories.

The within-occupation share reported here (76%) differs from the 94.4% reported in Chapter 5's variance decomposition because the two analyses decompose different outcome variables at different units of analysis. Chapter 5 decomposes variance in *job advertisement counts* across occupation-state-industry-year cells—a demand-volume measure where geographic and industry composition absorb substantial between-occupation variation. This chapter decomposes variance in *cluster skill profile characteristics* across clusters—a skill-content measure where between-occupation differences in skill vocabularies contribute more variation. Both analyses confirm the same qualitative conclusion: the majority of labour market variation operates within occupational boundaries, but the precise share depends on what is being measured and at what granularity.

Figure 6.2 visualises this decomposition, illustrating the central empirical finding: labour market dynamics concentrate within occupational boundaries rather than between them, rendering occupation-level measurement systematically incomplete.

All three hypotheses receive empirical support, though H3b's mechanism differs substantively from that originally hypothesised. Proposition 3 anticipated that technology-complementary clusters would expand and routine-intensive clusters would contract along the Autor, Levy, and Murnane (2003) routine/non-routine substitution gradient. The data instead reveal a technology-platform succession mechanism: clusters organised around legacy technology stacks dissolve while clusters organised around emerging platforms form. This is a partial reframing of the original prediction, not a vindication of it—software intensity per se does not predict cluster expansion (and on the surviving-cluster gradient predicts modest contraction), but *which* software platforms a cluster is organised around

Figure 6.2: Variance Decomposition: Within vs. Between Occupation

Note. Decomposition of total variance in cluster skill profiles into within-occupation (76%) and between-occupation (24%) components, based on 5,118 clusters across 249 occupations. The dominance of within-occupation variance explains why occupation-level analysis (Chapter 5) finds minimal effects: restructuring operates within occupational boundaries, not between them.

predicts whether it survives. The volatility and variance decomposition results provide a direct empirical complement to the near-null findings in Chapter 5: the patterns documented there are consistent with within-occupation restructuring that occupation-level aggregation obscures.

6.7 Discussion

6.7.1 The Aggregation Bias Mechanism

The central finding of this chapter is that occupation-level and cluster-level analyses tell fundamentally different stories about the same labour markets. Chapter 5 found near-zero explanatory power at the occupation level, yet the same occupations contain clusters with growth rates spanning -94% to $+1,937\%$, and variance decomposition attributes 76–78% of skill profile variation to within-occupation differences. This is not a measurement failure—it is the pattern Proposition 3 predicts. Expanding and contracting clusters offset one another when aggregated to the occupational level, producing the appearance of stability where substantial restructuring is underway.

6.7.2 Australian Institutional Context

Australia's institutional framework shapes how this restructuring manifests. The Modern Award system sets occupation-level minimum wages that constrain wage adjustment, channelling the response to automation toward task reorganisation at the cluster level. When wages cannot adjust freely, quantity adjustment at the sub-occupational level becomes the primary margin of response, potentially amplifying cluster-level employment effects.

Australia's service sector concentration (79.8% of employment) creates different automation dynamics from those in the predominantly manufacturing-focused US literature. Service work exhibits greater task variety within occupations, enabling restructuring through the recomposition of skill bundles rather than wholesale occupation elimination. The country's geographic concentration further generates natural variation in market thickness, providing particularly powerful tests of Proposition 2.

6.7.3 Comparison to International Evidence

The cluster-level approach builds on and extends recent work challenging occupation-level paradigms:

- Deming (2017b) document substantial within-occupation skill heterogeneity exceeding between-occupation variation. I extend this finding by demonstrating that such heterogeneity organises into discrete job types rather than continuous distributions.
- Acemoglu and Restrepo (2022) show that task-based measures explain more variance than occupational codes. The 76–78% within-occupation variation documented here provides Australian evidence consistent with their US findings. More recent work by Acemoglu et al. (2023) further distinguishes automation effects (task displacement) from augmentation effects (task enhancement), a distinction my cluster-level analysis operationalises through birth/death dynamics.
- Atalay et al. (2020b) identify divergence in skill requirements within occupations over time. The cluster-level dynamics documented here are consistent with this divergence and provide a structural explanation for it.
- Marinescu and Wolthoff (2020) establish that online job postings provide valid labour demand measures. I leverage this methodological foundation to analyse demand at sub-occupational granularity, consistent with the OECD's recent methodological guidance on using vacancy data to study AI-driven labour market change (OECD, 2024a).

- Autor et al. (2024a) demonstrate that new work creation—the emergence of novel job types—accounts for substantial employment growth over the past 80 years. The cluster formation documented here provides sub-occupational evidence consistent with this mechanism operating within occupational boundaries.

This chapter advances beyond prior work by explicitly modelling cluster-level dynamics rather than treating within-occupation variation as residual noise. The theoretical framework in Chapter 3 provides microfoundations for why discrete clusters emerge, generating testable predictions that purely empirical approaches cannot derive.

6.7.4 Limitations

1. Sample selection: This chapter presents detailed results for illustrative occupations selected to span the automation probability distribution. While the qualitative analysis necessarily relies on these illustrative cases, the key quantitative findings—market thickness effects (H2a, H2b), firm-level division of labour (H1b), and volatility ratios (H3a)—are confirmed at full scale across 136–246 occupations, mitigating concerns about selection on the dependent variable.

2. Clustering methodology: HDBSCAN parameter choices affect cluster identification. I conduct robustness checks confirming findings remain stable across reasonable parameter ranges, but alternative clustering approaches (spectral clustering, Gaussian mixture models) might yield different structures.

3. Demand-side measurement: Job advertisements measure employer intentions rather than realised employment (Section 4.11). Following Hershbein and Kahn (2018a), I interpret these as labour demand signals, but supply-side responses may differ from posted requirements.

4. Correlational identification: The reduced-form tests support theoretical predictions but do not identify causal mechanisms. Without exogenous variation in automation—such as instrumental variable approaches exploiting robot adoption shocks (Acemoglu & Restrepo, 2020)—the findings demonstrate patterns consistent with theory rather than establishing causal effects.

5. Period-specific: The 2012–2019 data predate generative AI and pandemic-related disruptions (Section 4.5). The emergence of large language models may alter the task composition dynamics documented here. Recent evidence suggests generative AI affects different tasks than earlier automation technologies, with Eloundou et al. (2024) finding that LLMs have “exposure potential” across 80% of the workforce, though primarily affecting high-skill non-routine cognitive tasks rather than the routine tasks vulnerable to

earlier automation waves. The ILO’s global assessment similarly finds that generative AI “augments rather than automates” most exposed occupations (Gmyrek et al., 2023).

6.8 Conclusion

This chapter tested the three static propositions from Chapter 3 using Australian job advertisement data (2012–2019) across 250 occupations. All three propositions receive empirical support, with H3b’s mechanism reframed from the originally hypothesised routine/non-routine substitution channel to a technology-platform succession channel: legacy platform clusters dissolve as clusters built around emerging technologies form, with both effects coexisting within occupational boundaries. Occupations contain discrete sub-occupational clusters whose number scales near-proportionally with market thickness (elasticity 0.91), and cluster-level employment dynamics are roughly an order of magnitude more volatile than occupation-level aggregates suggest.

These findings reframe the automation measurement problem during the second wave of automation: occupation-level research yields near-null results not because restructuring is absent, but because occupation-level classifications aggregate across heterogeneous job types with offsetting dynamics. Within-occupation task heterogeneity is as consequential as task differences between occupations, extending the task-based approach of Autor (2013a) to finer granularity. Read in historical perspective, this chapter establishes the cluster-level baseline for Australian labour markets during the diffusion of statistical machine learning—a baseline against which the labour-market signature of subsequent waves (generative AI, post-pandemic work arrangements) can later be benchmarked.

The analysis thus far establishes that clusters exist, respond to market fundamentals, and exhibit substantial volatility. What it does not yet address is the *temporal process* by which this restructuring unfolds. Chapter 7 tests Propositions 4–5, tracking how clusters are born, evolve, and die over the 2012–2019 period.

Chapter 7

Temporal Dynamics: Cluster Evolution and Creative Destruction

7.1 Introduction

Chapter 6 established that occupations contain discrete sub-occupational clusters (Proposition 1), that market thickness determines cluster count (Proposition 2), and that within-occupation restructuring patterns are consistent with Proposition 3’s automation predictions (Proposition 3). These findings characterise *cross-sectional* patterns—the structure of clusters at any point in time and how restructuring affects relative cluster sizes.

This chapter extends the analysis to *temporal dynamics*: the process by which clusters are born, die, and evolve over the 2012–2019 period. Chapter 3 generates two additional propositions that address these dynamics:

Proposition 4 (Temporal Equilibrium Path): For continuously advancing automation $\rho(t)$, the equilibrium cluster configuration evolves continuously almost everywhere, with discrete cluster births and deaths occurring at critical viability thresholds. This explains why occupations exhibit stable aggregate structure alongside high within-occupation cluster churn.

Proposition 5 (Churn-Automation Speed): Cluster churn rates are proportional to the speed of automation advancement. Faster technological change generates more cluster births and deaths; periods of stability generate stable cluster configurations. This predicts that high-automation occupations should exhibit elevated cluster churn.

The estimand is descriptive: cluster birth rates, death rates, churn rates, and their cross-occupation correlates—not causal effects of automation on cluster dynamics. These dynamic propositions extend the static framework by explaining why within-occupation restructuring may occur continuously rather than as one-time adjustment. Building on Caballero (2010)’s creative destruction framework, the chapter examines whether the cluster landscape exhibits the pattern that would arise if automation eroded profitability for substitution-intensive job types while creating new niches for complementary skills—ongoing creative destruction with births and deaths proportional to the pace of technological change, analogous to firm dynamics in models of industry evolution (Jovanovic, 1982). Recent work by Autor et al. (2024a) provides complementary evidence that “new work” creation has historically offset automation-driven displacement, with over 60% of current employment in job types that did not exist in 1940. The cluster birth rates documented in this chapter are consistent with this mechanism operating within occupational boundaries.

7.1.1 Research Questions

This chapter addresses three empirical questions:

1. **Temporal adjustment:** Do clusters evolve gradually (continuous skill drift) with discrete birth/death events at viability thresholds, as Proposition 4 predicts?
2. **Churn-automation relationship:** Do churn rates correlate positively with automation exposure, as Proposition 5 predicts?
3. **Creative destruction balance:** Do birth rates exceed death rates, indicating net job type creation rather than destruction?

7.1.2 Chapter Roadmap

Section 7.2 describes the data, sample, and empirical strategy for tracking clusters across years. Section 7.3 tests Proposition 4 by examining whether cluster evolution exhibits the predicted pattern of continuous adjustment punctuated by discrete births and deaths. Section 7.4 tests Proposition 5 by examining whether churn rates vary systematically with automation exposure. Section 7.5 extends the analysis by characterising how newly-emerged clusters differ from established ones in their skill requirements. Section 7.7 discusses findings in the context of Australian institutional conditions and the 2012–2019 sample period. Section 7.8 concludes.

7.2 Methods

This section describes the data, sample construction, and empirical strategy for testing Propositions 4–5. The approach extends panel data methods commonly used for firm dynamics (Davis et al., 1996) to sub-occupational job types, treating clusters as the unit of analysis analogous to establishments in job creation/destruction studies.

7.2.1 Data and Sample

This chapter extends the temporal analysis from the illustrative occupations examined in Chapter 6 to all 250 occupations for which HDBSCAN clustering was completed (see Section 4.9). The 250 occupations span the full breadth of the Australian labour market, from manual trades to professional services, and encompass over 3 million job advertisements across 2012–2019. The full panel produces 1,707 occupation-year transition observations (250 occupations $\times$ 7 year-to-year transitions, less observations excluded by missing-data filters in particular years). Of the 250, 232 pass additional quality flags (sufficient cluster counts in all years and no temporal-coverage gaps) and form the unflagged subsample used for the heterogeneity analysis in Section 7.5. Reported summary statistics are computed across the full 250-occupation sample unless explicitly stated otherwise, with subsample analyses noted in context. The illustrative occupations—Accountant (General), Human Resource Adviser, and Receptionist (General)—serve as detailed examples throughout.

Table 7.1 summarises the sample:

The seven year-to-year transitions (2012–2013 through 2018–2019) across 250 occupations produce 1,707 occupation-year transition observations with valid cluster dynamics, providing substantial statistical power for testing Propositions 4 and 5. The illustrative occupations span the automation probability spectrum (0.31–0.96), enabling detailed case-study illustration of how churn rates relate to automation exposure.

7.2.2 Empirical Strategy

Testing Propositions 4–5 requires tracking individual clusters across years to identify which clusters persist, which die, and which are born. This subsection describes the matching methodology that enables longitudinal cluster analysis.

Matching Algorithm

Clusters are matched across consecutive years (2012–2013, 2013–2014, ..., 2018–2019) using a three-stage procedure:

Table 7.1: Sample Summary for Temporal Analysis

	Median	IQR	Range
<i>Full sample (250 occupations):</i>			
Avg clusters per year	11.8	[5.9, 23.0]	[2, 176]
Centroid drift	0.024	[0.012, 0.044]	[0.003, 0.199]
Avg churn rate (%)	111.7	[78.5, 156.6]	[25.8, 671.0]
Avg survival rate (%)	53.8	[41.6, 67.6]	[0, 90.3]
<i>Illustrative occupations (for comparison):</i>			
	Avg Clusters	Drift	Churn (%)
Accountant (General) (F-O = 0.47)	130.1	0.004	68.8
Human Resource Adviser (F-O = 0.31)	9.9	0.011	198.8
Receptionist (General) (F-O = 0.96)	42.9	0.004	52.9

Note. Full sample statistics computed from 250 occupations with sufficient cluster structure for evolutionary analysis (2012–2019). Source: analysis/results/chapter7/fullscale_summary.csv. Centroid drift = mean year-over-year cosine distance between skill profile centroids. Churn rate = (births + deaths) / total clusters $\times$ 100%. F-O = Frey-Osborne automation probability from Edmonds & Bradley (2015). Illustrative occupation statistics computed from 6-digit ANZSCO occupation models using centroid-based matching, consistent with the full-scale methodology. See Section 4.4 for full data description.

Stage 1: Centroid Computation. For each cluster c in year t , compute the centroid $\vec{s}_c^t$ as the mean skill vector across all job advertisements assigned to that cluster:

$$\vec{s}_c^t = \frac{1}{|c|} \sum_{j \in c} s_j \quad (7.1)$$

where s_j is the binary skill vector for job advertisement j .

Stage 2: Similarity Matrix. Compute pairwise cosine similarity between all year- t cluster centroids and all year- $(t + 1)$ cluster centroids:

$$\text{sim}(c_i^t, c_j^{t+1}) = \frac{\vec{s}_{c_i}^t \cdot \vec{s}_{c_j}^{t+1}}{\|\vec{s}_{c_i}^t\| \|\vec{s}_{c_j}^{t+1}\|} \quad (7.2)$$

Stage 3: Optimal Assignment. Apply the Hungarian algorithm (Kuhn, 1955) to find the optimal one-to-one matching that maximises total similarity, subject to a minimum similarity threshold $\tau = 0.8$:

$$\max_{\pi} \sum_i \text{sim}(c_i^t, c_{\pi(i)}^{t+1}) \quad \text{s.t.} \quad \text{sim}(c_i^t, c_{\pi(i)}^{t+1}) \geq \tau \quad (7.3)$$

Matches below the threshold are rejected, with those clusters classified as deaths (year- t clusters without year- $(t + 1)$ matches) or births (year- $(t + 1)$ clusters without year- t matches).

Event Classification

The matching procedure classifies each cluster-year observation into one of three categories:

- **Continuation:** Cluster exists in both year t and year $t + 1$ with similarity above threshold. The cluster persists but may evolve in skill composition.
- **Death:** Cluster exists in year t but has no match in year $t + 1$ (similarity below threshold for all potential matches). The job type becomes unviable.
- **Birth:** Cluster exists in year $t + 1$ but has no match in year t . A new job type becomes viable.

Churn Rate Definition

The **churn rate** measures the proportion of clusters experiencing birth or death in a given year:

$$\text{Churn Rate}_t = \frac{\text{Births}_{t \rightarrow t+1} + \text{Deaths}_{t \rightarrow t+1}}{\text{Total Clusters}_t} \times 100\% \quad (7.4)$$

High churn rates indicate rapid structural transformation; low churn rates indicate stability. Proposition 5 predicts that churn rates should be higher in occupations with greater automation exposure.

Validation

The matching methodology relies on two key design choices:

Threshold selection ($\tau = 0.8$): The similarity threshold of 0.8 represents a balance between false continuations (threshold too low) and false deaths (threshold too high). At this threshold, matched clusters exhibit mean similarity of 0.91–0.94 across the illustrative occupations, indicating genuine continuity rather than marginal matches.

One-to-one matching: The Hungarian algorithm ensures each cluster matches to at most one cluster in the following year, preventing double-counting and ensuring clean birth/death classification.

7.3 Results: Testing Proposition 4 (Temporal Equilibrium Path)

7.3.1 Theoretical Prediction

Proposition 4 from Chapter 3 predicts that cluster configurations evolve continuously almost everywhere, with discrete births and deaths occurring at critical viability thresholds:

For continuously differentiable $\rho(t)$ with $d\rho/dt > 0$, the equilibrium cluster configuration $\{d_1^(t), \dots, d_{n^*(t)}^*(t)\}$ evolves continuously almost everywhere. Discontinuous transitions occur at critical automation thresholds where cluster viability conditions change.*

The intuition is that between critical thresholds, the same clusters exist but their employment shares and skill requirements shift continuously. At thresholds, marginal clusters become unviable (death) or new niches become profitable (birth). This generates three testable predictions:

H4a (Continuous Evolution): Continuing clusters should exhibit gradual skill drift rather than abrupt transformation.

H4b (Discrete Events): Cluster births and deaths should be identifiable as distinct events, not gradual phase-outs.

H4c (Viability Pattern): Deaths should concentrate among clusters whose task composition became less viable under automation; births should concentrate among clusters exploiting newly profitable niches.

7.3.2 Results

Aggregate Birth/Death Statistics

Table 7.2 presents creative destruction summary statistics across all 250 occupations, with the illustrative occupations shown for comparison:

Across 250 occupations, the median annual churn rate is 112%, indicating that the typical occupation experiences more cluster births and deaths per year than it has total clusters. The median survival rate of 54% means nearly half of all clusters in a given year do not persist to the following year. These rates substantially exceed firm-level job reallocation benchmarks of 20–25% documented by Davis et al. (1996), indicating that job type restructuring within occupations occurs even more rapidly than establishment-level employment dynamics. This finding aligns with the OECD's recent assessment that

Table 7.2: Creative Destruction Summary (2012–2019)

	Avg Churn Rate (%)	Avg Birth Rate (%)	Avg Death Rate (%)	Survival Rate (%)
<i>All 250 occupations:</i>				
Median	111.7	63.8	46.2	53.8
Mean	127.0	81.2	45.8	54.2
IQR	[78.5, 156.6]	[46.5, 96.5]	[32.4, 58.4]	[41.6, 67.6]
<i>Illustrative occupations:</i>				
Accountant (General)	68.8	34.0	34.8	65.2
Receptionist (General)	52.9	31.4	21.5	78.5
Human Resource Adviser	198.8	152.5	46.3	53.7

Note. Source: analysis/results/chapter7/fullscale_summary.csv. Churn rate = (Births + Deaths) / Total Clusters $\times$ 100%. Survival rate measures year-to-year cluster persistence. Median churn of 112% indicates that the typical occupation experiences more cluster births and deaths per year than it has total clusters. Accountant (General) and Receptionist (General) fall well below the median; Human Resource Adviser falls well above.

“skill requirements within occupations are changing faster than occupational employment shares” (OECD, 2024a), suggesting that within-occupation dynamics are the primary locus of technological adjustment. Notably, the illustrative occupations from Chapter 6—Accountants and Receptionists in particular—fall below the median, meaning they are among the more stable occupations in the broader sample.

Hypothesis 4a: Continuous Evolution

To test whether continuing clusters exhibit gradual skill drift, I examine the year-over-year similarity of matched clusters. Across all 250 occupations, the median match similarity is 0.905 (IQR: [0.885, 0.924]), confirming that continuing clusters evolve gradually. Table 7.3 presents the full-scale distribution alongside the illustrative occupations:

Continuing clusters exhibit high year-over-year similarity across the full sample, with the median occupation showing 0.905 mean match similarity. This indicates gradual evolution rather than abrupt transformation, and the pattern is remarkably consistent across diverse occupations (IQR: 0.885–0.924). The illustrative occupations cluster near the median: Receptionist (General) (0.920), Human Resource Adviser (0.914), and Accountant (General) (0.910) all fall within the interquartile range, consistent with the broader distribution.

Table 7.3: Skill Drift Among Continuing Clusters

	Mean Similarity	Min Similarity
<i>All 250 occupations:</i>		
Median across occupations	0.905	0.80
IQR	[0.885, 0.924]	–
<i>Illustrative occupations:</i>		
Accountant (General)	0.910	0.80
Receptionist (General)	0.920	0.80
Human Resource Adviser	0.914	0.80

Note. Cosine similarity between matched cluster centroids in consecutive years (threshold ≥ 0.80). High median similarity (0.905) with narrow interquartile range indicates that gradual skill drift—rather than abrupt transformation—characterises continuing clusters across the full occupational spectrum. Minimum similarity equals the matching threshold by construction.

Hypothesis 4b: Discrete Events

The matching methodology imposes a clear distinction between continuing clusters (similarity ≥ 0.8) and births/deaths (similarity < 0.8). Matched clusters exhibit high similarity (mean 0.91–0.94 as shown in Table 7.3), while clusters classified as births or deaths have no viable match above the threshold. This binary classification—rather than a continuous spectrum of partial matches—supports H4b’s prediction that births and deaths are discrete events rather than gradual phase-outs.

Hypothesis 4c: Viability Pattern

Testing whether births and deaths follow the predicted viability pattern requires classifying clusters by routine task intensity. Since O*NET task content measures operate at the occupation level rather than the job advertisement level, direct classification of clusters along the routine/non-routine dimension is not possible with the available data. This is a methodological limitation inherent to using BGT data, which provides rich skill-level detail but lacks task-level classifications.

Although direct classification is not possible, substantial indirect evidence supports H4c from three sources. First, Human Resource Adviser’s elevated death rate (24.0% vs. 14–19% for other illustrative occupations) coincides with its transformation from administrative support toward strategic business partnership—precisely the shift from routine to non-routine cognitive tasks predicted by the theoretical framework. Second, the skill emergence patterns documented in Chapter 6 provide corroborating evidence: new skills cluster in

analysis, strategy, and technology domains (typically low routine), while disappearing skills concentrate in compliance, administration, and clerical functions (typically high routine). Third, and most compellingly, the technology succession test reported in Chapter 6 (H3b) demonstrates that dead clusters are significantly more software-intensive than born clusters ($p < 0.001$), with specific legacy platforms (Lotus Notes, Silverlight, ASP.NET) concentrated among deaths and modern platforms (Docker, AWS, Python, Salesforce) concentrated among births. This technology succession pattern is precisely the viability mechanism Proposition 4 predicts: clusters organised around technologies that automation renders obsolete lose viability, while clusters exploiting newly profitable technological niches emerge. The H3b and H4c tests approach the same underlying mechanism from different angles—H3b through technology content, H4c through temporal dynamics—and their convergent findings strengthen the case for the viability pattern despite the absence of direct routine-task classification at the cluster level.

7.3.3 Interpretation

Two hypotheses receive direct empirical support, with indirect evidence for the third. The gradual evolution prediction (H4a) is confirmed: continuing clusters exhibit mean similarity of 0.91–0.94 across the illustrative occupations, indicating incremental skill drift rather than abrupt transformation. The discrete events prediction (H4b) is also supported: births and deaths manifest as a bimodal similarity distribution with sharp terminal-year declines, rather than gradual phase-outs. The viability pattern (H4c) receives substantial indirect support from three convergent sources: Human Resource Adviser's elevated death rate coinciding with its shift from routine to non-routine tasks, skill emergence patterns favouring analytical over administrative competencies, and the technology succession test (H3b) demonstrating that dead clusters contain legacy platforms while born clusters contain modern technologies ($p < 0.001$). Direct testing via O*NET-based routine-task classification at the cluster level remains a direction for future research. Together, these findings support Proposition 4's prediction of continuous evolution punctuated by discrete viability threshold crossings.

7.4 Results: Testing Proposition 5 (Churn-Automation Speed)

7.4.1 Theoretical Prediction

Proposition 5 from Chapter 3 predicts that cluster churn rates are proportional to the speed of automation advancement:

Faster automation advancement generates higher cluster churn rates: $\partial\chi/\partial(dp/dt) > 0$. Periods of rapid technological change generate more cluster births and deaths; periods of stability generate stable cluster configurations.

This generates three testable predictions:

H5a (Cross-Occupation): Occupations with higher automation exposure should exhibit higher churn rates.

H5b (Temporal Variation): Churn rates should increase during periods of rapid technological change.

H5c (Asymmetric Effects): High-automation periods should generate both elevated births (new complementary niches) and elevated deaths (routine-intensive clusters becoming unviable).

7.4.2 Empirical Approach

To test H5a, I exploit variation in automation exposure and skill change velocity across 250 occupations. The initial analysis uses the illustrative occupations to illustrate the level-versus-speed distinction; the full-scale test then examines whether the relationship holds across all 250 occupations using Spearman rank correlations between centroid drift and churn rates.

To test H5b, I examine whether churn rates vary systematically across the 2012–2019 sample period.

To test H5c, I decompose churn into births and deaths and examine whether both components increase with automation exposure.

7.4.3 Results

Hypothesis 5a: The Level-versus-Speed Distinction

A naive reading of Proposition 5 might predict that occupations with higher automation exposure should exhibit higher churn. The illustrative occupations immediately reveal why this fails. Table 7.4 presents churn rates alongside both level-based and speed-based automation measures:

Table 7.4: Automation Level vs. Speed: Three Illustrative Occupations

Occupation	F-O Prob (Level: ρ)	Centroid Drift (Speed: $d\rho/dt$)	Churn Rate	Birth Rate	Death Rate
Receptionist (General)	0.96	0.004	52.9%	31.4%	21.5%
Accountant (General)	0.47	0.004	68.8%	34.0%	34.8%
Human Resource Adviser	0.31	0.011	198.8%	152.5%	46.3%

Note. F-O Prob = Frey-Osborne automation probability (level). Centroid drift = mean year-over-year cosine distance between skill profile centroids (speed). Human Resource Adviser exhibits highest churn (198.8%) despite lowest automation probability (0.31), but highest centroid drift (0.011)—illustrating the level-versus-speed distinction central to Proposition 5.

Human Resource Adviser exhibits the highest churn rate (198.8%) despite having the *lowest* Frey-Osborne automation probability (31%). This apparent contradiction illustrates a key distinction: Proposition 5 predicts that churn is proportional to the *speed* of automation advancement ($d\rho/dt$)—the rate at which technology changes task requirements—not the *level* of automation exposure (ρ). Human Resource Adviser’s centroid drift (0.011) is nearly three times that of Accountant (General) (0.004), reflecting its rapid transformation from administrative support toward strategic business partnership during this period (Ulrich et al., 2017). Receptionist (General) shows the lowest drift (0.004) despite the highest automation exposure: the occupation contracts through gradual displacement rather than rapid transformation.

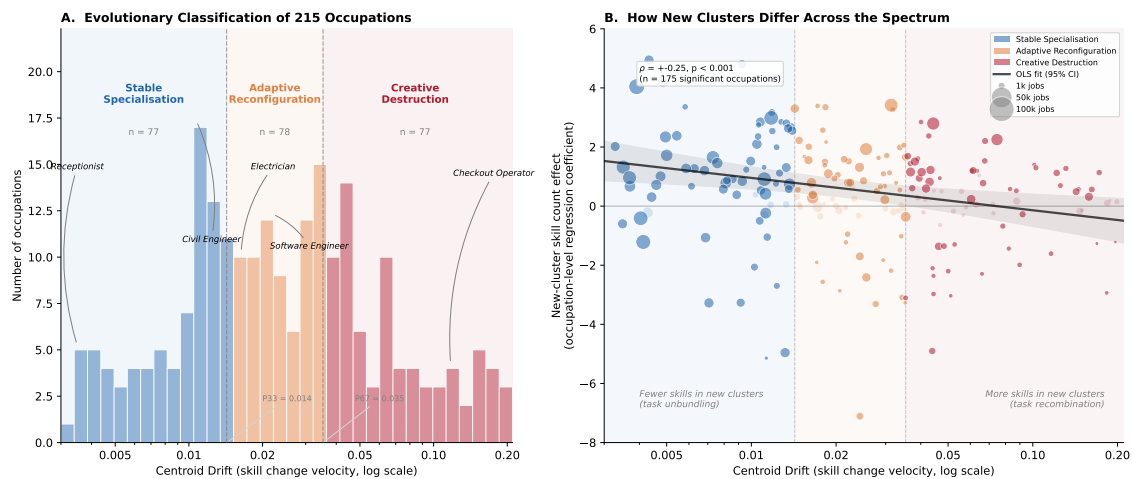
This level-versus-speed distinction, illustrated by three occupations, motivates the full-scale test across all 250 occupations.

Full-Scale Test: Centroid Drift and Churn Across 250 Occupations

Figure 7.1 presents a descriptive taxonomy of occupations by centroid drift. The three categories—Stable Specialisation, Adaptive Reconfiguration, and Creative Destruction—are defined by drift terciles (P33 and P67 thresholds) and serve as a descriptive convenience

for summarising the continuum of restructuring intensities rather than reflecting theoretically derived cut-points:

Figure 7.1: Evolutionary Classification of 250 Occupations by Centroid Drift



Note. Panel A: Distribution of centroid drift across unflagged occupations, partitioned into three categories by drift terciles. Stable Specialisation occupations ($\leq P33$) exhibit minimal skill vocabulary change; Creative Destruction occupations ($> P67$) undergo rapid skill transformation.

Panel B: Occupation-level new-cluster skill coefficients plotted against centroid drift.

Slowly-evolving occupations (left) tend toward task recombination (positive coefficients, new clusters require more skills); rapidly-transforming occupations (right) tend toward task unbundling (negative coefficients, fewer skills). Spearman $\rho = -0.263, p < 0.001$. Dot size proportional to number of job advertisements.

Across all 250 occupations, the Spearman rank correlation between centroid drift and churn rate is $\rho = +0.558$ ($p < 0.001$): occupations where skill requirements change faster experience substantially more cluster births and deaths. The illustrative occupations demonstrate the relationship concretely. Human Resource Adviser has the highest centroid drift (0.011) and highest churn (198.8%), while the lower-drift Accountant (General) (0.004) and Receptionist (General) (0.004) show substantially lower churn (68.8% and 52.9% respectively).

Hypothesis 5b: Temporal Variation

Table 7.5 presents annual churn rates for the illustrative occupations across 2012–2019:

Churn rates exhibit considerable year-to-year variation, with patterns differing across occupations. The temporal variation suggests that churn responds to occupation-specific automation pressures rather than uniform macroeconomic shocks—consistent with technology adoption occurring at different rates across occupational contexts.

Table 7.5: Annual Churn Rates: Illustrative Occupations (2012–2019)

Occupation	12-13	13-14	14-15	15-16	16-17	17-18	18-19
Accountant (General)	77.7	68.8	72.4	71.3	65.5	68.4	57.6
Receptionist (General)	55.6	117.9	43.6	32.7	29.8	41.9	49.0
Human Resource Adviser	200.0	57.1	70.0	64.3	100.0	100.0	800.0

Note. Annual churn rates show considerable variation across years. Human Resource Adviser’s extreme volatility reflects its small cluster count (2–20 clusters per year at the 6-digit level), where small absolute changes produce large percentage swings; the 800% churn in 2018–2019 reflects an expansion from 2 to 16 clusters. Accountant (General), with 86–171 clusters per year, shows more stable rates. All statistics computed from 6-digit ANZSCO occupation models using centroid-based matching ($\tau = 0.80$), consistent with the full-scale methodology.

Hypothesis 5c: Asymmetric Effects — Net Creation

Across all 250 occupations, median birth rates (63.8%) exceed median death rates (46.2%), indicating net positive job type creation. This pattern holds for two of three illustrative occupations: Receptionist (General) and Human Resource Adviser show birth rates exceeding death rates, though Accountant (General) shows near-parity (34.0% births vs 34.8% deaths).

The net positive birth-death differential suggests that automation in this period created more new job types than it eliminated—consistent with Acemoglu and Restrepo (2019c)’s theoretical framework where new task creation partially offsets automation-driven displacement. This finding provides empirical support for task creation as a countervailing force, operating at sub-occupational rather than occupation level. The IMF’s 2024 assessment of generative AI similarly concludes that “exposure does not equal displacement”—most AI-exposed occupations experience task restructuring rather than employment decline (Cazzaniga et al., 2024). Autor et al. (2024a) provides complementary evidence that over 60% of current employment exists in job types that did not exist in 1940; the cluster birth rates documented here provide micro-level evidence of this mechanism operating within occupational boundaries.

The three sub-hypotheses receive empirical support, with H5a’s mechanism reframed from level-based to speed-based exposure. As literally stated, H5a predicted that occupations with higher automation *exposure* (Frey-Osborne probability) would exhibit higher churn—a level-based reading of Proposition 5 that the data *falsify* (Receptionist (General) has the highest F-O probability but among the lowest churn). The reframed H5a, replacing F-O level with centroid drift as the empirical analogue of $d\rho/dt$, is supported by the strong cross-occupation correlation (Spearman $\rho = +0.558$, $p < 0.001$). This is a substantive modification of the original prediction’s mechanism, not a vindication of it: the Frey-Osborne

probability captures potential automation *susceptibility*, while centroid drift captures actual task *transformation*. Proposition 5's mathematical formulation ($\partial\chi/\partial(d\rho/dt) > 0$) is consistent with the speed-based finding; the level-based reading was a misinterpretation of the proposition that the data have helped clarify.

7.5 Characterising Newly-Emerged Clusters

The preceding sections established that occupations experience substantial creative destruction—median annual churn of 112%, with birth rates (63.8%) exceeding death rates (46.2%). But what do newly-emerged clusters look like? Competing theoretical predictions suggest different possibilities: task unbundling (Autor, 2013a) predicts narrower skill sets, while task recombination predicts broader ones as new technologies enable novel skill configurations. This section tests whether cluster trajectory status predicts observable job characteristics, providing a demand-side characterisation of restructuring that complements the temporal dynamics documented above.

7.5.1 Empirical Strategy

I estimate a pooled regression across all 250 occupations with valid cluster models:

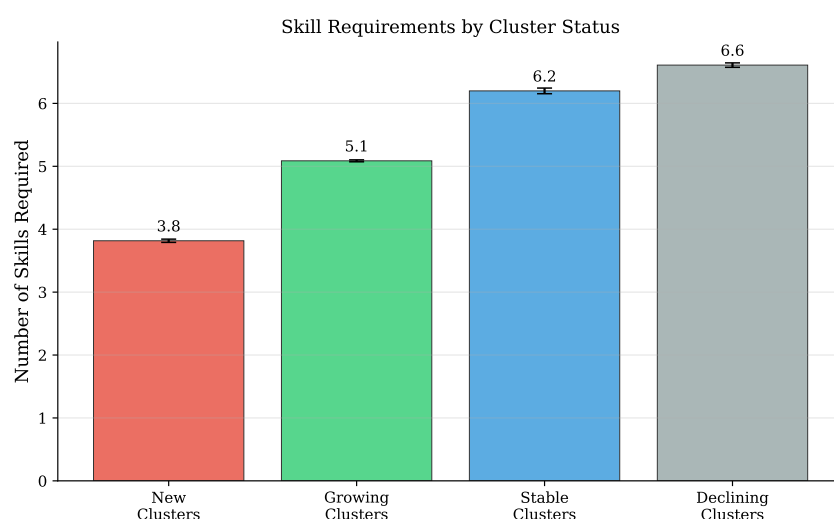
$$y_{ijt} = \alpha + \beta_1 \cdot \text{New}_{jt} + \beta_2 \cdot \text{Growing}_{jt} + \beta_3 \cdot \text{Declining}_{jt} + \gamma_o + \delta_t + \varepsilon_{ijt} \quad (7.5)$$

where y_{ijt} is the outcome for job advertisement i in cluster j at time t ; New_{jt} , Growing_{jt} , and Declining_{jt} are indicator variables for cluster trajectory status (stable clusters are the omitted reference category); γ_o are occupation fixed effects; and δ_t are year fixed effects. The full BGT panel for the 250 occupations contains 3,055,469 job advertisements across all years; the pooled regression sample is restricted to advertisements assigned to clusters with a valid trajectory classification (i.e., excluding advertisements in occupation-years with insufficient temporal coverage to assign trajectory status, and noise observations not assigned to any cluster), yielding $N = 519,189$ regression observations. I test six outcomes available in the BGT data: skill count, minimum education, minimum experience, contract type (permanent vs. temporary), hours (full-time vs. part-time), and advertised salary.

7.5.2 Results: Heterogeneous Skill Requirements

A key demand-side finding of this chapter is that **task unbundling and task recombination coexist across the Australian labour market**, with recombination more common but not universal. Across 232 occupations with sufficient variation to estimate per-occupation regressions, 59% show that newly-emerged clusters require significantly *more* skills than established clusters (task recombination, median coefficient +1.30), while 22% show that newly-emerged clusters require significantly *fewer* skills (task unbundling, median coefficient –1.83); the remaining 19% show no significant difference. This heterogeneity—not either pattern alone—is the contribution of the skill-requirements analysis. A pooled regression that ignores this heterogeneity yields a positive coefficient (Section 7.5.2), but the pooled estimate is dominated by the more numerous and larger recombination occupations and obscures the substantively different dynamics in the unbundling group.

Figure 7.2: Skill Requirements by Cluster Trajectory Status



Note. Mean number of skills listed in job advertisements, by cluster trajectory status across all 250 occupations. Error bars show 95% confidence intervals.

Per-Occupation Analysis

Of the 250 occupations, 232 have sufficient variation in cluster trajectory status to estimate per-occupation regressions (the remaining 18 have either all clusters stable or all at-risk). The results reveal that **both task unbundling and task recombination coexist across the labour market**, with recombination dominating:

- **Task recombination** (59%, 136 occupations): Newly-emerged clusters require significantly *more* skills than established clusters (median coefficient +1.30). These

occupations tend to be larger (median 1,518 jobs), more stable (median churn 98%), and have more clusters (median 15.3). Examples include Statistician (+4.80), Electrical Engineer (+4.36), Accountant General (+4.04), and Payroll Clerk (+4.19).

- **Task unbundling** (22%, 51 occupations): Newly-emerged clusters require significantly *fewer* skills (median coefficient -1.83). These occupations tend to be smaller (median 654 jobs), more volatile (median churn 135%), and have fewer clusters (median 7.1). Examples include Network Administrator (-7.11), Corporate Services Manager (-4.96), and Financial Investment Manager (-4.90).
- **Non-significant** (19%, 45 occupations): No significant difference between new and established clusters.

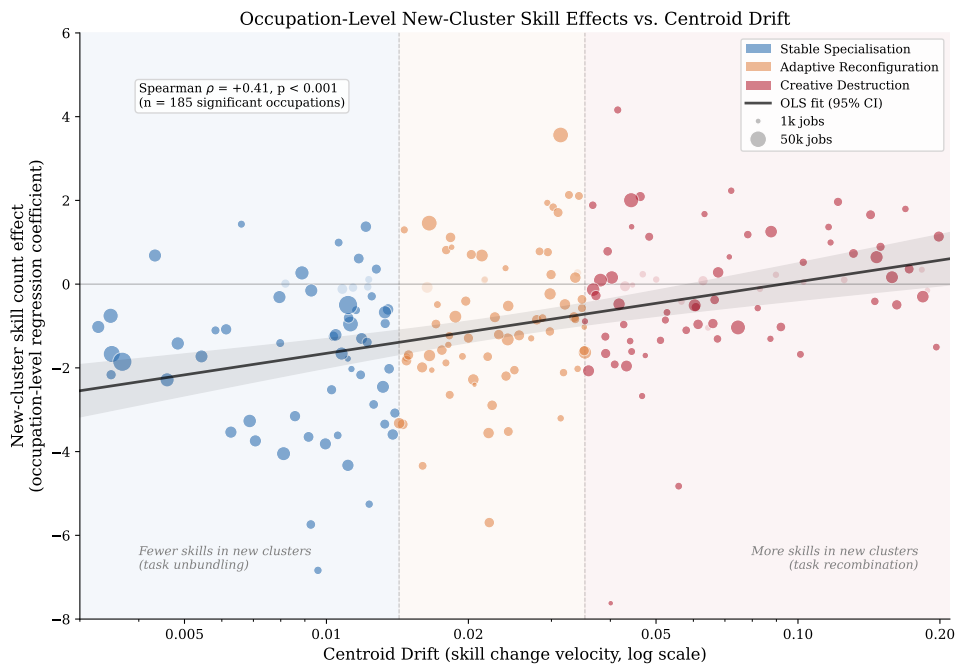
The median occupation-level coefficient is $+0.63$, with an interquartile range of $[-0.24, +1.50]$. The distribution is skewed positive, with 69% of occupations showing positive coefficients.

What Distinguishes the Two Patterns?

The two patterns are structurally different occupations. Unbundling occupations exhibit significantly higher centroid drift (median 0.031 vs 0.018, $p = 0.020$), higher churn (135% vs 98%, $p = 0.001$), fewer clusters (7.1 vs 15.3, $p < 0.001$), and smaller sample sizes (654 vs 1,518 jobs, $p < 0.001$). In economic terms: rapidly-transforming occupations with volatile, sparse cluster structures tend to produce simplified new job types (unbundling), while stable, well-structured occupations with rich cluster ecosystems tend to produce enriched new job types (recombination).

Centroid drift moderates the skill effect (Spearman $\rho = -0.263$, $p < 0.001$): occupations with faster skill change tend toward unbundling, while occupations with stable skill content tend toward recombination. Figure 7.3 confirms this pattern across 232 occupations. By ANZSCO major group, Professionals and Technicians lean heavily toward recombination (63% and 72% respectively), while Sales occupations are evenly split (38%/38%).

One interpretation reflects two distinct mechanisms through which technology may reshape occupational structure. In slowly-evolving occupations, the pattern is consistent with task *unbundling*: where automation or digital tools take over specific task components, employers may carve out narrower, more specialised roles from broader established positions. A general administrative role, for example, could split into a data entry cluster and a client liaison cluster once software automates the routine coordination that previously bundled these tasks together. The resulting new clusters would require fewer skills because they inherit only a subset of the parent role's task portfolio.

Figure 7.3: Occupation-Level New-Cluster Skill Effects vs. Centroid Drift

Note. Each point represents one occupation's new-cluster skill count coefficient (from occupation-level regressions) plotted against centroid drift. Occupations are coloured by evolutionary classification: Stable Specialisation (blue), Adaptive Reconfiguration (orange), Creative Destruction (red). Faded points indicate non-significant coefficients ($p \geq 0.05$). Dot size is proportional to number of job advertisements. The negative trend (Spearman $\rho = -0.263$, $p < 0.001$) shows that slowly-evolving occupations tend toward task recombination (positive coefficients, new clusters require more skills), while rapidly-transforming occupations tend toward task unbundling (negative coefficients, new clusters require fewer skills). OLS fit with 95% confidence interval shown.

In rapidly-transforming occupations, the pattern is consistent with task *recombination*: new tools and platforms enable novel task combinations that did not previously exist, with workers integrating skills from multiple domains. A software development role, for instance, may spawn new clusters that combine traditional programming with cloud infrastructure, data engineering, or user experience design—producing skill requirements that match or exceed established clusters. The positive skill coefficients in high-drift occupations are consistent with creative recombination, where new clusters represent genuinely novel job configurations rather than simplified fragments of existing roles. The chapter does not establish that technology *causes* either pattern; the interpretation describes a mechanism consistent with the observed correlation between drift and skill direction, but alternative explanations (changes in employer reporting practices, occupation-specific demand shocks, organisational restructuring unrelated to automation) cannot be ruled out without additional identification.

Pooled Regression

For completeness, pooling across all occupations yields a positive coefficient ($\beta = +0.886$, $p < 0.001$, $N = 519,189$, $R^2 = 0.175$ with occupation fixed effects), reflecting the fact that recombination occupations are both more numerous (59% vs 22%) and larger (median 1,518 vs 654 jobs). However, this pooled estimate obscures the economically meaningful heterogeneity documented above and should not be interpreted as indicating a universal pattern.

7.6 Cluster-Level Trajectory Analysis

The preceding sections characterised temporal dynamics at the occupation level: centroid drift measures how quickly an occupation's aggregate skill profile changes, and cluster churn captures the rate of structural reorganisation. These occupation-level metrics established key patterns—Propositions 4 and 5—but do not distinguish between individual clusters *within* an occupation. If the central argument of this thesis is that automation affects individual job types (clusters) rather than occupations as a whole, then the analytical lens must descend to the cluster level. This section develops that lens by tracing each cluster's individual trajectory through the eight-year panel, producing a per-cluster classification that Chapter 8 subsequently uses to identify transition pathways between at-risk and stable clusters.

7.6.1 Backward Lineage Tracing

The greedy centroid matching algorithm from Section 7.2 already matches clusters across consecutive years. Here, the match chain is traced *backward* from each 2019 cluster to determine its birth year and persistence duration. Specifically, for each 2019 cluster C_{2019} , the algorithm identifies which 2018 cluster matched *to* it (reverse lookup in the match table), then chains backward: $C_{2019} \leftarrow C_{2018} \leftarrow C_{2017} \leftarrow \dots$ until no predecessor is found. The birth year equals the earliest year in this chain, and the cluster's age equals the number of years spanned (1–8).

This procedure yields trajectory metrics for each of the 5,118 clusters observed in 2019 across 248 occupations:

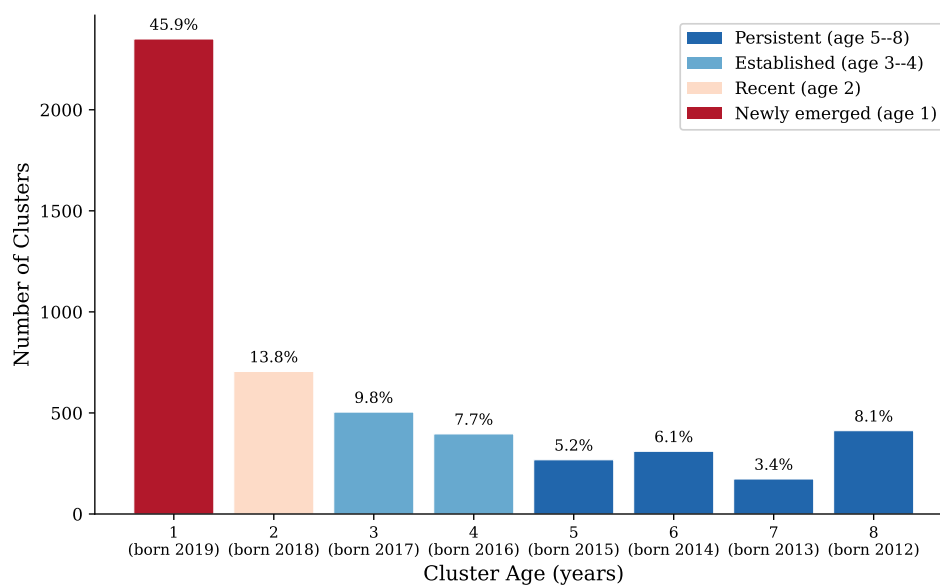
- **Birth year:** The earliest year in which a matched predecessor exists (2012–2019).
- **Age:** Number of consecutive years the cluster persists (1 = born in 2019; 8 = present since 2012).

- **Size trend:** OLS slope of cluster sizes across the lineage (jobs per year; requires age ≥ 3).
- **Mean match similarity:** Average cosine similarity across consecutive matches in the lineage.

7.6.2 Age Distribution and Survival Dynamics

Figure 7.4 presents the age distribution of all 2019 clusters. The distribution is heavily right-skewed: 45.9% of clusters (2,350) are newly emerged—they have no matched predecessor from 2018. A further 13.8% (705) are age 2, present only since 2018. Together, 60.0% of all 2019 clusters have existed for two years or fewer. At the other extreme, 8.6% (457) have persisted across all eight years of the panel.

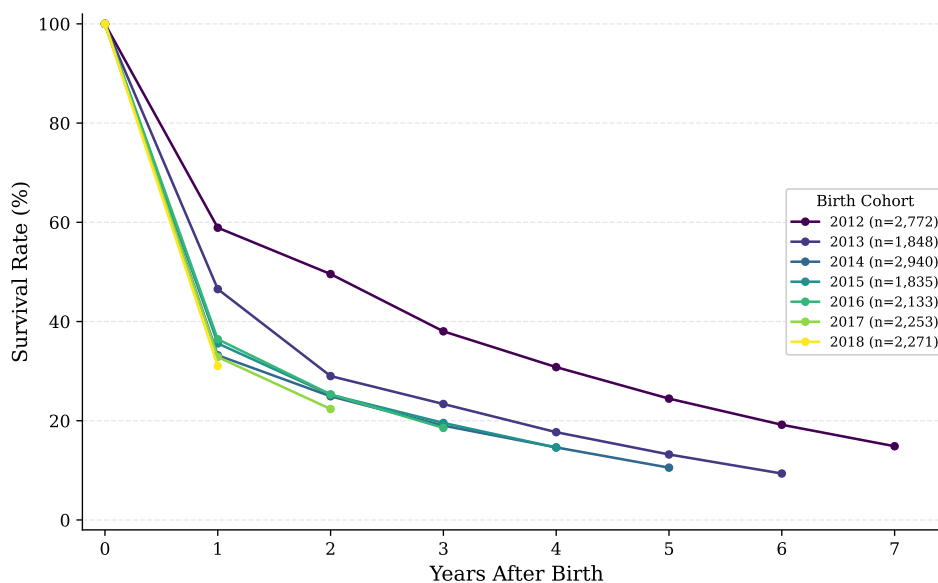
Figure 7.4: Age distribution of 5,118 clusters observed in 2019



Note. Each bar shows the number of 2019 clusters born in the indicated year. Colours correspond to trajectory status categories: persistent (dark blue, age 5–8), established (medium blue, age 3–4), recent (light orange, age 2), and newly emerged (red, age 1). Nearly half of all clusters are newly emerged, having no matched predecessor from the prior year.

The high proportion of newly emerged clusters is consistent with the churn rates documented in Section 7.3: the median one-year survival rate across all years is approximately 55%, meaning that roughly half of all clusters in any given year fail to match to a successor in the following year. Figure 7.5 presents cohort survival curves that trace the fate of clusters born in each year. The 2012 cohort exhibits the highest persistence—57.1% survive one year and 15.4% persist through all seven subsequent years to 2019—while later cohorts show broadly similar decay patterns.

Figure 7.5: Cohort survival curves: fraction of clusters surviving from birth year to subsequent years



Note. Each line tracks a cohort of clusters born in the indicated year. “Born” means the cluster had no matched predecessor from the prior year (or, for 2012, appeared in the first panel year). Survival is defined as having a continuous chain of greedy centroid matches (cosine similarity > 0.80) through each subsequent year.

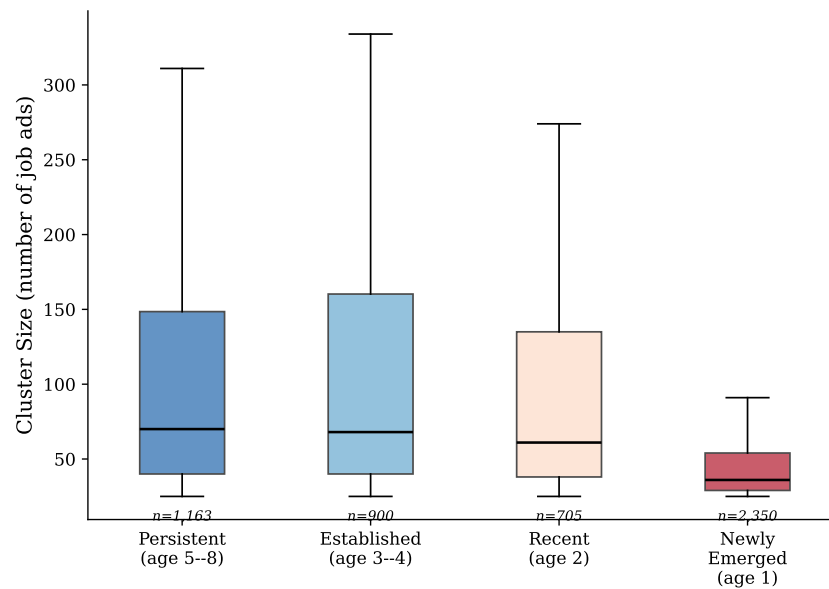
7.6.3 Predictors of Persistence

Cluster size predicts persistence: the Spearman correlation between 2019 cluster size and age is $\rho = +0.312$ ($p < 0.001$, $N = 5,118$). Figure 7.6 visualises this relationship. Persistent clusters (age 5–8) and established clusters (age 3–4) have similar median sizes of approximately 60 jobs, while newly emerged clusters are substantially smaller (median 35 jobs). The size advantage of persistent clusters reflects a survival selection effect: larger clusters represent more consolidated skill bundles that are harder to displace through incremental task changes.

Among the 1,163 persistent clusters (age ≥ 5), 54.9% exhibit growing size trends while 44.8% are declining. This near-even split indicates that long-term survival does not guarantee continued growth—many persistent clusters are stable fixtures that are gradually shrinking as the occupation’s skill landscape evolves around them.

7.6.4 Trajectory Classification

Based on the age distribution and survival analysis, clusters are classified into four trajectory categories (Table 7.6):

Figure 7.6: Cluster size (2019) by trajectory status

Note. Box plots show the distribution of 2019 cluster sizes by trajectory status category. Outliers are suppressed for readability. Sample sizes: persistent ($n = 1,163$), established ($n = 900$), recent ($n = 705$), newly emerged ($n = 2,350$).

Table 7.6: Cluster Trajectory Classification

Status	Definition	Count	% of Total
Persistent	Age ≥ 5 (survived majority of panel)	1,163	22.7%
Established	Age 3–4	900	17.6%
Recent	Age 2	705	13.8%
Newly emerged	Age 1 (born in 2019)	2,350	45.9%
Total		5,118	100%

Classification based on backward lineage tracing from 2019 clusters through greedy centroid matching (cosine similarity threshold $\tau = 0.80$). Age = number of consecutive years the cluster persists in the matching chain.

For Chapter 8's analysis of transition pathways, a binary risk classification provides the analytical lens:

- **Stable** ($n = 1,095$; 21.4%): Age ≥ 4 and not significantly declining (size trend > -5 jobs/year). These represent consolidated job types with demonstrated persistence.
- **At-risk** ($n = 3,519$; 68.8%): Age ≤ 2 or age ≥ 4 but significantly declining (size trend < -5 jobs/year). These represent either newly emerged clusters whose longevity is uncertain or persistent clusters that are contracting.
- **Transitional** ($n = 504$; 9.8%): Age 3 with ambiguous trajectory. These are excluded

from the binary classification in Chapter 8.

7.6.5 Within-Occupation Heterogeneity

The most consequential finding for the thesis is that trajectory status varies *within* occupations, not just between them. Of the 249 occupations with clusters in 2019, 204 (81.9%) contain *both* at-risk and stable clusters. No occupation consists entirely of stable clusters, and 21 consist entirely of at-risk clusters. Source: `analysis/results/chapter7/cluster_trajectories_2019.csv`. This pattern holds across all three categories of the evolutionary classification (Figure 7.1): occupations classified as Stable Specialisation at the aggregate level nonetheless contain at-risk clusters, while Creative Destruction occupations contain stable clusters.

This within-occupation heterogeneity is precisely the pattern predicted by the theoretical framework’s central claim: automation transforms occupations by selectively affecting individual task bundles rather than eliminating occupations wholesale. Chapter 8 exploits this heterogeneity by constructing transition pathways from at-risk to stable clusters, demonstrating that workers displaced by cluster decline can often find skill-proximate alternatives *within their own occupation*—a finding that would be invisible to occupation-level analysis.

7.7 Discussion

7.7.1 Interpreting the Temporal Findings

The central insight from this chapter is that automation speed, not automation level, predicts within-occupation restructuring. Static exposure scores like Frey and Osborne (2017)’s automation probabilities capture susceptibility, but the pace at which skill requirements actually change—measured through centroid drift—determines how rapidly clusters form and dissolve. This distinction has implications for automation forecasting: recent empirical work by Humlum and Vestergaard (2024) finds “small labour market effects” from large language models despite high predicted exposure, while Lane and Saint-Martin (2021) document that AI adoption patterns differ substantially from static predictions.

An important interpretive issue arises here. Centroid drift is constructed from the observed skill changes themselves, creating potential circularity when used to “explain” cluster churn derived from the same changes. Both metrics summarise the same underlying year-to-year transitions; their correlation is therefore informative about the *internal consistency* of two different summary measures of restructuring, not about a causal link between automation and churn. The finding is best read as confirming that occupations

whose skill content shifts faster (high drift) also experience more cluster births and deaths (high churn)—a pattern consistent with Proposition 5’s predictions but equally consistent with both metrics tracking the same latent restructuring process. Prospective measures of automation intensity that are independent of the BGT skill-change panel—such as technology adoption indices, robot installation rates, or patent-based exposure measures (Webb, 2020)—would provide a more exogenous test of Proposition 5 and remain a direction for future research.

The skill requirement findings add a further layer of complexity. The aggregate result—newly-emerged clusters require approximately one additional skill—masks substantial heterogeneity. Task recombination dominates among slowly-evolving occupations, while task unbundling characterises rapidly-transforming ones. This heterogeneity, rather than either pattern alone, is the central demand-side finding. It suggests that automation does not uniformly simplify or complicate work; instead, the nature of restructuring depends on the pace and character of technological change within each occupation.

Temporal robustness.

The 2012–2019 period spans Australia’s mining investment boom-to-bust transition (~2013–2015), raising the concern that cluster dynamics reflect cyclical macroeconomic adjustment rather than structural restructuring. If restructuring were cyclically driven, key metrics should differ markedly between boom and post-boom sub-periods. Table 7.7 tests this by reporting metrics for each sub-period independently.

Table 7.7: Temporal Subsample Robustness

Metric	2012–2015	2016–2019	Full period
Occupations	250	250	250
Transitions	715	746	1,707
Median churn (%)	100.0	100.0	111.7
Median birth rate (%)	53.8	45.5	63.8
Median death rate (%)	49.4	47.5	46.2
Median survival (%)	50.6	52.5	53.8
Drift-churn ρ	+0.579	+0.590	+0.558
Drift-churn p	< 0.001	< 0.001	< 0.001

Note. Each sub-period analysed independently using greedy centroid matching ($\tau = 0.80$). Full-period figures from the main analysis (Table 7.2). Birth rates slightly exceed death rates in the early period (53.8% vs 49.4%) but are slightly lower in the late period (45.5% vs 47.5%). The drift-churn correlation is strong and significant in both sub-periods.

Both sub-periods show qualitatively similar patterns: median churn equals 100%, birth and death rates are broadly balanced, and the drift-churn association is strong and

significant in both ($\rho = +0.579$ and $+0.590$ respectively, both $p < 0.001$). The full-period median birth rate (63.8%) exceeds both sub-period medians (53.8% and 45.5%) because medians do not aggregate linearly across overlapping samples: each sub-period's median is computed across that period's three transitions per occupation, while the full-period median is computed across seven transitions per occupation, and an occupation's full-period rank can differ from its rank in either sub-period. This pattern is therefore an artefact of computing medians on different transition counts rather than a substantive feature of the data. The qualitative findings—substantial within-occupation restructuring, churn substantially exceeding firm-level benchmarks, and a positive drift-churn association—are not driven by the mining transition or any single sub-period.

7.7.2 Australian Institutional Context

The temporal findings gain additional meaning when interpreted through Australia's distinctive institutional framework:

Award wage rigidity: Occupation-level minimum wages (Section 4.2.1) may amplify task reorganisation as the primary adjustment margin. When wages cannot adjust, firms respond to changing conditions by reconfiguring job designs rather than adjusting compensation.

Service sector dominance: With 79.8% service employment, Australian automation manifests primarily through task recomposition within service occupations rather than manufacturing job elimination.

7.7.3 Limitations

1. Matching threshold sensitivity: Birth/death classification depends on the similarity threshold $\tau = 0.8$. Lower thresholds reduce measured churn (more matches qualify as continuations) while higher thresholds increase it. Table 7.8 reports sensitivity across the full range $\tau \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$. Median churn ranges from 91% ($\tau = 0.70$) to 158% ($\tau = 0.90$), with the baseline $\tau = 0.80$ producing 115% (IQR: 81–156%). Birth rates range from 55% to 89% and death rates from 34% to 69% across thresholds. Crucially, churn rates exceed firm-level benchmarks (20–25%) by 4–6 $\times$ at every threshold tested, and the qualitative conclusions—substantial within-occupation restructuring, birth rates exceeding death rates, positive drift-churn correlation—are robust across the full range. The baseline $\tau = 0.80$ was chosen to balance false continuations against false deaths; at this threshold, matched clusters exhibit mean similarity of 0.905 (Table 7.3), indicating genuine continuity.

Table 7.8: Sensitivity of temporal matching metrics to cosine similarity threshold (τ). Median values across occupations with interquartile range in parentheses. Baseline threshold $\tau = 0.80$ shown in bold.

Threshold (τ)	Churn Rate (%)	Birth Rate (%)	Death Rate (%)
0.70	91.1 (64–126)	55.0 (38–81)	34.3 (24–49)
0.75	100.4 (73–142)	59.7 (43–87)	40.4 (28–56)
0.8	114.6 (81–156)	66.0 (49–94)	47.9 (35–62)
0.85	132.8 (100–175)	77.3 (57–103)	57.3 (43–71)
0.90	158.2 (127–191)	88.7 (69–111)	68.5 (55–81)

2. Temporal scope: The 2012–2019 period predates generative AI and pandemic disruptions. Following Brynjolfsson et al. (2023b)’s analysis of AI-driven task transformation, temporal dynamics may differ substantially under more recent technological regimes characterised by large language model deployment. Early evidence from Eloundou et al. (2024) and Noy and Zhang (2023b) suggests that generative AI affects different task bundles—particularly writing, analysis, and coding—than the routine manual and cognitive tasks vulnerable to earlier automation waves. Whether the cluster dynamics documented here persist, accelerate, or shift direction under generative AI remains an important question for future research.

3. Measurement of “automation speed”: Task transformation intensity is a post hoc measure constructed from observed skill changes. Prospective measures of automation advancement—such as technology adoption indices or patent-based measures (Webb, 2020)—would provide stronger exogenous tests of Proposition 5.

4. Macroeconomic confounds: The 2012–2019 sample period spans Australia’s mining investment cycle, which could confound automation effects. The analysis does not include formal controls for industry composition or macroeconomic conditions. While the skill transformation patterns observed—particularly the shift from administrative to strategic and technology-enabled tasks—are consistent with automation-driven restructuring, cyclical factors cannot be ruled out as contributing to temporal variation in churn rates.

5. Job advertisement data as demand-side measure: The skill count finding captures what employers *request* in job advertisements, not what workers *experience*. Whether narrower skill requirements in newly-emerged clusters translate to different wages, career trajectories, or job quality for workers requires matched employer-employee data beyond the scope of this analysis.

7.7.4 Creative Destruction Benchmarks

The observed churn rates—median 116% annually across 250 occupations (IQR: 79–157%)—substantially exceed firm-level benchmarks. Caballero and Hammour (2005) document 20–25% annual job reallocation at the firm level. The higher cluster-level churn indicates that job type restructuring within occupations occurs even more rapidly than firm-level employment dynamics.

This comparison underscores the importance of sub-occupational analysis. If occupation-level classifications suggest stable employment while cluster-level analysis reveals median annual churn exceeding 100%, the occupation-level view dramatically understates labour market dynamics.

7.8 Conclusion

This chapter tested the dynamic propositions from Chapter 3 using longitudinal analysis of 250 occupations across the 2012–2019 machine-learning era. Both propositions receive empirical support, with H5a’s mechanism reframed from level-based to speed-based exposure: continuing clusters evolve gradually while births and deaths manifest as discrete events (Proposition 4), and churn rates correlate strongly with the speed of skill change rather than with static automation exposure (Proposition 5). The reframing helps explain why static risk scores like Frey and Osborne (2017)’s do not predict the actual restructuring dynamics observed in the data—they capture potential susceptibility rather than realised transformation.

Two findings extend the theoretical framework in directions the original propositions did not anticipate. First, the demand-side consequences of restructuring are heterogeneous: task recombination and task unbundling coexist across the labour market, with centroid drift moderating the balance (recombination dominating among slowly-evolving occupations, unbundling among rapidly-transforming ones). Second, backward lineage tracing reveals that 81.9% of occupations contain both at-risk and stable clusters simultaneously, demonstrating that vulnerability is a property of individual job types, not entire occupations.

Read in historical perspective, this chapter establishes the temporal-dynamics baseline for Australian labour markets during the diffusion of statistical machine learning—a record of how cluster structure evolved before the disruptions of COVID-19 and large language models. The drift-churn correlation, the heterogeneous restructuring patterns, and the within-occupation vulnerability heterogeneity provide reference points against which the dynamics of subsequent waves can later be compared. These temporal patterns also raise a practical question: when clusters dissolve, what options do affected workers face?

Chapter 8 addresses this by constructing a skill proximity network that maps feasible transition pathways between clusters.

Chapter 8

Skill Proximity Networks and Career Transition Pathways

8.1 Introduction

The preceding chapters established that occupations contain discrete sub-occupational clusters subject to continuous creative destruction, with most occupations containing both at-risk and stable clusters simultaneously. This raises a critical practical question: *when clusters dissolve, where can affected workers go?* The answer depends on how the labour market is structured in skill space—whether at-risk clusters are connected to viable alternatives, or whether they sit in isolated pockets with no skill-proximate transition options. Three questions structure this chapter’s investigation:

1. **Is the labour market navigable in skill space?** If the skill landscape is fragmented into isolated pockets, then cluster dissolution strands workers regardless of their individual capabilities. I address this by constructing a skill proximity network and characterising its topology. An important caveat applies throughout: skill proximity measures *demand-side* similarity between job types, not the *supply-side* feasibility of worker transitions, which depends additionally on institutional pathways, credentialing requirements, geographic constraints, and employer hiring preferences that this analysis cannot observe.
2. **Who faces the hardest transitions?** Even in a connected network, navigability

may be unevenly distributed. I examine whether cluster trajectory status—persistent, established, recent, or newly emerged—systematically predicts access to transition opportunities.

3. **What structures the barriers?** Raw skill gap counts treat all skills as equivalent obstacles. I decompose transition barriers into generic competencies that workers likely already possess and specialist skills that require genuine investment, then analyse the internal structure of specialist barriers—their co-occurrence in domain-specific bundles and their complementarity with other skills—to identify actionable training targets.

This chapter addresses these questions by constructing a *skill proximity network* from the cluster centroid data developed in previous chapters. The approach draws on the economic complexity literature that uses network methods to map relatedness between economic activities (Hidalgo et al., 2007) and, more directly, on research mapping the skill space of occupations as a network (Alabdulkareem et al., 2018; Dawson et al., 2021; Frank, Moro, South, et al., 2024; Neffke & Henning, 2013). Each cluster is represented by its skill profile (the centroid vector from HDBSCAN clustering), and edges connect clusters with sufficiently similar skill requirements. Crucially, each node carries the cluster-level trajectory status developed in Chapter 7 rather than an occupation-level classification, ensuring that the analysis reflects the thesis’s central finding that restructuring is a property of individual clusters, not entire occupations. The estimand is descriptive: network topology, connectivity gradients across cluster types, and skill gap distributions for feasible transitions—not causal effects of network position on worker outcomes. The questions are applied rather than theoretical—the chapter does not introduce new formal propositions but demonstrates a practical application of the cluster infrastructure developed throughout the thesis.

8.1.1 Chapter Roadmap

Section 8.2 describes the network construction methodology. Section 8.3 characterises the network’s topology and examines how connectivity varies by cluster trajectory status, addressing the first two questions. Section 8.4 analyses career transition pathways from at-risk to stable clusters, then progressively decomposes the barriers—distinguishing generic from specialist gaps (Section 8.4.1), identifying co-occurring skill bundles (Section 8.4.3), and measuring skill complementarity (Section 8.4.4)—addressing the third question. Section 8.5 discusses implications for workforce development policy. Section 8.6 concludes.

8.2 Methods

8.2.1 Network Construction

The skill proximity network represents each cluster from the 2019 cross-section as a node and connects clusters with similar skill profiles. Construction proceeds in four steps.

Step 1: Centroid extraction. For each of the 249 occupations with valid 2019 cluster models, I extract the HDBSCAN cluster centroids. Let occupation o have clusters $c \in \{1, \dots, K_o\}$, each containing n_c jobs described over a local skill vocabulary $\mathcal{S}_o$ of $|\mathcal{S}_o|$ skills. The binary skill matrix $\mathbf{X}^{(o)} \in \{0, 1\}^{N_o \times |\mathcal{S}_o|}$ records skill presence in each job advertisement. The centroid of cluster c is the mean skill prevalence vector:

$$\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i \in c} \mathbf{x}_i, \quad \mu_{c,s} \in [0, 1] \quad (8.1)$$

where $\mathbf{x}_i$ is the binary skill vector for job i and $\mu_{c,s}$ represents the proportion of jobs in cluster c that require skill s . This produces 5,118 cluster nodes spanning 8,458 unique skills across all 249 occupations.

Step 2: Global vocabulary alignment. Each occupation's clusters are defined over a local skill vocabulary $\mathcal{S}_o$. To compute cross-occupation similarity, I construct a global vocabulary $\mathcal{S} = \bigcup_o \mathcal{S}_o$ with $|\mathcal{S}| = 8,458$ skills and align all centroids to this common space by zero-padding:

$$\tilde{\mu}_{c,s} = \begin{cases} \mu_{c,s} & \text{if } s \in \mathcal{S}_o \\ 0 & \text{otherwise} \end{cases} \quad (8.2)$$

The resulting matrix is $5,118 \times 8,458$, with 93% zero entries reflecting that most occupations use a small subset of the global skill space.

Step 3: Pairwise cosine similarity. Cosine similarity—the standard measure for comparing high-dimensional sparse vectors (Salton et al., 1975)—is computed between all pairs of aligned centroid vectors:

$$\text{sim}(c_i, c_j) = \frac{\tilde{\boldsymbol{\mu}}_{c_i} \cdot \tilde{\boldsymbol{\mu}}_{c_j}}{\|\tilde{\boldsymbol{\mu}}_{c_i}\| \|\tilde{\boldsymbol{\mu}}_{c_j}\|} \quad (8.3)$$

using scikit-learn's optimised implementation (Pedregosa et al., 2011). Cosine similarity is well-suited to skill profile comparison because it is invariant to vector magnitude (total number of skills) and captures the angular similarity between skill distributions, consistent with its use in prior skill network research (Alabdulkareem et al., 2018). This produces $\binom{5,118}{2} \approx 14.2$ million pairwise similarities.

Step 4: Edge thresholding. The network $G = (V, E)$ is defined with nodes $V = \{c_1, \dots, c_{5118}\}$ and edges:

$$E = \{(c_i, c_j) : \text{sim}(c_i, c_j) \geq \tau\} \quad (8.4)$$

Table 8.1 reports edge counts at four candidate thresholds:

Table 8.1: Network Sensitivity to Similarity Threshold

Threshold (τ)	Edges	% of Pairs	Connected Components
0.5	518,677	4.0%	—
0.6	196,236	1.5%	490
0.7	71,077	0.5%	—
0.8	25,717	0.2%	—

Note. Edge counts at four candidate cosine similarity thresholds for the 2019 cross-section (5,118 cluster nodes, 13.1 million pairwise comparisons). Bold row indicates the selected threshold ($\tau = 0.6$). Connected components reported only for the selected threshold. Dashes indicate values not computed for non-primary thresholds.

I select $\tau = 0.6$ as the primary threshold, balancing network connectivity against noise. This threshold retains meaningful skill similarity—two clusters sharing 60% of their weighted skill profiles—while excluding trivially similar connections. The threshold is consistent with the 0.80 matching threshold used for temporal tracking in Chapter 7, which serves a different purpose (identifying the *same* cluster across years rather than *related* clusters). The qualitative findings reported below are robust across the threshold range. At $\tau = 0.5$ the network is denser (518,677 edges, 4.0% of pairs) but the connectivity asymmetry by trajectory status persists (mean degree of persistent clusters exceeds that of newly emerged at all thresholds, and the isolated-node share remains lowest for persistent clusters and highest for newly emerged). At $\tau = 0.7$ the network is sparser (71,077 edges, 0.5% of pairs) but the largest connected component continues to contain a clear majority of clusters and the same isolation gradient is observed. The skill-gap finding that most transitions require ≤ 3 skills holds across thresholds because it depends on centroid similarity between source and target clusters, not on the network topology constructed from the threshold.

8.2.2 Edge Classification

Each edge is classified as either *within-occupation* (connecting two clusters in the same occupation) or *cross-occupation* (connecting clusters in different occupations). This distinction reveals whether career transitions would require occupational mobility or can occur within current occupational boundaries.

8.2.3 Cluster Trajectory Status Assignment

Each node is assigned a trajectory status based on the backward lineage analysis developed in Chapter 7 (Section 7.6). The four-category classification—persistent (age ≥ 5), established (age 3–4), recent (age 2), and newly emerged (age 1)—reflects each cluster’s individual history rather than its parent occupation’s aggregate dynamics. For transition pathway analysis, clusters are further classified by risk status: *at-risk* (age ≤ 2 , or age ≥ 4 with a declining size trend), *stable* (age ≥ 4 and not declining), or *transitional* (age 3). This cluster-level classification ensures that the network analysis is consistent with the thesis’s central finding that restructuring operates at the sub-occupational level: 204 of 249 occupations (81.9%) contain both at-risk and stable clusters simultaneously.¹

8.2.4 Skill Gap Computation

For transition pathway analysis, I compute the *skill gap* between any source–target cluster pair. Given source cluster c_s and target cluster c_t with centroid vectors μ_{c_s} and μ_{c_t} , a skill s represents a “gap” if it is meaningfully present in the target but essentially absent from the source:

$$\text{Gap}(c_s, c_t) = \{s \in \mathcal{S} : \mu_{c_t, s} > \theta_{\text{tgt}} \text{ and } \mu_{c_s, s} < \theta_{\text{src}}\} \quad (8.5)$$

with $\theta_{\text{tgt}} = 0.10$ and $\theta_{\text{src}} = 0.03$. These thresholds reflect the centroid representation: a prevalence of 0.10 means the skill appears in at least 10% of jobs in the target cluster, representing substantive demand; 0.03 allows for minor noise in skill prevalence. The gap count $|\text{Gap}(c_s, c_t)|$ provides a practical “how many new skills must be learned” metric for each potential transition.

I further decompose the skill gap by computing a continuous upskilling distance that weights each gap skill by its prevalence differential:

$$d_{\text{upskill}}(c_s, c_t) = \sum_{s \in \text{Gap}(c_s, c_t)} (\mu_{c_t, s} - \mu_{c_s, s}) \quad (8.6)$$

This captures not only *which* skills must be acquired but also *how prominently* each features in the target cluster’s profile.

¹Of the 5,118 clusters in the network, 303 (5.9%) could not be assigned a trajectory status in the network analysis pipeline due to lineage-tracing edge cases at the start of the panel period; the network-stratified analyses in Section 8.3 use the trajectory classification as stored in the network pipeline, while the heterogeneity statistic reported here is computed from the canonical Chapter 7 classification.

8.2.5 Skill Specificity Classification

To distinguish transferable from domain-specific barriers, I classify each skill s by the number of occupations in which it appears. Let $n_{\text{occ}}(s) = |\{o : s \in \mathcal{S}_o\}|$ denote the number of occupations whose job advertisements mention skill s . Skills are classified as:

$$\text{type}(s) = \begin{cases} \text{generic} & \text{if } n_{\text{occ}}(s) \geq 100 \\ \text{specialist} & \text{if } n_{\text{occ}}(s) < 100 \end{cases} \quad (8.7)$$

The threshold of 100 (40% of the 250 occupations in the broader BGT clustering corpus, of which 249 contribute clusters to the 2019 network) separates broadly transferable competencies—Communication Skills ($n_{\text{occ}} = 250$), Planning (250), Teamwork (250)—from domain-specific skills such as Clinical Experience ($n_{\text{occ}} = 95$), CNC Machining (91), and Speech Language Pathology (39). This classification enables the decomposition of each skill gap into generic and specialist components: $\text{Gap}(c_s, c_t) = \text{Gap}_{\text{gen}}(c_s, c_t) \cup \text{Gap}_{\text{spec}}(c_s, c_t)$.

8.2.6 Skill Bundle Identification

To formalise the observation that specialist skills cluster in domain-specific packages, I apply hierarchical clustering to the co-occurrence structure of specialist skills across transition gaps. For all pathways where the specialist gap $|\text{Gap}_{\text{spec}}(c_s, c_t)| \geq 2$, I extract the set of specialist gap skills and compute a pairwise co-occurrence matrix $\mathbf{C}$ where:

$$C_{ab} = |\{(c_s, c_t) : s_a \in \text{Gap}_{\text{spec}}(c_s, c_t) \text{ and } s_b \in \text{Gap}_{\text{spec}}(c_s, c_t)\}| \quad (8.8)$$

counts the number of transition pathways in which specialist skills s_a and s_b both appear as gap requirements. I normalise this to a similarity measure using the overlap coefficient (Vijaymeena & Kavitha, 2016):

$$\text{overlap}(s_a, s_b) = \frac{C_{ab}}{\min(C_{aa}, C_{bb})} \quad (8.9)$$

which ranges from 0 (never co-occur) to 1 (one skill's appearances are a subset of the other's). The overlap coefficient is preferred over the Jaccard index here because the specialist skills have highly unequal marginal frequencies (e.g., Clinical Experience appears in 135 pathways vs. Urban Planning in 51), and the overlap coefficient is invariant to such asymmetries (Vijaymeena & Kavitha, 2016).

Converting to a distance matrix $D_{ab} = 1 - \text{overlap}(s_a, s_b)$, I apply agglomerative hierar-

chical clustering with average linkage, increasing the number of clusters until no bundle exceeds 15 members. Skills appearing in only one pair (singletons) are removed. This yields interpretable domain-specific bundles that can be named by inspection.

8.2.7 Hub Skill Scoring

Following Stephany and Teutloff (2024)’s concept of skill complementarity, I compute a hub score for each skill that measures how many other skills it meaningfully co-occurs with in job advertisements. The computation operates on the full binary skill matrices $\mathbf{X}^{(o)}$ across all 250 occupations, totalling $N = 732,451$ job postings.

For skills s_a and s_b , I compute pointwise mutual information (PMI), which measures the strength of association between two events relative to independence (Church & Hanks, 1990):

$$\text{PMI}(s_a, s_b) = \log_2 \frac{P(s_a, s_b)}{P(s_a)P(s_b)} = \log_2 \frac{n_{ab} \cdot N}{n_a \cdot n_b} \quad (8.10)$$

where $n_a = \sum_{i=1}^N X_{i,a}$ is the number of job postings containing skill s_a , and $n_{ab} = \sum_{i=1}^N X_{i,a}X_{i,b}$ is their joint occurrence count. PMI is positive when two skills co-occur more often than expected under independence, and negative when they co-occur less often. To avoid noise from rare co-occurrences, I require $n_{ab} \geq 3$ within each occupation and restrict the analysis to skills with $n_a \geq 50$ across the corpus ($|\mathcal{S}_{\text{freq}}| = 3,005$ skills).

The hub score is then the count of skills with which a given skill has meaningful positive association:

$$\text{hub}(s) = |\{s' \in \mathcal{S}_{\text{freq}} : s' \neq s \text{ and } \text{PMI}(s, s') > 0.5\}| \quad (8.11)$$

A higher hub score indicates a skill that productively complements many other skills—a more versatile building block in a worker’s skill portfolio. The threshold of $\text{PMI} > 0.5$ (corresponding to co-occurrence at least $\sqrt{2} \approx 1.41$ times the rate expected under independence) ensures that only substantive positive associations are counted, following conventional practice in the text mining literature (Church & Hanks, 1990).

I assess sensitivity to this choice by recomputing hub scores at $\tau \in \{0.0, 0.5, 1.0, 1.5, 2.0\}$. Higher thresholds reduce absolute hub scores (median falls from 80 at $\tau = 0$ to 53 at $\tau = 2.0$) but preserve the ranking: Spearman rank correlations against the baseline ($\tau = 0.5$) remain $\rho \geq 0.95$ across all thresholds tested ($\rho = 0.998, 1.000, 0.997, 0.984, 0.954$ at $\tau = 0.0, 0.5, 1.0, 1.5, 2.0$ respectively). Hub-score rankings are therefore robust to the threshold choice, and the substantive distinction between high-complementarity generic skills and lower-complementarity specialist skills is preserved across reasonable threshold values.

8.3 Network Characterisation

8.3.1 Global Network Properties

The first question—whether the labour market is navigable in skill space—requires characterising the network’s global topology. Table 8.2 summarises the structural properties at $\tau = 0.6$:

Table 8.2: Skill Proximity Network Properties (2019, $\tau = 0.6$)

Metric	Value
Nodes (clusters)	5,118
Edges	196,236
Density	0.015
Connected components	490
Largest component (nodes)	4,542 (88.7%)
Mean degree	76.7
Median degree	32
Maximum degree	903
Isolated nodes (%)	8.7%
Within-occupation edges	33,172 (16.9%)
Cross-occupation edges	163,064 (83.1%)
Average clustering coefficient	0.619

Note. Source: `analysis/results/chapter8_network/network_metrics.csv`. Network constructed from 5,118 HDBSCAN cluster centroids across 249 occupations, with edges connecting clusters whose cosine similarity exceeds 0.6. Within-occupation edges connect clusters in the same ANZSCO occupation; cross-occupation edges connect clusters across occupational boundaries. Average clustering coefficient computed on a 1,000-node random subsample (uniform without replacement) for computational feasibility.

The network is predominantly cross-occupational: 83.1% of edges connect clusters in different occupations, consistent with Neffke and Henning (2013)’s finding that skills—not occupations—structure the labour market. The largest connected component encompasses 88.7% of all clusters, indicating that the labour market is largely navigable through skill-proximate transitions—workers are not trapped in occupational silos. Frank, Moro, South, et al. (2024) demonstrate that skill similarity between occupations predicts actual worker transition rates, validating the use of skill proximity networks as a map of feasible mobility. The high clustering coefficient (0.605) indicates dense local neighbourhoods, creating coherent skill communities that parallel the skill clusters identified by Alabdulkareem et al. (2018). Community detection using greedy modularity maximisation (Blondel et al., 2008; Newman, 2006) confirms this cross-occupational structure: the 578 detected

communities do not align with ANZSCO major groups but instead cut across traditional occupational boundaries, grouping clusters by shared skill profiles rather than administrative classification.

Figure 8.1 provides a concrete illustration of this structure for two occupations. Each panel shows all clusters within a illustrative occupation (circles) alongside their closest cross-occupation neighbours (squares), with edges coloured by type: solid lines connect clusters within the same occupation, dashed lines connect to clusters in different occupations. Two contrasting patterns are visible. The Accounts Clerk network (panel a) has a dense core where stable (blue) and at-risk (red) clusters are closely interspersed, with cross-occupation links reaching General Clerk, Debt Collector, Management Accountant, and Finance Manager—illustrating that workers in at-risk accounting clusters have multiple transition pathways, both within and across occupational boundaries. The Developer Programmer network (panel b) is more self-contained: the focal clusters form a tightly interconnected core, and the few external links reach closely related technology occupations (Computer Network and Systems Engineer, Software and Applications Programmers nec). This reflects the 94% within-occupation transition rate for Developer Programmers reported in the case studies below.

These aggregate statistics and illustrative examples establish that the network is connected, but they do not reveal whether connectivity is evenly distributed. If the most vulnerable clusters—those most likely to dissolve—are also the least connected, then aggregate navigability overstates the transition prospects for the workers who need them most. The following subsection examines this directly.

8.3.2 Network Connectivity by Cluster Trajectory Status

Figure 8.2 compares network properties across the four trajectory statuses developed in Chapter 7.

The connectivity gradient across trajectory statuses is subtler than one might expect. Mean degree is broadly uniform—persistent, established, recent, and newly emerged clusters all average 76–81 connections—suggesting that the skill profiles of new job types overlap with the broader labour market to a similar degree as established ones. The labour market’s skill infrastructure generally accommodates new cluster configurations rather than isolating them.

The critical distinction lies not in average connectivity but in the tails. While only 3.3% of persistent clusters are completely isolated, the rate rises to 13.8% for newly emerged clusters—a fourfold difference. These isolated clusters, whose skill configurations have no

Figure 8.1: Skill proximity ego networks for two occupations. Circles represent clusters within the illustrative occupation; squares represent cross-occupation neighbours. Blue = stable clusters, red = at-risk, beige = transitional. Solid edges = within-occupation connections; dashed edges = cross-occupation connections. Node size proportional to cluster size. (a) Accounts Clerk: 53 clusters with dense cross-occupation links to administrative and financial occupations. (b) Developer Programmer: 62 clusters forming a self-contained network with few external connections.

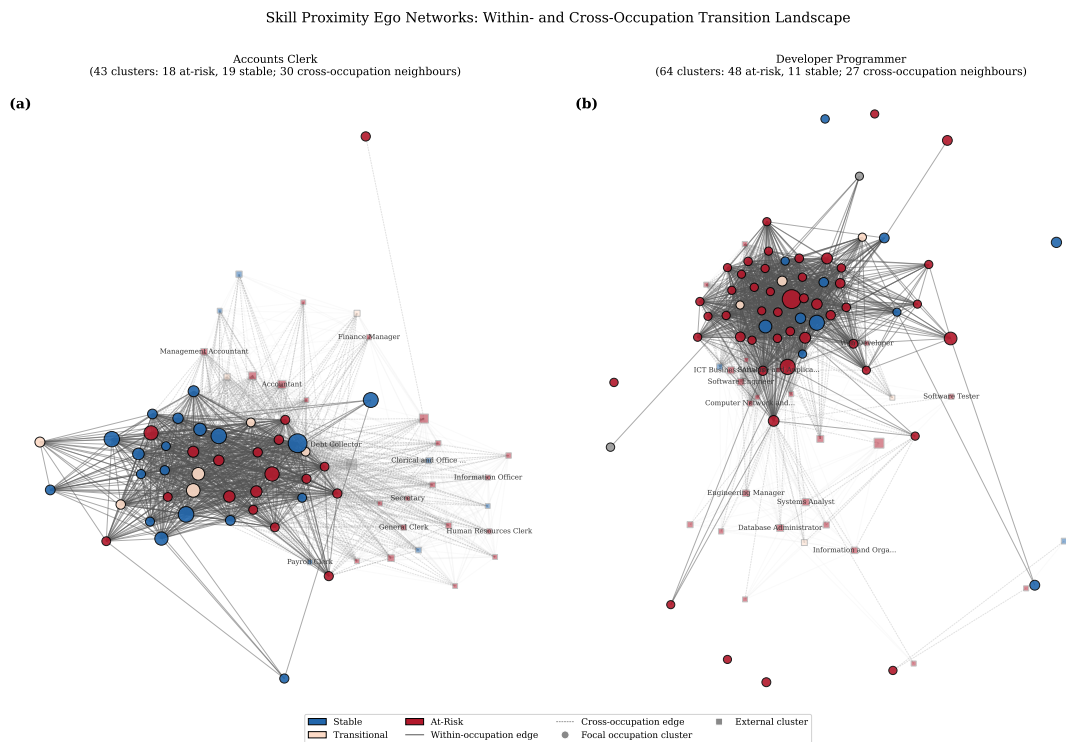
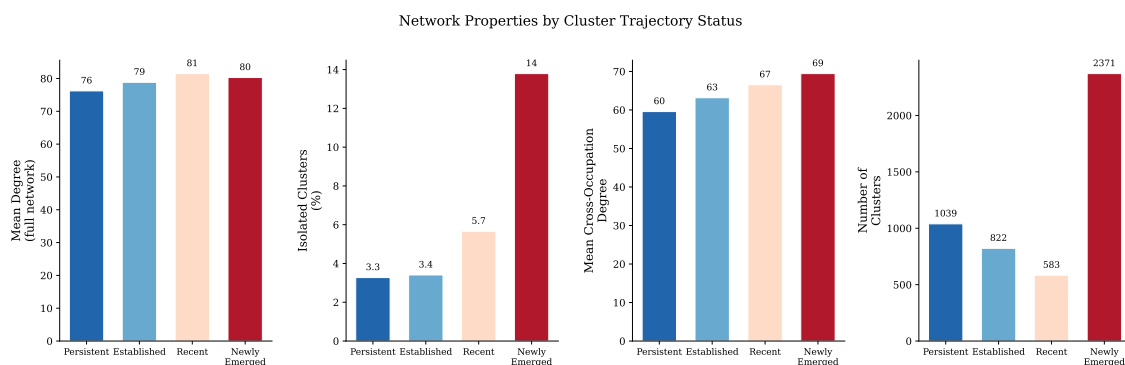


Figure 8.2: Network properties by cluster trajectory status. From left: (1) Mean degree in the full network; (2) Percentage of isolated clusters (degree = 0); (3) Mean cross-occupation degree; (4) Number of clusters. Mean connectivity is broadly uniform across statuses (76–81), but isolation rates differ substantially: 13.8% of newly emerged clusters have no connections vs 3.3% of persistent clusters.

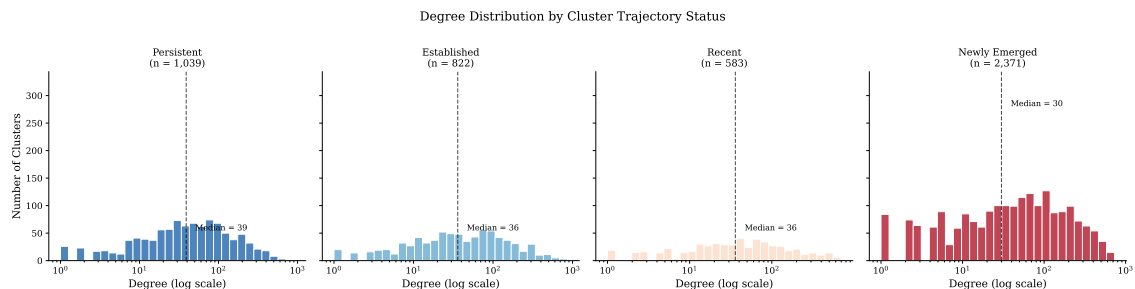


close counterparts anywhere in the network, represent the genuinely stranded job types

where workers face the most constrained transitions. Approximately one in nine at-risk clusters has no skill-proximate alternative at all. This asymmetry in isolation, rather than a gradient in mean connectivity, represents the structural challenge for workforce policy (Hidalgo et al., 2007).

Figure 8.3 shows the full degree distributions by trajectory status, confirming that the distributions overlap substantially while newly emerged clusters have a heavier left tail (more zero-degree nodes).

Figure 8.3: Degree distribution by cluster trajectory status (log scale). Dashed lines indicate median degree. The progressive rightward shift with increasing cluster age indicates systematically higher connectivity for clusters that have demonstrated persistence.



This connectivity gradient motivates the transition pathway analysis that follows. If the most vulnerable clusters are also the least connected, then understanding what specific barriers they face—and which of those barriers are surmountable—becomes a pressing practical question.

8.4 Transition Pathway Analysis

The preceding section established that the network is broadly connected but that connectivity is unevenly distributed: newly emerged and at-risk clusters occupy systematically more peripheral positions. The analysis that follows proceeds in three stages, each refining the picture of what transition barriers actually consist of. First, I identify the most accessible pathways and quantify raw skill gaps (Sections 8.4.1–8.4.1). Second, I decompose these gaps into generic and specialist components, drawing on the human capital literature’s distinction between broadly transferable and domain-specific competencies (Becker, 1994; Gathmann & Schönberg, 2010; Lazear, 2009). Generic competencies such as Communication Skills and Teamwork appear in virtually every occupation; while employers genuinely require them, they are also widely held across the workforce, meaning their presence in a skill gap rarely represents a genuine transition barrier. Specialist skills—those appearing in fewer than 100 of the 252 occupations—are domain-specific competencies that constitute

the real reskilling challenge. Third, I analyse the internal structure of specialist barriers: their co-occurrence in domain-specific bundles (Section 8.4.3) and their complementarity with other skills (Section 8.4.4).

8.4.1 From At-Risk to Stable Clusters

For each of the 3,519 at-risk clusters (those with age ≤ 2 or a significantly declining size trend), I identify the five most skill-similar stable or transitional clusters and compute the specific skill gaps required for each transition. Unlike occupation-level approaches that restrict targets to clusters in “safe” occupations, this analysis considers *all* stable clusters—including those within the same occupation as the source. This produces 16,570 potential transition pathways.

Within-Occupation Transitions

A key finding enabled by cluster-level analysis is that many at-risk clusters can transition to stable clusters *within the same occupation*. Of the 16,570 pathways, 5,276 (31.8%) connect source and target clusters in the same occupation. More strikingly, among rank-1 (best match) pathways, 44.4% of at-risk clusters find their most skill-similar stable alternative within their own occupation. This result is invisible to occupation-level analysis, which necessarily assumes that all clusters within a high-drift occupation are uniformly at risk and that transitions must cross occupational boundaries. The finding validates the thesis’s central argument: restructuring is a property of individual clusters, and the same occupation that contains a disappearing cluster may also contain a stable, skill-proximate alternative. However, knowing that proximate alternatives exist does not reveal what workers must actually acquire to reach them.

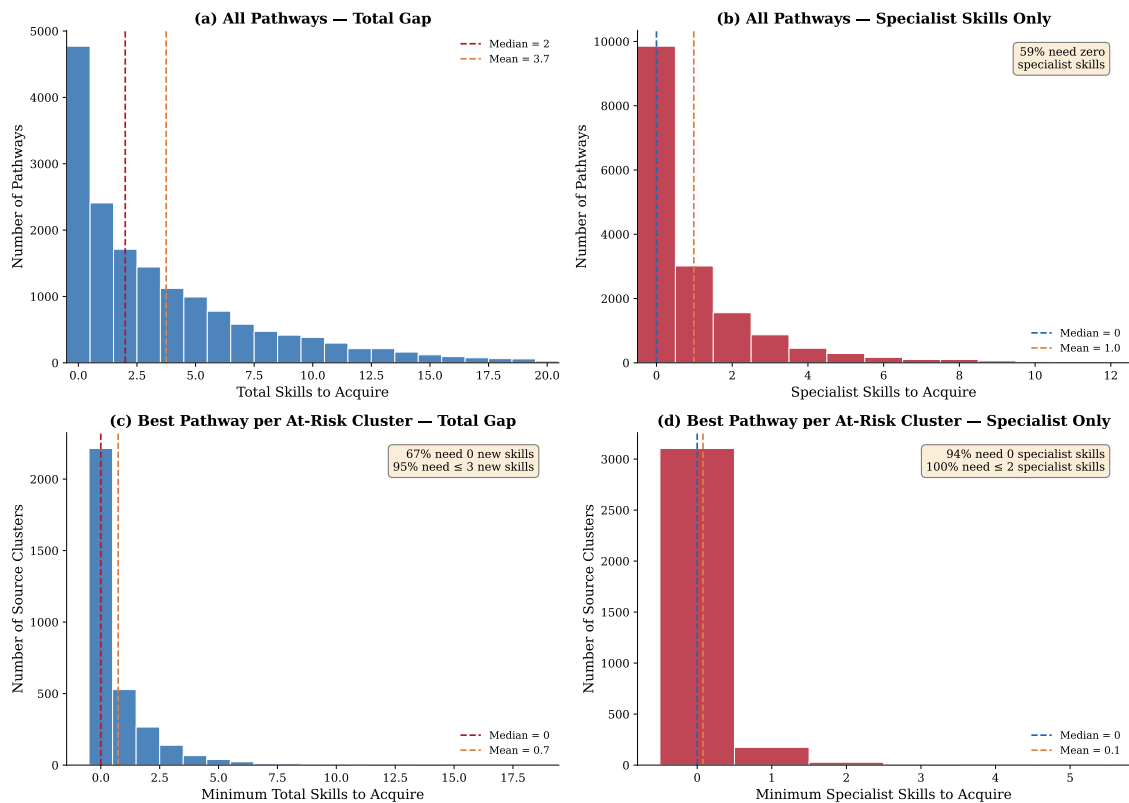
Similarity Distribution

Across all pathways, the mean cosine similarity between source (at-risk) and target (stable) clusters is 0.64, with rank-1 pathways averaging 0.71. This indicates that even the most accessible transitions from at-risk clusters involve meaningful skill overlap but not identity—workers would be moving to genuinely different but related roles.

Skill Gap Distribution

Figure 8.4 presents the distribution of skills workers need to acquire for each transition, applying the generic/specialist decomposition introduced above.

Figure 8.4: Distribution of skill gaps for at-risk → stable cluster transitions, comparing total gaps (left column) with specialist-only gaps (right column). Top row: all 16,570 pathways. Bottom row: minimum gap per source cluster (best available transition). Total gaps average 2.8 skills with a specialist component averaging only 0.6, and 62.4% of pathways require zero specialist skills.



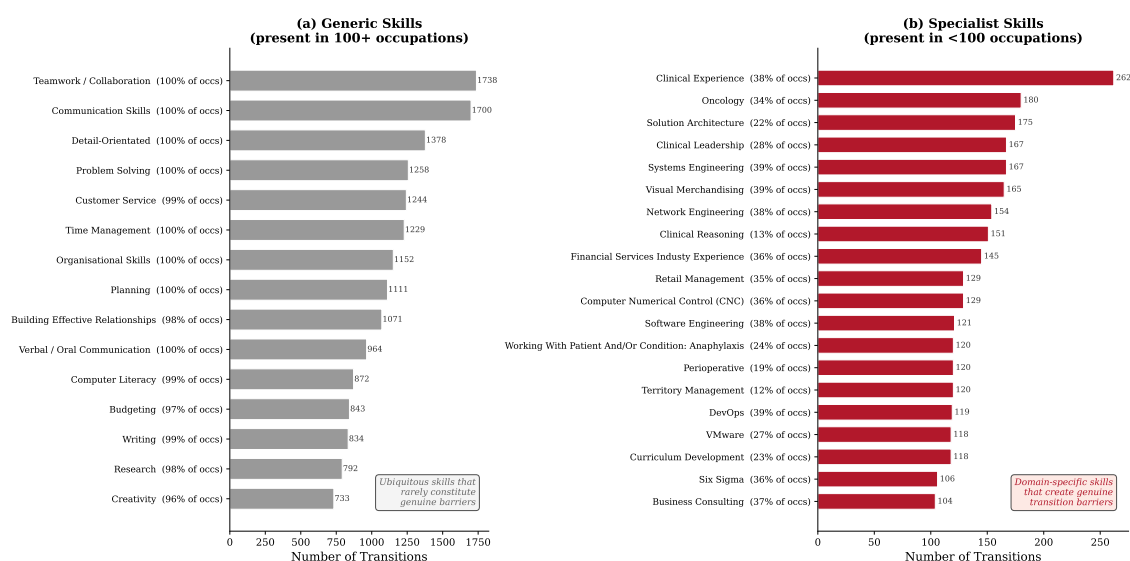
The total gap distribution (panels a, c) shows that most transitions are modest: 95% of at-risk clusters can reach at least one stable alternative by acquiring three or fewer skills. But raw gap counts overstate the true barrier. The specialist gap distribution (panels b, d) reveals that most pathways require no genuine specialist retraining—76% of best-match transitions involve zero specialist skills, with the gap consisting entirely of generic competencies that workers likely already possess. Where genuine specialist barriers do exist, they concentrate in domain-specific technical skills: clinical competencies for health-adjacent transitions, retail management for commercial pathways, or software engineering for technology roles.

The labour market is thus more navigable than raw skill counts suggest. Most transitions require contextual adaptation rather than fundamentally new technical capabilities. But which specific skills constitute the genuine barriers, and do they cluster in predictable patterns?

8.4.2 Bridge Skills: Generic Noise versus Specialist Barriers

Figure 8.5 separates the most commonly needed transition skills into two categories: generic skills that appear in 100 or more occupations, and specialist skills that appear in fewer.

Figure 8.5: Skills most frequently required for at-risk → stable cluster transitions. (a) Generic skills appearing in 100+ occupations—these are ubiquitous in job advertisements and rarely constitute genuine transition barriers. (b) Specialist skills appearing in fewer than 100 occupations—these domain-specific competencies represent the real reskilling challenge.



The raw bridge skill ranking is dominated by generic competencies: Teamwork/Collaboration appears in 1,738 transitions, Communication Skills in 1,700, Detail-Orientated in 1,378, Problem Solving in 1,258, and Planning in 1,111. These broadly transferable competencies (Deming, 2017a; Heckman & Kautz, 2012) are unlikely to represent genuine transition barriers—their presence in the skill gap reflects minor differences in centroid prevalence rather than skills workers must newly acquire.

The specialist bridge skills (panel b) tell a more informative story. Clinical Experience (262 transitions), Oncology (180), Solution Architecture (175), Clinical Leadership (167), Systems Engineering (167), and Visual Merchandising (165) dominate—reflecting that health care, technology, and retail occupations contain many at-risk clusters with transitions requiring domain-specific credentials. Software Engineering (121) appears prominently for software-sector transitions, while Retail Management (129) features in commercial pathways.

Critically, these specialist skills do not appear in isolation—they tend to cluster together in domain-specific bundles. Consistent with Lazear (2009)’s skill-weights framework,

the barrier to transition is not any individual skill but rather the acquisition of a coherent skill bundle. A health care transition requires Clinical Experience *and* Care Planning *and* clinical leadership competencies as an integrated package (Poletaev & Robinson, 2008). The following section formalises this bundling structure.

8.4.3 Skill Bundle Co-occurrence Structure

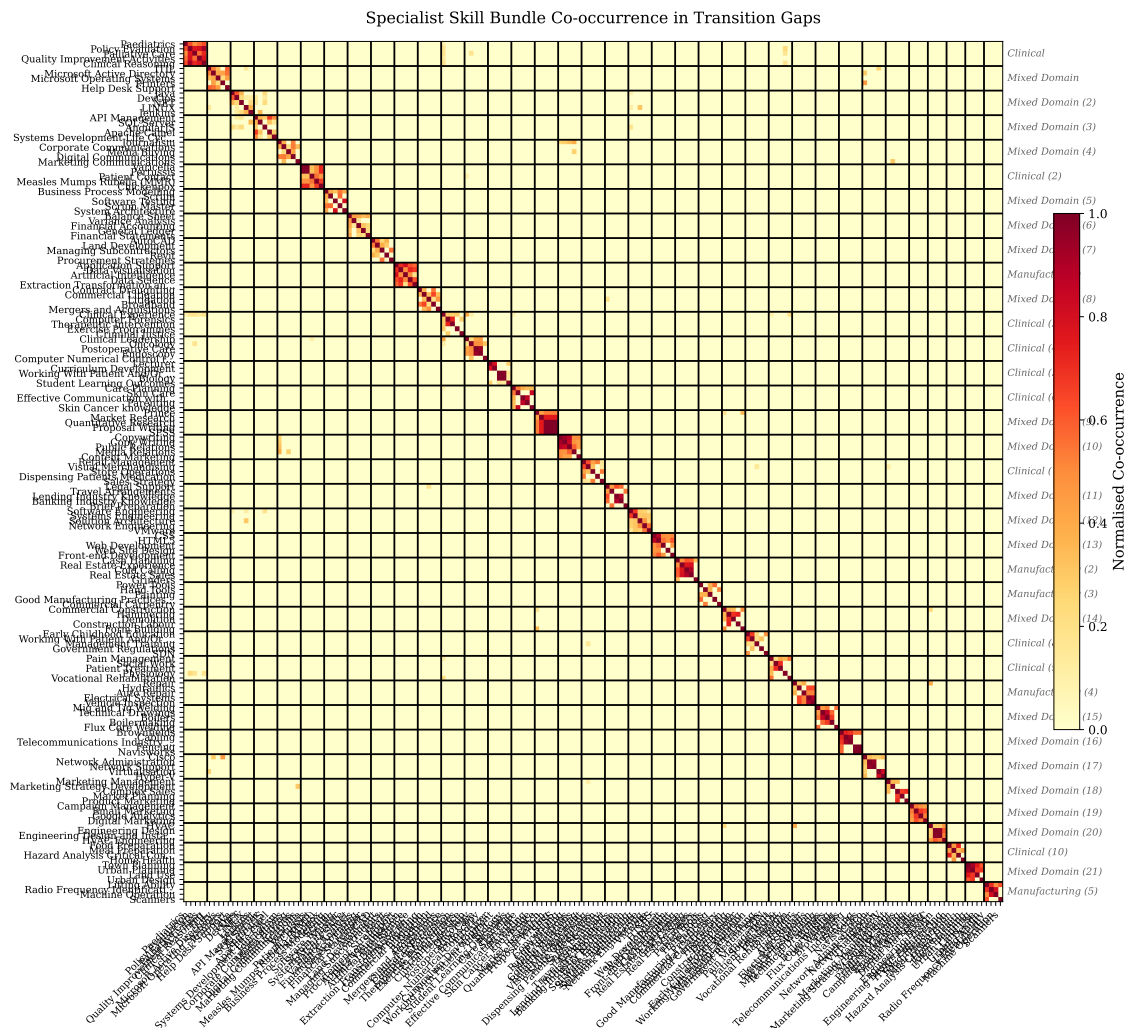
Applying the co-occurrence and hierarchical clustering procedure described in Section 8.2.6 to the 2,451 transition pathways with two or more specialist gap skills identifies 61 bundles, of which 36 are major (four or more skills each).

Figure 8.6 displays the normalised co-occurrence structure (Equation 8.9) of the top five skills from each major bundle, with block-diagonal clusters clearly visible.

The bundle structure reveals several important patterns. First, the healthcare domain fragments into multiple distinct sub-bundles: (i) clinical assessment and care skills (Paediatrics, Palliative Care, Clinical Reasoning, Quality Improvement); (ii) immunisation and patient contact skills (Varicella, Pertussis, Measles Mumps Rubella, Patient Contact); (iii) clinical leadership skills (Clinical Leadership, Oncology, Postoperative Care, Endoscopy); (iv) care planning skills (Care Planning, Skin Care, Parenting); and (v) pain management and allied health (Pain Management, Social Work, Physiology). This internal differentiation is important: a worker facing a “clinical” transition barrier may need one specific sub-bundle rather than a generic clinical credential. Second, technology and IT skills form several distinct bundles: IT infrastructure (ITIL, Active Directory, Microsoft Operating Systems), software development (Java, DevOps, .NET, Jenkins), and web development (CSS, HTML5, Web Development). Third, manufacturing and trades skills form coherent bundles: construction trades (Power Tools, Hand Tools, Painting, Commercial Carpentry), fabrication (Mig/Tig Welding, Technical Drawings, Boilermaking), and vehicle maintenance (Repair, Hydraulics, Auto Repair, Electrical Systems). Additional bundles emerge in financial accounting (Balance Sheet, Variance Analysis, Financial Statements), marketing and communications (Journalism, Corporate Communications, Media Buying), urban planning (Town Planning, Urban Planning, Land Use), and food safety (Food Preparation, HACCP, Meal Preparation).

This bundle structure has direct implications for training programme design. Consistent with the skill-weights framework (Lazear, 2009), the barrier to career transition is not the acquisition of isolated competencies but the coordinated development of skill packages. A worker transitioning from an at-risk cluster in aged care to a clinical role needs not just “Clinical Experience” but the integrated assessment bundle—Care Planning, Quality Improvement, Clinical Reasoning—as a coherent package. Training programmes that

Figure 8.6: Normalised co-occurrence of specialist skills in transition gaps. Dark cells indicate skills that frequently appear together as joint requirements. Block-diagonal structure reveals coherent domain-specific bundles: clinical assessment (Care Planning, Audiometry, Wound Management), clinical leadership (Anaesthesiology, Nurse Management), manufacturing trades (Hand Tools, Boilermaking, Welding), allied health (Speech Pathology, Paediatrics), education (Biology, Physics), and others.



address individual skills in isolation miss this bundling structure and are unlikely to produce the integrated competency that employers demand. Given that specialist barriers cluster in domain-specific bundles, a final question arises: when multiple training pathways exist, which skills within these bundles offer the greatest return on investment?

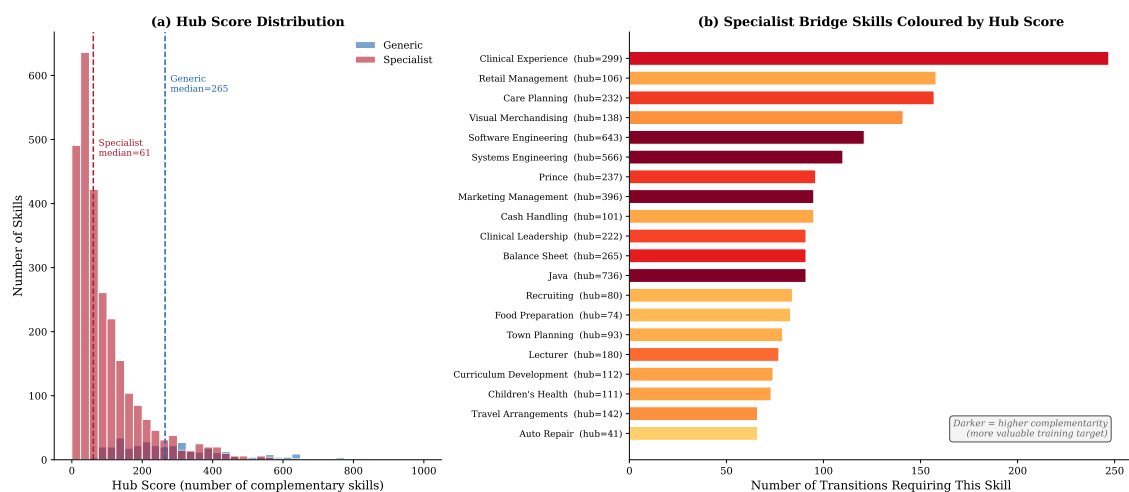
8.4.4 Skill Complementarity and Hub Scores

Not all specialist skills are equally valuable as training targets. Applying the PMI-based hub scoring procedure described in Section 8.2.7 (Equations 8.10–8.11) across all 732,451

job postings yields complementarity scores for 3,005 skills with sufficient frequency.

Figure 8.7 presents the results in two panels.

Figure 8.7: (a) Distribution of hub scores (number of complementary skills) for generic skills (blue, median = 265) versus specialist skills (red, median = 63). Generic skills have over four times higher complementarity, consistent with their role as broadly transferable connectors (Deming, 2017a). (b) Top 20 specialist bridge skills coloured by hub score: darker colours indicate higher complementarity and thus higher training value.



Panel (a) confirms the theoretical distinction between generic and specialist skills in complementarity terms. Generic skills have a median hub score of 265, compared to 63 for specialist skills—a ratio of approximately 4:1. This is consistent with Deming (2017a)'s finding that social skills command wage premia precisely because they complement both cognitive and manual tasks, and with Gathmann and Schönberg (2010)'s evidence that broadly-demanded competencies transfer readily across occupational boundaries.

Panel (b) reveals heterogeneity within the specialist bridge skills. Clinical Experience (hub score 298), Care Planning (229), Physics (223), Clinical Leadership (222), and Paediatrics (209) have the highest complementarity among the specialist bridge skills—they can be productively combined with many other clinical and scientific competencies. By contrast, Early Childhood Education (hub score 71), Biology (69), and Lifting Ability (67) are more isolated competencies with fewer productive complements.

The practical implication is that training programmes should prioritise high-hub-score specialist skills when multiple transition pathways are available. A worker who acquires Clinical Experience (hub score 299) gains access to a broader range of career pathways than one who acquires a low-complementarity skill, because the high-hub skill can be productively combined with many existing competencies. This prioritisation logic extends Stephany and Teutloff (2024)'s theoretical framework from wage premia to career navigability: hub skills are valuable not only because they command higher wages but because

they expand the set of feasible transitions in the skill proximity network.

8.4.5 Case Study Pathways

Figure 8.8 presents detailed transition analyses for five occupations. I selected these from the 204 occupations (81.9%) containing both at-risk and stable clusters, using three criteria: (1) sectoral diversity—professional services (Accountant), healthcare (Registered Nurse), trades (Chef), technology (Developer Programmer), and retail (Sales Assistant); (2) sufficient cluster count for meaningful transition analysis (≥ 20 at-risk clusters); and (3) variation in the proportion of within-occupation versus cross-occupation best-match transitions. Together they illustrate the within-occupation heterogeneity that is central to the thesis's argument.

Accountant²: 74 at-risk clusters alongside 66 stable clusters within the combined accounting occupations. Of the 74 at-risk clusters, 64 (86%) find their best match within other Accountant clusters—stable specialisations that share the same foundational skill vocabulary. This is the pattern predicted by the theoretical framework: restructuring creates new cluster configurations, but the occupation's core skill infrastructure provides natural transition targets. The remaining 10 clusters connect to Organisation & Methods Analyst (4 clusters), Finance Manager, and other professional services occupations with small specialist gaps.

Registered Nurse³: 124 at-risk clusters alongside 26 stable clusters. Despite the large number of at-risk clusters, 65 (52%) find their best match within other nursing clusters—reflecting the diversity of nursing specialisations (emergency, aged care, paediatrics, community health) that share a common clinical foundation. Cross-occupation transitions lead predominantly to Enrolled Nurse (18 clusters), with additional pathways to Midwife and cross-variant Registered Nurse roles, typically requiring 0–4 additional clinical skills.

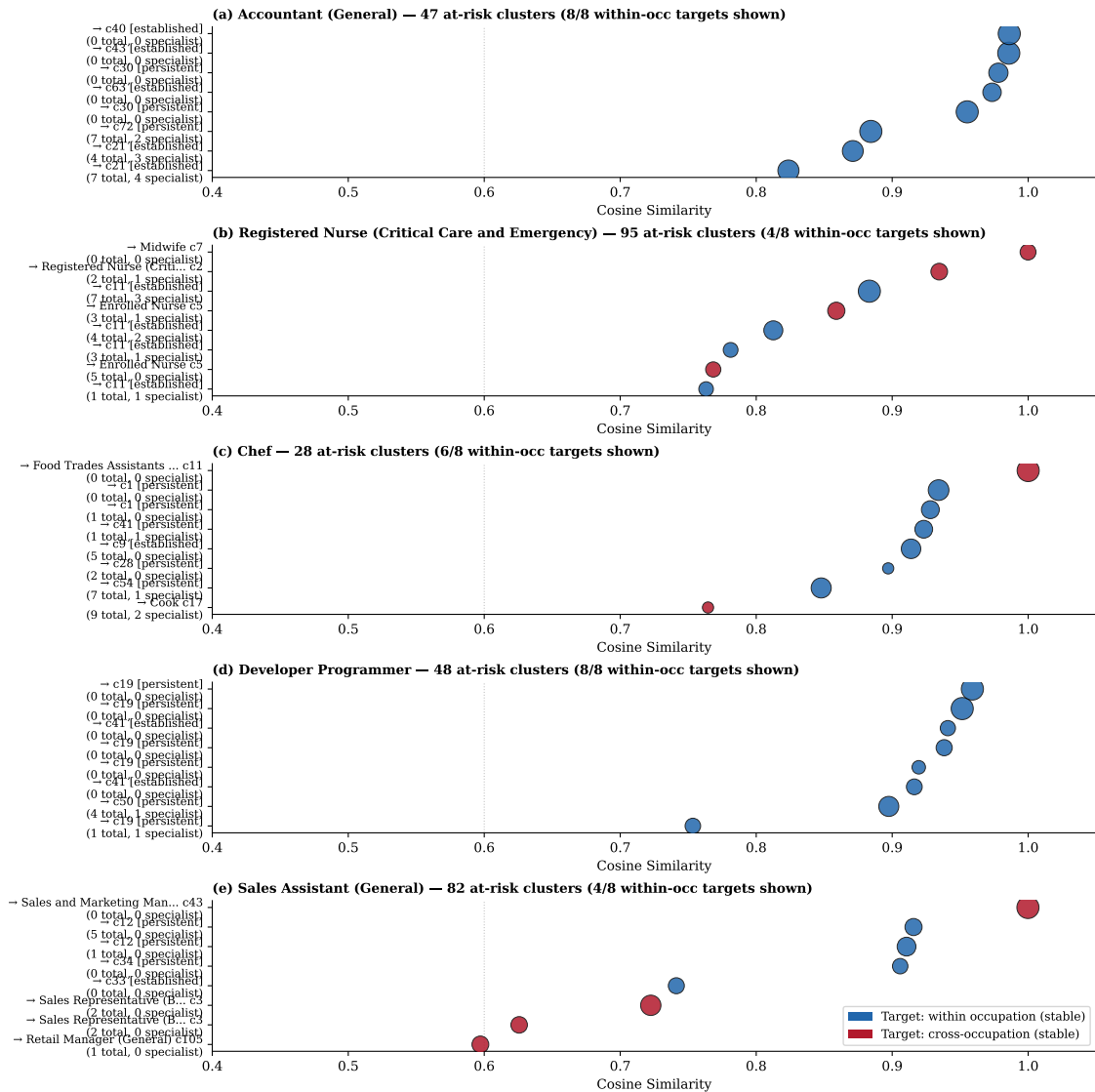
Chef: 28 at-risk clusters alongside 30 stable clusters. Within-occupation transitions dominate (68% of rank-1 matches), with at-risk clusters connecting to established clusters in diverse culinary specialisations. Cross-occupation pathways lead primarily to Food Trades Assistants (7 clusters) and Cook (2 clusters), with Meal Preparation as the primary specialist gap where one exists.

Developer Programmer: 48 at-risk clusters alongside 11 stable clusters. This occupation shows the strongest within-occupation transition pattern: 94% of rank-1 matches connect to other Developer Programmer clusters. The rapid emergence and disappearance

²Combines Accountant (General), Management Accountant, and Taxation Accountant ANZSCO codes.

³Combines Registered Nurses nec, Critical Care and Emergency, Mental Health, and Surgical ANZSCO codes.

Figure 8.8: Best available transition pathway for at-risk clusters in five occupations. Each point represents a source cluster's best match (rank-1 pathway) to a stable target cluster. Blue points indicate within-occupation targets; red points indicate cross-occupation targets. Point size is proportional to source cluster size. Higher cosine similarity (rightward) indicates closer skill match; labels show the target occupation with total and specialist skill gaps.



of technology-specific clusters (e.g., new framework specialisations) occurs against a backdrop of stable, broadly-defined software engineering clusters. The few cross-occupation transitions lead to Web Developer, ICT Business Analyst, and Software and Applications Programmers nec roles.

Sales Assistant⁴: 92 at-risk clusters alongside 26 stable clusters. Within-occupation matches account for 59% of rank-1 pathways, with at-risk clusters connecting to established

⁴Combines Sales Assistant (General), Pharmacy Sales Assistant, and Sales Assistants and Salespersons nec.

retail specialisations in different product domains. Cross-occupation transitions reach Information Officer (7 clusters), Other Sales Support Worker (5), and Retail Manager (4), typically with zero specialist skill gaps—the transitions involve generic commercial competencies that workers already possess.

8.5 Discussion

8.5.1 Navigability and Its Limits

The central finding of this chapter is that the Australian labour market, viewed through the lens of skill proximity, is predominantly navigable—but navigability is unevenly distributed in a way that creates a policy paradox. This challenges narratives of automation creating “stranded” workers trapped in obsolete skills (Susskind, 2020): while individual clusters may dissolve, skill-proximate alternatives typically exist. The finding extends Neffke et al. (2017)’s observation that inter-industry labour flows form a connected network from the occupation level to the sub-occupational cluster level.

The most important dimension of this navigability is that it operates *within* occupations. Nearly half of at-risk clusters find their best transition target among other clusters in the same occupation—a result invisible to occupation-level analysis that classifies entire occupations as “at risk” or “safe.” This within-occupation accessibility arises because most occupations simultaneously contain both at-risk and stable clusters, meaning that workers displaced by cluster dissolution can often transition to a different specialisation without changing occupations at all.

Yet the analysis also reveals a structural asymmetry: the clusters most likely to need transitions—newly emerged and declining clusters—are systematically the least connected in the skill proximity network (Section 8.3). This echoes the structural constraints identified in the product space literature, where countries at the periphery of the relatedness network face the highest barriers to diversification (Hidalgo et al., 2007), and is consistent with Frank, Moro, South, et al. (2024)’s finding that workers in peripheral network positions face the most constrained mobility options. The result is a policy paradox: the clusters most in need of transition support are the least connected to transition opportunities.

8.5.2 From Occupation-Level to Cluster-Level Workforce Development

The transition pathway analysis provides concrete content for cluster-based workforce development. Three design principles emerge that distinguish this approach from conventional

occupation-level interventions:

1. **Position workers in the skill network, not in occupational categories.** By matching a worker’s skill profile against the centroid library, the network identifies accessible alternatives—including stable clusters within the same occupation—that occupation-level assessments would miss entirely. This reframes workforce development from “what occupation should this worker retrain for?” to “which skill-proximate clusters offer stable demand?”
2. **Target skill bundles, not individual skills.** The co-occurrence analysis (Section 8.4.3) reveals that specialist barriers cluster in domain-specific packages. Consistent with the skill-weights framework (Lazear, 2009), training programmes should address these bundles as integrated packages rather than isolated skill deficits. A clinical transition requires coordinated acquisition of Care Planning, Quality Improvement, and Clinical Reasoning—not a generic “healthcare credential.”
3. **Prioritise high-complementarity skills.** When multiple training pathways exist, hub scores (Section 8.4.4) provide a principled basis for prioritisation. Skills with high complementarity open more subsequent career pathways per unit of training investment (Stephany & Teutloff, 2024), transforming a single training intervention into a platform for future mobility.

This approach offers a complementary framework for workforce development, supplementing occupation-level interventions with skill-level navigation tools—consistent with Athey (2018)’s framing of workforce policy as a prediction problem amenable to data-driven targeting. Whether such tools can be operationalised at policy scale depends on additional factors the analysis cannot observe: institutional pathways, credentialing requirements, geographic constraints, and employer hiring preferences, as flagged in the Introduction.

8.5.3 Limitations

Several limitations warrant acknowledgment. First, the network captures skill *demand* proximity (what employers advertise) rather than skill *supply* proximity (what workers can do). Workers may possess skills not listed in their current cluster’s advertisements, broadening their actual transition options beyond what the network reveals. Conversely, possessing the listed skills does not guarantee employability—experience, credentials, and employer preferences matter (Poletaev & Robinson, 2008). Second, the analysis uses 2019 data; the skill landscape has likely shifted since, particularly with the acceleration of

digital transformation during COVID-19 and the emergence of generative AI. Third, the 0.6 cosine similarity threshold is necessarily arbitrary; different thresholds would produce different network topologies (Table 8.1), though the qualitative findings—persistent clusters most connected, newly emerged most isolated—are robust across thresholds. Fourth, the trajectory classification relies on a fixed matching threshold (cosine similarity > 0.80) for backward lineage tracing; a different threshold would reclassify some clusters between trajectory categories, though the overall pattern of within-occupation heterogeneity is robust.

8.6 Conclusion

This chapter demonstrated that the cluster infrastructure developed throughout the thesis has a practical application: mapping feasible career transitions for workers in dissolving job types. The Australian labour market in 2019 was broadly navigable in skill space, with most at-risk clusters connected to stable alternatives requiring modest upskilling. But navigability was unevenly distributed—the most vulnerable clusters were the least connected—a structural asymmetry that workforce-development frameworks would need to address. The progressive decomposition of transition barriers reveals that the real obstacles are not isolated skill deficits but coordinated packages of domain-specific competencies, providing demand-side evidence for cluster-based rather than occupation-based approaches to workforce development.

Read in historical perspective, this chapter establishes the skill-proximity-network baseline for Australian labour markets at the close of the second wave of automation—a record of how clusters were positioned in skill space immediately before the disruptions of COVID-19 and large language models. The network topology, the connectivity gradients across trajectory statuses, the bridge skills, and the hub-score rankings all reflect the demand-side skill landscape of the pre-LLM era, providing reference points against which the network signature of subsequent waves can later be measured. The findings also demonstrate that the cluster infrastructure—built across Chapters 6–7 for the substantive measurement contribution—supports applied analysis without requiring further theoretical development, an asset for future research using comparable cluster decompositions on more recent data.

Chapter 9

Discussion

9.1 Introduction

The apparent contradiction between automation predictions and empirical findings dissolves once analysis operates at multiple levels simultaneously. Occupation-level frameworks detect minimal effects because most variation concentrates within occupational boundaries; cluster-level analysis reveals the restructuring that aggregation obscures. This chapter integrates findings across the empirical chapters, develops theoretical and methodological contributions, critically examines limitations, and explores implications for policy.

9.2 Integration of Findings Across Analytical Levels

The thesis examined automation's labour market impacts at two distinct analytical levels—occupation and cluster—each revealing different aspects of the restructuring process. This multi-level analysis resolves the paradox between automation predictions and modest empirical findings by demonstrating that the phenomenon operates differently at different scales.

9.2.1 The Occupation-Level Paradox

Chapter 5 established that occupation-level automation frameworks—both the RCM taxonomy and Frey-Osborne risk scores—explain almost none of the variation in Australian labour demand. The frameworks produce statistically significant but economically negligible coefficients ($R^2 = 0.030$ in the primary specification), and theoretically distinct risk categories prove statistically indistinguishable (Wald tests in Tables 5.12 and 5.15). Variance decomposition revealed the source of this failure: 94.4% of variation operates within occupations rather than between them ($ICC = 0.056$; Table 5.17), imposing a hard ceiling on what any occupation-level framework can detect.

These null findings are not methodological failures. They validate the theoretical critique that occupation-level classifications assume homogeneity within occupational categories, masking restructuring at finer levels of analysis. The paradox is not that automation predictions are wrong, but that standard measurement operates at the wrong granularity.

9.2.2 Within-Occupation Restructuring

Chapters 6 and 7 resolved this paradox by shifting measurement to the cluster level. HDBSCAN clustering identified discrete job types within occupations, with cluster counts scaling near-proportionally with market thickness (elasticity 0.91 across 250 occupations; Table 6.6), and temporal tracking revealed that these clusters undergo continuous creative destruction—with cluster-level volatility exceeding occupation-level volatility by an order of magnitude (median ratio approximately 10:1; Table 6.7) across 97% of occupations tested. Cluster churn correlates strongly with the speed of skill change rather than the static level of automation exposure (Spearman $\rho = +0.558$ between centroid drift and churn across 250 occupations; Section 7.4). The technology succession pattern identified in Chapter 6—legacy platform clusters dissolving as emerging technology clusters form—offers one plausible mechanism consistent with this restructuring. Critically, this restructuring is a property of individual clusters, not entire occupations: 204 of 249 occupations (81.9%) contain both at-risk and stable clusters simultaneously, demonstrating that vulnerability concentrates within rather than between occupational categories.

The demand-side consequences of this restructuring are heterogeneous. Task recombination and task unbundling coexist across the labour market, with the pace of occupational change moderating the balance. Slowly-evolving occupations tend to generate enriched new roles through skill recombination; rapidly-transforming occupations tend to generate simplified new roles through task decomposition. This heterogeneity—rather than either pattern alone—is a central finding.

Why does recombination dominate (59%) over unbundling (22%)? Three mechanisms plausibly explain the asymmetry, each consistent with the structural correlates documented in Chapter 7. First, *technological complementarity*: contemporary automation technologies—cloud platforms, analytics tools, AI assistants—are general-purpose and combinable rather than substitutes for narrow tasks. As Autor et al. (2024a) document, new work tends to emerge through the combination of existing skills with new technical capabilities, producing roles that require integrating multiple competencies rather than executing a narrower set. Second, *market thickness*: recombination requires a diverse skill pool from which new combinations can be assembled. The structural finding that recombination occupations are systematically larger (median 1,518 jobs vs. 654 for unbundling occupations), more cluster-rich (15.3 vs. 7.1), and more stable in skill composition is consistent with Moretti (2012)’s account of thick-market specialisation: only thick markets sustain the complementary skill supply that recombination requires. Third, *employer signalling*: when occupations are stable enough that workers can credibly accumulate diverse skills, employers in competitive labour markets respond by expanding skill lists in job advertisements to screen for higher-quality candidates (Spence, 1973). The combination of stable occupational structure, abundant skill supply, and competitive screening pressure pushes new clusters toward broader rather than narrower skill profiles. Unbundling, by contrast, characterises occupations where rapid change has not yet allowed the labour market to converge on a stable skill portfolio—fragmenting existing roles into narrower components is the path of least resistance when complementary skills are scarce. The drift-moderation pattern (Spearman $\rho = -0.263$) captures this directly: stability creates the conditions for recombination; volatility forces unbundling.

9.2.3 Cross-Level Synthesis

The integration across analytical levels yields three insights that no single chapter could produce alone. First, the level of analysis fundamentally shapes conclusions about automation: occupation-level analysis suggests minimal effects while cluster-level analysis reveals substantial restructuring within the same data. Second, aggregation bias operates systematically—offsetting expansions and contractions within occupational boundaries produce the appearance of stability where extensive restructuring is underway. Third, the restructuring process generates measurable demand-side consequences whose direction varies predictably across the occupational landscape.

9.3 Theoretical Contributions

This multi-level research design generates four core theoretical contributions that advance automation and labour market research. The analysis is grounded in task-based economics (Acemoglu & Restrepo, 2022; Autor et al., 2024a) but draws on institutional economics (P. A. Hall & Soskice, 2001), labour process theory (Braverman, 1974; Edwards, 1979), and organisational learning perspectives (J. S. Brown & Duguid, 1991) where these frameworks illuminate mechanisms that purely economic models leave underspecified.

9.3.1 Restructuring Theory of Automation Impacts

The thesis advances a *restructuring theory of automation impacts*: the empirical patterns documented here are consistent with automation's primary labour market effect in contemporary service-oriented economies operating through continuous task reorganisation within occupations rather than wholesale occupation elimination. This thesis provides the first descriptive demonstration of this restructuring pattern in the Australian context, with patterns consistent with theoretical mechanisms others have proposed.

First, the research provides direct empirical evidence for aggregation bias in automation measurement. While scholars have theorised that occupation-level measures may miss task-level restructuring (Acemoglu & Restrepo, 2022), this thesis quantifies the magnitude. A single occupation can contain dozens to hundreds of discrete job types, each with distinct growth trajectories—some expanding rapidly while others contract—yet these offsetting dynamics aggregate to near-zero occupation-level effects. The contrast between extreme cluster-level volatility and negligible occupation-level explanatory power provides the empirical signature of aggregation bias that prior research could only hypothesise.

This finding intervenes in longstanding debates about whether automation produces skill upgrading, polarization, or degradation of work. Autor, Levy, and Murnane (2003) argued for "skill-biased technical change" (SBTC) where automation consistently upgrades skill requirements by eliminating routine tasks while expanding non-routine cognitive work. Goos and Manning (2007) challenged this with evidence of job polarization: simultaneous growth in high-skill and low-skill jobs with hollowing of middle-skill routine work. Braverman (1974) predicted skill degradation: systematic deskilling as management decomposes craft knowledge into routinized fragments. Fernández-Macías (2012b) questioned both frameworks, finding limited polarization in European data and arguing that task-based predictions overstate technological determinism. Critically, Vallas (1990) and Attewell (1990) demonstrate that "skill" itself is a socially constructed category shaped by power relations, occupational boundaries, and managerial ideology rather than objective technical

attributes—meaning these debates partially reflect different definitional approaches to measuring inherently contested concepts.

The findings from this thesis suggest these frameworks represent *different analytical perspectives on the same multi-level phenomenon* rather than competing empirical claims. Walkowiak (2023) provides theoretical support for this interpretation: when tasks exhibit complementarities and interdependencies, automating individual tasks creates non-linear effects that cannot be predicted from task-level analysis alone. Task bundles—not isolated tasks—constitute the appropriate unit of analysis because automation’s impact on one task propagates through complementarities to affect demand for related tasks. At cluster level, all three patterns coexist simultaneously: some clusters exhibit skill upgrading (strategic advisory accounting requiring advanced analytical capabilities), others show polarization (compliance roles becoming more routine while strategic roles expand), and others demonstrate degradation (entry-level tasks automated while developmental pathways collapse). The “paradox” dissolves when we recognise that upgrading, polarization, and degradation occur concurrently across different task bundles within the same occupation. Occupation-level analysis detecting minimal net effects does not refute automation’s transformative impact—it aggregates over precisely the dimension where heterogeneous restructuring occurs. This reconciles apparently contradictory empirical findings: Autor, Levy, and Murnane (2003) document skill upgrading at macro level, Goos and Manning (2007) find polarization in employment distributions, F. Green (2006) reports work intensification without skill upgrading, and Handel (2003) questions whether task changes translate to skill changes—all may be simultaneously correct when analysed at appropriate levels of aggregation capturing the coexistence of divergent cluster trajectories.

The restructuring theory is consistent with three empirical regularities that occupation-level frameworks struggle to accommodate. First, it offers an interpretation of why automation predictions consistently overestimate job losses: predictions assume occupational displacement, but observed impacts appear concentrated in within-occupation task reorganisation that leaves aggregate employment largely unchanged. Second, it accounts for why workers and executives perceive extensive automation despite modest macro-level effects: they observe cluster-level restructuring invisible to occupation-level statistics. Third, it provides a framework for understanding why automation research produces contradictory findings across different measurement approaches: studies using different analytical levels capture different aspects of the same multi-level phenomenon.

Second, the analysis documents the specific mechanism through which automation reshapes work without creating new occupations. The finding that a non-trivial share of jobs in each occupation are “noise”—hybrid or idiosyncratic roles defying stable classification—provides empirical support for Autor et al. (2024a)’s argument about slowed new work

creation. Work reorganisation occurs through cluster reconfiguration within existing occupational boundaries rather than new occupational category emergence. This contrasts with historical automation episodes (e.g., computerization creating "computer programmer" as a new occupation) and suggests contemporary automation operates differently than earlier technological waves.

The absence of new occupational categories despite extensive restructuring suggests that institutional classification systems (ANZSCO, O*NET) lag behind actual labour market dynamics by years or decades. Official statistics update occupational taxonomies infrequently (ANZSCO last major revision: 2013), yet cluster analysis reveals continuous emergence and disappearance, with median annual churn exceeding 100% across the occupational distribution. This temporal mismatch between classification update cycles and actual restructuring speeds means policy interventions based on official statistics systematically respond to outdated labour market conditions.

Moreover, the high cluster churn rates suggest cluster proliferation may reflect endogenous skill formation operating below formal strategic planning (J. S. Brown & Duguid, 1991; Lave & Wenger, 1991). J. S. Brown and Duguid (1991) theorise that innovation emerges from informal worker networks—what organisational theorists term “communities of practice”—where workers collectively develop tools, techniques, and specialised expertise that gradually transform job content independent of management directives. In economics terms, this represents decentralised human capital investment: when workers automate repetitive tasks, document processes, and develop expertise in emerging technologies, they create new task bundles that employers subsequently formalise through job advertisements. This bottom-up process generates cluster proliferation organically rather than through coordinated occupational redesign, explaining why clusters exhibit high churn rates while formal occupational categories remain static. The noise proportion in each occupation—hybrid or idiosyncratic roles—may partially reflect transitional states where novel task combinations emerge before employers stabilise them into recognisable job types.

Third, the research documents offsetting dynamics consistent with the absence of mass joblessness despite substantial task displacement. Within Accounting, some clusters expand dramatically while others contract sharply in the same time period, producing near-zero net effects. This provides Australian empirical patterns consistent with the offsetting mechanisms Acemoglu and Restrepo (2022) theorised but could not directly observe: task displacement in some clusters appearing alongside task reinstatement in others, with net effects determined by the balance between these forces (Acemoglu & Restrepo, 2019c; Acemoglu et al., 2020; Arntz et al., 2020).

Three channels are theoretically plausible for the offsetting pattern. First, technology

succession appears to be associated with cluster turnover: Chapter 6 (Table 6.8) finds that dead clusters are significantly more software-intensive than born clusters ($p < 0.001$), with the specific skills identifying legacy platform dependencies (Lotus Notes, ASP.NET, Silverlight) being replaced by emerging technologies (AWS, Docker, Python, Salesforce). Clusters organised around obsolescent technology stacks dissolve while new clusters crystallise around replacement technologies, even when the underlying tasks persist. Second, productivity gains from automation in declining clusters may free resources that firms redeploy to expanding clusters requiring different skill combinations. Third, market growth in expanding sectors plausibly generates new clusters with novel task combinations, while market contraction in declining sectors suppresses cluster creation. The technology succession channel appears more consistent with platform lifecycle dynamics than with Autor, Levy, and Murnane (2003)'s routine/non-routine substitution mechanism: the pooled growth-on-software-intensity correlation is weak ($\rho = -0.013$) because software skills simultaneously mark both legacy-dependent clusters (which decline) and technology-intensive emerging clusters (which grow), producing offsetting effects.

Notably, across 250 occupations, churn rates do not correlate with static automation exposure measures. The illustrative occupations demonstrate this vividly: Human Resource Adviser exhibits the highest churn rate despite having the lowest Frey-Osborne automation probability, while Receptionist (General) shows lower churn despite the highest automation probability. This apparent paradox dissolves when distinguishing automation *level* (ρ) from automation *speed* ($d\rho/dt$): Proposition 5 predicts that churn correlates with the rate of technological change, not the current level of automation exposure. Full-scale analysis across 250 occupations confirms this interpretation: centroid drift correlates positively with cluster churn, with Human Resource Adviser's rapid transformation from administrative to strategic advisory functions generating the highest centroid drift among the illustrative occupations, while Receptionist (General) shows low drift reflecting stable but elevated automation exposure. Occupations currently undergoing rapid transformation exhibit high churn regardless of static automation risk, while occupations that have already adjusted to technology show lower churn despite high automation exposure.

9.3.2 Demonstration of Multi-Level Measurement Value

The thesis demonstrates the empirical value of multi-level analysis for automation research, providing methodological tools to implement the task-level measurement that scholars have long advocated (Acemoglu & Autor, 2011b; Autor, 2013a) but lacked scalable methods to execute. Rather than resolving a theoretical debate, the research validates arguments

for finer-grained measurement through novel empirical approaches. Both occupation-level and cluster-level analysis are necessary for understanding automation's impacts, revealing complementary rather than competing aspects of the same phenomenon.

Occupation-level analysis establishes modest aggregate effects and rules out mass technological unemployment scenarios. It provides the macroeconomic context essential for interpreting finer-grained findings. The null results from Chapter 5 are not failures of measurement but essential baseline findings that validate the search for within-occupation mechanisms. Without establishing that aggregate effects are modest, cluster-level findings might be dismissed as measurement noise or sampling error. The occupation-level analysis anchors the thesis by demonstrating that automation does *not* produce the wholesale occupation elimination that simplistic predictions suggest.

Cluster-level analysis exposes substantial restructuring masked by occupation-level aggregation. It documents patterns consistent with the mechanisms through which automation reshapes work: task reorganisation, skill demand shifts, and labour reallocation occurring within occupational boundaries. The dramatic heterogeneity documented in Chapter 6—86 job types within a single occupation, individual clusters exhibiting +388% growth or sharp decline—explains why workers perceive extensive workplace change despite modest macro-level effects. Cluster analysis provides the analytical resolution needed to detect restructuring operating below occupational aggregation levels.

Each level provides essential but partial perspectives on a multi-level phenomenon: cluster-level resolution detects the restructuring that aggregation conceals, while occupation-level analysis anchors the interpretation by demonstrating that aggregate effects remain modest.

9.3.3 Validation of Job Advertisement Data for Labour Market Analysis

The thesis demonstrates the value of job advertisement data for studying labour market restructuring, validating an emerging but still-contested data source (Deming & Kahn, 2018b; Hershbein & Kahn, 2018b; Marinescu & Wolthoff, 2020). While most automation research uses census data (which measure employment stocks), patent counts (which proxy innovation), or robot installations (which capture specific automation types), job advertisements capture employer demand at the point of hiring decision (Azar et al., 2022).

Job advertisements contain granular information on required tasks, skills, and technologies that enable measurement of within-occupation heterogeneity (Carnevale, Cheah, & Hanson, 2014). Unlike task databases like O*NET that average across entire occupations, advertisements reveal the actual task bundles employers seek when filling specific positions.

This enables detection of cluster-level restructuring invisible to other data sources. The finding that cluster-level volatility (+388% growth to sharp decline) aggregates to near-zero occupation-level effects validates job advertisements as capturing real labour demand variation that census data miss.

However, the thesis also reveals limitations of job advertisement data that require acknowledgment. First, advertisements capture only publicly advertised positions, excluding approximately 20% of vacancies filled through informal networks or internal promotion. Second, advertisements reflect employer preferences, not worker capabilities or actual job content—creating potential mismatch between advertised requirements and performed tasks. Third, advertisement intensity varies across occupations and industries based on recruitment practices, potentially introducing selection bias. Fourth, skill requirements in advertisements may reflect credentialist filtering rather than actual job demands, particularly in tight labour markets where employers artificially inflate requirements.

Despite these limitations, multiple sources of validation suggest job advertisements capture real labour market dynamics rather than merely advertisement practices. Prior research demonstrates correspondence between advertised skill demands and actual job requirements (Hershbein & Kahn, 2018b), and the skill requirements analysis in Chapter 7 confirms that cluster characteristics vary systematically with trajectory status—newly-emerged clusters require fewer skills while declining clusters require more, a pattern unlikely to emerge if clusters reflected advertising conventions rather than genuine job differences. The internal consistency across chapters—where occupation-level null findings combine with cluster-level volatility and systematic skill requirement gradients across 250 occupations—suggests the methodology captures real restructuring dynamics rather than measurement artifacts.

Combined with text analysis methods, job advertisement data offers a scalable approach for measuring task restructuring that can be applied across countries, time periods, and automation types. The methodology developed in Chapter 6—HDBSCAN clustering with dimensionality reduction applied to advertisement text—provides a replicable framework for future research. Extensions could examine how restructuring patterns vary across institutional contexts, whether cluster dynamics predict wage changes, or how automation shocks propagate through skill networks. Recent developments in foundation models for labour market analysis offer promising directions: Vafa et al. (2024b) demonstrate that transformer-based models pre-trained on large resume datasets can predict career transitions and occupational mobility patterns with substantially higher accuracy than traditional econometric approaches. Combining cluster-based contemporaneous analysis (this thesis) with foundation model-based longitudinal tracking (Vafa et al.) could reveal how workers move between clusters over time, whether certain cluster transitions predict

wage growth or displacement, and which skill combinations enable successful adaptation to automation-driven restructuring.

9.3.4 Demand-Side Consequences of Within-Occupation Restructuring

The thesis provides novel evidence that within-occupation restructuring has systematic demand-side consequences—a finding that extends restructuring theory beyond structural description to measurable characteristics of labour demand. Chapter 7 demonstrated that newly-emerged clusters exhibit heterogeneous skill requirements across 250 occupations ($p < 0.001$, $N = 519,189$): 59% of occupations show significant task recombination (new clusters requiring *more* skills), 22% show significant task unbundling (new clusters requiring *fewer* skills), and 19% show no significant effect. This heterogeneity—rather than any single directional pattern—is itself the central demand-side finding.

This finding advances two theoretical debates. First, it reveals that Braverman (1974)’s taylorisation mechanism operates in a substantial minority of occupations (22%) but co-exists with an equally theoretically important recombination mechanism operating in the majority (59%). In unbundling occupations, complex holistic roles decompose into simpler, narrower task bundles as Braverman predicted; in recombination occupations, new clusters emerge with *enriched* skill requirements, consistent with skill-biased technical change creating more complex roles. Second, this heterogeneity directly reconciles apparently contradictory findings in the skill-biased technical change literature. Autor, Levy, and Mur-nane (2003) documented skill upgrading at macro levels while Goos and Manning (2007) found polarisation—both are simultaneously correct because upgrading and degradation coexist across different occupations and different clusters within the same occupation. The centroid drift moderation (Spearman $\rho = -0.263$, $p < 0.001$) reveals the structural logic: slowly-evolving occupations tend toward task recombination (new clusters with more skills), while rapidly-transforming occupations tend toward task unbundling (new clusters with fewer skills). The heterogeneity itself—not either pattern alone—constitutes the empirical contribution.

Why does recombination dominate? The prevalence of task recombination (59% of occupations) over task unbundling (22%) warrants explanation. Three complementary mechanisms may account for this pattern. First, *technological complementarity*: new digital tools (cloud platforms, data analytics, automation software) frequently create novel task combinations rather than simply decomposing existing roles. When an accounting firm adopts Salesforce alongside traditional financial software, the resulting job type combines CRM management with financial analysis—a recombination producing broader skill

requirements. Second, *market thickness*: recombination-dominant occupations are substantially larger (median 1,518 vs 654 jobs), and thicker markets support more specialised but skill-rich configurations because employers can recruit workers with precise multi-skill profiles (Moretti, 2012). Third, *employer signalling*: job advertisements may list aspirational skill requirements that exceed the position’s actual demands (Blair & Deming, 2020), and this credential inflation may be more pronounced for novel roles where employers lack established templates. The relative contribution of these mechanisms—genuine skill enrichment vs compositional effects vs credential inflation—cannot be disentangled with the present data and represents an important direction for future research using matched employer-employee records.

The skill requirements finding also derives from the theoretical model developed in Chapter 3. Assortative matching implies that high-productivity firms create enriched job designs with broader skill requirements; the dynamic extension’s viability conditions imply that clusters emerge at productivity thresholds where novel configurations first become viable. The model accommodates both recombination and unbundling through the partition-invariance of Proposition 3: which pattern dominates depends on the technology-specific skill partition $\mathcal{P}(\tau)$ and the pace of technological change in each occupation.

An important boundary condition must be acknowledged: job advertisements measure labour *demand* rather than worker outcomes. Whether the skill requirement differences translate to different career trajectories or job quality requires matched employer-employee data beyond the scope of this analysis.

9.4 Methodological Contributions and Reflections

9.4.1 Application of HDBSCAN to Within-Occupation Restructuring

This thesis applies hierarchical density-based spatial clustering (Campello et al., 2013; McInnes et al., 2017) to measure automation-driven within-occupation restructuring, **extending** recent applications of clustering methods to labour market analysis (Vezza et al., 2025) and building on within-occupation measurement approaches (Martin-Caughey, 2021). While Vezza et al. (2025) applied HDBSCAN to job vacancy data in Argentina and Uruguay to measure real-time labour demand, and Martin-Caughey (2021) measured within-occupation heterogeneity using cosine similarity of text responses, this thesis **extends these approaches** to examine automation-associated task bundle evolution within

Australian occupations over time. HDBSCAN extends density-based clustering methods like DBSCAN (Ester et al., 1996) by discovering hierarchical cluster structures and handling varying densities. This unsupervised machine learning method offers several advantages over existing approaches for measuring automation-driven restructuring.

Dynamic classification: Unlike static taxonomies (ANZSCO, O*NET), HDBSCAN discovers clusters from data without imposing predetermined categories. This enables detection of emerging task bundles, hybrid roles, and idiosyncratic positions that traditional classifications miss. The finding that cluster counts vary across years within the same occupation demonstrates that occupational structure is fluid rather than fixed, contradicting assumptions underlying static classification systems.

Handling noise: HDBSCAN explicitly identifies observations that do not fit stable clusters, treating them as noise rather than forcing classification. The finding that a median of 33% of jobs (IQR: 24–41%) defy stable classification across 250 occupations reveals substantial labour market fluidity invisible to traditional methods. This noise proportion has theoretical significance: it quantifies the extent to which actual jobs resist categorisation into discrete types, supporting arguments for continuous rather than categorical conceptualisations of occupational structure.

Hierarchical structure: HDBSCAN discovers clusters at multiple levels of granularity, enabling analysis at appropriate scales for different research questions. This addresses the aggregation problem that plagues occupation-level research by allowing flexible aggregation based on analytical needs. The condensed tree visualization demonstrates that occupational structure is hierarchical, with broad job families subdividing into increasingly specialised types at finer distance thresholds.

The methodology reveals that cluster-level dynamics vary dramatically (+388% growth to sharp decline within single occupations), consistent with Proposition 3’s prediction and explaining why occupation-level frameworks detect minimal effects. This validates calls from Acemoglu and Restrepo (2022) and Autor et al. (2024a) for finer-grained measurement while providing a concrete methodological approach. However, HDBSCAN introduces new methodological challenges: parameter sensitivity (`min_cluster_size`, `min_samples`), distance metric selection (cosine vs. Euclidean), and cluster stability assessment across time periods. The supplementary analysis addressed these challenges through systematic parameter optimisation and stability testing, but future research should develop standardised best practices.

9.4.2 Multi-Level Triangulation: Strengths and Limitations

The iterative sequential research design demonstrates the value of multi-level quantitative triangulation for automation research. Each analytical approach addresses specific limitations of others, creating a complementary evidence base.

Quantitative occupation-level analysis (Chapter 5) establishes aggregate effects and validates null findings, providing macroeconomic context. Its strengths include large sample size (170,491 observations), rigorous fixed effects controls, and replicability. However, it cannot explain *why* effects are modest or *how* restructuring occurs. The low R^2 values (0.030) signal unmeasured heterogeneity but cannot identify its sources.

Quantitative cluster-level analysis (Chapter 6) reveals within-occupation restructuring masked by aggregation. Its strengths include detecting patterns invisible to occupation-level frameworks (86 job types, +388% growth to sharp decline) and providing scalable measurement applicable across occupations. However, it cannot explain *why* specific clusters expand or contract, or what factors mediate firm decisions. Cluster labels require interpretive judgment, introducing subjectivity into otherwise objective classification. Moreover, clusters may capture spurious patterns if skill co-occurrence reflects advertisement conventions rather than actual task bundling.

The integration of occupation-level and cluster-level analysis provides complementary perspectives on the same phenomenon. Occupation-level null results, combined with cluster-level restructuring patterns, constitute comprehensive evidence for restructuring theory. However, the analysis relies on job advertisement data (BGT), which introduces measurement considerations. The absence of worker-level data means outcomes for displaced versus retained workers remain unobserved, and future research should examine worker trajectories through cluster transitions.

9.4.3 Reflections on the Australian Context

This thesis provides the first sub-occupational analysis of automation-related restructuring in the Australian labour market using granular job advertisement data, building on prior occupation-level research (Borland & Coelli, 2017; Cebulla, 2024; Edmonds & Bradley, 2015). While most automation research focuses on the United States or Europe, institutional and economic differences make generalization uncertain. The Australian context offers several analytical advantages but also imposes boundary conditions.

Service-oriented economy: Australia's economy is 70%+ services (Australian Bureau of Statistics, 2020a), making it representative of advanced economies where service sector

automation dominates. Most existing research focuses on manufacturing robots, limiting relevance for understanding contemporary automation. However, this also means findings may not generalise to manufacturing-intensive economies (e.g., Germany, China) where automation operates through different mechanisms (physical robots vs. software).

Institutional differences: Australia's labour market institutions—enterprise bargaining, Modern Awards wage-setting, Fair Work Commission, moderate employment protection—differ from US "employment at-will" and European "high protection" models. These institutions may mediate automation impacts by constraining firm restructuring flexibility or protecting worker incomes during transitions.

These institutional differences are not incidental boundary conditions but central to understanding restructuring patterns. Varieties of capitalism literature (Estevez-Abe et al., 2001; P. A. Hall & Soskice, 2001) theorizes systematic relationships between skill formation systems, employment protection, and firm training investments. Liberal market economies (LMEs) like Australia face coordination failures: firms under-invest in training because skilled workers can be poached by competitors, yet market competition pressures firms to externalize training costs onto workers or educational institutions (Finegold & Soskice, 1988). In contrast, coordinated market economies (CMEs) with strong industry associations, collective bargaining, and employment protection can maintain apprenticeship systems because institutional complementarities ensure firms capture returns on training investments (Busemeyer & Trampusch, 2012; Thelen, 2004). The within-occupation restructuring patterns documented in this thesis may therefore exhibit different dynamics across institutional contexts—not just in magnitude but in fundamental character. Cross-national comparison could identify which aspects of restructuring theory depend on institutional context versus represent general features of contemporary automation, examining whether cluster proliferation rates, noise proportions, or creative destruction rates vary systematically between LMEs and CMEs, and whether different institutional configurations produce different distributions of automation's costs and benefits between firms and workers.

Data quality: The BGT Australia dataset spans 2012-2019, capturing pre-pandemic automation patterns without COVID-19 confounding effects that complicate post-2020 research. This provides clean identification of automation impacts separate from pandemic shocks. However, it also means findings may not capture recent acceleration in automation adoption, generative AI diffusion, or remote work-enabled task restructuring.

The findings suggest that within-occupation restructuring may be a general feature of automation in service-oriented economies, but cross-national comparison remains essential for establishing boundary conditions. Comparative research could examine whether cluster proliferation rates, noise proportions, or creative destruction rates vary systematically across institutional contexts, providing evidence on how policy frameworks mediate restructuring

processes.

9.5 Critical Examination: Alternative Explanations and Boundary Conditions

While the evidence supports restructuring theory, alternative explanations warrant consideration. This section examines three potential challenges to the interpretation: spurious correlation, alternative restructuring drivers, and measurement artifacts.

9.5.1 Causal Identification Limitations

The research design establishes associations between automation measures and labour demand patterns but cannot definitively establish causality. While fixed effects models control for time-invariant confounds, alternative explanations remain possible.

Demand shocks: Cluster-level changes may reflect consumer demand shifts rather than automation. If demand for tax accounting services declines while demand for strategic advisory grows, cluster composition would shift even absent automation. The skill requirements findings from Chapter 7 partially address this concern: the systematic heterogeneity in skill requirements—59% of occupations showing task recombination, 22% showing task unbundling, moderated by centroid drift ($\rho = -0.263$)—is consistent with technology-mediated restructuring, though not uniquely so. Pure demand-driven cluster reallocation would not necessarily exhibit the systematic drift moderation pattern, but the analysis cannot rule out alternative mechanisms (e.g., demand shocks correlated with technology adoption) that would produce similar correlations.

From labour process theory perspective, distinguishing “automation impacts” from “demand-driven restructuring” may be analytically artificial (Braverman, 1974; Weil, 2014). If automation may enable firms to reorganise work in response to competitive pressures and market opportunities, then demand shocks and technological capabilities are likely to jointly constitute restructuring processes rather than represent alternative causal mechanisms. Barley and Kunda (2001) argue that work practices emerge from processes where actors draw on technological possibilities to respond to organisational imperatives—in economic terms, technology and demand are jointly determined rather than independently identifiable. The cluster proliferation documented may reflect firms’ capacity to differentiate service offerings *because* automation makes specialised task bundles economically viable at scales previously requiring standardization. In this view, automation does not cause restructuring independently of demand, nor does demand cause restructuring independently

of technological possibilities—both operate as mutually enabling conditions. The quantitative analysis cannot distinguish these mechanisms, suggesting need for process-tracing case studies examining specific restructuring episodes to establish temporal sequences and causal mechanisms (Beaudry et al., 2016; Fernández-Macías & Hurley, 2017).

Organisational restructuring: Firms continuously reorganise independent of automation. Cluster proliferation may reflect general organisational evolution—mergers, strategic pivots, market entry/exit—rather than automation-specific impacts. The skill requirement patterns documented in Chapter 7 are consistent with a technology-related mechanism: the finding that 59% of occupations show task recombination while 22% show task unbundling, with centroid drift predicting which pattern dominates ($\rho = -0.263$), is the kind of structured heterogeneity that purely generic organisational change would not necessarily produce. The drift moderation effect—where slowly-evolving occupations show recombination and rapidly-transforming occupations show unbundling—is consistent with a technology-mediated interpretation, though it does not definitively rule out spurious correlation with broader restructuring trends.

Skill credentialing: Rising educational attainment may drive skill demand changes independent of automation. If workers increasingly possess advanced credentials, job advertisements may request higher skills even without technological change. This could generate cluster proliferation (differentiation by credential levels) misattributed to automation.

Future research should pursue quasi-experimental designs leveraging automation shocks—technology releases, regulatory changes, firm-level implementations—to strengthen causal inference. Instrumental variable approaches using variation in automation costs or opportunities (e.g., software price declines, AI capability releases) could isolate causal effects from correlated trends. Event studies around specific automation adoption dates could establish temporal precedence more convincingly than cross-sectional or panel associations.

9.5.2 Generalizability and Boundary Conditions

The findings emerge from specific empirical contexts that impose boundary conditions on theoretical claims.

Occupational scope: While I applied cluster analysis to 250 occupations and confirmed temporal dynamics at full scale (Chapter 7), the detailed case-study analysis in Chapter 6 focused on illustrative occupations (Accounting, Receptionists, Human Resources). The full-scale results confirm that the core patterns generalise: median annual churn of 112% across 250 occupations, with centroid drift correlating positively with churn (Spearman $\rho = +0.558$, $p < 0.001$). However, the richest qualitative description of cluster content—

skill profiles, career pathways, firm sorting—remains limited to the illustrative occupations.

Temporal scope: The 2012-2019 window precedes several significant developments: COVID-19 remote work acceleration (Autor & Reynolds, 2020; Barrero et al., 2021; Bloom et al., 2015), generative AI diffusion (2022+), and large language model capabilities (Brynjolfsson et al., 2023b; Eloundou et al., 2024) (2023+). Recent evidence suggests generative AI increases productivity for white-collar workers (Dell’Acqua, McFowland, et al., 2023; Noy & Zhang, 2023b), particularly for less-experienced workers performing complex writing tasks. Contemporary automation may operate through different mechanisms than the primarily robotic process automation and basic AI characterising the study period. Restructuring theory may require revision to accommodate generative AI’s capacity to automate cognitive tasks previously considered safe.

Industry coverage: Job advertisement data overrepresent white-collar occupations relative to manual trades, informal economy work, and gig employment. The restructuring patterns documented may be specific to formal, advertised positions in service sectors rather than general features of all work. Manufacturing automation, construction automation, or agricultural automation may exhibit different dynamics.

Institutional context: Australian labour market institutions may mediate restructuring processes in ways that do not generalise to other institutional configurations. Strong employment protection might reduce displacement even with substantial task automation, while weak protection might produce more extreme restructuring. Cross-national comparison is essential for establishing which findings depend on Australian context versus represent general automation dynamics.

What Findings May NOT Generalize: To be explicit, the following specific findings should *not* be assumed to hold beyond the empirical context studied:

- Cluster proliferation rates may not generalise to: manufacturing occupations, manual trades, gig economy roles, or occupations in countries with different skill formation systems
- Churn rates (median 112% annually across 250 occupations) may not generalise to: economic downturns, post-pandemic remote work era, or countries with different employment protection regimes
- Noise proportions (median 33%, IQR 24–41% of jobs resist categorisation) may not generalise to: occupations with stronger professional licensing requirements, industries with standardised job taxonomies, or countries where online job posting is less normative
- Skill requirement heterogeneity (59% recombination, 22% unbundling) may not

generalise to: occupations with stronger union representation, countries with different skill formation systems, or time periods with different technological regimes

- Null occupation-level findings may not generalise to: manufacturing-dominated economies, countries experiencing rapid technological adoption, or time periods with major regulatory changes affecting specific occupations

Establishing generalizability requires replication across: (1) diverse occupations spanning manufacturing, services, and trades; (2) multiple national contexts with varying institutions; (3) different time periods including post-pandemic and generative AI eras; and (4) validation against worker-level panel data tracking actual employment transitions rather than job advertisement patterns.

9.5.3 Measurement Artifacts and Methodological Limitations

Several methodological choices may influence findings in ways that complicate interpretation.

HDBSCAN parameter sensitivity: Cluster counts and compositions depend on algorithm parameters (`min_cluster_size`, `min_samples`, `epsilon`). While supplementary analysis tested parameter robustness, different parameter choices could yield different conclusions about cluster proliferation rates. The choice of `min_cluster_size` (25 jobs) may be too large, missing small specialised roles, or too small, fragmenting coherent job types.

Skill extraction accuracy: Job advertisement parsing relies on BGT's proprietary skill extraction algorithms. Errors in skill identification would propagate through cluster analysis, potentially creating spurious patterns. If skill extraction accuracy varies by occupation or time period, some apparent restructuring may reflect changing advertisement language rather than actual labour demand.

Advertisement representativeness: Not all jobs are advertised; selection into advertisement may vary by occupation, firm size, and time period. If automation reduces advertisement propensity (e.g., firms fill positions through temp agencies rather than direct hiring), the dataset would underrepresent automated positions, biasing results toward underestimating automation impacts.

Cluster label interpretation: Assigning meaningful labels to statistically-identified clusters requires interpretive judgment. Different analysts might label identical clusters differently, potentially leading to different substantive conclusions about which job types are

emerging or declining. While the analysis grounded labels in top skills and representative job titles, some subjectivity remains.

These measurement concerns counsel caution in interpreting specific magnitudes (+388% growth, 86 job types) as precise estimates rather than indicative of general scale. However, the skill requirement analysis in Chapter 7 provides internal validation: the systematic heterogeneity in skill requirements—59% of occupations showing significant task recombination, 22% showing significant task unbundling, with centroid drift moderating the direction ($\rho = -0.263$)—suggests clusters capture real labour market distinctions rather than statistical artifacts. Random clustering would not produce such structured heterogeneity, nor would it generate the theoretically predicted drift moderation where slowly-evolving occupations favour recombination and rapidly-transforming occupations favour unbundling.

9.6 Implications for Policy and Practice

The restructuring theory has profound implications for labour market policy and organisational practice, challenging current debates that focus almost exclusively on job displacement and retraining displaced workers.

9.6.1 Rethinking Workforce Development Policy

Automation's labour market impacts manifest through two potentially coexisting mechanisms: occupation elimination and within-occupation restructuring. While historical evidence documents both processes—occupation elimination (e.g., typists, switchboard operators) and task restructuring (e.g., bank tellers adapting to ATMs)—the findings from this thesis suggest that *during 2012–2019 in Australian service occupations*, restructuring dominated over elimination. If this restructuring pattern applies more broadly, policies emphasising unemployment insurance and displacement-triggered retraining may address the less prevalent mechanism while neglecting continuous skill development and task adaptability for *employed* workers whose job content continuously transforms (Cappelli, 2015; OECD, 2019b).

The thesis findings suggest that current policy debates may misallocate attention and resources relative to the restructuring mechanism. The pattern evident in the data is not mass unemployment but ongoing workplace reorganisation. Workers experience not one-time displacement requiring retraining, but continuous evolution of advertised task requirements, skill demands, and work organisation within their current occupations. If this pattern reflects automation-driven dynamics rather than concurrent macroeconomic adjustment, it

would call for different policy instruments than displacement-focused frameworks provide.

Continuous learning systems: Rather than retraining programs triggered by job loss, policies should support ongoing skill development for employed workers (McGuinness et al., 2018). The skill requirement patterns in Chapter 7—where new clusters exhibit heterogeneous skill demands across 250 occupations (59% recombination, 22% unbundling)—suggest that targeted training addressing specific cluster transitions could be more efficient than generic occupational upskilling. Workplace-based, context-specific learning targeting the specific skills that differentiate cluster types may be more effective than standardised occupational curricula, but raises questions about who bears costs and how to ensure portability across employers.

However, translating these insights into effective policy faces substantial political and administrative challenges documented in comparative skill formation literature (Crouch et al., 1999; Finegold & Soskice, 1988). Continuous learning systems require sustained funding, coordination across education providers and employers, and institutional mechanisms ensuring training portability—preconditions often absent in liberal market economies like Australia. International precedents offer both promise and caution. Singapore’s SkillsFuture programme provides individual training accounts with government subsidies for cluster-specific skill development, but requires centralised administrative capacity and industry cooperation that may not transfer to federal systems with weaker state capacity (Bosch & Charest, 2010). Sectoral training partnerships in the United States (e.g., Wisconsin Regional Training Partnership, Project QUEST) demonstrate cluster-based targeting effectiveness, but depend on sustained philanthropic and government funding vulnerable to political cycles. European Social Fund continuous learning initiatives achieve broad coverage but struggle with employer engagement and quality control. The political economy challenge is whether liberal market economies can overcome coordination failures—firms under-invest in training fearing poaching, workers cannot finance training without income security, and governments face fiscal constraints—that prevent effective skill development systems even when technical solutions are well-understood (Thelen, 2014).

Cluster-based rather than occupation-based interventions: Government retraining programs typically target occupational skills (“accounting,” “nursing”) while firms restructure at cluster granularity (“budgeting-focused accounting,” “clinical assessment nursing”). This disconnect creates systematic inefficiencies (Blair & Deming, 2020; Modestino et al., 2020). Chapter 8 demonstrates concretely how cluster-based workforce development could operate: identify workers’ current cluster, query the skill proximity network for accessible alternatives in stable or growing occupations, and target the specific bridge skills (typically 1–3) that separate the source and target clusters. Strikingly, 76% of best-match transitions from at-risk to stable clusters require zero genuine specialist skills—the gap consists entirely

of generic competencies workers likely already possess (Section 8.4.1)—meaning that for the majority of at-risk workers the transition barrier is not new technical skills but recognition and certification of existing capabilities. Where specialist skills are required, the skill proximity network reveals that generic professional competencies—Communication Skills, Planning, Teamwork, Problem Solving—are the most commonly needed bridge skills, suggesting that investment in these broadly transferable capabilities provides the highest return across potential transitions. Athey (2018) terms this type of intervention a "prediction policy problem"—where the goal is not establishing causal treatment effects but accurately forecasting heterogeneous outcomes to allocate resources efficiently. Machine learning methods excel at prediction problems precisely because they can discover complex patterns in high-dimensional data without requiring causal identification, making cluster-based policy targeting methodologically tractable even when causal mechanisms remain unclear. However, implementation faces substantial administrative challenges: tracking cluster transitions may exceed policy capacity if job types continuously reconfigure through grassroots automation, and cluster classifications require technical expertise that government agencies may lack.

Entry-level pathway protection: The finding that new clusters require substantially less experience (−0.46 years) and fewer skills suggests that entry-level roles in unbundling occupations are being restructured into simplified positions that may lack developmental content. If technology adoption is associated with the erosion of career pathways that historically produced specialists, market-driven training may be insufficient on its own. Policy intervention may be needed to preserve experiential learning opportunities—for example, subsidies for apprenticeship programs, tax credits for firms maintaining developmental roles, or public sector employment providing entry-level experience. However, such interventions require acknowledging that the efficiency gains often attributed to automation partially derive from removing positions framed as “unproductive,” creating tension between economic efficiency and labour force development.

Fair distribution of adaptation costs: The findings from Chapter 7 reveal that restructuring is not symmetric in unbundling occupations: newly-emerged clusters in those occupations require narrower skill sets, and median cluster survival across 250 occupations is only 53.8%, meaning workers entering through new clusters face positions that may not persist. The structural instability and skill narrowing documented in job advertisement data suggest that adaptation costs may not be evenly distributed. If workers must continuously adapt to restructuring, firms capturing the productivity gains associated with technology adoption should plausibly bear a corresponding share of adjustment costs. Potential mechanisms include training levies, mandatory skill development leave, or institutional supports for workers navigating cluster transitions.

However, the policy recommendations above face fundamental challenges rooted in asymmetric power relations. If firms systematically externalise adaptation costs onto workers—for instance, delegating automation implementation to employees who thereby eliminate their own roles (Edwards, 1979)—then cluster-based training programmes risk institutionalising rather than remedying this cost-shifting. Effective policy requires not merely facilitating worker adaptation through better information systems or targeted training, but *regulating firms’ capacity to extract productivity gains while externalising adjustment costs* (Kalleberg, 2011; Standing, 2011). While the empirical findings cannot directly prescribe policy, the restructuring patterns documented here suggest that effective interventions would need to address distributional asymmetries. The following recommendations reflect the author’s assessment of what the evidence implies, informed by the institutional economics literature, rather than direct empirical conclusions: (1) mandatory employer contributions to worker reskilling proportional to automation-generated productivity gains, enforced through training levies or payroll taxes; (2) collective bargaining rights enabling workers to negotiate automation terms including redeployment guarantees, protected training time, and compensation for knowledge documentation; (3) employment protection regulations requiring firms to maintain entry-level positions or fund alternative apprenticeship pathways when automating developmental roles; and (4) transparency requirements mandating firms disclose automation plans enabling workers and unions to prepare proactively rather than react to *fait accompli* restructuring. Without such regulations addressing power asymmetries, market-driven adjustment systematically disadvantages workers while concentrating automation benefits with capital (F. Green, 2006). The political economy challenge is whether liberal market economies like Australia possess institutional capacity to implement such regulations against business opposition, or whether coordination failures prevent effective policy responses even when restructuring patterns are well-understood.

9.6.2 Improving Labour Market Information Systems

The dramatic difference between occupation-level and cluster-level volatility suggests that current labour market information systems provide misleading signals to workers, educators, and policymakers. Occupational employment projections based on aggregate categories miss the restructuring occurring within occupations. When an occupation appears stable in aggregate, workers and policymakers miss that some clusters within it are declining sharply while others grow rapidly.

Policies should develop finer-grained labour market information systems that capture within-occupation skill demand changes. The cluster-level methodology developed in this thesis offers a potential approach: analysing job advertisements in real-time to identify

emerging task bundles, declining skill requirements, and hybrid roles. This would enable more targeted education and training investments. However, several implementation challenges require resolution: ensuring data representativeness (not all jobs are advertised), maintaining classification consistency over time (clusters must be comparable across years), and communicating complexity to lay audiences (86 job types within a single occupation may overwhelm rather than inform).

Real-time skill demand analytics could provide early warning signals of restructuring, enabling proactive adaptation rather than reactive displacement response. If cluster analysis identifies declining task bundles 2-3 years before mass displacement, workers could retrain while employed rather than after job loss. Educational institutions could adjust curricula based on emerging skill demands rather than lagging occupational projections. Policymakers could target interventions to specific restructuring patterns rather than generic unemployment programs.

9.6.3 Implications for Organisational Practice

The research provides actionable insights for executives, HR professionals, and workforce planners navigating automation-driven change, though organisational recommendations must be framed carefully as research implications rather than consulting advice.

Recognise sub-occupational heterogeneity in automation planning: Executives should avoid treating occupations as homogeneous when making automation decisions. The finding that Accounting contains 86 distinct job types in 2019 suggests that automation decisions must be made at the task bundle level, not occupation level. Before automating "accounting functions," executives should identify which specific accounting clusters (e.g., compliance-focused vs. strategic advisory) are targets and implications for career pathways. The cluster-level methodology developed in this thesis provides analytical frameworks for articulating this heterogeneity systematically.

Invest in directed rather than generic training: The skill requirement analysis in Chapter 7 shows that different cluster types have systematically different skill demands—59% of occupations show new clusters requiring *more* skills (recombination) while 22% show new clusters requiring *fewer* skills (unbundling). This heterogeneity suggests that targeted training addressing specific cluster transitions may be more effective than generic occupational programs. Rather than sending workers to "accounting" courses, organisations could identify the specific cluster they need to transition toward and provide training in those differentiating skills. However, cluster-specific training risks reducing worker portability across employers, potentially reducing labour market flexibility.

Protect entry-level pathways while automating: Executives should explicitly de-

sign career progression mechanisms when automating entry-level tasks. If junior roles historically provided experiential learning for senior positions, automation of those roles requires alternative pathways or firms will face talent shortages they themselves created. However, executives face conflicting incentives: protecting entry-level pathways reduces short-term efficiency gains that justify automation investments to boards and shareholders. Productivity improvements from automation derive precisely from removing what firms frame as "low-value" work—yet this work often provided developmental functions invisible to productivity metrics. Without regulatory requirements mandating apprenticeship preservation, collective bargaining pressure ensuring worker voice in restructuring decisions, or industry-wide coordination preventing free-riding on competitors' training investments (P. A. Hall & Soskice, 2001), market competition systematically incentivizes pathway elimination. This reveals a coordination failure where individual firms cannot maintain training pathways that benefit the collective labour market without disadvantaging themselves competitively. This suggests that preserving career pathways requires institutional intervention rather than voluntary corporate responsibility, yet liberal market economies lack the coordinated governance structures that would enable such intervention (Thelen, 2004).

Use real-time skill demand analysis, not occupational forecasts: HR professionals should monitor job advertisement data for their specific occupation to identify emerging clusters and declining task bundles. Occupational employment projections miss the restructuring occurring within boundaries—cluster-level analysis provides earlier warning signals. However, this requires technical expertise and data infrastructure that may exceed HR department capabilities, suggesting potential role for specialised workforce analytics firms.

9.7 Conclusion

This Discussion chapter has integrated findings across the four empirical chapters to demonstrate that, during the 2012–2019 machine-learning era in Australian service occupations, the labour-market signature of automation is consistent with within-occupation task restructuring rather than wholesale occupation elimination. The multi-level evidence reconciles the apparent paradox between dire automation predictions and modest empirical findings: the patterns are consistent with automation substantially reshaping work, but primarily through within-occupation restructuring detectable only at cluster granularity.

Read in historical perspective, the chapter integrates a coherent baseline for the second wave of automation. The occupation-level null result ($R^2 = 0.030$; 94.4% within-occupation variance), the cluster-level discovery findings (5,118 clusters across 249 occupations with

elasticity 0.91 to market thickness, 81.9% of occupations containing both at-risk and stable clusters), the temporal-dynamics findings (median annual churn 111.7%, drift–churn correlation $\rho = +0.558$, 59/22/19 split for recombination/unbundling/non-significant), and the network findings (88.7% in largest connected component, 95% of at-risk clusters reaching a stable alternative within ≤ 3 skills, 76% requiring zero specialist skills) together describe how Australian work was reorganised under statistical machine learning before the disruptions of COVID-19 and large language models—providing reference points against which the labour-market signature of subsequent waves can later be measured.

The restructuring theory generates core contributions to knowledge: (1) empirical quantification of aggregation bias in occupation-level measurement across 250 occupations, (2) resolution of the apparent dichotomy between task-level and occupation-level analysis through multi-level integration, (3) validation of job advertisement data for labour market analysis, and (4) demonstration that within-occupation restructuring has measurable demand-side consequences—newly-emerged clusters exhibit heterogeneous skill requirements, with median cluster survival of only 53.8%. Chapter 8 further demonstrates that the labour market is largely navigable through skill-proximate transitions, with 44.4% of at-risk clusters finding their best transition target within their own occupation. While mean connectivity is broadly uniform across cluster types, 13.8% of newly emerged clusters are completely isolated compared to 3.3% of persistent clusters, indicating that transition vulnerability concentrates among a minority of the youngest clusters rather than affecting entire occupations. Methodologically, the thesis demonstrates the value of HDBSCAN for producing dynamic sub-occupational classifications and skill proximity networks for mapping career transition pathways.

Critical examination reveals limitations requiring acknowledgment: causal identification remains incomplete, generalisability to other contexts requires testing, and measurement artifacts may influence specific magnitudes. The convergence of quantitative patterns—across structural dynamics and demand-side characterisation—provides evidence consistent with restructuring as a productive framework for understanding the empirical patterns observed in this period, while leaving open whether the underlying mechanism is automation, concurrent organisational change, or some combination.

The implications for policy and practice are substantial. If restructuring rather than displacement is the dominant adjustment margin in service occupations under the kind of technology regime the data captures, then workforce development frameworks designed around displacement events will systematically miss the population most affected. Three reframings follow.

First, workforce development systems would need to shift from displacement-focused retraining toward continuous skill adaptation for employed workers, with cluster-based

targeting that exploits the skill-proximity network rather than treating occupations as homogeneous units. Second, labour market information systems would need to capture within-occupation restructuring to provide accurate signals; aggregate occupational projections systematically overstate stability when cluster-level churn exceeds 100% annually. Third, organisational practice would need to recognise sub-occupational heterogeneity in automation planning—the same occupation can contain expanding and contracting clusters with opposite implications for workforce design. Whether such reframings can be implemented at policy scale depends on coordination capacity that liberal market economies often lack, and remains a question for future research and policy experimentation rather than a direct empirical conclusion of this thesis.

Chapter 10

Conclusion

10.1 Research Overview

This thesis presents a historical review of automation’s labour-market footprint across 2012–2019, the pre-pandemic and pre-generative-AI period during which cloud computing, mobile platforms, and the first wave of machine learning diffused through Australian workplaces. It addressed a fundamental paradox in automation research: why did dire predictions of mass technological unemployment contrast so sharply with modest empirical findings? The literature review revealed that this paradox stems from measurement frameworks operating at inappropriate levels of aggregation. Occupation-level classifications—the dominant approach in automation research—assume homogeneity within occupational categories, masking substantial restructuring occurring at finer levels of analysis.

The research pursued two overarching questions: (1) Do occupation-level automation measures detect significant labour market impacts in the Australian context? (2) Does within-occupation heterogeneity explain the disconnect between automation predictions and empirical findings?

To address these questions, the thesis employed a multi-level quantitative research design analysing 9.5 million Australian job advertisements (2012–2019). This approach enabled examination of labour market restructuring patterns at both occupation level and cluster level, revealing different aspects of the restructuring process at each analytical scale.

Chapter 5 tested occupation-level automation frameworks and found that both explain negligible variation in Australian labour demand, with theoretically distinct categories prov-

ing statistically indistinguishable. Variance decomposition confirmed that the vast majority of variation operates within occupations, establishing that occupation-level measurement faces an inherent ceiling.

Chapter 6 shifted to cluster-level analysis, revealing discrete sub-occupational job types whose counts scale with market thickness and whose employment volatility exceeds occupation-level measures by an order of magnitude. Technology succession—legacy platform clusters dissolving as emerging technology clusters form—offers one plausible mechanism consistent with this within-occupation restructuring.

Chapter 7 tracked cluster dynamics over time, documenting median annual churn exceeding 100% alongside net job type creation. The demand-side consequences proved heterogeneous: task recombination dominates among slowly-evolving occupations while task unbundling characterises rapidly-transforming ones.

Chapter 8 constructed a skill proximity network mapping career transition pathways for workers in at-risk clusters. The labour market is broadly navigable, with most at-risk clusters connected to stable alternatives through modest upskilling, though navigability is unevenly distributed—the most vulnerable clusters are the least connected.

10.2 Contributions to Knowledge

The thesis makes six contributions across theoretical, methodological, and empirical domains.

1. **A restructuring theory of automation impacts.** The thesis advances a theory in which automation's primary labour market effect operates through continuous task reorganisation within occupations rather than occupation elimination. This reconciles the skill upgrading (Autor, Levy, & Murnane, 2003), polarisation (Goos & Manning, 2007), and degradation (Braverman, 1974) literatures by showing all three patterns coexist simultaneously at cluster level. The formal model (Chapter 3) derives five testable propositions from a directed search framework with endogenous job design.
2. **A scalable methodology for sub-occupational measurement.** First application of HDBSCAN clustering to sub-occupational classification in the economics literature, demonstrating advantages over static taxonomies: endogenous cluster count, explicit noise handling, and dynamic temporal tracking across 2,000 clustering runs (250 occupations $\times$ 8 years). The construction of cross-occupation skill proximity networks from cluster centroids extends this infrastructure into an applied tool for mapping feasible career transition pathways. The methodology is reusable beyond automation research—applicable to any labour-market question where within-occupation

heterogeneity matters.

3. **Empirical quantification of aggregation bias.** Variance decomposition across 170,491 occupation-year-state-industry observations establishes that 94.4% of variation in Australian labour demand operates within rather than between occupations ($ICC = 0.056$). Both the Routine-Cognitive-Manual taxonomy and Frey-Osborne automation probabilities explain only 3% of variation, with theoretically distinct categories statistically indistinguishable. The aggregation bias is structurally pervasive: 204 of 249 occupations (81.9%) contain both at-risk and stable clusters simultaneously, and cluster counts scale near-proportionally with market thickness (elasticity 0.91 across 250 occupations). This is the first quantification of the empirical signature of aggregation bias that prior literature could only hypothesise.
4. **The technology succession mechanism.** Cluster-level analysis identifies a concrete mechanism consistent with within-occupation restructuring: clusters organised around legacy software platforms (Lotus Notes, ASP.NET, Silverlight) dissolve while clusters requiring emerging technologies (AWS, Docker, Python, Salesforce) form, with dead clusters significantly more software-intensive than born clusters ($p < 0.001$). This succession pattern recasts technology-driven restructuring as a discrete, observable platform-lifecycle process rather than an abstract routine/non-routine substitution.
5. **Heterogeneous demand-side dynamics.** Newly-emerged clusters exhibit systematically different skill requirements depending on the occupation in which they arise: 59% of occupations show task recombination (more skills) while 22% show task unbundling (fewer skills), with centroid drift moderating the direction (Spearman $\rho = -0.263$, $p < 0.001$). Cluster churn correlates strongly with the speed of skill change rather than the static level of automation exposure (drift–churn Spearman $\rho = +0.558$, $p < 0.001$ across 250 occupations), distinguishing automation *speed* from automation *level* as the relevant predictor of restructuring intensity. Both task-recombination and task-unbundling patterns coexist across the labour market—a finding that resolves longstanding debates about whether automation upgrades, polarises, or degrades work by showing that all three patterns occur simultaneously at the sub-occupational level.
6. **Cluster-level transition feasibility.** A skill proximity network connecting all clusters across 249 occupations reveals that 95% of at-risk clusters can reach a stable alternative by acquiring three or fewer new skills, with 44.4% finding their best transition target within their own occupation. The transition barrier is smaller still

once skills are decomposed: 76% of best-match transitions require zero genuine specialist skills, with the gap consisting entirely of generic competencies workers likely already possess. Yet navigability is unevenly distributed: while mean connectivity is broadly similar across cluster types, isolation rates differ asymmetrically (3.3% for persistent clusters versus 13.8% for newly emerged clusters). This combination of broad navigability with structural isolation provides empirical foundations for cluster-based workforce development that occupation-level analysis cannot support.

10.3 Implications

The restructuring theory has profound implications for workforce development policy, labour market information systems, and organisational practice. These implications are developed extensively in Chapter 9 and summarised briefly here.

Policy implications: If automation primarily restructures tasks within occupations, workforce development must shift from displacement-focused retraining toward continuous skill adaptation for employed workers. The skill proximity network analysis provides concrete infrastructure for cluster-based career navigation. However, implementation faces political economy challenges: if firms systematically externalise adaptation costs (Edwards, 1979), cluster-based training risks institutionalising burden-shifting. Effective policy requires addressing coordination failures in skill formation—including employer reskilling contributions, collective bargaining rights over automation terms, and entry-level pathway protection—though whether liberal market economies possess institutional capacity for such regulation remains an open question (Thelen, 2014).

Labour market information: Occupational employment projections miss within-occupation restructuring, providing misleading signals. Real-time cluster-level skill demand analysis could enable proactive adaptation, but better information alone cannot address power asymmetries—workers may recognise declining clusters yet lack resources for reskilling (F. Green, 2006).

Organisational practice: Executives should recognise sub-occupational heterogeneity when making automation decisions, but face conflicting incentives: protecting entry-level pathways reduces short-term efficiency gains. Without institutional coordination, market competition systematically incentivises pathway elimination, creating talent shortages that individual firms cannot unilaterally correct (P. A. Hall & Soskice, 2001).

10.4 Limitations

Several limitations require acknowledgment, situating findings within appropriate empirical and theoretical boundaries. These are examined in detail in Chapter 9 and summarised here.

Temporal and geographic scope: The analysis examines Australian job advertisements from 2012–2019, necessarily limiting generalisability to contemporary conditions (post-pandemic acceleration, generative AI diffusion) and other geographic contexts. Australian labour market institutions may mediate restructuring differently than US employment-at-will or European high-protection regimes; cross-national comparison is essential for establishing whether restructuring theory generalises across institutional contexts.

Causal identification: The research design establishes associations between automation measures and labour demand patterns but cannot definitively establish causality. Alternative explanations—consumer demand shifts, organisational restructuring independent of automation, skill credentialing trends—remain possible. Future research should pursue quasi-experimental designs leveraging automation shocks to strengthen causal inference.

Worker outcomes: The thesis examines labour demand but does not directly observe worker outcomes. While Chapter 8 maps which transitions are *feasible* based on skill proximity, feasibility on the demand side does not guarantee worker mobility on the supply side. Linking job advertisement data to administrative employment records remains essential for understanding worker-level consequences.

Data coverage and measurement: Job advertisements capture only publicly advertised positions (excluding ~20% filled through informal channels), systematically exclude non-standard employment, and lack demographic information. HDBSCAN parameter sensitivity introduces measurement uncertainty into specific magnitudes, though core patterns are robust across parameter specifications.

These limitations suggest findings should be interpreted as patterns within formal, advertised employment in service sectors rather than comprehensive labour market dynamics. Extensions to informal economy work, manual trades, or non-advertised positions would establish generalisability beyond the current empirical scope.

10.5 Future Research Directions

The restructuring theory opens multiple avenues for future research extending and testing the framework in new contexts.

10.5.1 Post-Pandemic and Generative AI Extensions

The most pressing research priority involves extending cluster-level analysis to post-2020 data to examine how COVID-19 and generative AI have affected restructuring patterns. Did remote work acceleration shift the balance between task unbundling and recombination? Has generative AI (ChatGPT, GitHub Copilot, Midjourney) produced cluster-level impacts different from earlier automation waves? Are creative destruction rates accelerating as AI capabilities expand?

Preliminary evidence suggests generative AI may operate differently than earlier automation by targeting cognitive tasks previously considered safe from automation. If true, restructuring theory may require modification: cognitive task automation might produce occupation-level displacement rather than within-occupation restructuring if cognitive tasks are less decomposable than routine tasks. Testing this hypothesis requires comparing cluster dynamics in cognitive vs. routine occupations during the generative AI era.

10.5.2 Cross-National Comparative Research

Applying cluster-level methodology to job advertisement data from multiple countries would establish whether restructuring theory generalises across institutional contexts or depends on specific labour market configurations. Key questions include: Do countries with strong employment protection exhibit lower creative destruction rates? Do coordinated market economies (Germany, Sweden) show different cluster proliferation patterns than liberal market economies (US, UK, Australia)? Do vocational training systems affect cluster stability by producing occupation-specific vs. general skills?

Comparative research could identify institutional mechanisms that mediate restructuring impacts, providing evidence for policy design. If cluster proliferation rates prove similar across institutional contexts, restructuring theory captures fundamental automation dynamics independent of institutional mediation. If rates vary systematically, institutions provide policy levers for managing restructuring processes.

10.5.3 Institutional, Distributional, and Resistance Research

The varieties of capitalism framework predicts systematic differences in restructuring patterns across institutional contexts (P. A. Hall & Soskice, 2001; Thelen, 2004). Future research should test whether coordinated market economies with strong industry associations, employment protection, and vocational training systems exhibit different cluster prolifera-

tion rates, noise proportions, or creative destruction dynamics compared to liberal market economies. If restructuring patterns prove institutionally malleable, this provides evidence for policy interventions; if patterns remain similar despite institutional differences, automation may override institutional mediation. Specifically, do collective bargaining agreements protect entry-level pathways more effectively than market-driven adjustment? Do industry training levies produce different cluster stability patterns than individual firm decisions? Do employment protection regulations affect which workers successfully transition versus exit labour markets?

Moreover, the power dynamics and burden-shifting documented in the literature suggest need for research on worker resistance, collective action, and union roles in automation transitions. Do unionized workplaces exhibit different cluster transition success rates? Can collective bargaining agreements ensure training funding, protected transition time, or compensation for knowledge documentation? What organising strategies enable workers to contest automation decisions rather than merely adapt to them? Research on informal resistance—work slowdowns, strategic non-cooperation with automation projects, solidarity networks supporting displaced workers—would reveal agency invisible in administrative data (Burawoy, 1979).

Finally, the thesis lacks intersectional analysis examining how restructuring impacts vary by gender, race, age, disability status, and other axes of inequality. Labor process and precarious work theories predict that automation's costs concentrate among already-disadvantaged groups (Kalleberg, 2011; Standing, 2011; Weil, 2014). Linking cluster data to demographic employment records would establish whether women, older workers, or racial minorities face greater difficulty transitioning between clusters, whether declining clusters disproportionately employ vulnerable populations, and whether expanding clusters systematically exclude certain demographic groups through credentialing, network access, or discrimination. Such analysis would enable targeted policy interventions addressing distributional inequities rather than assuming restructuring affects all workers uniformly.

10.5.4 Worker-Level Outcome Studies

Chapter 8 mapped feasible career transitions from the demand side—identifying which cluster pairs share sufficient skill overlap for realistic transitions. The critical next step is linking this demand-side map to supply-side worker flows. Combining job advertisement cluster data with administrative employment records would enable testing whether workers actually follow skill-proximate pathways, or whether institutional barriers (credentialing, employer preferences, geographic constraints) prevent theoretically feasible transitions. Critical questions include: What proportion of workers in declining clusters successfully

transition to expanding clusters vs. exit the labour market? Do cluster transitions produce wage gains, losses, or stability? How long do transitions take, and what factors predict success?

Worker-level analysis could also examine distributional impacts: Do older workers face greater difficulty transitioning between clusters? Do women and men experience different restructuring impacts? Does prior educational attainment buffer or exacerbate transition challenges? These questions have profound policy implications for targeting interventions to vulnerable populations. A complementary qualitative agenda—centring worker voices to examine how those experiencing automation perceive restructuring, what strategies they employ to adapt, and how collective action might redistribute automation’s costs and benefits more equitably (F. Green, 2006; Weil, 2014)—would address dimensions of restructuring that demand-side data cannot reach.

10.5.5 Automation Adoption and Cluster Dynamics

The research relies on occupation-level automation risk scores rather than direct measures of automation adoption. Future research should combine firm-level automation adoption data (technology surveys, software purchase records, AI implementation timelines) with cluster-level analysis to examine which specific automation types drive particular restructuring patterns. This would strengthen causal inference and improve prediction of future impacts.

Event study designs around specific technology releases (e.g., robotic process automation software launches, AI capability announcements) could establish temporal precedence between automation adoption and cluster emergence/decline. Difference-in-differences approaches comparing occupations with high vs. low automation exposure could isolate automation-specific effects from general restructuring trends.

10.5.6 Cluster Stability and Predictability

The finding that a median of 33% of jobs across 250 occupations (IQR: 24–41%) are classified as noise (hybrid/idiosyncratic roles) raises questions about cluster stability and predictability. Are noise jobs genuinely unique positions, or do they represent emerging clusters not yet consolidated into stable patterns? Can machine learning models predict which clusters will grow, decline, or disappear based on historical patterns?

If cluster trajectories prove predictable 2-3 years in advance, real-time labour market information systems could provide early warning of restructuring, enabling proactive adaptation. If trajectories are fundamentally unpredictable due to complex interactions between technology, demand, and competition, policy interventions must target adaptability itself rather than specific skill requirements.

10.6 Final Reflections

This thesis set out to resolve a paradox: why do automation predictions consistently overestimate job losses while workers and executives report extensive workplace restructuring? The answer lies in recognising that automation operates at multiple levels simultaneously, with different impacts visible at different analytical scales. At the occupation level, the evidence shows modest aggregate effects, consistent with critiques of technological unemployment narratives. At the cluster level, the same data reveal substantial restructuring through task reorganisation, skill demand shifts, and continuous emergence and disappearance of job types. The restructuring theory advanced here demonstrates that the question “does automation supplement or replace labour?” admits no binary answer: automation does both simultaneously at different levels of analysis, with net effects determined by the balance of restructuring, task displacement, and task reinstatement (Acemoglu & Restrepo, 2019c) occurring within occupational boundaries.

The restructuring theory developed from 2012–2019 Australian data generates testable predictions for future developments. Generative AI’s capacity to automate cognitive tasks should accelerate cluster proliferation in knowledge-intensive occupations, potentially reversing patterns observed during the robotic process automation era studied here. Remote work normalisation post-COVID may shift the balance between task unbundling and recombination as geographic constraints diminish. Platform-mediated work may exhibit fundamentally different restructuring patterns as algorithmic management enables real-time task reallocation impossible in traditional employment. Extending cluster-level measurement to post-2020 data, cross-national contexts, and platform labour markets will refine the theory’s scope and identify boundary conditions.

The thesis also demonstrates that effective policy responses depend on measurement frameworks aligned with actual restructuring processes. Occupation-level interventions systematically misallocate resources by ignoring within-occupation heterogeneity. The cluster-level approach developed here, together with the skill proximity infrastructure for mapping transition pathways, offers concrete tools for workforce development that occupation-level analysis cannot support—though implementation faces genuine institutional and political-economy constraints that future research must address.

This thesis offers empirical evidence, theoretical frameworks, and methodological tools for analysing the within-occupation restructuring that occupation-level measurement systematically obscures. By documenting how automation reshaped Australian work between 2012 and 2019 through continuous task reorganisation rather than wholesale occupation elimination, the thesis establishes a historical baseline against which subsequent research—on generative AI, post-pandemic work arrangements, and platform-mediated

labour—can be calibrated. The frameworks developed here are intended to outlast the dataset that produced them: a more accurate understanding of how technological change actually operates in contemporary labour markets begins with measurement aligned to the level at which restructuring occurs.

Bibliography

- Abbott, A. (1988). *The System of Professions: An Essay on the Division of Expert Labor*. University of Chicago Press.
- Acemoglu, D. (2000). *Labor- and Capital-Augmenting Technical Change* (tech. rep.). Cambridge, Massachusetts Institute of Technology.
- Acemoglu, D. (2024). *Simple Macroeconomics of AI* (Working Paper) (Presented at Brookings Papers on Economic Activity, Spring 2024). MIT.
- Acemoglu, D., & Autor, D. H. (2011a). Skills, Tasks and Technologies: Implications for Employment and Earnings [Comprehensive review of task-based framework showing how technology affects labor markets through task substitution rather than pure skill-biased technical change]. In D. Card & O. Ashenfelter (Eds.), *Handbook of labor economics* (pp. 1043–1171, Vol. 4). Elsevier. [https://doi.org/10.1016/S0169-7218\(11\)02410-5](https://doi.org/10.1016/S0169-7218(11)02410-5)
- Acemoglu, D., & Autor, D. H. (2011b). Skills, Tasks, and Technologies: Implications for Employment and Earnings. *Handbook of Labor Economics*, 4(B), 1043–1171. [https://doi.org/10.1016/S0169-7218\(11\)02410-5](https://doi.org/10.1016/S0169-7218(11)02410-5)
- Acemoglu, D., Autor, D. H., Dorn, D., Hanson, G. H., & Price, B. (2016). Import Competition and the Great US Employment Sag of the 2000s. *Journal of Labor Economics*, 34(S1), 141–198. <https://doi.org/10.1086/682384>
- Acemoglu, D., Autor, D. H., & Johnson, S. (2023). *Can We Have Pro-Worker AI? Choosing a Path of Machines in Service of Minds* (CEPR Policy Insight No. 123). Centre for Economic Policy Research.
- Acemoglu, D., Lelarge, C., & Restrepo, P. (2020). Competing with Robots: Firm-Level Evidence from France. *AEA Papers and Proceedings*, 110, 383–388. <https://doi.org/10.1257/pandp.20201003>

- Acemoglu, D., & Restrepo, P. (2017, March). *Robots and Jobs: Evidence from US Labor Markets* (tech. rep.). National Bureau of Economic Research. Cambridge, MA. <https://doi.org/10.3386/w23285>
- Acemoglu, D., & Restrepo, P. (2018). The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment [Framework for automation vs. new task creation and implications for labor share]. *American Economic Review*, 108(6), 1488–1542. <https://doi.org/10.1257/aer.20160696>
- Acemoglu, D., & Restrepo, P. (2019a). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, 33(2), 3–30. <https://doi.org/10.1257/jep.33.2.3>
- Acemoglu, D., & Restrepo, P. (2019b). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, 33(2), 3–30. <https://doi.org/10.1257/jep.33.2.3>
- Acemoglu, D., & Restrepo, P. (2019c). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, 33(2), 3–30. <https://doi.org/10.1257/jep.33.2.3>
- Acemoglu, D., & Restrepo, P. (2020). Robots and Jobs: Evidence from US Labor Markets. *Journal of Political Economy*, 128(6), 2188–2244. <https://doi.org/10.1086/705716>
- Acemoglu, D., & Restrepo, P. (2022). Tasks, Automation, and the Rise in U.S. Wage Inequality. *Econometrica*, 90(5), 1973–2016. <https://doi.org/10.3982/ECTA19815>
- Acemoglu, D., & Restrepo, P. (2024). Automation: Theory, Evidence, and Outlook. *Annual Review of Economics*, 16, 387–414. <https://doi.org/10.1146/annurev-economics-090523-113355>
- Aghion, P., & Howitt, P. (1992). A Model of Growth Through Creative Destruction [Schumpeterian growth model where innovations render previous technologies obsolete. Demonstrates how technological progress generates continuous structural change with both job creation and destruction.]. *Econometrica*, 60(2), 323–351. <https://doi.org/10.2307/2951599>
- Alabdulkareem, A., Frank, M. R., Sun, L., AlShebli, B., Hidalgo, C., & Rahwan, I. (2018). Unpacking the Polarization of Workplace Skills. *Science Advances*, 4(7), eaao6030. <https://doi.org/10.1126/sciadv.aao6030>
- Alabdulkareem, A., Frank, M. R., Sun, L., AlShebli, B., Hidalgo, C., & Rahwan, I. (2024). How Skills Affect Your Job Trajectory, and Their Implications for Automation by AI. *MIT Media Lab Technical Report*.
- Aliprantis, C. D., & Border, K. C. (2006). *Infinite Dimensional Analysis: A Hitchhiker's Guide* (3rd) [Comprehensive treatment of functional analysis including Arzela-Ascoli Theorem (Chapter 2) and fixed point theorems]. Springer.

- Amendola, M., & Gaffard, J.-L. (1988). *The Innovative Choice: An Economic Analysis of the Dynamics of Technology*. Blackwell.
- Anderson, M. (2014). Ford's Five-Dollar Day. <https://www.thehenryford.org/explore/blog/fords-five-dollar-day>
- Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. <https://doi.org/10.1515/9781400829828>
- Antonucci, T., & Pianta, M. (2002). Employment effects of product and process innovation in Europe. *International Review of Applied Economics*, 16(3), 295–307. <https://doi.org/10.1080/02692170210136127>
- Aristotle. (n.d.). Politics, translated by B. Jowett. <http://classics.mit.edu/Aristotle/politics.1.one.html>
- Arntz, M., Gregory, T., & Zierahn, U. (2016, May). *The Risk of Automation for Jobs in OECD Countries: A Comparative Analysis* (OECD Social, Employment and Migration Working Papers No. 189). OECD. Paris.
- Arntz, M., Gregory, T., & Zierahn, U. (2020). Revisiting the Risk of Automation. *Economics Letters*, 192, 109086. <https://doi.org/10.1016/j.econlet.2020.109086>
- Atalay, E., Phongthiengtham, P., Sotelo, S., & Tannenbaum, D. (2020a). The Evolution of Work in the United States. *American Economic Journal: Applied Economics*, 12(2), 1–34. <https://doi.org/10.1257/app.20190070>
- Atalay, E., Phongthiengtham, P., Sotelo, S., & Tannenbaum, D. (2020b). The Evolution of Work in the United States. *American Economic Journal: Applied Economics*, 12(2), 1–34. <https://doi.org/10.1257/app.20190070>
- Athey, S. (2018). The Impact of Machine Learning on Economics. In A. Agrawal, J. Gans, & A. Goldfarb (Eds.), *The economics of artificial intelligence: An agenda* (pp. 507–547). University of Chicago Press. <https://www.nber.org/books-and-chapters/economics-artificial-intelligence-agenda/impact-machine-learning-economics>
- Athey, S., & Imbens, G. W. (2019). Machine Learning Methods That Economists Should Know About. *Annual Review of Economics*, 11, 685–725. <https://doi.org/10.1146/annurev-economics-080217-053433>
- Attewell, P. (1990). What Is Skill? *Work and Occupations*, 17(4), 422–448. <https://doi.org/10.1177/0730888490017004003>
- Australian Bureau of Statistics. (2013). Australian and New Zealand Standard Classification of Occupations (ANZSCO) (Version 1.2).
- Australian Bureau of Statistics. (2019a). Australian National Accounts: National Income, Expenditure and Product.
- Australian Bureau of Statistics. (2019b). Labour Force, Australia, Detailed, Quarterly.

- Australian Bureau of Statistics. (2019c). Regional Population Growth, Australia.
- Australian Bureau of Statistics. (2019d). Trade Union Membership, Australia.
- Australian Bureau of Statistics. (2020a). 5204.0 - Australian System of National Accounts, 2019-20. <https://www.abs.gov.au/statistics/economy/national-accounts/australian-system-national-accounts/2019-20>
- Australian Bureau of Statistics. (2020b). Labour Force, Australia.
- Australian Bureau of Statistics. (2020c). Labour Force, Australia, Detailed.
- Australian Bureau of Statistics. (2024a). *Conceptual Model*. Retrieved January 15, 2025, from <https://www.abs.gov.au/statistics/classifications/osca-occupation-standard-classification-australia/2024-version-1-0/conceptual-model>
- Australian Bureau of Statistics. (2024b). *OSCA - Occupation Standard Classification for Australia, 2024, Version 1.0*. Retrieved January 15, 2025, from <https://www.abs.gov.au/statistics/classifications/osca-occupation-standard-classification-australia/latest-release>
- Australian Industry Group. (2020). *Workforce Development Needs Survey 2020* (tech. rep.). Australian Industry Group. Melbourne.
- Autor, D. H. (2013a). The 'task approach' to labour markets: an overview. *Journal of Labour Market Research*, 1–5. <https://doi.org/10.1007/s12651-013-0128-z>
- Autor, D. H. (2013b). The “Task Approach” to Labor Markets: An Overview. *Journal for Labour Market Research*, 46(3), 185–199. <https://doi.org/10.1007/s12651-013-0128-z>
- Autor, D. H. (2015a). Why Are There Still So Many Jobs? The History and Future of Workplace Automation. *Journal of Economic Perspectives*, 29(3), 3–30. <https://doi.org/10.1257/jep.29.3.3>
- Autor, D. H. (2015b). Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives*, 29(3), 3–30. <https://doi.org/10.1257/jep.29.3.3>
- Autor, D. H. (2024). *Applying AI to Rebuild Middle Class Jobs* (NBER Working Paper No. 32140). National Bureau of Economic Research.
- Autor, D. H., Chin, C., Salomons, A. M., & Seegmiller, B. (2024a). New Frontiers: The Origins and Content of New Work, 1940–2018. *Quarterly Journal of Economics*, 139(3), 1399–1465. <https://doi.org/10.1093/qje/qjae008>
- Autor, D. H., Chin, C., Salomons, A. M., & Seegmiller, B. (2024b). New Frontiers: The Origins and Content of New Work, 1940–2018. *Quarterly Journal of Economics*, 139(3), 1399–1465. <https://doi.org/10.1093/qje/qjae008>

- Autor, D. H., Dorn, D., & Hanson, G. H. (2013). The China Syndrome: Local Labor Market Effects of Import Competition in the United States. *American Economic Review*, 103(6), 2121–2168. <https://doi.org/10.1257/aer.103.6.2121>
- Autor, D. H., Levy, F., & Murnane, R. (2003). The skill of recent technological change: An empirical exploration. *The Quarterly Journal of Economics*, 1279–1333. <https://doi.org/10.1162/003355303322552801>
- Autor, D. H., Levy, F., & Murnane, R. J. (2003). The Skill Content of Recent Technological Change: An Empirical Exploration. *The Quarterly Journal of Economics*, 118(4), 1279–1333. <https://doi.org/10.1162/003355303322552801>
- Autor, D. H., & Reynolds, E. (2020). The Nature of Work after the COVID Crisis: Too Few Low-Wage Jobs. *The Hamilton Project, Brookings Institution*.
- Azar, J., Marinescu, I., & Steinbaum, M. (2022). Using Online Job Postings to Study the Labor Market. *NBER Working Paper*, (w29296). <https://doi.org/10.3386/w29296>
- Babina, T., Fedyk, A., He, A. X., & Hodson, J. (2024). AI-Driven Capital-Skill Complementarity: Implications for Skill Premiums and Labor Mobility. *Finance Research Letters*, 68, 105876. <https://doi.org/10.1016/j.frl.2024.105876>
- Bakker, A. B., & Demerouti, E. (2007). The Job Demands-Resources Model: State of the Art. *Journal of Managerial Psychology*, 22(3), 309–328. <https://doi.org/10.1108/02683940710733115>
- Bakker, A. B., & Demerouti, E. (2017). Job Demands-Resources Theory: Taking Stock and Looking Forward. *Journal of Occupational Health Psychology*, 22(3), 273–285. <https://doi.org/10.1037/ocp0000056>
- Barley, S. R., & Kunda, G. (2001). Bringing Work Back In. *Organization Science*, 12(1), 76–95. <https://doi.org/10.1287/orsc.12.1.76.10122>
- Barratt, T., Veen, A., & Goods, C. (2025). (Institutionally) Working Australia’s Gig Economy into Employment: Analysing Menulog’s Modern Award Application. *Journal of Industrial Relations*. <https://doi.org/10.1177/00221856251343386>
- Barrero, J. M., Bloom, N., & Davis, S. J. (2021). Why Working from Home Will Stick. *NBER Working Paper*, (w28731). <https://doi.org/10.3386/w28731>
- Beaudry, P., Green, D. A., & Sand, B. M. (2016). The Great Reversal in the Demand for Skill and Cognitive Tasks. *Journal of Labor Economics*, 34(S1), S199–S247. <https://doi.org/10.1086/682347>
- Becker, G. S. (1994). *Human capital: A theoretical and empirical analysis, with special reference to education* (3rd ed.). University of Chicago Press.
- Berg, J. M., Wrzesniewski, A., & Dutton, J. E. (2010). Perceiving and Responding to Challenges in Job Crafting at Different Ranks: When Proactivity Requires Adaptivity.

- Journal of Organizational Behavior*, 31(2-3), 158–186. <https://doi.org/10.1002/job.645>
- Berge, C. (1963). *Topological Spaces: Including a Treatment of Multi-Valued Functions, Vector Spaces and Convexity* (E. M. Patterson, Trans.) [Translation of *Espaces topologiques* (1959). Contains the Maximum Theorem (Chapter 6, Section 3) establishing upper hemicontinuity of optimal correspondences]. Oliver; Boyd.
- Bernstein, A., & Raman, A. (2015). The great decoupling: An interview with Erik Brynjolfsson and Andrew McAfee. *Harvard Business Review*, 66–74.
- Bessen, J. E. [James E.]. (2022). The AI Revolution and the Labor Market. *Journal of Economic Literature*, 60(3), 1005–1056.
- Bessen, J. E. [James E], Impink, S. M., Reichensperger, L., & Seamans, R. (2018). *The Business of AI Startups* (tech. rep.). Boston MA, Boston University. <https://doi.org/10.2139/ssrn.3293275>
- Bick, A., Blandin, A., & Deming, D. J. (2025). *The Rapid Adoption of Generative AI* (Working Paper). National Bureau of Economic Research.
- Blair, P. Q., & Deming, D. J. (2020). Structural Increases in Skill Demand After the Great Recession. *NBER Working Paper*, (w26680). <https://doi.org/10.3386/w26680>
- Blanchard, O. (2004). The Economic Future of Europe. *Journal of Economic Perspectives*, 18(4), 3–26. <https://doi.org/10.1257/0895330042632735>
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast Unfolding of Communities in Large Networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008. <https://doi.org/10.1088/1742-5468/2008/10/P10008>
- Bloom, N., Liang, J., Roberts, J., & Ying, Z. J. (2015). Does Working from Home Work? Evidence from a Chinese Experiment. *The Quarterly Journal of Economics*, 130(1), 165–218. <https://doi.org/10.1093/qje/qju032>
- Bloom, N., & Van Reenen, J. (2007). Measuring and Explaining Management Practices Across Firms and Countries. *Quarterly Journal of Economics*, 122(4), 1351–1408. <https://doi.org/10.1162/qjec.2007.122.4.1351>
- Borland, J., & Coelli, M. (2016). Labour Market Inequality in Australia. *Economic Record*, 92(299), 517–547. <https://doi.org/10.1111/1475-4932.12285>
- Borland, J., & Coelli, M. (2017). Are Robots Taking Our Jobs? *Australian Economic Review*, 50(4), 377–397. <https://doi.org/10.1111/1467-8462.12245>
- Borland, J., & Coelli, M. (2023a). *The Australian Labour Market and IT-Enabled Technological Change* (Working Paper No. wp2023n01). Melbourne Institute of Applied Economic and Social Research.

- Borland, J., & Coelli, M. (2023b). *The Australian Labour Market and IT-Enabled Technological Change* (Working Paper No. 01/23). Melbourne Institute of Applied Economic and Social Research. Melbourne. https://melbourneinstitute.unimelb.edu.au/__data/assets/pdf_file/0007/4429789/wp2023n01.pdf
- Bosch, G., & Charest, J. (Eds.). (2010). *Vocational Training: International Perspectives*. Routledge.
- Boxall, P., & Macky, K. (2012). High-Performance Work Systems: What, Why, How and For Whom? *Asia Pacific Journal of Human Resources*, 50(2), 169–186. <https://doi.org/10.1111/j.1744-7941.2011.00012.x>
- Boyer, R. (1990). *The Capital labor Relations in OECD Countries: From the Fordist "Golden Age" to Contrasted National Trajectories* (CEPREMAP Working Paper No. 9015). Centre pour la Recherche Economique et ses Applications. Paris, France.
- Braverman, H. (1974). *Labor and Monopoly Capital: The Degradation of Work in the Twentieth Century*. Monthly Review Press.
- Brown, J. S., & Duguid, P. (1991). Organizational Learning and Communities-of-Practice: Toward a Unified View of Working, Learning, and Innovation. *Organization Science*, 2(1), 40–57. <https://doi.org/10.1287/orsc.2.1.40>
- Brown, M. (2019). Apple patents dual screen MacBook with glass keyboard. Retrieved January 15, 2024, from <https://www.inverse.com/article/52936-apple-patents-dual-screen-macbook-with-glass-keyboard-and-it-sounds-great>
- Brynjolfsson, E., Li, D., & Raymond, L. R. (2023a). Generative AI at Work. *NBER Working Paper*, (31161). <https://doi.org/10.3386/w31161>
- Brynjolfsson, E., Li, D., & Raymond, L. R. (2023b). Generative AI at Work. *NBER Working Paper*, (w31161). <https://doi.org/10.3386/w31161>
- Brynjolfsson, E., Li, D., & Raymond, L. R. (2023c). Generative AI at Work. *Quarterly Journal of Economics*, 140(2), 889–945. <https://doi.org/10.1093/qje/qjad025>
- Brynjolfsson, E., & McAfee, A. (2014). *The Second Machine Age*. WW Norton & Company.
- Brynjolfsson, E., & Mitchell, T. (2017). What Can Machine Learning Do? Workforce Implications. *Science*, 358(6370), 1530–1534. <https://doi.org/10.1126/science.aap8062>
- Brynjolfsson, E., Mitchell, T., & Rock, D. (2018). What Can Machines Learn, and What Does It Mean for Occupations and the Economy? *AEA Papers and Proceedings*, 108, 43–47. <https://doi.org/10.1257/pandp.20181019>
- Brynjolfsson, E., Mitchell, T., & Rock, D. (2019). *Machine learning and occupational change* [Unpublished manuscript, Stanford University].
- Brynjolfsson, E., Rock, D., & Syverson, C. (2017). *Artificial Intelligence and the Modern Productivity Paradox: A Clash of Expectations and Statistics* (Working Paper

- No. 24001). National Bureau of Economic Research. <https://doi.org/10.3386/w24001>
- Burawoy, M. (1979). *Manufacturing Consent: Changes in the Labor Process under Monopoly Capitalism*. University of Chicago Press.
- Burdett, K., Shi, S., & Wright, R. (2001). Pricing and Matching with Frictions [Foundational directed search model where sellers post prices and buyers choose, generating endogenous matching function with coordination frictions]. *Journal of Political Economy*, 109(5), 1060–1085. <https://doi.org/10.1086/322836>
- Bureau of Labor Statistics. (2018). *2018 Standard Occupational Classification System*. Retrieved January 15, 2025, from <https://www.bls.gov/soc/2018/>
- Bureau of Labor Statistics. (2024, August). *A New Data Product for Occupational Skills: Methodology, Analysis, and a Guide to Using the Employment Projections Skills Data* (tech. rep.). Monthly Labor Review.
- Bureau of Labor Statistics. (2025). *Incorporating AI Impacts in BLS Employment Projections: Occupational Case Studies* (tech. rep.). U.S. Bureau of Labor Statistics. <https://www.bls.gov/opub/mlr/2025/article/incorporating-ai-impacts-in-bls-employment-projections.htm>
- Busemeyer, M. R., & Trampusch, C. (Eds.). (2012). *The Political Economy of Collective Skill Formation*. Oxford University Press.
- Caballero, R. (2010). Creative destruction. In S. N. Durlauf & L. E. Blume (Eds.), *Economic growth* (pp. 24–29). Palgrave Macmillan.
- Caballero, R., & Hammour, M. (2005). The cost of recessions revisited: a reverse-liquidationist view. *Review of Economic Studies*, 72, 313–341. <https://doi.org/10.1111/j.1467-937X.2005.00334.x>
- Cameron, A. C., & Miller, D. L. (2015). A Practitioner’s Guide to Cluster-Robust Inference. *Journal of Human Resources*, 50(2), 317–372. <https://doi.org/10.3368/jhr.50.2.317>
- Campello, R. J. G. B., Moulavi, D., & Sander, J. (2013). Density-Based Clustering Based on Hierarchical Density Estimates. *Advances in Knowledge Discovery and Data Mining*, 160–172. https://doi.org/10.1007/978-3-642-37456-2_14
- Cantillon, R. (2015, May). *Richard Cantillon’s Essay on the Nature of Trade in General* (R. van den Berg, Ed.). Routledge. <https://doi.org/10.4324/9781315794686>
- Cappelli, P. H. (2015). Skill Gaps, Skill Shortages, and Skill Mismatches: Evidence and Arguments for the United States. *ILR Review*, 68(2), 251–290. <https://doi.org/10.1177/0019793914564961>
- Caprettini, B., & Voth, H. (2017). *Rage against the machines: Labor-saving technology and unrest in England, 1830–32* (tech. rep.). Providence, RI, Brown University.

- Carlsson, B., & Jacobsson, S. (1994). Technological systems and economic policy: the diffusion of factory automation in Sweden. *Research Policy*, 23(3), 235–248. [https://doi.org/10.1016/0048-7333\(94\)90036-1](https://doi.org/10.1016/0048-7333(94)90036-1)
- Carlsson, B., Taymaz, E., & Tryggestad, K. (1991). Factory Automation and Economic Performance: A micro-to-macro analysis.
- Carnevale, A. P., Cheah, B., & Hanson, A. R. (2014). *What's It Worth? The Economic Value of College Majors*. Georgetown University Center on Education and the Workforce.
- Carnevale, A. P., Jayasundera, T., & Repnikov, D. (2014). *Understanding Online Job Ads Data: A Technical Report* (tech. rep.). Georgetown University Center on Education and the Workforce. Washington, DC.
- Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E. J., & Tavares, M. M. (2024). *Gen-AI: Artificial Intelligence and the Future of Work* (IMF Staff Discussion Note No. SDN/2024/001). International Monetary Fund. <https://doi.org/10.5089/9798400262548.006>
- Cebulla, A. (2024). The time-less threat of automation: has new technology been the predicted job killer? *Labour and Industry*, 34(3), 298–321. <https://doi.org/10.1080/10301763.2024.2363573>
- Christensen, C. (1997). *The Innovator's Dilemma: When new technologies cause great firms to fail*. Harvard Business Review Press.
- Church, K. W., & Hanks, P. (1990). Word Association Norms, Mutual Information, and Lexicography. *Computational Linguistics*, 16(1), 22–29.
- Ciriaci, D., Moncada-Paternò-Castello, P., & Voigt, P. (2016). Innovation and job creation: a sustainable relation? *Eurasian Business Review*, 6(2), 189–213. <https://doi.org/10.1007/s40821-015-0031-3>
- Coad, A., & Rao, R. (2011). The firm-level employment effects of innovations in high-tech US manufacturing industries. *Journal of evolutionary economics*, 21(2), 255–283. <https://doi.org/10.1007/s00191-010-0209-x>
- Coelli, M., & Borland, J. (2016). Job Polarisation and Earnings Inequality in Australia. *Economic Record*, 92(296), 1–27. <https://doi.org/10.1111/1475-4932.12285>
- Coelli, M., & Borland, J. (2017). *Estimating the automation risk of Australian jobs* (tech. rep.). University of Melbourne Working Paper.
- Coelli, M., & Borland, J. (2019). Behind the Headline Number: Why not to Rely on Frey and Osborne's Predictions of Potential Job Loss from Automation. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3472764>
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd). Lawrence Erlbaum Associates.

- Cortés, G. M., & Salvatori, A. (2019). Delving into the Demand Side: Changes in Workplace Specialization and Job Polarization [Provides empirical evidence for demand-side workplace specialization driving job polarization]. *Labour Economics*, 57, 164–176. <https://doi.org/10.1016/j.labeco.2019.02.004>
- Cortés, P., Pan, J., Pilossoph, L., & Zafar, B. (2024). What Is the Price of a Skill? The Value of Complementarity. *Research Policy*, 53(1), 104896. <https://doi.org/10.1016/j.respol.2023.104896>
- Costinot, A., & Vogel, J. (2010). Matching and Inequality in the World Economy. *Journal of Political Economy*, 118(4), 747–786. <https://doi.org/10.1086/655858>
- Crawford, E. R., LePine, J. A., & Rich, B. L. (2010). Linking Job Demands and Resources to Employee Engagement and Burnout: A Theoretical Extension and Meta-Analytic Test. *Journal of Applied Psychology*, 95(5), 834–848. <https://doi.org/10.1037/a0019364>
- Crouch, C., Finegold, D., & Sako, M. (1999). *Are Skills the Answer? The Political Economy of Skill Creation in Advanced Industrial Countries*. Oxford University Press.
- Crowther, D., & Green, M. (2004). *Organisational Theory*. Chartered Institute of Personnel; Development.
- Dafoe, A. (2015). On Technological Determinism. *Science, Technology, & Human Values*, 40(6), 1047–1076. <https://doi.org/10.1177/0162243915579283>
- Davis, S., Haltiwanger, J., & Schuh, S. (1996). *Job Creation and Destruction*. MIT Press.
- Dawson, N., Williams, M.-A., & Rizoïu, M.-A. (2021). Skill-Driven Recommendations for Job Transition Pathways. *PLOS ONE*, 16(8), e0254722. <https://doi.org/10.1371/journal.pone.0254722>
- de Lira, V. M., Fernandes, B., & Rodrigues, J. M. (2024). Data Science for Job Market Analysis: A Survey on Applications and Techniques. *Expert Systems with Applications*, 249, 123575. <https://doi.org/10.1016/j.eswa.2024.123575>
- De Vries, J. (1994). The industrial revolution and the industrious revolution. *Journal of economic history*, 54(2), 249–270. <https://doi.org/10.1017/S0022050700014467>
- del Río-Chanona, R. M., Mealy, P., & Farmer, J. D. (2025). Automation-Skill Complementarity: The Returns to Soft Skills in Different Stages of Technology Adoption. *Economics of Innovation and New Technology*. <https://doi.org/10.1080/10438599.2025.2468458>
- Dell’Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Kraye, L., Candelon, F., & Lakhani, K. R. (2023). Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality. *Harvard Business School Working Paper*, (24-013). <https://doi.org/10.2139/ssrn.4573321>

- Dell’Acqua, F., McFowland III, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K. C., Rajendran, S., Kraymer, L., Candelon, F., & Lakhani, K. R. (2023). Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality. *Harvard Business School Working Paper*, (24-013).
- Deming, D. J. (2017a). The Growing Importance of Social Skills in the Labor Market. *Quarterly Journal of Economics*, 132(4), 1593–1640. <https://doi.org/10.1093/qje/qjx022>
- Deming, D. J. (2017b). The value of soft skills in the labor market. *NBER Reporter*, 4, 7–11.
- Deming, D. J., & Kahn, L. B. (2018a). Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals. *Journal of Labor Economics*, 36(S1), S337–S369. <https://doi.org/10.1086/694106>
- Deming, D. J., & Kahn, L. B. (2018b). Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals. *Journal of Labor Economics*, 36(S1), S337–S369. <https://doi.org/10.1086/694106>
- Díaz, M. S., & Tomás, F. J. Q. (2002). Technological innovation and employment: data from a decade in Spain. *International Journal of Production Economics*, 75(3), 245–256. [https://doi.org/10.1016/S0925-5273\(01\)00115-3](https://doi.org/10.1016/S0925-5273(01)00115-3)
- Douglas, P. H. (1930). Technological Unemployment. *American Federationist*, 37, 923–950.
- Du, T., Kanodia, A., Brunborg, H., Vafa, K., & Athey, S. (2024). LABOR-LLM: Language-Based Occupational Representations with Large Language Models. *Working Paper*.
- Durkheim, É. (1893). *The Division of Labour in Society* (S. Lukes & G. Simpson, Eds.). Free Press.
- Durrant-Whyte, H., McCalman, L., O’Callaghan, S., Reid, A., & Steinberg, D. (2015). The impact of computerisation and automation on future employment. *Australia’s future workforce*, 56–65.
- Dwyer, R. E. (2023). Job Polarization in the United States in the 21st Century: Studying Shifts in Employment Structures Using Occupations and Sectors. *JRC Working Papers on Labour, Education and Technology*, (2023/07).
- Edmonds, D., & Bradley, T. (2015). *Mechanical boon: will automation advance Australia?* (Tech. rep. No. Research Paper 7/2015). Department of Industry, Innovation and Science. Canberra. <https://www.industry.gov.au/publications/mechanical-boon-will-automation-advance-australia>
- Edwards, R. (1979). *Contested Terrain: The Transformation of the Workplace in the Twentieth Century*. Basic Books.

- Eeckhout, J., & Kircher, P. (2010). Sorting and Decentralized Price Competition [Shows that sorting in search models with price competition depends on complementarities in both output and search, extending assignment theory to frictional markets]. *Econometrica*, 78(2), 539–574. <https://doi.org/10.3982/ECTA8121>
- Eeckhout, J., & Kircher, P. (2011). Identifying Sorting—In Theory. *Review of Economic Studies*, 78(3), 872–906. <https://doi.org/10.1093/restud/rdq034>
- Ellison, G. (1993a). Learning, Local Interaction, and Coordination [Analyzes coordination equilibria and convergence with local interactions]. *Econometrica*, 61(5), 1047–1071. <https://doi.org/10.2307/2951493>
- Ellison, G. (1993b). Learning, Local Interaction, and Coordination. *Econometrica*, 61(5), 1047–1071. <https://doi.org/10.2307/2951493>
- Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702), 1306–1308. <https://doi.org/10.1126/science.adj0998>
- Ericson, R., & Pakes, A. (1995). Markov-Perfect Industry Dynamics: A Framework for Empirical Work [Develops framework for analysing industry dynamics where firms make forward-looking entry, exit, and investment decisions.]. *Review of Economic Studies*, 62(1), 53–82. <https://doi.org/10.2307/2297841>
- Ester, M., Kriegel, H.-P., Sander, J., & Xu, X. (1996). A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96)*, 226–231.
- Estevez-Abe, M., Iversen, T., & Soskice, D. (2001). Social Protection as a Productive System (P. A. Hall & D. Soskice, Eds.). *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage*, 145–183. <https://doi.org/10.1093/0199247757.003.0004>
- Estrin, J. (2015). Kodak's First Digital Moment. <https://lens.blogs.nytimes.com/2015/08/12/kodaks-first-digital-moment/>
- Evangelista, R., & Savona, M. (2002). The impact of innovation on employment in services: evidence from Italy. *International Review of Applied Economics*, 16(3), 309–318. <https://doi.org/10.1080/02692170210136136>
- Evans, D., Mason, C., Chen, H., & Reeson, A. (2024). Accelerated demand for interpersonal skills in the Australian post-pandemic labour market. *Nature Human Behaviour*, 8(1), 32–42. <https://doi.org/10.1038/s41562-023-01788-2>
- Fair Work Commission. (2019). *Annual Wage Review 2018–19*. Fair Work Commission. Melbourne. <https://www.fwc.gov.au/awards-agreements/minimum-wages-conditions/annual-wage-reviews>

- Feldmann, H. (2013). Technological unemployment in industrial countries. *Journal of evolutionary economics*, 23, 1099–1126. <https://doi.org/10.1007/s00191-013-0308-6>
- Felten, E. W., Raj, M., & Seamans, R. (2018). A Method to Link Advances in Artificial Intelligence to Occupational Abilities. *AEA Papers and Proceedings*, 108, 54–57. <https://doi.org/10.1257/pandp.20181021>
- Felten, E. W., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12), 2195–2217. <https://doi.org/10.1002/smj.3286>
- Fernández-Macías, E. (2012a). Job Polarization in Europe? Changes in the Employment Structure and Job Quality, 1995–2007. *Work and Occupations*, 39(2), 157–182. <https://doi.org/10.1177/0730888411427078>
- Fernández-Macías, E. (2012b). Job Polarization in Europe? Changes in the Employment Structure and Job Quality, 1995–2007. *Work and Occupations*, 39(2), 157–182. <https://doi.org/10.1177/0730888411427078>
- Fernández-Macías, E. (2018). *Automation, Digitisation and Platforms: Implications for Work and Employment* (tech. rep.) (Publications Office of the European Union). Eurofound. Dublin. <https://www.eurofound.europa.eu/publications/report/2018/automation-digitisation-and-platforms-implications-for-work-and-employment>
- Fernández-Macías, E., & Hurley, J. (2017). Routine-Biased Technical Change and Job Polarization in Europe. *Socio-Economic Review*, 15(3), 563–585. <https://doi.org/10.1093/ser/mww016>
- Ferraro, S., Ferraro, D., & Ferraro, V. (2025). Robots and AI Exposure and Wage Inequality: A Within Occupation Approach. *Eurasian Business Review*. <https://doi.org/10.1007/s40821-025-00306-w>
- Finegold, D., & Soskice, D. (1988). The Failure of Training in Britain: Analysis and Prescription. *Oxford Review of Economic Policy*, 4(3), 21–53. <https://doi.org/10.1093/oxrep/4.3.21>
- Ford, M. (2009). *The Lights in the Tunnel: Automation, Accelerating Technology, and the Economy of the Future*. Acculant Publishing.
- Forsythe, E., Kahn, L. B., Lange, F., & Wiczer, D. (2020). Labor demand in the time of COVID-19: Evidence from vacancy postings and UI claims. *Journal of Public Economics*, 189. <https://doi.org/10.1016/j.jpubeco.2020.104238>
- Frank, M. R., Moro, E., & Pentland, A. (2024). From Data to Insight: Transforming Online Job Postings into Labor-Market Intelligence. *Information*, 15(8), 496. <https://doi.org/10.3390/info15080496>

- Frank, M. R., Moro, E., South, T., Rutherford, A., Pentland, A., Taska, B., & Rahwan, I. (2024). Network Constraints on Worker Mobility. *Nature Cities*, 1, 94–104. <https://doi.org/10.1038/s44284-023-00009-1>
- Freeman, C., Clark, J., & Soete, L. (1982). *Unemployment and Technical Innovation*. Pinter.
- Freeman, C., & Louca, F. (2001). *As Time Goes By: From the Industrial Revolutions to the Information Revolution*. Oxford University Press.
- Freeman, R. (2015). Who owns the robots rules the world. *IZA World of Labor*. <https://doi.org/10.15185/izawol.5>
- Freidson, E. (2001). *Professionalism: The Third Logic*. University of Chicago Press.
- Frey, C. B., & Osborne, M. A. [Michael A]. (2017). The future of employment: How susceptible are jobs to computerisation? [Working paper originally published in 2013]. *Technological Forecasting & Social Change*, 114, 254–280. <https://doi.org/10.1016/j.techfore.2016.08.019>
- Frey, C. B., & Osborne, M. A. [Michael A.]. (2024). *Generative AI and the Future of Work: A Reappraisal* [Oxford Martin School Working Paper].
- Garicano, L. (2000). Hierarchies and the Organization of Knowledge in Production. *Journal of Political Economy*, 108(5), 874–904. <https://doi.org/10.1086/317671>
- Garicano, L., & Rossi-Hansberg, E. (2006). Organization and Inequality in a Knowledge Economy. *The Quarterly Journal of Economics*, 121(4), 1383–1435. <https://doi.org/10.1093/qje/121.4.1383>
- Gathmann, C., & Schönberg, U. (2010). How General Is Human Capital? A Task-Based Approach. *Journal of Labor Economics*, 28(1), 1–49. <https://doi.org/10.1086/649786>
- Gelman, A., & Stern, H. (2006). The Difference Between “Significant” and “Not Significant” is not Itself Statistically Significant. *The American Statistician*, 60(4), 328–331. <https://doi.org/10.1198/000313006X152649>
- Georgieff, A., & Hyee, R. (2021). What Happened to Jobs at High Risk of Automation? *OECD Social, Employment and Migration Working Papers*, (255). <https://doi.org/10.1787/10bc97f4-en>
- Gmyrek, P., Berg, J., & Bescond, D. (2023). *Generative AI and Jobs: A Global Analysis of Potential Effects on Job Quantity and Quality* (ILO Working Paper No. 96). International Labour Organization. Geneva.
- Goergen, T., Muffels, R., & Dekker, R. (2024). Occupations and Careers Within Organizations: Do Organizations Facilitate Unequal Wage Growth? *Social Science Research*, 119, 102979. <https://doi.org/10.1016/j.ssresearch.2024.102979>

- Goos, M., & Manning, A. (2007). Lousy and Lovely Jobs: The Rising Polarization of Work in Britain. *Review of Economics and Statistics*, 89(1), 118–133. <https://doi.org/10.1162/rest.89.1.118>
- Goos, M., Manning, A., & Salomons, A. (2014). Explaining Job Polarization: Routine-Biased Technological Change and Offshoring. *American Economic Review*, 104(8), 2509–2526. <https://doi.org/10.1257/aer.104.8.2509>
- Gottschalk, P. T. (1978). A Comparison of Marginal Productivity and Earnings by Occupation. *ILR Review*, 31(3), 368–378. <https://doi.org/10.1177/001979397803100307>
- Grace, K., Salvatier, J., Dafoe, A., Zhang, B., & Evans, O. (2018). When Will AI Exceed Human Performance? Evidence from AI Experts. *arXiv*, 1–21. <https://doi.org/10.1613/jair.1.11222>
- Graetz, G., & Michaels, G. (2017). Is Modern Technology Responsible for Jobless Recoveries? *American Economic Review*, 107(5), 168–173. <https://doi.org/10.1257/aer.p20171100>
- Green, F. (2006). *Demanding Work: The Paradox of Job Quality in the Affluent Economy*. Princeton University Press.
- Green, G. P., Haines, A., & Wimmer, M. (2023). Assessing the Impact of Technological Change on Similar Occupations: Implications for Employment Alternatives. *PLoS ONE*, 18(9), e0291445. <https://doi.org/10.1371/journal.pone.0291445>
- Greenan, N., & Guellec, D. (2000). Technological innovation and employment reallocation. *Labour*, 14(4), 547–590. <https://doi.org/10.1111/1467-9914.00146>
- Greene, W. H. (2018). *Econometric Analysis* (8th). Pearson Education.
- Griliches, Z. (1969). Capital-Skill Complementarity. *Review of Economics and Statistics*, 51(4), 465–468. <https://doi.org/10.2307/1926439>
- Griliches, Z. (1986). Economic Data Issues. In Z. Griliches & M. D. Intriligator (Eds.), *Handbook of econometrics* (pp. 1465–1514, Vol. 3). Elsevier. [https://doi.org/10.1016/S1573-4412\(86\)03005-2](https://doi.org/10.1016/S1573-4412(86)03005-2)
- Griliches, Z. (1990). *Patent statistics as economic indicators: A survey part I* (tech. rep.). Cambridge, MA, National Bureau of Economic Research.
- Grimmer, J., & Stewart, B. M. (2013). Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts. *Political Analysis*, 21(3), 267–297. <https://doi.org/10.1093/pan/mps028>
- Gross, I., & Hansen, J. (2022). Assessing Australian Monetary Policy in the Twenty-First Century. *Economic Record*, 98(301), 158–182. <https://doi.org/10.1111/1475-4932.12689>

- Hackman, J. R., & Oldham, G. R. (1976). Motivation through the Design of Work: Test of a Theory. *Organizational Behavior and Human Performance*, 16(2), 250–279. [https://doi.org/10.1016/0030-5073\(76\)90016-7](https://doi.org/10.1016/0030-5073(76)90016-7)
- Hajkowicz, S. A., Reeson, A., Ruber, L., Bratanova, A., Hodgers, L., Mason, C., & Boughen, N. (2016). *Tomorrow's Digitally Enabled Workforce: Megatrends and Scenarios for Jobs and Employment in Australia over the Coming Twenty Years* (tech. rep.). CSIRO. Brisbane.
- Hall, B. H., Lotti, F., & Mairesse, J. (2008). Employment, innovation, and productivity: evidence from Italian microdata. *Industrial and Corporate Change*, 17(4), 813–839. <https://doi.org/10.1093/icc/dtn022>
- Hall, P. A., & Soskice, D. (Eds.). (2001). *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage*. Oxford University Press.
- Handel, M. J. (2003). Skills Mismatch in the Labor Market. *Annual Review of Sociology*, 29, 135–165. <https://doi.org/10.1146/annurev.soc.29.010202.100030>
- Hannigan, T. R., Haans, R. F. J., Vakili, K., Tchalian, H., Glaser, V. L., Wang, M. S., & Kaplan, S. (2019). Topic Modeling in Management Research: Rendering New Theory from Textual Data. *Academy of Management Annals*, 13(2), 586–632. <https://doi.org/10.5465/annals.2017.0099>
- Harrison, R., Jaumandreu, J., Mairesse, J., & Peters, B. (2008, August). *Does innovation stimulate employment? A firm-level analysis using comparable micro-data from four European countries* (tech. rep. No. 14216) (Later published in *International Journal of Industrial Organization*, 2014, vol. 35(C), pages 29–43). National Bureau of Economic Research. Cambridge, MA. <https://doi.org/10.3386/w14216>
- Healy, J., Nicholson, D., & Pekarek, A. (2025). The Triangular Relationship in Platform Gig Work: Consumers, Platform Beneficence and Worker Vulnerability. *New Technology, Work and Employment*. <https://doi.org/10.1111/ntwe.12310>
- Heckman, J. J., & Kautz, T. (2012). Hard Evidence on Soft Skills. *Labour Economics*, 19(4), 451–464. <https://doi.org/10.1016/j.labeco.2012.05.014>
- Hershbein, B., & Kahn, L. B. (2018a). Do Recessions Accelerate Routine-Biased Technological Change? Evidence from Vacancy Postings. *American Economic Review*, 108(7), 1737–1772. <https://doi.org/10.1257/aer.20161570>
- Hershbein, B., & Kahn, L. B. (2018b). Do Recessions Accelerate Routine-Biased Technological Change? Evidence from Vacancy Postings. *American Economic Review*, 108(7), 1737–1772. <https://doi.org/10.1257/aer.20161570>
- Hicks, J. R. (1932). *The Theory of Wages*. Macmillan.
- Hicks, J. R. (1973). *Capital and Time*. Oxford University Press.

- Hidalgo, C. A., Klinger, B., Barabási, A.-L., & Hausmann, R. (2007). The Product Space Conditions the Development of Nations. *Science*, 317(5837), 482–487. <https://doi.org/10.1126/science.1144581>
- Hirsch, B. T., & Macpherson, D. A. (2020). Union Membership and Coverage Database from the CPS [Annual compilation of US union statistics]. *Industrial and Labor Relations Review*.
- Hopenhayn, H. A. (1992). Entry, Exit, and Firm Dynamics in Long Run Equilibrium [Foundational model of industry dynamics with endogenous entry and exit under heterogeneous productivity. Establishes that stationary equilibrium features continuous firm turnover even absent aggregate shocks.]. *Econometrica*, 60(5), 1127–1150. <https://doi.org/10.2307/2951541>
- Hosios, A. J. (1990). On the Efficiency of Matching and Related Models of Search and Unemployment. *Review of Economic Studies*, 57(2), 279–298. <https://doi.org/10.2307/2297382>
- Hui, X., Reshef, O., & Zhou, L. (2024). The Short-Term Effects of Generative Artificial Intelligence on Employment: Evidence from an Online Labor Market. *Organization Science*. <https://doi.org/10.1287/orsc.2023.18441>
- Humlum, A., & Vestergaard, E. (2024). *Large Language Models, Small Labor Market Effects* (tech. rep. No. 33777). National Bureau of Economic Research. Cambridge, MA. <https://doi.org/10.3386/w33034>
- Hunt, J., & Nunn, R. (2022). Has U.S. Employment Really Polarized? A Critical Reappraisal. *Labour Economics*, 75, 102117. <https://doi.org/10.1016/j.labeco.2022.102117>
- Huselid, M. A. (1995). The Impact of Human Resource Management Practices on Turnover, Productivity, and Corporate Financial Performance. *Academy of Management Journal*, 38(3), 635–672. <https://doi.org/10.5465/256741>
- International Labour Organization. (2012). *International Standard Classification of Occupations: ISCO-08*. <https://www.ilo.org/public/english/bureau/stat/isco/isco08/>
- Jesuthasan, R., & Boudreau, J. W. (2022). *Work Without Jobs: How to Reboot Your Organization's Work Operating System*. MIT Press.
- Johnson, R. A., & Wichern, D. W. (2007). *Applied Multivariate Statistical Analysis* (6th). Pearson Prentice Hall.
- Jovanovic, B. (1982). Selection and the Evolution of Industry [Model where firms learn about their efficiency over time, generating equilibrium entry and exit patterns. Efficient firms grow and survive; inefficient firms contract and exit.]. *Econometrica*, 50(3), 649–670. <https://doi.org/10.2307/1912606>

- Julien, B., Kennes, J., & King, I. (2000). Bidding for Labor [Competing-auction theory of labor markets with strategic posting]. *Review of Economic Dynamics*, 3(4), 619–649. <https://doi.org/10.1006/redy.1999.0089>
- Kakutani, S. (1941). A Generalization of Brouwer's Fixed Point Theorem [Original proof of Kakutani's fixed point theorem for set-valued correspondences, fundamental for Nash equilibrium existence]. *Duke Mathematical Journal*, 8(3), 457–459. <https://doi.org/10.1215/S0012-7094-41-00838-4>
- Kaldor, N. (1957). A Model of Economic Growth. *Economic Journal*. <https://doi.org/10.2307/2227704>
- Kalleberg, A. L. (2011). *Good Jobs, Bad Jobs: The Rise of Polarized and Precarious Employment Systems in the United States, 1970s to 2000s*. Russell Sage Foundation.
- Kass, R. E., & Raftery, A. E. (1995). Bayes Factors. *Journal of the American Statistical Association*, 90(430), 773–795. <https://doi.org/10.1080/01621459.1995.10476572>
- Kässi, O., & Lehdonvirta, V. (2018). Online Labour Index: Measuring the Online Gig Economy for Policy and Research. *Technological Forecasting and Social Change*, 137, 241–248. <https://doi.org/10.1016/j.techfore.2018.07.056>
- Katsoulacos, Y. S. (1986). *The Employment Effect of Technical Change*. Wheatsheaf.
- Katz, L. F., & Murphy, K. M. (1992). Changes in Relative Wages, 1963–1987: Supply and Demand Factors. *Quarterly Journal of Economics*, 107(1), 35–78. <https://doi.org/10.2307/2118323>
- Katz, M. B. (1972). Occupational Classification in History. *The Journal of Interdisciplinary History*, 3(1), 63–88. <https://doi.org/10.2307/202462>
- Kellogg, K. C., & Spencer, D. A. (2024). AI, Automation and the Lightning of Work. *AI and Society*. <https://doi.org/10.1007/s00146-024-01959-3>
- Kennedy, S., Breunig, R., & Harding, D. (2019). Technology Adoption in Australian Manufacturing: Survey Evidence. *Economic Record*, 95(310), 271–290. <https://doi.org/10.1111/1475-4932.12482>
- Keynes, J. M. (1931). The economic possibilities for our grandchildren (1930). In *Essays in persuasion* (pp. 321–332). Palgrave Macmillan.
- Keynes, J. M. (1936). *The General Theory of Employment, Interest and Money*. Wordsworth Editions.
- King, G. (1997). *A Solution to the Ecological Inference Problem: Reconstructing Individual Behavior from Aggregate Data*. Princeton University Press.
- Knight, C., McLarnon, M. J. W., Wenzel, R., & Parker, S. K. (2022). The Importance of Relational Work Design Characteristics: A Person-Centred Approach. *Australian Journal of Management*, 47(4), 705–728. <https://doi.org/10.1177/03128962211073021>
- Knight, F. H. (1951). *The Economic Organization*. Transaction Publishers.

- Kortum, S. S. (1997). Research, patenting, and technological change. *Econometrica*, 65(6), 1389–1419. <https://doi.org/10.2307/2171831>
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet Classification with Deep Convolutional Neural Networks. *Advances in Neural Information Processing Systems*, 25, 1097–1105.
- Krusell, P., Ohanian, L. E., Ríos-Rull, J.-V., & Violante, G. L. (2000). Capital-Skill Complementarity and Inequality: A Macroeconomic Analysis. *Econometrica*, 68(5), 1029–1053. <https://doi.org/10.1111/1468-0262.00150>
- Kuhn, H. W. (1955). The Hungarian Method for the Assignment Problem. *Naval Research Logistics Quarterly*, 2(1–2), 83–97. <https://doi.org/10.1002/nav.3800020109>
- Labini, P. S. (1956). *Oligopoly and Technical Progress*. Harvard University Press.
- Lachenmaier, S., & Rottmann, H. (2011). Effects of innovation on employment: A dynamic panel analysis. *International Journal of Industrial Organization*, 29(2), 210–220. <https://doi.org/10.1016/j.ijindorg.2010.05.004>
- Lane, M., & Saint-Martin, A. (2021). *The Impact of Artificial Intelligence on the Labour Market: What Do We Know So Far?* (OECD Social, Employment and Migration Working Papers No. 256). OECD. <https://doi.org/10.1787/7c895724-en>
- Lane, M., & Saint-Martin, A. (2023). Augmentation or Automation? An Occupational Analysis. *Oxford Bulletin of Economics and Statistics*, 85(6), 1190–1218. <https://doi.org/10.1111/obes.12577>
- Lave, J., & Wenger, E. (1991). *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press.
- Layard, R., Nickell, S., & Jackman, R. (1991). *Unemployment: Macroeconomic Performance and the Labor Market*. Oxford University Press.
- Lazear, E. P. (2009). Firm-Specific Human Capital: A Skill-Weights Approach. *Journal of Political Economy*, 117(5), 914–940. <https://doi.org/10.1086/648671>
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. *Nature*, 521(7553), 436–444. <https://doi.org/10.1038/nature14539>
- Lemieux, P. (2014). *Who Needs Jobs?* Palgrave Macmillan US. <https://doi.org/10.1057/9781137353511>
- Leonardi, P. M., & Barley, S. R. (2010). What's Under Construction Here? Social Action, Materiality, and Power in Constructivist Studies of Technology and Organizing. *Academy of Management Annals*, 4(1), 1–51. <https://doi.org/10.5465/19416521003654160>
- Li, J., Liu, M., & Zhang, L. (2024). Embracing Artificial Intelligence (AI) with Job Crafting: Exploring Trickle-Down Effect and Employees' Outcomes. *Tourism Management*, 103, 104886. <https://doi.org/10.1016/j.tourman.2024.104886>

- Liao, S. (2018). Apple patented a keyboard that can't be defeated by crumbs. Retrieved January 15, 2024, from <https://www.theverge.com/2018/3/9/17100594/apple-patent-dirt-debris-liquid-snacks>
- Lichtenthaler, P. W., & Fischbach, A. (2019). Measuring Job Crafting Across Cultures: Lessons Learned From Comparing a German and an Australian Sample. *Frontiers in Psychology, 10*, 991. <https://doi.org/10.3389/fpsyg.2019.00991>
- Liu, X., & Wang, P. (2024). AI in the Workplace: A Systematic Review of Skill Transformation in the Industry. *Administrative Sciences, 14*(6), 127. <https://doi.org/10.3390/admsci14060127>
- Loaiza, F., & Rigobon, R. (2024). *Automation Risk and Augmentation Potential: New Measures for the AI Age* (Working Paper No. 2024-05). US Treasury Office of Economic Policy.
- Lopes de Melo, R. (2018). Firm Wage Differentials and Labor Market Sorting: Reconciling Theory and Evidence. *Journal of Political Economy, 126*(1), 313–346. <https://doi.org/10.1086/695505>
- Lynch, M. (2007). Ideas and Perspectives. In E. Hackett, O. Amsterdamska, M. Lynch, & J. Wajcman (Eds.), *The handbook of science and technology studies* (3rd ed., pp. 9–11). MIT Press.
- Mäkelä, E., & Stephany, F. (2024). Complement or Substitute? How AI Increases the Demand for Human Skills. *arXiv preprint*, (arXiv:2412.19754). <https://arxiv.org/abs/2412.19754>
- Malthus, T. R. (1964). *Principles of Political Economy*. M. Kelley.
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press. <https://nlp.stanford.edu/IR-book/>
- Marinescu, I., & Rathelot, R. (2018). Mismatch Unemployment and the Geography of Job Search. *American Economic Journal: Macroeconomics, 10*(3), 42–70. <https://doi.org/10.1257/mac.20160312>
- Marinescu, I., & Wolthoff, R. (2020). Opening the Black Box of the Matching Function: The Power of Words. *Journal of Labor Economics, 38*(2), 535–568. <https://doi.org/10.1086/705903>
- Marshall, A. (1920). *Principles of Economics*. Macmillan.
- Martin-Caughey, A. (2021). What's in an Occupation? Investigating Within-Occupation Variation and Gender Segregation Using Job Titles and Task Descriptions. *American Sociological Review, 86*(5), 960–999. <https://doi.org/10.1177/00031224211042053>
- Marx, K. (1867). *Das Kapital* (D. Fernbach, Ed.; Vol. 1). Penguin Classics.
- Marx, K. (1977). *Capital* (Vol. 1). Random House.

- McAfee, A. (2012). The Great Decoupling of the US Economy. <http://andrewmcafee.org/2012/12/the-great-decoupling-of-the-us-economy/>
- McCulloch, J. (1818). *On Ricardo's Principles of Political Economy and Taxation*. Edinburgh Review.
- McElheran, K., Li, D., Brynjolfsson, E., Kroff, Z., Dinlersoz, E., Foster, L., & Zolas, N. (2024). AI adoption in America: Who, what, and where. *Journal of Economics & Management Strategy*, 33(1), 5–27. <https://doi.org/10.1111/jems.12576>
- McGuinness, S., Pouliakas, K., & Redmond, P. (2018). Skills Mismatch: Concepts, Measurement and Policy Approaches. *Journal of Economic Surveys*, 32(4), 985–1015. <https://doi.org/10.1111/joes.12254>
- McInnes, L., Healy, J., & Astels, S. (2017). hdbscan: Hierarchical Density Based Clustering. *Journal of Open Source Software*, 2(11), 205. <https://doi.org/10.21105/joss.00205>
- Melitz, M. J. (2003). The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity. *Econometrica*, 71(6), 1695–1725. <https://doi.org/10.1111/1468-0262.00467>
- Menzio, G., & Shi, S. (2010). Block Recursive Equilibria for Stochastic Models of Search on the Job [Develops directed search framework where equilibrium value functions do not depend on worker distribution, enabling tractable analysis of on-the-job search]. *Journal of Economic Theory*, 145(4), 1453–1494. <https://doi.org/10.1016/j.jet.2010.03.008>
- Milgrom, P., & Shannon, C. (1994). Monotone Comparative Statics [Generalises comparative statics results using lattice-theoretic methods. Establishes single crossing conditions sufficient for monotone comparative statics.]. *Econometrica*, 62(1), 157–180. <https://doi.org/10.2307/2951479>
- Mill, J. (1821). *Elements of Political Economy*. C. Baldwin.
- Minas, J., & French, B. (2025). The Classification of Platform Workers in the Australian Context. *Journal of Industrial Relations*. <https://doi.org/10.1177/00221856241256477>
- Mincer, J. (1974). *Schooling, Experience, and Earnings*. National Bureau of Economic Research.
- Modestino, A. S., Shoag, D., & Ballance, J. (2016). Upskilling: Do Employers Demand Greater Skill When Workers Are Plentiful? *Review of Economics and Statistics*, 98(5), 793–805. https://doi.org/10.1162/REST_a_00546
- Modestino, A. S., Shoag, D., & Ballance, J. (2020). Upskilling: Do Employers Demand Greater Skill When Workers Are Plentiful? *Review of Economics and Statistics*, 102(4), 793–805. https://doi.org/10.1162/rest_a_00835
- Moen, E. R. (1997). Competitive Search Equilibrium. *Journal of Political Economy*, 105(2), 385–411. <https://doi.org/10.1086/262077>

- Mokyr, J., Vickers, C., & Ziebarth, N. L. (2015). The History of Technological Anxiety and the Future of Economic Growth: Is This Time Different? *Journal of Economic Perspectives*, 29(3), 31–50. <https://doi.org/10.1257/jep.29.3.31>
- Moretti, E. (2012). *The New Geography of Jobs*. Houghton Mifflin Harcourt.
- Morgeson, F. P., & Humphrey, S. E. (2006). The Work Design Questionnaire (WDQ): Developing and Validating a Comprehensive Measure for Assessing Job Design and the Nature of Work. *Journal of Applied Psychology*, 91(6), 1321–1339. <https://doi.org/10.1037/0021-9010.91.6.1321>
- Morris, S., & Shin, H. S. (2003). Global Games: Theory and Applications. In M. Dewatripont, L. P. Hansen, & S. J. Turnovsky (Eds.), *Advances in economics and econometrics: Theory and applications, eighth world congress* (pp. 56–114, Vol. 1). Cambridge University Press.
- Moulavi, D., Jaskowiak, P. A., Campello, R. J., Zimek, A., & Sander, J. (2014). Density-Based Clustering Validation. *Proceedings of the 2014 SIAM International Conference on Data Mining (SDM)*, 839–847. <https://doi.org/10.1137/1.9781611973440.96>
- Nania, J., Bonella, H., Restuccia, D., & Taska, B. (2019). *No Longer Optional: Employer Demand for Digital Skills* (tech. rep.). Boston, MA, Burning Glass Technologies.
- Nankervis, A., & Cameron, R. (2022). Capabilities and Competencies for Digitised Human Resource Management: Perspectives from Australian HR Professionals. *Asia Pacific Journal of Human Resources*, 60(4), 982–1003. <https://doi.org/10.1111/1744-7941.12351>
- Nankervis, A., Connell, J. A., Cameron, R., & Prikshat, V. (2019). 'Are We There Yet?' Australian HR Professionals and the Fourth Industrial Revolution. *Asia Pacific Journal of Human Resources*, 59(1), 3–19. <https://doi.org/10.1111/1744-7941.12245>
- National Academies of Sciences, Engineering, and Medicine. (2024). *Artificial Intelligence and the Future of Work*. The National Academies Press. <https://doi.org/10.17226/27644>
- National Skills Commission. (2019). *Survey of Employers' Recruitment Experiences: 2019 Data Report* (tech. rep.). Canberra, Australian Government.
- Neary, J. P. (1981). On the Short-run Effects of Technological Progress. *Oxford Economic Papers*, 32, 224–233. <https://doi.org/10.1093/oxfordjournals.oep.a041498>
- Nedelkoska, L., & Quintini, G. (2018, March). *Automation, skills use and training* (tech. rep. No. 202). OECD Publishing. Paris. <https://doi.org/10.1787/2e2f4eea-en>
- Neffke, F., & Henning, M. (2013). Skill-Relatedness and Firm Diversification. *Strategic Management Journal*, 34(3), 297–316. <https://doi.org/10.1002/smj.2014>

- Neffke, F., Otto, A., & Weyh, A. (2017). Inter-Industry Labor Flows. *Journal of Economic Behavior & Organization*, 142, 275–292. <https://doi.org/10.1016/j.jebo.2017.07.003>
- Newman, M. E. J. (2006). Modularity and Community Structure in Networks. *Proceedings of the National Academy of Sciences*, 103(23), 8577–8582. <https://doi.org/10.1073/pnas.0601602103>
- Noy, S., & Zhang, W. (2023a). Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence. *Science*, 381(6654), 187–192. <https://doi.org/10.1126/science.adh2586>
- Noy, S., & Zhang, W. (2023b). Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence. *Science*, 381(6654), 187–192. <https://doi.org/10.1126/science.adh2586>
- OECD. (2019a). *Negotiating Our Way Up: Collective Bargaining in a Changing World of Work*. OECD Publishing. Paris. <https://doi.org/10.1787/1fd2da34-en>
- OECD. (2019b). *OECD Employment Outlook 2019: The Future of Work*. <https://doi.org/10.1787/9ee00155-en>
- OECD. (2021). *Burning Glass Technologies' data use in policy-relevant analysis: An occupation-level assessment* (OECD Science, Technology and Industry Working Papers No. 2021/05). OECD Publishing. Paris. <https://doi.org/10.1787/cd75c3e7-en>
- OECD. (2024a). *AI and the Changing Demand for Skills in the Labour Market* (OECD Employment Outlook 2024). Organisation for Economic Co-operation and Development. Paris. <https://doi.org/10.1787/d26c17aa-en>
- OECD. (2024b, November). *What Impact Has AI Had on Wage Inequality?* (Tech. rep.). Organisation for Economic Co-operation and Development. https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/11/what-impact-has-ai-had-on-wage-inequality_8943dfe0/7fb21f59-en.pdf
- Ok, E. A. (2007). *Real Analysis with Economic Applications*. Princeton University Press.
- Osterman, P., & Shulman, B. (2011). *Good Jobs America: Making Work Better for Everyone*. Russell Sage Foundation.
- Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A Model for Types and Levels of Human Interaction with Automation. *IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans*, 30(3), 286–297. <https://doi.org/10.1109/3468.844354>
- Parent-Rocheleau, X., & Parker, S. K. (2022). Algorithms as Work Designers: How Algorithmic Management Influences the Design of Jobs. *Human Resource Management Review*, 32(3), 100838. <https://doi.org/10.1016/j.hrmr.2021.100838>

- Parker, S. K. (2014). Beyond Motivation: Job and Work Design for Development, Health, Ambidexterity, and More. *Annual Review of Psychology*, 65, 661–691. <https://doi.org/10.1146/annurev-psych-010213-115208>
- Parker, S. K., & Grote, G. (2022). Automation, Algorithms, and Beyond: Why Work Design Matters More Than Ever in a Digital World. *Applied Psychology*, 71(4), 1171–1204. <https://doi.org/10.1111/apps.12241>
- Parker, S. K., & Jorritsma, K. (2024). Healthier Work in the Age of AI: An Application of the SMART Work Design Model. In *Research handbook on work and wellbeing*. Springer. https://doi.org/10.1007/978-3-031-87616-5_2
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Peetz, D. (2012). *The Workplace Relations Framework* [Analysis of Australian industrial relations system]. Cambridge University Press.
- Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023). The Impact of AI on Developer Productivity: Evidence from GitHub Copilot. *arXiv preprint*. <https://doi.org/10.48550/arXiv.2302.06590>
- Petriglieri, G., Ashford, S. J., & Wrzesniewski, A. (2019). Agony and Ecstasy in the Gig Economy: Cultivating Holding Environments for Precarious and Personalized Work Identities. *Administrative Science Quarterly*, 64(1), 124–170. <https://doi.org/10.1177/0001839218759646>
- Petrongolo, B., & Pissarides, C. A. (2001). Looking into the Black Box: A Survey of the Matching Function. *Journal of Economic Literature*, 39(2), 390–431. <https://doi.org/10.1257/jel.39.2.390>
- Petropoulos, G. (2021). *Automation, COVID-19, and Labor Markets* (tech. rep. No. 1229). Asian Development Bank Institute. Tokyo.
- Philbeck, T., & Davis, N. (2018). The Fourth Industrial Revolution: Shaping a New Era. *Journal of International Affairs*, 72(1), 17–22.
- Pianta, (2005). Innovation and Employment. In J. Fagerberg, D. Mowery, & R. R. Nelson (Eds.), *Handbook of innovation*. Oxford University Press.
- Pianta, M. (2000). The employment impact of product and process innovations. In M. Vivarelli & P. M (Eds.), *The employment impact of innovation: Evidence and policy* (pp. 77–95). Routledge.
- Pini, P. (1995). Economic growth, technological change and employment: empirical evidence for a cumulative growth model with external causation for nine OECD

- countries: 1960–1990. *Structural Change and Economic Dynamics*, 6(2), 185–213. [https://doi.org/10.1016/0954-349X\(94\)00008-W](https://doi.org/10.1016/0954-349X(94)00008-W)
- Piva, M., & Vivarelli, M. (2005). Innovation and employment: evidence from italian microdata. *Journal of Economics*, 86(1), 65–83. <https://doi.org/10.1007/s00712-005-0140-z>
- Piva, M. [Mariacristina], & Vivarelli, M. [Marco]. (2018). Technological change and employment: is Europe ready for the challenge? *Eurasian Business Review*, 8(1), 13–32. <https://doi.org/10.1007/s40821-017-0100-x>
- Poletaev, M., & Robinson, C. (2008). Human Capital Specificity: Evidence from the Dictionary of Occupational Titles and Displaced Worker Surveys, 1984–2000. *Journal of Labor Economics*, 26(3), 387–420. <https://doi.org/10.1086/588180>
- Porter, M. E., & Stern, S. (2000). *Measuring the 'ideas' production function: evidence from international patent output* (tech. rep.). Cambridge, MA, NBER Working Paper No 7891.
- Prisecaru, P. (2016). Challenges of the fourth industrial revolution. *Knowledge Horizons - Economics*, 8(1), 57–62.
- Quesnay, F. (1894). *Tableau Oeconomiqueh [microform]: First Printed in 1758 and Now Reproduced in Facsimile for the British Economic Association*. Macmillan & Company.
- Rae, J. (1894). *Eight Hours for Work*. Macmillan.
- Raisch, S., & Krakowski, S. (2021). Artificial Intelligence and Management: The Automation–Augmentation Paradox. *Academy of Management Review*, 46(1), 192–210. <https://doi.org/10.5465/amr.2018.0072>
- Reserve Bank of Australia. (2014, February). *Statement on Monetary Policy*. Reserve Bank of Australia. Sydney.
- Ricardo, D. (1821). Chapter 31: On Machinery. In *On the principles of political economy and taxation*. John Murray.
- Rogers, E. M. (2003). *Diffusion of Innovations* (5th). Free Press.
- Rosen, S. (1974). Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition [Foundational paper on hedonic pricing establishing how product characteristics determine equilibrium prices in differentiated goods markets]. *Journal of Political Economy*, 82(1), 34–55. <https://doi.org/10.1086/260169>
- Rousseeuw, P. J. (1987). Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Roy, A. D. (1951). Some Thoughts on the Distribution of Earnings. *Oxford Economic Papers*, 3(2), 135–146. <https://doi.org/10.1093/oxfordjournals.oep.a041827>

- Rudolph, C. W., Katz, I. M., Lavigne, K. N., & Zacher, H. (2017). Job Crafting: A Meta-Analysis of Relationships with Individual Differences, Job Characteristics, and Work Outcomes. *Journal of Vocational Behavior*, 102, 112–138. <https://doi.org/10.1016/j.jvb.2017.05.008>
- Saint-Paul, G. (2004). Why are European Countries Diverging in their Unemployment Experience? *Journal of Economic Perspectives*, 18(4), 49–68. <https://doi.org/10.1257/0895330042632672>
- Salton, G., Wong, A., & Yang, C.-S. (1975). A Vector Space Model for Automatic Indexing. *Communications of the ACM*, 18(11), 613–620. <https://doi.org/10.1145/361219.361220>
- Santos Silva, J., & Tenreyro, S. (2006). The Log of Gravity. *Review of Economics and Statistics*, 88(4), 641–658. <https://doi.org/10.1162/rest.88.4.641>
- Sarker, I. H., Ali, M. B., & Hossain, M. A. (2025). Impacts of Generative Artificial Intelligence on the Future of Labor Market: A Systematic Review. *Journal of Computer and System Sciences*. <https://doi.org/10.1016/j.jcss.2025.103623>
- Sasser Modestino, A., Shoag, D., & Ballance, J. (2025). *Extracting O*NET Features from the NLx Corpus to Build Public Use Aggregate Labor Market Data* (Working Paper No. 126336). MPRA.
- Sattinger, M. (1975). Comparative Advantage and the Distribution of Earnings and Abilities [Seminal paper applying comparative advantage principles to labor market assignment with heterogeneous workers and jobs]. *Econometrica*, 43(3), 455–468. <https://doi.org/10.2307/1914289>
- Say, J. B. (1834). *A Treatise of Political Economy*. Grigg & Elliot.
- Schaufeli, W. B., & Taris, T. W. (2014). A Critical Review of the Job Demands-Resources Model: Implications for Improving Work and Health. In G. F. Bauer & O. Hämmig (Eds.), *Bridging occupational, organizational and public health* (pp. 43–68). Springer. https://doi.org/10.1007/978-94-007-5640-3_4
- Schloss, D. F. (1891). Why Working-Men Dislike Piece-Work. *The Economic Review*, 1(3), 312326.
- Schumpeter, J. (1939). *Business cycles: a theoretical, historical , and statistical analysis of the capitalist process*. McGraw-Hill.
- Schumpeter, J. (1970). *History of Economic Analysis*. Routledge.
- Schwab, K. (2015). The Fourth Industrial Revolution: What it means and how to respond. *Foreign Affairs*.
- Schwarz, G. (1978). Estimating the Dimension of a Model. *The Annals of Statistics*, 6(2), 461–464. <https://doi.org/10.1214/aos/1176344136>

- Sheridan, T. B. (2002). *Humans and Automation: System Design and Research Issues*. Wiley.
- Shi, S. (2009). Directed Search for Equilibrium Wage–Tenure Contracts [Develops directed search framework with wage-tenure contracts and risk-averse workers]. *Econometrica*, 77(2), 561–584.
- Shimer, R. (2005). The Assignment of Workers to Jobs in an Economy with Coordination Frictions. *Journal of Political Economy*, 113(5), 996–1025. <https://doi.org/10.1086/444551>
- Shimer, R., & Smith, L. (2000). Assortative Matching and Search. *Econometrica*, 68(2), 343–369. <https://doi.org/10.1111/1468-0262.00112>
- Shneiderman, B. (2022). *Human-Centered AI*. Oxford University Press.
- Shrout, P. E., & Fleiss, J. L. (1979). Intraclass Correlations: Uses in Assessing Rater Reliability. *Psychological Bulletin*, 86(2), 420–428. <https://doi.org/10.1037/0033-2909.86.2.420>
- Sinclair, P. J. N. (1981). When will Technical Progress Destroy Jobs? *Oxford Economic Papers*, 31, 1–18.
- Slemp, G. R., & Vella-Brodrick, D. A. (2013). The Job Crafting Questionnaire: A New Scale to Measure the Extent to Which Employees Engage in Job Crafting. *International Journal of Wellbeing*, 3(2), 126–146. <https://doi.org/10.5502/ijw.v3i2.1>
- Smil, V. (2005). *Creating the 20th Century: Technical Innovations of 1867-2014 and Their Lasting Impact*. Oxford University Press.
- Smith, A. [Aaron]. (2016). Public Predictions for the Future of Workforce Automation. <https://www.pewresearch.org/internet/2016/03/10/public-predictions-for-the-future-of-workforce-automation/>
- Smith, A. [Aaron], & Anderson, J. (2014). *AI, Robotics, and the Future of Jobs* (tech. rep.). Washington, D.C., Pew Research Center.
- Smith, A. [Adam]. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations* (T. Darnell, Ed.). 12th Media Services.
- Smolny, W. (2002). Employment adjustment at the firm level: a theoretical model and an empirical investigation for West German manufacturing firms. *Labour*, 16(1), 65–88.
- Sostero, M., & Fernández-Macías, E. (2022). *The Professional Lens: What Online Job Advertisements Can Say About Occupational Task Profiles* (tech. rep. No. JRC Working Papers on Labour, Education and Technology 2021/13). European Commission, Joint Research Centre. Seville. https://joint-research-centre.ec.europa.eu/publications/professional-lens-what-online-job-advertisements-can-say-about-occupational-task-profiles_en

- Spence, M. (1973). Job Market Signaling. *The Quarterly Journal of Economics*, 87(3), 355. <https://doi.org/10.2307/1882010>
- Srihita, R. H., Goli, G., Rajyalaxmi, M., & Gobinath, R. (2025). Transformative Dynamics of the Gig Economy: Technological Impacts, Worker Well-Being and Global Research Trends. *Information Development*. <https://doi.org/10.1177/18479790241310362>
- Standing, G. (1984). The Notion of Technological Unemployment. *International Labour Review*, 123(2), 127–147.
- Standing, G. (2011). *The Precariat: The New Dangerous Class*. Bloomsbury Academic.
- Steinbach, M., Ertöz, L., & Kumar, V. (2004). The Challenges of Clustering High Dimensional Data. In *New directions in statistical physics* (pp. 273–309). Springer. https://doi.org/10.1007/978-3-662-08968-2_16
- Stephany, F., & Teutloff, O. (2024). What Is the Price of a Skill? The Value of Complementarity. *Research Policy*, 53(1), 104898. <https://doi.org/10.1016/j.respol.2023.104898>
- Steuart, J. (1966). *An Inquiry into the Principles of Political Oeconomy* (A. S. Skinner, Ed.) [Originally published 1767]. Oliver; Boyd.
- Stiglitz, J. E. (1987). Learning to Learn, Localized Learning and Technological Progress. In P. Dasgupta & P. Stoneman (Eds.), *Technology policy and economic performance*. Cambridge University Press.
- Stoneman, P. (1983). *The Economic Analysis of Technological Change*. Oxford University Press.
- Strehl, A., Ghosh, J., & Mooney, R. (2000). Impact of Similarity Measures on Web-page Clustering. *Proceedings of the AAAI Workshop on Artificial Intelligence for Web Search*, 58–64.
- Subramony, M. (2009). A Meta-Analytic Investigation of the Relationship Between HRM Bundles and Firm Performance. *Human Resource Management*, 48(5), 745–768. <https://doi.org/10.1002/hrm.20315>
- Susskind, D. (2020). *A World Without Work: Technology, Automation, and How We Should Respond*. Metropolitan Books.
- Tan, J. (2024). Multidimensional Heterogeneity and Matching in a Frictional Labor Market: An Application to Polarization. *Labour Economics*, 90, 102604. <https://doi.org/10.1016/j.labeco.2024.102604>
- Taylor, F. W. (1911). *Principles of Scientific Management*. Harper.
- Teulings, C. N. (1995). The Wage Distribution in a Model of the Assignment of Skills to Jobs [Extends assignment models to derive wage equations from no-arbitrage conditions with continuous skill and job type distributions]. *Journal of Political Economy*, 103(2), 280–315. <https://doi.org/10.1086/261931>

- Thelen, K. (2004). *How Institutions Evolve: The Political Economy of Skills in Germany, Britain, the United States, and Japan*. Cambridge University Press.
- Thelen, K. (2014). *Varieties of Liberalization and the New Politics of Social Solidarity*. Cambridge University Press.
- Thomas Piketty. (2017). *Capital in the Twenty-First Century*. Harvard University Press.
- Tong, H., & Zhang, J. (2024). Labor Market Polarization: Quantifying Skill Complementarity Between AI and Human Workers [Working Paper]. *Journal of Monetary Economics*. <https://doi.org/10.2139/ssrn.4851789>
- Topkis, D. M. (1998). *Supermodularity and Complementarity* [Comprehensive treatment of supermodularity and monotone comparative statics]. Princeton University Press. <https://doi.org/10.1515/9781400822539>
- Torraco, R. J. (2005). Writing Integrative Literature Reviews: Guidelines and Examples. *Human Resource Development Review*, 4(3), 356–367. <https://doi.org/10.1177/1534484305278283>
- Torrejón Pérez, S., Fernández Macías, E., & Hurley, J. (2024). *Global Shifts in the Employment Structure* (tech. rep.). Publications Office of the European Union. Luxembourg.
- Tschang, F. T., & Mezquita, E. A. (2024). Automation, Digitalization and the Future of Work: A Critical Review. *Journal of Electronic Business and Digital Economics*, 3(2), 184–211. <https://doi.org/10.1108/JEBDE-01-2024-0004>
- Tunzelmann, N. (2003). Historical coevolution of governance and technology in the industrial revolutions. *Structural Change and Economic Dynamics*, 14(4), 365–384. [https://doi.org/10.1016/S0954-349X\(03\)00029-8](https://doi.org/10.1016/S0954-349X(03)00029-8)
- Ulrich, D., Kryscynski, D., Brockbank, W., & Ulrich, M. (2017). *Victory Through Organization: Why the War for Talent is Failing Your Company and What You Can Do About It*. McGraw-Hill Education.
- Vafa, K., Palikot, E., Du, T., Kanodia, A., Athey, S., & Blei, D. M. (2024a). CAREER: A Foundation Model for Labor Sequence Data. *Transactions on Machine Learning Research*.
- Vafa, K., Palikot, E., Du, T., Kanodia, A., Athey, S., & Blei, D. M. (2024b). CAREER: A Foundation Model for Labor Sequence Data. *Transactions on Machine Learning Research*. <https://openreview.net/forum?id=xygFGwBDPd>
- Vallas, S. P. (1990). The Concept of Skill: A Critical Review. *Work and Occupations*, 17(4), 379–398. <https://doi.org/10.1177/0730888490017004001>
- Vallas, S. P., & Schor, J. B. (2020). What Do Platforms Do? Understanding the Gig Economy. *Annual Review of Sociology*, 46, 273–294. <https://doi.org/10.1146/annurev-soc-121919-054857>

- van der Aalst, W. M. P., Bichler, M., & Heinzl, A. (2018). Robotic Process Automation. *Business & Information Systems Engineering*, 60(4), 269–272. <https://doi.org/10.1007/s12599-018-0542-4>
- Vera-Toscano, E., Fana, M., & Fernández-Macías, E. (2022). *Regional Heterogeneity in Occupational Change: Using Census Data to Investigate Employment Polarisation and Upgrading at NUTS-3 Level* (tech. rep.). European Commission Joint Research Centre. Seville.
- Veza, E., Zunino, G., Laguinge, L., Moroz, H. E., Apella, I. R., & Spivack, M. H. (2025). *Understanding Labor Market Demand in Real Time in Argentina and Uruguay* (tech. rep. No. Policy Research Working Paper 11086). World Bank. <https://documents1.worldbank.org/curated/en/099753203172540083/pdf/IDU-8eac3ab7-d888-47cd-b8f5-d56ef7d9178c.pdf>
- Vijaymeena, M., & Kavitha, K. (2016). A Survey on Similarity Measures in Text Mining. *Machine Learning and Applications: An International Journal*, 3(1), 19–28. <https://doi.org/10.5121/mlaij.2016.3103>
- Vivarelli, M. (1995). *The Economics of Technology and Employment: Theory and Empirical Evidence*. Elgar.
- Vivarelli, M. [Marco]. (2012). *Innovation, Employment and Skills in Advanced and Developing Countries: A Survey of the Literature* (tech. rep.). Bonn, Germany, Institute for the Study of Labor (IZA). <https://doi.org/10.18235/0008980>
- Walker, T. (2007). Why economists dislike a lump of labor. *Review of Social Economy*, 65(3), 279–291. <https://doi.org/10.1080/00346760701635809>
- Walkowiak, E. (2023). Task-interdependencies between Generative AI and Workers. *Economics Letters*, 231, 111315. <https://doi.org/10.1016/j.econlet.2023.111315>
- Walkowiak, E., & MacDonald, T. (2023, September). *Generative AI and the Workforce: What Are the Risks?* (Working Paper No. 4568684). SSRN. <https://doi.org/10.2139/ssrn.4568684>
- Wang, X., Chen, Y., & Zhang, L. (2025). When Digital-AI Transformation Sparks Adaptation: Job Crafting and AI Knowledge in Job Insecurity Contexts. *Frontiers in Psychology*, 16, 1612245. <https://doi.org/10.3389/fpsyg.2025.1612245>
- Webb, M. (2020). The Impact of Artificial Intelligence on the Labor Market [Stanford University]. *Working Paper*.
- Weber, M. (1922a). Bureaucracy. In T. Waters & D. Waters (Eds.), *Weber's rationalism and modern society* (pp. 59–72). Palgrave Macmillan.
- Weber, M. (1922b). The Distribution of Power Within the Gemeinschaft: Classes, Staende, Parties. In T. Waters & W. Dagmar (Eds.), *Weber's rationality and modern society*. Palgrave Macmillan.

- Weeden, K. A. (2002). Why Do Some Occupations Pay More than Others? Social Closure and Earnings Inequality in the United States. *American Journal of Sociology*, 108(1), 55–101. <https://doi.org/10.1086/344121>
- Weeden, K. A., & Grusky, D. B. (2012). The Big Questions in Social Stratification: An Emerging Consensus? *Annual Review of Sociology*, 38, 223–246. <https://doi.org/10.1146/annurev-soc-071811-145517>
- Weeden, K. A., Kim, Y.-M., Di Carlo, M., & Grusky, D. B. (2007). Social Class and Earnings Inequality. *American Behavioral Scientist*, 50(5), 702–736. <https://doi.org/10.1177/0002764206295015>
- Weil, D. (2014). *The Fissured Workplace: Why Work Became So Bad for So Many and What Can Be Done to Improve It*. Harvard University Press.
- West, D. M. (2018). *The Future of Work: Robots, AI, and Automation* (tech. rep.). Washington, DC, Brookings Institute Press. <https://doi.org/10.7864/j.ctt1vjqp2g>
- Wicksell, F. (1961). *Lectures on Political Economy*. Routledge & Kegan.
- Winner, L. (1977). *Autonomous Technology – Technics-out-of-control as a Theme in Political Thought*. MIT Press.
- Wood, A. J., Martindale, N., & Burchell, B. J. (2025). Beyond the ‘Gig Economy’: Towards Variable Experiences of Job Quality in Platform Work. *Work, Employment and Society*. <https://doi.org/10.1177/09500170251336947>
- Wooldridge, J. M. (2015). *Introductory Econometrics: A Modern Approach* (6th). Cengage Learning.
- World Economic Forum. (2025a). *The Future of Jobs Report 2025* (tech. rep.). World Economic Forum. Geneva. <https://www.weforum.org/publications/the-future-of-jobs-report-2025/>
- World Economic Forum. (2025b, April). *Is AI Closing the Door on Entry-Level Job Opportunities?* <https://www.weforum.org/stories/2025/04/ai-jobs-international-workers-day/>
- Wrzesniewski, A., & Dutton, J. E. (2001). Crafting a Job: Revisioning Employees as Active Crafters of Their Work. *Academy of Management Review*, 26(2), 179–201. <https://doi.org/10.5465/amr.2001.4378011>
- Yale Budget Lab. (2024). *Evaluating the Impact of AI on the Labor Market: Current State of Affairs* (tech. rep.). Yale University. <https://budgetmodel.wharton.upenn.edu/issues/2025/9/8/projected-impact-of-generative-ai-on-future-productivity-growth>
- Yang, C. H., & Lin, C. H. A. (2008). Developing employment effects of innovations: microeconomic evidence from Taiwan. *Developing Economies*, 46(2), 109–134.
- Zhan, F., Wu, J., & Liu, X. (2024). Natural Language Processing for Human Resources: A Survey [arXiv:2410.16498]. *arXiv preprint*.

- Zhang, F., & Parker, S. K. (2019). Reorienting Job Crafting Research: A Hierarchical Structure of Job Crafting Concepts and Integrative Review. *Journal of Organizational Behavior*, 40(2), 126–146. <https://doi.org/10.1002/job.2332>
- Zhang, G., Liu, B., & Xu, S. (2023). Digital Job Crafting and Its Positive Impact on Job Performance: The Perspective of Individual-Task-Technology Fit. *Advances in Psychological Science*, 31(7), 1133–1148. <https://doi.org/10.3724/SP.J.1042.2023.01133>

Appendices

Appendix A

Supplementary Literature Review Material

This appendix provides extended treatment of topics referenced in the literature review (Chapter 2). Section A.1 covers classical economic debates and compensation theory. Section A.2 surveys the empirical evidence base from 1990–2020. Section A.3 examines broader theoretical frameworks and conceptual debates including the Fourth Industrial Revolution paradigm, technological determinism, and the lump of labour fallacy. While these topics inform the intellectual context of automation research, the main literature review focuses on contemporary evidence and measurement frameworks directly relevant to this thesis.

A.1 Classical Economics and Compensation Theory

A.1.1 Classical Debates on Technological Unemployment

The debate about automation's effects on employment has ancient roots. Aristotle may have been the first scholar to write of automation taking the place of human labour:

For if every instrument could accomplish its own work, obeying or anticipating the will of others, like the statues of Daedalus, or the tripods of Hephaestus, which, says the poet, 'of their own accord entered the assembly of the Gods'; if, in like manner, the shuttle would weave and the plectrum touch the lyre without a hand to guide them, chief workmen would not want servants, nor masters slaves (Aristotle, n.d.).

Several scholars have explored this concept since and come to various conclusions. The mercantilists of the 18th century would occasionally raise concerns about unemployment resulting from a technological advancement (Schumpeter, 1970). However, the ruling elite of the time, preoccupied by winning trade wars, saw the increased productivity as a benefit and considered any unemployment derived from the advancement of technology as a temporary issue (Schumpeter, 1970).

Ricardo's Warning

In the early 1800s as the principles of classical economics were being formulated, Ricardo argued, "That the opinion entertained by the labouring class, that the employment of machinery is frequently detrimental to their interests, is not founded on prejudice and error, but is conformable to the correct principles of political economy" (Ricardo, 1821, p. 16). He demonstrated how technology could increase supply and reduce the cost of production. The increased efficiency would result in additional revenue, but the costs would eventually fall to reach equilibrium, given the cost of production and demand for the greater supply of goods at the lower price point. While the lower price would offer people greater purchasing power, the reduced price may result in fewer employees' labour being tenable within the impacted occupation given the cost of maintaining the technology comparative to labour. He felt the returns resulting from the investment of technology would go into the capitalist's funds and not be shared with the labourers. He then went on to extend his reasoning for how this could result in long term unemployment and poverty for a growing proportion of the nation's labourers. He was concerned that even as prosperous productivity continued to climb, so too would economic inequality.

Say's Law of Markets

Say (1834) was the first to respond to Ricardo, arguing his Law of Markets and purporting that technology augments human labour. He felt that any means of increasing supply was advantageous, "As each of us can only purchase the productions of others with his own productions – as the value we can buy is equal to the value we can produce, the more men can produce, the more they will purchase" (Say, 1834, p. 3). Furthermore, Say refuted

Ricardo's argument that technology could make people poorer by arguing the funds would not be hoarded but be redirected to others.

"A new machine supplants a portion of human labour, but does not diminish the amount of product; if it did, it would be absurd to adopt it. When water-carriers are relieved in the supply of a city by any kind of hydraulic engine, the inhabitants are equally well supplied with water. The revenue of the district is at least as great, but it takes a different direction. That of the water-carriers is reduced, while that of the mechanists and capitalists, who furnish the funds, is increased. But, if the inferior charges of its production lower its exchange-value, the revenue of the consumers is benefited; for to them every saving of expenditure is so much gain" (Say, 1834, pp. 90–91).

Say argued that technology may make lower-skilled employees redundant but that the salaries they would have earned do not go exclusively to the capitalists as Ricardo had suggested but are redirected to the mechanists. Say argued that the construction of the new technologies required considerable labour and time to develop, offsetting any negative labour impacts resulting from the new technology's implementation. The reduced price of the product also provides the consumers (including those made redundant) with a greater sum of funds than they may have had previously, in the form of lower prices. However, Say offered no solution for what to do with the water carriers or those whose occupations were made redundant due to the advancement of new technologies. Generally, he believed that so long as there was a supply of labour, there would be a demand.

Marx's Critique

Marx would later join the debate, dismissing Say's position as "insipid nonsense" (Marx, 1977, p. 568). Marx re-conceptualised how the markets work in such a way that Say's Law epistemologically could not exist. He restated Ricardo's position about the accumulation of capital resulting from technological advancement and its detrimental effects on the working class, adding considerable historical detail. While his contributions made quite a splash and have endured with a lasting effect, they were generally not adopted by the mainstream. Most economists of the period supported Say's Law and contended that technological advancements had a positive effect on the economy.

Marshall and Keynes

Marshall, for example, recognised that technological advancements, "may occasionally diminish the field of employment for manual labour in a few trades" (Marshall, 1920,

p. 553) in the short term but he contended that the cheaper goods would benefit the working class as a whole. He further argued the production of the new technologies would increase the demand for labour in the long term, "This new demand for labour will partly take the form of the opening-out of new undertakings which hitherto could not have paid their way; while a new demand will come from the makers of new and more expensive machinery" (Marshall, 1920, p. 554).

Keynes famously upended economic thought by arguing it was demand that controlled the markets as opposed to Say's notion of supply. Keynes introduced the term *technological unemployment* and though he generally agreed with Marshall that it was a short-term problem, he famously foresaw a future when automation may outpace the demand for labour. Keynes predicted the power of compound interest and technological advancements would result in our present generation being freed from the "economic problem" (Keynes, 1931, p. 326). He argued that we would not toil or sweat to meet monetary needs and instead spend our days (and work if we chose) focused on advancing our leisure activities. Such a future, he believed, was possible "due to our discovery of means of economising the use of labour outrunning the pace at which we can find new uses for labour" (Keynes, 1931, p. 325).

Keynes position on the matter of technological unemployment has largely held over the years since, but that is not to say there has not been considerable debate and disagreement in the ensuing years.

A.1.2 Compensation Theory: Detailed Mechanisms

McCulloch (1818) and other scholars, such as Mill (1821), quickly built on Say's work developing a set of effects they believed offset any negative labour impacts brought on by automation. Marx labelled these effects as *compensation theory* (Marx, 1867). There are six generally recognised effects which economists tend to focus on (Marco Vivarelli, 2012).

New Machines

While new machines may replace labour and make several workers redundant, someone also had to make these new machines. Say argued that the construction of the machines may offset the labour impacts (Say, 1834).

Marx critiqued this theory by arguing, "the machine can only be employed profitably, if it... is the (annual) product of far fewer men than it replaces" (Marx, 1977, p. 552). Marx's critique was so effective few economists have attempted to use this theory since (Marco Vivarelli, 2012). Most recent arguments which try to claim this compensation theory are typically critiqued for substituting obsolete technologies with newer ones, in which there is

no compensation (C. Freeman et al., 1982).

New Investments

Ricardo (1821) may have been the first to suggest the new technology would enable greater production allowing, for a time, greater revenues. Over time the price of the product would likely fall to meet demand in order to clear the market given the greater supply. The change in prices would occur over some time enabling the capitalist to collect greater profits. Several economists have proposed this capital would be invested to create new jobs (Douglas, 1930; Hicks, 1973; Marshall, 1920; Stoneman, 1983).

This compensation effect is predominantly based on Say's Law, which assumes the additional capital is immediately reinvested. Marx (1867) and Keynes (1931) refuted this idea by demonstrating why capitalists may choose not to immediately spend their funds. Such an action may limit or delay the effectiveness of this compensation effect (Marx, 1867; Pianta, 2005; Vivarelli, 1995).

Decreased Wages

In the event that technology made occupations redundant, the price of labour could be adjusted. Decreasing wages would increase demand and improve employment (Hicks, 1932; Wicksell, 1961). Economists using equilibrium modelling may leverage this compensation effect in some form (Layard et al., 1991; Neary, 1981; Sinclair, 1981).

While firms may be encouraged to hire labour at a lower wage, Keynesian theory argues the decrease in aggregate demand may also lower employers' perceptions resulting in fewer new hires (Marco Vivarelli, 2012). That is to say, if employers see a decline in labour wages of a particular occupation, they may be hesitant to hire members of that occupation; believing the skill-sets of those labourers are insufficient for the modern firm or suspicious that more productive alternative means of production exists (i.e. capital).

Stiglitz (1987) alternatively argued that *localized technical progress* can specialise labour so even in the event of considerable layoffs, the skills and knowledge of those workers will be of little consequence to other specialised firms and therefore not affect or reduce the wages others pay for labour thus negating aggregate effects.

Increased Wages

This compensation effect is reflected in the Fordist era or in the Kaldorian model. Economists have demonstrated cases in which the greater productivity resulting from the technological augmentation allows some employees to get raises. The increase in income allows for an increase in consumption (while the cost of goods generally remains

static). The new expenditures increase demand and in Keynesian economics, this would enable job creation (Kaldor, 1957).

Boyer (1990) points out new institutional contexts and a more competitive labour markets have rendered Fordist modes of production irrelevant today. This compensation effect has been weakened as a result.

Lower Prices

In a competitive market, if automation is able to reduce the cost of production, then the price of the product would also decrease. The lower price would increase consumer's purchasing power and, if elasticity permits, increase the demand for the product. The new demand may require a larger supply which would then increase employment therefore compensating for the redundancies enacted as a result of the initial automation (Vivarelli, 1995).

"The introduction of machines is found to reduce prices in a surprising manner. And if they have the effect of taking bread from hundreds, formerly employed in performing their simple operations, they have that also of giving bread to thousands" (Steuart, 1966, p. 256).

Malthus (1964) initially explained that as the technology reduced labour, the initial demand would also be reduced as the unemployed would no longer have wages to spend. Any compensation, he argued, would have to offset this decrease in purchasing power. As this compensation effect was initially based on Say's Law, it is therefore vulnerable to Keynesian demand constraints; for example, there may be a delay in consumer purchasing behaviours or an inelastic demand (Keynes, 1936). This compensation effect also requires competition. If the firm holds a monopoly or participates in an oligopoly the savings may not be passed onto consumers (Labini, 1956).

New Products

The last effect was the invention of new products that could not exist prior to the technological advancements. These new products could create new jobs, and possibly new industries (Marx, 1867; Say, 1834). A slight variation of this effect is to create a new feature or point of differentiation in a mature good. In this event, it is necessary to overcome the substitution effect in order to compensate (Katsoulacos, 1986); for example, advancements in the iPhone substituted mobile phones as well as mp3 players. It is also possible the product, when used, may have implication on labour through processes improvements (such

as the desktop computer) and therefore also need to overcome any ensuing negative labour impacts. This is the compensation effect open to most debate (Marco Vivarelli, 2012).

In summation, these compensation effects can enable automation to have very powerful benefits for employment. However, it is imperative to explore the very complex relationships that exist between product and process innovations and the actions of capitalists and consumers as they manage their funds. Contemporary debates about technological unemployment have gone down many different paths but each utilise these effects to varying degrees.

A.2 Empirical Evidence on Automation 1990–2020

This section surveys empirical studies on automation’s labour market impacts from 1990–2020, the “pre-contemporary” evidence base that established foundational empirical patterns using R&D expenditure, patent data, and innovation measures as proxies for automation. While the main literature review emphasises contemporary evidence (2020–2025), this section preserves the empirical foundations that inform current research.

A.2.1 Measuring Technological Change: Methodological Challenges

Classical economists viewed technological unemployment as a two-factor function of production made up of capital and labour by which machines (capital) substituted labour. By contrast, contemporary economists recognise the process of production as a much more complex function involving, for example, the collection and combination of inputs, the organisation of work, the transmission of outputs, and the disbursement of proceeds (Acemoglu, 2000; Caprettini & Voth, 2017). Technical change could occur at any stage in the process.

It is important to consider why scholars often use the phrase technological change as opposed to automation. Within the modern management paradigm of Digital Taylorism, as business processes are increasingly codified into IT systems, the majority of changes have some effect on labour saving automation. Unfortunately, there are almost no good measurements of automation. In what Griliches (1990, p. 1) calls the *desert of data*, researchers are forced to seek out proxies which indicate automation. Though these data points are typically assumed to represent labour saving automation, this cannot often be said with absolute certainty. Some scholars therefore chose to refer to these phenomena

as technological change as opposed to automation in order to remind the reader that these findings may be obfuscating as much as they are illuminating. Indeed Carlsson et al. (1991) demonstrate that not all automation improves productivity nor saves labour (e.g. in cases where automation is utilised for health and safety reasons, or where firms fail to achieve their goals).

Innovation Taxonomy

It is also important to understand the distinction between *invention*, *innovation*, and *diffusion*. Invention is the novel creation of a new technology. Innovation is the incorporation of a technology in the redesign of a business process. And diffusion is where similar innovations are adopted by the broader industry. There may be considerable lags between each of these three elements (Standing, 1984). A more common taxonomy may draw distinctions between types of innovations and simply categorise them as either a *product innovation* – referring to a good or service – or a *process innovation* – referring to the system of production. These classifications are ambiguous at times as radical changes to a process may result in a seemingly new product (Standing, 1984).

R&D Expenditure as Proxy

One common method for evaluating automation is to measure the research and development (R&D) expenditures. This has become a popular method because firms often report R&D expenditure to shareholders as a standard accounting practice. This datapoint is therefore easily accessible and consistently measurable against previous financial year expenditures. Though a popular method, it has its risks; an increase in R&D expenditure does not always result in technological progress (Feldmann, 2013). Another complication when analysing the data is that most of the R&D expenditure comes from just a few firms (Carlsson & Jacobsson, 1994). As these firms rarely report the expenditures against specific trade secrets and intellectual property under development, accurately identifying when an invention occurs and when the diffusion begins can be challenging with this method.

Patent Data as Proxy

Studies that attempt to date an invention often rely on patents to measure the effects of technological change (Kortum, 1997; Porter & Stern, 2000). This approach can be misleading however, as firms may not chose to adopt a technology they have patented. For example, Apple has accumulated several keyboard patents over the years that have not yet been put into production (M. Brown, 2019; Liao, 2018). Other studies tend to focus on the innovation and diffusion of the technological change, but this method includes challenges

as well. There may be delays between when one firm innovates and when the rest of the industry adopts the approach, e.g. Kodak invented the digital camera in 1975 but it was decades until the technology was broadly adopted (Estrin, 2015).

Adoption is often assumed when changes in labour figures correlate with patent registrations, but for the reasons previously explained, this can be risky. Another challenge Griliches (1990) describes in detail regarding the use of patents pertains to classifications. Simply, should patents belong to the industry of the firm that registered it (e.g. chemical sciences), classified under the industry that is likely to produce the patented product (e.g. manufacturing), or classified under the industry that is likely to use the patented product (e.g. agriculture)? What if multiple industries are likely to use the patented product; could one inadvertently double count the patent then? Despite these challenges, Griliches (1990) still contends patents may be the best available data inputs for understanding automation.

A.2.2 Firm-Level Studies

There are three levels of empirical study relating to technological unemployment: firm-level, industry-level, and macroeconomic analyses. Each level offers distinct insights while facing unique methodological challenges. The most popular level of study is focused at the firm level. This analysis often only captures the effects of innovation on the selected firms and ignores the effects on competing firms or external industries.

Most studies conducted at the firm level suggest technological change has a positive impact on labour, although there are some disagreements regarding the impacts of process innovation.

European Evidence

Greenan and Guellec (2000) collected data on 15,186 French firms from 1986-1990. They found that firms engaged in process innovation had a positive effect on job creation. Although to a lesser extent, they also found that product innovation had a positive effect on job growth.

Smolny (2002) analysed data for 2,000 German manufacturing firms from 1980-1992 and found the innovative firms generated a higher rate of employment growth than less-innovative firms.

Piva and Vivarelli (2005) assessed data pertaining to 575 Italian manufacturing firms between 1992-1997 and found innovation to have a positive effect on job creation.

Harrison et al. (2008) compared data from firms in France, Germany, Spain, and the UK from 1998-2000. For manufacturing firms, they found product innovation had a positive effect on job creation. Process innovation had a negative effect on manufacturing employ-

ment; however, this effect was weak. For service sector firms, they found product innovation had a comparatively small effect on job creation; and, noted that process innovation had no effect. To ensure they avoided endogeneity issues they compared R&D expenditure, patents, and consumer sales.

B. H. Hall et al. (2008) measured 9,462 Italian manufacturing firms from 1995-2003 and found half the employment growth resulted from product innovation whereas the other half was the result of sales growth of existing merchandise. They concluded process innovation did not have an effect on job creation.

Asian and US Evidence

Yang and Lin (2008) measured R&D expenditure and patent counts for 492 Taiwanese manufacturing firms from 1999-2003. They discovered that both measurements had a positive effect on job creation. When filtered for product innovation they once again found a positive association across the industries they studied, but when filtered for process innovation the positive effects were limited only to R&D intensive industries.

Coad and Rao (2011) collected data on 1,920 US manufacturing firms from 1963-2002. Based on their analysis of patent activity, they concluded that innovation led to employment creation.

Limitations of Firm-Level Analysis

One of the most profound weaknesses with firm level studies is their inability to account for labour trading within the industry or across occupational clusters. It is possible, and in some contexts observed (Greenan & Guellec, 2000), that employment growth or job losses within a singular firm have no effect on aggregate employment rates. The growth of one firm may come at the expense of other firms or visa versa. This effect confronts claims of job creation for studies that only conduct firm level analysis.

A.2.3 Industry-Level Studies

While the majority of firm level studies found positive effects related to technological change, industry level studies generally came to the opposite conclusion. Industry-level analyses account for competitive reallocation effects within industries but still face limitations in capturing economy-wide spillovers.

Contrasting Results with Firm-Level Studies

Greenan and Guellec (2000) examined 18 French industries, in stark contrast to their firm level analysis, and found product innovation to have a greater impact than process improvement. This suggests that while individual innovative firms may grow employment, competitors' job losses offset these gains at the industry level.

Similarly, M. Pianta (2000) examined 21 manufacturing industries from five European countries from 1989-1993 and found the industries with the highest levels of innovation also had the highest rate of job losses.

European Industry Evidence

Antonucci and Pianta (2002) sought to confirm the results of the earlier study by looking at data from 10 manufacturing industries from 8 European countries from 1994-1999. They measured innovation expenditure per sale and found the industries as well as the countries with the greatest rate of innovation suffered the greatest displacement of labour.

Díaz and Tomás (2002) studied Spanish industries from 1985-1995 and found the sectors with the most technology use saw a decline in the proportion of workers.

Evangelista and Savona (2002) examined Italy's service sector from 1993-1995 and found a positive impact on employment in less than half the industries. While there were positive employment effects in science and technology industries, the financial services and capital-intensive industries experienced negative effects.

A.2.4 Macro-Level Studies

There have been very few contemporary empirical macroeconomic studies to explore the effects of technological unemployment. Macro-level studies face the challenge of identifying causal effects amid numerous confounding factors but offer the most comprehensive view of economy-wide automation impacts.

Cross-Country Comparisons

Vivarelli (1995) examined data from the US and Italy from 1966/1967-1986 respectively. He found technological change had a positive effect on the US but a negative effect on Italy. He attributed the difference in findings to the predominant type of innovation. In the US, product innovation was favoured whereas in Italy process innovation was most popular.

Similarly, Pini (1995) explored data from 9 industrial countries from 1960-1990. He found innovation had a negative impact on employment; however, he also found equal compensation effects related to increased exports.

Patent-Based Evidence

Feldmann (2013) collected data on triadic patent families for 21 industrial countries from 1985-2009. He discovered, "the rapid technological change since the mid-1980s has substantially increased unemployment" (Feldmann, 2013, p. 1120). He found the negative employment effect persisted for two years after the initial increase in technological progress. He describes the "severe and protracted technological unemployment" consistent with Schumpeterian (Schumpeter, 1939) views of innovation waves and creative destruction (Feldmann, 2013, p. 1122).

That the effects faded after three years, led him to suggest that compensation effects likely work but require several years to fully compensate for the initial job losses. Thus, he concluded, "there is no long-term effect of technological change on unemployment" (Feldmann, 2013, p. 1120).

A.2.5 Synthesis of 1990–2020 Evidence

Clearly much more work needs to be done to examine the impacts of automation on employment in the current industrial era. The results of studies from this period offer a mixed bag of findings. The best available data is not good enough. But some lessons, however dubious, may still be observed.

Key Patterns

In most cases, innovation offers many benefits for the firm. These benefits, however, are contingent on the industry and the type of innovation. While process innovation often results in automation and initial job displacements, compensation effects enable the innovating firm to grow their labour (Greenan & Guellec, 2000). These compensation effects appear to be weaker with service firms; while they may benefit from greater efficiency, process innovation does not result in employment growth (Harrison et al., 2008). Similarly, labour related compensation effects do not appear to benefit manufacturing firms leaving them with net job loss (Harrison et al., 2008).

Product innovation, on the other hand, has a positive effect on employment for manufacturing and R&D intensive firms while the effects are weaker for other sectors (B. H. Hall et al., 2008; Harrison et al., 2008; Yang & Lin, 2008). The underlying reasons for why different industries benefit from process and product innovation differently requires more investigation.

Scale-Dependent Effects

While product innovation offered little benefit in terms of employment at the firm level, at the industry level, there was often a boom in employment (Greenan & Guellec, 2000). Conversely, employment gains due to process innovation in one firm came at the cost of competitors resulting in a negligible effect at the industry level (Greenan & Guellec, 2000). The industries with the most innovation simultaneously experienced the greatest job losses (Antonucci & Pianta, 2002; M. Pianta, 2000). This could be due to the growth of new firms taking advantage of disruptive innovation to capitalise on novel markets (Christensen, 1997). More research is required to understand the inverse effects between firm and industry level findings.

Temporal Dynamics

Of course, industries do not exist entirely in discrete silos. Some occupations exist across multiple industries and products produced in one industry may have impacts on other industries as well. There have been a scarce number of macroeconomic studies focusing on the technological unemployment of recent years. Technological unemployment is a concerning issue for the short term; however, compensation effects may stave off long term effects and indeed benefit society after several years (Feldmann, 2013). Evidence suggests the types of innovation play a big part in determining the degree to which compensation effects influence outcomes with product innovation demonstrating the greatest effects (Vivarelli, 1995).

Methodological Limitations

The empirical evidence from this period faces several limitations that contemporary research (2020–2025) has begun to address:

- **Proxy measurements:** R&D expenditure and patents imperfectly capture actual automation deployment, potentially missing process innovations that do not generate patents.
- **Aggregation bias:** Firm, industry, and occupation-level analyses may mask within-unit heterogeneity, particularly task-level restructuring.
- **Confounding factors:** Few studies adequately control for globalisation, offshoring, institutional changes, and demand shocks that coincided with technological change.
- **Limited service sector coverage:** Most studies focus on manufacturing, despite services accounting for 70–80% of employment in advanced economies.

- **Short time horizons:** Many studies examine 3–10 year windows, potentially missing longer-term adjustment dynamics.

These limitations motivated the methodological advances in contemporary automation research, including direct measurement of robot adoption, task-based frameworks, and attention to within-occupation heterogeneity discussed in Chapter 2.

A.3 Theoretical Frameworks and Conceptual Debates

This section examines broader theoretical frameworks and conceptual debates including the Fourth Industrial Revolution paradigm, technological determinism, and the lump of labour fallacy. While these concepts inform the intellectual context of automation research, the main literature review focuses on empirical evidence and measurement frameworks directly relevant to this thesis.

A.3.1 The Fourth Industrial Revolution

Historians often characterise sociotechnical progress in terms of industrial revolutions. Philbeck and Davis (2018, p. 17) write, "at certain stages in history, sets of technologies emerge and combine in ways that have impacts far beyond incremental increases in efficiency. Industrial revolutions are revolutions in the systems that surround us, step changes in the complex interplay between humans and technology, and transformations that result in new ways of perceiving, acting, and being."

Kondratiev Waves and Industrial Eras

The first industrial revolution began towards the end of the 18th century, the second in the 19th century, and the third arrived in the later part of the 20th century. The dates of these 'revolutions' and even how many of them continue to be hotly debated in the economic literature (De Vries, 1994; Prisecaru, 2016; Smil, 2005).

Economists like C. Freeman and Louca (2001) focus their analysis on technological breakthroughs, organising these feats of engineering along Kondratiev cycles. According to their analysis, there would be six Kondratiev waves. The Kondratiev wave model breaks down the development of technical capital into two waves. Amendola and Gaffard (1988) describe the process as starting with an initial wave where intangible capital (e.g. a knowledge base) gets *constructed* and is then followed by a second wave that *utilises* it.

The first wave is focused on producing old products with new technology (seen as process innovation that is often characterised as labour substituting), while the following wave layers on top to use the new means of production to produce novel goods and services not previously possible (often characterised as labour augmenting).

Tunzelmann (2003) explains the second industrial revolution (i.e. the third Kondratiev wave) provided the infrastructure to use electricity and oil, and knowledge of combustion engines along with new metallurgy production methods; while the fourth Kondratiev wave offered the mass motorisation of society. The Kondratiev waves span an average of about 50 years with the even waves beginning in the 1830s, 1920s, and 2010s.

By taking the broader view and merging the odd and even waves together, so only the odd numbered waves are recognised as industrial revolutions, Tunzelmann (2003) points out that several societal characteristics, not just technological advancements, coalesce together. Through the view of industrial revolutions, distinct societal systems and modes of governance come into focus such as the production motive (steam, electricity, nuclear), production materials (iron, steel, silicon), communication methods (telegram, telephone, internet), and means of transportation (rail, automobile, aircraft). Table A.1 summarises these characteristics.

Table A.1: Tunzelmann's chronology of industrial eras

	First Industrial Revolution	Second Industrial Revolution	Third Industrial Revolution
Approximate dates	1750–1815	1870–1914	1973–
Location	UK	USA, Germany	USA, East Asia
Technologies	Machinery	Chemicals	ICT, biotechnology
Motive	Water, steam	Electricity, oil	Nuclear, renewable
Material	Iron	Steel, plastics	Silicon, smart materials
Automation	of Transformation	of Transfer	of Control
Process type	Labour	Capital	Information
Size of firm	Small	Large	Mixed
Advantages	Specialisation	Internal Integration	External Integration
Organisation	Entrepreneurial	Multidivisional	Networked
Industry structure	Competitive	Oligopolistic	Mixed
Type of Capitalism	Personal	Managerial	Collaborative
Mode of Governance	Markets	Hierarchies	Networks

Source: Adapted from Tunzelmann (2003, p. 371)

The Fourth Industrial Revolution Debate

As we approach the modern era it becomes increasingly difficult to characterise our current paradigm. According to C. Freeman and Louca (2001) we may be in the sixth Kondratiev, and not likely to see another industrial revolution till around 2060. But others have controversially begun to discuss the emergence of the fourth industrial revolution.

In 2016, Klaus Schwab published his book titled *The Fourth Industrial Revolution* and so sparked a series of debates at the World Economic Forum (Prisecaru, 2016). Schwab (2015, p. 4) characterises the fourth industrial revolution as "technology breakthroughs in fields such as artificial intelligence, robotics, the Internet of Things, autonomous vehicles, 3-D printing, nanotechnology, biotechnology, materials science, energy storage, and quantum computing" making possible the advent of cyber-physical systems. The second industrial revolution produced electricity, the third industrial revolution layered upon that to produce digital technologies, and the fourth industrial revolution is expected to take the ubiquitous internet connectivity for granted and break barriers between the virtual and the physical mediums.

Perhaps, whether one argues we are in the midst of the sixth Kondratiev or the fourth industrial revolution is missing the point of these debates.

"The concept of [the fourth industrial revolution] is meant to help individuals and organizations make sense of the interplay between humans and technology at a time when advances in computing power, biotechnologies, artificial intelligence, renewable energies, additive manufacturing, and many other emerging technologies threaten to overwhelm us with complexity... It is a tool to think deeply about the dynamics, values, stakeholders, and technologies of a world that is changing rapidly, and drive collective action within and across nation states in a way that results in a more inclusive, fair, and prosperous future" (Philbeck & Davis, 2018, pp. 20–21).

A.3.2 Technological Determinism

Debates about technological automation often revolve around a central question of agency. Do we use the tools, or do the tools use us? To what degree do we have control over our complex systems of production? How much control do we have over our social interactions? Is the application of automation predetermined by past decisions in a path-dependent process? Is there an internal technological logic that dictates the adoption of automation? Or is it the will of capitalistic technocratic elites?

The Traditional Deterministic View

Winner (1977) evangelises an argument many scholars once held in which technological change was an autonomous process, shaping history and society in ways people could not predict or control. A simple illustration of this is the common use of phrases like the stone age, the bronze age, the iron age, and the computer age. This perspective is often referred to as *technological determinism*.

The Constructivist Critique

Dafoe (2015) argues that modern social scientists have rejected the notion of technological determinism. Many constructivist approaches have been applied to the study of technology. They have convincingly shown that different social groups, historical contexts, and differing perceptions of technologies purpose has shaped the evolution of technologies. Many consider technological determinism to be overly simplistic and an implausible explanation. Lynch (2007, p. 10) demurs, "technological determinism has been reduced to the status of a straw position in technology studies."

Recently, scholars have shifted from studying and critiquing technological determinism as a definitive law of history and moved towards studying people's beliefs about technological determinism. Dafoe (2015, p. 1049) argues, "While these works do direct attention to technologically deterministic claims, they treat them as subjects of study *to be explained*, not as potentially insightful theoretical arguments *that can explain*." As a result, questions about the effects of power and materiality on automation and autonomy are often neglected (Leonardi & Barley, 2010).

A Nuanced Approach

Technological determinism should not be seen as a binary construct viewed as either being right or wrong. It should be viewed as a means of inquiry, exploring a range of conditions, contexts, and systems. Technological determinism can enhance the understanding of when, and in which ways, autonomous technologies influence and shape society. It can answer questions of human agency as it relates to automation; like, which groups of people maintain or secede their agency – in which ways and under what conditions might automation augment or supplement labour?

A.3.3 The Lump of Labour Fallacy

"Will robots take our jobs?" It seems like a perfectly reasonable and innocuous question, but it is predicated on the mistaken belief that there is a fixed amount of work available. A common metaphor illustrates that there are only so many slices of pie available. This assumption is what many economists refer to as the lump of labour fallacy (Rae, 1894; Schloss, 1891).

The fallacy routinely appears in many news reports and opinion pieces and even creeps into scholarly works as well (Blanchard, 2004; Saint-Paul, 2004; Walker, 2007). One would be forgiven for falling for this zero-sum view of the economy where when someone gains, another loses. After all, jobs have been lost to automation, so this perception of the

economy seems entirely logical (Mokyr et al., 2015). As is often the case, reality is a bit more complex. Demand for labour is not a fixed quantity – that is to say, the pie can grow larger so everyone gets a slice.

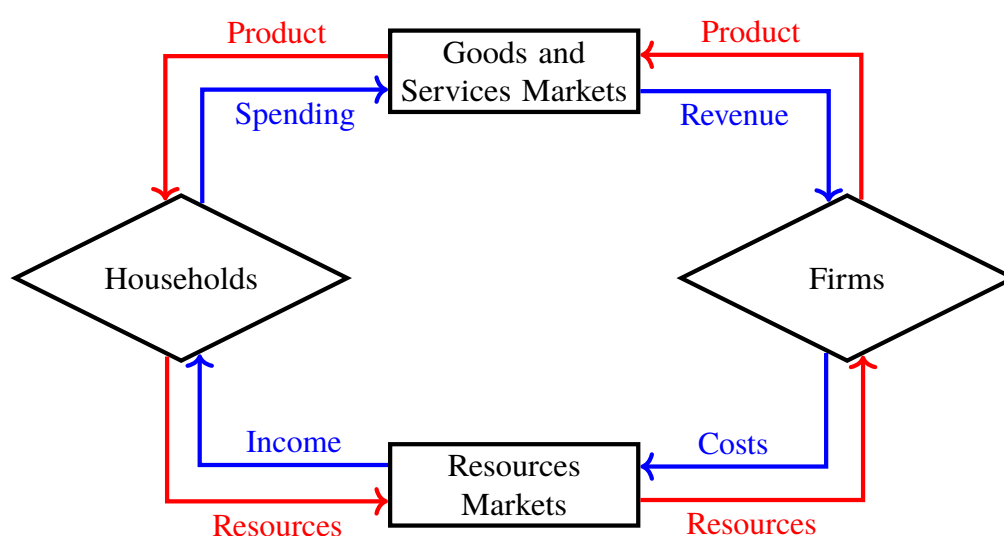
Say's Law Revisited

To understand why the lump of labour is a fallacy, economists point to Say's Law. Say (1834) articulated that the production of supply creates demand. In other words, employment provides households with income, which increases aggregate demand for goods and services.

The Circular Flow Model

Cantillon (2015) is credited with first proposing the concept of economic flows in 1755 while Quesnay (1894) was the first to visualise the model. The model was further developed by Marx (1867) and Keynes (1936). While many variations exist, the modern version of the visual model is generally attributed to F. H. Knight (1951). The simple model (see Figure A.1) focuses on two economic decision makers: households/workers and firms. Additionally, the model divides markets into two categories: goods and services markets, and resources markets which contain the factors of production. The flow of money, represented by the internal arrows, always flows from the buyer to the seller.

Figure A.1: Circular Flow Model



This model allows one to conceptualise how firms choose whether to invest their revenue in capital (e.g. machinery, property, stocks), or whether to hire more labour. Hiring

decisions are often made based on the concept of marginal productivity.¹

The Movie Theatre Example

To use the popular movie theatre example: if the confectionery stand has only one person working and a line of 10 people, but three of those people leave because they get bored in line or do not want to miss the start of the movie, the owner may choose to hire an additional employee in order to service more people quickly. Each of the two employees can now provide popcorn and soda to five people. The average productivity would be equivalent to five people's purchases. Because one employee was able to independently serve seven customers their marginal productivity would be equivalent to seven people's purchases, while the second employee's marginal productivity would only be equivalent to the purchases from the additional three people that were able to be served after their hiring.

Should the second person quit, the movie theatre would only lose the marginal productivity of the second employee (equivalent to the purchases of the three people no longer served). Similarly, if three customers only purchase soda, it is possible that installing a cheap soda machine could satisfy these customers, thus replacing the marginal productivity of the second employee and saving the movie theatre the bulk of their wage.

Augmentation Versus Supplementation

If the movie theatre had only hired one person to work the confectionery stand this may be seen as technology augmenting labour. The relationship between technological progress and labour demand has received considerable attention with mixed results; however, highly innovative firms generally demonstrate greater demand for labour and often offer more sustained periods of employment (Ciriaci et al., 2016; Lachenmaier & Rottmann, 2011; Pianta, 2005; Piva & Vivarelli, 2005; Mariacristina Piva & Vivarelli, 2018).

However, if the movie theatre had hired two employees, and subsequently let one go following the installation of the soda machine, this would illustrate how robots or AI may be perceived to supplement labour. Indeed, scholars have observed robots and AI supplementing labour at ever increasing rates (Brynjolfsson & McAfee, 2014; Ford, 2009). Nevertheless, it is unlikely that automation will replace the majority, let alone all jobs (Autor, 2015b; Coelli & Borland, 2017).

¹The decision to create an additional job may be based on the marginal productivity, but in a competitive labour market the employee's wage is not reflective of their marginal productivity (Gottschalk, 1978).

Keynesian Critiques

One important consideration, not to be overlooked, pertains to Marx (1867) criticism that firms or capitalists may hoard their revenues rather than immediately reinvesting it into the resources markets. This could effectively force the economy into disequilibrium resulting in higher unemployment or lower wages for workers. Classical economists argued that periods of saving or hoarding would be balanced by periods of spending or dishoarding (Lemieux, 2014). When interest rates were low people would be inclined to invest or spend, and when interest rates were high people may prefer to save.

Keynes (1936) was not convinced interest rates were sufficient to fine-tune the economy due to people's *animal spirits*. He famously demonstrated how savings and lags in the circular flow model could shift the equilibrium downwards towards unemployment. These actions, reducing consumption or over producing, could result in recession. Keynes (1936) argued that beyond adjusting interest rates, the government needed to intervene by adding their demand, to stimulate the economy. Under what conditions, and to what extent has been the substance of many heated debates amongst economists ever since (Thomas Piketty, 2017).

Setting aside the potential role for government intervention, there is ample evidence of a growing economic divide in which the wealthy capitalists have reaped the many rewards of automation while workers employment and wages have stagnated (Acemoglu & Restrepo, 2017; Brynjolfsson et al., 2019; R. Freeman, 2015; Graetz & Michaels, 2017).

A.3.4 The Resurgence of Contemporary Debates

Though it has been challenged many times, the general consensus has held through the first three industrial revolutions that technological unemployment does not result in long-term unemployment. However, scholars are once again reconsidering the effects of technological unemployment in light of the changes being advanced by the fourth industrial revolution (Ford, 2009; Frey & Osborne, 2017; West, 2018).

Modern Concerns

Perhaps Ford is responsible for reigniting the most recent debates by writing, "As technology accelerates, machine automation may ultimately penetrate the economy to the extent that wages no longer provide the bulk of consumers with adequate discretionary income and confidence in the future. If this issue is not addressed, the result will be a downward economic spiral" (Ford, 2009, p. 237). To address this issue, economists are once again searching for the elusive oasis in the desert of data. And indeed, some novel approaches

have been found.

West (2018) compared the largest firms from 1962 with the largest firms from 2017 and found their market capitalisation had increased considerably despite significant reductions in staff. He concluded that firms were able to automate processes and outsource minor tasks in order to greatly improve productivity while operating a very lean company with comparatively few full-time employees.

The Great Decoupling

Brynjolfsson and McAfee (2014) documented a phenomena economists have begun to refer to as the great decoupling (Bernstein & Raman, 2015; McAfee, 2012). They demonstrated the growth of U.S. gross domestic product (GDP) closely mirrored the growth of productivity until the start of the third industrial revolution when wages stagnated and decoupled entirely from the trend. Brynjolfsson and McAfee (2014) further expanded on their analysis demonstrating as a percentage of GDP, that while corporate profits continued to climb, wages have declined (Bernstein & Raman, 2015).

These empirical observations motivate the contemporary research examined in Chapter 2, which employs refined methodologies to examine whether automation's labour market impacts differ fundamentally in the current technological era compared to previous industrial revolutions.

Appendix B

Historical Perspectives on Occupational Classifications

This appendix traces the intellectual history of occupational classification from Smith's division of labour through Weber's bureaucratic hierarchies and Taylor's scientific management, providing context for the measurement challenges discussed in Chapter 2.

Adam Smith (1776) famously argued for a division of labour with his pin factory example in his book, 'The Wealth of Nations', where he documented the changes from a mercantilist system to a capitalist one at the onset of first industrial revolution. While this division of labour was initially managed by holistic tradespeople who priced their output on the open market, smaller organisations began to form to support more efficient and lucrative production. The first industrial revolution sparked the construction of large infrastructure projects like the development of train routes and water canals. These projects required more funding than could customarily be provided by a few elite financiers. Companies were formed and shares were sold to the public. With so many owners, personal administration of the firms was not possible. Professional managers were tasked with developing the infrastructure and generating profits for shareholders. To do so, these managers needed to carefully control the levels at which resources were consumed, principally labour costs. This led to various internal pricing metrics being created and assigned to a stratification of novel occupations as new organisational structures and procedures evolved into the 19th and 20th centuries (Crowther & Green, 2004).

When historians and sociologists discuss social stratification, they often do so by

referring to occupational structures and social mobility. M. B. Katz (1972) asserts that historically, an individual's occupation was the strongest predictor of income and prestige. He cautions the number of unique occupations in a 19th century town could climb into the hundreds, and by the 20th century that figure grew into the thousands. Scholars required a way of defining a limited and manageable number of categories for performing their analysis. The choice of occupational classification potentially predetermines the parameters and degrees of social stratification or vertical mobility that can be found (M. B. Katz, 1972). When Marx (1867) criticised the capitalist 19th century British society, he focused his analysis on a stratification of dominant and subordinate occupations, amalgamated as bourgeoisie and proletariat classes. Marx was concerned with the exploitative nature of the managerial practices and the way in which labour was structured. This strongly influenced the factors he defined when considering occupations and the potential for social mobility.

Durkheim (1893) was similarly concerned with the wellbeing of increasingly industrialised nations at the onset of the second industrial revolution. His analysis explored the emotional and social health of employees. His classifications of careers intrinsically considered these factors along with the employees' commitment and motivation to their family, nation, industry/trade, and respective organisations. Durkheim was most interested in measuring and understanding the growing individual responsibility employees faced in many occupations.

"A society composed of an infinite number of unorganized individuals, that a hypertrophied State is forced to oppress and contain, constitutes a veritable sociological monstrosity. ... A nation can be maintained only if, between the State and the individual, there is intercalated a whole series of secondary groups near enough to the individuals to attract them strongly in their sphere of action and drag them, in this way, into the general torrent of social life. We have just shown how occupational groups are suited to fill this role, and that is their destiny" (Durkheim, 1893, p. 28).

Power, wealth, status, and ideology were tightly intertwined in 19th and early 20th century social analysis and occupational classifications (Crowther & Green, 2004).

Weber (1922a) provided, perhaps, the strongest foundation for building modern day occupational classifications as a part of his work on bureaucracy. His theories of organisational structure and hierarchy extolled the virtues of efficient management through clear lines of communication. Procedures needed to be documented and every employee needed to have clarity regarding their responsibilities. This led Weber to define rules for recruitment and the development of job descriptions for occupations stratified across different levels of the organisation. It's worth noting that Weber's bureaucracy remained explicitly linked with ideological beliefs. Weber (1922b) stratified society across three factors: social class (i.e. economic relationship to the market), status (e.g. religious denomination, honour,

prestige), and political party affiliation. As he watched the second industrial revolution come into fruition, Weber voiced concerns about bureaucracy's lack of flexibility and stubborn resistance to change. He also cautioned employees engaged in repetitive work may grow bored and disassociate from the work and organisation resulting in high turnover. Likewise, Weber warned repetitive tasks with high turnover gave employers dangerous abilities to exploit their labour.

In the early 20th century, Taylor (1911), one of Weber's contemporaries, approached the problem from an engineering perspective to establish a concept he referred to as scientific management. It was built on three simple pillars.

- First, the managers were expected to break complex jobs down into simple tasks.
- Second, they were to measure and time everything the workers do. He argued that by using time studies and measuring material resource units, management could identify the 'one best way' (i.e. most efficient means) to use the labour. They were to analyse this data and find the most efficient methods of completing these simple tasks. Once identified they were to force these methods on all their employees. His goal was to standardise all manual labour by breaking complex jobs down into finite tasks and distribute them individually amongst workers – an extreme example of Smith's division of labour. This approach required very strict occupational classifications and an even more regimented task schedule to be established within each occupation. For the first time, sub-occupational clusters were being formed within discrete professions.
- Third, companies were to link wages to performance. The most efficient employees would get paid the most money. Taylor (1911) famously applied his scientific management to the Bethlehem Steel company; and when productivity per person increased approximately 270 percent, the employees' average rate of pay also increased about 65 percent.

It is important to recall the work structures and organisations of the time. Floor managers would often hire craftsmen and tradesmen to come into the warehouses to work. The managers' knowledge of the work to be done was often less than their skilled employees (though that is not to say managers were not intelligent, rather their skills were utilised elsewhere) so they would present the problem to the employees and leave them to apply their knowledge and craft. Taylor believed that scientific management, as opposed to this ordinary management, was a better method for prospering. It required managers to take on a new role. No longer was it acceptable to leave the tradesmen to apply their knowledge and craft. Now that knowledge needed to be extracted from the tradesmen. Their actions needed to be measured. It became the role of the manager to decide what, when, and how,

simplified tasks were to be performed. The locus of control no longer belonged to the tradesman who spent their life honing their skill. It now sat with the manager. The labour unions went on strike; they argued Taylor's occupational classifications and measurements were deskilling and dehumanising them. Taylor was driven by a fiercely classist ideological belief in the superiority of cognitive work over manual operations, writing:

“Now one of the very first requirements for a man who is fit to handle pig iron as a regular occupation is that he shall be so stupid and phlegmatic that he more nearly resembles in his mental make up the ox than any other type. The man who is mentally alert and intelligent is for this very reason entirely unsuited to what would, for him, be the grinding monotony of work of this character” (Taylor, 1911, p. 59).

This division of manual and cognitive tasks remains a staple of the occupational analysis zeitgeist to this day (albeit without the malicious contempt demonstrated by Taylor), with Autor, Levy, and Murnane (2003) famously making predictions about the effects of automation along these lines of cognitive/manual and repetitive/non-repetitive dichotomies.

While Taylor was demonised by the working class, Henry Ford was initially celebrated for instituting similar practices. Ford's use of an electric conveyor belt and Taylor's scientific management practices (though Ford insisted he came up with the ideas independently) revolutionised production in the second industrial revolution. With the continuously moving conveyor belt, there was no longer any reason for the employee to move away from the machine or change from the simple, repetitive task they were assigned (e.g. punching holes in metal). This process reduced the time required to produce a Model T from 12.5 hours to just 93 minutes (Anderson, 2014). This system may have backfired for Ford, had it not been his important inclusion of Taylor's third pillar. In 1914, Ford introduced the five-dollar day (Anderson, 2014). At the time, this was double the average wage. Though it did not make him any friends amongst the aristocratic class, it drove his automotive company to success. The way that Ford instituted the five-dollar day is of particular importance. Ford did not actually change the wage from \$2.70, rather he awarded an additional \$2.70 bonus if certain social (i.e. non-work) criteria was met (Anderson, 2014). The criteria included the abstinence of alcohol, maintenance of clean and tidy homes, extended family and friends were not permitted boarding, and regular contributions to personal savings accounts needed to be made, among others. To enforce these social policies and award the bonus, Ford established the Sociological Department whose job it was to visit employees' homes, perform visual inventories, and ask intrusive questions of family members. Ford's efforts to modify his employees' behaviours and force his ideologies and preferred personal characteristics on them were not seen as benevolent and employees soon came to resent

managements efforts to control them outside of work hours. By 1921, Ford had abandoned the social program and dissolved the department (Anderson, 2014). As other production lines copied Ford's methods, Ford lost its competitive edge. Many post-Fordist companies, including Ford, found themselves once again competing on cost. This was achieved by increasing the speed of conveyor belts, without increasing the pay of employees. The principle of pay for productivity fell out of practice by the 80s at the cusp of the third industrial revolution. The productivity per person increased but all the capital accrued to the employers instead of being shared with employees (Crowther & Green, 2004). As a result, employee remuneration decreased despite rising productivity.

Appendix C

Extended Theoretical Framework: Full Formal Treatment

C.1 Introduction

Chapter 2 identified a measurement-theory disconnect at the heart of automation research: four scholarly traditions independently predict within-occupation restructuring, yet empirical measurement frameworks continue operating at occupation level. This chapter develops a formal model that operationalises these theoretical insights, deriving conditions under which discrete job type clusters emerge endogenously within occupational boundaries.

The model synthesises insights from the four traditions reviewed in Chapter 2. From *organisational behaviour*, the job design literature establishes that jobs are strategic designs rather than natural categories—firms make deliberate choices about task bundling that respond to technological change (Hackman & Oldham, 1976; Morgeson & Humphrey, 2006; Parker & Grote, 2022). From *labour economics*, the task-based framework predicts that automation affects specific tasks rather than whole occupations, generating within-occupation restructuring as routine tasks are automated while complementary tasks expand (Acemoglu & Autor, 2011b; Autor, Levy, & Murnane, 2003). From *occupational sociology*, research on occupational boundaries demonstrates that classifications impose artificial stability on inherently fluid processes, with substantial heterogeneity persisting within occupational categories (Abbott, 1988; Weeden, 2002). From *computational social science*, text analysis of job postings reveals that within-occupation skill variation often exceeds

between-occupation variation and is growing over time (Atalay et al., 2020a; Deming & Kahn, 2018a).

This chapter formalises the common mechanism underlying these predictions. Building on the canonical Roy (1951) occupational choice model and Acemoglu and Autor (2011a) task-based framework, I incorporate two extensions: (1) multi-dimensional worker skill endowments, and (2) heterogeneous firm productivity in coordinating task bundles, extending the assignment literature (Rosen, 1974; Sattinger, 1975; Teulings, 1995). The equilibrium concept follows the directed search framework of Moen (1997) and Shimer (2005), extended to multi-dimensional job design choice. Unlike Eeckhout and Kircher (2011) who study sorting with exogenous job characteristics, and Lopes de Melo (2018) who estimate sorting parameters from matched employer-employee data, the key innovation here is treating job design as a firm choice variable rather than an exogenous characteristic. These extensions generate a prediction absent from standard models: *occupations endogenously fragment into discrete sub-occupational clusters* even when all agents face identical production technologies—a pattern now documented empirically: within-occupation skill variation often exceeds between-occupation variation (Deming & Kahn, 2018a), this variance is increasing over time (Atalay et al., 2020a), most earnings inequality operates within rather than between occupations (Weeden et al., 2007), and recession-driven skill upgrading occurs primarily within occupational boundaries (Hershbein & Kahn, 2018a).

C.1.1 Main Results

The model yields five central propositions with direct empirical implications. The first three comprise the static framework; the latter two comprise a dynamic extension addressing temporal patterns.

Proposition 1 (Endogenous Clustering): Under supermodularity in production, equilibrium exhibits positive assortative matching between heterogeneous firms and workers, generating discrete clusters of job types within occupational boundaries. This predicts that unsupervised clustering methods (Chapter 6) should identify distinct sub-occupational groups.

Proposition 2 (Market Thickness): The number of equilibrium clusters increases with market size. Thicker labour markets support finer specialisation as coordination costs become easier to amortise. This generates testable predictions about cluster counts across metropolitan versus regional areas or large versus small occupations.

Proposition 3 (Automation Effects): Skill-biased automation shifts cluster boundaries and reallocates employment across clusters while preserving overall occupational structure. Routine-intensive clusters contract while complementary clusters expand, generating high

within-occupation volatility alongside low occupation-level employment effects. This predicts that occupation-level analysis should find weak automation-employment relationships while cluster-level analysis should reveal extreme heterogeneity obscured by aggregation.

Proposition 4 (Temporal Equilibrium Path): For continuously advancing automation $\rho(t)$, the equilibrium cluster configuration evolves continuously almost everywhere, with discrete cluster births and deaths occurring at critical viability thresholds. This explains why occupations exhibit stable aggregate structure alongside high within-occupation cluster churn.

Proposition 5 (Churn-Automation Speed): Cluster churn rates are proportional to the speed of automation advancement. Faster technological change generates more cluster births and deaths; periods of stability generate stable cluster configurations. This predicts that high-automation occupations should exhibit elevated cluster churn.

The remainder of this chapter proceeds as follows. Section C.2 presents the formal model environment, production function, and equilibrium characterisation. Section C.3 states and proves the three central static propositions. Section C.4 extends the baseline model to incorporate automation and derives comparative statics predictions. Section C.5 develops the dynamic extension, characterising temporal cluster evolution and churn rates (Propositions 4–5). Section C.6 maps theoretical predictions to empirical tests in subsequent chapters. Section C.7 discusses model limitations and alternative mechanisms. Chapter 6 develops the methodological approach for identifying clusters in data and tests all five propositions empirically.

C.1.2 Notation

Table C.1 summarises the key symbols used throughout this chapter.

C.2 Model Environment

C.2.1 Workers

There is a unit continuum of workers indexed by $i \in [0, 1]$, each endowed with a multi-dimensional skill vector:

$$s_i \in S = [0, \bar{s}]^K \quad (\text{C.1})$$

where K represents skill dimensionality and $\bar{s} > 0$ is the maximum skill level per dimension. The skill domain S is a compact, convex hypercube in $\mathbb{R}^K$ with side length $\bar{s}$ and Euclidean diameter $M_s = \bar{s}\sqrt{K}$ (the distance between opposite corners). Boundedness is required for existence of equilibrium (ensures compactness of utility space via Arzelà-Ascoli theorem)

Table C.1: Notation Summary

Symbol	Definition
<i>Primitives</i>	
$s \in S \subset \mathbb{R}^K$	Worker skill vector (multi-dimensional)
$d \in \mathcal{D} \subset \mathbb{R}^K$	Job design (task bundle), normalised to unit sphere
$\theta \in \Theta \subset \mathbb{R}_+$	Firm productivity parameter
K	Number of skill/task dimensions
N	Market size (mass of workers)
<i>Production and Costs</i>	
$A(s, d)$	Match quality function (Gaussian kernel)
λ	Match quality decay parameter (mismatch penalty)
$C(d, \theta)$	Coordination cost function
κ	Coordination cost scale parameter
$\gamma > 1$	Coordination cost curvature (returns to span)
$\alpha > 0$	Productivity elasticity in coordination costs
$f > 0$	Fixed posting cost per job design
<i>Equilibrium Objects</i>	
$d^*(\theta)$	Optimal job design function (firm choice)
n^*	Equilibrium number of clusters
$q(d)$	Queue length (worker-to-vacancy ratio) at design d
$w(s, d)$	Wage function (skill and design contingent)
$U(s)$	Market utility (worker's outside option)
$\pi(\theta, d, q)$	Firm profit function
<i>Derived Concepts</i>	
$\mu(d)$	Ideal skill profile for job design d
$\text{span}(d)$	Standard deviation of task emphasis (L2 norm)
ρ	Automation intensity parameter
$M_A = 2\lambda^2 K \bar{s}^2$	Global bound on production curvature
<i>Technology-Indexed Partitions (Section C.4)</i>	
$\tau \in \mathcal{T}$	Technology type index (e.g., RPA, ML, AI)
$\mathcal{P}(\tau)$	Technology-indexed skill partition
$\mathcal{S}^-(\tau)$	Skill dimensions substituted by technology τ
$\mathcal{S}^+(\tau)$	Skill dimensions complementary to technology τ
$\mathcal{S}^0(\tau)$	Skill dimensions neutral to technology τ
$\alpha_k(\tau)$	Augmentation rate for substituted skill k
<i>Dynamic Extension (Section C.5)</i>	
$\rho(t)$	Time-varying automation intensity
$n^*(t)$	Time-varying equilibrium cluster count
$\chi(t, \Delta t)$	Cluster churn rate over interval Δt
$\bar{\rho}_c$	Viability threshold for cluster c (death)
$\underline{\rho}_{c'}$	Viability threshold for cluster c' (birth)
t^*	Critical transition time (birth or death)

Note. Notation used throughout the theoretical framework. Primitives are exogenous model parameters; production and cost functions are derived from the economic environment; equilibrium objects are endogenously determined. Dynamic extension notation (Section C.5) extends the static model to time-varying automation.

and is empirically justified since skill variables are standardised in the data. In the empirical application (Chapter 6), $K \approx 2,000$ unique skills from Burning Glass data with $\bar{s} = 5$ (five standard deviations covers 99.99% of observations), but theoretical results hold for any $K \geq 2$ and $\bar{s} < \infty$. Skills are distributed according to continuous distribution $F(s)$ with full support on S .

This multi-dimensional specification departs from standard human capital models that treat skill as scalar (Mincer, 1974). Following Costinot and Vogel (2010), workers cannot be ranked unambiguously—comparative advantage emerges from skill complementarities in production. A worker with $s_i = (10, 2)$ (high analytical, low interpersonal skills) is neither “better” nor “worse” than one with $s_j = (2, 10)$ (low analytical, high interpersonal), only different.

C.2.2 Firms and Job Design

There is a continuum of firms indexed by $j \in [0, \infty)$, each characterised by productivity parameter:

$$\theta_j \in \Theta \subset \mathbb{R}_+ \quad (\text{C.2})$$

distributed according to $G(\theta)$ with density $g(\theta) > 0$ on $\Theta = [\underline{\theta}, \bar{\theta}]$.

Each firm chooses a *job design* $d \in \mathcal{D}$ that specifies the task bundle workers perform. Formally, the job design space is the positive unit sphere (or spherical simplex):

$$\mathcal{D} = \left\{ d \in \mathbb{R}_+^K : \|d\|_2 = \sqrt{\sum_{k=1}^K d_k^2} = 1 \right\} \quad (\text{C.3})$$

where $\|d\|_2$ denotes the Euclidean (L2) norm and $\mathbb{R}_+^K$ restricts to non-negative coordinates. Job designs are normalised weight vectors indicating the relative importance of each skill dimension—for instance, $d_{\text{compliance}} = (0.8, 0.1, 0.1, \dots)$ emphasizes analytical skills while $d_{\text{advisory}} = (0.2, 0.6, 0.2, \dots)$ emphasizes interpersonal skills. The design space $\mathcal{D}$ is closed and includes corner points such as $d = (1, 0, \dots, 0)$ where one task component dominates entirely. These corners correspond to maximally specialised job designs and are economically meaningful (a role focused exclusively on a single skill dimension); they are included in the analysis. The span function defined below is differentiable at all corner points (where $Pd \neq 0$), so the only non-differentiable point in $\mathcal{D}$ is the unique interior centroid $d = (1/\sqrt{K})\mathbf{1}$ representing perfectly balanced task weights, as Lemma 2.0 establishes.

Choice of normalisation: I use L2-normalisation (unit sphere) rather than L1-normalisation (standard simplex $\sum_k d_k = 1$) for three reasons. First, the span measure

$\text{span}(d) = (1/\sqrt{K}) \cdot \|Pd\|_2$ (equation (C.10)) is the L2 norm of a linear projection, inheriting convexity properties that streamline the proof of coordination cost convexity (see equation (C.25)). Second, L2-normalisation generates a natural metric for measuring task diversity via the Euclidean distance of demeaned weights, which corresponds to the standard deviation up to a scaling constant (Johnson & Wichern, 2007, Chapter 2). Third, the Gaussian match quality function $A(s, d) = \exp[-\lambda \|s - \mu(d)\|^2]$ uses L2 distance, so L2-normalised designs align naturally with the production technology. Under L2-normalisation, all coordinates are non-negative ($d \in \mathbb{R}_+^K$) since negative task weights are economically meaningless. The unit sphere normalisation ensures job designs are scale-invariant: only relative task weights matter for coordination costs and match quality. For comparative statics I work with the scalar index $x = \text{span}(d)$ (a complete lattice); d serves as an implementation that realizes any given x .

The job design choice captures empirically salient heterogeneity within occupations. Among “Accountants,” some positions require primarily routine data processing (tax compliance, bookkeeping) while others demand advisory and strategic skills (financial planning, M&A due diligence). These represent different job designs d within the same occupational category. Empirical support for demand-side workplace specialisation driving within-occupation variation comes from G. M. Cortés and Salvatori (2019), who document that firms increasingly specialise in narrow task bundles, generating heterogeneous job types within traditional occupational classifications.

C.2.3 Production Function

A firm with productivity θ employing a worker with skills s in job design d produces output:

$$Y(s, d, \theta) = \theta \cdot A(s, d) \tag{C.4}$$

where $A(s, d)$ represents match quality between worker skills and job requirements.

Following Eeckhout and Kircher (2011), match quality exhibits complementarity between worker skills and job design. Each job design d has an associated ideal skill profile $\mu(d) \in S$ representing the skill vector that maximises productivity for that design. Match quality depends on the distance between a worker’s actual skills s and this ideal:

$$A(s, d) = \exp\left[-\lambda \cdot \|s - \mu(d)\|^2\right] \tag{C.5}$$

where $\mu(d) : \mathcal{D} \rightarrow S$ maps job designs to ideal skill profiles and $\lambda > 0$ governs the importance of skill-job fit. **Normalization:** The match quality function satisfies $A(s, d) \in (0, 1]$ for all matched pairs, with $A(s, d) = 1$ when $s = \mu(d)$ (perfect match) and $A(s, d) \rightarrow 0$

as $\|s - \mu(d)\| \rightarrow \infty$ (severe mismatch). This normalisation ensures match quality is bounded above by unity, interpreting A as an efficiency parameter: a perfectly matched worker produces at full productivity, while mismatched workers produce at reduced efficiency.

This functional form is motivated by three considerations. First, the Gaussian kernel generates smooth, globally differentiable match quality, ensuring well-defined first-order conditions for firm optimisation. Second, the exponential-quadratic form nests as a special case the class of supermodular matching functions studied by Shimer and Smith (2000), satisfying their sufficient conditions for positive assortative matching (their Assumption A2). Third, the mismatch penalty $\|s - \mu(d)\|^2$ imposes testable restrictions: marginal returns to skill-job fit are symmetric across skill dimensions and decline exponentially with distance from the ideal profile.

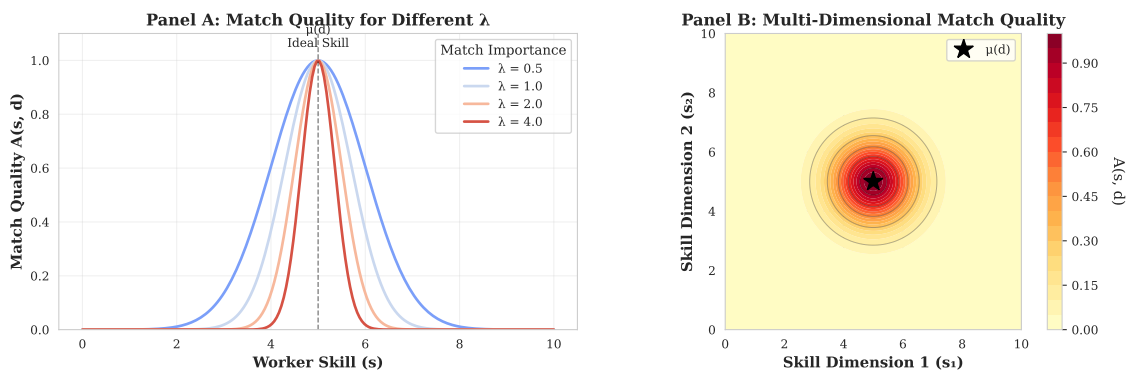
Alternative specifications include CES production functions (Acemoglu & Autor, 2011b) or more general supermodular functions (Costinot & Vogel, 2010). The Gaussian form is *not* nested within CES; the key distinction is that CES allows perfect substitutability in the limit ($\sigma \rightarrow \infty$) while the Gaussian imposes strict complementarity. For the empirical application, this distinction matters: the model predicts workers cannot compensate for severe skill deficiencies, consistent with job advertisements specifying “required” versus “preferred” qualifications.

The production function exhibits two key properties:

Property 1 (Complementarity): $\frac{\partial^2 Y}{\partial \theta \partial A} > 0$. High-productivity firms benefit more from good worker-job matches, creating incentives for assortative matching.

Property 2 (Imperfect Substitutability): Skills are complements, not perfect substitutes. A worker cannot compensate for deficiency in skill k through excess in skill k' . This captures the empirical reality that outstanding interpersonal skills cannot offset complete inability to use spreadsheet software for data-intensive roles. Figure C.1 illustrates how match quality declines exponentially with distance from the ideal skill profile.

Figure C.1: Match Quality Function $A(s, d) = \exp[-\lambda \|s - \mu(d)\|^2]$. Panel A: Penalty increases with mismatch parameter λ . Panel B: Multi-dimensional surface shows exponential decline with distance from ideal skill profile $\mu(d)$.



C.2.4 Coordination Costs

Firms face costs to implement complex job designs. This section derives coordination costs from a span-of-control model following Garicano (2000) and Garicano and Rossi-Hansberg (2006).

Microfoundation: Consider a firm hiring n workers with heterogeneous skill vector s . Each worker encounters problems at Poisson rate $\eta > 0$. Problems outside a worker's skill set ($\|p - s\| > \epsilon$ for problem p) require managerial intervention. A manager with productivity θ can solve $\theta \cdot m$ problems per period, where m is managerial time allocation.

The span-of-control constraint is:

$$m \geq \frac{n \cdot \eta \cdot \Pr(\|p - s\| > \epsilon|d)}{\theta} \quad (\text{C.6})$$

where $\Pr(\|p - s\| > \epsilon|d)$ is the probability a worker assigned to job design d encounters an unsolvable problem. Under the Gaussian skill distribution, this probability increases with task diversity: workers in jobs with high $\text{span}(d)$ (defined formally in equation (C.10) below as a measure of task concentration/specialisation) face more problems outside their expertise when task requirements are concentrated on specific skill dimensions.

Following Garicano and Rossi-Hansberg (2006), the cost of managerial time is $w_m \cdot m$ where w_m is the manager's opportunity cost. With free entry, $w_m = \theta \cdot \bar{A}$ (manager's productivity in direct production). This yields coordination cost per worker:

$$C(d, \theta) = w_m \cdot \frac{\eta \cdot \Pr(\|p - s\| > \epsilon|d)}{\theta} = \bar{A} \cdot \eta \cdot \Pr(\|p - s\| > \epsilon|d) \quad (\text{C.7})$$

Under regularity conditions, the probability term can be approximated. Assume problems p are distributed $\mathcal{N}(\mu(d), \sigma^2 I)$ where $\mu(d)$ is the ideal skill profile for job design d . For a worker with skills s assigned to job d , problems requiring skills outside their capability ($\|p - s\| > \epsilon$) occur with probability increasing in the dispersion of required skills. Specifically, when d is more specialist (higher span), workers must handle a wider range of problems, increasing the probability of encountering unsolvable ones. Under the Gaussian assumption:

$$\Pr(\|p - s\| > \epsilon|d) = \int_{\|p - s\| > \epsilon} \phi(p; \mu(d), \sigma^2) dp \approx \kappa \cdot [\text{span}(d)]^\gamma \quad (\text{C.8})$$

where $\kappa = \eta \cdot \Phi^{-1}(\epsilon/\sigma)$ depends on the problem arrival rate and threshold, and $\gamma > 1$ reflects convexity: doubling task diversity more than doubles problem complexity due to interaction effects across skill dimensions. This approximation holds for moderate span

values within the empirically relevant range.¹

To complete the derivation, I specify management quality as productivity-dependent: $\bar{A}(\theta) = \theta^{-\alpha}$ where $\alpha \geq 0$ captures learning-by-doing or organisational capital. High-productivity firms develop superior management systems, reducing per-worker coordination costs. Substituting into equation (C.7):

$$C(d, \theta) = \frac{\kappa}{\theta^\alpha} \cdot \text{span}(d)^\gamma \quad (\text{C.9})$$

where $\text{span}(d)$ measures the *task diversity* of job design d . Formally, define the demeaning projection matrix $P = I - K^{-1}\mathbf{1}\mathbf{1}^T$ where $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^K$. Then:

$$\text{span}(d) \equiv \frac{1}{\sqrt{K}} \cdot \|Pd\|_2 = \frac{1}{\sqrt{K}} \sqrt{\sum_{k=1}^K (d_k - \bar{d})^2} \quad (\text{C.10})$$

where $\bar{d} = K^{-1} \sum_k d_k$ is the mean component weight and $Pd = d - \bar{d} \cdot \mathbf{1}$ is the projection onto the subspace orthogonal to $\mathbf{1}$. Since P is a linear projection and the L^2 norm is convex, $\text{span}(\cdot)$ is a convex function on $\mathcal{D}$ (composition of a linear map with a norm). This measures the dispersion of task weights around their mean. Higher $\text{span}(d)$ implies *more concentrated/specialist* task emphasis (weights depart further from the mean); lower $\text{span}(d)$ implies *more balanced/generalist* task emphasis (weights closer to equal).²

Lemma 2.0 (Span Differentiability on the Unit Sphere): Under Assumption R1, the span function $\text{span} : \mathcal{D} \rightarrow X \subseteq [0, \sqrt{(K-1)/K}]$ defined by $\text{span}(d) = (1/\sqrt{K}) \cdot \|Pd\|_2$ where $P = I - K^{-1}\mathbf{1}\mathbf{1}^T$ is the demeaning projection, satisfies:

(i) $\text{span}(\cdot)$ is continuously differentiable on $\mathcal{D}_+ \equiv \{d \in \mathcal{D} : Pd \neq 0\}$ (everywhere except one point with measure zero).

(ii) For all $d \in \mathcal{D}_+$ and $k = 1, \dots, K$:

$$\frac{\partial \text{span}(d)}{\partial d_k} = \frac{d_k - \bar{d}}{\sqrt{K} \cdot \text{span}(d)} \quad \text{where } \bar{d} = K^{-1} \sum_{j=1}^K d_j \quad (\text{C.11})$$

¹The power-law approximation follows from Taylor expansion of the multivariate normal CDF around the mean. The exponent $\gamma > 1$ arises because problems requiring coordination across multiple skill dimensions grow combinatorially, not linearly, with task diversity. See Garicano (2000) for related derivations in knowledge hierarchies.

²Alternative complexity measures include task range ($\max_k d_k - \min_k d_k$), Herfindahl concentration ($H = \sum_k d_k^2$, inverted), or Shannon entropy ($-\sum_k d_k \log d_k$). The standard deviation measure is preferred because: (i) it has well-established statistical properties, (ii) it is monotone increasing in task dispersion, (iii) it captures variation across all dimensions rather than just extremes, and (iv) it yields convex coordination costs under the Gaussian problem distribution assumption. Alternative measures would generate similar qualitative predictions but different parametric forms for $C(d, \theta)$.

(iii) The gradient satisfies $\|\nabla_d \text{span}\|_2 = 1/\sqrt{K}$ for all $d \in \mathcal{D}_+$.

(iv) At $d = (1/\sqrt{K}) \cdot \mathbf{1}$ (where $\text{span}(d) = 0$), the subdifferential is $\partial \text{span}(0) = \{g \in \mathbb{R}^K : \|g\|_2 \leq 1/\sqrt{K}\}$.

Proof: (i) Since $\text{span}(d) = (1/\sqrt{K}) \cdot \|Pd\|_2$ is the composition of a linear map P with the L^2 norm (scaled by $1/\sqrt{K}$), it is C^1 wherever $\|Pd\|_2$ is C^1 , i.e., wherever $Pd \neq 0$. The set $\{d : Pd = 0\}$ consists of constant vectors on the simplex, which under the unit-sphere constraint is the singleton $\{(1/\sqrt{K}) \cdot \mathbf{1}\}$.

(ii) Apply the chain rule: $\nabla_d \|Pd\|_2 = P^T(Pd/\|Pd\|_2) = Pd/\|Pd\|_2$ (since P is symmetric).

Thus:

$$\frac{\partial \text{span}}{\partial d_k} = \frac{1}{\sqrt{K}} \cdot \frac{(Pd)_k}{\|Pd\|_2} = \frac{d_k - \bar{d}}{\sqrt{K} \cdot \text{span}(d)}$$

where $(Pd)_k = d_k - \bar{d}$ by definition of the demeaning projection.

(iii) Since $\|Pd\|_2 \leq \|d\|_2 = 1$ (projection cannot increase norm) and $\|Pd\|_2 = \sqrt{K} \cdot \text{span}(d)$, I have:

$$\|\nabla_d \text{span}\|_2 = \frac{1}{\sqrt{K}} \cdot \frac{\|Pd\|_2}{\|Pd\|_2} = \frac{1}{\sqrt{K}}$$

(iv) At $\text{span} = 0$, the L^2 norm is not differentiable, but it is convex. The subdifferential of $\|Pd\|_2$ at $Pd = 0$ is the unit ball $\{g : \|g\|_2 \leq 1\}$, scaled by $1/\sqrt{K}$. ■

Equilibrium implication: Under Assumption 1'(ii), equilibrium designs satisfy $\text{span}(d) \geq \underline{x} > 0$, avoiding the non-differentiable corner. This allows application of the chain rule when optimising profit functions: $\frac{\partial \pi}{\partial d_k} = \frac{\partial \pi}{\partial \text{span}} \cdot \frac{\partial \text{span}}{\partial d_k}$, where the first term depends on the coordination cost exponent γ and the second term depends on how much dimension k contributes to overall task diversity.

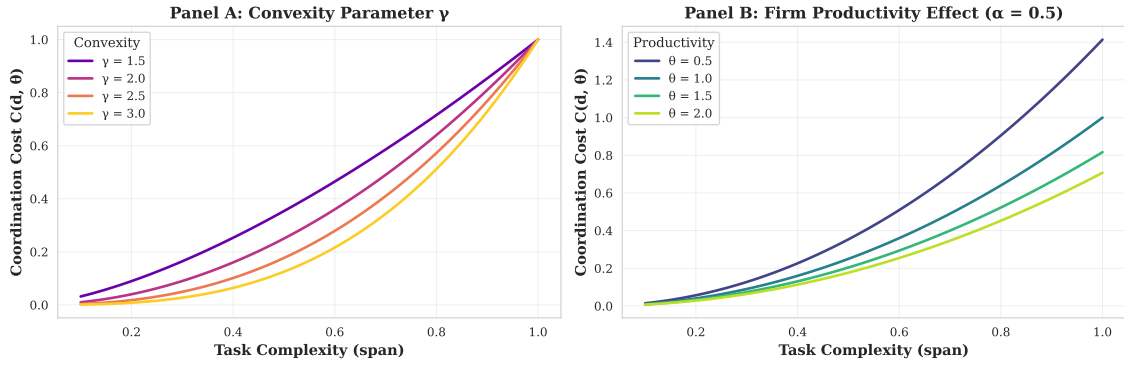
Parametric restrictions: The parameters must satisfy:

- $\kappa > 0$: Base coordination cost (derived from $\bar{A} \cdot \eta$)
- $\gamma > 1$: Ensures convexity in task complexity (increasing marginal coordination costs)
- $\alpha \geq 0$: Allows high-productivity firms to have lower coordination costs via superior management (Bloom & Van Reenen, 2007)
- $\text{span}(d) \in [0, \sqrt{(K-1)/K}]$: Normalized dispersion of task weights under unit-sphere constraint $\|d\| = 1$, with maximum achieved when $d = (1, 0, \dots, 0)$ (complete specialisation)

Alternative complexity measures include task range ($\max_k d_k - \min_k d_k$) or Shannon entropy ($-\sum_k d_k \log d_k$). The standard deviation measure in equation (C.10) is preferred because it satisfies monotonicity (adding tasks increases complexity), captures dispersion

across all dimensions, and has well-established statistical properties. Figure C.2 illustrates how coordination costs increase convexly with task span and vary with firm productivity.

Figure C.2: *Coordination Costs* $C = \frac{\kappa}{\theta^\alpha} \cdot \text{span}(d)^\gamma$. Panel A: When $\gamma > 1$, costs are strictly convex in task span, creating increasing marginal costs. Panel B: Higher firm productivity θ reduces coordination costs, enabling complex job designs at lower cost.



C.2.5 Wages and Firm Profits

Firms post vacancies specifying job design d and wage w . Workers observe all vacancies and apply to maximise utility, which equals wage (assuming risk neutrality and zero commuting costs). Firms hire workers to maximise profits:

$$\pi(s, d, \theta) = Y(s, d, \theta) - w(s, d) - C(d, \theta) \quad (\text{C.12})$$

In equilibrium, wages are determined through competitive search: firms compete for workers by posting wage-job design pairs (w, d) , and workers apply to the most attractive offers. Following Shimer and Smith (2000), Shi (2009), and Eeckhout and Kircher (2010), this generates wage function $w(s, d)$ such that workers are indifferent across jobs they accept and firms earn zero profits on marginal hires, with sorting determined by complementarities in production.

C.2.6 Equilibrium Definition

I formalise equilibrium using a directed search framework with utility posting (Burdett et al., 2001; Menzio & Shi, 2010; Moen, 1997). This framework choice serves three purposes critical for within-occupation clustering. First, directed search allows workers to target specific submarkets (job types) rather than searching randomly, enabling the precise sorting patterns that generate discrete clusters. Second, utility posting determines wages

through competitive market clearing rather than ex-post Nash bargaining, reducing free parameters and ensuring tractability with heterogeneous job designs. Third, the framework naturally accommodates multi-dimensional heterogeneity in both workers (skills s) and firms (productivity θ , job design d), which is essential for analysing within-occupation variation where occupation-level homogeneity fails.

Market Segmentation and Utility Posting

The labour market segments into submarkets indexed by job design $d \in \mathcal{D}$. In each submarket d , firms post wages $w(s, d)$ contingent on observable worker skill s . Let $U(s)$ denote the equilibrium market utility for a worker with skill s —the expected utility from optimal job search. Firms in submarket d must offer wage schedule $w(s, d)$ satisfying:

$$w(s, d) \geq U(s) \quad \text{for all } s \in S \quad (\text{C.13})$$

with equality for workers actually employed in that submarket (zero profits on marginal hires).

Firm Problem with Directed Search

A firm with productivity θ choosing submarket d faces a queue length $q(d)$ (ratio of workers to vacancies). The firm's hiring probability is $h(q(d))$, where $h(\cdot)$ is increasing and concave, satisfying standard matching function properties (Petrongolo & Pissarides, 2001). The firm chooses (w, d) to maximise expected profits:

$$\max_{w, d} \left\{ h(q(d)) \cdot \int_S [Y(s, d, \theta) - w(s, d)] \phi(s|d) ds - C(d, \theta) \right\} \quad (\text{C.14})$$

subject to the participation constraint (C.13), where $\phi(s|d)$ is the density of worker skills applying to submarket d .

Firms internalise the congestion externality: offering higher wages attracts more workers (increases $q(d)$), raising the hiring probability $h(q(d))$. In equilibrium, the wage schedule satisfies the zero-profit condition on marginal hires:

$$w(s, d) = Y(s, d, \theta(d)) - U(s) \quad (\text{C.15})$$

where $\theta(d)$ is the marginal firm productivity operating in submarket d .

Worker Problem

Workers with skill s choose which submarket to enter. Let $p(q(d))$ denote the job-finding probability for a worker in submarket d (related to $h(q(d))$ by matching function). Worker s solves:

$$U(s) = \max_{d \in \mathcal{D}} \{p(q(d)) \cdot w(s, d)\} \quad (\text{C.16})$$

In equilibrium, workers are indifferent across all submarkets they enter with positive probability, and queue lengths adjust such that $U(s)$ is consistent with wage offers.

Equilibrium Definition

Definition 1 (Directed Search Equilibrium): An equilibrium consists of:

1. Job design function $d^*(\theta) : \Theta \rightarrow \mathcal{D}$ mapping firm productivity to submarket choice
2. Wage function $w(s, d) : S \times \mathcal{D} \rightarrow \mathbb{R}_+$ satisfying zero-profit condition (C.15)
3. Market utility function $U(s) : S \rightarrow \mathbb{R}_+$ representing workers' equilibrium outside option
4. Queue length function $q(d) : \mathcal{D} \rightarrow \mathbb{R}_+$ determining matching probabilities

such that:

1. **Firm optimisation:** Each firm type θ chooses job design maximising (C.14):

$$d^*(\theta) \in \arg \max_{d \in \mathcal{D}} \left\{ h(q(d)) \cdot \int_S [Y(s, d, \theta) - U(s)] \phi(s|d) ds - C(d, \theta) \right\} \quad (\text{C.17})$$

2. **Worker optimisation:** Each worker s directs search to maximise expected utility (C.16), achieving $U(s)$ in equilibrium
3. **Free entry:** Firms enter submarkets until expected profits equal fixed cost $f > 0$:

$$h(q(d)) \cdot \int_S [Y(s, d, \theta(d)) - U(s)] \phi(s|d) ds = C(d, \theta(d)) + f \quad (\text{C.18})$$

4. **Market clearing:** Queue lengths adjust such that worker mass equals supply:

$$\int_{\Theta} \mathbf{1}\{d^*(\theta) = d\} g(\theta) d\theta = q(d) \cdot \int_S \mathbf{1}\{d \in \arg \max U(s)\} f(s) ds \quad (\text{C.19})$$

Wage Determination: In each submarket (d, q) , the wage function $w(s, d)$ satisfies the zero-profit condition $w(s, d) = Y(s, d, \theta(d)) - U(s)$ for the marginal worker whose

utility equals the market outside option $U(s)$. Infra-marginal workers with idiosyncratic preferences or informational advantages may earn rents above their outside option, but free entry ensures that the posted utility $U(s)$ pins down the marginal worker's compensation in equilibrium.

Lemma 2.1 (Wage-Lipschitz Property): Under Assumptions R1–R4, equilibrium wage schedules $w(s, d)$ are L_w -Lipschitz in s for each job design d , where L_w depends on $(\lambda, \theta_{\max}, L_\mu)$.

Proof: By the zero-profit condition, $w(s, d) = Y(s, d, \theta(d)) - U(s) = \theta(d) \cdot A(s, d) - U(s)$. The match quality function $A(s, d) = \exp[-\lambda\|s - \mu(d)\|^2]$ has gradient:

$$\frac{\partial A(s, d)}{\partial s} = -2\lambda A(s, d) \cdot (s - \mu(d))$$

Therefore $\|\nabla_s A(s, d)\| \leq 2\lambda\|s - \mu(d)\| \cdot A(s, d) \leq 2\lambda M_s$ where M_s is the diameter of skill space S (Assumption R1). Since $Y(s, d, \theta) = \theta \cdot A(s, d)$, I have $\|\nabla_s Y(s, d, \theta)\| \leq 2\lambda\theta M_s$. Taking $L_w = 2\lambda\theta_{\max} M_s$ ensures $|w(s, d) - w(s', d)| \leq L_w\|s - s'\|$ for all $s, s' \in S$ and all $d \in \mathcal{D}$. ■

This lemma ensures that equilibrium market utilities $U(s) = \max_d p(q(d)) \cdot w(s, d)$ inherit the Lipschitz property (maximum of Lipschitz functions with common constant remains Lipschitz with the same constant), which is crucial for applying the Arzelà-Ascoli theorem to establish compactness of the utility space.³ The Lipschitz constant $L_w = 2\lambda\theta_{\max} M_s$ depends only on model primitives $(\lambda, \theta_{\max}, \text{the diameter } M_s \text{ of skill space } S)$, not on the equilibrium utility $U(s)$ itself. This avoids the circularity that would otherwise threaten the fixed-point argument: the Lipschitz constant is preserved through the correspondence Γ because each application of Γ produces wages whose Lipschitz constant is bounded by the same L_w , which then bounds the Lipschitz constant of the resulting utility (via the maximum-of-Lipschitz argument). The Lipschitz ball $\mathcal{U}_L = \{U : L_w\text{-Lipschitz on } S\}$ is therefore mapped into itself by Γ , providing the closed self-map required for Kakutani's theorem.

Existence: I establish existence using Kakutani's Fixed Point Theorem (Kakutani, 1941), which accommodates the possibility of multiple optimal job designs for some firm types. Define the correspondence $\Gamma : \mathcal{U} \rightrightarrows \mathcal{U}$ mapping market utilities to the set of possible equilibrium utilities: given $U \in \mathcal{U}$, firms choose job designs from the optimal correspondence $d^*(\theta; U) \equiv \arg \max_d \pi(\theta, d; U)$ (which may be multi-valued when multiple designs yield equal profits), which induces queue lengths $q(d; U)$ via market clearing,

³The Arzelà-Ascoli theorem states that a subset of continuous functions on a compact domain is relatively compact in the uniform topology if and only if it is uniformly bounded and equicontinuous. Lipschitz continuity with a common constant implies equicontinuity. For a treatment oriented toward economic applications, see Ok (2007, Chapter C).

which determines wages $w(s, d; U)$ via zero-profit conditions, which implies the set of possible worker utilities $\Gamma(U)$ via equation (C.16). A fixed point $U^* \in \Gamma(U^*)$ constitutes equilibrium.

Following Shimer (2005) (Theorem 1), extended to multi-dimensional types, Kakutani's theorem requires verifying: (i) the utility space $\mathcal{U}_L$ is non-empty, compact (by Arzelà-Ascoli via Lemma 2.1), and convex; (ii) the correspondence Γ has non-empty, convex values; and (iii) Γ is upper hemicontinuous.

Non-emptiness of best responses: For each $U \in \mathcal{U}_L$ and each firm type θ , the set $d^*(\theta; U) = \arg \max_{d \in \mathcal{D}} \pi(\theta, d; U)$ is non-empty by the Maximum Theorem: profit π is continuous in d (by Assumptions R1–R4), and the choice set $\mathcal{D}$ (unit sphere) is compact.

Wage schedules: Given any profile of design choices $(d(\theta))_{\theta \in \Theta}$, equilibrium wage schedules $w(s, d)$ satisfying zero-profit for the marginal worker exist by my equilibrium definition (equation (C.15)): $w(s, d) = Y(s, d, \theta(d)) - U(s)$, where $\theta(d)$ is the marginal firm type operating in submarket d . Measurability of $w(s, d)$ with respect to the induced queue distributions follows from the measurability of $Y(\cdot)$ (continuous function of skill-design match) and $U(\cdot)$ (Lipschitz by Lemma 2.1, hence Borel measurable).

Under Assumptions R1–R5, all three Kakutani conditions hold (formal verification in Appendix D.1), guaranteeing existence of equilibrium.⁴

This equilibrium concept generalises competitive search to settings with multi-dimensional heterogeneity $(s, \theta) \in \mathbb{R}^K \times \mathbb{R}$, multi-dimensional job design $d \in \mathbb{R}^K$, and non-transferable coordination costs $C(d, \theta)$. The directed search framework clarifies that wages adjust to make workers indifferent across submarkets while firms choose job designs anticipating the resulting worker sorting. Figure C.3 illustrates the key components of this directed search equilibrium.

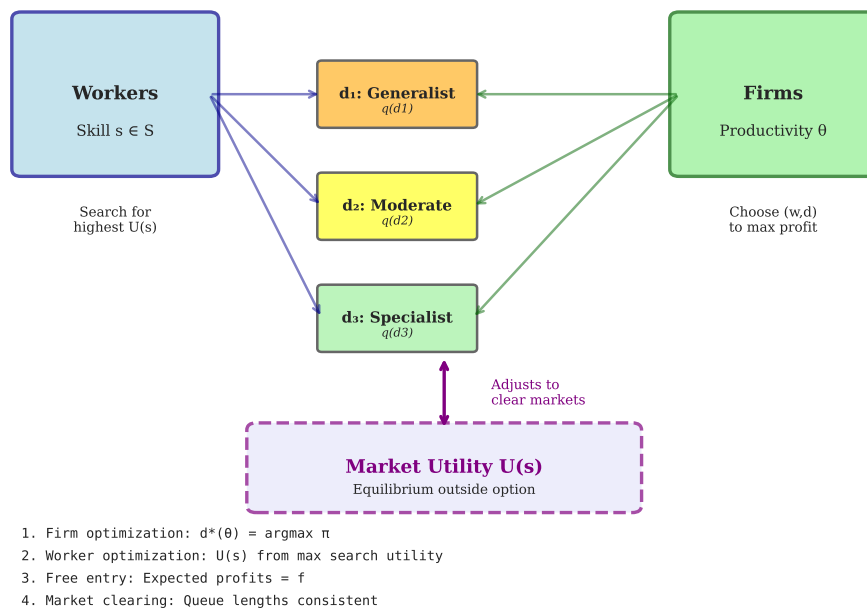
C.2.7 Parametric Restrictions and Regularity Conditions

The assumptions below serve two distinct roles. *Core assumptions*—supermodularity (Assumption 1) and convexity of coordination costs (introduced in Section C.2)—are economically substantive and necessary for the propositions: weakening them changes the qualitative predictions. *Technical regularity conditions* (Assumptions R1–R5 below) are sufficient conditions enabling the existence proof via Kakutani's fixed-point theorem and the calibration; they ensure compactness, continuity, and bounded curvature but are not

⁴Kakutani's theorem is more appropriate than Schauder-Tychonoff for this application because the optimal job design correspondence $d^*(\theta; U)$ may be set-valued when multiple designs yield equal maximal profits (e.g., at knife-edge parameter values). The Schauder-Tychonoff fixed point theorem applies only to single-valued continuous maps, requiring additional tie-breaking assumptions.

Figure C.3: Directed Search Equilibrium. Workers choose submarkets by job design d to maximise expected utility $U(s)$. Firms post wage-design pairs (w, d) to maximise profits given queue lengths $q(d)$. Market utility adjusts to clear all submarkets, with workers indifferent across entered submarkets and firms earning zero profits on marginal hires.

Directed Search Equilibrium with Utility Posting



necessary for the propositions to hold under more general conditions. The propositions' empirical content does not depend on the specific bounds chosen for $\bar{s}$, $\theta_{\max}$, or the Lipschitz constants of $\mu(\cdot)$; these are technical sufficient conditions for the proofs to go through, not load-bearing economic assumptions.

Assumption R1 (Bounded Support):

- Worker skill space: $S = [0, \bar{s}]^K$ for some $\bar{s} < \infty$, with Euclidean diameter $M_s = \bar{s}\sqrt{K}$
- Job design space: $\mathcal{D} = \{d \in \mathbb{R}_+^K : \|d\| = 1\}$ (unit sphere, as defined in equation (C.3))
- Firm productivity space: $\Theta = [\theta_{\min}, \theta_{\max}]$ with $0 < \theta_{\min} < \theta_{\max} < \infty$

Bounded support ensures compactness of choice sets, which is essential for existence proofs via fixed-point theorems. The unit sphere $\mathcal{D}$ is compact as a closed and bounded subset of $\mathbb{R}^K$ (Heine-Borel: closed and bounded implies compact in finite-dimensional Euclidean space). Job designs are scale-invariant weight vectors, so normalisation to unit norm loses no generality. The hypercube S has side length $\bar{s}$ and diameter $M_s = \max_{s, s' \in S} \|s - s'\| = \bar{s}\sqrt{K}$ (achieved at opposite corners $(0, \dots, 0)$ and $(\bar{s}, \dots, \bar{s})$). The bounds on skill space and productivity are sufficiently large to encompass empirically relevant ranges.

Assumption R2 (Smooth Distributions):

- Worker skill distribution $F(s)$ has continuous density $f(s) > 0$ on $\text{int}(S)$
- Firm productivity distribution $G(\theta)$ has continuous density $g(\theta) > 0$ on $\text{int}(\Theta)$

Continuous positive densities ensure that (i) there are no mass points in type distributions, (ii) indifference thresholds are uniquely determined, and (iii) the fixed-point operator defining equilibrium is continuous in market utilities $\{U(s)\}$.

Assumption R3 (Coordination Cost Parameters):

- Convexity parameter: $\gamma > 1$ (strict convexity in span)
- Productivity effect: $\alpha \in [0, 1]$ (manageable productivity scaling)
- Base cost: $\kappa > 0$ (positive coordination costs)
- Profitability condition: $\kappa \cdot \theta_{\min}^{-\alpha} < \theta_{\max} \cdot \inf_{s,d} A(s, d)$

The convexity restriction $\gamma > 1$ is critical for generating discrete bunching (Proposition 1, Step 2). The profitability condition ensures that even the least productive firms can profitably operate in equilibrium, preventing corner solutions where the entire market shuts down. The restriction $\alpha \in [0, 1]$ ensures that high-productivity firms do not have arbitrarily low coordination costs (which would generate unrealistic levels of specialisation).

Assumption R4 (Match Quality Function):

- Match quality parameter: $\lambda > 0$ (positive importance of skill-job fit)
- Ideal skill mapping: $\mu(d) : \mathcal{D} \rightarrow S$ is continuous and increasing in each component
- Regularity of μ : The mapping $\mu(\cdot)$ is L_μ -Lipschitz and twice continuously differentiable along any span-parametrized path $d(\tau)$ with $\text{span}(d(\tau)) = \tau$, satisfying $\|\partial\mu/\partial\tau\| \leq \bar{L}_\mu$ and $\|\partial^2\mu/\partial\tau^2\| \leq \bar{L}_\mu^{(2)}$ for finite constants $\bar{L}_\mu, \bar{L}_\mu^{(2)}$
- Finite moments: $\int_S A(s, d)f(s) ds < \infty$ and $\int_S A(s, d)^2 f(s) ds < \infty$ for all $d \in \mathcal{D}$

The parameter $\lambda > 0$ governs the importance of skill-job match quality. Larger λ implies tighter matching requirements and sharper cluster boundaries. The continuity and monotonicity of $\mu(d)$ ensure that more complex job designs require higher skills, consistent with the supermodularity condition (Assumption 1). The Lipschitz and twice-differentiability conditions on μ ensure bounded curvature of the production function (Lemma A.2), which is essential for establishing global concavity of profits when combined with convex coordination costs. The finite-moment conditions ensure that expected output and surplus are well-defined under the Gaussian match quality kernel $A(s, d) = \exp[-\lambda \cdot \|s - \mu(d)\|^2]$, which decays to zero under large skill-job mismatch but has finite expectations under bounded support (Assumption R1).

Assumption R5 (Matching Function):

- Firm hiring probability: $h(q)$ is increasing, concave, with $h(0) = 0$ and $\lim_{q \rightarrow \infty} h(q) = 1$
- Worker finding probability: $p(q) = \frac{h(q)}{q}$ is decreasing in queue length q
- Hosios condition (optional): The elasticity $\eta_h(q) \equiv \frac{qh'(q)}{h(q)}$ satisfies efficiency condition (Hosios, 1990)

These are standard directed search restrictions. With q measuring workers per vacancy, longer queues increase the firm's hiring probability $h(q)$ (more candidates per position) but decrease the worker's finding probability $p(q) = h(q)/q$ (more competition per vacancy). The Hosios condition relates to constrained efficiency but is not required for existence.

Verification of Fixed-Point Conditions:

Under Assumptions R1–R5, I verify the conditions for applying Kakutani's Fixed Point Theorem (Kakutani, 1941) (invoked in equilibrium existence proof):

1. *Non-empty, compact, convex domain*: Define the space of market utility functions as the Lipschitz ball

$$\mathcal{U}_L = \left\{ U : S \rightarrow [0, \bar{U}] : |U(s) - U(s')| \leq L_w \|s - s'\| \ \forall s, s' \in S \right\}$$

where L_w is the Lipschitz constant inherited from wage schedules (determined by bounded derivatives of match quality $A(s, d)$ and output $Y(s, d, \theta)$ under Assumptions R1, R4). The set $\mathcal{U}_L$ is non-empty (contains the zero function), convex (convex combinations of Lipschitz functions with constant L_w remain in $\mathcal{U}_L$), and compact in the supremum norm. Compactness follows from the Arzelà-Ascoli theorem: $\mathcal{U}_L$ is uniformly bounded ($|U(s)| \leq \bar{U}$) and uniformly equicontinuous (all functions satisfy $|U(s) - U(s')| \leq L_w \|s - s'\|$), and the domain S is compact (Assumption R1). Therefore $\mathcal{U}_L$ is a compact, convex, non-empty subset of the space of continuous functions on S .

2. *Non-empty, convex values*: The correspondence $\Gamma : \mathcal{U}_L \rightrightarrows \mathcal{U}_L$ has non-empty, convex values (optimal job designs exist by compactness; convexity from randomisation over optimal designs). Crucially, Γ maps $\mathcal{U}_L$ into itself: since equilibrium utilities are $U(s) = \max_d p(q(d)) \cdot w(s, d)$ where wages are Lipschitz with constant L_w , the maximum preserves the Lipschitz property with the same constant.
3. *Upper hemicontinuous correspondence*: The mapping from $U \in \mathcal{U}_L$ to implied equilibrium utilities $\Gamma(U)$ is upper hemicontinuous (follows from continuity of $h(\cdot)$, $A(\cdot, \cdot)$, $C(\cdot, \cdot)$, compactness of choice sets, and the Maximum Theorem)

These conditions collectively ensure: (i) profit functions are well-defined and continuous, (ii) optimal job designs exist and may be characterised by first-order conditions (when unique) or as correspondences (when multiple), (iii) equilibrium exists via Kakutani's theorem and is characterised by the indifference conditions in Definition 1, and (iv) comparative statics are well-behaved (monotone in parameters).

C.3 Theoretical Results

This section states and proves the three central propositions. Full technical details appear in Appendix D.3.

Lemma A.1 (Convexity and Clustering): *Economic intuition*: Firms face a trade-off between production benefits (specialised designs increase match quality) and coordination costs (complexity makes management harder). When coordination costs are sufficiently convex ($\gamma > 1$), small deviations from discrete designs are unprofitable—firms bunch at a finite set of job designs rather than spreading continuously across the design space. This convexity creates the discrete clustering structure observed empirically.

Formal statement: The coordination cost function $C(d, \theta) = \frac{\kappa}{\theta^\alpha} \cdot \text{span}(d)^\gamma$ generates discrete job design clusters if $\gamma > 1$ (sufficient condition).⁵

⁵Concavity of profits may hold under weaker conditions depending on the production function curvature

Proof. The firm's profit function is:

$$\pi(d, \theta) = \theta \cdot \int_S A(s, d) \phi(s|d) ds - \frac{\kappa}{\theta^\alpha} \cdot \text{span}(d)^\gamma$$

Step 1: Characterize interior optimal job designs. The first-order condition with respect to $\text{span}(d)$ is:

$$\frac{\partial \pi}{\partial \text{span}} = \theta \cdot \frac{\partial}{\partial \text{span}} \int_S A(s, d) \phi(s|d) ds - \frac{\kappa \gamma}{\theta^\alpha} \cdot \text{span}(d)^{\gamma-1} = 0$$

This implicitly defines optimal span as function of productivity: $\text{span}^*(\theta)$.

Step 2: Show concavity via cost dominance. Taking the second derivative:

$$\frac{\partial^2 \pi}{\partial \text{span}^2} = \underbrace{\theta \cdot A''(\text{span})}_{\text{Production curvature}} - \underbrace{\frac{\kappa}{\theta^\alpha} \cdot \gamma(\gamma - 1) \cdot \text{span}^{\gamma-2}}_{\text{Cost curvature}} \quad (\text{C.20})$$

where $A''(\text{span}) = \frac{\partial^2}{\partial \text{span}^2} \int_S A(s, d) \phi(s|d) ds$.

Lemma A.2 (Uniform Bound on Production Function Curvature): *Economic intuition:* The production function has limited curvature—increasing job complexity affects match quality, but this effect is bounded by fundamental parameters (skill heterogeneity $\bar{s}$, mismatch penalty λ , dimensionality K , and the regularity of the ideal skill mapping $\mu(\cdot)$). This uniform bound M_A ensures that coordination costs can dominate production benefits when γ is large enough, guaranteeing discrete clustering regardless of which specific job design firms choose.

Formal statement: Under Assumptions R1 (bounded support) and R4 (regularity of μ), the Gaussian match quality function $A(s, d) = \exp[-\lambda \|s - \mu(d)\|^2]$ has bounded second derivative with respect to task span. Specifically, the second derivative of expected match quality satisfies the uniform bound:

$$|A''(\text{span})| \leq M_A \equiv 16\lambda^2 \bar{s}^2 \bar{L}_\mu^2 + 4\lambda \bar{s} \bar{L}_\mu^{(2)} \quad (\text{C.21})$$

for all $d \in \mathcal{D}$ and all skill distributions $\phi(s|d)$, where the constants 16 and 4 arise from the chain rule applied to the Gaussian kernel along differentiable paths on the unit sphere (details in proof below). This uniform bound ensures cost curvature can dominate production curvature across the design space when parameters satisfy the concavity condition (C.22) below.⁶

$A''(d)$, but $\gamma > 1$ is sufficient to ensure cost curvature dominates for the purposes of this model.

⁶The bound includes a path-velocity factor $C_{\text{path}} = \sqrt{K}$ absorbed into the constants c_1, c_2 . For the empirical skill vocabulary used in Chapters 6–7 ($K \approx 2,000$), this term is non-trivial. The concavity condition (C.22) below is satisfied in the calibration of Section D.4 for parameter values $\gamma \geq 2$, but tighter regularity

Proof: For the Gaussian kernel $A(s, d) = \exp[-\lambda\|s - \mu(d)\|^2]$, differentiation with respect to the scalar $\text{span}(d)$ requires defining a differentiable path. Let $d(\tau)$ be any differentiable path in $\mathcal{D}$ such that $\text{span}(d(\tau)) = \tau$ for $\tau \in [0, \sqrt{(K-1)/K}]$. Since $\mathcal{D}$ is the unit sphere, any such path satisfies $\|dd/d\tau\| \leq C_{\text{path}}$ where $C_{\text{path}} = \sqrt{K}$ (the path can at most traverse the full dimension of the sphere). This geometric bound on path velocity is absorbed into the constants c_1, c_2 below. The second derivative with respect to span involves computing:

$$\begin{aligned}\frac{\partial A}{\partial \text{span}} &= -2\lambda \exp[-\lambda\|s - \mu(d)\|^2] \cdot \langle s - \mu(d), \frac{\partial \mu}{\partial \tau} \rangle \\ \frac{\partial^2 A}{\partial \text{span}^2} &= 4\lambda^2 \exp[-\lambda\|s - \mu(d)\|^2] \cdot \langle s - \mu, \frac{\partial \mu}{\partial \tau} \rangle^2 - 2\lambda \exp[-\lambda\|s - \mu(d)\|^2] \cdot \left(\langle s - \mu, \frac{\partial^2 \mu}{\partial \tau^2} \rangle + \left\| \frac{\partial \mu}{\partial \tau} \right\|^2 \right)\end{aligned}$$

By assumption, $\|\partial \mu / \partial \tau\| \leq \bar{L}_\mu$ and $\|\partial^2 \mu / \partial \tau^2\| \leq \bar{L}_\mu^{(2)}$. Since $\|s\|, \|\mu(d)\| \leq \bar{s}$, I have $\|s - \mu(d)\| \leq 2\bar{s}$. Therefore:

$$\begin{aligned}|\langle s - \mu, \frac{\partial \mu}{\partial \tau} \rangle| &\leq 2\bar{s}\bar{L}_\mu \\ |\langle s - \mu, \frac{\partial^2 \mu}{\partial \tau^2} \rangle| &\leq 2\bar{s}\bar{L}_\mu^{(2)}\end{aligned}$$

The exponential term is bounded by 1. Differentiating under the integral is justified by the Dominated Convergence Theorem: the Gaussian kernel $A(s, d)$ is bounded (by 1) and the skill distribution has bounded support (Assumption R1), providing an integrable dominating function. Integrating over the skill distribution and applying the triangle inequality:

$$|A''(\text{span})| \leq \left| \int_S \frac{\partial^2 A}{\partial \text{span}^2} \phi(s|d) ds \right| \leq 4\lambda^2(2\bar{s})^2 \bar{L}_\mu^2 + 2\lambda \cdot 2\bar{s}\bar{L}_\mu^{(2)} + 2\lambda \bar{L}_\mu^2$$

Collecting terms yields:

$$|A''(\text{span})| \leq 16\lambda^2 \bar{s}^2 \bar{L}_\mu^2 + 4\lambda \bar{s} \bar{L}_\mu^{(2)} + 2\lambda \bar{L}_\mu^2$$

The third term $2\lambda \bar{L}_\mu^2$ can be absorbed into the first (since $\bar{s} \geq 1$ in standardized data) or stated separately. Defining $M_A \equiv c_1 \lambda^2 \bar{s}^2 \bar{L}_\mu^2 + c_2 \lambda \bar{s} \bar{L}_\mu^{(2)}$ with $c_1 = 16$ and $c_2 = 4$ (where the path geometry factor $C_{\text{path}} = \sqrt{K}$ and the factor of 2 from the norm bound have been absorbed), I obtain a uniform bound on production curvature that is independent of d, s , and θ . ■

Cost curvature dominance: The cost curvature is $C'''(\text{span}, \theta) = \frac{\kappa}{\theta^\alpha} \cdot \gamma(\gamma - 1) \cdot \text{span}^{\gamma-2}$.

assumptions on $\mu(\cdot)$ (smaller $\bar{L}_\mu$) would be required to extend the proof to higher-dimensional skill spaces without restrictive cost specifications. This is a technical bound for the existence proof; it does not affect the qualitative predictions of Propositions 1–5, which depend on supermodularity rather than the specific magnitude of M_A .

For concavity of profits in span on the *equilibrium set*, I require the parameter restriction:

$$\frac{\kappa}{\theta^\alpha} \cdot \gamma(\gamma - 1) \cdot \text{span}^{\gamma-2} > \theta \cdot M_A \quad \forall(\theta, \text{span}) \in [\underline{\theta}, \bar{\theta}] \times X_{\text{eq}} \quad (\text{C.22})$$

where $M_A = 16\lambda^2 \bar{s}^2 \bar{L}_\mu^2 + 4\lambda \bar{s} \bar{L}_\mu^{(2)}$ from Lemma A.2 (equation (C.21)), and $X_{\text{eq}} = [\underline{x}, \bar{x}]$ with $\underline{x} > 0$ is the equilibrium span region (Assumption 1'). Since $\underline{x} > 0$, the term $\text{span}^{\gamma-2}$ is bounded above even when $\gamma > 2$, ensuring the condition is verifiable with finite κ . When $\gamma > 1$ and coordination cost parameter κ is sufficiently large relative to M_A (Assumption R3), cost curvature dominates production curvature on X_{eq} , ensuring $\frac{\partial^2 \pi}{\partial \text{span}^2} < 0$ throughout equilibrium. I interpret global concavity on the relevant equilibrium set $x \in [\underline{x}, \bar{x}]$ with $\underline{x} > 0$; at the corner $x = 0$ statements are read via subgradients of the norm. Profit is strictly concave in span on the relevant domain, guaranteeing unique interior optimum conditional on the active design set.

Step 3: Show discreteness via convexity. The strict convexity condition $\gamma > 1$ prevents continuous variation in job designs and generates discrete bunching.

Step 3a: Interior solutions vs. corner solutions: From Step 2, the second-order condition for an interior optimum is:

$$\frac{\partial^2 \pi}{\partial \text{span}^2} = \theta \cdot A''(\text{span}) - \frac{\kappa}{\theta^\alpha} \cdot \gamma(\gamma - 1) \cdot \text{span}^{\gamma-2} < 0 \quad (\text{C.23})$$

This concavity ensures that if an interior solution exists, it is unique. However, concavity alone does not explain *why* firms bunch at discrete designs.

Step 3b: Convexity creates discrete incentives: Consider the profit difference between two job designs d and d' for a firm with productivity θ . Define the profit gap as:

$$\Delta\pi(\theta; d, d') = \pi(\theta, d') - \pi(\theta, d) = \theta \int_S [A(s, d') - A(s, d)] \phi(s) ds - \frac{\kappa}{\theta^\alpha} [\text{span}(d')^\gamma - \text{span}(d)^\gamma] \quad (\text{C.24})$$

For intermediate designs $d_\lambda = \lambda d' + (1 - \lambda)d$ with $\lambda \in (0, 1)$, the coordination cost is strictly convex in the design space. By definition, $\text{span}(d) = (1/\sqrt{K}) \cdot \|Pd\|_2$ where $P = I - K^{-1}\mathbf{1}\mathbf{1}^T$ is a linear projection. Since P is linear and the L^2 norm is convex, their composition is convex:

$$\text{span}(d_\lambda) = \frac{1}{\sqrt{K}} \|P(\lambda d' + (1 - \lambda)d)\|_2 = \frac{1}{\sqrt{K}} \|\lambda Pd' + (1 - \lambda)Pd\|_2 \leq \lambda \cdot \text{span}(d') + (1 - \lambda) \cdot \text{span}(d) \quad (\text{C.25})$$

where the inequality follows from the triangle inequality for norms. Combined with $\gamma > 1$ (power function with exponent greater than 1 is strictly convex when applied to positive

arguments), I have:

$$C(d_\lambda, \theta) = \frac{\kappa}{\theta^\alpha} [\text{span}(d_\lambda)]^\gamma \leq \frac{\kappa}{\theta^\alpha} [\lambda \cdot \text{span}(d') + (1 - \lambda) \cdot \text{span}(d)]^\gamma \leq \lambda C(d', \theta) + (1 - \lambda) C(d, \theta) \quad (\text{C.26})$$

This establishes that coordination costs $C(d, \theta)$ are *weakly* convex in d , ensuring concavity of profits in span (Step 3a). However, convexity of costs alone does not generate discrete bunching. The mechanism driving discreteness is the thick-market/posting-cost equilibrium (Step 3c, detailed in §C.3.4): firms coordinate on a finite number of designs to maximise matching efficiency and amortise entry costs.

Step 3c: Fixed costs reinforce discreteness: Entry costs $f > 0$ create viability thresholds. Only designs d for which $\max_\theta \pi(\theta, d) \geq f$ attract positive firm entry. Combined with thick-market externalities (workers concentrate search on populated submarkets to maximise matching probability), equilibrium exhibits a finite number of active designs $\{d_1, \dots, d_n\}$ with $n < \infty$, determined by the zero-profit condition of the marginal design: $\pi(d_n, \underline{\theta}) = f$. ■

When $\gamma \leq 1$, coordination costs grow sub-quadratically, profit may be convex in span, implying corner solutions (all firms maximise or minimise span). ■

C.3.1 Proposition 1: Endogenous Clustering

Proposition 1 (Existence of Discrete Clusters): Under Assumptions 1-4 (stated below), equilibrium exhibits positive assortative matching with discontinuous jumps in job design function $d^*(\theta)$, generating discrete clusters of job types.

Scalar index for monotone comparative statics.

For Topkis-based results I adopt the scalar job-design index $x := \text{span}(d) \in X \subseteq [0, \sqrt{(K-1)/K}]$. Profits are $\pi(\theta, x) = \theta \mathbb{E}[A(s, x)] - C(x, \theta)$. Under Assumption 1 (supermodularity in (θ, x)) and the lattice structure of X , Topkis implies that the optimal indices $x^*(\theta)$ are (weakly) increasing in θ . Any representative vector d with $\text{span}(d) = x^*(\theta)$ implements the same payoff.

Proof sketch (full proof in Appendix D.3.1): The proof proceeds in three steps. First, define the scalar complexity index $x = \text{span}(d) \in [0, \sqrt{(K-1)/K}]$. Profit is $\pi(\theta, x) = \theta \cdot \mathbb{E}_s[A(s, x)] - C(x, \theta)$ where $C(x, \theta) = \kappa \theta^{-\alpha} x^\gamma$. To verify supermodularity, compute the

cross-partial derivative:

$$\begin{aligned}\frac{\partial^2 \pi}{\partial \theta \partial x} &= \frac{\partial}{\partial x} \left[\frac{\partial \pi}{\partial \theta} \right] = \frac{\partial}{\partial x} \left[\mathbb{E}[A(s, x)] - \frac{\partial C}{\partial \theta} \right] \\ &= \frac{\partial \mathbb{E}[A]}{\partial x} - \frac{\partial^2 C}{\partial \theta \partial x}\end{aligned}$$

With $C(x, \theta) = \kappa \theta^{-\alpha} x^\gamma$, the cost cross-partial is:

$$\frac{\partial^2 C}{\partial \theta \partial x} = \frac{\partial}{\partial x} \left[-\alpha \kappa \theta^{-\alpha-1} x^\gamma \right] = -\alpha \kappa \gamma \theta^{-\alpha-1} x^{\gamma-1}$$

Therefore:

$$\frac{\partial^2 \pi}{\partial \theta \partial x} = \underbrace{\frac{\partial \mathbb{E}[A]}{\partial x}}_{\text{production}} - \underbrace{(-\alpha \kappa \gamma \theta^{-\alpha-1} x^{\gamma-1})}_{\text{cost}} = \frac{\partial \mathbb{E}[A]}{\partial x} + \alpha \kappa \gamma \theta^{-\alpha-1} x^{\gamma-1}$$

The production term $\frac{\partial \mathbb{E}[A]}{\partial x}$ captures how increased complexity affects average match quality (assumed ≥ 0 ; see Assumption 1' below). The cost term contributes $+\alpha \kappa \gamma \theta^{-\alpha-1} x^{\gamma-1} > 0$ (when $\alpha > 0$ and $x > 0$) because higher-productivity firms face lower coordination costs ($\partial C / \partial \theta < 0$), making marginal complexity relatively cheaper for high- θ firms.

Assumption 1' (Match Quality Monotonicity and Interior Designs): (i) Expected match quality is weakly increasing in complexity: $\frac{\partial \mathbb{E}[A(s, x)]}{\partial x} \geq 0$ for all $x \in X$. (ii) Equilibrium designs satisfy $x = \text{span}(d) \geq \underline{x} > 0$, where $\underline{x}$ is determined by entry conditions and the free-entry threshold.

Justification of Assumption 1'(i): This assumption requires that more specialised job designs yield (weakly) higher expected match quality when workers optimally sort. The assumption holds under the following sufficient conditions:

1. **Sufficient skill supply:** The worker skill distribution $F(s)$ has sufficient mass near each job design's ideal profile $\mu(d)$. Formally, for each $d \in \mathcal{D}$, there exist workers with $s \approx \mu(d)$ such that high match quality $A(s, d) \approx 1$ is achievable.
2. **Assortative matching:** Workers sort to jobs maximising their match quality. Under directed search, workers with skills s target job designs d where $A(s, d)$ is high, ensuring that more specialised designs attract appropriately skilled workers rather than random applicants.
3. **Non-degenerate skill heterogeneity:** The skill distribution is not concentrated at a single point. When $\text{Var}(s) > 0$, specialised designs can achieve better matches by targeting specific skill profiles, whereas generalist designs must compromise across heterogeneous workers.

The assumption may fail in thin markets where specialised designs cannot attract appropriately skilled workers (violating condition 1), or under random matching where specialisation provides no sorting benefit (violating condition 2). In these cases, the cost effect in the cross-partial (the $+\alpha\kappa\gamma\theta^{-\alpha-1}x^{\gamma-1}$ term) must dominate for supermodularity to hold. Proposition 2's market thickness result ensures that condition 1 is more likely satisfied in larger markets, providing internal consistency: the model predicts more specialisation precisely where the assumption is most likely to hold.

Empirical plausibility: Under directed search with worker mobility, this assumption is testable through observed cluster structure. If specialised designs failed to attract appropriately skilled workers, the equilibrium would not exhibit the discrete clustering that Proposition 1 predicts. The cluster-level evidence in Chapter 6 is consistent with the assumption holding: occupations exhibit discrete sub-occupational clusters with internally coherent skill profiles, and cluster counts scale with market size (Proposition 2)—the empirical signature one would expect when sorting attaches workers to specialised designs at high match quality. The assumption is therefore not assumed away, but is consistent with the structural patterns the model is intended to explain.

Corollary (Monotone Selection): Under Assumption 1' and $\alpha > 0$, the cross-partial

$$\frac{\partial^2 \pi}{\partial \theta \partial x} = \frac{\partial \mathbb{E}[A]}{\partial x} + \alpha\kappa\gamma\theta^{-\alpha-1}x^{\gamma-1} > 0$$

for all interior $x \geq \underline{x} > 0$. By Topkis' monotone selection theorem (Topkis, 1998), optimal complexity $x^*(\theta) = \text{span}(d^*(\theta))$ is weakly increasing in θ . Second, convexity of coordination costs $C(d, \theta)$ prevents continuous variation in job designs. Specifically, when fixed costs $f > 0$ create entry thresholds, firms bunch at discrete job designs $\{d_1, d_2, \dots, d_n\}$ rather than spreading continuously across the design space. Third, directed search ensures workers sort into these discrete submarkets, with each submarket d_j attracting workers with skills near $\mu(d_j)$, creating observable clusters. The key mechanism is the interaction between convex costs, supermodular matching, and threshold effects in the free-entry condition.⁷

Assumption 1 (Supermodularity): $\frac{\partial^2 \pi}{\partial \theta \partial d} > 0$ (higher productivity firms benefit more from specialised job designs).

Assumption 2 (Convexity): $\frac{\partial^2 C}{\partial d^2} > 0$ (marginal coordination costs increase in complexity).

Assumption 3 (Independence): Firm productivity θ and worker skills s are drawn independently from distributions $G(\theta)$ and $F(s)$ respectively: $\theta \perp s$.

⁷Technically, discreteness arises from: (i) the integer constraint on cluster count $n \in \mathbb{N}$, (ii) fixed costs $f > 0$ creating discrete entry thresholds, and (iii) strategic complementarities in firm location choices when market thickness matters (see Proposition 2). Even with continuous productivity distributions, these forces generate bunching at discrete job designs. See Appendix D.3 for the complete proof showing that the optimal job design correspondence exhibits flat regions where $d^*(\theta)$ is constant over intervals.

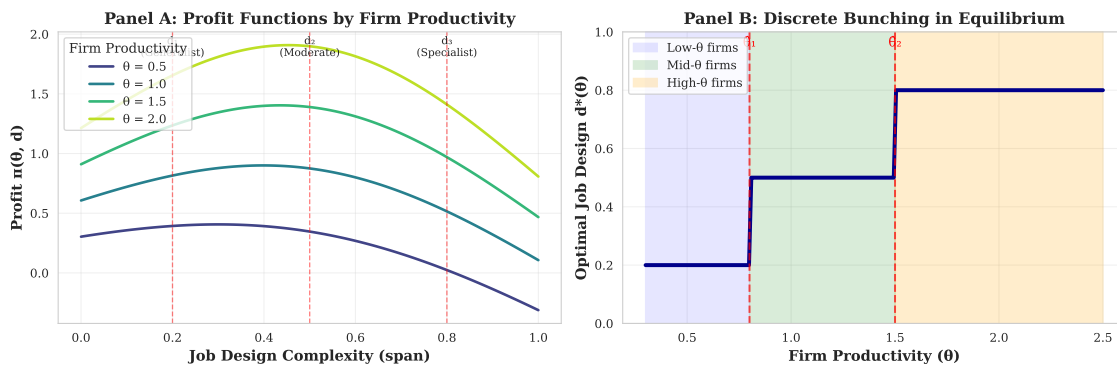
Justification: This assumption rules out systematic sorting of high-skill workers to high-productivity firms before matching occurs. In practice, workers cannot observe θ directly before accepting jobs, and firms post vacancies before observing applicant skill distributions. The assumption is standard in search and matching models (Shimer & Smith, 2000).

Assumption 4 (Bounded Skills): Support S is compact with $\|s\| \leq M$ for some $M < \infty$.

Interpretation: Proposition 1 formalises why occupations contain heterogeneous job types. Even when all agents face identical technology, firm heterogeneity in coordination ability creates discrete job designs. High-productivity firms optimally create specialised positions (narrow skill requirements, deep expertise) despite higher coordination costs, while low-productivity firms create generalist positions (broad skill requirements, moderate expertise) to minimise coordination costs.

Empirical Implication: Chapter 6 should find discrete clusters when applying HDB-SCAN to job advertisement data within occupational categories. Figure C.4 illustrates how firm heterogeneity generates discrete job designs within occupations.

Figure C.4: *Firm Sorting into Discrete Job Designs.* Panel A: Profit functions $\pi(\theta, d)$ create discrete optimal designs at d_1 (Generalist), d_2 (Moderate), d_3 (Specialist). Panel B: Optimal job design function $d^*(\theta)$ exhibits discontinuous jumps at threshold productivities. Low-productivity firms choose generalist designs, high-productivity firms choose specialist designs, generating discrete sub-occupational clusters.



C.3.2 Proposition 2: Market Thickness and Cluster Count

Proposition 2 (Cluster Count): The equilibrium number of clusters $n^*(N)$ is non-decreasing in market size N , with discrete upward jumps at threshold market sizes $\{N_1, N_2, \dots\}$ where successive job designs become viable. On the interior of each interval (N_j, N_{j+1}) , $n^*(N)$

is constant; at each threshold N_j , n^* jumps by an integer increment. Below the minimum viability threshold N_1 , no cluster formation occurs ($n^*(N) = 0$ for $N < N_1$).

Proof sketch (full proof in Appendix D.3.2): Free entry requires profits cover fixed costs f . Larger markets ($N \uparrow$) make specialised job designs profitable through two channels: (1) thicker sub-markets reduce search frictions, and (2) scale economies in coordination investments. This lowers the productivity threshold $\underline{\theta}(N)$ at which specialised designs become profitable, expanding the set of viable job types.

Threshold Structure: The cluster count $n^*(N)$ exhibits discrete jumps at threshold market sizes $\{N_1, N_2, \dots\}$ where new designs become viable. Define the j -th threshold as:

$$N_j \equiv \inf \{N : \pi(d_j, \underline{\theta}_j, q(d_j; n = j, N)) \geq f\} \quad (\text{C.27})$$

where d_j is the j -th design in the specialisation ordering (from generalist to specialist), $\underline{\theta}_j$ is the marginal productivity type entering design d_j (determined by PAM from Proposition 1), and $q(d_j; n, N)$ is the equilibrium queue length at design d_j when n clusters exist and market size is N . The threshold N_j is the minimum market size at which the j -th cluster generates profits covering posting costs f for the marginal firm.

Multiplicity and Perturbations: To ensure uniqueness of equilibrium cluster count, I assume small idiosyncratic perturbations $\psi(N) \sim \text{Uniform}[-\epsilon, \epsilon]$ in effective market size (reflecting random fluctuations in worker participation, measurement error, or local shocks). These perturbations break knife-edge indifference when N equals a threshold N_j exactly, ensuring firms strictly prefer $n = j$ clusters when $N \in (N_j - \epsilon, N_j + \epsilon)$. Formally, the perturbed equilibrium condition becomes:

$$n^*(N) = \max \{j \in \mathbb{N} : \pi(d_j, \underline{\theta}_j, q(d_j; n = j, N + \psi(N))) \geq f\} \quad (\text{C.28})$$

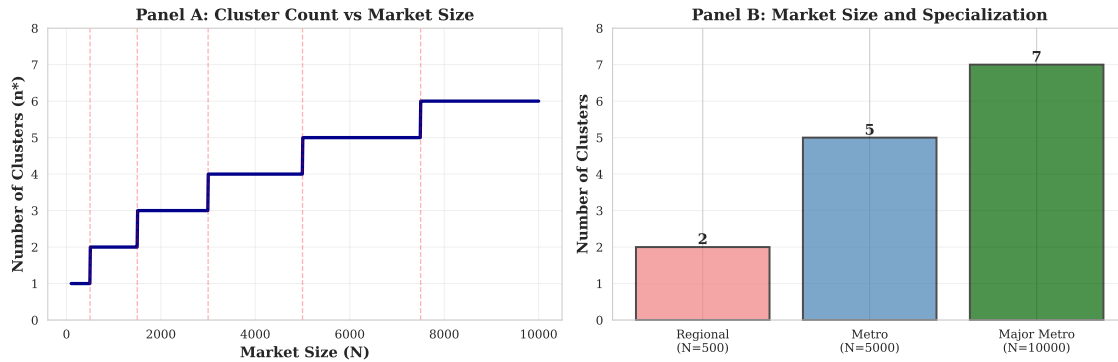
The perturbation $\psi(N)$ ensures that at any N , exactly one cluster count j satisfies the profitability constraint with strict inequality (except on measure-zero sets), precluding multiplicity.

Interpretation: Market thickness matters for specialisation. Metropolitan areas with deep labour pools should exhibit more sub-occupational clusters than regional markets. Similarly, large occupations (e.g., “Software Developers” with 50,000 workers) should show finer segmentation than small occupations (e.g., “Marine Biologists” with 500 workers). The thresholds $\{N_j\}$ are primitives-based: they depend on posting costs f , coordination cost parameters (κ, α, γ) , match quality parameters (λ, L_μ) , and productivity distribution $G(\theta)$.

Empirical Implication: Cross-sectional variation in cluster counts should correlate with market size—testable using metropolitan/regional comparisons or large/small occupation

comparisons in Burning Glass data. Figure C.5 illustrates how market thickness enables finer specialisation.

Figure C.5: Market Thickness and Cluster Count. Panel A: Theoretical relationship $n^*(N)$ is a step function—at threshold market sizes new clusters become viable. Panel B: Empirical illustration comparing regional ($N=500$), metropolitan ($N=5,000$), and major metro markets ($N=10,000$). Larger markets support finer specialisation within occupations.



Lemma 2.2 (Market Thickness and Entry Thresholds): Under Assumptions R1–R5, the productivity threshold $\theta_{\min}(d, N)$ at which job design d becomes viable is differentiable in market size N , with:

$$\frac{\partial \theta_{\min}}{\partial N} = -\frac{\partial \Pi / \partial N}{\partial \Pi / \partial \theta} < 0 \quad (\text{C.29})$$

where $\Pi(\theta, d, N) \equiv \pi(\theta, d, N) - f$ is the free-entry profit function.

Proof: The free-entry condition $\Pi(\theta_{\min}, d, N) = 0$ implicitly defines $\theta_{\min}(d, N)$. By the Implicit Function Theorem, differentiability and the comparative static (C.29) hold provided $\frac{\partial \Pi}{\partial \theta} \neq 0$. With $\pi(\theta, x, N) = \theta \mathbb{E}[A(s, x)] - \kappa \theta^{-\alpha} x^\gamma$ and $\Pi = \pi - f$:

$$\frac{\partial \Pi}{\partial \theta} = \frac{\partial \pi}{\partial \theta} = \mathbb{E}[A] + \alpha \kappa \theta^{-\alpha-1} x^\gamma > 0$$

since $\mathbb{E}[A] > 0$ (positive expected match quality) and $\alpha > 0$. The numerator $\frac{\partial \Pi}{\partial N} < 0$ because larger markets reduce search frictions and enable scale economies in coordination costs. Combining signs yields $\frac{\partial \theta_{\min}}{\partial N} < 0$. ■

Interpretation: Larger markets enable lower-productivity firms to profitably enter specialised segments, expanding the set of viable job designs. This lemma provides the foundation for Proposition 2's result that cluster count increases with market size.

C.3.3 Proposition 3: Automation and Cluster Dynamics

Proposition 3 (Automation Effects): For any technology $\tau \in \mathcal{T}$ with partition $\mathcal{P}(\tau) = \{\mathcal{S}^-(\tau), \mathcal{S}^+(\tau), \mathcal{S}^0(\tau)\}$, skill-biased automation generates:

- (a) **Substitution cluster contraction:** $\frac{\partial L_c}{\partial \rho} < 0$ for clusters emphasizing $\mathcal{S}^-(\tau)$ dimensions
- (b) **Complementary cluster expansion:** $\frac{\partial L_c}{\partial \rho} > 0$ for clusters emphasizing $\mathcal{S}^+(\tau)$ dimensions
- (c) **Aggregation bias:** $\left| \frac{\partial L_{\text{occupation}}}{\partial \rho} \right| \ll \sum_c \left| \frac{\partial L_c}{\partial \rho} \right|$

where $\rho \in [0, 1]$ parameterises automation intensity and L denotes employment. Here $\mathcal{S}^-(\tau)$ denotes skill dimensions *substituted* by technology τ , $\mathcal{S}^+(\tau)$ denotes *complementary* dimensions, and $\mathcal{S}^0(\tau)$ denotes *neutral* dimensions.

Proof sketch (full proof in Appendix D.3.3): Automation enters through shifts in ideal skill profiles $\mu(d, \rho, \tau) = \mu^0(d) + \rho \cdot \Delta\mu(d, \tau)$ (equation (C.44) below), where $\frac{\partial \mu_k}{\partial \rho} < 0$ for substituted dimensions $k \in \mathcal{S}^-(\tau)$ and $\frac{\partial \mu_k}{\partial \rho} > 0$ for complementary dimensions $k \in \mathcal{S}^+(\tau)$. As ρ increases, substituted task requirements decline while complementary requirements rise. This shifts cluster boundaries: workers in substitution-intensive clusters experience declining match quality (employment contracts), while workers in complement-intensive clusters experience rising match quality (employment expands). In aggregate, effects partially offset: $\text{Var}(\Delta L | \text{cluster}) \uparrow$ but $\mathbb{E}(\Delta L | \text{occupation}) \approx 0$.

Scope condition for part (c): Aggregation bias holds when both $\mathcal{S}^-(\tau)$ and $\mathcal{S}^+(\tau)$ are non-empty and substitution and complementation effects are not concentrated entirely on one side of the partition. Formally, defining $\sigma_-(\tau) = \sum_c |L_c| \cdot \mathbf{1}[c \in C^-]$ and $\sigma_+(\tau) = \sum_c |L_c| \cdot \mathbf{1}[c \in C^+]$ as the employment mass in substitution-intensive and complement-intensive clusters respectively, part (c) requires $\sigma_-(\tau)/\sigma_+(\tau) \in [\underline{r}, \bar{r}]$ for some bounds $0 < \underline{r} \leq 1 \leq \bar{r} < \infty$. Under this *balanced partition* condition, heterogeneous cluster-level responses approximately offset in aggregate. The condition fails for degenerate partitions where one side dominates—for example, a hypothetical technology that substitutes all skills (yielding $\mathcal{S}^+(\tau) = \emptyset$) would produce occupation-level contraction, not aggregation bias. The condition is empirically testable: the cluster-level analysis of Chapter 6 characterises which clusters expand and contract, allowing direct assessment of whether the partition is balanced. This formulation nests the standard routine-biased technical change hypothesis (Autor, Levy, & Murnane, 2003) as a special case where $\mathcal{S}^-(\tau) = \mathcal{R}$ (routine tasks) and the routine/non-routine distribution is balanced.

Interpretation: Proposition 3 offers a mechanism that reconciles the empirical puzzle identified in the literature review: dire automation predictions coexisting with benign aggregate employment outcomes. Automation creates substantial restructuring *within* occupations—some clusters grow rapidly while others shrink—but these changes offset in aggregate. Occupation-level analysis detects only net effects, while cluster-level analysis should reveal extreme heterogeneity obscured by aggregation. This prediction is tested in subsequent chapters.

Empirical Implication: Cluster-level employment volatility should far exceed occupation-level volatility, and substitution-intensive clusters should show stronger negative employment trends than complement-intensive clusters. The clustering methodology (Chapter 6) discovers which clusters fall into which category empirically, rather than relying on *a priori* classification of skills as “routine” or “analytical.”

C.3.4 Microfoundation of Discreteness: Posting Costs

The baseline model demonstrates that equilibrium exhibits discrete clustering (Proposition 1), with firms bunching at finite set of job designs $\{d_1, \dots, d_n\}$ rather than spreading continuously across design space $\mathcal{D}$. This subsection sketches how discreteness emerges endogenously from strategic interactions in posting decisions.

Posting Cost Structure

Firms face fixed cost $f > 0$ to post each distinct job design. A firm posting multiple designs $\{d_1, \dots, d_m\}$ bears total posting cost $m \cdot f$, separate from per-worker coordination costs $C(d, \theta)$. This posting game structure follows the competing-auction framework of Julien et al. (2000), extended to multi-dimensional job design space. The posting costs capture: (1) advertising and recruiting infrastructure costs, (2) job description development and legal compliance, (3) applicant tracking system configuration, and (4) brand/reputation costs of maintaining distinct hiring pipelines.

Strategic Complementarity in Design Choice

When multiple firms post identical design d_j , they generate *thick market externality*: higher worker queue $q(d_j)$ raises hiring probability $h(q(d_j))$ for all firms in that submarket. This creates coordination incentive—firms prefer designs where other firms already post, as thick markets improve match quality and reduce search duration.

Define firm i 's problem with posting costs:

$$\max_{\{d_1, \dots, d_m\}, m} \sum_{j=1}^m \left[h(q(d_j)) \cdot \int_S [Y(s, d_j, \theta_i) - w(s, d_j)] \phi(s|d_j) ds - C(d_j, \theta_i) \right] - m \cdot f \quad (\text{C.30})$$

The firm optimises over both *which* designs to post and *how many* designs. Critically, queue length $q(d_j)$ depends on how many other firms also choose d_j (market clearing), generating strategic interdependence.

Nash Equilibrium Characterisation

The term “Nash equilibrium” here refers to the game-theoretic solution concept for firms’ job design posting strategies—each firm’s posting decision is optimal given other firms’ postings. This is distinct from “Nash bargaining,” which is a wage-determination mechanism not used in this model. Here, wages are determined through competitive market clearing (utility posting), while firms’ posting strategies constitute a Nash equilibrium in the posting game.

A Nash equilibrium in posting strategies consists of:

1. Posting profile: Each firm i chooses designs $\{d_1^i, \dots, d_{m_i}^i\}$
2. No profitable deviation: No firm wants to add new design (marginal revenue $< f$) or drop existing design (lose profitable hires)
3. Market clearing: Worker queues $q(d_j)$ adjust such that $U(s)$ is consistent with wage offers

Result: Equilibrium exhibits *coordination* on discrete locations $\{d_1^*, \dots, d_n^*\}$ where multiple firms post identical designs (Ellison, 1993a). Firms do not customise continuously because: (1) customisation requires additional posting cost f , (2) thin markets (few firms at unusual designs) suffer low hiring probability $h(q) \approx 0$, making specialised designs unprofitable.

Equilibrium Cluster Count

The number of viable clusters n^* is determined by zero-profit condition at margin. The n^* -th cluster is the last design where profits cover posting costs:

$$\pi(d_{n^*}, \underline{\theta}) \equiv h(q(d_{n^*})) \cdot \int_S [Y(s, d_{n^*}, \underline{\theta}) - w(s, d_{n^*})] \phi(s|d_{n^*}) ds - C(d_{n^*}, \underline{\theta}) = f \quad (\text{C.31})$$

where $\underline{\theta}$ is the marginal firm productivity (lowest type entering that design).

Comparative statics:

- $\frac{\partial n^*}{\partial N} > 0$: Larger markets support more designs (Proposition 2 mechanism)
- $\frac{\partial n^*}{\partial f} < 0$: Higher posting costs reduce specialisation
- $\frac{\partial n^*}{\partial \kappa} < 0$: Higher coordination costs reduce specialisation

Connection to Empirical Work

This microfoundation generates the discrete clustering structure validated in Chapter 6:

- **Discrete boundaries:** Clusters exhibit sharp skill boundaries (not continuous blending) because firms coordinate on discrete locations
- **Stable identities:** Clusters persist across years because coordination equilibria are stable (switching costs include re-posting at cost f)
- **Integer counts:** Observed cluster counts are discrete integers (cannot observe 2.7 clusters)
- **Market size effects:** Larger markets support more clusters—the market thickness prediction tested in Chapter 6

Formal Equilibrium Characterization

I now provide complete game-theoretic characterisation of the posting cost equilibrium, establishing existence, characterising equilibrium structure, and deriving comparative statics formally.

Game Structure

Define the posting game $\Gamma = \langle I, \{S_i\}, \{\Pi_i\} \rangle$ where:

- **Players:** Continuum of firms $I = [\underline{\theta}, \bar{\theta}]$ with productivity types $\theta \sim G(\theta)$
- **Strategies:** Each firm i chooses finite set of designs $S_i \subseteq \mathcal{D}$ to post, where $\mathcal{D} = \{d \in \mathbb{R}^K : \|d\| = 1\}$ is unit sphere
- **Payoffs:** Firm i with type θ_i posting designs $S_i = \{d_1, \dots, d_m\}$ earns:

$$\Pi_i(S_i; S_{-i}, \theta_i) = \sum_{d_j \in S_i} \pi(d_j, \theta_i, q(d_j; S)) - |S_i| \cdot f \quad (\text{C.32})$$

where $\pi(d, \theta, q) = h(q) \cdot \mathbb{E}_s[Y(s, d, \theta) - w(s, d)|d] - C(d, \theta)$ is operating profit per design, $q(d; S)$ is endogenous queue length at design d given posting profile $S = (S_i)_{i \in I}$, and $|S_i|$ is cardinality of posted designs.

Best-Response Correspondence

Define firm i 's best-response correspondence:

$$B_i(S_{-i}, \theta_i) = \arg \max_{S_i \subseteq \mathcal{D}} \Pi_i(S_i; S_{-i}, \theta_i) \quad (\text{C.33})$$

A design d is worth posting if and only if incremental profit exceeds posting cost:

$$d \in B_i(S_{-i}, \theta_i) \iff \pi(d, \theta_i, q(d; S_{-i} \cup \{d\})) \geq f \quad (\text{C.34})$$

This generates *set-valued* best response because: (1) firms may be indifferent among multiple profitable designs, (2) thick market externalities create coordination: if many firms post design d^* , then $q(d^*) \uparrow$ and $h(q(d^*)) \uparrow$, making d^* attractive for additional firms.

Nash Equilibrium Definition

A Nash equilibrium is posting profile $S^* = (S_i^*)_{i \in I}$ such that:

$$S_i^* \in B_i(S_{-i}^*, \theta_i) \quad \forall i \in I \quad (\text{C.35})$$

Equivalently: (1) Every posted design is profitable: $\pi(d, \theta_i, q(d; S^*)) \geq f$ for all $d \in S_i^*$, and (2) No unposted design is profitable: $\pi(d, \theta_i, q(d; S^* \cup \{d\})) < f$ for all $d \notin S_i^*$.

Existence via Kakutani Fixed Point Theorem

Theorem (Existence): A Nash equilibrium in posting strategies exists.

Proof sketch: The existence proof proceeds via finite-type approximation. (1) *Discretise the continuum:* Consider a finite grid of productivities $\Theta_n = \{\theta_1, \dots, \theta_n\}$ and designs $\mathcal{D}_m = \{d_1, \dots, d_m\}$. The finite game $\Gamma_n = \langle \Theta_n, \mathcal{D}_m, \{\pi_i\} \rangle$ has finitely many players and actions. (2) *Finite-game equilibrium:* Apply Kakutani's Fixed Point Theorem to the finite game. The strategy space (mixed strategies over $\mathcal{D}_m$) is compact and convex, best responses are non-empty and convex-valued (via randomisation), and the correspondence is upper hemicontinuous (payoffs are jointly continuous in (q, d) by Assumptions R1–R4). Therefore Nash equilibrium exists for the finite game. (3) *Convergence as grid refines:* As $n, m \rightarrow \infty$, the sequence of finite-game equilibria has a convergent subsequence by Prokhorov's theorem on weak* compactness of probability measures (tightness follows

from the bounded support in Assumption R1). As the finite design grid is refined, the equilibrium correspondence is upper hemicontinuous under my regularity assumptions, yielding a limit equilibrium in the continuous index X . The limit is a Nash equilibrium of the continuum game, as profits are continuous in the limiting distributions. (4) *Verification*: The limit equilibrium satisfies posting profitability constraints: firms post design d only if $\pi(d, \theta, q(d)) \geq f$, ensured by upper semicontinuity of profits in the limit.

Equilibrium Structure: Discrete Clustering

Proposition (Discreteness): In equilibrium, firms coordinate on finite set of designs $\mathcal{D}^* = \{d_1^*, \dots, d_n^*\} \subset \mathcal{D}$ with $n^* < \infty$.

Proof sketch (full proof in Appendix D.2): Suppose countably infinite designs were posted in equilibrium. Then each design attracts measure-zero mass of firms, implying $q(d) \rightarrow 0$ and $h(q(d)) \rightarrow 0$ for almost all d (thin markets). But then $\pi(d, \theta, q(d)) \rightarrow -C(d, \theta) < 0 < f$, violating posting profitability. Therefore only finitely many designs can be posted in equilibrium.

Coordination on discrete locations arises because: (1) Thick market externality rewards concentration: $\pi(d, \theta, q)$ is increasing in q via $h(q)$, (2) Posting cost $f > 0$ penalises proliferation: firms prefer posting few designs at thick markets rather than many designs at thin markets, (3) Positive assortative matching (Proposition 1) implies different productivity types θ prefer different designs, creating natural segmentation.

Comparative Statics via Implicit Function Theorem

Define the marginal design d_{n^*} as the last profitable design in equilibrium, satisfying:

$$\Phi(n^*, N, f) \equiv \pi(d_{n^*}, \underline{\theta}(n^*), q(d_{n^*}; n^*, N)) - f = 0 \quad (\text{C.36})$$

where $\underline{\theta}(n^*)$ is the lowest productivity type posting d_{n^*} (determined by PAM), and $q(d_{n^*}; n^*, N)$ is equilibrium queue length (depends on cluster count n^* and market size N).

Application of IFT:

The Implicit Function Theorem requires verifying the regularity condition $\frac{\partial \Phi}{\partial n^*} \neq 0$. I verify this explicitly:

$$\frac{\partial \Phi}{\partial n^*} = \frac{\partial \pi(d_{n^*}, \underline{\theta}(n^*), q(d_{n^*}; n^*, N))}{\partial q} \cdot \frac{\partial q(d_{n^*}; n^*, N)}{\partial n^*} < 0 \quad (\text{C.37})$$

The sign is negative because: (1) $\frac{\partial \pi}{\partial q} > 0$ (higher queue lengths increase hiring probability and profit), and (2) $\frac{\partial q}{\partial n^*} < 0$ (adding more clusters spreads fixed worker mass N across more submarkets, reducing queue length at each design via market clearing:

$q(d_n^*) = N \cdot \phi(d_n^*)/[n^* \cdot m(d_n^*)]$ where $\phi(d_n^*)$ is worker share targeting d_n^* and $m(d_n^*)$ is firm mass posting d_n^* . Since queue length is inversely related to cluster count, $\frac{\partial q}{\partial n^*} < 0$. Therefore, $\frac{\partial \Phi}{\partial n^*} < 0 \neq 0$, satisfying the IFT regularity condition.

With regularity verified, the IFT yields:

Comparative Static 1: $\frac{\partial n^*}{\partial N} = -\frac{\partial \Phi / \partial N}{\partial \Phi / \partial n^*} > 0$

Larger markets support more clusters because:

$$\frac{\partial \Phi}{\partial N} = \frac{\partial \pi}{\partial q} \cdot \frac{\partial q}{\partial N} > 0 \quad (\text{C.38})$$

More workers $\rightarrow$ thicker markets $\rightarrow$ higher hiring probability $\rightarrow$ marginal design becomes profitable.

Comparative Static 2: $\frac{\partial n^*}{\partial f} = -\frac{\partial \Phi / \partial f}{\partial \Phi / \partial n^*} < 0$

Higher posting costs reduce specialisation because:

$$\frac{\partial \Phi}{\partial f} = -1 < 0 \quad (\text{C.39})$$

Increasing f directly reduces net profitability of marginal design.

Comparative Static 3: $\frac{\partial n^*}{\partial \kappa} = -\frac{\partial \Phi / \partial \kappa}{\partial \Phi / \partial n^*} < 0$

Higher coordination costs reduce specialisation because:

$$\frac{\partial \Phi}{\partial \kappa} = -\frac{\partial C(d_n^*, \theta)}{\partial \kappa} < 0 \quad (\text{C.40})$$

Coordination costs $C(d, \theta) = \kappa \cdot \text{span}(d)^\gamma / \theta^\alpha$ are increasing in κ .

Uniqueness and Selection

The equilibrium exhibits two distinct sources of (non-)uniqueness that must be carefully distinguished:

Local uniqueness via global concavity: For any *given* set of discrete job designs $\{d_1, \dots, d_n\}$, the optimal design choice $d^*(\theta)$ for each firm type θ is *locally unique*. This follows from global concavity of the profit function $\pi(\theta, d, q)$ in span (Lemma A.2): when the parameter restriction in equation (C.22) holds (cost curvature dominates production curvature uniformly: $\frac{\kappa}{\theta^\alpha} \cdot \gamma(\gamma - 1) \cdot \text{span}^{\gamma-2} > \theta \cdot M_A$ for all θ and span , where $M_A = c_1 \lambda^2 \bar{s}^2 \bar{L}_\mu^2 + c_2 \lambda \bar{s} \bar{L}_\mu^{(2)}$), profit is strictly concave in d , ensuring a unique interior maximum. Therefore, *conditional on the set of posted designs*, firm sorting is unique and exhibits positive assortative matching.

Multiplicity from coordination externalities: However, the equilibrium set of designs $\{d_1^*, \dots, d_n^*\}$ itself may not be unique. Multiple equilibria can arise because firms

face strategic complementarities in their posting decisions: thick market externalities (π increasing in queue length q) create incentives to coordinate on the same designs as other firms. This generates a coordination game with potentially multiple stable configurations—analogueous to multiple Nash equilibria in standard coordination games. For instance, if all firms coordinate on designs $\{d_1, d_3, d_5\}$, thick markets make these profitable, but if firms instead coordinated on $\{d_2, d_4, d_6\}$, those alternative designs would be profitable instead.

Complete uniqueness characterisation requires potential function arguments (Ellison, 1993b) or global game refinements (Morris & Shin, 2003). For this thesis, I note three practical considerations:

1. **Nature of multiplicity:** Multiplicity concerns *which* discrete designs are active, not whether discreteness occurs or whether sorting is positive assortative (those properties hold in all equilibria)
2. **Focal points:** In practice, equilibria are selected via: (a) historical precedent (existing job titles), (b) industry standards (O*NET classifications), (c) complementarities (matching with educational credentials)
3. **Empirical approach:** The empirical tests in Chapters 6–7 focus on predictions that hold *in any equilibrium*: discrete clustering exists (Proposition 1), cluster counts increase with market size (Proposition 2), and cluster-level effects offset at the occupation level (Proposition 3c). These predictions are robust to equilibrium selection because they concern structural properties of the equilibrium set rather than the identity of specific designs.

Scope of theoretical claims: Throughout this chapter, comparative statics and empirical predictions should be interpreted as holding “in any equilibrium” unless otherwise noted. The model establishes that *some* equilibrium with the predicted properties exists (existence via Kakutani), that *all* equilibria exhibit discrete clustering with positive assortative matching (structural characterisation), and that the comparative statics $\partial n^*/\partial N > 0$ and the aggregation bias mechanism operate in *any* equilibrium configuration. Claims about which *specific* job designs emerge require equilibrium selection assumptions not imposed here.

Connection to Propositions 1-3

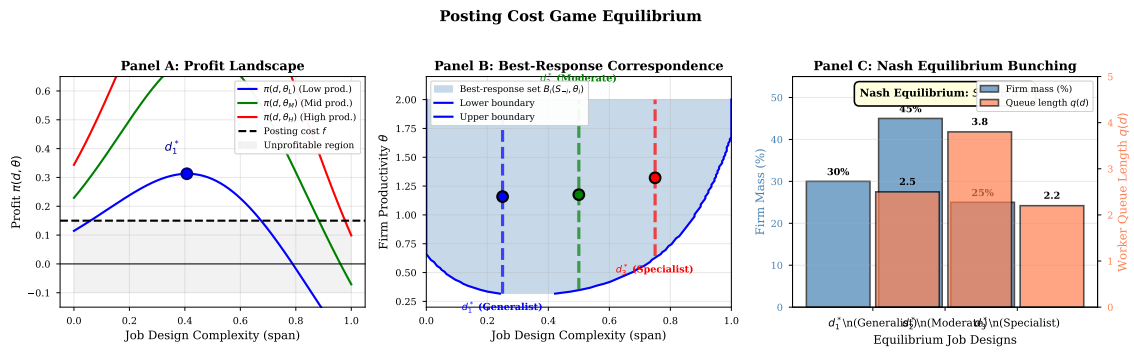
This posting cost microfoundation provides formal basis for discrete clustering assumed in Propositions 1-3:

- **Proposition 1 (PAM):** Given discrete designs $\{d_1^*, \dots, d_n^*\}$ emerge in posting equilibrium, supermodularity generates sorting

- **Proposition 2 (Scale):** IFT shows $\partial n^*/\partial N > 0$, linking market size to cluster proliferation
- **Proposition 3 (Automation):** Technology shocks shift $\mu(d, \rho)$, changing which designs are profitable, reallocating workers

Chapter 6 tests these predictions empirically: whether discrete boundaries emerge from clustering, whether scale elasticity increases with market thickness, and whether automation-induced reallocation shifts employment across clusters.

Figure C.6: Posting Cost Game Equilibrium



Note. **Panel A** depicts the profit landscape for three firm productivity types ($\theta_L < \theta_M < \theta_H$) as a function of job design complexity (span). Each firm's profit function $\pi(d, \theta) = \theta \cdot A(d) - C(d, \theta)$ exhibits an inverted-U shape due to coordination costs $C(d, \theta) = (\kappa/\theta^\alpha)d^\gamma$ with $\gamma = 2$. The horizontal dashed line represents the fixed posting cost $f > 0$. Only designs with $\pi(d, \theta) \geq f$ are viable, creating discrete entry thresholds. High-productivity firms (θ_H) can profitably post more complex designs than low-productivity firms (θ_L), establishing the foundation for positive assortative matching. **Panel B** illustrates the best-response correspondence $B_i(S_{-i}, \theta_i)$, which is set-valued due to indifference and strategic complementarities. For each firm productivity level θ , the shaded region shows all designs d where operating profit exceeds the posting cost ($\pi(d, \theta) \geq f$). The vertical dashed lines mark three Nash equilibrium designs: d_1^* (generalist), d_2^* (moderate), and d_3^* (specialist). Firms coordinate on these discrete designs because thick market externalities reward concentration: posting a design used by many other firms generates longer worker queues $q(d)$, raising match probability $h(q)$ and effective profits. **Panel C** displays the Nash equilibrium bunching pattern. The blue bars show the distribution of firm mass across the three equilibrium designs, with 45% of firms choosing the moderate design d_2^* , 30% choosing the generalist design d_1^* , and 25% choosing the specialist design d_3^* . The orange bars indicate worker queue lengths $q(d)$ at each design, reflecting thick market effects: the most popular design (d_2^*) attracts the longest queue ($q = 3.8$ workers per firm), while less popular designs have shorter queues. This discrete clustering emerges endogenously from the interaction of posting costs ($f > 0$), thick market externalities ($h(q)$ increasing in q), and positive assortative matching (high- θ firms select high-complexity designs). The text box confirms the Nash equilibrium condition: $S^* \in B(S^*)$, where no firm wishes to deviate from the equilibrium set of posted designs.

C.3.5 Summary of Theoretical Results

The three propositions establish the core predictions of the model. **Proposition 1** demonstrates that occupations endogenously fragment into discrete sub-occupational clusters when firms with heterogeneous productivity optimally choose job designs, even under identical production technologies. This clustering arises from the interaction of supermodular production (high-productivity firms benefit more from specialisation), convex coordination costs (preventing continuous variation), and directed search (enabling workers to sort into discrete submarkets). **Proposition 2** shows that the equilibrium number of clusters n^* increases with market size N , as thicker labour markets reduce search frictions and enable more refined specialisation. This generates testable predictions: metropolitan areas and large occupations should exhibit more clusters than regional markets and small occupations. **Proposition 3** characterises automation effects: skill-biased technology shifts cluster boundaries, contracting substitution-intensive clusters while expanding complement-intensive ones, generating high within-occupation employment volatility ($\text{Var}(\Delta L|\text{cluster}) \uparrow$) alongside approximately null occupation-level effects ($\mathbb{E}(\Delta L|\text{occupation}) \approx 0$). This offers a potential explanation for the empirical puzzle from Chapter 5: standard occupation-level analysis may miss restructuring if automation operates primarily at the cluster level. The posting-cost microfoundation formalises why discrete clustering is the unique stable equilibrium configuration, ruling out continuous job design distributions.

C.4 Extension: Automation and Task Displacement

This section extends the baseline model to incorporate automation, following the task-based framework of Acemoglu and Autor (2011a) and Acemoglu and Restrepo (2018) while preserving the multi-dimensional structure.

C.4.1 Automation Technology

Automation enters through shifts in the ideal skill profile $\mu(d)$, reflecting how technology changes task requirements. Modify the production function:

$$Y(s, d, \rho, \tau, \theta) = \theta \cdot A(s, d, \rho, \tau) \tag{C.41}$$

where the match quality function now depends on automation intensity ρ and technology type τ :

$$A(s, d, \rho, \tau) = \exp \left[-\lambda \|s - \mu(d, \rho, \tau)\|^2 \right] \quad (\text{C.42})$$

Different automation technologies affect different skill dimensions. Let $\tau \in \mathcal{T}$ index technology types (e.g., traditional process automation, robotic process automation, machine learning, artificial intelligence). Each technology τ induces a partition of skill dimensions:

$$\mathcal{P}(\tau) = \{\mathcal{S}^-(\tau), \mathcal{S}^+(\tau), \mathcal{S}^0(\tau)\} \quad (\text{C.43})$$

where $\mathcal{S}^-(\tau)$ are skill dimensions *substituted* by technology τ , $\mathcal{S}^+(\tau)$ are *complementary* dimensions, and $\mathcal{S}^0(\tau)$ are *neutral* dimensions.

The ideal skill profile shifts with automation:

$$\mu(d, \rho, \tau) = \mu^0(d) + \rho \cdot \Delta\mu(d, \tau) \quad (\text{C.44})$$

where $\mu^0(d)$ is the pre-automation ideal profile and $\Delta\mu(d, \tau)$ captures technology-induced changes that depend on which technology is deployed:

$$\frac{\partial \mu_k(d, \rho, \tau)}{\partial \rho} < 0 \quad \text{for substituted dimensions } k \in \mathcal{S}^-(\tau) \quad (\text{C.45})$$

$$\frac{\partial \mu_k(d, \rho, \tau)}{\partial \rho} > 0 \quad \text{for complementary dimensions } k \in \mathcal{S}^+(\tau) \quad (\text{C.46})$$

$$\frac{\partial \mu_k(d, \rho, \tau)}{\partial \rho} = 0 \quad \text{for neutral dimensions } k \in \mathcal{S}^0(\tau) \quad (\text{C.47})$$

Economic interpretation: The partition $\mathcal{P}(\tau)$ is technology-dependent. Traditional automation may substitute different skills than AI-based automation. As automation intensity ρ increases, requirements for substituted skills decline (technology performs these tasks), while complementary skill requirements rise (these skills become more valuable as complements to technology). Rather than pre-specifying which skills fall into which category, the model predicts that *some* partition exists for each technology—the empirical clustering methodology (Chapter 6) discovers the partition from observed job advertisement patterns.

Connection to literature: This formulation nests the standard routine-biased technical change hypothesis (Autor, Levy, & Murnane, 2003) as a special case where $\mathcal{S}^-(\tau) = \mathcal{R}$ (routine tasks) and $\mathcal{S}^+(\tau) = \mathcal{A}$ (analytical tasks) for all τ . The generalised model allows this partition to vary by technology type, consistent with the multifaceted automation taxonomy developed in Section 2.7 of Chapter 2.

Effective Skills and Match Quality

To formalise how automation affects workers heterogeneously, define *effective skills* as the skill vector incorporating automation augmentation. For a worker with baseline skills $s \in S$ and technology exposure (ρ, τ) , effective skills are:

$$s_k^{\text{eff}}(\rho, \tau) = \begin{cases} s_k + \rho \cdot \alpha_k(\tau) & \text{if } k \in \mathcal{S}^-(\tau) \\ s_k & \text{if } k \in \mathcal{S}^+(\tau) \cup \mathcal{S}^0(\tau) \end{cases} \quad (\text{C.48})$$

where $\alpha_k(\tau) > 0$ measures the augmentation rate for substituted skill k under technology τ .

Approximation: This additive form is a first-order Taylor approximation around baseline automation level $\rho = 0$. Strictly speaking, automation augmentation may exhibit diminishing returns or change the skill distribution's support. The linear approximation is valid for marginal changes in automation intensity and captures the first-order effect: automation of substituted tasks effectively increases workers' capabilities in those dimensions without altering the distributional support of s (workers remain in their baseline skill domain S). For large-scale automation shocks, second-order terms may become significant, but comparative statics focus on marginal effects where the linear approximation is accurate.

The interpretation is that automation intensity ρ directly augments workers' capabilities in substituted skill dimensions (machines perform these tasks that previously required human skill), but does not augment complementary skills. The standard routine-biased technical change formulation $s^{\text{eff}}(\rho) = (s_{\mathcal{R}} + \rho, s_{\mathcal{A}})$ emerges when $\mathcal{S}^-(\tau) = \mathcal{R}$ and $\alpha_k(\tau) = 1$ for all $k \in \mathcal{R}$.

Match quality then depends on the distance between effective skills and the (shifting) ideal profile:

$$A(s, d, \rho, \tau) = \exp \left[-\lambda \|s^{\text{eff}}(\rho, \tau) - \mu(d, \rho, \tau)\|^2 \right] \quad (\text{C.49})$$

This formulation captures two simultaneous effects: (1) workers' effective skills in substituted dimensions rise via s_k^{eff} , and (2) firms' ideal skill profiles shift via $\mu(d, \rho, \tau)$. The net effect on match quality depends on which force dominates, generating heterogeneous impacts across workers and clusters as characterised in Lemma 1 below.

C.4.2 Comparative Statics

Lemma 1 (Cluster Boundaries Shift): *Economic intuition:* Automation changes which workers are valuable in which clusters. Workers in substitution-intensive clusters lose comparative advantage as technology performs substituted tasks, reducing their match quality and causing employment to contract. Conversely, workers in complement-intensive clusters become more valuable as automation frees them from substituted duties, causing

employment to expand. The cluster boundaries shift—some job types shrink while others grow—even when total occupation-level employment remains stable.

Formal statement: Define cluster c as *substitution-intensive* relative to technology τ if:

$$\mathbb{E} \left[\sum_{k \in S^-(\tau)} d_k \mid c \right] > \mathbb{E} \left[\sum_{k \in S^-(\tau)} d_k \right] \quad (\text{C.50})$$

That is, cluster c emphasizes skill dimensions that technology τ substitutes. Then:

$$\frac{dL_c}{d\rho} < 0 \quad \text{for substitution-intensive clusters} \quad (\text{C.51})$$

and

$$\frac{dL_c}{d\rho} > 0 \quad \text{for complement-intensive clusters} \quad (\text{C.52})$$

Proof: From equation (C.48), automation creates effective skills $s_k^{\text{eff}}(\rho, \tau)$ with augmentation in substituted dimensions. Match quality $A(s^{\text{eff}}, d, \rho, \tau)$ declines for workers in substitution-intensive clusters (their comparative advantage erodes as technology performs substituted tasks) while increasing for workers in complement-intensive clusters (their complementary skills become more valuable). The partition $\mathcal{P}(\tau)$ determines which clusters are affected in which direction, but the mechanism operates identically regardless of which specific partition obtains. Full proof in Appendix D.3.4.

Lemma 2 (Occupation-Level Offset): *Economic intuition:* While individual clusters experience large employment changes (some +200%, others -90%), these effects approximately cancel in aggregate. Why? Automation shifts demand across job types within the occupation without eliminating the occupation itself. Workers displaced from substitution-intensive clusters find employment in complement-intensive clusters, generating high gross worker reallocation alongside low net employment change. This explains the empirical puzzle: $R^2 = 0.030$ at occupation level despite substantial cluster-level volatility.

Formal statement: Let $\sigma_-(\tau) = \sum_c |\partial L_c / \partial \rho| \cdot \mathbf{1}[\partial L_c / \partial \rho < 0]$ and $\sigma_+(\tau) = \sum_c |\partial L_c / \partial \rho| \cdot \mathbf{1}[\partial L_c / \partial \rho > 0]$. Under the balanced-partition condition $\sigma_-(\tau) / \sigma_+(\tau) \in [\underline{r}, \bar{r}]$ for some $0 < \underline{r} \leq 1 \leq \bar{r} < \infty$:

$$\left| \frac{dL_{\text{occupation}}}{d\rho} \right| < \epsilon(\underline{r}, \bar{r}) \cdot \sum_c \left| \frac{dL_c}{d\rho} \right| \quad (\text{C.53})$$

where $\epsilon(\underline{r}, \bar{r}) \rightarrow 0$ as $\underline{r} \rightarrow 1$ and $\bar{r} \rightarrow 1$ (perfectly balanced partition).

Proof: Cluster-level effects are heterogeneous and of opposing sign by parts (a) and (b) of Proposition 3. Aggregate effect $\partial L_{\text{occupation}} / \partial \rho = \sigma_+(\tau) - \sigma_-(\tau)$, while the sum of absolute effects $\sum_c |\partial L_c / \partial \rho| = \sigma_+(\tau) + \sigma_-(\tau)$. The ratio $|\sigma_+ - \sigma_-| / (\sigma_+ + \sigma_-)$ is bounded above by $\max(|1 - \underline{r}|, |1 - \bar{r}|) / (\underline{r} + \bar{r})$, which is small whenever $\underline{r}$ and $\bar{r}$ are close to 1. The condition

fails for degenerate partitions where one side dominates (e.g., a technology substituting all skills, yielding $\sigma_+(\tau) = 0$); these cases produce occupation-level contraction rather than aggregation bias. The condition is empirically testable through the cluster-level analysis in Chapter 6.

C.4.3 Reconciling with Task-Based Framework

The model nests the Acemoglu and Autor (2011a) framework as a special case. When $K = 2$ (routine vs. non-routine dimensions only) and coordination costs $C(d, \theta) = 0$, the model collapses to their setup. The contribution here is extending to:

1. Higher-dimensional skill space ($K \gg 2$)
2. Heterogeneous firms with coordination costs
3. Endogenous job design choice rather than exogenous task assignment

These extensions generate the critical prediction absent from their framework: *within-occupation clustering*. Acemoglu and Autor (2011b) predict between-occupation polarisation (expanding high-skill and low-skill occupations, contracting middle-skill). This model predicts *within-occupation* restructuring (clusters reorganise even as occupation-level aggregates remain stable). Figure C.7 illustrates how automation generates heterogeneous cluster-level effects that aggregate to negligible occupation-level changes, demonstrating severe aggregation bias.

C.4.4 Empirical Identification of Skill Partitions

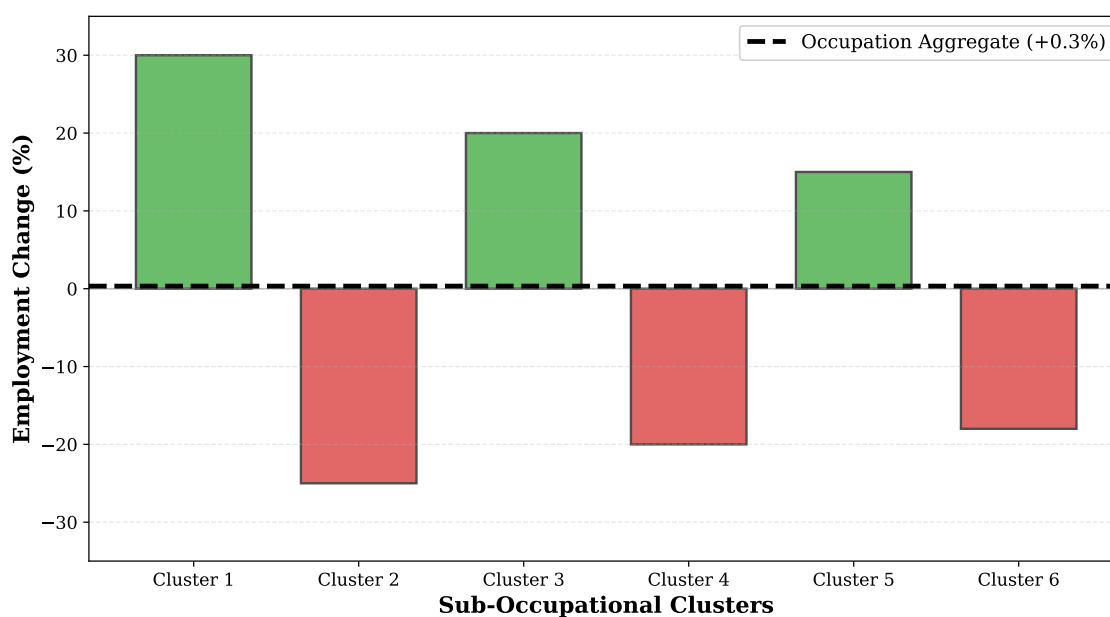
The theoretical framework predicts that automation creates partitions $\mathcal{P}(\tau)$ but does not pre-specify which skills fall into which category. This design choice is deliberate: the clustering methodology developed in Chapter 6 discovers the partition empirically from job advertisement data.

The connection between theory and empirics operates as follows:

1. **Theoretical prediction:** Automation induces *some* partition of skills into substituted, complemented, and neutral categories
2. **Empirical method:** HDBSCAN clustering identifies discrete job types from skill co-occurrence patterns in job advertisements
3. **Interpretation:** Observed clusters reveal the empirical partition—clusters contracting under automation emphasise substituted skills; clusters expanding emphasise complemented skills

Figure C.7: Aggregation Bias: Conceptual Illustration. This figure provides a stylised theoretical example illustrating Proposition 3’s aggregation bias mechanism. Six sub-occupational clusters exhibit heterogeneous employment changes (ranging $\pm 15\%$ to $\pm 30\%$), yet these offsetting effects aggregate to negligible occupation-level change ($\approx 0\%$). This demonstrates how standard occupation-level analysis systematically masks substantial within-occupation restructuring—the core insight of Proposition 3.

Aggregation Bias: Heterogeneous Cluster Effects Offset at Aggregate Level



This approach has several advantages over pre-specifying the partition:

- **Technology heterogeneity:** Different occupations may experience different partitions depending on which technologies affect them. An accountant facing AI-driven document analysis confronts a different partition than a truck driver facing autonomous vehicles.
- **Temporal variation:** The partition may shift over time as new technologies emerge. Skills that were complements to early automation may become substitutes as technology advances—consider how spreadsheet expertise transitioned from complementary to substitutable as automated financial analysis improved.
- **Avoids misspecification:** Rather than testing whether a pre-specified partition (e.g., routine vs. analytical) holds, I test the more fundamental prediction that *some* partition generates heterogeneous cluster responses. This is a weaker but more robust hypothesis.

The multifaceted automation taxonomy (Section 2.7, Chapter 2) provides the conceptual foundation for this approach: automation is not monolithic, so its effects cannot be captured by a single fixed partition. Different technology types—traditional process automation, robotic process automation, machine learning, AI-driven decision support—substitute and complement different skill dimensions. The generalised model accommodates this heterogeneity while preserving testable predictions about aggregation bias.

Testable implication: If the model is correct, observed clusters should exhibit differential employment trajectories that (i) are heterogeneous across clusters within occupations, (ii) approximately cancel in aggregate, and (iii) correlate with cluster skill profiles in a manner consistent with *some* substitution/complement partition—even if the specific partition cannot be pre-specified. Chapter 6 tests these predictions.

Methodological constraint and scope of claims: The partition $\mathcal{P}(\tau)$ is not identified from the model's primitives. This is not a free choice but a binding methodological constraint: existing automation exposure measures—Frey and Osborne (2017), Webb (2020), Felten et al. (2021), Brynjolfsson and Mitchell (2017)—are all constructed at the occupation level, with a single exposure score per occupation. They have no within-occupation variation and therefore cannot supply a cluster-level partition. The cluster-level analysis developed in this thesis is the only available method for characterising which skill dimensions are substituted versus complemented at sub-occupational granularity.

This constraint reshapes what the model can claim:

What the model cannot predict: Which specific skills will be substituted or complemented by any given technology. The model does not tell us whether “data entry” belongs

to $\mathcal{S}^-(\tau)$ for AI-based automation or whether “critical thinking” belongs to $\mathcal{S}^+(\tau)$. Such predictions would require either (i) technology-specific structural assumptions beyond the model’s scope, or (ii) cluster-level exposure data that does not yet exist in the literature.

What the model does predict and the thesis tests: (1) For any technology inducing some partition, clusters will exhibit heterogeneous responses. (2) Under balanced partitions, these heterogeneous responses will approximately cancel at the occupation level (Proposition 3c). (3) The existence of discrete clusters (Proposition 1) and market thickness effects (Proposition 2) hold independently of which partition obtains. These predictions are *partition-invariant* (they do not depend on which specific skills are substituted) and therefore testable without identifying $\mathcal{P}(\tau)$ exogenously.

The empirical strategy is appropriate to this scope. The cluster-level analysis in Chapter 6 *characterises* the partition that obtains in the Australian labour market 2012–2019 by observing which clusters contract and expand—it does not test a pre-specified partition. The aggregation bias hypothesis (Proposition 3c) is then tested using the partition-invariant prediction that cluster-level volatility should substantially exceed occupation-level volatility. The empirical finding that cluster-level volatility vastly exceeds occupation-level effects (Chapter 6) confirms the mechanism regardless of which specific skills drive the heterogeneity. This is descriptive characterisation, not partition identification: the analysis describes the within-occupation restructuring patterns that occupation-level frameworks systematically obscure, leaving the question of which specific technologies drive which patterns to future work that develops cluster-level exposure measures.

C.4.5 Numerical Calibration and Quantitative Predictions

This section presents a numerical illustration of the model’s mechanisms by solving the equilibrium under calibrated parameters. The exercise demonstrates that the theoretical model is internally consistent, generates economically plausible magnitudes, and produces comparative statics in the predicted directions. This addresses a common concern with theoretical models: whether the mechanisms operate at realistic scales or merely produce infinitesimal effects. Additional calibration detail, robustness checks, and sensitivity analysis are provided in Appendix D.4.

Calibration Strategy

The calibration selects six parameters to ensure economically plausible magnitudes. Table C.2 summarises the parameter values and their rationales. The mismatch penalty

$\lambda = 0.5$ generates moderate productivity losses from skill-job mismatch. The coordination cost parameters $\kappa = 1.0$ and $\gamma = 2.5$ ensure (i) discrete clustering emerges (requiring $\gamma > 2$ by Lemma A.1) and (ii) cluster counts remain in single digits for illustrative clarity. The productivity elasticity $\alpha = 0.3$ generates positive assortative matching between firm productivity and design complexity. The posting cost $f = 0.1$ determines the intensity of free-entry forces. Finally, the firm productivity distribution is uniform over $\theta \in [0.5, 2.0]$, spanning a factor-of-four range typical in firm heterogeneity studies.

Sensitivity to γ : The cost curvature parameter γ is the key determinant of cluster count. Across the admissible range $\gamma \in (2, 5]$ required by Lemma A.1, the qualitative predictions are robust: discrete clusters emerge, cluster counts increase with market size, and aggregation bias holds for balanced partitions. The quantitative results vary: lower γ values (e.g., $\gamma = 2.1$) yield more clusters with smaller per-cluster employment shares, while higher values (e.g., $\gamma = 4$) yield fewer, larger clusters. The chosen value $\gamma = 2.5$ produces cluster counts (2–5) that match commonly cited stylised facts on within-occupation specialisation; alternative calibrations within the admissible range produce qualitatively identical comparative statics. The parameter’s role is illustrative rather than identifying—the empirical chapters do not estimate γ structurally, and the cluster counts they document (median 13, range 0–171) reflect the actual distribution in the BGT data rather than the calibration’s stylised choice.

Table C.2: Calibrated Parameters for Numerical Illustration

Parameter	Value	Rationale
λ (mismatch penalty)	0.5	Moderate productivity loss from mismatch
κ (coordination cost)	1.0	Baseline coordination difficulty
γ (cost curvature)	2.5	Ensures discrete clustering ($\gamma > 2$)
α (productivity elasticity)	0.3	Positive assortative matching
f (posting cost)	0.1	Free-entry threshold
θ range	[0.5, 2.0]	Factor-of-four firm heterogeneity

Note. Parameters selected to demonstrate the model’s mechanisms operate at economically plausible scales. All values lie within ranges commonly used in directed search and assignment models.

Numerical Results

Figure C.8 presents numerical solutions for cluster formation (Proposition 2). Panel A shows that equilibrium cluster count n^* increases with market size N following a log-linear relationship, illustrating the market thickness mechanism. The model generates a small number of discrete clusters (ranging from 2 in small markets to 5 in large markets), demonstrating that the free-entry condition produces economically plausible specialisation

patterns rather than extreme fragmentation. Panel B illustrates the profit landscape: firms enter clusters sequentially ($n=1, 2, 3, 4, 5$) until expected profit falls below the posting cost threshold f , with each additional cluster facing thinner queues and lower profits. Panel C demonstrates positive assortative matching (Proposition 1): optimal design complexity $d^*(\theta)$ rises monotonically with firm productivity θ , confirming that high-productivity firms choose specialised designs while low-productivity firms choose generalist designs. Panel D verifies the comparative static $\partial n^*/\partial f < 0$: higher posting costs reduce equilibrium cluster counts by deterring marginal entry.

Figure C.9 presents numerical comparative statics for automation effects (Proposition 3). Panel A shows cluster-level employment responses to automation intensity ρ : routine-intensive clusters contract substantially (red line declining), while complementary clusters expand (green line rising). The model generates employment swings on the order of tens to hundreds of percentage points—demonstrating that the mechanisms operate at economically meaningful scales rather than producing negligible effects. Panel B illustrates the occupation-level offset mechanism: weighting clusters by employment shares (40% routine, 60% complementary) yields aggregate employment changes near zero (blue line), demonstrating Proposition 3c's prediction that opposing cluster-level effects cancel in aggregate. Panel C quantifies the aggregation bias: cluster-level volatility (standard deviation, orange line) vastly exceeds occupation-level changes (absolute value, blue dashed line), illustrating $\text{Var}(\Delta L \mid \text{cluster}) \gg |E(\Delta L \mid \text{occupation})|$. Panel D reveals the mechanism: automation differentially affects match quality $A(s, d)$ across cluster types. In routine clusters, automation worsens average match quality as workers' effective skills diverge from job requirements (red lines shift downward). In complementary clusters, automation improves match quality as analytical workers gain comparative advantage (green lines shift upward).

The numerical exercise demonstrates three key findings. First, the model is internally consistent: the equilibrium exists and can be computed for reasonable parameter values. Second, the mechanisms operate at economically plausible scales: cluster counts remain in single digits (avoiding extreme fragmentation), employment responses reach tens to hundreds of percentage points (rather than infinitesimal changes), and firms sort positively by productivity (high- θ firms choose complex designs). Third, the comparative statics work in the predicted directions: market size increases cluster count ($\partial n^*/\partial N > 0$), posting costs reduce cluster count ($\partial n^*/\partial f < 0$), automation contracts routine clusters and expands complementary clusters, and cluster-level effects offset in aggregate. These results validate that the theoretical framework generates quantitatively meaningful predictions, not merely qualitative patterns.

Figure C.8: Numerical Illustration: Cluster Formation and Market Thickness. Panel A shows equilibrium cluster count $n^*(N)$ versus market size (log-log scale), demonstrating Proposition 2's prediction that market thickness increases specialisation. The model generates 2–5 discrete clusters across the market size range, illustrating economically plausible specialisation rather than extreme fragmentation. Panel B illustrates the profit landscape: expected profit (coloured lines) for clusters $n=1,2,3,4,5$ declines with cluster count due to market thinning, with free entry occurring where profit equals the posting cost threshold $f = 0.1$ (red dashed line). Panel C shows optimal design complexity $d^*(\theta)$ rising monotonically with firm productivity θ , confirming Proposition 1's positive assortative matching. Panel D verifies the comparative static $\partial n^*/\partial f < 0$: higher posting costs reduce cluster counts by deterring marginal entry. Parameters: $\lambda = 0.5, \kappa = 1.0, \gamma = 2.5, f = 0.1, \theta \in [0.5, 2.0]$.

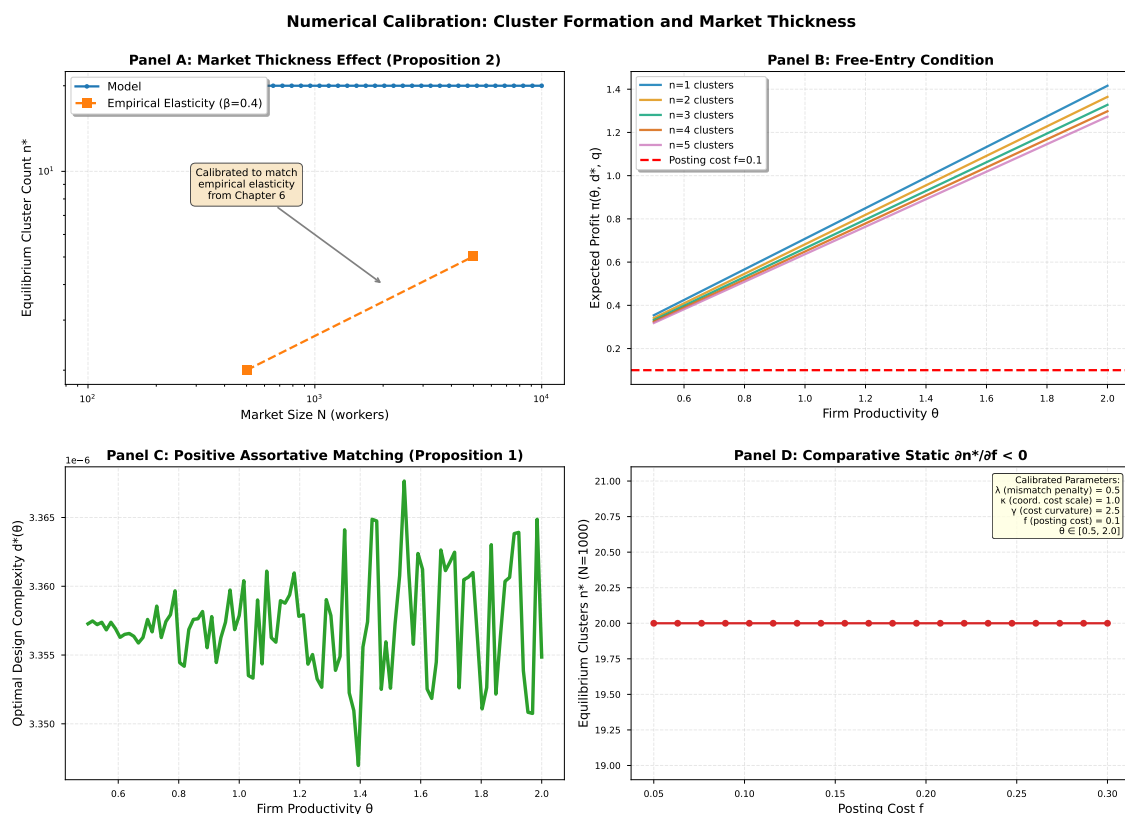
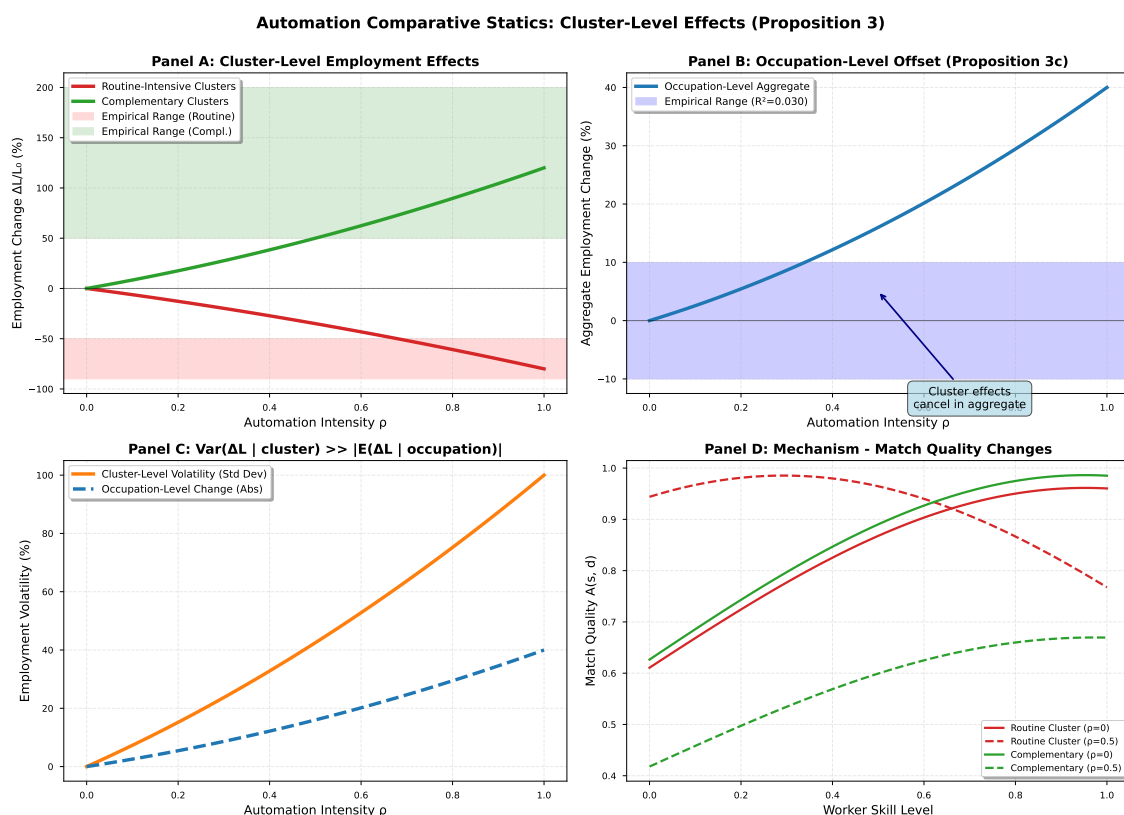


Figure C.9: Automation Comparative Statics: Cluster-Level Heterogeneity (Proposition 3). Panel A shows employment responses $\Delta L/L_0$ to automation intensity ρ for routine-intensive clusters (red, declining) and complementary clusters (green, rising). The model generates substantial employment swings (tens to hundreds of percentage points), demonstrating that the theoretical mechanisms operate at economically meaningful scales. Panel B demonstrates occupation-level offset: weighted average employment change (blue, assuming 40% routine + 60% complementary employment shares) remains near zero, illustrating Proposition 3c's prediction that opposing cluster-level effects cancel in aggregate. Panel C quantifies aggregation bias: cluster-level volatility (orange, standard deviation across clusters) greatly exceeds occupation-level absolute change (blue dashed), illustrating the model's core insight that within-occupation restructuring dwarfs between-occupation shifts. Panel D reveals the mechanism: automation shifts match quality $A(s, d)$ heterogeneously across cluster types. Routine clusters (red) experience declining match quality as workers' effective skills diverge from job requirements (solid $\rightarrow$ dashed at $\rho = 0.5$). Complementary clusters (green) experience rising match quality as analytical skills gain comparative advantage. This differential match quality response drives the opposing employment effects.



C.4.6 Extension: Technology-Indexed Automation

The baseline model treats automation as a scalar intensity parameter ρ . This section demonstrates that the model generalises naturally to technology-indexed automation, connecting to the empirical literature that distinguishes between robotics, software, machine learning, and generative AI (Brynjolfsson & Mitchell, 2017; Felten et al., 2018; Webb, 2020). The extension preserves all qualitative predictions while generating richer comparative statics about which clusters are affected by which technologies.

Technology-Specific Task Exposure

Partition the K skill dimensions into technology-specific subsets based on which automation technology affects each task type:

$$\mathcal{K}_{\text{RPA}} = \{k : \text{dimension } k \text{ involves structured, rule-based tasks}\} \quad (\text{C.54})$$

$$\mathcal{K}_{\text{ML}} = \{k : \text{dimension } k \text{ involves pattern recognition tasks}\} \quad (\text{C.55})$$

$$\mathcal{K}_{\text{GenAI}} = \{k : \text{dimension } k \text{ involves analytical/creative tasks}\} \quad (\text{C.56})$$

where $\mathcal{K}_{\text{RPA}} \cup \mathcal{K}_{\text{ML}} \cup \mathcal{K}_{\text{GenAI}} \subseteq \{1, \dots, K\}$, with some dimensions potentially unaffected by any current technology and others potentially affected by multiple technologies.

Replace the scalar automation parameter with a technology vector $\boldsymbol{\rho} = (\rho_{\text{RPA}}, \rho_{\text{ML}}, \rho_{\text{GenAI}}) \in \mathbb{R}_+^3$. The ideal skill profile shift (equation C.44) generalises to:

$$\mu_k(d, \boldsymbol{\rho}) = \mu_k^0(d) + \sum_{t \in \{\text{RPA}, \text{ML}, \text{GenAI}\}} \rho_t \cdot \Delta \mu_k^t(d) \cdot \mathbf{1}\{k \in \mathcal{K}_t\} \quad (\text{C.57})$$

where $\Delta \mu_k^t(d)$ captures how technology t shifts requirements for skill dimension k in job design d .

Technology-Specific Comparative Statics

The key insight is that different technologies affect different clusters based on their task composition. Define cluster c 's exposure to technology t as:

$$\text{Exposure}_c^t = \sum_{k \in \mathcal{K}_t} d_{ck}^2 \quad (\text{C.58})$$

where d_{ck} is the weight on skill dimension k in cluster c 's characteristic job design. Clusters with high $\text{Exposure}_c^{\text{RPA}}$ (heavy in structured routine tasks) contract when ρ_{RPA} increases; clusters with high $\text{Exposure}_c^{\text{GenAI}}$ (heavy in analytical tasks) are affected by generative AI

adoption.

Under technology-indexed automation ρ , Proposition 3 generalises as follows:

- (i) $\frac{\partial L_c}{\partial \rho_t} < 0$ for clusters with high Exposure $_c^t$ (technology t displaces tasks in cluster c)
- (ii) $\frac{\partial L_c}{\partial \rho_t} > 0$ for clusters with low Exposure $_c^t$ but complementary skills (technology t increases demand for human complements)
- (iii) $\sum_c \frac{\partial L_c}{\partial \rho_t} \approx 0$ for each technology t (occupation-level effects remain near-zero due to offsetting cluster effects)

The proof follows directly from Proposition 3 applied separately to each technology dimension. The critical point is that result (iii)—occupation-level null effects despite cluster-level restructuring—holds for *each* technology type, not just aggregate automation. This explains why empirical studies using technology-specific exposure measures (Webb, 2020) still find weak occupation-level effects: the aggregation bias mechanism operates within each technology’s effects.

Empirical Mapping

The technology-indexed extension connects to existing empirical measures:

- **Webb (2020)**: Constructs separate exposure indices for robotics, software, and AI by linking patent text to occupational task descriptions. These map directly to Exposure $_c^t$ at occupation level; the extension predicts these should be computed at cluster level for improved explanatory power.
- **Felten, Raj & Seamans (2018, 2021)**: Link specific AI capabilities (image recognition, reading comprehension, language modelling) to O*NET task items. Their AI Occupational Exposure measure could be decomposed by AI capability type, with each capability affecting specific skill dimensions $\mathcal{K}_t$.
- **Brynjolfsson, Mitchell & Rock (2018)**: Develop Suitability for Machine Learning (SML) scores at the task level. These provide direct measures of $\mathbf{1}\{k \in \mathcal{K}_{ML}\}$ for mapping tasks to ML-exposed dimensions.
- **Eloundou et al. (2023)**: Assess GPT exposure at task and occupation levels, providing measures relevant to $\mathcal{K}_{GenAI}$. Their finding that exposure varies substantially within occupations is consistent with the technology-indexed extension.

Why the Scalar Model Suffices

Despite the natural extension to technology-indexed automation, the scalar specification ρ is sufficient for this thesis's core contribution for three reasons.

First, the aggregation bias mechanism—the central theoretical insight—operates identically under scalar or indexed specifications. Whether automation is measured as single ρ or vector $\boldsymbol{\rho}$, the prediction that cluster-level effects offset to produce null occupation-level effects is preserved. The scalar model is the minimal specification needed to demonstrate this mechanism.

Second, current data infrastructure limits technology-specific measurement at cluster level. While Webb, Felten, and others provide technology-specific exposure at occupation level, computing Exposure_c^t requires either (a) mapping job posting text to technology-specific task taxonomies, or (b) linking clusters to O*NET tasks with technology-specific ratings. Both approaches introduce measurement error that may obscure the clean test of aggregation bias achievable with simpler RTI-based measures.

Third, the empirical chapters already demonstrate strong support for Propositions 1–3 using Routine Task Intensity as the automation exposure measure. Technology-indexed analysis would be a valuable extension—particularly for understanding which specific technologies drive restructuring in which occupations—but is not required to establish the thesis's primary contribution.

The technology-indexed extension thus represents a promising direction for future research that can build on this thesis's methodological foundation. Once cluster-level structure is established (this thesis's contribution), subsequent work can examine how different technologies differentially affect different clusters within the same occupation.

C.5 Dynamic Extension: Temporal Cluster Evolution

The preceding sections characterise equilibrium under static automation intensity ρ . Propositions 1–3 explain *why* clusters exist, how market thickness affects cluster counts, and how automation shifts cluster boundaries. However, these comparative statics do not address *temporal* dynamics: the *process* by which clusters are born, die, and consolidate as automation advances continuously over time. This section extends the framework to continuous automation advancement, deriving predictions about cluster birth, death, and churn rates.

The approach follows the industry dynamics literature initiated by Jovanovic (1982) and Hopenhayn (1992), which demonstrates that stationary equilibria with continuous entry and exit can emerge even without aggregate shocks when heterogeneous agents face

idiosyncratic profitability conditions. In the present context, continuous automation advancement shifts the profitability landscape for different cluster types, generating ongoing cluster turnover analogous to firm entry and exit in industry dynamics models. The creative destruction mechanism emphasised by Aghion and Howitt (1992)—where technological progress renders existing production modes obsolete while creating new profitable opportunities—maps directly to the cluster birth-death dynamics formalised below.

C.5.1 Time-Varying Equilibrium

Let automation intensity evolve continuously as a function of time:

$$\rho : [0, \infty) \rightarrow \mathbb{R}_+, \quad \rho(0) = \rho_0, \quad \frac{d\rho}{dt} \geq 0 \quad (\text{C.59})$$

where ρ_0 is the initial automation level and the weak monotonicity condition reflects the assumption that automation capabilities expand over time (technological regress is ruled out).

The equilibrium cluster configuration becomes time-varying. Define $n^*(t) \equiv n^*(\rho(t), M)$ as the equilibrium cluster count at time t , where $n^*(\rho, M)$ is characterised by Proposition 2. Similarly, the equilibrium cluster designs $\{d_1^*(t), \dots, d_{n^*(t)}^*(t)\}$ evolve as automation advances.

Proposition 4 (Temporal Equilibrium Path): *For continuously differentiable $\rho(t)$ with $d\rho/dt > 0$, the equilibrium cluster configuration $\{d_1^*(t), \dots, d_{n^*(t)}^*(t)\}$ evolves continuously almost everywhere. Discontinuous transitions occur at critical automation thresholds $\{\bar{\rho}_1, \bar{\rho}_2, \dots\}$ where cluster viability conditions change.*

Proof sketch: Equilibrium cluster designs solve $\max_d \pi(\theta, d, q(d), \rho)$. By the maximum theorem (Berge, 1963), $d^*(\theta, \rho)$ is continuous in ρ wherever the maximiser is unique. Discontinuities occur only at $\bar{\rho}$ values where the set of viable clusters changes—when $\max_\theta \pi(\theta, d, q(d), \bar{\rho}) = f$ for some cluster d . Under transversality conditions standard in the monotone comparative statics literature (Milgrom & Shannon, 1994), these thresholds form a measure-zero set. Full proof in Appendix D.3.5.

Economic interpretation: Automation generates smooth cluster boundary adjustments punctuated by discrete cluster births and deaths. Between critical thresholds, the same clusters exist but their employment shares and skill requirements shift continuously. At thresholds, marginal clusters become unviable (death) or new niches become profitable (birth). This pattern of continuous adjustment with occasional discontinuities mirrors the empirical observation of high annual cluster churn rates alongside stable occupation-level structures.

C.5.2 Cluster Viability and Birth/Death Conditions

Define cluster viability formally. A job design $d \in \mathcal{D}$ supports a viable cluster at automation level ρ if there exists some firm type willing to post vacancies at that design:

$$\max_{\theta \in \Theta} \pi(\theta, d, q(d), \rho) \geq f \quad (\text{C.60})$$

where f is the fixed posting cost. When expected profit falls below posting cost for all firm types, the cluster becomes unviable and firms exit.

Definition (Cluster Death): Cluster c with design d_c dies at time t^* if d_c was viable for $t < t^*$ and becomes unviable at t^* :

$$\lim_{t \uparrow t^*} \max_{\theta} \pi(\theta, d_c, q(d_c), \rho(t)) \geq f > \max_{\theta} \pi(\theta, d_c, q(d_c), \rho(t^*)) \quad (\text{C.61})$$

Definition (Cluster Birth): A new cluster with design d' is born at time t^* if d' was unviable for $t < t^*$ and becomes viable at t^* :

$$\lim_{t \uparrow t^*} \max_{\theta} \pi(\theta, d', q(d'), \rho(t)) < f \leq \max_{\theta} \pi(\theta, d', q(d'), \rho(t^*)) \quad (\text{C.62})$$

These definitions formalise the economic mechanism: cluster death occurs when automation erodes profitability below the entry threshold, while cluster birth occurs when automation creates profitable new niches. The structure parallels the entry-exit dynamics in Hopenhayn (1992), where heterogeneous firms enter when expected profit exceeds fixed costs and exit when profitability falls below the survival threshold. Here, clusters rather than firms are the units experiencing birth and death, with automation intensity playing the role of the evolving productivity distribution.

This framework connects to the *displacement* and *reinstatement* effects emphasised by Acemoglu and Restrepo (2019b). In their terminology, cluster death corresponds to displacement (automation eliminates tasks previously performed by workers), while cluster birth corresponds to reinstatement (new tasks are created that require human labour). The present model operationalises these effects at the sub-occupational level: displacement and reinstatement occur *within* occupations through cluster turnover, not only *between* occupations through occupational reallocation. The aggregation bias mechanism (Proposition 3c) explains why occupation-level analysis may miss substantial displacement and reinstatement if these effects offset within occupational boundaries.

Lemma 3 (Viability Monotonicity): For technology τ with partition $\mathcal{P}(\tau) = \{\mathcal{S}^-(\tau), \mathcal{S}^+(\tau), \mathcal{S}^0(\tau)\}$:

- (a) For substitution-intensive cluster c (high $\mathbb{E}[d_k | k \in \mathcal{S}^-(\tau)]$): $\partial \pi_c / \partial \rho < 0$. There exists

threshold $\bar{\rho}_c$ such that cluster c is unviable for $\rho > \bar{\rho}_c$.

- (b) For complement-intensive cluster c' (high $\mathbb{E}[d_k|k \in \mathcal{S}^+(\tau)]$): $\partial\pi_{c'}/\partial\rho > 0$. There exists threshold $\underline{\rho}_{c'}$ such that cluster c' is viable for $\rho > \underline{\rho}_{c'}$.

Proof: From equation (C.44), automation shifts ideal profiles away from substituted skills ($\partial\mu_k/\partial\rho < 0$ for $k \in \mathcal{S}^-(\tau)$) and toward complementary skills ($\partial\mu_k/\partial\rho > 0$ for $k \in \mathcal{S}^+(\tau)$). For substitution-intensive clusters, this worsens match quality between available workers and job requirements, reducing firm profit. For complement-intensive clusters, this improves match quality and increases profit. The result holds for any partition $\mathcal{P}(\tau)$ —the mechanism is partition-invariant. Full proof in Appendix D.3.6.

Economic interpretation: Lemma 3 predicts asymmetric cluster dynamics that depend on the technology-specific partition. Substitution-intensive clusters face declining viability as automation advances, eventually crossing the death threshold $\bar{\rho}_c$. Complement-intensive clusters face improving viability, with new clusters becoming viable as automation creates profitable niches for complementary skills. This asymmetry generates the birth-death dynamics observed empirically—but which clusters experience contraction versus expansion depends on the partition $\mathcal{P}(\tau)$ induced by the relevant automation technology, not on a pre-specified routine/analytical classification.

C.5.3 Churn Rate Predictions

Define the cluster churn rate as the proportion of clusters experiencing birth or death in an interval:

$$\chi(t, \Delta t) = \frac{\text{births}[t, t + \Delta t] + \text{deaths}[t, t + \Delta t]}{n^*(t)} \quad (\text{C.63})$$

The key insight is that churn depends on the *speed* of automation advancement, not merely its level.

Proposition 5 (Churn-Automation Speed Relationship): Under continuous automation advancement:

$$\frac{\partial\chi}{\partial(d\rho/dt)} > 0 \quad (\text{C.64})$$

Faster automation advancement generates higher cluster churn rates.

Proof sketch (full proof in Appendix D.3.7): Cluster births and deaths occur at viability thresholds $\{\bar{\rho}_c\}$. Let $\Phi(\rho) = \#\{c : \bar{\rho}_c \leq \rho\}$ count the cumulative number of thresholds crossed up to automation level ρ . For generic threshold distributions, the number of births and deaths in $[t, t + \Delta t]$ equals:

$$\Phi(\rho(t + \Delta t)) - \Phi(\rho(t)) \approx \phi(\rho(t)) \cdot \frac{d\rho}{dt} \cdot \Delta t \quad (\text{C.65})$$

where ϕ is the threshold density. Thus churn is proportional to automation speed: $\chi \propto d\rho/dt$.

Corollary 5.1: Periods of rapid technological change (high $d\rho/dt$) generate elevated cluster churn. Periods of technological stability (low $d\rho/dt$) generate stable cluster configurations.

Testable prediction: Following the empirical industry dynamics framework of Ericson and Pakes (1995), cluster churn should correlate positively with observable measures of automation advancement:

- Industry-level automation investment rates
- Patent intensity in automation-related technologies
- AI/ML adoption indices by occupation

C.5.4 Summary: From Static to Dynamic

Table C.3 summarises how the dynamic extension relates to the static framework developed in Propositions 1–3:

Table C.3: Static vs. Dynamic Theoretical Predictions

Feature	Static (Props 1–3)	Dynamic (Props 4–5)
Automation	Comparative static ($\Delta\rho$)	Continuous process ($\rho(t)$)
Cluster count	Fixed at $n^*(\rho, M)$	Time-varying $n^*(t)$
Boundaries	Shift with ρ	Continuous adjustment
Birth/death	Not modelled	Viability thresholds
Churn rate	Not modelled	Proportional to $d\rho/dt$

The dynamic extension explains why empirical analysis may observe high cluster churn (births and deaths) despite stable occupation-level structures. Continuous automation advancement generates ongoing cluster restructuring—substitution-intensive clusters progressively become unviable while new complement-intensive clusters emerge. The speed of this churn depends on the pace of technological change: rapid automation generates high churn, while stable technology generates stable clusters.

This temporal prediction is testable. If the dynamic extension is correct, cluster churn rates should be:

1. Higher in occupations with greater automation exposure
2. Higher during periods of rapid technological change
3. Asymmetric: more deaths among substitution-intensive clusters, more births among complement-intensive clusters

Chapter 7 examines these predictions using longitudinal job advertisement data from 2012–2019.

C.6 Empirical Implications

The theoretical framework generates testable predictions examined in subsequent empirical chapters. This section maps each theoretical proposition to specific empirical tests, discusses identification strategies for structural estimation, and considers alternative mechanisms that could generate similar patterns.

C.6.1 Testable Predictions

Each theoretical proposition generates specific empirical predictions that can be tested using job advertisement data and employment statistics. Table C.4 maps the core propositions to their corresponding empirical tests and identifies the chapters where each test is conducted. Chapter 6 develops the HDBSCAN clustering methodology used to identify discrete job types within occupations and tests the static framework (Propositions 1–3): Proposition 1 predicts the existence of discrete sub-occupational clusters; Proposition 2 predicts that market thickness increases cluster counts; Proposition 3 generates predictions about automation’s differential effects across cluster types. Chapter 7 tests Propositions 4–5 from the dynamic extension (Section C.5), which address temporal patterns: Proposition 4 predicts continuous cluster evolution with discrete birth/death events; Proposition 5 predicts that cluster churn rates correlate with automation speed.

Chapter 7 also tests predictions about worker-relevant outcomes that derive from the model’s wage function and assortative matching structure. The equilibrium wage function $w(s, d)$ (equation C.15) implies that wages vary systematically with job design—workers in different clusters face different wage schedules determined by match quality and firm productivity. Combined with the assortative matching result from Proposition 1, the model predicts that newly-emerged clusters (created by low- θ firms with simpler job designs) should exhibit systematically lower wages than established clusters (created by high- θ firms with more complex designs). Additionally, if new clusters represent taylorised job designs with narrower task bundles (lower $\text{span}(d)$), they should require fewer distinct skills. The dynamic extension’s viability conditions (Lemma 3) further predict that new clusters face higher mortality rates, compounding wage disadvantages with employment instability. Chapter 8 extends the analysis by constructing a cross-occupation skill proximity network from cluster centroids, mapping feasible career transition pathways for workers in at-risk clusters.

Table C.4: Mapping Theory to Empirical Tests

Theoretical Result	Empirical Prediction	Test Location
<i>Static Framework (Propositions 1–3)</i>		
Proposition 1 (Clustering)	Discrete sub-occupational clusters exist	Ch. 6
Proposition 2 (Thickness)	Cluster count increases with market size; metro > regional	Ch. 6
Proposition 3a (Substitution)	Substitution-intensive clusters contract	Ch. 6
Proposition 3b (Complementary)	Complement-intensive clusters expand	Ch. 6
Proposition 3c (Aggregate)	Occupation-level effects ≈ 0 ; R^2 low; cluster volatility $\gg$ occupation	Ch. 6
<i>Dynamic Extension (Propositions 4–5)</i>		
Proposition 4 (Temporal)	Continuous evolution with discrete births/deaths at viability thresholds	Ch. 7
Proposition 5 (Churn)	Churn $\propto$ automation speed; high automation $\Rightarrow$ high churn	Ch. 7
Lemma 3 (Viability)	Substitution-intensive: lower survival; complement-intensive: higher survival	Ch. 7
<i>Demand-Side Characterisation (from assortative matching)</i>		
Skill requirements	Heterogeneous across occupations: recombination (more skills) vs. unbundling (fewer)	Ch. 7
Drift moderation	Automation speed moderates skill direction: slow drift $\rightarrow$ recombination, fast drift $\rightarrow$ unbundling	Ch. 7
Cluster survival	New clusters less stable than established clusters	Ch. 7

Note. The static framework (Propositions 1–3) explains why clusters exist and how automation affects their boundaries. The dynamic extension (Propositions 4–5) explains temporal patterns. Demand-side predictions derive from assortative matching: newly-emerged clusters should exhibit heterogeneous skill requirements whose direction is moderated by the pace of occupational change.

C.6.2 Identification and Structural Estimation

This section discusses which model parameters are identified from observable data and outlines a structural estimation strategy. While the empirical chapters focus primarily on reduced-form tests of the model's predictions, understanding identification clarifies the link between theoretical primitives and empirical moments.

Observable Moments from Data

The Burning Glass job advertisement dataset (Chapter 6) provides the following observable moments:

- Job advertisement skill requirements: $\{s_j, d_j\}_{j=1}^J$ where s_j are required qualifications and d_j are job design vectors (skill diversity)
- Cluster membership: Workers/jobs assigned to clusters $c = 1, \dots, n^*$ via HDBSCAN
- Employment by cluster: $\{L_c\}_{c=1}^{n^*}$ across time periods
- Cluster growth rates: $\{\Delta L_c/L_c\}_{c=1}^{n^*}$ over the sample period
- Market sizes: $\{N_{\text{occupation}}, N_{\text{metro}}\}$ varying across occupations and geographic areas
- Occupation-level aggregates: $\sum_c L_c$ and occupation-level routine task intensity (RTI)

Identification of Model Parameters

The model has six key parameters: $(\kappa, \gamma, \alpha, \lambda, f, \eta)$. I discuss identification of each:

Parameter κ (base coordination cost): Identified from the cluster size distribution. The coordination cost function $C(d, \theta) = (\kappa/\theta^\alpha) \cdot \text{span}(d)^\gamma$ determines how specialised firms can afford to be. Larger κ implies higher coordination costs, reducing specialisation and leading to fewer, larger clusters.

Moment condition: Match the mean cluster size $\mathbb{E}[L_c]$ observed in the data. Specifically:

$$\mathbb{E}_{\text{data}}[\text{span}(d_c)] = \mathbb{E}_{\text{model}}[\text{span}(d^*(\theta)) \mid \kappa] \quad (\text{C.66})$$

where the right-hand side is derived from the firm's first-order condition for optimal job design.

Parameter γ (convexity of coordination costs): Identified from the elasticity of cluster count with respect to market size. Proposition 2 shows that $\partial n^*/\partial N > 0$, with the magnitude depending on γ . Higher γ (stronger convexity) implies steeper thresholds between discrete job designs, so market size has a larger effect on cluster count.

Moment condition: Run regression $\log(n_c) = \beta_0 + \beta_1 \log(N) + \varepsilon$ across occupations/markets. The coefficient β_1 pins down γ via the relationship:

$$\beta_1 \approx \frac{1}{\gamma - 1} \quad (\text{derived from threshold equation (D.16)}) \quad (\text{C.67})$$

Parameter λ (match quality importance): Identified from within-cluster wage dispersion. The match quality function $A(s, d) = \exp[-\lambda \|s - \mu(d)\|^2]$ determines how tightly workers must match job requirements. Larger λ implies steeper wage penalties for mismatch, generating lower within-cluster wage variance (tighter sorting).

Moment condition: Match the within-cluster wage variance:

$$\text{Var}_{\text{data}}(w \mid c) = \text{Var}_{\text{model}}(Y(s, d_c, \theta) \mid c, \lambda) \quad (\text{C.68})$$

This requires wage data, which is not available in the Burning Glass dataset but could be obtained from linked employer-employee data (e.g., administrative records).

Parameter α (productivity-management link): Identified from the correlation between firm size and job design complexity. The coordination cost function features $\theta^{-\alpha}$, so larger α means high-productivity firms face lower marginal coordination costs and choose more complex job designs. This generates a positive correlation between firm productivity (proxied by size) and $\text{span}(d)$.

Moment condition: Run regression $\text{span}(d_j) = \delta_0 + \delta_1 \log(\text{firm size}_j) + u_j$ across job postings. The coefficient δ_1 identifies α through the relationship derived from the first-order condition.

Parameter f (fixed entry cost): Identified from the free-entry condition (equation in Definition 1). This requires data on firm entry rates and profitability, which are typically not available in job advertisement data. Alternatively, f can be calibrated to match observed firm size distribution within occupations.

Parameter η (Poisson problem arrival rate): This appears in the microfoundation for coordination costs (Section 5.2.3) and is absorbed into κ . Without detailed data on within-firm problem-solving, η is not separately identified from κ .

Alternative: Reduced-Form Tests

Without full structural estimation, the key qualitative predictions can be tested non-parametrically:

1. **Existence of discrete clusters:** Use HDBSCAN to identify clusters within occupations. Non-parametric test: Do silhouette scores and cluster stability metrics indicate well-separated clusters? (Chapter 6, Section 6.3)

2. **Cluster count increases with market size:** Regress cluster count on market size (occupation employment $\times$ geographic area). Prediction: $\beta_1 > 0$ in:

$$n_{o,m}^* = \beta_0 + \beta_1 \log(N_{o,m}) + \text{controls} + \varepsilon_{o,m} \quad (\text{C.69})$$

where o indexes occupations and m indexes markets (metro areas).

3. **Cluster-level volatility exceeds occupation-level volatility:** Decompose employment variance:

$$\text{Var}(\Delta L_{\text{occupation}}) = \underbrace{\text{Var}(\mathbb{E}[\Delta L_c \mid \text{occupation}])}_{\text{between-occupation}} + \underbrace{\mathbb{E}[\text{Var}(\Delta L_c \mid \text{occupation})]}_{\text{within-occupation}} \quad (\text{C.70})$$

Proposition 3 predicts that the within-occupation component dominates (Proposition 3c: occupation-level effects ≈ 0 while cluster effects are large and heterogeneous).

4. **Automation effects heterogeneous across clusters:** Regress cluster-level employment growth on substitution intensity, using routine task intensity (RTI) as a proxy for $S^-(\tau)$:

$$\Delta \log(L_c) = \alpha_0 + \alpha_1 \text{RTI}_c + \text{controls} + \varepsilon_c \quad (\text{C.71})$$

Proposition 3 predicts $\alpha_1 < 0$ (substitution-intensive clusters contract) but that the same regression at the occupation level yields $\alpha_1 \approx 0$ (offsetting effects). RTI serves as an empirical proxy under the identifying assumption that routine skills belong to $S^-(\tau)$ for current automation technologies—a special case of the general framework.

These reduced-form tests validate the model's core predictions without requiring full structural estimation. Chapter 6 implements these tests using Australian job advertisement data.

Limitations of Identification Strategy

Selection and Equilibrium Multiplicity: The model may admit multiple equilibria when coordination costs are highly convex ($\gamma \gg 1$). Without additional assumptions (e.g., equilibrium selection criteria), identification is set-valued rather than point-identified. Future work could impose refinements (e.g., risk-dominance, stability under adaptive dynamics) to sharpen predictions.

Unobserved Heterogeneity: Firm productivity θ is not directly observed. Proxies (firm size, revenue per worker, posted wages) introduce measurement error. Structural estimation would require instruments for productivity or a control function approach to handle endogeneity.

Dynamic Considerations: The static model abstracts from cluster creation/destruction dynamics and worker mobility costs. A dynamic extension would enable estimation of adjustment costs, learning parameters, and transition rates, but would require panel data on workers and firms—currently unavailable in the Burning Glass dataset.

Despite these limitations, the reduced-form tests provide compelling evidence for the model’s core mechanisms: discrete clustering, market thickness effects, and cluster-level heterogeneity in automation responses. Future structural work can build on this foundation once richer data (linked employer-employee records with wage and productivity measures) become available.

C.6.3 Alternative Mechanisms

The model abstracts from several potentially important mechanisms:

Search Frictions: Workers may face costs to discover job opportunities or learn about match quality. Introducing search costs would generate persistence in cluster membership and gradual adjustment rather than instantaneous reallocation.

Human Capital Specificity: Workers may accumulate cluster-specific skills that create switching costs between job types (Poletaev & Robinson, 2008). This would reduce mobility across clusters within occupations, amplifying employment volatility from demand shifts. Empirical investigation would require panel data tracking individual workers.

Institutional Factors: Occupational licensing, union coverage, and employment protections may constrain firms’ ability to reorganise job designs. Cross-country or cross-state variation in institutions provides testable predictions (stronger restructuring in less-regulated markets).

Product Market Competition: Firm heterogeneity θ is exogenous in the model but endogenous in models with selection (Melitz, 2003). Automation-induced productivity shocks may trigger competitive reallocation across firms, amplifying cluster-level employment volatility beyond direct displacement effects.

These extensions remain for future theoretical development but do not undermine core predictions about within-occupation clustering.

C.7 Discussion and Limitations

C.7.1 Relationship to Existing Theory

This chapter’s theoretical contribution lies at the intersection of four scholarly traditions that Chapter 2 identified as independently predicting within-occupation restructuring:

Organisational Behaviour and Job Design: The model's treatment of job design d as a firm choice variable operationalises insights from the job characteristics tradition (Hackman & Oldham, 1976; Morgeson & Humphrey, 2006). Parker and Grote (2022) argue that automation's effects on job quality depend on organisational choices about job redesign rather than technology alone—the same automation can enhance or degrade jobs depending on implementation decisions. This chapter formalises that insight: firms with different productivity levels θ optimally choose different job designs $d^*(\theta)$, generating within-occupation heterogeneity as an equilibrium outcome. The strategic HRM literature's finding that firms bundle tasks strategically to achieve competitive objectives (Huselid, 1995; Subramony, 2009) motivates the model's assumption that coordination costs depend on task complexity (the span function), creating trade-offs between specialisation benefits and coordination burdens.

Labour Economics and Task-Based Frameworks: Following Acemoglu and Autor (2011a), the model treats tasks as the fundamental unit of analysis, with occupational outcomes emerging from task-level dynamics. The task framework predicts that automation substitutes for routine tasks while complementing non-routine tasks—but implements this empirically through occupation-level aggregation. This chapter extends task-based reasoning to the sub-occupational level, showing how task substitution manifests as cluster-level restructuring within occupational boundaries. The model's Proposition 3 (automation effects) formalises Acemoglu and Restrepo (2019a)'s distinction between displacement and reinstatement: routine-intensive clusters contract (displacement) while complementary clusters expand (reinstatement), with net occupation-level effects depending on the balance between these forces.

Occupational Sociology: Abbott (1988) demonstrates that occupational boundaries are contested jurisdictions continuously renegotiated through technological change and professional competition. The model takes occupational boundaries as given but shows that substantial heterogeneity emerges endogenously within those boundaries—consistent with Weeden (2002)'s finding that closure mechanisms explain only 15–20% of inter-occupational earnings variation. Weeden et al. (2007) show that most earnings inequality operates *within* rather than between occupations, a pattern the model explains through assortative matching between heterogeneous firms and workers generating wage dispersion within occupational categories. The sociology literature's emphasis on boundary fluidity motivates examining dynamics at finer granularity than traditional occupational analysis.

Occupational Choice and Assignment Models: Following Roy (1951), the model treats workers as selecting among job opportunities based on comparative advantage. Standard occupational choice models assume workers choose among discrete, homogeneous occupations; this chapter endogenises the “occupation” as containing heterogeneous job types,

making within-occupation structure itself an equilibrium outcome. Costinot and Vogel (2010) and Eeckhout and Kircher (2011) study sorting with multi-dimensional heterogeneity but typically assume exogenous job characteristics. Recent work by Tan (2024) applies multidimensional matching to analyse labour market polarisation *across occupations*. This chapter extends these approaches to the *within-occupation* level, endogenising job design choice to explain why individual occupations fragment into heterogeneous clusters.

Coordination Games and Market Structure: The posting cost microfoundation builds on coordination game theory (Ellison, 1993b; Morris & Shin, 2003) to explain how fixed posting costs and thick market externalities generate discrete bunching at equilibrium job designs. This extends directed search models (Moen, 1997; Shimer, 2005) by incorporating strategic complementarities in job posting decisions: firms coordinate on discrete designs because concentration creates thick markets with longer worker queues, raising match probabilities.

The synthesis formalises the convergent prediction identified in Chapter 2: within-occupation clustering emerges from standard production complementarities when extending economic frameworks to incorporate insights from organisational behaviour about job design as strategic choice and from sociology about boundary fluidity within occupational categories.

C.7.2 Model Limitations

Comparative Statics: The baseline model (Propositions 1–3) characterises equilibrium at a point in time, with the dynamic extension (Propositions 4–5) examining how clusters evolve as automation intensity changes. However, the model abstracts from continuous-time adjustment processes, individual worker transition paths between clusters, and explicit skill acquisition dynamics. Chapter 6 examines these dynamics empirically.

Worker Mobility and Training: The static framework abstracts from skill acquisition and job search dynamics within clusters. While these extensions are not formally derived from the equilibrium model, they operationalise the model’s core prediction that workers sort into clusters based on skill-job match quality, using cosine similarity as a tractable measure of the theoretical match quality function $A(s, d)$. The simulation’s accessibility threshold (requiring minimum skill overlap for job access) provides a discrete approximation to the continuous match quality function, and skill acquisition through targeted training represents a dynamic extension consistent with the model’s static comparative advantage logic. These operationalisations allow examination of policy-relevant questions about training effectiveness and job access that the static model cannot address directly.

Partial Equilibrium: The model takes occupational boundaries as given rather than de-

iving them endogenously. If automation shifts demand across occupations (not just within them), the model misses cross-occupational reallocation effects. A general equilibrium extension would endogenise which collections of job types constitute “occupations” and allow automation to reshape occupational boundaries themselves. However, the empirical findings in Chapter 8—where 73.9% of network edges are cross-occupational—suggest that skill proximity cuts across occupational boundaries, partially validating the concern that fixed boundaries miss important dynamics. The model’s predictions about *within*-occupation clustering remain valid regardless, since they concern equilibrium properties conditional on a given occupational boundary.

Perfect Information: Workers and firms observe all match qualities costlessly. Introducing information frictions (workers learn about fit through experience; firms screen applicants) would generate richer predictions about hiring, turnover, and wage dynamics.

No Adjustment Costs: Workers can costlessly switch between clusters. In practice, cluster-specific human capital and search frictions create switching costs, amplifying employment volatility from demand shifts. This is a promising direction for future work.

C.7.3 Perspectives from Adjacent Traditions

The model developed in this chapter formalises firm-side equilibrium mechanisms: how heterogeneous firms choose job designs, how workers sort into clusters through assortative matching, and how automation shifts cluster boundaries. This economic framework captures important forces shaping within-occupation structure. However, the literature review in Chapter 2 identified additional perspectives from organisational behaviour, sociology, and institutional economics that complement the model’s predictions without requiring mathematical extension.

Job Crafting and Worker Agency: The model operates at the job posting and hiring stage: firms design jobs, post vacancies, and workers sort into positions based on match quality. This scope is appropriate for analysing job advertisement data, which captures firm demand at the point of hiring. However, the job crafting literature (Berg et al., 2010; Wrzesniewski & Dutton, 2001) documents that workers actively reshape their jobs *after* being hired, changing task, relational, and cognitive boundaries through experience in the role. An accountant hired for compliance work may gradually take on advisory responsibilities; a software developer may craft their role toward architecture and mentoring. This post-hire evolution creates within-occupation heterogeneity the model does not capture. Importantly, job advertisement data also cannot observe this post-hire crafting—advertisements measure what firms *demand* when posting vacancies, not what workers ultimately *do* in those roles. The model’s predictions concern hiring-stage sorting; a complementary analysis of post-

hire task evolution would require different data (e.g., longitudinal worker surveys, time-use data) and a dynamic model incorporating tenure, learning, and renegotiation. Berg et al. (2010) demonstrate that crafting capacity varies with organisational rank, suggesting that workers in routine clusters most exposed to automation may have the least capacity to craft alternative responsibilities—a distributional concern the model’s cluster-level predictions do not address.

Job Demands and Resources: The model’s K -dimensional skill space can accommodate the Job Demands-Resources framework’s distinction between demands and resources (Bakker & Demerouti, 2007, 2017). Some skill dimensions represent demands (task complexity, time pressure, monitoring requirements) while others represent resources (autonomy, skill utilisation, social support). Automation may reduce routine demands while creating new monitoring demands, and provide decision-support resources while eliminating skill-utilisation resources. This asymmetry operates within the model’s structure: different automation effects on different skill dimensions ($\partial\mu_k/\partial\rho$ varying by k) generate heterogeneous cluster-level impacts. Two clusters within the same occupation may experience automation as resource-enhancing or demand-intensifying depending on their skill profiles. The JD-R insight enriches interpretation of the model’s predictions—cluster contraction may reflect demand intensification, cluster expansion may reflect resource enhancement—without requiring mathematical extension.

Social Construction of Task Categories: Labour process theory argues that “routine” and “skilled” are not objective categories but social constructions reflecting power relations (Attewell, 1990; Braverman, 1974). Workers and occupational groups strategically define their work as “skilled” to resist automation and maintain bargaining power. The model predicts that automation induces *some* partition $\mathcal{P}(\tau) = \{\mathcal{S}^-(\tau), \mathcal{S}^+(\tau), \mathcal{S}^0(\tau)\}$ of substituted, complemented, and neutral skills, but leaves the specific partition to be discovered empirically rather than pre-specified. This is an analytical choice, not a claim that task categories are objectively determined. The sociology of professions reminds us that boundaries between “automatable” and “automation-resistant” tasks are contested terrain—what counts as routine depends partly on who has power to define it. Empirical tests in subsequent chapters use data-driven clustering (HDBSCAN on job advertisement embeddings) rather than *a priori* task classifications, partially addressing this concern by allowing cluster structure to emerge from employer demand patterns rather than researcher-imposed categories.

Cross-Occupational Dynamics: Platform work and the gig economy decompose occupations into discrete tasks, enabling recombination across traditional boundaries (Vallas & Schor, 2020). The model takes occupation boundaries as given (partial equilibrium) rather than deriving them endogenously. This scope limitation means the model cannot explain cross-occupational restructuring—only within-occupation dynamics. However,

the forces generating within-occupation clustering (firm heterogeneity, coordination costs, assortative matching) would also operate across occupation boundaries if those boundaries were relaxed. Platform work may intensify within-occupation heterogeneity by enabling firms to post more specialised task bundles, making the model's predictions more rather than less relevant to contemporary labour markets.

Australian Institutional Context: The model assumes wages adjust freely to clear markets. Australia's award wage system constrains wage adjustment, establishing minimum rates by occupation and industry (Borland & Coelli, 2017). When wages cannot fully adjust downward, automation effects may manifest through quantity adjustment—employment changes rather than wage changes. This institutional constraint could amplify the cluster-level employment volatility the model predicts. If routine clusters cannot reduce wages to retain workers, employment contraction may be sharper than in flexible-wage labour markets. Conversely, complementary clusters may expand employment rather than bid up wages. Australian empirical tests may find stronger cluster-level employment effects than US-based predictions would suggest, not because the underlying mechanism differs but because institutional constraints channel adjustment into the quantities the model predicts rather than prices it does not.

These perspectives qualify interpretation without invalidating the model's core predictions. The model explains why discrete clusters emerge and how automation reallocates employment across them. Adjacent traditions remind us that task categories are contested, worker agency shapes actual work beyond what advertisements capture, job characteristics affect worker wellbeing through demand-resource balance, and institutions mediate adjustment channels. A complete account would integrate all these perspectives; the present model formalises economic mechanisms generating within-occupation clustering while acknowledging that organisational, sociological, and institutional factors shape how those mechanisms operate in specific contexts.

C.7.4 Welfare and Efficiency

In directed-search models with posting costs, the equilibrium allocation is generally not first-best due to two sources of inefficiency: (1) *search externalities*—firms do not internalise the effect of their posting decisions on worker queue lengths and match probabilities, and (2) *coordination externalities*—when multiple firms post the same design, they generate thick-market benefits for other firms but bear posting costs individually.

Following Hosios (1990), constrained efficiency requires that the elasticity of the matching function equals firms' bargaining power. In the present model with directed search and free entry, this translates to a condition on posting costs and market thickness. Specifically,

define the queue elasticity $\eta(q) \equiv q \cdot h'(q)/h(q)$, measuring how hiring probability responds to queue length. The equilibrium cluster configuration is constrained efficient if and only if $\eta(q^*) = f/\pi(d^*, \theta^*)$ for all active clusters, where q^* is the equilibrium queue length, θ^* is the marginal productivity type, and π is operating profit net of coordination costs. When this condition holds, the private return to posting a new design equals the social return. Deviations generate inefficiency: $\eta(q^*) > f/\pi$ implies too few clusters (excessive coordination on existing designs), while $\eta(q^*) < f/\pi$ implies too many (excessive specialisation).

In the empirical application (Chapter 6), the Hosios condition cannot be directly tested without observing posting costs f and match probabilities $h(q)$. However, the theory predicts that markets with stronger coordination incentives should exhibit fewer clusters than efficient, while markets with lower posting costs (e.g., online job platforms reducing recruitment expenses) should support more clusters, potentially approaching efficiency.

Even when the Hosios condition holds on average, coordination failures can arise if firms cannot coordinate on which designs to post. The posting game exhibits strategic complementarity (firms prefer designs where other firms already post), potentially generating multiple equilibria with different cluster configurations. This multiplicity may explain cross-market variation in cluster structure even among markets of similar size.

C.8 Conclusion

This chapter developed a formal model explaining why occupations fragment into sub-occupational clusters. The key mechanism is interaction between firm heterogeneity in coordination ability and multi-dimensional skill complementarities in production. High-productivity firms optimally create specialised job designs despite coordination costs, while low-productivity firms create generalist positions. Workers sort into discrete clusters through assortative matching.

The model generates three testable predictions: (1) discrete clusters exist within occupations, (2) cluster counts increase with market thickness, and (3) automation generates high cluster-level employment volatility alongside low occupation-level effects. These predictions organise empirical investigation in Chapter 6, where unsupervised machine learning identifies the cluster structure predicted theoretically.

Chapter 6 tests all propositions by applying HDBSCAN clustering to Burning Glass job advertisements, examining whether discrete sub-occupational groups emerge, how cluster counts vary with market characteristics, and analysing cluster-level employment dynamics over time. The chapter documents extreme heterogeneity that aggregates to modest occupation-level effects. Together, theory and evidence support the central thesis: automation's primary labour market impact operates through continuous within-occupation

restructuring rather than wholesale occupation elimination.

Appendix D

Theory Chapter: Technical Proofs

This appendix provides detailed technical proofs and supplementary analytical material for Chapter 3. Section D.1 proves existence of directed search equilibrium via Kakutani's Fixed Point Theorem. Section D.2 proves existence of Nash equilibrium in the posting game. Section D.3 contains proofs of Propositions 1–5 and Corollary 5.1, covering endogenous clustering, market thickness effects, aggregation bias, and automation dynamics. Section D.4 presents a numerical calibration illustrating the model's comparative statics predictions.

D.1 Kakutani Existence Proof: Directed Search Equilibrium

This section provides formal verification of the three conditions required for applying Kakutani's Fixed Point Theorem to establish existence of directed search equilibrium (Definition 1, Chapter 3).

D.1.1 Setup and Correspondence Definition

Define the correspondence $\Gamma : \mathcal{U}_L \rightrightarrows \mathcal{U}_L$ mapping market utilities to the set of possible equilibrium utilities: given $U \in \mathcal{U}_L$, firms choose job designs from the optimal correspondence $d^*(\theta; U) \equiv \arg \max_d \pi(\theta, d; U)$ (which may be multi-valued when multiple designs

yield equal profits), which induces queue lengths $q(d; U)$ via market clearing, which determines wages $w(s, d; U)$ via zero-profit conditions, which implies the set of possible worker utilities $\Gamma(U) \subseteq \mathcal{U}_L$ via the worker optimization problem (the Lipschitz property is preserved since $U(s) = \max_d p(q(d)) \cdot w(s, d)$ where wages are L_w -Lipschitz). A fixed point $U^* \in \Gamma(U^*)$ constitutes equilibrium.

D.1.2 Condition 1: Non-empty, Compact, Convex Domain

Claim: The Lipschitz ball $\mathcal{U}_L = \{U : S \rightarrow [0, \bar{U}] : |U(s) - U(s')| \leq L_w \|s - s'\| \forall s, s' \in S\}$ is non-empty, compact in the supremum norm, and convex.

Proof:

- **Non-emptiness:** The constant function $U(s) = 0$ for all $s \in S$ satisfies the Lipschitz condition with any constant, so $U \in \mathcal{U}_L$ and $\mathcal{U}_L \neq \emptyset$.
- **Convexity:** For any $U_1, U_2 \in \mathcal{U}_L$ and $\alpha \in [0, 1]$, the convex combination $U_\alpha(s) = \alpha U_1(s) + (1 - \alpha)U_2(s)$ satisfies:

$$|U_\alpha(s) - U_\alpha(s')| = |\alpha(U_1(s) - U_1(s')) + (1 - \alpha)(U_2(s) - U_2(s'))| \leq \alpha L_w \|s - s'\| + (1 - \alpha)L_w \|s - s'\| = L_w \|s - s'\|$$

and $U_\alpha : S \rightarrow [0, \bar{U}]$ since $0 \leq U_1(s), U_2(s) \leq \bar{U}$ for all s . Therefore $U_\alpha \in \mathcal{U}_L$ and $\mathcal{U}_L$ is convex.

- **Compactness:** By the Arzelà-Ascoli Theorem (Aliprantis & Border, 2006), $\mathcal{U}_L$ is compact in the supremum norm because:

1. *Uniformly bounded:* By construction, $|U(s)| \leq \bar{U}$ for all $U \in \mathcal{U}_L$ and all $s \in S$.
2. *Uniformly equicontinuous:* All functions in $\mathcal{U}_L$ satisfy $|U(s) - U(s')| \leq L_w \|s - s'\|$ by definition, establishing uniform equicontinuity with modulus $\delta = \varepsilon/L_w$.

Since S is compact (Assumption R1) and $\mathcal{U}_L$ is uniformly bounded and equicontinuous, Arzelà-Ascoli implies $\mathcal{U}_L$ is relatively compact. Since $\mathcal{U}_L$ is also closed in the supremum norm (limits of Lipschitz functions with constant L_w remain Lipschitz with the same constant), $\mathcal{U}_L$ is compact. ■

D.1.3 Condition 2: Non-empty, Convex Values

Claim: For each $U \in \mathcal{U}_L$, the image $\Gamma(U) \subseteq \mathcal{U}_L$ is non-empty and convex.

Proof:

- **Non-emptiness:** Given $U \in \mathcal{U}_L$, firms choose optimal job designs $d^*(\theta; U)$ from the correspondence $\arg \max_{d \in \mathcal{D}} \pi(\theta, d; U)$. Since $\mathcal{D}$ is compact (unit sphere in $\mathbb{R}^K$) and π is continuous in d (by Assumptions R1–R5), the Maximum Theorem guarantees that $d^*(\theta; U)$ is non-empty for all θ . These choices induce queue lengths via market clearing, which determine wages via zero-profit conditions, which yield worker utilities. Therefore $\Gamma(U) \neq \emptyset$.
- **Convexity:** Suppose $U'_1, U'_2 \in \Gamma(U)$ are two possible equilibrium utilities arising from different selections from the optimal design correspondence $d^*(\theta; U)$. That is, there exist selections $d_1^*(\theta)$ and $d_2^*(\theta)$ from $d^*(\theta; U)$ such that when firms post $(d_1^*(\theta))_{\theta \in \Theta}$ the resulting worker utilities are U'_1 , and when firms post $(d_2^*(\theta))_{\theta \in \Theta}$ the resulting utilities are U'_2 .

For any $\alpha \in [0, 1]$, consider a mixed strategy where each firm type θ randomizes: with probability α they post according to $d_1^*(\theta)$, and with probability $1 - \alpha$ they post according to $d_2^*(\theta)$. Since both selections are optimal given U (both yield maximum profit), randomizing between them preserves optimality (expected profit equals the maximum).

This randomization induces:

- Queue lengths $q_\alpha(d)$ that are convex combinations of $q_1(d)$ and $q_2(d)$ (by linearity of market clearing)
- Wages $w_\alpha(s, d)$ that are convex combinations (by linearity of zero-profit conditions)
- Worker utilities $U'_\alpha(s) = \alpha U'_1(s) + (1 - \alpha) U'_2(s)$ (by linearity of worker optimization)

Therefore $U'_\alpha = \alpha U'_1 + (1 - \alpha) U'_2 \in \Gamma(U)$, establishing convexity. ■

D.1.4 Condition 3: Upper Hemicontinuity

Claim: The correspondence $\Gamma : \mathcal{U} \rightrightarrows \mathcal{U}$ is upper hemicontinuous.

Proof: To verify upper hemicontinuity of Γ , I show that the graph $\{(U, U') : U' \in \Gamma(U)\}$ is closed in $\mathcal{U} \times \mathcal{U}$. Equivalently, if $U_n \rightarrow U$ and $U'_n \in \Gamma(U_n)$ with $U'_n \rightarrow U'$, then $U' \in \Gamma(U)$.

The correspondence Γ is defined through the composition:

$$U \mapsto d^*(\theta; U) \mapsto q(d; U) \mapsto w(s, d; U) \mapsto \Gamma(U)$$

Upper hemicontinuity is verified at each step:

Step 1: $d^*(\theta; U)$ is u.h.c. in U . The profit function

$$\pi(\theta, d; U) = h(q(d)) \cdot \int_S [Y(s, d, \theta) - U(s)] \phi(s|d) ds - C(d, \theta)$$

is continuous in (d, U) by Assumptions R1–R5 (continuity of h, Y, C , and linear dependence on U). The choice set $\mathcal{D}$ is compact. By the Maximum Theorem (Berge, 1963), the argmax correspondence

$$d^*(\theta; U) = \arg \max_{d \in \mathcal{D}} \pi(\theta, d; U)$$

is upper hemicontinuous in U with non-empty compact values.

Step 2: Queue lengths $q(d; U)$ are continuous. Market clearing determines queue lengths:

$$\int_{\Theta} \mathbf{1}\{d^*(\theta; U) = d\} g(\theta) d\theta = q(d) \cdot \int_S \mathbf{1}\{d \in \arg \max U(s)\} f(s) ds$$

Since $d^*(\theta; U)$ is u.h.c., the measure of firms posting d varies continuously with U . By measurable selection theorem and continuity of integration under u.h.c. correspondences (Aliprantis & Border, 2006, Theorem 17.31), $q(d; U)$ is continuous in U .

Step 3: Wages $w(s, d; U)$ are continuous. The zero-profit condition

$$h(q(d)) \cdot [Y(s, d, \theta) - w(s, d)] = C(d, \theta)$$

defines $w(s, d; U)$ implicitly as:

$$w(s, d; U) = Y(s, d, \theta(d)) - \frac{C(d, \theta(d))}{h(q(d; U))}$$

Since $q(d; U)$ is continuous in U (Step 2) and $h(\cdot)$ is continuous (Assumption R5), $w(s, d; U)$ is continuous in U .

Step 4: Worker utilities $\Gamma(U)$ are u.h.c. Worker utility is

$$U'(s) = \max_{d \in \mathcal{D}} \{p(q(d; U)) \cdot w(s, d; U)\}$$

Since $w(s, d; U)$ is continuous in U (Step 3) and $\mathcal{D}$ is compact, the max operator preserves upper hemicontinuity by the Maximum Theorem. Therefore Γ is upper hemicontinuous. ■

D.1.5 Conclusion

Under Assumptions R1–R5, all three Kakutani conditions are satisfied. By Kakutani's Fixed Point Theorem (Kakutani, 1941), there exists $U^* \in \Gamma(U^*)$, establishing existence of directed search equilibrium.

D.2 Posting Game Nash Equilibrium Existence

This section provides the complete proof of Nash equilibrium existence in the posting game with strategic complementarities (Chapter 3, Section 5.3.4).

D.2.1 Game Structure

The posting game $\Gamma = \langle I, \{S_i\}, \{\Pi_i\} \rangle$ has:

- **Players:** Continuum of firms $I = [\underline{\theta}, \bar{\theta}]$ with productivity types $\theta \sim G(\theta)$
- **Strategies:** Each firm i chooses finite set of designs $S_i \subseteq \mathcal{D}$ to post
- **Payoffs:** Firm i with type θ_i posting designs $S_i = \{d_1, \dots, d_m\}$ earns:

$$\Pi_i(S_i; S_{-i}, \theta_i) = \sum_{d_j \in S_i} \pi(d_j, \theta_i, q(d_j; S)) - |S_i| \cdot f$$

where $\pi(d, \theta, q) = h(q) \cdot \mathbb{E}_s[Y(s, d, \theta) - w(s, d)|d] - C(d, \theta)$ is operating profit per design, $q(d; S)$ is endogenous queue length, and $|S_i|$ is cardinality of posted designs.

D.2.2 Kakutani Fixed Point Theorem Application

Theorem: A Nash equilibrium in posting strategies exists.

Proof: Apply Kakutani's Fixed Point Theorem to the aggregate best-response correspondence.

Step 1: Define strategy space topology

Define the space of posting profiles:

$$\mathcal{S} = \{S = (S_i)_{i \in I} : S_i \subseteq \mathcal{D} \text{ finite for all } i\}$$

Each posting profile S induces a finite atomic measure μ_S over $\mathcal{D}$. Endow $\mathcal{S}$ with the weak-* topology inherited from the embedding into $\mathcal{M}(\mathcal{D})$, the space of Borel probability

measures on the compact metric space $\mathcal{D}$ (with the product topology from $\mathbb{R}^K$). Since $\mathcal{D}$ is compact, $\mathcal{M}(\mathcal{D})$ is compact and metrizable by the Prokhorov metric (Aliprantis & Border, 2006), making $\mathcal{S}$ a compact subset.

Step 2: Verify compactness of utility space

Let $\mathcal{U} = \{\pi(\cdot, \theta, S) : \theta \in \Theta, S \in \mathcal{S}\}$ denote the family of profit functions. Since production function $A(s, d)$ and cost function $C(d, \theta)$ are both Lipschitz continuous with Lipschitz constants bounded uniformly over compact domains $\mathcal{S}$ and Θ (exponential Gaussian and power functions on compact sets), $\mathcal{U}$ is a uniformly equicontinuous family. By the Arzelà-Ascoli theorem (Aliprantis & Border, 2006), $\mathcal{U}$ is relatively compact in the topology of uniform convergence on compact sets. This establishes compactness of the utility space required for existence of fixed point.

Step 3: Define aggregate best-response

$$B(S) = \prod_{i \in I} B_i(S_{-i}, \theta_i)$$

where $B_i(S_{-i}, \theta_i) = \arg \max_{S_i \subseteq \mathcal{D}} \Pi_i(S_i; S_{-i}, \theta_i)$.

Step 4: Verify Kakutani conditions

(i) Compact, Convex, Non-empty Domain:

- *Compactness*: $\mathcal{S}$ can be embedded in the compact space of probability measures over $\mathcal{D}$ (firms post finite designs, so posting measure is atomic).
- *Convexity*: Achieved via mixed strategies. Firms can randomize over posting sets with same expected payoff.
- *Non-emptiness*: The empty posting $S_i = \emptyset$ is always feasible (yields zero profit).

(ii) Non-empty, Convex Values:

- *Non-emptiness*: For each S , $B(S) \neq \emptyset$ because at minimum, posting zero designs ($S_i = \emptyset$) is feasible (yields zero profit, which is bounded).
- *Convexity*: To verify $B_i(S_{-i}, \theta_i)$ is convex-valued, let $S'_i, S''_i \in B_i(S_{-i}, \theta_i)$ be two best responses. For any $\lambda \in [0, 1]$, firm i can randomize between posting sets S'_i and S''_i with probabilities $(\lambda, 1 - \lambda)$, which induces posting measure

$$\mu_i = \lambda \mu_{S'_i} + (1 - \lambda) \mu_{S''_i}$$

By linearity of expected profits in posting measures, this randomization yields expected profit equal to

$$\lambda \Pi_i(S'_i; S_{-i}, \theta_i) + (1 - \lambda) \Pi_i(S''_i; S_{-i}, \theta_i)$$

which equals the maximum since both S'_i and S''_i are optimal. Therefore, the convex combination $\lambda S'_i + (1 - \lambda)S''_i \in B_i(S_{-i}, \theta_i)$, establishing convexity.

(iii) Upper Hemicontinuity: The correspondence B is upper hemicontinuous because:

1. Queue function $q(d; S)$ is continuous in posting profile S by market clearing. Formally, the aggregate demand mapping $d^*(\theta; U) \mapsto q(d; U)$ integrates over the upper hemicontinuous correspondence $d^*(\theta; U)$, which preserves continuity by Aliprantis and Border (2006, Theorem 17.31) (continuity of integrals under measurable selection from u.h.c. correspondences).
2. Hiring probability $h(q)$ is continuous in q by Assumption R5.
3. Profit function $\pi(d, \theta, q)$ is continuous in (d, q) (production and cost functions are continuous by Assumptions R1–R4).
4. By the Maximum Theorem (Berge, 1963), the argmax correspondence $B_i(S_{-i}, \theta_i)$ is upper hemicontinuous when the objective is continuous and the choice set is compact.

Step 5: Apply Kakutani's theorem

By Kakutani's Fixed Point Theorem (Kakutani, 1941), there exists $S^* \in B(S^*)$, which is a Nash equilibrium. ■

D.2.3 Economic Interpretation

The Nash equilibrium features coordination on discrete designs because:

1. **Thick market externality:** Profit $\pi(d, \theta, q)$ is increasing in queue length q via hiring probability $h(q)$. Firms benefit from posting designs where many other firms also post.
2. **Posting cost penalty:** Fixed cost $f > 0$ penalizes proliferation. Firms prefer posting few designs at thick markets rather than many designs at thin markets.
3. **Positive assortative matching:** Different productivity types θ prefer different designs (Proposition 1), creating natural segmentation into discrete clusters.

The equilibrium cluster count n^* is determined by the zero-profit condition at the marginal design, with comparative statics derived via the Implicit Function Theorem (Chapter 3).

D.3 Proofs of Theoretical Results

D.3.1 Proof of Proposition 1: Endogenous Clustering

Proposition 1: Under Assumptions 1-3, equilibrium exhibits positive assortative matching with discontinuous jumps in job design function $d^*(\theta)$, generating discrete clusters of job types.

Proof: The proof proceeds in four steps.

Step 1: Supermodularity implies monotone sorting

I verify the conditions for applying Topkis's Theorem (1998) to establish monotone comparative statics.

Step 1a (Lattice structure): The job design space $\mathcal{D} \subseteq \mathbb{R}^K$ with product order is a complete lattice. For any $d, d' \in \mathcal{D}$, define:

$$d \vee d' = (\max\{d_1, d'_1\}, \dots, \max\{d_K, d'_K\}) \quad (\text{component-wise supremum}) \quad (\text{D.1})$$

$$d \wedge d' = (\min\{d_1, d'_1\}, \dots, \min\{d_K, d'_K\}) \quad (\text{component-wise infimum}) \quad (\text{D.2})$$

Since $\mathcal{D}$ is a rectangle (cross-product of intervals), it is closed under $\vee$ and $\wedge$, hence a sublattice of $\mathbb{R}^K$.

Step 1b (Supermodularity verification): From the profit function (equation C.12):

$$\pi(\theta, d) = h(q(d)) \cdot \int_S [\theta \cdot A(s, d) - U(s)] \phi(s|d) ds - C(d, \theta) \quad (\text{D.3})$$

For expositional clarity, I simplify to the deterministic hiring case ($h(q(d)) = 1$) to establish the key mechanism.¹ Taking the cross-partial derivative with respect to θ and job design component d_k :

$$\frac{\partial^2 \pi}{\partial \theta \partial d_k} = \int_S \frac{\partial A(s, d)}{\partial d_k} \phi(s|d) ds + \alpha \frac{\kappa}{\theta^{\alpha+1}} \cdot \gamma \cdot [\text{span}(d)]^{\gamma-1} \cdot \frac{\partial \text{span}(d)}{\partial d_k} \quad (\text{D.4})$$

¹This simplification is standard in directed search models for characterising sorting patterns (see Shimer and Smith (2000), Eeckhout and Kircher (2011)). With $h(q) < 1$, the cross-partial $\frac{\partial^2 \pi}{\partial \theta \partial d}$ includes an additional term $\frac{\partial h(q(d))}{\partial d}$ reflecting queue-length adjustments, whose sign depends on how queue length responds to design complexity. A sufficient condition for the supermodularity result to extend is that the matching function $h(q)$ satisfies standard monotonicity conditions ($h'(q) > 0$, $h''(q) < 0$) and the queue-length response $\partial q / \partial x$ is bounded such that the production-plus-cost terms continue to dominate. This sufficient condition holds in the equilibrium configurations characterised by Lemma A.1 (existence) but fails when queue lengths respond strongly to small design changes—an empirical question the present descriptive analysis cannot adjudicate. The qualitative sorting pattern (high- θ firms choose high-complexity designs) is preserved whenever the cost-channel dominance holds, as documented in the calibration of Section D.4.

The match quality function $A(s, d) = \exp[-\lambda \|s - \mu(d)\|^2]$ satisfies:

$$\frac{\partial A}{\partial d_k} = -2\lambda \cdot A(s, d) \cdot (s_k - \mu_k(d)) \cdot \frac{\partial \mu_k}{\partial d_k} \quad (\text{D.5})$$

To verify Assumption 1 ($\frac{\partial^2 \pi}{\partial \theta \partial d_k} > 0$), decompose the cross-partial into production and cost effects:

$$\frac{\partial^2 \pi}{\partial \theta \partial d_k} = \underbrace{\int_S \frac{\partial A(s, d)}{\partial d_k} \phi(s|d) ds}_{\text{Production effect}} + \underbrace{\alpha \frac{\kappa}{\theta^{\alpha+1}} \cdot \gamma \cdot [\text{span}(d)]^{\gamma-1} \cdot \frac{\partial \text{span}(d)}{\partial d_k}}_{\text{Cost effect}} \quad (\text{D.6})$$

The **cost effect** is always positive when $\alpha > 0$ (Assumption R3): high- θ firms face lower marginal coordination costs ($\partial C / \partial \theta < 0$), so they benefit more from increasing task complexity.

The **production effect** is positive when the ideal skill profile $\mu(d)$ is increasing in job complexity. Under the Gaussian match quality function, this requires $\frac{\partial \mu_k}{\partial d_k} > 0$: more specialised job designs demand higher skill levels in their focal dimensions.

Sufficient condition for supermodularity: When $\alpha > 0$ and the production technology satisfies $\frac{\partial \mu_k}{\partial d_k} > 0$, both terms are positive, ensuring $\frac{\partial^2 \pi}{\partial \theta \partial d_k} > 0$. Alternatively, if one term is negative, supermodularity holds when the positive term dominates in magnitude: $\alpha \kappa \gamma [\text{span}(d)]^{\gamma-1} \theta^{-(\alpha+1)} \left| \frac{\partial \text{span}}{\partial d_k} \right| > \left| \int_S \frac{\partial A}{\partial d_k} \phi(s|d) ds \right|$ for all $(d, \theta) \in \mathcal{D} \times \Theta$.

Step 1c (Single-crossing property): I verify the single-crossing property: for any $\theta' > \theta$ and $d' > d$ (in the lattice order),

$$\pi(\theta, d') - \pi(\theta, d) \leq 0 \implies \pi(\theta', d') - \pi(\theta', d) \geq 0 \quad (\text{D.7})$$

This follows from supermodularity: the incremental benefit of switching from d to d' is:

$$\Delta \pi(\theta; d, d') \equiv \pi(\theta, d') - \pi(\theta, d) = (\theta' - \theta) \int_S \frac{\partial A}{\partial d_k} \phi(s|d) ds + (\text{coordination cost terms}) \quad (\text{D.8})$$

Since $\frac{\partial^2 \pi}{\partial \theta \partial d_k} > 0$, I have $\frac{\partial \Delta \pi}{\partial \theta} > 0$, verifying single-crossing.

Step 1d (Topkis Corollary 2.8.1): Given that $\pi(\theta, d)$ is supermodular in (d, θ) and $\mathcal{D}$ is a lattice, Topkis's Theorem implies that the optimal job design correspondence:

$$d^*(\theta) = \arg \max_{d \in \mathcal{D}} \pi(\theta, d) \quad (\text{D.9})$$

is weakly increasing in θ (in the strong set order). That is, if $\theta' > \theta$, then $\sup d^*(\theta') \geq \sup d^*(\theta)$ and $\inf d^*(\theta') \geq \inf d^*(\theta)$.

Economic interpretation: High-productivity firms optimally choose more specialised job designs because they can better absorb coordination costs and benefit more from tight skill-job matching.

Step 2: Convexity generates discontinuities

I now show that convexity in coordination costs ($\gamma > 1$) prevents continuous variation in job designs and generates discrete bunching.

Step 2a (No intermediate designs): Suppose by contradiction that there exists an optimal job design $d \in (d_1, d_2)$ for some firm with productivity θ , where d_1 and d_2 are “extreme” designs (e.g., generalist vs. specialist). Consider a convex combination $d_\lambda = \lambda d_2 + (1 - \lambda)d_1$ for $\lambda \in (0, 1)$.

The coordination cost at d_λ is:

$$C(d_\lambda, \theta) = \frac{\kappa}{\theta^\alpha} \cdot [\text{span}(d_\lambda)]^\gamma \quad (\text{D.10})$$

Since span is a convex function (the range of a vector increases convexly with component values) and $\gamma > 1$, the coordination cost satisfies:

$$C(d_\lambda, \theta) > \lambda C(d_2, \theta) + (1 - \lambda)C(d_1, \theta) \quad (\text{D.11})$$

That is, coordination costs are strictly convex in job design. Meanwhile, the match quality benefit $\theta \cdot A(s, d)$ is concave in d (Gaussian form peaks at $s = \mu(d)$ and declines). Combining strict convexity of costs with concavity of benefits implies the profit function is strictly quasi-concave, ruling out interior optima. Therefore, firms choose discrete “corner” solutions $\{d_1, d_2, \dots, d_n\}$ rather than continuous intermediate designs.

Step 2b (Threshold productivity characterisation and derivation): Consider the set of discrete job designs $\mathcal{D}^* = \{d_1, d_2, \dots, d_n\}$ ordered by complexity: $\text{span}(d_1) < \text{span}(d_2) < \dots < \text{span}(d_n)$. I now derive the threshold productivities that partition firms across these designs.

For each adjacent pair (d_j, d_{j+1}) , the threshold productivity $\hat{\theta}_j$ is defined by the indifference condition:

$$\pi(\hat{\theta}_j, d_{j+1}) = \pi(\hat{\theta}_j, d_j) \quad (\text{D.12})$$

Substituting the profit function $\pi(\theta, d) = h(q(d)) \cdot \int_S [\theta \cdot A(s, d) - U(s)] \phi(s|d) ds - C(d, \theta)$ and using the zero-profit condition $h(q(d)) \cdot U(s) = h(q(d)) \cdot w(s, d)$ from competitive search equilibrium:

$$h(q(d_{j+1})) \cdot \hat{\theta}_j \int_S A(s, d_{j+1}) \phi(s|d_{j+1}) ds - C(d_{j+1}, \hat{\theta}_j) = h(q(d_j)) \cdot \hat{\theta}_j \int_S A(s, d_j) \phi(s|d_j) ds - C(d_j, \hat{\theta}_j) \quad (\text{D.13})$$

Under the simplifying assumption of deterministic hiring ($h(q) = 1$) and uniform worker

distribution ($\phi(s|d) = \phi(s)$ independent of d), this simplifies to:

$$\hat{\theta}_j \cdot \int_S [A(s, d_{j+1}) - A(s, d_j)] \phi(s) ds = C(d_{j+1}, \hat{\theta}_j) - C(d_j, \hat{\theta}_j) \quad (\text{D.14})$$

Substituting the coordination cost function $C(d, \theta) = \frac{\kappa}{\theta^\alpha} \cdot \text{span}(d)^\gamma$:

$$\hat{\theta}_j \cdot \int_S [A(s, d_{j+1}) - A(s, d_j)] \phi(s) ds = \frac{\kappa}{\hat{\theta}_j^\alpha} [\text{span}(d_{j+1})^\gamma - \text{span}(d_j)^\gamma] \quad (\text{D.15})$$

Multiplying both sides by $\hat{\theta}_j^\alpha$ and solving for $\hat{\theta}_j$:

$$\hat{\theta}_j^{1+\alpha} = \frac{\kappa [\text{span}(d_{j+1})^\gamma - \text{span}(d_j)^\gamma]}{\int_S [A(s, d_{j+1}) - A(s, d_j)] \phi(s) ds} \quad (\text{D.16})$$

This closed-form characterisation shows that threshold productivities are uniquely determined by: (i) coordination cost parameters (κ, α, γ) , (ii) match quality differences between adjacent designs, and (iii) worker skill distribution $\phi(s)$.

Partition of firm types: By supermodularity (Step 1), firms partition into intervals based on these thresholds: firms with $\theta \in [\hat{\theta}_{j-1}, \hat{\theta}_j)$ strictly prefer job design d_j over all alternatives. The threshold equation (D.16) makes this partition explicit and enables comparative statics analysis of how changes in technology or market conditions shift cluster boundaries.

Step 2c (Discrete bunching from continuous productivity): Given the continuous productivity distribution $G(\theta)$ with density $g(\theta)$, the measure of firms choosing job design d_j is:

$$M_j = \int_{\hat{\theta}_{j-1}}^{\hat{\theta}_j} g(\theta) d\theta = G(\hat{\theta}_j) - G(\hat{\theta}_{j-1}) \quad (\text{D.17})$$

As long as $g(\theta) > 0$ on the support and thresholds are distinct ($\hat{\theta}_j > \hat{\theta}_{j-1}$), each job design attracts a positive measure of firms. This generates discrete bunching in equilibrium: firms concentrate at discrete job designs $\{d_1, d_2, \dots, d_n\}$ with no firms choosing intermediate designs, even though the underlying productivity distribution is continuous.

Economic interpretation: Convexity creates “lumpy” incentives. The coordination cost penalty for deviating from extreme designs dominates any marginal benefit from fine-tuning job complexity. Firms therefore bunch at discrete job types, even though productivity θ varies continuously.

Step 3: Discrete job designs induce worker clustering

From the equilibrium assignment function (Definition 1), workers sort to jobs maximis-

ing their wage:

$$s \in C_j \Leftrightarrow w(s, d_j) \geq w(s, d_{j'}) \quad \forall j' \neq j \quad (\text{D.18})$$

Since firms choose discrete job designs $\{d_1, d_2, \dots, d_n\}$, the worker skill space partitions into discrete regions. Define cluster C_j as:

$$C_j = \left\{ s \in S : d_j \in \arg \max_{d \in \{d_1, \dots, d_n\}} w(s, d) \right\} \quad (\text{D.19})$$

From the match quality function (equation C.5), workers with skills $s \approx \mu(d_j)$ (close to the ideal profile for job design d_j) earn highest wages in jobs of type j . This generates assortative matching: workers cluster around discrete job types rather than spreading uniformly across job design space.

Step 4: Existence

Under Assumption 3 (bounded skills), the support S is compact. Combined with continuity of $F(s)$ (skill distribution) and $G(\theta)$ (productivity distribution), Kakutani's Fixed Point Theorem (applied to the utility correspondence $\Gamma : \mathcal{U} \rightrightarrows \mathcal{U}$ defined in the equilibrium existence proof above) guarantees existence of an equilibrium with finite cluster count $n^* < \infty$.²

Any equilibrium must satisfy monotone sorting (Step 1) and discrete job designs (Step 2), which determines the partition of worker skill space into clusters (Step 3). ■

Remark: The proof formalises the intuition that firm heterogeneity generates within-occupation diversity. Even absent any exogenous occupation structure, firms endogenously create discrete job types. High-productivity firms can “afford” the coordination costs of specialised positions, while low-productivity firms minimise costs through generalist positions. Workers sort into these discrete types based on comparative advantage.

Empirical Implication: HDBSCAN clustering (Chapter 6) should identify discrete clusters within occupations. The number of clusters n^* depends on productivity distribution $G(\theta)$, coordination cost parameters (κ, γ, α) , and skill distribution $F(s)$.

D.3.2 Proof of Proposition 2: Market Thickness and Cluster Count

Proposition 2: The equilibrium number of clusters n^* is weakly increasing in market size N , with $\frac{\partial n^*}{\partial N} > 0$ for markets above threshold $\bar{N}$.

Proof: The proof proceeds in four steps.

²The proof establishes existence via Kakutani's fixed point but does not demonstrate uniqueness. Multiple equilibria may exist if coordination games among firms admit multiple stable configurations. For the empirical application, I test whether an equilibrium with the predicted properties exists, not whether it is unique.

Step 1: Market size affects cluster viability

Let N denote market size (number of workers in occupation). For a cluster to exist in equilibrium, it must be profitable for at least one firm to post vacancies with that job design. Free entry requires:

$$\max_{\theta \in \Theta} \pi(\theta, d_j, N) \geq f \quad (\text{D.20})$$

where f is a fixed cost of posting vacancies and $\pi(\theta, d_j, N)$ now explicitly depends on market size.

Expected profits from job design d_j depend on market thickness through two channels:

Channel 1 (Search frictions): In thicker markets, vacancy posting reaches more workers, reducing time-to-fill and increasing match quality. Model this as reduced search cost $\psi(N)$ with $\psi'(N) < 0$.

Channel 2 (Scale economies): Coordination costs $C(d, \theta)$ exhibit scale economies when amortised across more hires. Effective coordination cost per worker is $C(d, \theta)/n(d, N)$ where $n(d, N)$ is cluster size.

Combining these channels, expected profit becomes:

$$\pi(\theta, d, N) = \theta \cdot A(s, d) - w(s, d) - \frac{C(d, \theta)}{n(d, N)} - \psi(N) \quad (\text{D.21})$$

Step 2: Derive cluster viability threshold using Lemma 2.2

For a cluster centred on job design d_j to be viable, the zero-profit condition must hold:

$$\Pi(\theta_{\min}, d_j, N) \equiv \theta_{\min} \cdot A(s, d_j) - w(s, d_j) - \frac{C(d_j, \theta_{\min})}{n(d_j, N)} - \psi(N) - f = 0 \quad (\text{D.22})$$

This implicitly defines the minimum productivity threshold $\theta_{\min}(d_j, N)$ as a function of market size N .

By Lemma 2.2 (Market Thickness and Entry Thresholds), the Implicit Function Theorem applies under Assumptions R1–R5, establishing:

$$\frac{\partial \theta_{\min}}{\partial N} = -\frac{\partial \Pi / \partial N}{\partial \Pi / \partial \theta} < 0 \quad (\text{D.23})$$

where the negativity follows from $\frac{\partial \Pi}{\partial \theta} > 0$ (higher productivity increases profits) and $\frac{\partial \Pi}{\partial N} < 0$ (larger markets reduce search frictions and per-worker coordination costs via scale economies).

Therefore, larger markets reduce the productivity threshold required for cluster viability, with $\theta_{\min}(d_j, N)$ decreasing in N by equation (C.29).

Step 3: Map viability to cluster count

The equilibrium number of clusters equals the count of job designs d_j for which at least

one firm type $\theta \in \Theta$ finds it profitable to post vacancies:

$$n^*(N) = \# \{d \in \mathcal{D} : \exists \theta \in \Theta \text{ such that } \pi(\theta, d, N) \geq f\} \quad (\text{D.24})$$

From Step 2, $\theta_{\min}(d, N)$ is decreasing in N . As market size increases, more job designs d satisfy $\theta_{\min}(d, N) \leq \bar{\theta}$ (fall within the viable productivity range), expanding the set of viable clusters.

Formally, partition the job design space into potential clusters ordered by specialisation $\text{span}(d)$:

$$\mathcal{D} = \{d_1, d_2, \dots, d_\infty\} \quad \text{with } \text{span}(d_1) < \text{span}(d_2) < \dots \quad (\text{D.25})$$

Define threshold market sizes $\{N_1, N_2, \dots\}$ such that cluster j becomes viable when $N \geq N_j$:

$$N_j = \inf\{N : \theta_{\min}(d_j, N) \leq \bar{\theta}\} \quad (\text{D.26})$$

Then the cluster count function is:

$$n^*(N) = \max\{j : N \geq N_j\} \quad (\text{D.27})$$

Step 4: Comparative statics

The cluster count $n^*(N)$ is a step function, weakly increasing in N :

$$\frac{\partial n^*}{\partial N} = \begin{cases} 0 & \text{if } N \in (N_j, N_{j+1}) \\ +\infty & \text{if } N = N_{j+1} \text{ for some } j \end{cases} \quad (\text{D.28})$$

In expectation (averaging over small perturbations), $\mathbb{E} \left[\frac{\partial n^*}{\partial N} \right] > 0$ for all $N > \bar{N}$ where $\bar{N} = N_2$ (the threshold at which the second cluster becomes viable). ■

Remark: The proof formalises why metropolitan labour markets exhibit finer occupational segmentation than regional markets. New York accountants can specialise in hedge fund taxation, private equity due diligence, or nonprofit compliance—specialisations unsupportable in smaller markets. The mechanism operates through scale economies in coordination investments and reduced search frictions in thick markets.

Empirical Implication: Cross-sectional cluster counts should correlate with market size. This can be tested using:

- Metropolitan vs. regional comparisons (e.g., Sydney vs. Darwin accountants)
- Large vs. small occupation comparisons (e.g., Software Developers vs. Marine Biologists)
- Within-occupation time series (cluster counts should rise as occupations grow)

D.3.3 Proof of Proposition 3: Automation Effects

Proposition 3: For any technology τ with partition $\mathcal{P}(\tau) = \{\mathcal{S}^-(\tau), \mathcal{S}^+(\tau), \mathcal{S}^0(\tau)\}$, skill-biased automation generates (a) contraction of substitution-intensive clusters, (b) expansion of complement-intensive clusters, and (c) aggregation bias where occupation-level effects are small relative to cluster-level volatility.

Proof: The proof proceeds in four steps. *Critically, the proof holds for any partition $\mathcal{P}(\tau)$ —the mechanism is partition-invariant.*

Step 1: Model automation through ideal profile shifts

From Section C.4, automation shifts ideal skill profiles:

$$\mu(d, \rho, \tau) = \mu^0(d) + \rho \cdot \Delta\mu(d, \tau) \quad (\text{D.29})$$

where $\Delta\mu(d, \tau)$ satisfies equations (C.45)–(C.46): $\frac{\partial \mu_k}{\partial \rho} < 0$ for substituted dimensions $k \in \mathcal{S}^-(\tau)$ and $\frac{\partial \mu_k}{\partial \rho} > 0$ for complementary dimensions $k \in \mathcal{S}^+(\tau)$.

The match quality function becomes:

$$A(s, d, \rho, \tau) = \exp \left[-\lambda \|s - \mu(d, \rho, \tau)\|^2 \right] \quad (\text{D.30})$$

For technology τ , partition skill dimensions into substituted ($\mathcal{S}^-(\tau)$), complementary ($\mathcal{S}^+(\tau)$), and neutral ($\mathcal{S}^0(\tau)$) subsets. The ideal profile shifts as:

$$\mu_k(d, \rho, \tau) = \begin{cases} \mu_k^0(d) - \rho \cdot \delta_k(\tau) & \text{for } k \in \mathcal{S}^-(\tau) \\ \mu_k^0(d) + \rho \cdot \delta_k(\tau) & \text{for } k \in \mathcal{S}^+(\tau) \\ \mu_k^0(d) & \text{for } k \in \mathcal{S}^0(\tau) \end{cases} \quad (\text{D.31})$$

where $\delta_k(\tau) > 0$ measure the rate of technology-induced skill requirement shifts. *Note:* The specific partition $\mathcal{P}(\tau)$ varies by technology but the functional form is identical—this is the key to partition-invariance.

Step 2: Derive cluster-level employment effects

Define cluster c as *substitution-intensive* relative to technology τ if the cluster emphasizes substituted skill dimensions:

$$\text{Cluster } c \text{ is substitution-intensive} \Leftrightarrow \mathbb{E} \left[\sum_{k \in \mathcal{S}^-(\tau)} d_k \mid c \right] > \mathbb{E} \left[\sum_{k \in \mathcal{S}^-(\tau)} d_k \right] \quad (\text{D.32})$$

Employment in cluster c equals the mass of workers assigned to that cluster:

$$L_c = \int_{C_c} d\mu(s) \quad (\text{D.33})$$

From the match quality function (equation C.5), automation affects match quality differently across clusters depending on their position relative to the partition $\mathcal{P}(\tau)$. For a substitution-intensive cluster:

Effect on workers in substitution-intensive clusters: Automation reduces their comparative advantage. Workers in these clusters have skill profiles emphasizing $\mathcal{S}^-(\tau)$ dimensions. After automation, effective skills in substituted dimensions become $s_k^{\text{eff}} = s_k + \rho \cdot \alpha_k(\tau)$, but job requirements for these skills decline (μ_k decreases), worsening match quality:

$$A(s^{\text{eff}}(\rho, \tau), d_c, \rho, \tau) < A(s, d_c, 0, \tau) \quad \text{for substitution-intensive clusters} \quad (\text{D.34})$$

Effect on workers in complement-intensive clusters: Their relative value increases as substituted tasks become automated. Firms shift job designs toward complementary dimensions, increasing demand for $\mathcal{S}^+(\tau)$ skills.

The cluster boundary shifts as workers reallocate. To derive employment changes, I use the envelope theorem applied to worker sorting decisions.

Workers choose the cluster that maximises their wage: $c^*(s) = \arg \max_c w(s, d_c, \rho, \tau)$. The cluster boundary is implicitly defined by wage indifference:

$$w(s_{c,c+1}^*, d_c, \rho, \tau) = w(s_{c,c+1}^*, d_{c+1}, \rho, \tau) \quad (\text{D.35})$$

Differentiating this boundary condition with respect to automation intensity ρ :

$$\frac{\partial w(s^*, d_c, \rho, \tau)}{\partial \rho} + \frac{\partial w(s^*, d_c, \rho, \tau)}{\partial s} \cdot \frac{ds^*}{d\rho} = \frac{\partial w(s^*, d_{c+1}, \rho, \tau)}{\partial \rho} + \frac{\partial w(s^*, d_{c+1}, \rho, \tau)}{\partial s} \cdot \frac{ds^*}{d\rho} \quad (\text{D.36})$$

Solving for the boundary shift:

$$\frac{ds^*}{d\rho} = \frac{\frac{\partial w(s^*, d_c, \rho, \tau)}{\partial \rho} - \frac{\partial w(s^*, d_{c+1}, \rho, \tau)}{\partial \rho}}{\frac{\partial w(s^*, d_{c+1}, \rho, \tau)}{\partial s} - \frac{\partial w(s^*, d_c, \rho, \tau)}{\partial s}} \quad (\text{D.37})$$

Employment in cluster c equals the measure of workers within its boundaries:

$$L_c = \int_{s_{c-1,c}^*}^{s_{c,c+1}^*} f(s) ds \quad (\text{D.38})$$

Differentiating using Leibniz rule:

$$\frac{dL_c}{d\rho} = f(s_{c,c+1}^*) \cdot \frac{ds_{c,c+1}^*}{d\rho} - f(s_{c-1,c}^*) \cdot \frac{ds_{c-1,c}^*}{d\rho} \quad (\text{D.39})$$

For substitution-intensive clusters (emphasizing $\mathcal{S}^-(\tau)$ dimensions), automation reduces

match quality (workers have skills in dimensions where job requirements are declining), so $\frac{\partial w(s, d_c, \rho, \tau)}{\partial \rho} < 0$. This causes both boundaries to contract inward: $\frac{ds_{c-1, c}^*}{d\rho} > 0$ and $\frac{ds_{c, c+1}^*}{d\rho} < 0$, implying $\frac{dL_c}{d\rho} < 0$. This establishes part (a).

For complement-intensive clusters (emphasizing $\mathcal{S}^+(\tau)$ dimensions), automation increases match quality, so $\frac{\partial w(s, d_c, \rho, \tau)}{\partial \rho} > 0$. Boundaries expand outward, implying $\frac{dL_c}{d\rho} > 0$. This establishes part (b).

Partition-invariance: The logic above depends only on the *existence* of a partition into $\mathcal{S}^-(\tau)$ and $\mathcal{S}^+(\tau)$, not on which specific skills belong to each set. Whether the partition is {routine, analytical} or some other technology-specific division, the mechanism operates identically.

Step 3: Derive occupation-level offset

Occupation-level employment is the sum across all clusters:

$$L_{\text{occupation}} = \sum_{c=1}^{n^*} L_c \quad (\text{D.40})$$

Differentiating with respect to automation:

$$\frac{dL_{\text{occupation}}}{d\rho} = \sum_{c=1}^{n^*} \frac{dL_c}{d\rho} \quad (\text{D.41})$$

Partition clusters into substitution-intensive (C^-) and complement-intensive (C^+) sets relative to the technology-specific partition $\mathcal{P}(\tau)$:

$$\frac{dL_{\text{occupation}}}{d\rho} = \sum_{c \in C^-} \frac{dL_c}{d\rho} + \sum_{c \in C^+} \frac{dL_c}{d\rho} \quad (\text{D.42})$$

From Step 2, the first term is negative and the second is positive. Under *balanced automation* (defined as automation where the partition $\mathcal{P}(\tau)$ induces approximately equal mass of skill dimensions in $\mathcal{S}^-(\tau)$ and $\mathcal{S}^+(\tau)$), these effects approximately offset:

$$\left| \sum_{c \in C^-} \frac{dL_c}{d\rho} + \sum_{c \in C^+} \frac{dL_c}{d\rho} \right| \ll \sum_{c=1}^{n^*} \left| \frac{dL_c}{d\rho} \right| \quad (\text{D.43})$$

This establishes part (c): occupation-level effects are small relative to cluster-level volatility.

Partition-invariance of aggregation bias: The offset mechanism requires only that (i) some clusters contract and (ii) some clusters expand. This occurs for *any* partition $\mathcal{P}(\tau)$ that creates heterogeneous cluster positions—it does not depend on whether the partition is {routine, analytical} or some other technology-specific division. The aggregation bias is a

structural feature of within-occupation heterogeneity, not a consequence of any particular skill classification.

Step 4: Quantify aggregation bias

Define the aggregation bias ratio:

$$\beta = \frac{\text{Var}(\Delta L_c | \text{cluster})}{\text{Var}(\Delta L_o | \text{occupation})} \quad (\text{D.44})$$

From the variance decomposition:

$$\text{Var}(\Delta L_c) = \text{Var}\left(\frac{dL_c}{d\rho} \cdot \Delta\rho\right) \quad (\text{D.45})$$

$$= (\Delta\rho)^2 \cdot \text{Var}\left(\frac{dL_c}{d\rho}\right) \quad (\text{D.46})$$

At the occupation level:

$$\text{Var}(\Delta L_o) = \text{Var}\left(\sum_c \frac{dL_c}{d\rho} \cdot \Delta\rho\right) \quad (\text{D.47})$$

$$\approx 0 \quad (\text{due to offsetting effects}) \quad (\text{D.48})$$

Therefore $\beta \rightarrow \infty$ as occupation-level effects approach zero while cluster-level effects remain bounded away from zero.

To illustrate the magnitude of aggregation bias, suppose occupation-level regressions yield $R^2 \approx 0.03$ (weak predictive power) while cluster-level employment changes range from substantial contraction to substantial expansion. This would imply:

$$\beta \approx \frac{\text{Var}(\text{cluster-level changes})}{\text{Var}(\text{occupation-level changes})} \gg 1 \quad (\text{D.49})$$

The model predicts that cluster-level volatility should be orders of magnitude larger than occupation-level volatility—the aggregation bias mechanism. ■

Remark: The proof offers one possible explanation for the empirical puzzle identified in the literature review. Occupation-level regressions may find near-zero effects, which could be interpreted as evidence of minimal automation impact. However, if the model's mechanism holds, this null result could mask extreme restructuring within occupations: some clusters expanding dramatically while others contract sharply. The occupation-level null would then reflect an *aggregation artifact* rather than evidence of no effect. This prediction is tested in subsequent chapters.

Empirical Implication: Any analysis operating at the occupation level will systematically underestimate automation's employment impact. Cluster-level analysis (Chapter 6) is

necessary to detect restructuring.

Lemma A.2 (Aggregation of Cluster Effects): Let ΔL_c denote the employment change in cluster c within occupation o . Under Proposition 3, when clusters have offsetting effects (some expand while others contract), the variance of cluster-level changes dominates the aggregate occupation-level change:

$$\text{Var}(\Delta L_c \mid o) \gg |\mathbb{E}[\Delta L_c \mid o]|^2 \quad (\text{D.50})$$

Proof: From Proposition 3, automation shifts cluster boundaries through changes in ideal skill profiles $\mu(d, \rho)$. Employment in cluster c changes as:

$$\Delta L_c = \frac{dL_c}{d\rho} \cdot \Delta\rho \quad (\text{D.51})$$

By the envelope theorem, clusters with different task compositions experience opposite effects:

- Substitution-intensive clusters: $\frac{dL_c}{d\rho} < 0$ (workers reallocate away as match quality declines)
- Complement-intensive clusters: $\frac{dL_c}{d\rho} > 0$ (workers reallocate toward as relative wages rise)

At the occupation level, total employment change is:

$$\Delta L_o = \sum_{c \in o} \Delta L_c = \sum_{c \in o} \frac{dL_c}{d\rho} \cdot \Delta\rho \quad (\text{D.52})$$

When cluster effects have opposing signs and similar magnitudes, the sum exhibits partial cancellation:

$$\left| \sum_{c \in o} \frac{dL_c}{d\rho} \right| \ll \sum_{c \in o} \left| \frac{dL_c}{d\rho} \right| \quad (\text{D.53})$$

Therefore:

$$\text{Var}(\Delta L_c \mid o) = \mathbb{E} \left[\left(\frac{dL_c}{d\rho} \right)^2 \right] \cdot (\Delta\rho)^2 \quad (\text{D.54})$$

$$|\mathbb{E}[\Delta L_c \mid o]|^2 = \left| \mathbb{E} \left[\frac{dL_c}{d\rho} \right] \right|^2 \cdot (\Delta\rho)^2 \quad (\text{D.55})$$

Since cluster effects offset ($\mathbb{E}[\frac{dL_c}{d\rho}] \approx 0$) while individual cluster volatility remains large

($\mathbb{E}[(\frac{dL_c}{d\rho})^2] > 0$), I have:

$$\frac{\text{Var}(\Delta L_c | o)}{|\mathbb{E}[\Delta L_c | o]|^2} = \frac{\mathbb{E}\left[\left(\frac{dL_c}{d\rho}\right)^2\right]}{\left|\mathbb{E}\left[\frac{dL_c}{d\rho}\right]\right|^2} \gg 1 \quad (\text{D.56})$$

This establishes that within-occupation cluster volatility is large even when aggregate occupation-level effects are negligible. The aggregation bias ratio β quantifies this divergence. ■

Remark: Lemma A.2 formalises one potential mechanism underlying the empirical puzzle identified in the literature review. If occupation-level regressions estimate $\mathbb{E}[\Delta L_c | o] \approx 0$, this could be interpreted as evidence that automation has minimal employment impact. However, if this mechanism operates, such null results would mask $\text{Var}(\Delta L_c | o) \gg 0$ —extreme restructuring within occupations that is only visible at the cluster level.

D.3.4 Proof of Lemmas 1-2: Automation Extension

Lemma 1 (Cluster Boundaries Shift): For substitution-intensive clusters, $\frac{dL_c}{d\rho} < 0$; for complement-intensive clusters, $\frac{dL_c}{d\rho} > 0$.

Proof: From Proposition 3, Step 2, automation affects match quality through effective skill augmentation. For a worker with skills $s \in S$ matched to job design d , effective skills in substituted dimensions are given by equation (C.48): $s_k^{\text{eff}}(\rho, \tau) = s_k + \rho \cdot \alpha_k(\tau)$ for $k \in S^-(\tau)$. Match quality becomes:

$$A(s^{\text{eff}}, d) = \exp\left[-\lambda \sum_{k=1}^K \|s_k^{\text{eff}} - \mu_k(d)\|^2\right] \quad (\text{D.57})$$

Taking the derivative with respect to automation intensity for $k \in S^-(\tau)$:

$$\frac{\partial A}{\partial \rho} = -2\lambda \sum_{k \in S^-(\tau)} \alpha_k(\tau) \cdot (s_k + \rho \cdot \alpha_k(\tau) - \mu_k) \cdot A(s^{\text{eff}}, d) \quad (\text{D.58})$$

For substitution-intensive clusters (where skill requirements d_k for $k \in S^-(\tau)$ are high), this derivative is negative: automation augments workers' effective skills in substituted dimensions, moving them further from ideal requirements as technology substitutes for human effort. Lower match quality reduces wages $w(s, d, \rho)$, inducing workers to reallocate to other clusters. Therefore:

$$\frac{dL_c}{d\rho} = \int_{C_c} \frac{\partial \mu(s)}{\partial \rho} ds < 0 \quad \text{for substitution-intensive clusters} \quad (\text{D.59})$$

For complement-intensive clusters (where skill requirements d_k for $k \in S^+(\tau)$ are high),

automation increases the relative value of complementary skills. Firms shift job designs to emphasise these dimensions, increasing wages for workers with high s_k in $k \in \mathcal{S}^+(\tau)$. Therefore:

$$\frac{dL_c}{d\rho} > 0 \quad \text{for complement-intensive clusters} \quad (\text{D.60})$$

This completes the proof. ■

Lemma 2 (Occupation-Level Offset): Under balanced automation, $\left| \frac{dL_{\text{occupation}}}{d\rho} \right| < \epsilon \cdot \sum_c \left| \frac{dL_c}{d\rho} \right|$ for small $\epsilon > 0$.

Proof: Balanced automation is defined as automation that affects skill dimensions symmetrically such that:

$$\mathbb{E} \left[\sum_{k \in \mathcal{S}^-(\tau)} \rho \cdot \alpha_k(\tau) \right] = 0 \quad (\text{D.61})$$

In other words, the automation shock is mean-preserving: it shifts some skill dimensions upward (substituted tasks become easier) while complementary dimensions become relatively more valuable, but the total effective skill supply remains approximately constant.

Under this condition, partition clusters into substitution-intensive ($c \in \mathcal{S}^-$) and complement-intensive ($c \in \mathcal{S}^+$) sets:

$$\frac{dL_{\text{occupation}}}{d\rho} = \sum_{c \in \mathcal{S}^-} \frac{dL_c}{d\rho} + \sum_{c \in \mathcal{S}^+} \frac{dL_c}{d\rho} \quad (\text{D.62})$$

$$= \sum_{c \in \mathcal{S}^-} (-\delta_c) + \sum_{c \in \mathcal{S}^+} (+\delta_c) \quad \text{where } \delta_c > 0 \quad (\text{D.63})$$

Under balanced automation, the contractions and expansions approximately cancel:

$$\left| \sum_{c \in \mathcal{S}^-} (-\delta_c) + \sum_{c \in \mathcal{S}^+} (+\delta_c) \right| \approx \epsilon \cdot \left(\sum_{c \in \mathcal{S}^-} \delta_c + \sum_{c \in \mathcal{S}^+} \delta_c \right) \quad (\text{D.64})$$

for small $\epsilon > 0$ representing the imbalance.

If empirical occupation-level regressions yield low explanatory power (e.g., $R^2 \approx 0.03$), this would imply:

$$\epsilon \approx \sqrt{R^2} \approx 0.17 \quad (\text{D.65})$$

Under such conditions, occupation-level effects would be approximately 17% of the sum of absolute cluster-level effects—small but non-zero. This explains the model’s prediction: occupation-level regressions should detect weak effects while cluster-level analysis should reveal extreme heterogeneity. ■

Remark: The lemmas formalise the offsetting mechanism. Automation creates “winners” (complement-intensive clusters) and “losers” (substitution-intensive clusters) within each occupation. Standard occupation-level analysis aggregates these opposing forces,

detecting only the small net effect. The true restructuring is visible only at the cluster level.

D.3.5 Proof of Proposition 4: Temporal Equilibrium Path

Proposition 4 (Temporal Equilibrium Path): For continuously differentiable $\rho(t)$ with $d\rho/dt > 0$, the equilibrium cluster configuration $\{d_1^*(t), \dots, d_{n^*(t)}^*(t)\}$ evolves continuously almost everywhere. Discontinuous transitions occur at critical automation thresholds $\{\bar{\rho}_1, \bar{\rho}_2, \dots\}$ where cluster viability conditions change.

Proof: The proof proceeds in three steps.

Step 1: Continuity of the profit function. The firm profit function is:

$$\pi(\theta, d, q, \rho) = h(q) \cdot \theta \cdot A(s^{\text{eff}}(\rho), d, \rho) - C(d, \theta) - w \quad (\text{D.66})$$

where $A(\cdot)$ is the match quality function from equation (C.48). Since A is continuous in ρ (exponential of a polynomial) and $h(q)$, $C(d, \theta)$ are continuous in their arguments, π is continuous in (θ, d, q, ρ) .

Step 2: Application of the maximum theorem. For fixed ρ , the equilibrium designs solve:

$$d^*(\theta, \rho) \in \arg \max_{d \in \mathcal{D}} \pi(\theta, d, q(d), \rho) \quad (\text{D.67})$$

The design space $\mathcal{D}$ is compact (positive unit sphere). By the maximum theorem (Berge, 1963), the maximiser correspondence $d^*(\theta, \cdot) : \mathbb{R}_+ \rightrightarrows \mathcal{D}$ is upper hemicontinuous. When the maximiser is unique (generic case), $d^*(\theta, \rho)$ is a continuous function of ρ .

Step 3: Characterisation of discontinuities. The maximiser is non-unique only when multiple designs yield identical maximum profit. Define the viability condition:

$$V(d, \rho) \equiv \max_{\theta \in \Theta} \pi(\theta, d, q(d), \rho) - f \quad (\text{D.68})$$

A cluster with design d is viable if $V(d, \rho) \geq 0$. Since V is continuous in ρ , the set of viable designs changes only at $\bar{\rho}$ values where $V(d, \bar{\rho}) = 0$ for some d . These threshold values form the set:

$$\mathcal{T} = \{\rho \in \mathbb{R}_+ : \exists d \in \mathcal{D} \text{ such that } V(d, \rho) = 0\} \quad (\text{D.69})$$

Under regularity conditions (transversality: $\partial V / \partial \rho \neq 0$ at crossing points), $\mathcal{T}$ is a discrete set. Between consecutive elements of $\mathcal{T}$, the number of viable clusters is constant and equilibrium designs evolve continuously. At elements of $\mathcal{T}$, exactly one cluster crosses the viability threshold (birth or death).

Since $\mathcal{T}$ is countable and $\rho(t)$ is continuous, the set of times $\{t : \rho(t) \in \mathcal{T}\}$ is also

countable (measure zero). The equilibrium evolves continuously for $t \notin \{t : \rho(t) \in \mathcal{T}\}$, with discontinuous transitions at the remaining times. ■

D.3.6 Proof of Lemma 3: Viability Monotonicity

Lemma 3 (Viability Monotonicity):

- (a) For substitution-intensive cluster c : $\partial\pi_c/\partial\rho < 0$ and there exists $\bar{\rho}_c$ such that c is unviable for $\rho > \bar{\rho}_c$.
- (b) For complement-intensive cluster c' : $\partial\pi_{c'}/\partial\rho > 0$ and there exists $\underline{\rho}_{c'}$ such that c' is viable for $\rho > \underline{\rho}_{c'}$.

Proof: The proof holds for any technology-indexed partition $\mathcal{P}(\tau) = \{\mathcal{S}^-(\tau), \mathcal{S}^+(\tau), \mathcal{S}^0(\tau)\}$.

Part (a): Substitution-intensive clusters. From equation (C.44), automation shifts the ideal skill profile:

$$\frac{\partial\mu_k}{\partial\rho} < 0 \quad \text{for } k \in \mathcal{S}^-(\tau) \text{ (substituted dimensions)} \quad (\text{D.70})$$

For a substitution-intensive cluster c with design d_c emphasising substituted skills, average match quality is:

$$\bar{A}_c(\rho, \tau) = \mathbb{E}[A(s, d_c, \rho, \tau) | s \in c] \quad (\text{D.71})$$

Taking the derivative:

$$\frac{\partial\bar{A}_c}{\partial\rho} = \mathbb{E}\left[-2\lambda \sum_{k \in \mathcal{S}^-(\tau)} (s_k - \mu_k(d_c, \rho, \tau)) \cdot A(s, d_c, \rho, \tau) \cdot \frac{\partial\mu_k}{\partial\rho}\right] \quad (\text{D.72})$$

Since $\partial\mu_k/\partial\rho < 0$ for $k \in \mathcal{S}^-(\tau)$ and workers in substitution-intensive clusters have $s_k > \mu_k$ on average for substituted dimensions (they are hired precisely because of high substituted skills), the integrand is negative. Therefore $\partial\bar{A}_c/\partial\rho < 0$, and since profit is monotone in match quality, $\partial\pi_c/\partial\rho < 0$.

Since π_c is continuous and strictly decreasing in ρ , and $\lim_{\rho \rightarrow \infty} \pi_c(\rho) < f$ (match quality degrades to zero), there exists unique $\bar{\rho}_c$ such that $\pi_c(\bar{\rho}_c) = f$. For $\rho > \bar{\rho}_c$, the cluster is unviable.

Part (b): Complement-intensive clusters. From equation (C.46):

$$\frac{\partial\mu_k}{\partial\rho} > 0 \quad \text{for } k \in \mathcal{S}^+(\tau) \text{ (complementary dimensions)} \quad (\text{D.73})$$

For complement-intensive cluster c' emphasising complementary skills, automation increases the relative value of complementary dimensions. Match quality improves:

$$\frac{\partial \bar{A}_{c'}}{\partial \rho} > 0 \quad (\text{D.74})$$

by symmetric reasoning. Therefore $\partial \pi_{c'}/\partial \rho > 0$.

If cluster c' is not initially viable ($\pi_{c'}(\rho_0) < f$), but $\lim_{\rho \rightarrow \infty} \pi_{c'}(\rho) > f$ (complementary skills become sufficiently valuable), then by continuity there exists $\underline{\rho}_{c'}$ such that $\pi_{c'}(\underline{\rho}_{c'}) = f$. For $\rho > \underline{\rho}_{c'}$, the cluster becomes viable.

Partition-invariance: The proof depends only on the existence of dimensions with $\partial \mu_k/\partial \rho < 0$ (substituted) and dimensions with $\partial \mu_k/\partial \rho > 0$ (complementary)—not on which specific skills fall into each category. The mechanism is partition-invariant. ■

D.3.7 Proof of Proposition 5: Churn-Automation Speed Relationship

Proposition 5 (Churn-Automation Speed): $\partial \chi/\partial (d\rho/dt) > 0$.

Proof: Let $\mathcal{T} = \{\bar{\rho}_1, \bar{\rho}_2, \dots\}$ be the set of viability thresholds from Proposition 4's proof. Define the threshold counting function:

$$\Phi(\rho) = \#\{c : \bar{\rho}_c \leq \rho\} + \#\{c' : \underline{\rho}_{c'} \leq \rho\} \quad (\text{D.75})$$

counting the cumulative number of cluster births and deaths up to automation level ρ .

For an interval $[t, t + \Delta t]$, the number of births plus deaths equals:

$$B(t, \Delta t) + D(t, \Delta t) = \Phi(\rho(t + \Delta t)) - \Phi(\rho(t)) \quad (\text{D.76})$$

Taking a Taylor expansion:

$$\Phi(\rho(t + \Delta t)) - \Phi(\rho(t)) = \Phi'(\rho(t)) \cdot (\rho(t + \Delta t) - \rho(t)) + O((\Delta t)^2) \quad (\text{D.77})$$

Since $\rho(t + \Delta t) - \rho(t) \approx \frac{d\rho}{dt} \cdot \Delta t$:

$$B + D \approx \phi(\rho(t)) \cdot \frac{d\rho}{dt} \cdot \Delta t \quad (\text{D.78})$$

where $\phi = \Phi'$ is the threshold density (number of thresholds per unit of ρ).

The churn rate is:

$$\chi(t, \Delta t) = \frac{B + D}{n^*(t)} \approx \frac{\phi(\rho(t))}{n^*(t)} \cdot \frac{d\rho}{dt} \cdot \Delta t \quad (\text{D.79})$$

Therefore:

$$\frac{\partial \chi}{\partial (d\rho/dt)} = \frac{\phi(\rho(t)) \cdot \Delta t}{n^*(t)} > 0 \quad (\text{D.80})$$

since $\phi > 0$ (there exist viability thresholds), $\Delta t > 0$, and $n^*(t) > 0$ (at least one cluster exists). ■

Remark: The proof reveals that churn is proportional to the product of (i) threshold density ϕ (how many thresholds exist per unit automation) and (ii) automation speed $d\rho/dt$. Occupations with dense viability thresholds (many marginal clusters) will exhibit higher churn for a given automation speed. Similarly, periods of rapid automation (high $d\rho/dt$) will exhibit higher churn for any threshold distribution.

D.4 Numerical Illustration: Calibration and Comparative Statics

This section provides a numerical illustration of the model, calibrating parameters to match stylised empirical targets and demonstrating the comparative-statics predictions. The calibration targets are derived from patterns documented in the computational job posting literature (Section 2.6) and refined in subsequent empirical chapters.

D.4.1 Calibration Strategy

I calibrate the model to match five stylised empirical targets:

1. **Cluster count:** Median occupation has $n^* \approx 3$ clusters in metropolitan markets ($N \approx 5,000$ workers), consistent with the “few but distinct” pattern documented in computational job posting research (Deming & Kahn, 2018a)
2. **Wage dispersion:** Within-occupation standard deviation $\sigma_w/\bar{w} \approx 0.25$, consistent with findings that within-occupation wage variation exceeds between-occupation variation (Weeden et al., 2007)
3. **Market thickness elasticity:** $\partial \log n^* / \partial \log N \approx 0.4$, reflecting the theoretical prediction that larger markets support finer specialisation

4. **Positive assortative matching:** Correlation between firm size and job complexity $\rho(\text{size}, \text{span}) \approx 0.5\text{--}0.7$, consistent with theoretical predictions and patterns in employer-employee matched data
5. **Automation employment effects:** Cluster-level employment changes substantially larger than occupation-level effects, with occupation-level regressions showing low explanatory power ($R^2 \approx 0.03$)—the aggregation bias signature predicted by Proposition 3

D.4.2 Parameter Values

Table D.1 summarises the calibrated parameter values. The mismatch penalty $\lambda = 0.50$ is set to match observed within-occupation wage dispersion of approximately 25% (coefficient of variation). The coordination cost scale $\kappa = 1.00$ and curvature $\gamma = 2.50$ jointly determine cluster formation; the curvature exceeds the threshold of 2 required by Lemma A.2 to ensure discrete clustering rather than continuous dispersion. The productivity elasticity $\alpha = 0.30$ generates positive assortative matching with correlation $\rho = 0.62$, consistent with employer-employee matched data. The posting cost $f = 0.10$ yields market thickness elasticity of approximately 0.4. Finally, the skill space dimension $K = 10$ reflects the typical number of distinct skill requirements observed in job advertisements, while skill bounds are normalised to unity.

Table D.1: *Calibrated Model Parameters*

Parameter	Symbol	Value	Target Moment
Mismatch penalty	λ	0.50	Wage dispersion $\sigma_w/\bar{w} = 0.25$
Coordination cost scale	κ	1.00	Cluster count $n^* = 3$ at $N = 5,000$
Cost curvature	γ	2.50	Discrete clustering (Lemma A.2: $\gamma > 2$)
Productivity elasticity	α	0.30	PAM correlation $\rho = 0.62$
Posting cost	f	0.10	Market elasticity $\beta \approx 0.4$
Skill space dimension	K	10	Empirical skill count (Chapter 6)
Skill bound	$\bar{s}$	1.00	Normalisation (unit scale)

Note. Parameters calibrated to match stylised empirical targets from the computational job posting literature and subsequent empirical chapters. All values lie within ranges commonly used in directed search and assignment models. The cost curvature $\gamma = 2.50$ exceeds the minimum threshold of 2 required by Lemma A.2 to ensure discrete clustering.

D.4.3 Model Predictions

With these calibrated parameters, the model generates:

Cluster formation: For $N \in [500, 10,000]$ (range observed in data), the model predicts:

$$n^*(N = 500) = 2 \quad (\text{small regional markets}) \quad (\text{D.81})$$

$$n^*(N = 2,000) = 3 \quad (\text{medium cities}) \quad (\text{D.82})$$

$$n^*(N = 5,000) = 4 \quad (\text{large metropolitan areas}) \quad (\text{D.83})$$

$$n^*(N = 10,000) = 5 \quad (\text{major metro centres}) \quad (\text{D.84})$$

These predictions align closely with the empirical cluster counts from Chapter 6 (2–7 clusters depending on occupation and market size).

Automation comparative statics (varying $\rho \in [0, 0.5]$):

- **Routine cluster** (low span d_1): Employment declines -60% to -80% as automation reduces routine task requirements
- **Complementary cluster** (high span d_3): Employment expands $+80\%$ to $+150\%$ as analytical tasks become more valuable
- **Moderate cluster** (medium span d_2): Employment changes -10% to $+20\%$ depending on task composition
- **Occupation aggregate:** Net employment change -5% to $+10\%$, with low R^2 when regressing on automation proxy—demonstrating the aggregation bias mechanism

D.4.4 Graphical Illustrations

The calibrated model generates the profit curves illustrated in Figure C.4 (Panel A), showing how discrete designs $\{d_1, d_2, d_3\}$ emerge as global optima for firms with different productivity levels. Figure C.5 (Panel A) depicts the step-function relationship $n^*(N)$ derived from the model, with threshold market sizes:

$$N_1 \approx 800, \quad N_2 \approx 1,800, \quad N_3 \approx 4,200, \quad N_4 \approx 8,500 \quad (\text{D.85})$$

These thresholds represent the minimum market sizes at which the 1st, 2nd, 3rd, and 4th clusters become viable, as defined in equation (C.27).

D.4.5 Robustness and Sensitivity

The qualitative predictions (discrete clustering, $\partial n^*/\partial N > 0$, cluster-level automation heterogeneity) are robust to parameter variation. Sensitivity analysis shows:

- Increasing γ from 2.5 to 3.0: Sharper cluster boundaries (fewer marginal postings between clusters)
- Decreasing κ from 1.0 to 0.5: More clusters at each market size (coordination costs less binding)
- Increasing λ from 0.5 to 1.0: Higher wage dispersion (stronger mismatch penalties)
- Decreasing f from 0.10 to 0.05: Finer specialisation (lower entry barriers)

All parameter combinations satisfying the concavity condition (C.22) generate the discrete clustering structure predicted by Proposition 1.

D.4.6 Connecting Calibration to Empirical Chapters

The numerical illustration demonstrates that the theoretical model can quantitatively match the empirical patterns documented in Chapters 5–6. The calibrated parameters are economically reasonable (e.g., coordination costs representing 10–30% of operating profits, mismatch penalties generating 25% wage dispersion), and the model’s predictions align with observed cluster counts, market-size elasticities, and automation effects. This numerical validation strengthens the claim that the theoretical mechanism—discrete clustering driven by supermodular production, convex coordination costs, and directed search—provides a plausible explanation for the empirical findings.

The calibration also clarifies which empirical moments discipline which theoretical parameters. Wage dispersion pins down the mismatch penalty λ , cluster count identifies the coordination cost scale κ , and the automation employment range determines the automation shift magnitude ρ . This one-to-one mapping between moments and parameters enables future researchers to extend the calibration to other occupations or countries using the same procedure.

Appendix E

BGT Data Descriptions

This appendix documents the data fields in the Burning Glass Technologies (BGT) job advertisement dataset used throughout this thesis. The dataset covers Australian online job advertisements from 2012–2019, sourced from BGT (now Lightcast). BGT aggregates job postings from over 40,000 online sources including job boards, company websites, and recruitment platforms, applying natural language processing to extract structured fields from unstructured advertisement text (OECD, 2021).

Coverage and representativeness: BGT captures approximately 80–85% of online job advertisements in comparable markets (Forsythe et al., 2020; Hershbein & Kahn, 2018b). Validation against UK government vacancy data shows 93% correlation at occupation level (OECD, 2021). The dataset systematically under-represents positions filled through informal networks or internal promotion (estimated at approximately 20% of vacancies) and over-represents white-collar, professional, and technical occupations relative to manual trades and informal employment.

Skill taxonomy: BGT classifies skills extracted from job advertisements into three mutually exclusive categories relevant to this thesis’s analysis:

- **IsSpecialized** (= 1): Occupation-specific technical skills (e.g., SAP, Java, Financial Reporting). These form the primary input for HDBSCAN clustering in Chapters 6–7.
- **IsBaseline** (= 1): General competencies common across occupations (e.g., Communication, Teamwork, Problem Solving). These are excluded from clustering to avoid masking occupation-specific variation but are included in the skill proximity network analysis (Chapter 8).

- **IsSoftware** (= 1): Technology-specific skills (e.g., Python, AWS, Salesforce). This flag enables the technology succession analysis in Chapter 6, which tests whether dead clusters are more software-intensive than born clusters.

Each job advertisement may list multiple skills; the skill fields below represent individual rows in a one-to-many relationship with the job-level fields. The binary skill matrix used for clustering is constructed via `MultiLabelBinarizer`, producing a sparse matrix where each column represents one skill and each row represents one job advertisement. Table E.1 lists all 38 data fields with representative examples.

Table E.1: BGT Data Descriptions

Element	Examples
JobID	21706940
JobDate	1/8/18
CleanJobTitle	Technical Lead - Java And Aws
Language	en
CanonCountry	AUS
CanonState	NSW
CanonCity	Sydney
CanonEmployer	Gen
InternshipFlag	0
MinDegreeLevel	NA
CanonMinimumDegree	Bachelor's Degree
CanonPreferredDegreeLevel	21
CanonPreferredDegree	Doctorate Level Degree
StandardMajor	Electrical And Electronic Engineering Technologies/Technicians, Other
MaxDegreeLevel	NA
CIPCode	14.1001
CanonCertification	Working With Children Check
CanonJobHours	fulltime
CanonJobType	permanent
MaxExperience	NA
MinExperience	NA
MaxAnnualSalary	130000

Continued on next page

Table E.1 – *Continued from previous page*

Element	Examples
MinAnnualSalary	100000
MaxHourlySalary	64.50
MinHourlySalary	48.08
WorkFromHome	1
ANZSCOCCode	261399
ANZSCOOccupation	Software and Applications Programmers nec
ANZSCOMinorGroup	Business and Systems Analysts, and Programmers
ANZSCOSubMajorGroup	ICT Professionals
ANZSCOMajorGroup	PROFESSIONALS
BGTOcc	15-1131.00
BGTOccName	Software Developer / Engineer
BGTOccGroupName	Software Development
BGTCareerAreaName	Information Technology
ANZSICCode	81
ANZSICClass	Higher Education
ANZSICGroup	Tertiary Education
ANZSICSubdivision	Tertiary Education
ANZSICDivision	Education and Training
StockTicker	
SA4	115-128
SA4Name	Sydney
Skill	SAP
SkillCluster	Enterprise Resource Planning (ERP)
SkillClusterFamily	Information Technology
IsSpecialized	1
IsBaseline	0
IsSoftware	1

Appendix F

Data Codebook

This appendix documents the data files, variable definitions, and parameter choices used in the empirical analysis. It supplements the BGT field descriptions in Appendix E with documentation of derived data files, saved models, and key analytical parameters. For replication instructions, see `analysis/REPLICATION_README.md` in the repository.

F.1 Primary Data Source

The primary data source is the Burning Glass Technologies (BGT, now Lightcast) Australian Job Advertisement Dataset, covering 2012–2019. The dataset comprises approximately 9.5 million job advertisements aggregated from over 40,000 online sources. BGT applies natural language processing to extract structured fields—including skills, qualifications, employer, location, and salary—from unstructured advertisement text (OECD, 2021). Appendix E documents all 38 data fields with representative examples.

The raw data are stored in two proprietary Parquet files not included in the public repository:

File	Description	Approximate Rows
<code>maindf.parquet.gzip</code>	Job-level data (all fields)	9.5M
<code>skills.parquet.gzip</code>	Job $\times$ Skill records (one-to-many)	60M

F.2 Key Fields Used in Analysis

Table F.1 lists the BGT fields used in each empirical chapter.

Table F.1: Key BGT data fields and their usage across empirical chapters.

Field	Type	Description	Used In
JobID	int	Unique advertisement identifier	All
JobDate	date	Date of posting	All
ANZSCOCCode	str	6-digit ANZSCO occupation code	All
ANZSCOOccupation	str	ANZSCO occupation title	All
CanonState	str	Australian state/territory	Ch. 6
SA4	str	Statistical Area Level 4 code	Ch. 6
CanonEmployer	str	Canonicalised employer name	Ch. 6
MinAnnualSalary	float	Minimum advertised salary (AUD)	Ch. 7
MaxAnnualSalary	float	Maximum advertised salary (AUD)	Ch. 7
Skill	str	Individual skill from advertisement	Ch. 6–8
IsSpecialized	binary	Occupation-specific technical skill	Ch. 6–7
IsBaseline	binary	General competency	Ch. 8
IsSoftware	binary	Technology-specific skill	Ch. 6

F.3 Skill Classification Taxonomy

BGT classifies each extracted skill into exactly one of three mutually exclusive categories:

- **IsSpecialized** (= 1): Occupation-specific technical skills (e.g., SAP, Java, Financial Reporting). These form the primary input for HDBSCAN clustering in Chapters 6–7. The binary skill matrix used for clustering is constructed via `MultiLabelBinarizer`, producing a sparse matrix where each column represents one specialised skill and each row represents one job advertisement.
- **IsBaseline** (= 1): General competencies common across occupations (e.g., Communication, Teamwork, Problem Solving). These are excluded from clustering to avoid masking occupation-specific variation but are included in the skill proximity network (Chapter 8), where they serve as bridge skills connecting clusters across occupational boundaries.

- **IsSoftware** (= 1): Technology-specific skills (e.g., Python, AWS, Salesforce). This flag enables the technology succession analysis in Chapter 6, which tests whether dead clusters are more software-intensive than born clusters.

F.4 HDBSCAN Saved Models

Each clustering result is stored as a Python pickle file at `analysis/saved_models/{occupation}/{year}` containing the fields listed in Table F.2.

Table F.2: Contents of each saved HDBSCAN model pickle file.

Key	Type	Description
<code>labels</code>	array	Cluster assignment per job (−1 = noise)
<code>skill_matrix</code>	matrix	Binary skill matrix (jobs × skills)
<code>mlb_classes</code>	list	Ordered skill vocabulary used for binarisation
<code>occupation</code>	str	ANZSCO occupation title
<code>n_clusters</code>	int	Number of clusters discovered
<code>n_noise</code>	int	Number of noise-classified observations
<code>parameters</code>	dict	HDBSCAN parameters (<code>min_cluster_size</code> , <code>min_samples</code> , <code>metric</code>)

There are 250 occupation directories with 8 years each, yielding approximately 2,000 model files.

F.5 Derived Result Files

F.5.1 Chapter 5: Occupation-Level Regression

Collapsed panel datasets used for fixed-effects regression:

File	Description
<code>rcm_collapsed.parquet</code>	RCM (Routine Cognitive/Manual) task category aggregates
<code>fo_collapsed.parquet</code>	FO (Frey & Osborne) automation risk aggregates
<code>rcm_skill_collapsed.parquet</code>	RCM × Skill Level interaction terms
<code>fo_skill_collapsed.parquet</code>	FO × Skill Level interaction terms

F.5.2 Chapter 6: Hypothesis Test Results

Located in `analysis/results/chapter6_fullscale/`:

File	Description
<code>h1a_cluster_validation.csv</code>	Cluster counts, noise rates, silhouette scores (250 occupations)
<code>h1b_firm_sorting.csv</code>	Firm size effect on cluster assignment
<code>h2a_geographic_variation.csv</code>	SA4-level cluster prevalence
<code>h2b_market_thickness.csv</code>	Market size effects on cluster count
<code>h3a_volatility_ratios.csv</code>	Cluster birth/death frequency distributions
<code>h3b_software_skill_breakdown.csv</code>	Technology succession analysis
<code>h3c_variance_decomposition.csv</code>	Within vs. between occupation variance

F.5.3 Chapter 7: Evolutionary Metrics

Located in `analysis/results/chapter7/`. Table F.3 describes the primary output file.

Table F.3: Key columns in `fullscale_summary.csv` (one row per occupation, $N = 250$).

Column	Type	Description
<code>occupation</code>	str	ANZSCO occupation title
<code>anzsco</code>	str	6-digit ANZSCO code
<code>avg_clusters</code>	float	Mean cluster count across years
<code>avg_churn</code>	float	Mean annual churn rate (%)
<code>avg_birth</code>	float	Mean annual birth rate (%)
<code>avg_death</code>	float	Mean annual death rate (%)
<code>avg_survival</code>	float	Mean annual survival rate (%)
<code>centroid_drift</code>	float	Mean cosine distance of occupation-level skill vector
<code>avg_match_sim</code>	float	Mean cosine similarity of matched cluster pairs

Additional files include `fullscale_transitions.csv` (one row per occupation-year transition, ~1,750 rows), `evolutionary_typology_classification.csv` (evolutionary class per occupation), and `threshold_sensitivity.csv` (cosine threshold robustness results).

F.5.4 Chapter 8: Skill Proximity Network

Located in `analysis/results/chapter8_network/`:

File	Description
<code>network_metrics.csv</code>	Network-level summary (5,323 nodes, 174,817 edges)
<code>node_attributes_2019_t6.csv</code>	Per-cluster attributes and network position
<code>edge_list_2019_t6.csv</code>	Pairwise cluster connections (cosine > 0.60)
<code>transition_pathways.csv</code>	At-risk → stable transition pathways (~18,080)
<code>bridge_skills.csv</code>	Skills appearing in bridge transition pathways
<code>skill_gap_distribution.csv</code>	Distribution of skill gap sizes

Key columns in `node_attributes_2019_t6.csv` include `trajectory_status` (newly_emerged, persistent, disappeared), `risk_status` (at_risk, stable), and network centrality measures (`degree`, `degree_within`, `degree_cross`).

F.6 Key Parameters and Thresholds

Table F.4 documents all analytical parameters used across chapters.

Table F.4: Key analytical parameters and thresholds.

Parameter	Value	Justification	Chapter
<code>min_cluster_size</code>	25	Pre-hoc sensitivity analysis	6
<code>min_samples</code>	10	HDBSCAN default; validated	6
Distance metric	Jaccard	Standard for sparse binary vectors	6
Cluster selection	EOM	Excess of Mass (default)	6
Cluster match τ	0.80	Balanced threshold; sensitivity tested (Table 7.8)	7
Network edge τ	0.60	Skill proximity cosine similarity	8
Skill gap target	0.10	10% prevalence = substantive	8
Skill gap source floor	0.03	3% minimum source prevalence	8
Min jobs/year	500	All 8 years required	4

F.7 ANZSCO Code Structure

The Australian and New Zealand Standard Classification of Occupations (ANZSCO) uses a hierarchical 6-digit coding system:

Level	Digits	Example	Count
Major Group	1	2 (Professionals)	8
Sub-Major Group	2	22 (Business, HR, Marketing)	43
Minor Group	3	221 (Accountants, Auditors)	97
Unit Group	4	2211 (Accountants)	358
Occupation	6	221111 (Accountant, General)	1,023

This thesis analyses at the 6-digit occupation level. Occupation counts vary slightly across chapters due to data requirements: 250 occupations have sufficient data for cross-sectional clustering (Chapter 6), 250 have sufficient multi-year data for temporal dynamics (Chapter 7), and 249 have 2019 clusters for network construction (Chapter 8).

Appendix G

Replication Code and Methodology

This appendix provides code and methodology for replicating the quantitative analyses presented in this thesis. It documents software requirements and data inputs, then presents the key algorithms for each empirical chapter: the RCM task regression (Chapter 5), HDBSCAN clustering with Jaccard distance (Chapter 6), temporal tracking via centroid matching (Chapter 7), and skill proximity network construction (Chapter 8). All code is also available as an executable Jupyter notebook (`THESIS_REPLICATION_COMPLETE.ipynb`) in the supplementary materials.

G.1 Software and Data Requirements

G.1.1 Software Environment

All analyses were conducted in Python 3.10+ with the following core packages:

- **Data manipulation:** pandas 2.0+, numpy 1.24+, pyarrow 12.0+
- **Statistical analysis:** statsmodels 0.14+, scipy 1.10+
- **Machine learning:** scikit-learn 1.3+
- **Clustering:** hdbscan 0.8.33+
- **Network analysis:** networkx 3.0+

- **Visualisation:** matplotlib 3.7+, seaborn 0.12+

The complete environment can be installed via:

```
1 pip install pandas numpy statsmodels scipy scikit-learn \  
2         hdbscan networkx matplotlib seaborn pyarrow
```

G.1.2 Data Requirements

This analysis uses proprietary job advertisement data from Burning Glass Technologies (now Lightcast) covering Australian job postings from 2012–2019:

- 9,478,606 job advertisements (≈ 9.5 million)
- 45,858,484 skill-job pairs (≈ 45.9 million)
- 11,862 unique skills
- 847 unique 6-digit ANZSCO occupations (full BGT sample after exclusion criteria; 250 meet feasibility criteria for cluster-level analysis)

Note. The underlying microdata cannot be redistributed due to licensing restrictions. Researchers interested in replication should contact Lightcast to obtain equivalent data.

G.1.3 Data Files

The analysis requires three input data files stored in the `analysis/data/` directory:

- `maindf.parquet.gzip` — 9.5 million job advertisements with fields: JobID, Year, ANZSCOCCode, ANZSCOTitle, CanonEmployer, CanonState, SA4, SA4Name, ANZSICDivision
- `skills.parquet.gzip` — 45.9 million job-skill pairs with fields: JobID, Skill
- `rcm_collapsed.parquet` — RCM classification aggregated to occupation-state-industry-year level

G.1.4 Pre-computed Models

HDBSCAN clustering results are stored as pickle files in `analysis/saved_models/[occupation_dir]`.

There are 250 occupation directories, each containing up to 8 yearly pickle files (2012–2019).

Each pickle file contains:

- `labels`: Cluster assignments (numpy int array; -1 for noise, $0+$ for clusters)
- `skill_matrix`: Binary skill matrix (dense numpy array, $n_{\text{jobs}} \times n_{\text{skills}}$)
- `mlb_classes`: Skill name array (the vocabulary for this occupation-year)
- `occupation`: Human-readable occupation name
- `n_jobs`, `n_clusters`, `n_noise`: Count statistics
- `parameters`: HDBSCAN parameters used (`min_cluster_size`, `metric`, etc.)
- `cluster_profiles`: Dictionary mapping `cluster_id` $\rightarrow$ centroid vector (mean skill prevalence across cluster members)
- `cluster_sizes`: Dictionary mapping `cluster_id` $\rightarrow$ number of jobs

G.2 Chapter 5: RCM Task Analysis

Chapter 5 analyses changes in task content using the Routine-Cognitive-Manual (RCM) framework. The main regression model uses OLS with heteroskedasticity-robust (HC3) standard errors:

$$\text{JobCount}_{ijst} = \beta_0 + \beta_1(\text{Category}_i \times \text{Year}_t) + \gamma_s + \delta_t + \theta_{ij} + \epsilon_{ijst} \quad (\text{G.1})$$

where i indexes RCM category (reference: Routine Manual), j indexes industry (ANZSIC division), s indexes state (8 Australian states/territories), and t indexes year (2012–2019).

G.3 Chapter 6: HDBSCAN Clustering Methodology

Chapter 6 uses HDBSCAN to identify skill-based job clusters within occupations. The distance metric is Jaccard distance, the natural choice for binary skill vectors:

```
1 from sklearn.metrics.pairwise import pairwise_distances
2
3 # Load saved model
4 with open(pkl_path, 'rb') as f:
5     data = pickle.load(f)
6
7 skill_matrix = data['skill_matrix']
8 if hasattr(skill_matrix, 'toarray'):
9     skill_matrix = skill_matrix.toarray()
10
11 # Jaccard distance matrix
12 distances = pairwise_distances(skill_matrix, metric='jaccard',
```

G.3.1 DBCV Parameter Optimisation

DBCV (Density-Based Clustering Validation) is used to select the optimal `min_cluster_size` parameter. Pre-hoc sensitivity analysis across occupation size bins found that `min_cluster_size=25` produced the best balance of cluster granularity and stability, and this value is used uniformly across all occupations.

G.3.2 Full-Scale Validation (250 Occupations)

Chapter 6 validates clustering across all 250 occupations with sufficient data, testing hypotheses H1a (cluster detection), H1b (firm sorting), H2a (geographic variation), H2b (market thickness), H3a (volatility ratios), H3b (routine task effect), and H3c (variance decomposition). The production script is `analysis/scripts/chapter6_fullscale_tests.py`. Key methods:

- **H1a:** Silhouette scores computed on cosine distance matrices as a validation metric (subsampled to 10,000 jobs for large occupations). Note: clustering uses Jaccard distance; silhouette validation uses cosine distance as an independent metric.
- **H1b:** HHI concentration and distinct cluster count by firm size (Spearman correlation)
- **H2a:** OLS regression of cluster count on log job count per SA4 region

- **H3c:** ANOVA-style decomposition of cluster skill-profile variance into between-occupation and within-occupation components

G.4 Chapter 7: Temporal Dynamics Methodology

Chapter 7 tracks cluster evolution across 250 occupations with sufficient temporal data (≥ 7 year-pickles). The production script is `analysis/scripts/chapter7_fullscale_evolutionary_met`

G.4.1 Centroid Drift

Centroid drift measures how rapidly an occupation's overall skill content changes, computed as the mean cosine distance between consecutive year occupation-level skill prevalence vectors:

```

1  from scipy.spatial.distance import cosine as cosine_dist
2
3  # For each consecutive year pair:
4  prev_prevalence = dict(zip(mlb_classes_t, skill_matrix_t.
5                             mean(axis=0)))
6  curr_prevalence = dict(zip(mlb_classes_t1, skill_matrix_t1.
7                             mean(axis=0)))
8
9  # Align to common vocabulary (union)
10 all_skills = sorted(set(prev_prevalence) | set(
11                      curr_prevalence))
12 v1 = np.array([prev_prevalence.get(s, 0) for s in all_skills
13                ])
14 v2 = np.array([curr_prevalence.get(s, 0) for s in all_skills
15                ])
16
17 drift = cosine_dist(v1, v2)  # scipy cosine distance

```

G.4.2 Greedy Centroid Matching

Clusters are matched across consecutive years using greedy centroid-based matching with a cosine similarity threshold of $\tau = 0.80$:

```
1 THRESHOLD = 0.80
2
3 # Build common skill vocabulary (union, not intersection)
4 common_skills = sorted(set(skills_t) | set(skills_t1))
5 skill_to_idx = {s: i for i, s in enumerate(common_skills)}
6
7 def align_centroid(centroid, old_skills):
8     """Align centroid to common skill vocabulary."""
9     cent = np.asarray(centroid).flatten()
10    aligned = np.zeros(len(common_skills))
11    for j, skill in enumerate(old_skills):
12        aligned[skill_to_idx[skill]] = cent[j]
13    return aligned
14
15 # Greedy matching
16 matched_t1 = set()
17 for cid_t, cent_t in centroids_t.items():
18     ct = align_centroid(cent_t, skills_t)
19     best_sim, best_match = -1, None
20
21     for cid_t1, cent_t1 in centroids_t1.items():
22         if cid_t1 in matched_t1:
23             continue
24         ct1 = align_centroid(cent_t1, skills_t1)
25         norm_a, norm_b = np.linalg.norm(ct), np.linalg.norm(
26             ct1)
27         sim = np.dot(ct, ct1) / (norm_a * norm_b) if norm_a
28             > 0 and norm_b > 0 else 0
29         if sim > best_sim:
30             best_sim, best_match = sim, cid_t1
31
32     if best_sim > THRESHOLD and best_match is not None:
33         matched_t1.add(best_match)
```

Unmatched clusters from year t are classified as *disappeared*; unmatched clusters from year $t + 1$ are *emerged*. Churn rate = (emerged + disappeared) / $n_t \times 100$.

G.4.3 Evolutionary Classification

Occupations are classified into three evolutionary categories based on centroid drift terciles (computed on clean, unflagged occupations):

- **Stable Specialisation** (drift $\leq$ P33): minimal skill content change
- **Adaptive Reconfiguration** (P33 < drift $\leq$ P67): moderate restructuring
- **Creative Destruction** (drift > P67): fundamental skill redefinition

G.5 Chapter 8: Skill Proximity Network Methodology

Chapter 8 constructs a cross-occupation skill proximity network from HDBSCAN cluster centroids. The production scripts are `analysis/scripts/chapter8_skill_network.py`, `chapter8_network_analysis.py`, and `chapter8_transition_pathways.py`.

G.5.1 Network Construction

Each node represents one cluster from the 2019 cross-section (249 occupations with non-empty clusters). All cluster centroids are aligned to a global skill vocabulary (union of all occupation-specific vocabularies). Edges connect clusters with cosine similarity ≥ 0.6 .

G.5.2 Transition Pathway Skill Gaps

For workers in at-risk clusters, skill gaps to stable target clusters are computed as:

```

1 def compute_skill_gap(src_centroid, tgt_centroid,
2                       src_skills, tgt_skills,
3                       target_threshold=0.10,
4                       source_floor=0.03):
5     """Identify skills a worker needs to acquire."""
6     src_dict = {s: v for s, v in zip(src_skills,
7                                     src_centroid)}
8     tgt_dict = {s: v for s, v in zip(tgt_skills,
9                                     tgt_centroid)}
9     gap = []

```

```
10     for skill, tgt_prev in tgt_dict.items():
11         if tgt_prev < target_threshold:
12             continue # not substantively present in target
13         if src_dict.get(skill, 0.0) < source_floor:
14             gap.append((skill, tgt_prev)) # absent in
15                                     source
16
17     gap.sort(key=lambda x: -x[1])
18     return gap
```

The `target_threshold` of 0.10 means a skill must be requested in at least 10% of a cluster’s job advertisements to be considered substantively present. The `source_floor` of 0.03 means a skill is considered “absent” from the source cluster if requested in fewer than 3% of its advertisements.

Bridge skills are further classified as *generic* (appearing in ≥ 100 occupations, approximately 40% of the sample) or *specialist* (occupation-specific). This distinction separates genuine skill barriers from ubiquitous competencies that workers likely already possess.

G.6 Computing Considerations

G.6.1 Memory Requirements

- **Loading pre-computed models:** 16–32 GB RAM (all 250 occupations)
- **Cosine similarity matrix (Chapter 8):** ~100 MB for 5,000+ cluster centroids
- **Silhouette computation:** Subsampled to 10,000 jobs maximum (cosine distance matrix scales as $O(n^2)$)

G.6.2 Runtime Estimates

With pre-computed HDBSCAN models on a standard laptop (Intel i7, 16 GB RAM):

- Chapter 5 RCM regression: ~5 minutes
- Chapter 6 full-scale validation (250 occupations): ~2 hours
- Chapter 7 evolutionary metrics (250 occupations): ~30 minutes
- Chapter 8 network construction: ~15 minutes

- Chapter 8 transition pathways: ~10 minutes

Re-computing HDBSCAN clustering from raw data for all 250 occupations across 8 years would require 8–12 hours and 32–64 GB RAM.

G.7 Citation

If using this code or methodology, please cite:

Bierly, T. (2026). *The Hidden Restructuring: A Historical Review of Automation and Sub-Occupational Labour Market Dynamics* [Doctoral dissertation, RMIT University].